Table S1. Phantom construction checked against the ICRP Publication 145 and 156 reference values, all twelve phantoms. Masses are the mesh volumes times the assigned densities; red marrow by the Annex H mass-weighted method.

Phantom	Total mass	ICRP	Red marrow	ICRP
AM	73.0 kg	73 kg	1169 g	1170 g
AF	60.0 kg	60 kg	899 g	900 g
00M	3.5 kg	3.5 kg	50.4 g	50 g
00F	3.5 kg	3.5 kg	50.4 g	50 g
01M	10.0 kg	10 kg	156 g	155.5 g
01F	10.0 kg	10 kg	156 g	155.5 g
05M	19.0 kg	19 kg	377 g	374.9 g
05F	19.0 kg	19 kg	377 g	374.9 g
10M	32.0 kg	32 kg	757 g	754.1 g
10F	32.0 kg	32 kg	757 g	754.1 g
15M	56.0 kg	56 kg	1053 g	1053.5 g
15F	53.0 kg	53 kg	1050 g	1049.1 g

Table S2. Per-case agreement between FLUKA and each reference code. Organs is the number of organs where both codes report a statistical uncertainty of 10% or better; Median is the median FLUKA/reference dose ratio over those organs; Within 5% is how many of them have a ratio between 0.95 and 1.05; Within 1σ is how many differ from the reference by less than the combined statistical uncertainty of the two calculations.

Case	Reference	Organs	Median	Within 5%	Within 1σ
AM Internal	Geant4	139	1.0021	122	89
	MCNP6	138	1.0006	118	90
	PHITS	138	1.0025	122	90
AM External	Geant4	98	0.9925	65	69
	MCNP6	97	0.9948	65	68
	PHITS	97	0.9928	65	68
AF Internal	Geant4	144	1.0029	119	91
	MCNP6	141	0.9996	120	85
	PHITS	141	0.9994	121	84
AF External	Geant4	101	0.9999	61	67
	MCNP6	101	1.0062	61	68
	PHITS	100	0.9959	59	68
00M Internal	Geant4	136	0.9996	105	88
	MCNP6	136	0.9989	101	83
	PHITS	137	1.0000	104	92
00M External	Geant4	13	0.9881	5	10
	MCNP6	11	1.0401	5	6
	PHITS	15	0.9627	8	9
00F Internal	MCNP6	136	0.9956	113	90
	PHITS	132	0.9985	112	93
00F External	MCNP6	14	1.0158	5	8
	PHITS	15	1.0362	6	10
01M Internal	MCNP6	129	0.9991	103	93
	PHITS	129	0.9973	102	81
01M External	MCNP6	23	0.9904	12	17
	PHITS	21	1.0029	12	16
01F Internal	MCNP6	130	1.0000	95	82
	PHITS	131	0.9973	106	89
01F External	MCNP6	24	1.0019	13	17
	PHITS	22	1.0373	11	18

Case	Reference	Organs	Median	Within 5%	Within 1σ
05M Internal	MCNP6	128	0.9973	100	79
	PHITS	128	0.9969	100	76
05M External	MCNP6	44	0.9911	14	28
	PHITS	45	0.9988	23	31
05F Internal	MCNP6	129	0.9964	107	86
	PHITS	128	0.9973	103	80
05F External	MCNP6	47	1.0114	19	27
	PHITS	44	0.9885	21	29
10M Internal	MCNP6	133	0.9986	105	92
	PHITS	132	0.9995	103	83
10M External	MCNP6	66	0.9940	32	43
	PHITS	66	0.9926	35	41
10F Internal	MCNP6	132	0.9988	101	82
	PHITS	131	0.9973	98	78
10F External	MCNP6	65	1.0074	35	45
	PHITS	64	1.0028	35	45
15M Internal	MCNP6	125	0.9976	101	78
	PHITS	126	0.9955	101	75
15M External	MCNP6	75	0.9995	43	49
	PHITS	76	1.0156	42	51
15F Internal	MCNP6	128	0.9965	108	82
	PHITS	128	0.9962	117	89
15F External	MCNP6	87	0.9991	40	52
	PHITS	85	1.0054	47	64

Table S3. Adult male, internal exposure (uniform liver source). Absorbed dose per primary history, Gy. σ is the FLUKA statistical uncertainty. Ratios are FLUKA / reference.

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
100	Adrenal left	8.68	1.6096×10^{-15}	2.35	1.5010×10^{-15}	1.4943×10^{-15}	1.4514×10^{-15}	1.0724	1.0772	1.1090
200	Adrenal right	8.68	7.1194×10^{-15}	1.29	6.9800×10^{-15}	6.9341×10^{-15}	6.9990×10^{-15}	1.0200	1.0267	1.0172
300	ET1(0-8)	0.0223	6.5496×10^{-17}	34.90	5.4740×10^{-17}	5.4261×10^{-17}	2.3028×10^{-16}	1.1965	1.2071	0.2844
301	ET1(8-40)	0.0895	1.2118×10^{-16}	23.94	1.0840×10^{-16}	4.6291×10^{-17}	1.6950×10^{-16}	1.1179	2.6177	0.7149
302	ET1(40-50)	0.0281	9.7429×10^{-17}	26.34	8.0300×10^{-17}	6.1021×10^{-17}	9.1044×10^{-17}	1.2133	1.5966	1.0701
303	ET1(50-Surface)	11.29	1.1527×10^{-16}	10.70	1.0930×10^{-16}	9.9985×10^{-17}	1.0278×10^{-16}	1.0546	1.1528	1.1215
400	ET2(-15-0)	0.141	1.3283×10^{-16}	22.16	1.5790×10^{-16}	1.4892×10^{-16}	1.6047×10^{-16}	0.8412	0.8919	0.8277
401	ET2(0-40)	0.39	1.2473×10^{-16}	12.31	1.5580×10^{-16}	1.3367×10^{-16}	1.6621×10^{-16}	0.8006	0.9331	0.7504
402	ET2(40-50)	0.0978	1.0685×10^{-16}	18.71	2.0230×10^{-16}	1.2720×10^{-16}	1.6192×10^{-16}	0.5282	0.8400	0.6599
403	ET2(50-55)	0.049	1.4124×10^{-16}	49.46	1.5340×10^{-16}	1.2379×10^{-16}	2.8693×10^{-16}	0.9207	1.1409	0.4922
404	ET2(55-65)	0.0981	1.8127×10^{-16}	23.77	1.6090×10^{-16}	1.1111×10^{-16}	1.3459×10^{-16}	1.1266	1.6314	1.3468
405	ET2(65-Surface)	28.81	1.2858×10^{-16}	2.72	1.2610×10^{-16}	1.3156×10^{-16}	1.3021×10^{-16}	1.0197	0.9774	0.9875
500	Oral mucosa tongue	0.086	1.5205×10^{-16}	18.60	1.5100×10^{-16}	1.7837×10^{-16}	1.0301×10^{-16}	1.0069	0.8524	1.4760
501	Oral mucosa mouth floor	0.0235	4.0523×10^{-17}	17.98	1.3370×10^{-16}	8.2160×10^{-17}	1.4438×10^{-16}	0.3031	0.4932	0.2807
600	Oral mucosa lips and cheeks	0.0231	1.3152×10^{-16}	44.46	1.8500×10^{-16}	8.4059×10^{-17}	2.2841×10^{-16}	0.7109	1.5646	0.5758
700	Trachea	10.36	5.1781×10^{-16}	4.12	5.4070×10^{-16}	5.2765×10^{-16}	5.3186×10^{-16}	0.9577	0.9813	0.9736
800	BB(-11–6)	0.0252	1.2695×10^{-15}	16.04	1.1320×10^{-15}	1.3743×10^{-15}	1.5033×10^{-15}	1.1215	0.9237	0.8445
801	BB(-6-0)	0.0312	1.4022×10^{-15}	19.21	1.1750×10^{-15}	1.0727×10^{-15}	1.2890×10^{-15}	1.1933	1.3072	1.0878
802	BB(0-10)	0.052	1.2647×10^{-15}	12.25	9.6070×10^{-16}	1.3752×10^{-15}	1.5450×10^{-15}	1.3165	0.9197	0.8186
803	BB(10-35)	0.13	1.1736×10^{-15}	8.92	1.0210×10^{-15}	1.3112×10^{-15}	1.2447×10^{-15}	1.1495	0.8950	0.9429
804	BB(35-40)	0.0262	1.1709×10^{-15}	13.57	1.0240×10^{-15}	1.3359×10^{-15}	1.0863×10^{-15}	1.1435	0.8765	1.0779
805	BB(40-50)	0.0524	1.4380×10^{-15}	13.35	1.1090×10^{-15}	1.2578×10^{-15}	1.3736×10^{-15}	1.2967	1.1433	1.0469
806	BB(50-60)	0.0525	1.3062×10^{-15}	11.47	1.0610×10^{-15}	1.1524×10^{-15}	1.0133×10^{-15}	1.2311	1.1335	1.2891
807	BB(60-70)	0.0526	1.4994×10^{-15}	12.79	1.0310×10^{-15}	1.2712×10^{-15}	1.1727×10^{-15}	1.4543	1.1795	1.2786
808	BB(70-surface)	2.78	1.2748×10^{-15}	4.33	1.2370×10^{-15}	1.1636×10^{-15}	1.1936×10^{-15}	1.0306	1.0956	1.0680
900	Blood in large arteries head	1.50	2.3461×10^{-16}	11.06	1.8250×10^{-16}	1.7956×10^{-16}	2.0094×10^{-16}	1.2855	1.3066	1.1676
910	Blood in large veins head	6.94	1.8665×10^{-16}	6.40	2.0490×10^{-16}	2.0091×10^{-16}	1.8904×10^{-16}	0.9109	0.9290	0.9874
1000	Blood in large arteries trunk	193.20	1.4213×10^{-15}	0.33	1.4230×10^{-15}	1.4148×10^{-15}	1.4114×10^{-15}	0.9988	1.0045	1.0070
1010	Blood in large veins trunk	444.04	9.5526×10^{-16}	0.41	9.4920×10^{-16}	9.5213×10^{-16}	9.4963×10^{-16}	1.0064	1.0033	1.0059
1100	Blood in large arteries arms	32.46	5.8111×10^{-16}	1.35	6.0900×10^{-16}	6.0140×10^{-16}	5.9622×10^{-16}	0.9542	0.9663	0.9747
1110	Blood in large veins arms	167.27	4.7546×10^{-16}	1.17	4.6800×10^{-16}	4.7410×10^{-16}	4.7003×10^{-16}	1.0159	1.0029	1.0116
1200	Blood in large arteries legs	108.83	5.4561×10^{-18}	13.29	4.8590×10^{-18}	5.9281×10^{-18}	4.9311×10^{-18}	1.1229	0.9204	1.1065
1210	Blood in large veins legs	389.74	6.0021×10^{-18}	6.32	5.4580×10^{-18}	5.7212×10^{-18}	5.8207×10^{-18}	1.0997	1.0491	1.0312
1300	Humeri upper cortical	159.20	4.0444×10^{-16}	1.32	4.1220×10^{-16}	3.9971×10^{-16}	3.9935×10^{-16}	0.9812	1.0118	1.0127
1400	Humeri upper spogiosa	145.69	3.1155×10^{-16}	1.46	3.1250×10^{-16}	3.0231×10^{-16}	3.0696×10^{-16}	0.9969	1.0305	1.0149

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
1500	Humeri upper medullary cavity	34.24	5.7979×10^{-16}	1.52	5.6420×10^{-16}	5.6598×10^{-16}	5.5727×10^{-16}	1.0276	1.0244	1.0404
1600	Humeri lower cortical	106.72	6.5839×10^{-16}	0.78	6.5800×10^{-16}	6.5654×10^{-16}	6.5442×10^{-16}	1.0006	1.0028	1.0061
1700	Humeri lower spongiosa	50.89	5.8406×10^{-16}	1.00	5.7970×10^{-16}	5.8906×10^{-16}	5.7645×10^{-16}	1.0075	0.9915	1.0132
1800	Humeri lower medullary cavity	37.40	6.9578×10^{-16}	1.57	7.1040×10^{-16}	7.1218×10^{-16}	7.1526×10^{-16}	0.9794	0.9770	0.9728
1900	Ulnae and radii cotical	273.41	2.0328×10^{-16}	0.93	2.0400×10^{-16}	2.0437×10^{-16}	2.0280×10^{-16}	0.9965	0.9947	1.0024
2000	Ulnae and radii spongiosa	154.98	2.9666×10^{-16}	0.96	2.9310×10^{-16}	2.9233×10^{-16}	2.9505×10^{-16}	1.0121	1.0148	1.0055
2100	Ulnae and radii medullary cavity	23.00	1.5247×10^{-16}	4.46	1.5530×10^{-16}	1.6112×10^{-16}	1.5846×10^{-16}	0.9818	0.9463	0.9622
2200	Wrists and hand bones cortical	181.53	3.0551×10^{-17}	2.36	3.1460×10^{-17}	3.1331×10^{-17}	3.2140×10^{-17}	0.9711	0.9751	0.9506
2300	Wrists and hand bones spongiosa	118.93	3.5884×10^{-17}	3.98	2.9840×10^{-17}	3.1458×10^{-17}	3.3683×10^{-17}	1.2025	1.1407	1.0653
2400	Clavicles cortical	48.25	4.1751×10^{-16}	1.83	4.2720×10^{-16}	4.2182×10^{-16}	4.1536×10^{-16}	0.9773	0.9898	1.0052
2500	Clavicles spongiosa	45.06	4.2709×10^{-16}	2.36	4.0370×10^{-16}	4.1070×10^{-16}	4.0969×10^{-16}	1.0579	1.0399	1.0425
2600	Cranium cortical	568.47	5.7950×10^{-17}	1.03	5.5960×10^{-17}	5.6755×10^{-17}	5.5898×10^{-17}	1.0356	1.0211	1.0367
2700	Cranium spongiosa	382.07	5.7866×10^{-17}	1.72	5.7840×10^{-17}	5.7949×10^{-17}	5.5367×10^{-17}	1.0005	0.9986	1.0451
2800	Femora upper cortical	253.55	3.6749×10^{-17}	1.72	3.7920×10^{-17}	3.9095×10^{-17}	3.8569×10^{-17}	0.9691	0.9400	0.9528
2900	Femora upper spongiosa	413.23	5.3978×10^{-17}	1.33	5.4690×10^{-17}	5.4392×10^{-17}	5.3505×10^{-17}	0.9870	0.9924	1.0088
3000	Femora upper medullary cavity	26.05	9.0350×10^{-18}	11.30	1.6230×10^{-17}	1.5748×10^{-17}	1.1168×10^{-17}	0.5567	0.5737	0.8090
3100	Femora lower cortical	307.76	4.5243×10^{-18}	6.19	4.3780×10^{-18}	4.3755×10^{-18}	4.4764×10^{-18}	1.0334	1.0340	1.0107
3200	Femora lower spongiosa	373.65	1.9357×10^{-18}	8.18	2.0380×10^{-18}	1.7701×10^{-18}	2.0068×10^{-18}	0.9498	1.0935	0.9646
3300	Femora lower medullary cavity	82.18	6.9354×10^{-18}	9.56	5.2010×10^{-18}	5.5415×10^{-18}	6.7632×10^{-18}	1.3335	1.2515	1.0255
3400	Tibiae fibulae and patellae cortical	536.65	5.4614×10^{-19}	13.63	6.0220×10^{-19}	5.6098×10^{-19}	6.1329×10^{-19}	0.9069	0.9735	0.8905
3500	Tibiae fibulae and patellae spongiosa	621.41	8.9380×10^{-19}	9.06	8.1520×10^{-19}	6.5314×10^{-19}	8.7312×10^{-19}	1.0964	1.3685	1.0237
3600	Tibiae fibulae and patellae medullary ca	79.81	3.3191×10^{-19}	33.73	1.6990×10^{-19}	1.6237×10^{-19}	6.0636×10^{-20}	1.9536	2.0442	5.4739
3700	Ankles and foot cortical	234.88	5.5296×10^{-19}	23.79	1.0140×10^{-18}	8.2332×10^{-19}	1.3176×10^{-18}	0.5453	0.6716	0.4197
3800	Ankles and foot spongiosa	432.62	6.8980×10^{-19}	16.77	5.3570×10^{-19}	6.2033×10^{-19}	6.0361×10^{-19}	1.2877	1.1120	1.1428
3900	Mandible cortical	76.88	1.5366×10^{-16}	1.31	1.5200×10^{-16}	1.5741×10^{-16}	1.5694×10^{-16}	1.0109	0.9762	0.9791
4000	Mandible spongiosa	56.29	1.7292×10^{-16}	2.87	1.6730×10^{-16}	1.7857×10^{-16}	1.6525×10^{-16}	1.0336	0.9684	1.0464
4100	Pelvis cortical	402.59	1.8435×10^{-16}	0.64	1.8540×10^{-16}	1.8513×10^{-16}	1.8487×10^{-16}	0.9944	0.9958	0.9972
4200	Pelvis spongiosa	619.67	1.6929×10^{-16}	0.50	1.6850×10^{-16}	1.7131×10^{-16}	1.7008×10^{-16}	1.0047	0.9882	0.9954
4300	Ribs cortical	368.80	1.9686×10^{-15}	0.24	1.9690×10^{-15}	1.9693×10^{-15}	1.9594×10^{-15}	0.9998	0.9997	1.0047
4400	Ribs spongiosa	457.35	1.9957×10^{-15}	0.25	2.0020×10^{-15}	2.0088×10^{-15}	2.0099×10^{-15}	0.9969	0.9935	0.9930
4500	Scapulae cortical	223.33	4.9941×10^{-16}	0.57	5.0460×10^{-16}	5.0512×10^{-16}	5.0091×10^{-16}	0.9897	0.9887	0.9970
4600	Scapulae spongiosa	156.67	4.4362×10^{-16}	0.86	4.5050×10^{-16}	4.5105×10^{-16}	4.4475×10^{-16}	0.9847	0.9835	0.9975
4700	Cervical spine cortical	103.94	1.8572×10^{-16}	1.55	1.9290×10^{-16}	1.8272×10^{-16}	1.9081×10^{-16}	0.9628	1.0164	0.9733
4800	Cervical spine spongiosa	78.92	1.8858×10^{-16}	2.11	1.8580×10^{-16}	1.8999×10^{-16}	1.8669×10^{-16}	1.0150	0.9926	1.0101
4900	Thoracic spine cortical	289.44	1.5943×10^{-15}	0.51	1.5910×10^{-15}	1.5918×10^{-15}	1.5942×10^{-15}	1.0021	1.0016	1.0001
5000	Thoracic spine spongiosa	345.22	2.0722×10^{-15}	0.27	2.0650×10^{-15}	2.0666×10^{-15}	2.0693×10^{-15}	1.0035	1.0027	1.0014
5100	Lumbar spine cortical	188.05	1.0444×10^{-15}	0.55	1.0500×10^{-15}	1.0527×10^{-15}	1.0471×10^{-15}	0.9947	0.9921	0.9975

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
5200	Lumbar spine spongiosa	291.58	1.1732×10^{-15}	0.71	1.1790×10^{-15}	1.1818×10^{-15}	1.1657×10^{-15}	0.9951	0.9927	1.0064
5300	Sacrum cortical	110.32	2.4670×10^{-16}	1.37	2.4810×10^{-16}	2.5421×10^{-16}	2.4752×10^{-16}	0.9944	0.9705	0.9967
5400	Sacrum spongiosa	192.22	2.5491×10^{-16}	1.11	2.6220×10^{-16}	2.6749×10^{-16}	2.6085×10^{-16}	0.9722	0.9530	0.9772
5500	Sternum cortical	9.99	1.6916×10^{-15}	1.77	1.7170×10^{-15}	1.7125×10^{-15}	1.7119×10^{-15}	0.9852	0.9878	0.9881
5600	Sternum spongiosa	61.42	1.7058×10^{-15}	0.70	1.7250×10^{-15}	1.7341×10^{-15}	1.7208×10^{-15}	0.9889	0.9837	0.9913
5700	Cartilage costal	56.33	3.0774×10^{-15}	0.68	3.0150×10^{-15}	3.0236×10^{-15}	2.9930×10^{-15}	1.0207	1.0178	1.0282
5800	Cartilage discs	82.06	1.6405×10^{-15}	0.74	1.6450×10^{-15}	1.6600×10^{-15}	1.6434×10^{-15}	0.9973	0.9883	0.9982
6100	Brain	1517.39	4.8440×10^{-17}	1.05	4.7740×10^{-17}	4.9604×10^{-17}	4.8408×10^{-17}	1.0147	0.9765	1.0007
6200	Breast left adipose tissue	7.77	8.2448×10^{-16}	4.43	8.3200×10^{-16}	8.1834×10^{-16}	8.1665×10^{-16}	0.9910	1.0075	1.0096
6300	Breast left glandular tissue	5.18	6.9372×10^{-16}	2.00	7.1170×10^{-16}	6.7854×10^{-16}	7.2356×10^{-16}	0.9747	1.0224	0.9588
6400	Breast right adipose tissue	7.77	2.6501×10^{-15}	1.93	2.9160×10^{-15}	2.8025×10^{-15}	2.7605×10^{-15}	0.9088	0.9456	0.9600
6500	Breast right glandular tissue	5.18	2.4512×10^{-15}	1.73	2.4610×10^{-15}	2.5070×10^{-15}	2.5622×10^{-15}	0.9960	0.9778	0.9567
6600	Eye lens sensitive left	0.0389	4.7784×10^{-17}	86.91	6.0170×10^{-17}	1.1877×10^{-17}	5.2957×10^{-17}	0.7941	4.0232	0.9023
6601	Eye lens insensitive left	0.189	5.2277×10^{-17}	75.49	3.7090×10^{-17}	1.0937×10^{-16}	1.1874×10^{-17}	1.4095	0.4780	4.4026
6700	Cornea left	1.11	1.0287×10^{-16}	15.80	6.4090×10^{-17}	6.7903×10^{-17}	4.5523×10^{-17}	1.6051	1.5149	2.2597
6701	Aqueous left	0.308	7.3048×10^{-17}	50.16	4.0290×10^{-17}	6.3538×10^{-17}	2.7145×10^{-17}	1.8130	1.1497	2.6910
6702	Vitreous left	6.12	7.9236×10^{-17}	12.90	6.5490×10^{-17}	6.3704×10^{-17}	5.8032×10^{-17}	1.2099	1.2438	1.3654
6800	Eye lens sensitive right	0.0389	0.0000×10^{0}	0.00	7.9880×10^{-17}	2.4669×10^{-16}	1.6551×10^{-16}	–	–	–
6801	Eye lens insensitive right	0.189	3.4792×10^{-17}	89.94	6.9380×10^{-17}	6.8350×10^{-17}	9.3092×10^{-17}	0.5015	0.5090	0.3737
6900	Cornea right	1.11	9.1436×10^{-17}	24.29	1.1040×10^{-16}	8.6937×10^{-17}	9.0863×10^{-17}	0.8282	1.0517	1.0063
6901	Aqueous right	0.308	7.5215×10^{-17}	55.53	7.9190×10^{-17}	1.3808×10^{-16}	5.0158×10^{-17}	0.9498	0.5447	1.4996
6902	Vitreous right	6.12	1.0434×10^{-16}	13.03	9.4980×10^{-17}	9.3508×10^{-17}	9.1489×10^{-17}	1.0985	1.1158	1.1404
7000	Gall bladder wall	10.36	9.4476×10^{-15}	0.75	9.4870×10^{-15}	9.5037×10^{-15}	9.5601×10^{-15}	0.9958	0.9941	0.9882
7100	Gall bladder contents	58.00	9.1927×10^{-15}	0.33	9.1660×10^{-15}	9.1206×10^{-15}	9.1339×10^{-15}	1.0029	1.0079	1.0064
7200	Stomach wall(0-60)	1.78	3.1506×10^{-15}	0.76	3.0010×10^{-15}	3.0277×10^{-15}	2.9446×10^{-15}	1.0499	1.0406	1.0700
7201	Stomach wall(60-100)	1.19	3.1127×10^{-15}	1.29	3.0360×10^{-15}	3.0676×10^{-15}	2.9976×10^{-15}	1.0253	1.0147	1.0384
7202	Stomach wall(100-300)	6.01	3.0663×10^{-15}	1.03	3.0050×10^{-15}	3.0647×10^{-15}	3.0223×10^{-15}	1.0204	1.0005	1.0146
7203	Stomach wall(300-surface)	185.29	3.3257×10^{-15}	0.40	3.3030×10^{-15}	3.2986×10^{-15}	3.2751×10^{-15}	1.0069	1.0082	1.0154
7300	Stomach contents	250.00	2.6791×10^{-15}	0.26	2.6660×10^{-15}	2.6757×10^{-15}	2.6642×10^{-15}	1.0049	1.0013	1.0056
7400	Small intestine wall(0-130)	14.55	9.2192×10^{-16}	1.62	9.1160×10^{-16}	9.2355×10^{-16}	9.1728×10^{-16}	1.0113	0.9982	1.0051
7401	Small intestine wall(130-150)	2.26	9.3776×10^{-16}	2.75	9.0760×10^{-16}	9.6281×10^{-16}	9.3855×10^{-16}	1.0332	0.9740	0.9992
7402	Small intestine wall(150-200)	5.69	9.1102×10^{-16}	1.85	8.8180×10^{-16}	9.3355×10^{-16}	9.0773×10^{-16}	1.0331	0.9759	1.0036
7403	Small intestine wall(200-surface)	840.10	9.2817×10^{-16}	0.26	9.2400×10^{-16}	9.3066×10^{-16}	9.2681×10^{-16}	1.0045	0.9973	1.0015
7500	Small intestine contents(-500-0)	53.34	9.2599×10^{-16}	0.43	9.1810×10^{-16}	9.2476×10^{-16}	9.2187×10^{-16}	1.0086	1.0013	1.0045
7501	Small intestine contents(centre–500)	296.66	9.1681×10^{-16}	0.37	9.0360×10^{-16}	9.1193×10^{-16}	9.0964×10^{-16}	1.0146	1.0054	1.0079
7600	Ascending colon wall(0-280)	3.07	1.7869×10^{-15}	2.55	1.7660×10^{-15}	1.7760×10^{-15}	1.7723×10^{-15}	1.0118	1.0061	1.0082
7601	Ascending colon wall(280-300)	0.223	1.8410×10^{-15}	4.40	1.7840×10^{-15}	1.7767×10^{-15}	1.8265×10^{-15}	1.0320	1.0362	1.0080
7602	Ascending colon wall(300-surface)	116.63	1.7862×10^{-15}	0.40	1.7730×10^{-15}	1.7666×10^{-15}	1.7787×10^{-15}	1.0074	1.0111	1.0042

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
7700	Ascending colon content	55.00	1.6452×10^{-15}	0.98	1.6250×10^{-15}	1.6571×10^{-15}	1.6343×10^{-15}	1.0125	0.9929	1.0067
7800	Transverse colon wall right(0-280)	3.99	4.0899×10^{-15}	1.63	4.1500×10^{-15}	4.0802×10^{-15}	4.0470×10^{-15}	0.9855	1.0024	1.0106
7801	Transverse colon wall right(280-300)	0.289	4.2865×10^{-15}	2.71	4.2770×10^{-15}	4.3682×10^{-15}	4.0398×10^{-15}	1.0022	0.9813	1.0611
7802	Transverse colon wall right(300-surface)	75.67	4.1060×10^{-15}	0.46	4.0510×10^{-15}	4.0661×10^{-15}	4.1004×10^{-15}	1.0136	1.0098	1.0014
7900	Transverse colon contents right	95.00	3.9412×10^{-15}	0.56	3.9520×10^{-15}	3.9586×10^{-15}	3.9746×10^{-15}	0.9973	0.9956	0.9916
8000	Transverse colon wall left(0-280)	2.82	1.7098×10^{-15}	1.92	1.7110×10^{-15}	1.6504×10^{-15}	1.6132×10^{-15}	0.9993	1.0360	1.0599
8001	Transverse colon wall left(280-300)	0.205	1.7212×10^{-15}	4.03	1.6240×10^{-15}	1.6699×10^{-15}	1.6607×10^{-15}	1.0598	1.0307	1.0364
8002	Transverse colon wall left(300-surface)	76.92	1.5792×10^{-15}	0.94	1.6000×10^{-15}	1.5780×10^{-15}	1.6061×10^{-15}	0.9870	1.0007	0.9833
8100	Transverse colon content left	40.00	1.6904×10^{-15}	0.81	1.6810×10^{-15}	1.6874×10^{-15}	1.7240×10^{-15}	1.0056	1.0018	0.9805
8200	Descending colon wall(0-280)	2.78	7.5887×10^{-16}	4.52	7.4270×10^{-16}	8.2221×10^{-16}	7.5257×10^{-16}	1.0218	0.9230	1.0084
8201	Descending colon wall(280-300)	0.203	8.4984×10^{-16}	7.27	7.2600×10^{-16}	8.1481×10^{-16}	8.6219×10^{-16}	1.1706	1.0430	0.9857
8202	Descending colon wall(300-surface)	116.95	7.3220×10^{-16}	0.57	7.2890×10^{-16}	7.3371×10^{-16}	7.3576×10^{-16}	1.0045	0.9979	0.9952
8300	Descending colon content	35.00	8.4375×10^{-16}	1.31	8.4100×10^{-16}	8.5311×10^{-16}	8.2407×10^{-16}	1.0033	0.9890	1.0239
8400	Sigmoid colon wall(0-280)	4.45	1.6839×10^{-16}	6.78	1.6150×10^{-16}	1.6351×10^{-16}	1.6654×10^{-16}	1.0427	1.0299	1.0111
8401	Sigmoid colon wall(280-300)	0.324	1.8994×10^{-16}	10.15	1.3270×10^{-16}	1.9111×10^{-16}	1.4891×10^{-16}	1.4314	0.9939	1.2756
8402	Sigmoid colon wall(300-surface)	48.52	1.7617×10^{-16}	3.85	1.6580×10^{-16}	1.7130×10^{-16}	1.7201×10^{-16}	1.0626	1.0285	1.0242
8500	Sigmoid colon contents	75.00	1.6657×10^{-16}	2.37	1.7690×10^{-16}	1.7158×10^{-16}	1.7001×10^{-16}	0.9416	0.9708	0.9798
8600	Rectum wall	39.98	1.0422×10^{-16}	4.31	1.0630×10^{-16}	1.0308×10^{-16}	1.0676×10^{-16}	0.9804	1.0110	0.9762
8700	Heart wall	385.84	2.3932×10^{-15}	0.30	2.3940×10^{-15}	2.3965×10^{-15}	2.3877×10^{-15}	0.9997	0.9986	1.0023
8800	Blood in heart chamber	510.00	2.3393×10^{-15}	0.37	2.3400×10^{-15}	2.3351×10^{-15}	2.3353×10^{-15}	0.9997	1.0018	1.0017
8900	Kidney left cortex	162.34	1.0346×10^{-15}	0.69	1.0290×10^{-15}	1.0208×10^{-15}	1.0245×10^{-15}	1.0054	1.0135	1.0098
9000	Kidney left medulla	38.36	1.0740×10^{-15}	1.35	1.0680×10^{-15}	1.0854×10^{-15}	1.0636×10^{-15}	1.0056	0.9895	1.0098
9100	Kidney left pelvis	7.65	1.2441×10^{-15}	2.01	1.2100×10^{-15}	1.1840×10^{-15}	1.1750×10^{-15}	1.0282	1.0508	1.0588
9200	Kidney right cortex	166.54	3.9361×10^{-15}	0.48	3.9030×10^{-15}	3.9184×10^{-15}	3.9137×10^{-15}	1.0085	1.0045	1.0057
9300	Kidney right medulla	39.36	4.3191×10^{-15}	0.86	4.3350×10^{-15}	4.2986×10^{-15}	4.3564×10^{-15}	0.9963	1.0048	0.9914
9400	Kidney right pelvis	7.89	4.4979×10^{-15}	1.49	4.4440×10^{-15}	4.4193×10^{-15}	4.3252×10^{-15}	1.0121	1.0178	1.0399
9500	Liver	2360.00	1.1300×10^{-14}	0.05	1.1290×10^{-14}	1.1292×10^{-14}	1.1305×10^{-14}	1.0008	1.0006	0.9995
9700	Lung(AI) left	545.88	8.1240×10^{-16}	0.37	8.1290×10^{-16}	8.0575×10^{-16}	8.0442×10^{-16}	0.9994	1.0082	1.0099
9900	Lung(AI) right	652.86	2.7689×10^{-15}	0.19	2.7720×10^{-15}	2.7759×10^{-15}	2.7750×10^{-15}	0.9989	0.9975	0.9978
10000	Lymphatic nodes ET	15.95	9.7601×10^{-17}	5.89	1.0150×10^{-16}	9.9531×10^{-17}	1.0217×10^{-16}	0.9616	0.9806	0.9553
10100	Lymphatic nodes thoracic	15.95	2.5415×10^{-15}	1.56	2.4830×10^{-15}	2.4902×10^{-15}	2.5170×10^{-15}	1.0235	1.0206	1.0097
10200	Lymphatic nodes head	5.51	2.1129×10^{-16}	6.90	2.3110×10^{-16}	2.1516×10^{-16}	2.4397×10^{-16}	0.9143	0.9820	0.8660
10300	Lymphatic nodes trunk	130.20	4.5987×10^{-16}	0.51	4.7620×10^{-16}	4.6252×10^{-16}	4.6025×10^{-16}	0.9657	0.9943	0.9992
10400	Lymphatic nodes arms	11.02	7.0143×10^{-16}	3.41	7.2820×10^{-16}	6.9155×10^{-16}	6.8041×10^{-16}	0.9632	1.0143	1.0309
10500	Lymphatic nodes legs	11.02	1.3428×10^{-18}	35.18	2.3130×10^{-18}	1.9458×10^{-18}	2.3439×10^{-18}	0.5805	0.6901	0.5729
10600	Muscle head	1200.83	1.4781×10^{-16}	0.36	1.4720×10^{-16}	1.4824×10^{-16}	1.4689×10^{-16}	1.0042	0.9971	1.0063

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
10700	Muscle trunk	14844.73	5.6634×10^{-16}	0.09	5.6600×10^{-16}	5.6608×10^{-16}	5.6565×10^{-16}	1.0006	1.0005	1.0012
10800	Muscle arms	2843.51	3.4685×10^{-16}	0.30	3.4850×10^{-16}	3.4496×10^{-16}	3.4706×10^{-16}	0.9953	1.0055	0.9994
10900	Muscle legs	10887.73	6.6578×10^{-18}	0.89	6.6940×10^{-18}	6.6259×10^{-18}	6.6412×10^{-18}	0.9946	1.0048	1.0025
11000	Oesophagus wall(0-190)	1.92	1.9938×10^{-15}	2.40	2.0850×10^{-15}	1.9470×10^{-15}	2.0112×10^{-15}	0.9562	1.0240	0.9913
11001	Oesophagus wall(190-200)	0.103	2.2561×10^{-15}	6.11	2.1570×10^{-15}	1.9902×10^{-15}	2.1087×10^{-15}	1.0459	1.1336	1.0699
11002	Oesophagus wall(200-surface)	49.78	2.0441×10^{-15}	0.64	2.0450×10^{-15}	2.0327×10^{-15}	2.0305×10^{-15}	0.9996	1.0056	1.0067
11003	Oesophagus contents	22.87	1.9514×10^{-15}	1.52	1.9720×10^{-15}	1.9739×10^{-15}	1.9647×10^{-15}	0.9895	0.9886	0.9932
11300	Pancreas	173.63	4.5426×10^{-15}	0.34	4.5710×10^{-15}	4.5488×10^{-15}	4.5469×10^{-15}	0.9938	0.9986	0.9990
11400	Pituitary gland	0.622	1.1187×10^{-16}	36.17	1.9640×10^{-17}	1.0385×10^{-16}	6.8843×10^{-17}	5.6960	1.0772	1.6250
11500	Prostate	17.62	8.0114×10^{-17}	6.54	7.6900×10^{-17}	8.9019×10^{-17}	8.8024×10^{-17}	1.0418	0.9000	0.9101
11600	RST head	975.62	1.1349×10^{-16}	1.02	1.1490×10^{-16}	1.1553×10^{-16}	1.1457×10^{-16}	0.9878	0.9824	0.9906
11700	RST trunk	11176.90	9.3074×10^{-16}	0.07	9.2720×10^{-16}	9.2609×10^{-16}	9.3036×10^{-16}	1.0038	1.0050	1.0004
11800	RST arms	1549.84	4.3703×10^{-16}	0.46	4.3690×10^{-16}	4.3604×10^{-16}	4.3650×10^{-16}	1.0003	1.0023	1.0012
11900	RST legs	4510.13	7.2501×10^{-18}	1.07	7.3460×10^{-18}	7.3681×10^{-18}	7.3681×10^{-18}	0.9870	0.9840	0.9840
12000	Salivary glands left	44.05	1.0648×10^{-16}	5.78	1.0380×10^{-16}	1.6309×10^{-16}	1.5512×10^{-16}	1.0258	0.6529	0.6865
12100	Salivary glandss right	44.05	1.6372×10^{-16}	1.93	1.6890×10^{-16}	1.0675×10^{-16}	9.8078×10^{-17}	0.9693	1.5336	1.6692
12200	Skin head insensitive	259.23	1.0192×10^{-16}	1.98	9.9180×10^{-17}	1.0181×10^{-16}	1.0065×10^{-16}	1.0276	1.0011	1.0126
12201	Skin head sensitive(50-100)	8.47	9.1906×10^{-17}	4.39	8.5020×10^{-17}	8.2755×10^{-17}	9.1738×10^{-17}	1.0810	1.1106	1.0018
12300	Skin trunk insensitive	1271.01	6.1789×10^{-16}	0.30	6.1340×10^{-16}	6.1882×10^{-16}	6.1635×10^{-16}	1.0073	0.9985	1.0025
12301	Skin trunk sensitive(50-100)	38.42	5.6552×10^{-16}	0.81	5.4850×10^{-16}	5.5849×10^{-16}	5.5046×10^{-16}	1.0310	1.0126	1.0274
12400	Skin arms insensitive	575.71	3.0676×10^{-16}	0.44	3.0650×10^{-16}	3.0556×10^{-16}	3.0962×10^{-16}	1.0009	1.0039	0.9908
12401	Skin arms sensitive(50-100)	18.84	2.7989×10^{-16}	1.51	2.7920×10^{-16}	2.8323×10^{-16}	2.7927×10^{-16}	1.0025	0.9882	1.0022
12500	Skin legs insensitive	1259.98	5.7253×10^{-18}	2.57	6.0490×10^{-18}	6.1146×10^{-18}	6.0462×10^{-18}	0.9465	0.9363	0.9469
12501	Skin legs sensitive(50-100)	37.79	5.3964×10^{-18}	10.33	4.2620×10^{-18}	4.3506×10^{-18}	4.2147×10^{-18}	1.2662	1.2404	1.2804
12600	Spinal cord	37.95	1.2238×10^{-15}	1.25	1.2100×10^{-15}	1.1831×10^{-15}	1.2040×10^{-15}	1.0114	1.0344	1.0165
12700	Spleen	228.40	1.0391×10^{-15}	0.55	1.0360×10^{-15}	1.0354×10^{-15}	1.0360×10^{-15}	1.0030	1.0035	1.0030
12800	Teeth	50.73	1.2831×10^{-16}	2.43	1.3120×10^{-16}	1.3463×10^{-16}	1.3546×10^{-16}	0.9780	0.9530	0.9472
12801	Teeth retention region	0.0431	9.2282×10^{-17}	15.89	1.0970×10^{-16}	1.6350×10^{-16}	1.6158×10^{-16}	0.8412	0.5644	0.5711
12900	Testis left	18.62	3.0706×10^{-17}	13.59	2.8370×10^{-17}	2.6474×10^{-17}	2.4446×10^{-17}	1.0823	1.1598	1.2561
13000	Testis right	18.62	2.4376×10^{-17}	11.38	2.6870×10^{-17}	2.5431×10^{-17}	2.5546×10^{-17}	0.9072	0.9585	0.9542
13100	Thymus	25.91	6.8359×10^{-16}	1.67	6.8800×10^{-16}	6.8751×10^{-16}	7.0004×10^{-16}	0.9936	0.9943	0.9765
13200	Thyroid	23.35	3.9869×10^{-16}	2.81	3.8570×10^{-16}	3.8558×10^{-16}	3.7863×10^{-16}	1.0337	1.0340	1.0530
13300	Tongue upper(food)	20.99	1.2830×10^{-16}	4.78	1.1110×10^{-16}	1.1724×10^{-16}	1.3493×10^{-16}	1.1548	1.0943	0.9508
13301	Tongue lower	54.55	1.5463×10^{-16}	3.06	1.4350×10^{-16}	1.4602×10^{-16}	1.4770×10^{-16}	1.0775	1.0589	1.0469
13400	Tonsils	3.11	7.6767×10^{-17}	13.65	7.7160×10^{-17}	7.9500×10^{-17}	8.0546×10^{-17}	0.9949	0.9656	0.9531
13500	Ureter left	8.81	7.8048×10^{-16}	3.34	7.9940×10^{-16}	7.8208×10^{-16}	8.1024×10^{-16}	0.9763	0.9980	0.9633
13600	Ureter right	7.77	1.0084×10^{-15}	3.03	1.0150×10^{-15}	1.0096×10^{-15}	9.9452×10^{-16}	0.9935	0.9988	1.0140
13700	Urinary bladder wall insensitive	49.78	1.4739×10^{-16}	2.84	1.4710×10^{-16}	1.3656×10^{-16}	1.3692×10^{-16}	1.0020	1.0793*	1.0765*

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
13701	Urinary bladder wall sensitive(75-118)	1.32	1.4287×10^{-16}	11.21	1.5770×10^{-16}	1.3664×10^{-16}	1.6268×10^{-16}	0.9060	1.0456*	0.8783*
13800	Urinary bladder content	200.00	1.3458×10^{-16}	1.48	1.3990×10^{-16}	1.4382×10^{-16}	1.4105×10^{-16}	0.9619	0.9357	0.9541
14000	Air inside body	0.14	2.3201×10^{-16}	13.72	2.3820×10^{-16}	3.4279×10^{-16}	3.2932×10^{-16}	0.9740	0.6768	0.7045

* The urinary bladder wall partition was revised between the 2018-dated MCNP6 and PHITS reference calculations and the currently distributed phantom, which the 2020-dated Geant4 files match. The marked ratio therefore compares two different partitions of the same wall, and is left out of the summary statistics of Table S2.

Table S4. Adult male, external exposure (isotropic point source, AP). Absorbed dose per primary history, Gy. σ is the FLUKA statistical uncertainty. Ratios are FLUKA / reference.

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
100	Adrenal left	8.68	2.1818×10^{-17}	19.79	1.8230×10^{-17}	1.9800×10^{-17}	1.9506×10^{-17}	1.1968	1.1019	1.1185
200	Adrenal right	8.68	1.7980×10^{-17}	17.45	2.2260×10^{-17}	2.1070×10^{-17}	1.8710×10^{-17}	0.8077	0.8533	0.9610
300	ET1(0-8)	0.0223	7.6724×10^{-18}	78.84	1.2120×10^{-17}	1.1893×10^{-17}	5.1894×10^{-17}	0.6330	0.6451	0.1478
301	ET1(8-40)	0.0895	3.0378×10^{-17}	62.97	1.2980×10^{-17}	1.8086×10^{-17}	2.0581×10^{-17}	2.3403	1.6797	1.4760
302	ET1(40-50)	0.0281	1.2580×10^{-17}	57.47	1.6130×10^{-17}	3.0102×10^{-17}	1.6993×10^{-17}	0.7799	0.4179	0.7403
303	ET1(50-Surface)	11.29	2.6961×10^{-17}	14.47	2.8670×10^{-17}	3.2130×10^{-17}	4.1042×10^{-17}	0.9404	0.8391	0.6569
400	ET2(-15-0)	0.141	4.4237×10^{-17}	47.72	3.1890×10^{-17}	4.6303×10^{-17}	2.3364×10^{-17}	1.3872	0.9554	1.8934
401	ET2(0-40)	0.39	5.2691×10^{-17}	38.33	1.9160×10^{-17}	2.8940×10^{-17}	2.6969×10^{-17}	2.7501	1.8207	1.9538
402	ET2(40-50)	0.0978	5.3065×10^{-17}	52.82	1.7950×10^{-17}	5.1064×10^{-17}	7.3029×10^{-17}	2.9563	1.0392	0.7266
403	ET2(50-55)	0.049	2.5350×10^{-17}	37.37	2.2510×10^{-17}	2.8462×10^{-17}	4.6171×10^{-17}	1.1262	0.8907	0.5490
404	ET2(55-65)	0.0981	3.7013×10^{-17}	47.27	2.3370×10^{-17}	1.9871×10^{-17}	3.7874×10^{-17}	1.5838	1.8626	0.9773
405	ET2(65-Surface)	28.81	2.7022×10^{-17}	9.76	2.2320×10^{-17}	2.1431×10^{-17}	2.7765×10^{-17}	1.2107	1.2609	0.9732
500	Oral mucosa tongue	0.086	1.6954×10^{-17}	49.44	9.4380×10^{-18}	9.8786×10^{-18}	3.7375×10^{-17}	1.7964	1.7163	0.4536
501	Oral mucosa mouth floor	0.0235	4.1184×10^{-17}	81.51	4.2640×10^{-17}	1.4627×10^{-17}	7.4326×10^{-18}	0.9658	2.8157	5.5410
600	Oral mucosa lips and cheeks	0.0231	2.2774×10^{-18}	67.38	3.0640×10^{-17}	4.7147×10^{-17}	8.1115×10^{-18}	0.0743	0.0483	0.2808
700	Trachea	10.36	2.5058×10^{-17}	17.28	2.4250×10^{-17}	1.9930×10^{-17}	2.8613×10^{-17}	1.0333	1.2573	0.8757
800	BB(-11–6)	0.0252	1.2833×10^{-17}	42.87	7.1560×10^{-18}	3.9789×10^{-19}	2.4538×10^{-17}	1.7933	32.2519	0.5230
801	BB(-6-0)	0.0312	3.6761×10^{-17}	50.31	7.9560×10^{-18}	6.8081×10^{-18}	7.5126×10^{-17}	4.6205	5.3996	0.4893
802	BB(0-10)	0.052	6.3068×10^{-17}	41.36	1.6360×10^{-17}	2.3737×10^{-17}	2.1396×10^{-17}	3.8550	2.6570	2.9476
803	BB(10-35)	0.13	3.9751×10^{-17}	54.14	1.9120×10^{-17}	3.3561×10^{-17}	2.0810×10^{-17}	2.0790	1.1844	1.9102
804	BB(35-40)	0.0262	4.1359×10^{-17}	62.18	4.1340×10^{-17}	2.1412×10^{-17}	1.8949×10^{-17}	1.0005	1.9315	2.1826
805	BB(40-50)	0.0524	9.2070×10^{-17}	44.49	4.3200×10^{-17}	3.4064×10^{-17}	3.4846×10^{-17}	2.1312	2.7029	2.6422
806	BB(50-60)	0.0525	1.5250×10^{-17}	35.30	3.7890×10^{-17}	4.3422×10^{-17}	1.7733×10^{-17}	0.4025	0.3512	0.8600
807	BB(60-70)	0.0526	2.2460×10^{-17}	41.27	8.0850×10^{-18}	2.8933×10^{-17}	1.7397×10^{-17}	2.7780	0.7763	1.2910
808	BB(70-surface)	2.78	2.5115×10^{-17}	21.34	2.5500×10^{-17}	1.7085×10^{-17}	3.6051×10^{-17}	0.9849	1.4700	0.6967
900	Blood in large arteries head	1.50	3.0368×10^{-17}	34.65	1.5890×10^{-17}	4.3376×10^{-17}	1.5161×10^{-17}	1.9111	0.7001	2.0030
910	Blood in large veins head	6.94	3.1852×10^{-17}	16.24	2.9470×10^{-17}	2.7704×10^{-17}	2.6063×10^{-17}	1.0808	1.1497	1.2221
1000	Blood in large arteries trunk	193.20	3.0035×10^{-17}	5.40	2.9740×10^{-17}	3.1504×10^{-17}	2.9933×10^{-17}	1.0099	0.9534	1.0034
1010	Blood in large veins trunk	444.04	3.4793×10^{-17}	1.67	3.3100×10^{-17}	3.3832×10^{-17}	3.3672×10^{-17}	1.0511	1.0284	1.0333
1100	Blood in large arteries arms	32.46	3.4542×10^{-17}	7.96	3.2930×10^{-17}	3.4961×10^{-17}	2.9349×10^{-17}	1.0490	0.9880	1.1769
1110	Blood in large veins arms	167.27	2.9611×10^{-17}	4.10	2.8160×10^{-17}	3.0513×10^{-17}	2.7788×10^{-17}	1.0515	0.9704	1.0656
1200	Blood in large arteries legs	108.83	2.6606×10^{-17}	5.29	2.6820×10^{-17}	2.6423×10^{-17}	2.6429×10^{-17}	0.9920	1.0069	1.0067
1210	Blood in large veins legs	389.74	2.8027×10^{-17}	2.12	2.7650×10^{-17}	2.8350×10^{-17}	2.8229×10^{-17}	1.0136	0.9886	0.9928
1300	Humeri upper cortical	159.20	2.7555×10^{-17}	6.24	2.7840×10^{-17}	2.6306×10^{-17}	2.5730×10^{-17}	0.9898	1.0475	1.0709
1400	Humeri upper spogiosa	145.69	2.5926×10^{-17}	3.86	2.6830×10^{-17}	2.8561×10^{-17}	2.6409×10^{-17}	0.9663	0.9078	0.9817

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
1500	Humeri upper medullary cavity	34.24	2.5505×10^{-17}	9.36	2.6050×10^{-17}	2.8113×10^{-17}	2.8220×10^{-17}	0.9791	0.9072	0.9038
1600	Humeri lower cortical	106.72	2.6250×10^{-17}	5.39	2.8870×10^{-17}	2.6784×10^{-17}	2.7117×10^{-17}	0.9092	0.9801	0.9680
1700	Humeri lower spongiosa	50.89	2.6485×10^{-17}	7.10	2.6820×10^{-17}	2.8095×10^{-17}	2.7482×10^{-17}	0.9875	0.9427	0.9637
1800	Humeri lower medullary cavity	37.40	2.7372×10^{-17}	12.57	2.6290×10^{-17}	3.1662×10^{-17}	2.8300×10^{-17}	1.0412	0.8645	0.9672
1900	Ulnae and radii cotical	273.41	2.8431×10^{-17}	2.99	2.7680×10^{-17}	2.9512×10^{-17}	2.8171×10^{-17}	1.0271	0.9634	1.0092
2000	Ulnae and radii spongiosa	154.98	3.0301×10^{-17}	3.07	3.1020×10^{-17}	2.8820×10^{-17}	3.0667×10^{-17}	0.9768	1.0514	0.9881
2100	Ulnae and radii medullary cavity	23.00	2.5731×10^{-17}	5.72	2.8050×10^{-17}	3.0648×10^{-17}	2.7797×10^{-17}	0.9173	0.8395	0.9257
2200	Wrists and hand bones cortical	181.53	2.9412×10^{-17}	2.92	2.6800×10^{-17}	2.8447×10^{-17}	2.7601×10^{-17}	1.0974	1.0339	1.0656
2300	Wrists and hand bones spongiosa	118.93	2.9621×10^{-17}	2.54	2.9830×10^{-17}	3.1859×10^{-17}	2.9803×10^{-17}	0.9930	0.9298	0.9939
2400	Clavicles cortical	48.25	2.7245×10^{-17}	8.71	2.7470×10^{-17}	2.9026×10^{-17}	2.8320×10^{-17}	0.9918	0.9386	0.9620
2500	Clavicles spongiosa	45.06	2.5416×10^{-17}	6.93	2.9740×10^{-17}	2.6864×10^{-17}	2.8629×10^{-17}	0.8546	0.9461	0.8878
2600	Cranium cortical	568.47	1.5676×10^{-17}	2.73	1.5590×10^{-17}	1.6010×10^{-17}	1.5863×10^{-17}	1.0055	0.9792	0.9882
2700	Cranium spongiosa	382.07	1.5545×10^{-17}	5.51	1.6640×10^{-17}	1.6223×10^{-17}	1.6304×10^{-17}	0.9342	0.9582	0.9535
2800	Femora upper cortical	253.55	3.1408×10^{-17}	3.51	2.8690×10^{-17}	3.1768×10^{-17}	3.0305×10^{-17}	1.0947	0.9887	1.0364
2900	Femora upper spongiosa	413.23	3.1109×10^{-17}	2.86	3.1930×10^{-17}	3.1313×10^{-17}	3.1805×10^{-17}	0.9743	0.9935	0.9781
3000	Femora upper medullary cavity	26.05	2.9402×10^{-17}	9.34	3.2410×10^{-17}	3.9994×10^{-17}	2.9132×10^{-17}	0.9072	0.7352	1.0093
3100	Femora lower cortical	307.76	2.8939×10^{-17}	2.39	3.0120×10^{-17}	3.0203×10^{-17}	3.0912×10^{-17}	0.9608	0.9582	0.9362
3200	Femora lower spongiosa	373.65	2.8638×10^{-17}	3.33	2.9080×10^{-17}	2.8557×10^{-17}	2.9344×10^{-17}	0.9848	1.0028	0.9759
3300	Femora lower medullary cavity	82.18	3.1745×10^{-17}	6.21	3.5330×10^{-17}	3.5074×10^{-17}	3.3153×10^{-17}	0.8985	0.9051	0.9575
3400	Tibiae fibulae and patellae cortical	536.65	2.3456×10^{-17}	2.45	2.4920×10^{-17}	2.4577×10^{-17}	2.4637×10^{-17}	0.9412	0.9544	0.9520
3500	Tibiae fibulae and patellae spongiosa	621.41	2.6209×10^{-17}	2.26	2.4620×10^{-17}	2.5826×10^{-17}	2.5525×10^{-17}	1.0645	1.0148	1.0268
3600	Tibiae fibulae and patellae medullary ca	79.81	2.8471×10^{-17}	6.33	2.6370×10^{-17}	2.7808×10^{-17}	2.8547×10^{-17}	1.0797	1.0238	0.9973
3700	Ankles and foot cortical	234.88	1.9301×10^{-17}	4.72	1.9690×10^{-17}	1.9235×10^{-17}	1.9802×10^{-17}	0.9802	1.0034	0.9747
3800	Ankles and foot spongiosa	432.62	1.9149×10^{-17}	2.74	1.8790×10^{-17}	1.8760×10^{-17}	2.0070×10^{-17}	1.0191	1.0207	0.9541
3900	Mandible cortical	76.88	2.6290×10^{-17}	6.21	2.2290×10^{-17}	2.5516×10^{-17}	2.4982×10^{-17}	1.1795	1.0303	1.0524
4000	Mandible spongiosa	56.29	2.8315×10^{-17}	5.50	2.6430×10^{-17}	2.5636×10^{-17}	2.5944×10^{-17}	1.0713	1.1045	1.0914
4100	Pelvis cortical	402.59	3.0673×10^{-17}	2.67	3.1790×10^{-17}	3.0167×10^{-17}	3.1289×10^{-17}	0.9649	1.0168	0.9803
4200	Pelvis spongiosa	619.67	3.2549×10^{-17}	2.64	3.3110×10^{-17}	3.1558×10^{-17}	3.0252×10^{-17}	0.9830	1.0314	1.0759
4300	Ribs cortical	368.80	2.1629×10^{-17}	3.24	2.3070×10^{-17}	2.0571×10^{-17}	2.2765×10^{-17}	0.9375	1.0514	0.9501
4400	Ribs spongiosa	457.35	2.2137×10^{-17}	2.28	2.2780×10^{-17}	2.2227×10^{-17}	2.2402×10^{-17}	0.9718	0.9960	0.9882
4500	Scapulae cortical	223.33	1.7488×10^{-17}	4.98	1.7730×10^{-17}	1.7876×10^{-17}	1.6973×10^{-17}	0.9863	0.9783	1.0303
4600	Scapulae spongiosa	156.67	1.8938×10^{-17}	5.25	1.8810×10^{-17}	1.9327×10^{-17}	1.8158×10^{-17}	1.0068	0.9799	1.0429
4700	Cervical spine cortical	103.94	2.1058×10^{-17}	4.20	2.2620×10^{-17}	2.0449×10^{-17}	2.1287×10^{-17}	0.9310	1.0298	0.9893
4800	Cervical spine spongiosa	78.92	2.3614×10^{-17}	6.89	2.4850×10^{-17}	2.2754×10^{-17}	2.1443×10^{-17}	0.9503	1.0378	1.1013
4900	Thoracic spine cortical	289.44	1.6307×10^{-17}	3.57	1.5490×10^{-17}	1.6729×10^{-17}	1.5876×10^{-17}	1.0527	0.9748	1.0271
5000	Thoracic spine spongiosa	345.22	1.8926×10^{-17}	4.52	1.8080×10^{-17}	1.7001×10^{-17}	1.8288×10^{-17}	1.0468	1.1132	1.0349
5100	Lumbar spine cortical	188.05	2.1655×10^{-17}	2.52	2.1240×10^{-17}	2.2619×10^{-17}	2.1972×10^{-17}	1.0195	0.9574	0.9856

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
5200	Lumbar spine spongiosa	291.58	2.3811×10^{-17}	2.32	2.4280×10^{-17}	2.4412×10^{-17}	2.5398×10^{-17}	0.9807	0.9754	0.9375
5300	Sacrum cortical	110.32	2.3578×10^{-17}	7.23	2.1660×10^{-17}	2.3701×10^{-17}	2.1552×10^{-17}	1.0886	0.9948	1.0940
5400	Sacrum spongiosa	192.22	2.4753×10^{-17}	2.96	2.5430×10^{-17}	2.6035×10^{-17}	2.5670×10^{-17}	0.9734	0.9508	0.9643
5500	Sternum cortical	9.99	3.5496×10^{-17}	11.97	3.6760×10^{-17}	3.6339×10^{-17}	3.0852×10^{-17}	0.9656	0.9768	1.1505
5600	Sternum spongiosa	61.42	3.4153×10^{-17}	4.56	3.4500×10^{-17}	3.8728×10^{-17}	3.9548×10^{-17}	0.9899	0.8819	0.8636
5700	Cartilage costal	56.33	4.5586×10^{-17}	5.34	4.0120×10^{-17}	3.6990×10^{-17}	3.6906×10^{-17}	1.1362	1.2324	1.2352
5800	Cartilage discs	82.06	2.5025×10^{-17}	5.38	2.2930×10^{-17}	2.4848×10^{-17}	2.5066×10^{-17}	1.0914	1.0071	0.9984
6100	Brain	1517.39	1.4994×10^{-17}	3.19	1.4530×10^{-17}	1.5195×10^{-17}	1.5478×10^{-17}	1.0319	0.9868	0.9687
6200	Breast left adipose tissue	7.77	3.6705×10^{-17}	16.18	4.3720×10^{-17}	5.0392×10^{-17}	4.0923×10^{-17}	0.8395	0.7284	0.8969
6300	Breast left glandular tissue	5.18	4.5600×10^{-17}	23.21	2.7100×10^{-17}	4.6768×10^{-17}	4.0735×10^{-17}	1.6826	0.9750	1.1194
6400	Breast right adipose tissue	7.77	3.9317×10^{-17}	15.93	2.6130×10^{-17}	3.6038×10^{-17}	3.6629×10^{-17}	1.5047	1.0910	1.0734
6500	Breast right glandular tissue	5.18	2.9242×10^{-17}	24.18	3.5790×10^{-17}	5.3607×10^{-17}	4.0405×10^{-17}	0.8170	0.5455	0.7237
6600	Eye lens sensitive left	0.0389	0.0000×10^{0}	0.00	–	–	–	–	–	–
6601	Eye lens insensitive left	0.189	0.0000×10^{0}	0.00	–	1.9315×10^{-17}	2.4244×10^{-17}	–	–	–
6700	Cornea left	1.11	4.0096×10^{-17}	26.41	3.2140×10^{-17}	1.1666×10^{-17}	1.0733×10^{-17}	1.2475	3.4371	3.7357
6701	Aqueous left	0.308	0.0000×10^{0}	0.00	3.6530×10^{-17}	4.5831×10^{-19}	1.6792×10^{-17}	–	–	–
6702	Vitreous left	6.12	3.9540×10^{-17}	21.28	4.6990×10^{-17}	2.0953×10^{-17}	2.2452×10^{-17}	0.8414	1.8871	1.7611
6800	Eye lens sensitive right	0.0389	1.2443×10^{-16}	100.00	2.5590×10^{-16}	–	2.0242×10^{-17}	0.4862	–	6.1469
6801	Eye lens insensitive right	0.189	6.7108×10^{-17}	100.00	–	–	6.0503×10^{-17}	–	–	1.1092
6900	Cornea right	1.11	1.1034×10^{-17}	53.46	2.1810×10^{-17}	1.7536×10^{-17}	2.5365×10^{-17}	0.5059	0.6292	0.4350
6901	Aqueous right	0.308	2.1659×10^{-17}	74.79	3.2180×10^{-17}	4.2311×10^{-17}	2.1476×10^{-17}	0.6730	0.5119	1.0085
6902	Vitreous right	6.12	3.2039×10^{-17}	10.79	1.7650×10^{-17}	1.9181×10^{-17}	2.2884×10^{-17}	1.8152	1.6703	1.4000
7000	Gall bladder wall	10.36	2.5486×10^{-17}	10.94	3.1040×10^{-17}	3.2986×10^{-17}	2.9569×10^{-17}	0.8211	0.7726	0.8619
7100	Gall bladder contents	58.00	2.9673×10^{-17}	3.96	3.0210×10^{-17}	3.2486×10^{-17}	2.8399×10^{-17}	0.9822	0.9134	1.0449
7200	Stomach wall(0-60)	1.78	3.5148×10^{-17}	19.51	4.0740×10^{-17}	3.0457×10^{-17}	3.5051×10^{-17}	0.8627	1.1540	1.0028
7201	Stomach wall(60-100)	1.19	4.7093×10^{-17}	11.66	3.8880×10^{-17}	3.3259×10^{-17}	3.5665×10^{-17}	1.2112	1.4160	1.3204
7202	Stomach wall(100-300)	6.01	3.4933×10^{-17}	14.66	3.3300×10^{-17}	3.0014×10^{-17}	3.1999×10^{-17}	1.0490	1.1639	1.0917
7203	Stomach wall(300-surface)	185.29	3.1911×10^{-17}	4.33	3.1660×10^{-17}	2.9755×10^{-17}	3.2224×10^{-17}	1.0079	1.0724	0.9903
7300	Stomach contents	250.00	3.1240×10^{-17}	2.93	3.2140×10^{-17}	3.2459×10^{-17}	3.1693×10^{-17}	0.9720	0.9624	0.9857
7400	Small intestine wall(0-130)	14.55	3.4995×10^{-17}	7.92	3.3890×10^{-17}	3.3870×10^{-17}	3.6848×10^{-17}	1.0326	1.0332	0.9497
7401	Small intestine wall(130-150)	2.26	3.1207×10^{-17}	13.75	3.1020×10^{-17}	3.4998×10^{-17}	3.3128×10^{-17}	1.0060	0.8917	0.9420
7402	Small intestine wall(150-200)	5.69	3.6391×10^{-17}	9.75	3.2700×10^{-17}	3.8770×10^{-17}	3.4250×10^{-17}	1.1129	0.9386	1.0625
7403	Small intestine wall(200-surface)	840.10	3.5100×10^{-17}	1.19	3.3880×10^{-17}	3.4480×10^{-17}	3.4742×10^{-17}	1.0360	1.0180	1.0103
7500	Small intestine contents(-500-0)	53.34	3.7169×10^{-17}	5.85	3.4030×10^{-17}	3.7833×10^{-17}	3.8532×10^{-17}	1.0923	0.9825	0.9646
7501	Small intestine contents(centre–500)	296.66	3.5310×10^{-17}	2.37	3.3550×10^{-17}	3.4710×10^{-17}	3.5183×10^{-17}	1.0525	1.0173	1.0036
7600	Ascending colon wall(0-280)	3.07	2.2125×10^{-17}	22.11	3.1520×10^{-17}	3.0605×10^{-17}	3.0748×10^{-17}	0.7019	0.7229	0.7196
7601	Ascending colon wall(280-300)	0.223	1.1173×10^{-17}	63.13	2.8460×10^{-17}	3.0678×10^{-17}	1.7370×10^{-17}	0.3926	0.3642	0.6432
7602	Ascending colon wall(300-surface)	116.63	3.1362×10^{-17}	4.42	3.1910×10^{-17}	3.4795×10^{-17}	3.4129×10^{-17}	0.9828	0.9013	0.9189

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
7700	Ascending colon content	55.00	3.0680×10^{-17}	7.72	2.9060×10^{-17}	3.0369×10^{-17}	2.8839×10^{-17}	1.0558	1.0103	1.0639
7800	Transverse colon wall right(0-280)	3.99	3.9434×10^{-17}	12.34	4.4600×10^{-17}	4.0624×10^{-17}	3.6705×10^{-17}	0.8842	0.9707	1.0744
7801	Transverse colon wall right(280-300)	0.289	5.3927×10^{-17}	20.49	3.2020×10^{-17}	3.8615×10^{-17}	3.9916×10^{-17}	1.6842	1.3965	1.3510
7802	Transverse colon wall right(300-surface)	75.67	3.9978×10^{-17}	5.58	4.0220×10^{-17}	3.6102×10^{-17}	3.4551×10^{-17}	0.9940	1.1073	1.1571
7900	Transverse colon contents right	95.00	3.5607×10^{-17}	3.76	3.5940×10^{-17}	3.8587×10^{-17}	3.6261×10^{-17}	0.9907	0.9228	0.9820
8000	Transverse colon wall left(0-280)	2.82	3.0884×10^{-17}	13.12	4.4900×10^{-17}	3.5407×10^{-17}	4.8360×10^{-17}	0.6878	0.8723	0.6386
8001	Transverse colon wall left(280-300)	0.205	3.8870×10^{-17}	33.98	4.4070×10^{-17}	6.2405×10^{-17}	5.1445×10^{-17}	0.8820	0.6229	0.7556
8002	Transverse colon wall left(300-surface)	76.92	3.7161×10^{-17}	6.42	3.8670×10^{-17}	4.2738×10^{-17}	4.1046×10^{-17}	0.9610	0.8695	0.9053
8100	Transverse colon content left	40.00	4.2456×10^{-17}	9.15	3.8150×10^{-17}	3.8182×10^{-17}	4.2007×10^{-17}	1.1129	1.1120	1.0107
8200	Descending colon wall(0-280)	2.78	3.8212×10^{-17}	15.32	2.2460×10^{-17}	3.3356×10^{-17}	2.2496×10^{-17}	1.7014	1.1456	1.6986
8201	Descending colon wall(280-300)	0.203	3.1667×10^{-17}	32.66	2.3150×10^{-17}	2.7662×10^{-17}	1.6539×10^{-17}	1.3679	1.1448	1.9147
8202	Descending colon wall(300-surface)	116.95	3.0737×10^{-17}	5.53	3.0910×10^{-17}	3.0945×10^{-17}	3.0259×10^{-17}	0.9944	0.9933	1.0158
8300	Descending colon content	35.00	2.9600×10^{-17}	14.54	2.8030×10^{-17}	3.0677×10^{-17}	2.8054×10^{-17}	1.0560	0.9649	1.0551
8400	Sigmoid colon wall(0-280)	4.45	3.4078×10^{-17}	13.78	4.0110×10^{-17}	3.4665×10^{-17}	4.8096×10^{-17}	0.8496	0.9831	0.7085
8401	Sigmoid colon wall(280-300)	0.324	3.7575×10^{-17}	26.04	5.1020×10^{-17}	5.1838×10^{-17}	5.0489×10^{-17}	0.7365	0.7248	0.7442
8402	Sigmoid colon wall(300-surface)	48.52	3.3576×10^{-17}	7.57	3.9790×10^{-17}	3.4724×10^{-17}	3.5152×10^{-17}	0.8438	0.9669	0.9552
8500	Sigmoid colon contents	75.00	3.6133×10^{-17}	7.03	3.3360×10^{-17}	3.2302×10^{-17}	3.5937×10^{-17}	1.0831	1.1186	1.0055
8600	Rectum wall	39.98	2.5322×10^{-17}	10.82	2.3880×10^{-17}	2.2473×10^{-17}	2.5239×10^{-17}	1.0604	1.1267	1.0033
8700	Heart wall	385.84	2.6640×10^{-17}	3.04	2.7600×10^{-17}	2.7027×10^{-17}	2.7131×10^{-17}	0.9652	0.9857	0.9819
8800	Blood in heart chamber	510.00	2.7961×10^{-17}	1.72	2.7170×10^{-17}	2.7301×10^{-17}	2.6298×10^{-17}	1.0291	1.0242	1.0632
8900	Kidney left cortex	162.34	1.9513×10^{-17}	4.67	2.0380×10^{-17}	2.1034×10^{-17}	2.1316×10^{-17}	0.9575	0.9277	0.9154
9000	Kidney left medulla	38.36	2.2544×10^{-17}	11.21	2.1710×10^{-17}	2.3926×10^{-17}	2.2915×10^{-17}	1.0384	0.9423	0.9838
9100	Kidney left pelvis	7.65	2.4615×10^{-17}	19.41	2.1610×10^{-17}	2.1348×10^{-17}	2.1428×10^{-17}	1.1390	1.1530	1.1487
9200	Kidney right cortex	166.54	2.1802×10^{-17}	5.49	2.2840×10^{-17}	2.1076×10^{-17}	2.3557×10^{-17}	0.9546	1.0345	0.9255
9300	Kidney right medulla	39.36	2.0056×10^{-17}	7.63	2.4160×10^{-17}	1.8447×10^{-17}	2.1295×10^{-17}	0.8301	1.0872	0.9418
9400	Kidney right pelvis	7.89	2.3484×10^{-17}	17.46	2.4810×10^{-17}	2.1388×10^{-17}	2.1105×10^{-17}	0.9465	1.0980	1.1127
9500	Liver	2360.00	2.7757×10^{-17}	1.17	2.7880×10^{-17}	2.8149×10^{-17}	2.8352×10^{-17}	0.9956	0.9861	0.9790
9700	Lung(AI) left	545.88	2.3636×10^{-17}	2.16	2.3980×10^{-17}	2.3032×10^{-17}	2.4529×10^{-17}	0.9857	1.0262	0.9636
9900	Lung(AI) right	652.86	2.5517×10^{-17}	2.03	2.4390×10^{-17}	2.5065×10^{-17}	2.5452×10^{-17}	1.0462	1.0180	1.0026
10000	Lymphatic nodes ET	15.95	1.6222×10^{-17}	14.75	2.0460×10^{-17}	1.8312×10^{-17}	2.1334×10^{-17}	0.7929	0.8859	0.7604
10100	Lymphatic nodes thoracic	15.95	2.2323×10^{-17}	12.92	1.6230×10^{-17}	2.4146×10^{-17}	2.1427×10^{-17}	1.3754	0.9245	1.0418
10200	Lymphatic nodes head	5.51	2.5211×10^{-17}	24.16	3.3120×10^{-17}	3.8543×10^{-17}	2.9733×10^{-17}	0.7612	0.6541	0.8479
10300	Lymphatic nodes trunk	130.20	3.8145×10^{-17}	4.02	3.9190×10^{-17}	3.6973×10^{-17}	3.5470×10^{-17}	0.9733	1.0317	1.0754
10400	Lymphatic nodes arms	11.02	3.1438×10^{-17}	12.63	3.9400×10^{-17}	2.1428×10^{-17}	3.6692×10^{-17}	0.7979	1.4671	0.8568
10500	Lymphatic nodes legs	11.02	2.8086×10^{-17}	20.06	2.3220×10^{-17}	2.4797×10^{-17}	2.7470×10^{-17}	1.2096	1.1326	1.0224
10600	Muscle head	1200.83	2.2432×10^{-17}	1.15	2.3210×10^{-17}	2.2975×10^{-17}	2.3045×10^{-17}	0.9665	0.9764	0.9734

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
10700	Muscle trunk	14844.73	2.6531×10^{-17}	0.46	2.6530×10^{-17}	2.6441×10^{-17}	2.6559×10^{-17}	1.0000	1.0034	0.9989
10800	Muscle arms	2843.51	3.0312×10^{-17}	1.11	2.9190×10^{-17}	2.9700×10^{-17}	3.0119×10^{-17}	1.0385	1.0206	1.0064
10900	Muscle legs	10887.73	2.7169×10^{-17}	0.73	2.7390×10^{-17}	2.7229×10^{-17}	2.7049×10^{-17}	0.9919	0.9978	1.0045
11000	Oesophagus wall(0-190)	1.92	2.4268×10^{-17}	26.09	1.8630×10^{-17}	1.5243×10^{-17}	1.7136×10^{-17}	1.3026	1.5921	1.4162
11001	Oesophagus wall(190-200)	0.103	4.0547×10^{-17}	55.57	2.3510×10^{-17}	1.0366×10^{-17}	5.8194×10^{-18}	1.7247	3.9116	6.9675
11002	Oesophagus wall(200-surface)	49.78	1.8626×10^{-17}	11.15	2.2610×10^{-17}	2.5565×10^{-17}	2.2726×10^{-17}	0.8238	0.7286	0.8196
11003	Oesophagus contents	22.87	2.4061×10^{-17}	13.12	2.3230×10^{-17}	1.9022×10^{-17}	2.5929×10^{-17}	1.0358	1.2649	0.9279
11300	Pancreas	173.63	2.9174×10^{-17}	4.09	3.0880×10^{-17}	2.9601×10^{-17}	3.0284×10^{-17}	0.9448	0.9856	0.9633
11400	Pituitary gland	0.622	5.0986×10^{-18}	52.32	2.2960×10^{-18}	8.1912×10^{-18}	1.9681×10^{-17}	2.2206	0.6224	0.2591
11500	Prostate	17.62	2.7380×10^{-17}	15.16	3.2090×10^{-17}	2.8326×10^{-17}	2.8186×10^{-17}	0.8532	0.9666	0.9714
11600	RST head	975.62	2.1477×10^{-17}	1.80	2.1610×10^{-17}	2.1335×10^{-17}	2.1630×10^{-17}	0.9939	1.0067	0.9929
11700	RST trunk	11176.90	2.9279×10^{-17}	0.40	2.9260×10^{-17}	2.9026×10^{-17}	2.8981×10^{-17}	1.0006	1.0087	1.0103
11800	RST arms	1549.84	3.0536×10^{-17}	1.23	2.9700×10^{-17}	3.1135×10^{-17}	3.0089×10^{-17}	1.0281	0.9807	1.0148
11900	RST legs	4510.13	2.8261×10^{-17}	0.70	2.7540×10^{-17}	2.8096×10^{-17}	2.7791×10^{-17}	1.0262	1.0059	1.0169
12000	Salivary glands left	44.05	2.6855×10^{-17}	9.33	2.6080×10^{-17}	2.2932×10^{-17}	2.9503×10^{-17}	1.0297	1.1710	0.9102
12100	Salivary glandss right	44.05	2.2910×10^{-17}	7.56	2.4970×10^{-17}	2.4336×10^{-17}	2.4394×10^{-17}	0.9175	0.9414	0.9392
12200	Skin head insensitive	259.23	1.8267×10^{-17}	3.04	1.8850×10^{-17}	1.9265×10^{-17}	1.9596×10^{-17}	0.9691	0.9482	0.9322
12201	Skin head sensitive(50-100)	8.47	1.1088×10^{-17}	12.81	1.2450×10^{-17}	1.3254×10^{-17}	1.3442×10^{-17}	0.8906	0.8366	0.8249
12300	Skin trunk insensitive	1271.01	2.3032×10^{-17}	1.34	2.3330×10^{-17}	2.3064×10^{-17}	2.4261×10^{-17}	0.9872	0.9986	0.9493
12301	Skin trunk sensitive(50-100)	38.42	1.3599×10^{-17}	4.91	1.4570×10^{-17}	1.5447×10^{-17}	1.4220×10^{-17}	0.9334	0.8804	0.9563
12400	Skin arms insensitive	575.71	2.5965×10^{-17}	2.25	2.5840×10^{-17}	2.6676×10^{-17}	2.4844×10^{-17}	1.0048	0.9734	1.0451
12401	Skin arms sensitive(50-100)	18.84	1.9626×10^{-17}	3.84	1.8800×10^{-17}	1.7094×10^{-17}	1.5407×10^{-17}	1.0440	1.1481	1.2739
12500	Skin legs insensitive	1259.98	2.4212×10^{-17}	1.45	2.4130×10^{-17}	2.3748×10^{-17}	2.3829×10^{-17}	1.0034	1.0196	1.0161
12501	Skin legs sensitive(50-100)	37.79	1.4304×10^{-17}	5.11	1.4730×10^{-17}	1.5505×10^{-17}	1.4349×10^{-17}	0.9710	0.9225	0.9968
12600	Spinal cord	37.95	1.7127×10^{-17}	13.41	2.1690×10^{-17}	1.7017×10^{-17}	1.7609×10^{-17}	0.7896	1.0065	0.9727
12700	Spleen	228.40	1.8595×10^{-17}	3.71	1.8190×10^{-17}	1.8288×10^{-17}	1.8158×10^{-17}	1.0223	1.0168	1.0241
12800	Teeth	50.73	2.0712×10^{-17}	10.54	2.4460×10^{-17}	2.1367×10^{-17}	2.3542×10^{-17}	0.8468	0.9693	0.8798
12801	Teeth retention region	0.0431	1.2794×10^{-17}	61.81	3.3060×10^{-17}	1.2749×10^{-17}	1.7229×10^{-17}	0.3870	1.0036	0.7426
12900	Testis left	18.62	4.0542×10^{-17}	9.94	3.9790×10^{-17}	3.4110×10^{-17}	3.2451×10^{-17}	1.0189	1.1886	1.2493
13000	Testis right	18.62	3.0091×10^{-17}	9.91	3.5810×10^{-17}	3.9556×10^{-17}	4.3907×10^{-17}	0.8403	0.7607	0.6853
13100	Thymus	25.91	3.3035×10^{-17}	11.17	3.1630×10^{-17}	3.4284×10^{-17}	2.9608×10^{-17}	1.0444	0.9636	1.1157
13200	Thyroid	23.35	3.6760×10^{-17}	11.66	2.4010×10^{-17}	3.7146×10^{-17}	2.9398×10^{-17}	1.5310	0.9896	1.2504
13300	Tongue upper(food)	20.99	2.3291×10^{-17}	13.96	2.1920×10^{-17}	2.3976×10^{-17}	2.4930×10^{-17}	1.0625	0.9714	0.9342
13301	Tongue lower	54.55	2.4194×10^{-17}	7.88	2.7710×10^{-17}	2.2318×10^{-17}	2.6282×10^{-17}	0.8731	1.0841	0.9206
13400	Tonsils	3.11	7.2527×10^{-18}	40.97	1.9490×10^{-17}	1.8049×10^{-17}	2.4035×10^{-17}	0.3721	0.4018	0.3018
13500	Ureter left	8.81	2.1632×10^{-17}	17.41	2.8320×10^{-17}	2.4580×10^{-17}	2.7162×10^{-17}	0.7638	0.8801	0.7964
13600	Ureter right	7.77	4.0544×10^{-17}	11.01	3.7690×10^{-17}	3.8766×10^{-17}	3.8641×10^{-17}	1.0757	1.0459	1.0493
13700	Urinary bladder wall insensitive	49.78	3.3659×10^{-17}	3.29	3.5890×10^{-17}	3.3410×10^{-17}	3.7629×10^{-17}	0.9378	1.0075*	0.8945*

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
13701	Urinary bladder wall sensitive(75-118)	1.32	2.8254×10^{-17}	13.61	3.5230×10^{-17}	4.5725×10^{-17}	2.2419×10^{-17}	0.8020	0.6179*	1.2603*
13800	Urinary bladder content	200.00	3.6290×10^{-17}	2.91	3.5750×10^{-17}	3.6164×10^{-17}	3.5196×10^{-17}	1.0151	1.0035	1.0311
14000	Air inside body	0.14	4.2035×10^{-17}	35.50	2.1530×10^{-17}	3.2524×10^{-17}	1.5552×10^{-17}	1.9524	1.2924	2.7029

* The urinary bladder wall partition was revised between the 2018-dated MCNP6 and PHITS reference calculations and the currently distributed phantom, which the 2020-dated Geant4 files match. The marked ratio therefore compares two different partitions of the same wall, and is left out of the summary statistics of Table S2.

Table S5. Adult female, internal exposure (uniform liver source). Absorbed dose per primary history, Gy. σ is the FLUKA statistical uncertainty. Ratios are FLUKA / reference.

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
100	Adrenal left	6.82	5.0544×10^{-15}	1.26	5.1940×10^{-15}	5.1402×10^{-15}	5.2257×10^{-15}	0.9731	0.9833	0.9672
200	Adrenal right	8.65	8.4704×10^{-15}	1.22	8.4690×10^{-15}	8.3974×10^{-15}	8.3304×10^{-15}	1.0002	1.0087	1.0168
300	ET1(0-8)	0.00856	8.2482×10^{-18}	83.84	1.1060×10^{-16}	3.4963×10^{-17}	1.0245×10^{-16}	0.0746	0.2359	0.0805
301	ET1(8-40)	0.0346	3.8346×10^{-17}	60.89	8.7600×10^{-17}	5.4250×10^{-17}	2.3733×10^{-16}	0.4377	0.7068	0.1616
302	ET1(40-50)	0.0109	7.6251×10^{-17}	79.41	1.8480×10^{-16}	1.6772×10^{-16}	1.1230×10^{-16}	0.4126	0.4546	0.6790
303	ET1(50-surface)	4.37	1.3129×10^{-16}	9.22	1.1580×10^{-16}	1.1217×10^{-16}	1.6393×10^{-16}	1.1337	1.1705	0.8009
400	ET2(-15-0)	0.104	2.3070×10^{-16}	22.11	2.0010×10^{-16}	1.3304×10^{-16}	1.8346×10^{-16}	1.1529	1.7341	1.2575
401	ET2(0-40)	0.288	1.6361×10^{-16}	20.17	1.3210×10^{-16}	1.1728×10^{-16}	1.7216×10^{-16}	1.2386	1.3951	0.9504
402	ET2(40-50)	0.0723	1.8495×10^{-16}	18.80	1.1120×10^{-16}	9.7709×10^{-17}	1.6337×10^{-16}	1.6632	1.8928	1.1321
403	ET2(50-55)	0.0362	2.1725×10^{-16}	29.70	1.6000×10^{-16}	1.3028×10^{-16}	8.6234×10^{-17}	1.3578	1.6675	2.5193
404	ET2(55-65)	0.0725	1.5583×10^{-16}	20.83	1.2960×10^{-16}	1.2735×10^{-16}	9.6881×10^{-17}	1.2024	1.2236	1.6085
405	ET2(65-surface)	14.18	1.6231×10^{-16}	5.71	1.6030×10^{-16}	1.4610×10^{-16}	1.6816×10^{-16}	1.0126	1.1109	0.9652
500	Oral mucosa tongue	0.0661	2.1927×10^{-16}	33.01	1.7840×10^{-16}	1.7579×10^{-16}	1.6219×10^{-16}	1.2291	1.2473	1.3519
501	Oral mucosa mouth floor	0.0161	7.6654×10^{-17}	37.65	9.2480×10^{-17}	1.5122×10^{-16}	2.3069×10^{-16}	0.8289	0.5069	0.3323
600	Oral mucosa lips and cheeks	0.0188	1.8113×10^{-16}	57.25	1.2560×10^{-16}	2.5803×10^{-16}	1.0727×10^{-16}	1.4421	0.7019	1.6885
700	Trachea	8.20	5.4650×10^{-16}	3.66	5.8670×10^{-16}	5.5129×10^{-16}	6.0120×10^{-16}	0.9315	0.9913	0.9090
800	BB(-11–6)	0.0102	1.9738×10^{-15}	29.74	1.2770×10^{-15}	1.0573×10^{-15}	1.3012×10^{-15}	1.5457	1.8669	1.5169
801	BB(-6-0)	0.0126	1.0145×10^{-15}	21.75	1.0780×10^{-15}	1.3358×10^{-15}	9.1725×10^{-16}	0.9411	0.7594	1.1060
802	BB(0-10)	0.0211	1.6514×10^{-15}	18.91	1.3650×10^{-15}	1.1616×10^{-15}	7.9968×10^{-16}	1.2098	1.4216	2.0650
803	BB(10-35)	0.053	9.5840×10^{-16}	20.59	1.1800×10^{-15}	1.1744×10^{-15}	9.5204×10^{-16}	0.8122	0.8161	1.0067
804	BB(35-40)	0.0107	1.4880×10^{-15}	18.68	1.2110×10^{-15}	9.7234×10^{-16}	9.1775×10^{-16}	1.2287	1.5303	1.6213
805	BB(40-50)	0.0213	1.0164×10^{-15}	19.91	1.2470×10^{-15}	1.2576×10^{-15}	1.0207×10^{-15}	0.8150	0.8082	0.9957
806	BB(50-60)	0.0214	1.3764×10^{-15}	17.22	1.0590×10^{-15}	1.4500×10^{-15}	9.2857×10^{-16}	1.2997	0.9492	1.4823
807	BB(60-70)	0.0215	1.2473×10^{-15}	21.62	8.0440×10^{-16}	1.2246×10^{-15}	9.9978×10^{-16}	1.5506	1.0185	1.2476
808	BB(70-surface)	1.18	1.2157×10^{-15}	3.12	1.0950×10^{-15}	1.1074×10^{-15}	1.0401×10^{-15}	1.1102	1.0978	1.1688
900	Blood in large arteries head	1.91	2.1396×10^{-16}	15.18	1.9810×10^{-16}	1.9213×10^{-16}	1.9467×10^{-16}	1.0801	1.1137	1.0991
910	Blood in large veins head	3.01	1.4323×10^{-16}	11.49	2.0560×10^{-16}	1.6763×10^{-16}	1.8617×10^{-16}	0.6967	0.8544	0.7694
1000	Blood in large arteries trunk	117.87	1.6239×10^{-15}	0.48	1.6230×10^{-15}	1.6395×10^{-15}	1.6186×10^{-15}	1.0006	0.9905	1.0033
1010	Blood in large veins trunk	239.85	6.1117×10^{-16}	0.66	5.9730×10^{-16}	6.1118×10^{-16}	6.0894×10^{-16}	1.0232	1.0000	1.0037
1100	Blood in large arteries arms	46.31	7.8570×10^{-16}	1.47	7.8180×10^{-16}	7.8018×10^{-16}	7.9338×10^{-16}	1.0050	1.0071	0.9903
1110	Blood in large veins arms	139.53	8.0545×10^{-16}	0.73	8.0720×10^{-16}	8.1055×10^{-16}	8.0062×10^{-16}	0.9978	0.9937	1.0060
1200	Blood in large arteries legs	79.91	1.0596×10^{-17}	5.84	1.0920×10^{-17}	1.3050×10^{-17}	1.1468×10^{-17}	0.9703	0.8120	0.9240
1210	Blood in large veins legs	355.61	1.1000×10^{-17}	4.93	1.0750×10^{-17}	1.1030×10^{-17}	1.0660×10^{-17}	1.0233	0.9973	1.0319
1300	Humeri upper cortical	113.68	4.7888×10^{-16}	1.11	4.8470×10^{-16}	4.6948×10^{-16}	4.7082×10^{-16}	0.9880	1.0200	1.0171
1400	Humeri upper spongiosa	107.72	2.7242×10^{-16}	1.70	2.7390×10^{-16}	2.8219×10^{-16}	2.7512×10^{-16}	0.9946	0.9654	0.9902

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
1500	Humeri upper medullary cavity	20.52	5.5769×10^{-16}	2.01	5.9010×10^{-16}	5.8662×10^{-16}	6.0401×10^{-16}	0.9451	0.9507	0.9233
1600	Humeri lower cortical	103.30	8.4285×10^{-16}	0.48	8.3500×10^{-16}	8.4122×10^{-16}	8.4672×10^{-16}	1.0094	1.0019	0.9954
1700	Humeri lower spongiosa	50.26	6.9400×10^{-16}	1.23	7.0710×10^{-16}	7.1212×10^{-16}	7.2099×10^{-16}	0.9815	0.9746	0.9626
1800	Humeri lower medullary cavity	20.49	9.4567×10^{-16}	1.86	9.4740×10^{-16}	9.8904×10^{-16}	9.9363×10^{-16}	0.9982	0.9562	0.9517
1900	Ulnae and radii cortical	156.71	2.9708×10^{-16}	0.85	2.8760×10^{-16}	2.9516×10^{-16}	2.8865×10^{-16}	1.0330	1.0065	1.0292
2000	Ulnae and radii spongiosa	86.88	3.5796×10^{-16}	1.26	3.4200×10^{-16}	3.5696×10^{-16}	3.6140×10^{-16}	1.0467	1.0028	0.9905
2100	Ulnae and radii medullary cavity	34.07	2.8097×10^{-16}	2.41	2.6650×10^{-16}	2.8299×10^{-16}	2.9490×10^{-16}	1.0543	0.9928	0.9528
2200	Wrists and hand bones cortical	105.13	6.5360×10^{-17}	2.96	6.1620×10^{-17}	5.9713×10^{-17}	6.1541×10^{-17}	1.0607	1.0946	1.0620
2300	Wrists and hand bones spongiosa	69.36	6.4492×10^{-17}	2.69	6.4450×10^{-17}	6.6842×10^{-17}	6.9490×10^{-17}	1.0007	0.9648	0.9281
2400	Clavicles cortical	32.83	4.2928×10^{-16}	2.68	4.2260×10^{-16}	4.3310×10^{-16}	4.2219×10^{-16}	1.0158	0.9912	1.0168
2500	Clavicles spongiosa	38.80	4.5574×10^{-16}	1.20	4.5620×10^{-16}	4.3774×10^{-16}	4.7676×10^{-16}	0.9990	1.0411	0.9559
2600	Cranium cortical	407.67	6.2970×10^{-17}	1.70	6.4980×10^{-17}	6.3619×10^{-17}	6.3010×10^{-17}	0.9691	0.9898	0.9994
2700	Cranium spongiosa	391.31	7.2089×10^{-17}	1.08	7.2680×10^{-17}	7.2910×10^{-17}	7.4205×10^{-17}	0.9919	0.9887	0.9715
2800	Femora upper cortical	244.13	4.7009×10^{-17}	2.34	4.6680×10^{-17}	4.6599×10^{-17}	4.6712×10^{-17}	1.0070	1.0088	1.0064
2900	Femora upper spongiosa	232.80	8.7161×10^{-17}	1.23	8.5280×10^{-17}	8.8742×10^{-17}	8.7985×10^{-17}	1.0221	0.9822	0.9906
3000	Femora upper medullary cavity	39.52	3.2174×10^{-17}	4.68	3.5200×10^{-17}	3.5351×10^{-17}	3.2031×10^{-17}	0.9140	0.9101	1.0045
3100	Femora lower cortical	240.93	9.2370×10^{-18}	5.33	9.6430×10^{-18}	9.5945×10^{-18}	9.2257×10^{-18}	0.9579	0.9627	1.0012
3200	Femora lower spongiosa	166.33	4.3294×10^{-18}	9.94	4.7980×10^{-18}	4.7727×10^{-18}	3.8243×10^{-18}	0.9023	0.9071	1.1321
3300	Femora lower medullary cavity	56.76	1.0182×10^{-17}	7.68	9.3680×10^{-18}	1.1346×10^{-17}	9.4819×10^{-18}	1.0868	0.8974	1.0738
3400	Tibiae fibulae and patellae cortical	544.85	1.2857×10^{-18}	7.87	1.0870×10^{-18}	1.1232×10^{-18}	1.0433×10^{-18}	1.1828	1.1447	1.2324
3500	Tibiae fibulae and patellae spongiosa	558.53	1.5297×10^{-18}	8.91	1.6370×10^{-18}	1.2341×10^{-18}	1.6619×10^{-18}	0.9345	1.2396	0.9205
3600	Tibiae fibulae and patellae medullary ca	88.88	3.2061×10^{-19}	38.91	4.5220×10^{-19}	3.6222×10^{-19}	5.9231×10^{-19}	0.7090	0.8851	0.5413
3700	Ankles and foot cortical	173.48	1.3287×10^{-18}	17.73	9.3240×10^{-19}	9.9360×10^{-19}	1.1958×10^{-18}	1.4250	1.3373	1.1111
3800	Ankles and foot spongiosa	257.45	1.0506×10^{-18}	15.63	7.4190×10^{-19}	8.0873×10^{-19}	4.6956×10^{-19}	1.4160	1.2990	2.2373
3900	Mandible cortical	45.39	1.9738×10^{-16}	2.24	1.9650×10^{-16}	1.8118×10^{-16}	1.9534×10^{-16}	1.0045	1.0894	1.0104
4000	Mandible spongiosa	33.48	2.0267×10^{-16}	2.63	2.2590×10^{-16}	2.0997×10^{-16}	2.0750×10^{-16}	0.8972	0.9652	0.9767
4100	Pelvis cortical	262.46	2.4584×10^{-16}	1.15	2.4050×10^{-16}	2.4538×10^{-16}	2.4660×10^{-16}	1.0222	1.0019	0.9969
4200	Pelvis spongiosa	455.60	2.3470×10^{-16}	0.99	2.3210×10^{-16}	2.4103×10^{-16}	2.3802×10^{-16}	1.0112	0.9737	0.9860
4300	Ribs cortical	164.50	2.3326×10^{-15}	0.39	2.3420×10^{-15}	2.3251×10^{-15}	2.3368×10^{-15}	0.9960	1.0032	0.9982
4400	Ribs spongiosa	277.32	2.3610×10^{-15}	0.31	2.3390×10^{-15}	2.3364×10^{-15}	2.3465×10^{-15}	1.0094	1.0105	1.0062
4500	Scapulae cortical	121.66	4.3254×10^{-16}	1.16	4.3300×10^{-16}	4.3009×10^{-16}	4.2941×10^{-16}	0.9989	1.0057	1.0073
4600	Scapulae spongiosa	96.73	3.8945×10^{-16}	1.18	3.8200×10^{-16}	3.7477×10^{-16}	3.7597×10^{-16}	1.0195	1.0392	1.0358
4700	Cervical spine cortical	71.60	2.0305×10^{-16}	3.38	1.9560×10^{-16}	1.9692×10^{-16}	1.9417×10^{-16}	1.0381	1.0312	1.0458
4800	Cervical spine spongiosa	75.60	2.0549×10^{-16}	2.45	2.1780×10^{-16}	1.9821×10^{-16}	2.0707×10^{-16}	0.9435	1.0367	0.9924
4900	Thoracic spine cortical	205.83	2.2322×10^{-15}	0.46	2.2580×10^{-15}	2.2172×10^{-15}	2.2565×10^{-15}	0.9886	1.0068	0.9892
5000	Thoracic spine spongiosa	271.92	2.5996×10^{-15}	0.26	2.6000×10^{-15}	2.6165×10^{-15}	2.6139×10^{-15}	0.9998	0.9936	0.9945
5100	Lumbar spine cortical	156.17	2.0297×10^{-15}	0.31	2.0240×10^{-15}	2.0330×10^{-15}	2.0425×10^{-15}	1.0028	0.9984	0.9937

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
5200	Lumbar spine spongiosa	264.98	2.0616×10^{-15}	0.25	2.0520×10^{-15}	2.0888×10^{-15}	2.0662×10^{-15}	1.0047	0.9870	0.9978
5300	Sacrum cortical	80.24	3.2929×10^{-16}	1.70	3.3190×10^{-16}	3.3924×10^{-16}	3.3156×10^{-16}	0.9921	0.9707	0.9932
5400	Sacrum spongiosa	154.84	3.4753×10^{-16}	0.93	3.4650×10^{-16}	3.5634×10^{-16}	3.4758×10^{-16}	1.0030	0.9753	0.9998
5500	Sternum cortical	1.69	2.7218×10^{-15}	1.82	2.6560×10^{-15}	2.6605×10^{-15}	2.6740×10^{-15}	1.0248	1.0231	1.0179
5600	Sternum spongiosa	51.35	2.4066×10^{-15}	0.56	2.4190×10^{-15}	2.4170×10^{-15}	2.4304×10^{-15}	0.9949	0.9957	0.9902
5700	Cartilage costal	41.96	4.8083×10^{-15}	0.54	4.7500×10^{-15}	4.8182×10^{-15}	4.7225×10^{-15}	1.0123	0.9979	1.0182
5800	Cartilage discs	69.35	2.5266×10^{-15}	0.74	2.5060×10^{-15}	2.5321×10^{-15}	2.5420×10^{-15}	1.0082	0.9978	0.9939
6100	Brain	1349.57	6.0762×10^{-17}	1.52	6.0260×10^{-17}	6.0110×10^{-17}	6.0256×10^{-17}	1.0083	1.0108	1.0084
6200	Breast left adipose tissue	153.66	1.0507×10^{-15}	0.98	1.0450×10^{-15}	1.0431×10^{-15}	1.0333×10^{-15}	1.0054	1.0073	1.0168
6300	Breast left glandular tissue	102.49	1.0168×10^{-15}	0.82	9.9920×10^{-16}	1.0064×10^{-15}	1.0028×10^{-15}	1.0177	1.0104	1.0140
6400	Breast right adipose tissue	153.66	1.8462×10^{-15}	0.78	1.8720×10^{-15}	1.8638×10^{-15}	1.8720×10^{-15}	0.9862	0.9906	0.9862
6500	Breast right glandular tissue	102.49	2.1006×10^{-15}	0.65	2.0780×10^{-15}	2.1157×10^{-15}	2.1220×10^{-15}	1.0109	0.9928	0.9899
6600	Eye lens sensitive left	0.0389	3.4015×10^{-19}	100.00	3.3110×10^{-16}	1.1362×10^{-16}	9.8166×10^{-17}	0.0010	0.0030	0.0035
6601	Eye lens insensitive left	0.189	5.1933×10^{-17}	47.21	1.9710×10^{-16}	3.9897×10^{-17}	8.9866×10^{-17}	0.2635	1.3017	0.5779
6700	Cornea left	1.10	1.2670×10^{-16}	13.18	9.2320×10^{-17}	1.1090×10^{-16}	1.1690×10^{-16}	1.3724	1.1425	1.0838
6701	Aqueous left	0.304	1.1182×10^{-16}	46.82	2.1700×10^{-16}	2.3201×10^{-16}	7.7318×10^{-17}	0.5153	0.4820	1.4463
6702	Vitreous left	6.05	9.8981×10^{-17}	7.92	9.9700×10^{-17}	1.0484×10^{-16}	9.6328×10^{-17}	0.9928	0.9441	1.0275
6800	Eye lens sensitive right	0.0389	1.7311×10^{-16}	57.33	4.4820×10^{-17}	1.7781×10^{-16}	3.2409×10^{-17}	3.8624	0.9736	5.3415
6801	Eye lens insensitive right	0.189	1.8226×10^{-16}	37.26	2.4860×10^{-16}	1.1394×10^{-16}	9.6377×10^{-17}	0.7331	1.5996	1.8911
6900	Cornea right	1.10	9.9005×10^{-17}	17.77	1.4780×10^{-16}	1.0919×10^{-16}	1.0357×10^{-16}	0.6699	0.9067	0.9559
6901	Aqueous right	0.304	8.8345×10^{-17}	40.02	2.0810×10^{-16}	8.2128×10^{-17}	1.9671×10^{-16}	0.4245	1.0757	0.4491
6902	Vitreous right	6.05	1.3881×10^{-16}	12.43	1.4080×10^{-16}	1.1623×10^{-16}	1.0921×10^{-16}	0.9858	1.1943	1.2710
7000	Gall bladder wall	8.20	1.0152×10^{-14}	0.81	1.0250×10^{-14}	1.0172×10^{-14}	1.0169×10^{-14}	0.9904	0.9980	0.9983
7100	Gall bladder contents	48.00	9.7498×10^{-15}	0.37	9.7230×10^{-15}	9.7345×10^{-15}	9.8108×10^{-15}	1.0028	1.0016	0.9938
7200	Stomach wall(0-60)	1.56	3.6663×10^{-15}	1.70	3.6970×10^{-15}	3.6537×10^{-15}	3.5595×10^{-15}	0.9917	1.0035	1.0300
7201	Stomach wall(60-100)	1.04	3.6836×10^{-15}	2.03	3.6540×10^{-15}	3.7214×10^{-15}	3.5983×10^{-15}	1.0081	0.9898	1.0237
7202	Stomach wall(100-300)	5.26	3.6764×10^{-15}	1.23	3.6690×10^{-15}	3.6662×10^{-15}	3.5657×10^{-15}	1.0020	1.0028	1.0311
7203	Stomach wall(300-surface)	165.01	3.4730×10^{-15}	0.34	3.4780×10^{-15}	3.4897×10^{-15}	3.3496×10^{-15}	0.9986	0.9952	1.0368
7300	Stomach contents	230.00	3.4934×10^{-15}	0.25	3.5200×10^{-15}	3.5114×10^{-15}	3.3965×10^{-15}	0.9924	0.9948	1.0285
7400	Small intestine wall(0-130)	12.34	8.4147×10^{-16}	1.11	8.3090×10^{-16}	8.4896×10^{-16}	8.5673×10^{-16}	1.0127	0.9912	0.9822
7401	Small intestine wall(130-150)	1.92	8.4805×10^{-16}	2.09	8.3600×10^{-16}	9.1084×10^{-16}	8.4101×10^{-16}	1.0144	0.9311	1.0084
7402	Small intestine wall(150-200)	4.83	8.5373×10^{-16}	1.48	8.2380×10^{-16}	8.6102×10^{-16}	8.5871×10^{-16}	1.0363	0.9915	0.9942
7403	Small intestine wall(200-surface)	736.66	8.6656×10^{-16}	0.39	8.6240×10^{-16}	8.6560×10^{-16}	8.5848×10^{-16}	1.0048	1.0011	1.0094
7500	Small intestine contents(-500-0)	45.23	8.4454×10^{-16}	1.01	8.3990×10^{-16}	8.4784×10^{-16}	8.5443×10^{-16}	1.0055	0.9961	0.9884
7501	Small intestine contents(centre–500)	234.77	8.3394×10^{-16}	0.47	8.4570×10^{-16}	8.4220×10^{-16}	8.3436×10^{-16}	0.9861	0.9902	0.9995
7600	Ascending colon wall(0-280)	4.45	1.2621×10^{-15}	1.40	1.2270×10^{-15}	1.2282×10^{-15}	1.2038×10^{-15}	1.0286	1.0276	1.0484
7601	Ascending colon wall(280-300)	0.322	1.3388×10^{-15}	3.66	1.2780×10^{-15}	1.2571×10^{-15}	1.2107×10^{-15}	1.0475	1.0649	1.1058
7602	Ascending colon wall(300-surface)	107.78	1.1325×10^{-15}	0.63	1.1140×10^{-15}	1.1297×10^{-15}	1.1384×10^{-15}	1.0166	1.0025	0.9948

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
7700	Ascending colon content	100.01	1.1106×10^{-15}	0.71	1.1000×10^{-15}	1.0806×10^{-15}	1.1030×10^{-15}	1.0096	1.0278	1.0069
7800	Transverse colon wall right(0-280)	3.68	1.1977×10^{-15}	2.29	1.1410×10^{-15}	1.1088×10^{-15}	1.1701×10^{-15}	1.0497	1.0801	1.0236
7801	Transverse colon wall right(280-300)	0.266	1.2741×10^{-15}	8.65	1.0990×10^{-15}	1.1560×10^{-15}	1.1168×10^{-15}	1.1593	1.1021	1.1408
7802	Transverse colon wall right(300-surface)	64.85	1.1615×10^{-15}	1.25	1.1870×10^{-15}	1.1645×10^{-15}	1.1671×10^{-15}	0.9785	0.9974	0.9952
7900	Transverse colon contents right	59.99	1.0690×10^{-15}	1.02	1.0550×10^{-15}	1.0475×10^{-15}	1.0546×10^{-15}	1.0133	1.0205	1.0137
8000	Transverse colon wall left(0-280)	2.20	6.4418×10^{-16}	3.95	6.8800×10^{-16}	6.6519×10^{-16}	6.5137×10^{-16}	0.9363	0.9684	0.9890
8001	Transverse colon wall left(280-300)	0.16	6.3231×10^{-16}	7.94	6.4770×10^{-16}	7.4963×10^{-16}	4.8849×10^{-16}	0.9762	0.8435	1.2944
8002	Transverse colon wall left(300-surface)	66.43	6.6654×10^{-16}	1.44	6.8120×10^{-16}	6.4840×10^{-16}	6.7642×10^{-16}	0.9785	1.0280	0.9854
8100	Transverse colon content left	30.01	6.4146×10^{-16}	1.38	6.3660×10^{-16}	6.3299×10^{-16}	6.3714×10^{-16}	1.0076	1.0134	1.0068
8200	Descending colon wall(0-280)	3.02	3.7825×10^{-16}	6.17	4.4130×10^{-16}	4.0905×10^{-16}	4.2745×10^{-16}	0.8571	0.9247	0.8849
8201	Descending colon wall(280-300)	0.22	4.6525×10^{-16}	10.70	3.9490×10^{-16}	3.8371×10^{-16}	4.8850×10^{-16}	1.1781	1.2125	0.9524
8202	Descending colon wall(300-surface)	109.32	3.8526×10^{-16}	1.17	3.8190×10^{-16}	3.8745×10^{-16}	3.8167×10^{-16}	1.0088	0.9943	1.0094
8300	Descending colon content	50.00	4.1671×10^{-16}	1.35	4.0450×10^{-16}	4.0594×10^{-16}	4.1744×10^{-16}	1.0302	1.0265	0.9982
8400	Sigmoid colon wall(0-280)	4.22	1.5742×10^{-16}	8.03	1.4800×10^{-16}	1.5489×10^{-16}	1.5036×10^{-16}	1.0637	1.0164	1.0470
8401	Sigmoid colon wall(280-300)	0.306	1.3860×10^{-16}	18.53	1.5180×10^{-16}	1.7408×10^{-16}	1.6360×10^{-16}	0.9130	0.7962	0.8472
8402	Sigmoid colon wall(300-surface)	51.76	1.5308×10^{-16}	1.78	1.6220×10^{-16}	1.5717×10^{-16}	1.5884×10^{-16}	0.9438	0.9740	0.9637
8500	Sigmoid colon contents	79.99	1.5798×10^{-16}	1.88	1.4920×10^{-16}	1.5236×10^{-16}	1.5608×10^{-16}	1.0589	1.0369	1.0122
8600	Rectum wall	31.27	1.0867×10^{-16}	3.32	1.1290×10^{-16}	1.2067×10^{-16}	1.0531×10^{-16}	0.9625	0.9005	1.0319
8700	Heart wall	290.89	2.5369×10^{-15}	0.34	2.5320×10^{-15}	2.5433×10^{-15}	2.5091×10^{-15}	1.0019	0.9975	1.0111
8800	Blood in heart chamber	370.00	2.7146×10^{-15}	0.28	2.7100×10^{-15}	2.7243×10^{-15}	2.7165×10^{-15}	1.0017	0.9964	0.9993
8900	Kidney left cortex	149.09	1.7906×10^{-15}	0.65	1.7850×10^{-15}	1.7756×10^{-15}	1.7653×10^{-15}	1.0031	1.0084	1.0143
9000	Kidney left medulla	37.44	1.6444×10^{-15}	1.02	1.6420×10^{-15}	1.6445×10^{-15}	1.6316×10^{-15}	1.0015	0.9999	1.0078
9100	Kidney left pelvis	7.49	1.8188×10^{-15}	2.93	1.7730×10^{-15}	1.8431×10^{-15}	1.8346×10^{-15}	1.0258	0.9868	0.9914
9200	Kidney right cortex	125.15	4.9734×10^{-15}	0.38	4.9600×10^{-15}	4.9712×10^{-15}	4.9669×10^{-15}	1.0027	1.0004	1.0013
9300	Kidney right medulla	31.44	5.0290×10^{-15}	0.76	5.0120×10^{-15}	5.0272×10^{-15}	5.0502×10^{-15}	1.0034	1.0004	0.9958
9400	Kidney right pelvis	6.29	5.2963×10^{-15}	1.47	5.4070×10^{-15}	5.2446×10^{-15}	5.3779×10^{-15}	0.9795	1.0099	0.9848
9500	Liver	1810.00	1.3359×10^{-14}	0.08	1.3360×10^{-14}	1.3361×10^{-14}	1.3431×10^{-14}	1.0000	0.9999	0.9947
9700	Lung(AI) left	427.26	1.0170×10^{-15}	0.34	1.0150×10^{-15}	1.0111×10^{-15}	1.0078×10^{-15}	1.0019	1.0058	1.0091
9900	Lung(AI) right	522.52	2.9660×10^{-15}	0.29	2.9820×10^{-15}	2.9622×10^{-15}	2.9890×10^{-15}	0.9946	1.0013	0.9923
10000	Lymphatic nodes ET	12.70	1.3029×10^{-16}	6.13	1.2980×10^{-16}	1.3151×10^{-16}	1.3095×10^{-16}	1.0038	0.9907	0.9950
10100	Lymphatic nodes thoracic	12.70	3.2378×10^{-15}	0.62	3.1140×10^{-15}	3.1411×10^{-15}	3.1934×10^{-15}	1.0397	1.0308	1.0139
10200	Lymphatic nodes head	4.39	2.6633×10^{-16}	4.67	2.7410×10^{-16}	2.7303×10^{-16}	2.7255×10^{-16}	0.9717	0.9755	0.9772
10300	Lymphatic nodes trunk	103.64	4.7606×10^{-16}	1.23	4.8460×10^{-16}	4.8876×10^{-16}	4.7958×10^{-16}	0.9824	0.9740	0.9927
10400	Lymphatic nodes arms	8.77	8.7422×10^{-16}	3.35	8.0480×10^{-16}	8.3582×10^{-16}	8.3812×10^{-16}	1.0863	1.0459	1.0431
10500	Lymphatic nodes legs	8.77	3.0283×10^{-18}	38.24	3.3280×10^{-18}	7.3075×10^{-18}	5.9131×10^{-18}	0.9100	0.4144	0.5121
10600	Muscle head	445.02	1.4753×10^{-16}	1.01	1.4300×10^{-16}	1.4927×10^{-16}	1.4663×10^{-16}	1.0317	0.9884	1.0061

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
10700	Muscle trunk	8324.97	6.5443×10^{-16}	0.13	6.5320×10^{-16}	6.5479×10^{-16}	6.5513×10^{-16}	1.0019	0.9994	0.9989
10800	Muscle arms	1479.72	5.0329×10^{-16}	0.28	5.0090×10^{-16}	5.0066×10^{-16}	5.0506×10^{-16}	1.0048	1.0053	0.9965
10900	Muscle legs	7676.72	1.2607×10^{-17}	1.00	1.2750×10^{-17}	1.2885×10^{-17}	1.2823×10^{-17}	0.9888	0.9785	0.9832
11000	Oesophagus wall(0-190)	1.87	2.2055×10^{-15}	3.21	2.2480×10^{-15}	2.2327×10^{-15}	2.2840×10^{-15}	0.9811	0.9878	0.9656
11001	Oesophagus wall(190-200)	0.101	2.2455×10^{-15}	7.26	2.1180×10^{-15}	2.3368×10^{-15}	2.3730×10^{-15}	1.0602	0.9609	0.9463
11002	Oesophagus wall(200-surface)	41.25	2.0963×10^{-15}	0.88	2.1010×10^{-15}	2.0938×10^{-15}	2.1063×10^{-15}	0.9978	1.0012	0.9953
11003	Oesophagus contents	21.24	2.2193×10^{-15}	0.95	2.2730×10^{-15}	2.2823×10^{-15}	2.2487×10^{-15}	0.9764	0.9724	0.9869
11100	Ovary left	6.32	1.0911×10^{-16}	7.40	9.9430×10^{-17}	1.2814×10^{-16}	9.9813×10^{-17}	1.0974	0.8515	1.0932
11200	Ovary right	6.32	1.3912×10^{-16}	6.85	1.6500×10^{-16}	1.3809×10^{-16}	1.4329×10^{-16}	0.8431	1.0074	0.9709
11300	Pancreas	144.55	5.0180×10^{-15}	0.47	5.0190×10^{-15}	5.0274×10^{-15}	5.0479×10^{-15}	0.9998	0.9981	0.9941
11400	Pituitary gland	0.615	8.8293×10^{-17}	45.95	8.4200×10^{-17}	8.7938×10^{-17}	5.1307×10^{-17}	1.0486	1.0040	1.7209
11600	RST head	844.54	1.2329×10^{-16}	0.64	1.2240×10^{-16}	1.2236×10^{-16}	1.2390×10^{-16}	1.0072	1.0075	0.9950
11700	RST trunk	11513.36	9.3660×10^{-16}	0.07	9.3550×10^{-16}	9.3340×10^{-16}	9.3540×10^{-16}	1.0012	1.0034	1.0013
11800	RST arms	2171.59	6.2896×10^{-16}	0.22	6.2670×10^{-16}	6.3078×10^{-16}	6.2862×10^{-16}	1.0036	0.9971	1.0005
11900	RST legs	7795.95	1.1962×10^{-17}	1.21	1.2140×10^{-17}	1.2103×10^{-17}	1.2167×10^{-17}	0.9854	0.9884	0.9832
12000	Salivary gland left	35.88	1.3728×10^{-16}	6.21	1.4110×10^{-16}	1.8226×10^{-16}	1.7631×10^{-16}	0.9729	0.7532	0.7786
12100	Salivary gland right	35.88	1.7863×10^{-16}	2.35	1.7840×10^{-16}	1.3471×10^{-16}	1.3161×10^{-16}	1.0013	1.3260	1.3573
12200	Skin head insensitive	155.58	1.0236×10^{-16}	1.41	9.7250×10^{-17}	1.0034×10^{-16}	1.0001×10^{-16}	1.0526	1.0201	1.0235
12201	Skin head sensitive(50-100)	6.32	8.8133×10^{-17}	7.20	8.6740×10^{-17}	8.3928×10^{-17}	7.4755×10^{-17}	1.0161	1.0501	1.1790
12300	Skin trunk insensitive	871.60	7.8694×10^{-16}	0.20	7.8860×10^{-16}	7.9051×10^{-16}	7.8514×10^{-16}	0.9979	0.9955	1.0023
12301	Skin trunk sensitive(50-100)	32.37	7.3694×10^{-16}	0.68	7.1840×10^{-16}	7.3926×10^{-16}	7.2207×10^{-16}	1.0258	0.9969	1.0206
12400	Skin arms insensitive	380.91	4.7658×10^{-16}	0.52	4.6730×10^{-16}	4.7214×10^{-16}	4.6970×10^{-16}	1.0199	1.0094	1.0147
12401	Skin arms sensitive(50-100)	15.60	4.3747×10^{-16}	1.16	4.1890×10^{-16}	4.2058×10^{-16}	4.1557×10^{-16}	1.0443	1.0402	1.0527
12500	Skin legs insensitive	924.58	9.1851×10^{-18}	1.71	9.2400×10^{-18}	8.9407×10^{-18}	8.8798×10^{-18}	0.9941	1.0273	1.0344
12501	Skin legs sensitive(50-100)	35.02	7.0658×10^{-18}	8.61	6.6900×10^{-18}	7.4071×10^{-18}	7.5042×10^{-18}	1.0562	0.9539	0.9416
12600	Spinal cord	19.10	1.7708×10^{-15}	1.23	1.7440×10^{-15}	1.7297×10^{-15}	1.7339×10^{-15}	1.0154	1.0238	1.0213
12700	Spleen	187.40	1.6219×10^{-15}	0.53	1.6230×10^{-15}	1.6025×10^{-15}	1.5946×10^{-15}	0.9993	1.0121	1.0171
12800	Teeth	40.56	1.6799×10^{-16}	2.63	1.7750×10^{-16}	1.8180×10^{-16}	1.7773×10^{-16}	0.9464	0.9240	0.9452
12801	Teeth retention region	0.0361	2.4720×10^{-16}	26.85	2.2700×10^{-16}	2.2609×10^{-16}	1.9856×10^{-16}	1.0890	1.0934	1.2450
13100	Thymus	20.50	7.8293×10^{-16}	2.25	7.7790×10^{-16}	7.6169×10^{-16}	7.7573×10^{-16}	1.0065	1.0279	1.0093
13200	Thyroid	19.46	4.1564×10^{-16}	2.77	4.0530×10^{-16}	4.2226×10^{-16}	3.8921×10^{-16}	1.0255	0.9843	1.0679
13300	Tongue upper(food)	21.00	1.5820×10^{-16}	3.17	1.6470×10^{-16}	1.7038×10^{-16}	1.6060×10^{-16}	0.9606	0.9285	0.9851
13301	Tongue lower	40.42	2.0085×10^{-16}	2.75	1.9680×10^{-16}	2.0093×10^{-16}	2.0345×10^{-16}	1.0206	0.9996	0.9872
13400	Tonsils	3.08	1.2887×10^{-16}	9.78	1.2430×10^{-16}	1.4859×10^{-16}	1.7231×10^{-16}	1.0368	0.8673	0.7479
13500	Ureter left	7.69	9.7954×10^{-16}	2.61	9.5270×10^{-16}	9.7039×10^{-16}	9.6745×10^{-16}	1.0282	1.0094	1.0125
13600	Ureter right	7.69	1.5666×10^{-15}	2.37	1.5360×10^{-15}	1.4988×10^{-15}	1.5159×10^{-15}	1.0199	1.0453	1.0335
13700	Urinary bladder wall insensitive	39.46	1.4199×10^{-16}	2.94	1.4830×10^{-16}	1.4176×10^{-16}	1.4186×10^{-16}	0.9575	1.0016*	1.0009*

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
13701	Urinary bladder wall sensitive(69-116)	1.35	1.4377×10^{-16}	8.51	1.6730×10^{-16}	1.4316×10^{-16}	1.5429×10^{-16}	0.8593	1.0043*	0.9318*
13800	Urinary bladder content	200.00	1.4563×10^{-16}	1.31	1.4280×10^{-16}	1.4881×10^{-16}	1.4757×10^{-16}	1.0198	0.9787	0.9869
13900	Uterus	81.99	1.3248×10^{-16}	2.21	1.3380×10^{-16}	1.3177×10^{-16}	1.3853×10^{-16}	0.9902	1.0054	0.9564
14000	Air inside body	0.036	2.8020×10^{-16}	22.54	3.6890×10^{-16}	3.5574×10^{-16}	3.0109×10^{-16}	0.7596	0.7877	0.9306

* The urinary bladder wall partition was revised between the 2018-dated MCNP6 and PHITS reference calculations and the currently distributed phantom, which the 2020-dated Geant4 files match. The marked ratio therefore compares two different partitions of the same wall, and is left out of the summary statistics of Table S2.

Table S6. Adult female, external exposure (isotropic point source, AP). Absorbed dose per primary history, Gy. σ is the FLUKA statistical uncertainty. Ratios are FLUKA / reference.

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
100	Adrenal left	6.82	2.5237×10^{-17}	24.76	3.6610×10^{-17}	2.1838×10^{-17}	2.3098×10^{-17}	0.6893	1.1556	1.0926
200	Adrenal right	8.65	2.4303×10^{-17}	11.58	3.5530×10^{-17}	2.4268×10^{-17}	2.1105×10^{-17}	0.6840	1.0014	1.1515
300	ET1(0-8)	0.00856	0.0000×10^{0}	0.00	–	3.1364×10^{-16}	5.9813×10^{-17}	–	–	–
301	ET1(8-40)	0.0346	1.6669×10^{-17}	100.00	–	3.5230×10^{-17}	8.7728×10^{-17}	–	0.4731	0.1900
302	ET1(40-50)	0.0109	1.0526×10^{-17}	100.00	–	3.9480×10^{-17}	6.0162×10^{-17}	–	0.2666	0.1750
303	ET1(50-surface)	4.37	3.6778×10^{-17}	20.82	2.1950×10^{-17}	4.6086×10^{-17}	2.6276×10^{-17}	1.6755	0.7980	1.3997
400	ET2(-15-0)	0.104	5.1764×10^{-18}	52.73	8.4710×10^{-18}	3.9620×10^{-18}	1.5018×10^{-17}	0.6111	1.3065	0.3447
401	ET2(0-40)	0.288	7.1753×10^{-18}	49.96	2.2000×10^{-17}	2.1979×10^{-17}	2.8733×10^{-17}	0.3262	0.3265	0.2497
402	ET2(40-50)	0.0723	8.4871×10^{-18}	55.30	1.4230×10^{-17}	1.2355×10^{-17}	1.2681×10^{-17}	0.5964	0.6870	0.6693
403	ET2(50-55)	0.0362	1.2009×10^{-17}	47.27	1.8180×10^{-17}	1.3833×10^{-17}	1.2952×10^{-17}	0.6605	0.8681	0.9272
404	ET2(55-65)	0.0725	1.8403×10^{-17}	56.52	1.7360×10^{-17}	1.1359×10^{-17}	3.1380×10^{-17}	1.0601	1.6201	0.5865
405	ET2(65-surface)	14.18	2.4256×10^{-17}	16.08	2.1130×10^{-17}	2.0749×10^{-17}	2.3138×10^{-17}	1.1480	1.1690	1.0483
500	Oral mucosa tongue	0.0661	6.2292×10^{-17}	41.70	1.6960×10^{-17}	3.1378×10^{-17}	4.0470×10^{-17}	3.6729	1.9852	1.5392
501	Oral mucosa mouth floor	0.0161	1.8368×10^{-17}	81.13	5.2180×10^{-17}	2.8683×10^{-18}	7.4388×10^{-17}	0.3520	6.4039	0.2469
600	Oral mucosa lips and cheeks	0.0188	4.4347×10^{-18}	67.52	3.3130×10^{-17}	1.4630×10^{-17}	1.4746×10^{-17}	0.1339	0.3031	0.3007
700	Trachea	8.20	3.5329×10^{-17}	13.34	3.2990×10^{-17}	2.9576×10^{-17}	3.1488×10^{-17}	1.0709	1.1945	1.1220
800	BB(-11–6)	0.0102	1.7140×10^{-17}	87.12	1.6360×10^{-17}	2.5578×10^{-17}	3.0163×10^{-18}	1.0477	0.6701	5.6823
801	BB(-6-0)	0.0126	4.7667×10^{-18}	67.07	3.7240×10^{-17}	9.8182×10^{-17}	3.0200×10^{-18}	0.1280	0.0486	1.5784
802	BB(0-10)	0.0211	5.0636×10^{-18}	66.91	1.7880×10^{-17}	1.8229×10^{-17}	7.9544×10^{-17}	0.2832	0.2778	0.0637
803	BB(10-35)	0.053	5.5068×10^{-18}	66.70	5.3300×10^{-17}	1.3387×10^{-17}	7.2570×10^{-18}	0.1033	0.4113	0.7588
804	BB(35-40)	0.0107	6.1982×10^{-18}	68.25	7.9790×10^{-17}	1.1289×10^{-17}	2.9876×10^{-18}	0.0777	0.5491	2.0746
805	BB(40-50)	0.0213	6.1292×10^{-18}	66.95	8.6790×10^{-17}	1.0377×10^{-17}	3.1088×10^{-18}	0.0706	0.5906	1.9716
806	BB(50-60)	0.0214	8.2492×10^{-18}	68.76	5.9240×10^{-17}	8.8851×10^{-18}	8.0429×10^{-17}	0.1392	0.9284	0.1026
807	BB(60-70)	0.0215	8.3186×10^{-18}	59.18	4.7060×10^{-17}	1.7730×10^{-17}	7.0869×10^{-18}	0.1768	0.4692	1.1738
808	BB(70-surface)	1.18	3.4389×10^{-17}	36.94	2.8030×10^{-17}	2.6055×10^{-17}	7.0146×10^{-18}	1.2269	1.3199	4.9024
900	Blood in large arteries head	1.91	2.6312×10^{-17}	42.69	4.8580×10^{-17}	1.4868×10^{-17}	2.3242×10^{-17}	0.5416	1.7698	1.1321
910	Blood in large veins head	3.01	2.0257×10^{-17}	16.05	3.2010×10^{-17}	2.6991×10^{-17}	2.6009×10^{-17}	0.6328	0.7505	0.7788
1000	Blood in large arteries trunk	117.87	3.1317×10^{-17}	3.70	2.6830×10^{-17}	2.7763×10^{-17}	2.9482×10^{-17}	1.1673	1.1280	1.0623
1010	Blood in large veins trunk	239.85	3.5302×10^{-17}	3.57	3.4720×10^{-17}	3.3896×10^{-17}	3.4907×10^{-17}	1.0168	1.0415	1.0113
1100	Blood in large arteries arms	46.31	3.2374×10^{-17}	4.78	2.9980×10^{-17}	3.2106×10^{-17}	2.9171×10^{-17}	1.0798	1.0083	1.1098
1110	Blood in large veins arms	139.53	3.3118×10^{-17}	2.72	3.2470×10^{-17}	3.0606×10^{-17}	3.3253×10^{-17}	1.0200	1.0821	0.9959
1200	Blood in large arteries legs	79.91	2.8234×10^{-17}	7.61	2.1900×10^{-17}	2.5382×10^{-17}	2.7446×10^{-17}	1.2892	1.1123	1.0287
1210	Blood in large veins legs	355.61	2.6471×10^{-17}	2.07	2.8400×10^{-17}	2.6901×10^{-17}	2.7270×10^{-17}	0.9321	0.9840	0.9707
1300	Humeri upper cortical	113.68	2.6577×10^{-17}	5.03	2.8810×10^{-17}	2.6521×10^{-17}	2.7389×10^{-17}	0.9225	1.0021	0.9704
1400	Humeri upper spongiosa	107.72	2.7734×10^{-17}	6.33	2.9000×10^{-17}	2.7093×10^{-17}	2.7647×10^{-17}	0.9563	1.0236	1.0031

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
1500	Humeri upper medullary cavity	20.52	3.0678×10^{-17}	9.57	3.1160×10^{-17}	3.2671×10^{-17}	2.8656×10^{-17}	0.9845	0.9390	1.0706
1600	Humeri lower cortical	103.30	2.7141×10^{-17}	3.94	2.7590×10^{-17}	2.5529×10^{-17}	2.9059×10^{-17}	0.9837	1.0631	0.9340
1700	Humeri lower spongiosa	50.26	2.8143×10^{-17}	7.50	2.7270×10^{-17}	2.6155×10^{-17}	2.5622×10^{-17}	1.0320	1.0760	1.0984
1800	Humeri lower medullary cavity	20.49	2.6875×10^{-17}	13.95	2.6150×10^{-17}	2.8354×10^{-17}	3.0733×10^{-17}	1.0277	0.9478	0.8745
1900	Ulnae and radii cortical	156.71	2.8063×10^{-17}	4.99	2.9120×10^{-17}	3.0308×10^{-17}	2.7007×10^{-17}	0.9637	0.9259	1.0391
2000	Ulnae and radii spongiosa	86.88	2.8930×10^{-17}	6.06	3.0350×10^{-17}	3.0867×10^{-17}	2.9895×10^{-17}	0.9532	0.9373	0.9677
2100	Ulnae and radii medullary cavity	34.07	3.1746×10^{-17}	9.71	3.7560×10^{-17}	3.2727×10^{-17}	3.3995×10^{-17}	0.8452	0.9700	0.9338
2200	Wrists and hand bones cortical	105.13	2.6583×10^{-17}	6.72	2.7320×10^{-17}	2.9776×10^{-17}	3.0936×10^{-17}	0.9730	0.8928	0.8593
2300	Wrists and hand bones spongiosa	69.36	3.0304×10^{-17}	6.17	2.9010×10^{-17}	3.2293×10^{-17}	3.2839×10^{-17}	1.0446	0.9384	0.9228
2400	Clavicles cortical	32.83	2.6011×10^{-17}	10.81	2.4740×10^{-17}	2.8693×10^{-17}	2.8319×10^{-17}	1.0514	0.9065	0.9185
2500	Clavicles spongiosa	38.80	2.8237×10^{-17}	10.44	3.0130×10^{-17}	2.5960×10^{-17}	3.0931×10^{-17}	0.9372	1.0877	0.9129
2600	Cranium cortical	407.67	1.6996×10^{-17}	4.31	1.7120×10^{-17}	1.7681×10^{-17}	1.7811×10^{-17}	0.9927	0.9613	0.9542
2700	Cranium spongiosa	391.31	2.0422×10^{-17}	3.97	1.9390×10^{-17}	1.9901×10^{-17}	1.9986×10^{-17}	1.0532	1.0262	1.0218
2800	Femora upper cortical	244.13	3.1152×10^{-17}	2.87	3.2430×10^{-17}	3.1952×10^{-17}	3.3626×10^{-17}	0.9606	0.9750	0.9264
2900	Femora upper spongiosa	232.80	3.3982×10^{-17}	3.39	3.1740×10^{-17}	3.2948×10^{-17}	3.2268×10^{-17}	1.0706	1.0314	1.0531
3000	Femora upper medullary cavity	39.52	3.2849×10^{-17}	3.47	3.2190×10^{-17}	3.6159×10^{-17}	3.5645×10^{-17}	1.0205	0.9085	0.9216
3100	Femora lower cortical	240.93	3.2497×10^{-17}	2.08	3.1140×10^{-17}	3.1944×10^{-17}	3.2426×10^{-17}	1.0436	1.0173	1.0022
3200	Femora lower spongiosa	166.33	3.1315×10^{-17}	3.29	3.1240×10^{-17}	3.3360×10^{-17}	3.2394×10^{-17}	1.0024	0.9387	0.9667
3300	Femora lower medullary cavity	56.76	3.3186×10^{-17}	4.60	3.5020×10^{-17}	3.3905×10^{-17}	3.4723×10^{-17}	0.9476	0.9788	0.9557
3400	Tibiae fibulae and patellae cortical	544.85	2.6609×10^{-17}	1.41	2.6660×10^{-17}	2.7102×10^{-17}	2.6421×10^{-17}	0.9981	0.9818	1.0071
3500	Tibiae fibulae and patellae spongiosa	558.53	2.7821×10^{-17}	1.53	2.7790×10^{-17}	2.5844×10^{-17}	2.6218×10^{-17}	1.0011	1.0765	1.0611
3600	Tibiae fibulae and patellae medullary ca	88.88	3.0786×10^{-17}	4.37	3.0370×10^{-17}	3.0807×10^{-17}	2.8886×10^{-17}	1.0137	0.9993	1.0658
3700	Ankles and foot cortical	173.48	2.0836×10^{-17}	4.61	2.3540×10^{-17}	2.0788×10^{-17}	1.9551×10^{-17}	0.8851	1.0023	1.0657
3800	Ankles and foot spongiosa	257.45	2.0648×10^{-17}	3.38	2.0950×10^{-17}	2.0503×10^{-17}	1.9763×10^{-17}	0.9856	1.0071	1.0448
3900	Mandible cortical	45.39	2.4457×10^{-17}	8.24	3.0190×10^{-17}	2.5816×10^{-17}	2.4730×10^{-17}	0.8101	0.9474	0.9890
4000	Mandible spongiosa	33.48	2.6849×10^{-17}	11.64	2.5540×10^{-17}	2.4995×10^{-17}	3.1672×10^{-17}	1.0512	1.0742	0.8477
4100	Pelvis cortical	262.46	3.0919×10^{-17}	2.43	3.2310×10^{-17}	3.0660×10^{-17}	3.2519×10^{-17}	0.9570	1.0085	0.9508
4200	Pelvis spongiosa	455.60	3.0781×10^{-17}	1.90	3.1950×10^{-17}	3.1969×10^{-17}	3.1109×10^{-17}	0.9634	0.9628	0.9895
4300	Ribs cortical	164.50	2.2720×10^{-17}	5.24	2.3270×10^{-17}	2.2912×10^{-17}	2.4820×10^{-17}	0.9764	0.9916	0.9154
4400	Ribs spongiosa	277.32	2.5810×10^{-17}	3.46	2.4120×10^{-17}	2.4832×10^{-17}	2.4874×10^{-17}	1.0701	1.0394	1.0376
4500	Scapulae cortical	121.66	1.8934×10^{-17}	5.64	1.7970×10^{-17}	1.8706×10^{-17}	1.9014×10^{-17}	1.0537	1.0122	0.9958
4600	Scapulae spongiosa	96.73	1.8954×10^{-17}	7.69	2.2540×10^{-17}	2.1947×10^{-17}	2.1736×10^{-17}	0.8409	0.8636	0.8720
4700	Cervical spine cortical	71.60	2.1730×10^{-17}	5.49	1.7430×10^{-17}	2.0880×10^{-17}	2.3973×10^{-17}	1.2467	1.0407	0.9064
4800	Cervical spine spongiosa	75.60	2.3025×10^{-17}	6.57	2.1790×10^{-17}	2.3803×10^{-17}	2.3795×10^{-17}	1.0567	0.9673	0.9676
4900	Thoracic spine cortical	205.83	1.8959×10^{-17}	3.71	1.8810×10^{-17}	2.0212×10^{-17}	1.8884×10^{-17}	1.0079	0.9380	1.0040
5000	Thoracic spine spongiosa	271.92	1.7570×10^{-17}	4.22	1.8960×10^{-17}	1.8279×10^{-17}	1.9246×10^{-17}	0.9267	0.9612	0.9129
5100	Lumbar spine cortical	156.17	2.5248×10^{-17}	2.32	2.4750×10^{-17}	2.5653×10^{-17}	2.5328×10^{-17}	1.0201	0.9842	0.9968

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
5200	Lumbar spine spongiosa	264.98	2.5513×10^{-17}	2.66	2.8690×10^{-17}	2.7576×10^{-17}	2.6904×10^{-17}	0.8893	0.9252	0.9483
5300	Sacrum cortical	80.24	2.5715×10^{-17}	5.69	2.5970×10^{-17}	2.4346×10^{-17}	2.3920×10^{-17}	0.9902	1.0562	1.0750
5400	Sacrum spongiosa	154.84	2.6031×10^{-17}	4.85	2.6960×10^{-17}	2.5789×10^{-17}	2.4723×10^{-17}	0.9656	1.0094	1.0529
5500	Sternum cortical	1.69	3.4287×10^{-17}	16.05	2.9890×10^{-17}	3.0489×10^{-17}	3.5953×10^{-17}	1.1471	1.1246	0.9537
5600	Sternum spongiosa	51.35	3.9273×10^{-17}	7.46	3.7250×10^{-17}	3.7851×10^{-17}	3.3892×10^{-17}	1.0543	1.0376	1.1588
5700	Cartilage costal	41.96	3.9993×10^{-17}	7.17	3.3290×10^{-17}	3.9656×10^{-17}	3.3608×10^{-17}	1.2014	1.0085	1.1900
5800	Cartilage discs	69.35	2.6743×10^{-17}	4.99	2.4910×10^{-17}	2.5115×10^{-17}	2.7294×10^{-17}	1.0736	1.0648	0.9798
6100	Brain	1349.57	1.7063×10^{-17}	1.11	1.7850×10^{-17}	1.7810×10^{-17}	1.8101×10^{-17}	0.9559	0.9581	0.9427
6200	Breast left adipose tissue	153.66	3.7596×10^{-17}	3.96	3.7060×10^{-17}	3.7863×10^{-17}	3.6840×10^{-17}	1.0145	0.9929	1.0205
6300	Breast left glandular tissue	102.49	3.7041×10^{-17}	5.47	3.3590×10^{-17}	3.2964×10^{-17}	3.8714×10^{-17}	1.1027	1.1237	0.9568
6400	Breast right adipose tissue	153.66	3.7995×10^{-17}	5.80	3.7740×10^{-17}	4.0180×10^{-17}	3.6325×10^{-17}	1.0068	0.9456	1.0460
6500	Breast right glandular tissue	102.49	3.2749×10^{-17}	5.04	3.6250×10^{-17}	3.7243×10^{-17}	3.7273×10^{-17}	0.9034	0.8793	0.8786
6600	Eye lens sensitive left	0.0389	0.0000×10^{0}	0.00	1.3980×10^{-17}	–	9.9672×10^{-18}	–	–	–
6601	Eye lens insensitive left	0.189	7.6539×10^{-17}	82.62	1.0560×10^{-16}	4.9514×10^{-17}	5.3479×10^{-17}	0.7248	1.5458	1.4312
6700	Cornea left	1.10	1.9405×10^{-17}	36.79	3.1900×10^{-17}	1.7495×10^{-17}	2.9794×10^{-17}	0.6083	1.1092	0.6513
6701	Aqueous left	0.304	1.5107×10^{-17}	96.80	1.1070×10^{-17}	6.2187×10^{-18}	1.1122×10^{-19}	1.3647	2.4293	135.8311
6702	Vitreous left	6.05	2.8574×10^{-17}	28.99	3.4260×10^{-17}	3.2834×10^{-17}	2.7379×10^{-17}	0.8340	0.8703	1.0436
6800	Eye lens sensitive right	0.0389	0.0000×10^{0}	0.00	–	5.5033×10^{-17}	2.0063×10^{-16}	–	–	–
6801	Eye lens insensitive right	0.189	0.0000×10^{0}	0.00	8.3450×10^{-17}	–	2.3948×10^{-17}	–	–	–
6900	Cornea right	1.10	1.2054×10^{-17}	46.56	3.4620×10^{-17}	2.7647×10^{-17}	4.4964×10^{-17}	0.3482	0.4360	0.2681
6901	Aqueous right	0.304	7.2829×10^{-18}	67.27	2.4490×10^{-17}	2.5713×10^{-17}	2.2840×10^{-17}	0.2974	0.2832	0.3189
6902	Vitreous right	6.05	3.4173×10^{-17}	15.16	2.6980×10^{-17}	3.1000×10^{-17}	2.6066×10^{-17}	1.2666	1.1023	1.3110
7000	Gall bladder wall	8.20	3.1048×10^{-17}	18.62	2.4600×10^{-17}	3.3413×10^{-17}	3.3496×10^{-17}	1.2621	0.9292	0.9269
7100	Gall bladder contents	48.00	3.0859×10^{-17}	6.80	2.8020×10^{-17}	2.9431×10^{-17}	3.0224×10^{-17}	1.1013	1.0485	1.0210
7200	Stomach wall(0-60)	1.56	2.7620×10^{-17}	28.96	2.5850×10^{-17}	3.2081×10^{-17}	4.2572×10^{-17}	1.0685	0.8609	0.6488
7201	Stomach wall(60-100)	1.04	2.7708×10^{-17}	22.10	3.2520×10^{-17}	3.9535×10^{-17}	3.1189×10^{-17}	0.8520	0.7008	0.8884
7202	Stomach wall(100-300)	5.26	3.1724×10^{-17}	17.74	3.4060×10^{-17}	3.3726×10^{-17}	3.3557×10^{-17}	0.9314	0.9406	0.9454
7203	Stomach wall(300-surface)	165.01	3.2095×10^{-17}	2.87	3.1830×10^{-17}	3.3658×10^{-17}	3.5107×10^{-17}	1.0083	0.9536	0.9142
7300	Stomach contents	230.00	3.0558×10^{-17}	3.69	3.1380×10^{-17}	3.0513×10^{-17}	3.0659×10^{-17}	0.9738	1.0015	0.9967
7400	Small intestine wall(0-130)	12.34	3.1241×10^{-17}	6.87	3.7390×10^{-17}	3.9358×10^{-17}	3.5509×10^{-17}	0.8355	0.7938	0.8798
7401	Small intestine wall(130-150)	1.92	3.2780×10^{-17}	10.10	3.3980×10^{-17}	3.9617×10^{-17}	3.3139×10^{-17}	0.9647	0.8274	0.9892
7402	Small intestine wall(150-200)	4.83	2.9583×10^{-17}	7.32	3.8220×10^{-17}	3.5415×10^{-17}	3.1697×10^{-17}	0.7740	0.8353	0.9333
7403	Small intestine wall(200-surface)	736.66	3.5450×10^{-17}	1.17	3.4660×10^{-17}	3.4526×10^{-17}	3.4395×10^{-17}	1.0228	1.0268	1.0307
7500	Small intestine contents(-500-0)	45.23	3.5788×10^{-17}	4.91	3.8280×10^{-17}	3.6139×10^{-17}	3.7068×10^{-17}	0.9349	0.9903	0.9655
7501	Small intestine contents(centre–500)	234.77	3.4919×10^{-17}	4.03	3.4930×10^{-17}	3.2918×10^{-17}	3.5447×10^{-17}	0.9997	1.0608	0.9851
7600	Ascending colon wall(0-280)	4.45	4.7795×10^{-17}	11.85	4.1240×10^{-17}	2.7853×10^{-17}	3.8931×10^{-17}	1.1589	1.7160	1.2277
7601	Ascending colon wall(280-300)	0.322	3.3847×10^{-17}	25.16	5.8380×10^{-17}	5.2738×10^{-17}	3.5806×10^{-17}	0.5798	0.6418	0.9453
7602	Ascending colon wall(300-surface)	107.78	3.7300×10^{-17}	4.07	4.0890×10^{-17}	4.2950×10^{-17}	3.7254×10^{-17}	0.9122	0.8684	1.0012

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
7700	Ascending colon content	100.01	3.8730×10^{-17}	5.35	3.8220×10^{-17}	3.9659×10^{-17}	3.8132×10^{-17}	1.0133	0.9766	1.0157
7800	Transverse colon wall right(0-280)	3.68	4.7944×10^{-17}	15.35	5.1430×10^{-17}	2.9553×10^{-17}	3.6564×10^{-17}	0.9322	1.6223	1.3112
7801	Transverse colon wall right(280-300)	0.266	6.6539×10^{-17}	24.77	3.1300×10^{-17}	2.6721×10^{-17}	2.4982×10^{-17}	2.1259	2.4901	2.6635
7802	Transverse colon wall right(300-surface)	64.85	3.8996×10^{-17}	5.28	3.8830×10^{-17}	3.8241×10^{-17}	4.2685×10^{-17}	1.0043	1.0197	0.9136
7900	Transverse colon contents right	59.99	4.1110×10^{-17}	5.54	3.9720×10^{-17}	3.9478×10^{-17}	3.7182×10^{-17}	1.0350	1.0413	1.1056
8000	Transverse colon wall left(0-280)	2.20	4.1790×10^{-17}	17.40	4.3880×10^{-17}	4.0447×10^{-17}	2.7217×10^{-17}	0.9524	1.0332	1.5354
8001	Transverse colon wall left(280-300)	0.16	2.7270×10^{-17}	37.78	3.5570×10^{-17}	2.4547×10^{-17}	2.5101×10^{-17}	0.7667	1.1109	1.0864
8002	Transverse colon wall left(300-surface)	66.43	4.3498×10^{-17}	4.12	4.3330×10^{-17}	3.9167×10^{-17}	3.6101×10^{-17}	1.0039	1.1106	1.2049
8100	Transverse colon content left	30.01	4.1052×10^{-17}	7.83	3.8090×10^{-17}	3.7087×10^{-17}	4.1604×10^{-17}	1.0778	1.1069	0.9867
8200	Descending colon wall(0-280)	3.02	2.9527×10^{-17}	15.89	4.1310×10^{-17}	3.4073×10^{-17}	3.8740×10^{-17}	0.7148	0.8666	0.7622
8201	Descending colon wall(280-300)	0.22	2.4048×10^{-17}	30.29	3.5500×10^{-17}	3.1347×10^{-17}	2.9002×10^{-17}	0.6774	0.7672	0.8292
8202	Descending colon wall(300-surface)	109.32	3.7497×10^{-17}	2.76	3.9430×10^{-17}	3.9799×10^{-17}	4.2399×10^{-17}	0.9510	0.9422	0.8844
8300	Descending colon content	50.00	4.1747×10^{-17}	7.49	3.6080×10^{-17}	3.6789×10^{-17}	3.3690×10^{-17}	1.1571	1.1348	1.2392
8400	Sigmoid colon wall(0-280)	4.22	2.9029×10^{-17}	10.35	2.8860×10^{-17}	2.4012×10^{-17}	2.5616×10^{-17}	1.0058	1.2089	1.1332
8401	Sigmoid colon wall(280-300)	0.306	3.7199×10^{-17}	32.72	1.7610×10^{-17}	2.3439×10^{-17}	3.3794×10^{-17}	2.1124	1.5871	1.1008
8402	Sigmoid colon wall(300-surface)	51.76	2.8161×10^{-17}	6.49	2.8730×10^{-17}	3.5404×10^{-17}	2.9520×10^{-17}	0.9802	0.7954	0.9540
8500	Sigmoid colon contents	79.99	2.9000×10^{-17}	5.83	2.8960×10^{-17}	2.7339×10^{-17}	2.9186×10^{-17}	1.0014	1.0607	0.9936
8600	Rectum wall	31.27	2.9084×10^{-17}	12.80	2.6780×10^{-17}	3.1150×10^{-17}	2.5743×10^{-17}	1.0860	0.9337	1.1298
8700	Heart wall	290.89	2.8214×10^{-17}	3.81	2.9870×10^{-17}	2.8881×10^{-17}	2.9457×10^{-17}	0.9446	0.9769	0.9578
8800	Blood in heart chamber	370.00	2.8743×10^{-17}	3.42	2.9270×10^{-17}	2.8478×10^{-17}	2.8368×10^{-17}	0.9820	1.0093	1.0132
8900	Kidney left cortex	149.09	2.3708×10^{-17}	5.11	2.2570×10^{-17}	2.2778×10^{-17}	2.4080×10^{-17}	1.0504	1.0408	0.9845
9000	Kidney left medulla	37.44	2.3350×10^{-17}	5.89	2.7040×10^{-17}	2.8845×10^{-17}	2.5466×10^{-17}	0.8635	0.8095	0.9169
9100	Kidney left pelvis	7.49	1.6179×10^{-17}	25.94	2.6660×10^{-17}	2.3066×10^{-17}	2.4696×10^{-17}	0.6069	0.7014	0.6551
9200	Kidney right cortex	125.15	2.6798×10^{-17}	3.62	2.7780×10^{-17}	2.6522×10^{-17}	2.8078×10^{-17}	0.9646	1.0104	0.9544
9300	Kidney right medulla	31.44	2.7469×10^{-17}	7.23	2.4920×10^{-17}	2.6995×10^{-17}	2.6319×10^{-17}	1.1023	1.0176	1.0437
9400	Kidney right pelvis	6.29	2.9325×10^{-17}	17.25	3.4610×10^{-17}	2.0889×10^{-17}	1.9835×10^{-17}	0.8473	1.4038	1.4784
9500	Liver	1810.00	3.0607×10^{-17}	1.05	3.0090×10^{-17}	3.0144×10^{-17}	3.0810×10^{-17}	1.0172	1.0154	0.9934
9700	Lung(AI) left	427.26	2.3589×10^{-17}	3.80	2.2800×10^{-17}	2.3685×10^{-17}	2.4023×10^{-17}	1.0346	0.9959	0.9819
9900	Lung(AI) right	522.52	2.6161×10^{-17}	3.05	2.5390×10^{-17}	2.6530×10^{-17}	2.6354×10^{-17}	1.0304	0.9861	0.9927
10000	Lymphatic nodes ET	12.70	1.9383×10^{-17}	17.08	2.4020×10^{-17}	2.3986×10^{-17}	2.3449×10^{-17}	0.8070	0.8081	0.8266
10100	Lymphatic nodes thoracic	12.70	2.9898×10^{-17}	14.95	2.3070×10^{-17}	2.1370×10^{-17}	2.5622×10^{-17}	1.2960	1.3991	1.1669
10200	Lymphatic nodes head	4.39	2.4611×10^{-17}	27.63	2.6350×10^{-17}	1.4749×10^{-17}	1.5927×10^{-17}	0.9340	1.6687	1.5452
10300	Lymphatic nodes trunk	103.64	3.5371×10^{-17}	5.30	3.4130×10^{-17}	3.3522×10^{-17}	3.2797×10^{-17}	1.0364	1.0552	1.0785
10400	Lymphatic nodes arms	8.77	4.1618×10^{-17}	13.19	2.4550×10^{-17}	2.6625×10^{-17}	3.2235×10^{-17}	1.6952	1.5631	1.2911
10500	Lymphatic nodes legs	8.77	2.4851×10^{-17}	12.48	2.6800×10^{-17}	3.1272×10^{-17}	1.8602×10^{-17}	0.9273	0.7947	1.3359
10600	Muscle head	445.02	2.3419×10^{-17}	3.11	2.3690×10^{-17}	2.3909×10^{-17}	2.3305×10^{-17}	0.9886	0.9795	1.0049

continued

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10700	Muscle trunk	8324.97	2.7119×10^{-17}	0.39	2.7330×10^{-17}	2.6883×10^{-17}	2.7444×10^{-17}	0.9923	1.0088	0.9881
10800	Muscle arms	1479.72	3.0886×10^{-17}	1.13	3.0400×10^{-17}	2.9396×10^{-17}	2.9535×10^{-17}	1.0160	1.0507	1.0457
10900	Muscle legs	7676.72	2.8146×10^{-17}	0.72	2.8280×10^{-17}	2.8137×10^{-17}	2.8422×10^{-17}	0.9953	1.0003	0.9903
11000	Oesophagus wall(0-190)	1.87	2.8368×10^{-17}	32.40	2.7900×10^{-17}	2.4673×10^{-17}	2.2278×10^{-17}	1.0168	1.1498	1.2734
11001	Oesophagus wall(190-200)	0.101	3.5062×10^{-17}	47.32	1.9530×10^{-17}	2.5040×10^{-17}	1.4977×10^{-17}	1.7953	1.4002	2.3411
11002	Oesophagus wall(200-surface)	41.25	2.3384×10^{-17}	12.04	2.6040×10^{-17}	2.3951×10^{-17}	2.2109×10^{-17}	0.8980	0.9763	1.0576
11003	Oesophagus contents	21.24	2.6360×10^{-17}	9.22	2.4200×10^{-17}	2.1427×10^{-17}	2.2953×10^{-17}	1.0892	1.2302	1.1484
11100	Ovary left	6.32	2.9761×10^{-17}	15.38	3.5420×10^{-17}	1.9699×10^{-17}	3.5399×10^{-17}	0.8402	1.5108	0.8407
11200	Ovary right	6.32	2.4547×10^{-17}	22.83	3.2270×10^{-17}	3.5757×10^{-17}	3.7869×10^{-17}	0.7607	0.6865	0.6482
11300	Pancreas	144.55	3.4502×10^{-17}	5.04	3.3780×10^{-17}	3.4288×10^{-17}	3.4492×10^{-17}	1.0214	1.0062	1.0003
11400	Pituitary gland	0.615	5.5319×10^{-18}	98.43	2.2890×10^{-17}	1.0010×10^{-17}	1.9687×10^{-17}	0.2417	0.5526	0.2810
11600	RST head	844.54	2.3781×10^{-17}	2.34	2.4120×10^{-17}	2.4252×10^{-17}	2.4081×10^{-17}	0.9859	0.9806	0.9875
11700	RST trunk	11513.36	2.8457×10^{-17}	0.66	2.8460×10^{-17}	2.8254×10^{-17}	2.8415×10^{-17}	0.9999	1.0072	1.0015
11800	RST arms	2171.59	3.0059×10^{-17}	1.35	3.0080×10^{-17}	3.0671×10^{-17}	2.9643×10^{-17}	0.9993	0.9800	1.0140
11900	RST legs	7795.95	2.9149×10^{-17}	0.89	2.9360×10^{-17}	2.9228×10^{-17}	2.9208×10^{-17}	0.9928	0.9973	0.9980
12000	Salivary gland left	35.88	2.5379×10^{-17}	7.93	2.7950×10^{-17}	2.5891×10^{-17}	2.3813×10^{-17}	0.9080	0.9802	1.0657
12100	Salivary gland right	35.88	2.5548×10^{-17}	5.76	2.1020×10^{-17}	2.3514×10^{-17}	2.8138×10^{-17}	1.2154	1.0865	0.9080
12200	Skin head insensitive	155.58	2.0079×10^{-17}	4.46	2.0130×10^{-17}	1.8089×10^{-17}	1.8915×10^{-17}	0.9975	1.1100	1.0615
12201	Skin head sensitive(50-100)	6.32	1.5386×10^{-17}	13.42	9.3970×10^{-18}	1.3341×10^{-17}	1.2887×10^{-17}	1.6373	1.1532	1.1939
12300	Skin trunk insensitive	871.60	2.3471×10^{-17}	2.04	2.4140×10^{-17}	2.3213×10^{-17}	2.3869×10^{-17}	0.9723	1.0111	0.9833
12301	Skin trunk sensitive(50-100)	32.37	1.5873×10^{-17}	6.75	1.3880×10^{-17}	1.3819×10^{-17}	1.3780×10^{-17}	1.1436	1.1486	1.1519
12400	Skin arms insensitive	380.91	2.6494×10^{-17}	3.52	2.5870×10^{-17}	2.5727×10^{-17}	2.5078×10^{-17}	1.0241	1.0298	1.0565
12401	Skin arms sensitive(50-100)	15.60	2.1091×10^{-17}	9.39	2.0870×10^{-17}	1.9349×10^{-17}	1.8346×10^{-17}	1.0106	1.0901	1.1496
12500	Skin legs insensitive	924.58	2.5654×10^{-17}	0.75	2.4110×10^{-17}	2.4851×10^{-17}	2.4818×10^{-17}	1.0640	1.0323	1.0337
12501	Skin legs sensitive(50-100)	35.02	1.6397×10^{-17}	6.22	1.6170×10^{-17}	1.6118×10^{-17}	1.6610×10^{-17}	1.0140	1.0173	0.9872
12600	Spinal cord	19.10	2.3500×10^{-17}	8.78	2.2240×10^{-17}	2.1559×10^{-17}	1.9839×10^{-17}	1.0566	1.0900	1.1845
12700	Spleen	187.40	2.1191×10^{-17}	3.25	2.2480×10^{-17}	2.2048×10^{-17}	2.1742×10^{-17}	0.9427	0.9611	0.9746
12800	Teeth	40.56	2.5934×10^{-17}	7.34	2.8760×10^{-17}	3.0687×10^{-17}	2.4468×10^{-17}	0.9017	0.8451	1.0599
12801	Teeth retention region	0.0361	1.0771×10^{-17}	39.04	1.5150×10^{-17}	2.1686×10^{-17}	7.3278×10^{-17}	0.7109	0.4967	0.1470
13100	Thymus	20.50	2.8047×10^{-17}	12.87	2.5740×10^{-17}	3.1398×10^{-17}	3.1254×10^{-17}	1.0896	0.8933	0.8974
13200	Thyroid	19.46	2.5768×10^{-17}	13.09	3.0700×10^{-17}	3.0010×10^{-17}	3.0024×10^{-17}	0.8394	0.8587	0.8583
13300	Tongue upper(food)	21.00	2.4553×10^{-17}	18.59	2.5160×10^{-17}	2.8523×10^{-17}	2.4778×10^{-17}	0.9759	0.8608	0.9909
13301	Tongue lower	40.42	2.7640×10^{-17}	10.23	2.6510×10^{-17}	3.0102×10^{-17}	2.8081×10^{-17}	1.0426	0.9182	0.9843
13400	Tonsils	3.08	1.8110×10^{-17}	27.61	2.4720×10^{-17}	3.0218×10^{-17}	1.0454×10^{-17}	0.7326	0.5993	1.7324
13500	Ureter left	7.69	3.5204×10^{-17}	11.85	3.8420×10^{-17}	3.1769×10^{-17}	2.5028×10^{-17}	0.9163	1.1082	1.4066
13600	Ureter right	7.69	2.7904×10^{-17}	14.14	2.5610×10^{-17}	3.5505×10^{-17}	3.0265×10^{-17}	1.0896	0.7859	0.9220
13700	Urinary bladder wall insensitive	39.46	3.9259×10^{-17}	5.79	3.9360×10^{-17}	4.2336×10^{-17}	3.7994×10^{-17}	0.9974	0.9273*	1.0333*

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
13701	Urinary bladder wall sensitive(69-116)	1.35	3.7753×10^{-17}	22.93	3.9680×10^{-17}	4.0251×10^{-17}	3.7316×10^{-17}	0.9514	0.9380*	1.0117*
13800	Urinary bladder content	200.00	4.0863×10^{-17}	3.23	3.9000×10^{-17}	4.1898×10^{-17}	4.1366×10^{-17}	1.0478	0.9753	0.9879
13900	Uterus	81.99	3.3002×10^{-17}	5.54	3.5700×10^{-17}	3.1348×10^{-17}	3.0739×10^{-17}	0.9244	1.0528	1.0736
14000	Air inside body	0.036	1.1704×10^{-16}	45.99	1.5610×10^{-17}	1.4694×10^{-17}	1.5470×10^{-17}	7.4978	7.9654	7.5657

* The urinary bladder wall partition was revised between the 2018-dated MCNP6 and PHITS reference calculations and the currently distributed phantom, which the 2020-dated Geant4 files match. The marked ratio therefore compares two different partitions of the same wall, and is left out of the summary statistics of Table S2.

Table S7. 00M_Internal. Absorbed dose per primary history, Gy. σ is the FLUKA statistical uncertainty. Ratios are FLUKA / reference.

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
100	Adrenal left	3.60	1.3087×10^{-14}	1.44	1.2850×10^{-14}	1.3101×10^{-14}	1.3189×10^{-14}	1.0184	0.9989	0.9922
200	Adrenal right	3.60	2.6787×10^{-14}	0.82	2.6940×10^{-14}	2.6927×10^{-14}	2.6932×10^{-14}	0.9943	0.9948	0.9946
300	ET1(0-8)	0.000741	1.5077×10^{-15}	62.03	2.1450×10^{-15}	7.5284×10^{-16}	7.8639×10^{-16}	0.7029	2.0027	1.9173
301	ET1(8-40)	0.00304	1.7217×10^{-15}	57.07	3.0540×10^{-15}	1.2318×10^{-15}	1.7330×10^{-15}	0.5637	1.3976	0.9935
302	ET1(40-50)	0.000976	7.8885×10^{-16}	55.26	1.5280×10^{-15}	1.5766×10^{-15}	4.1161×10^{-15}	0.5163	0.5003	0.1917
303	ET1(50-Surface)	0.0968	1.3453×10^{-15}	23.63	1.7030×10^{-15}	1.9386×10^{-15}	1.6026×10^{-15}	0.7900	0.6940	0.8395
400	ET2(-15-0)	0.0327	1.9624×10^{-15}	13.12	1.7850×10^{-15}	1.8515×10^{-15}	1.7386×10^{-15}	1.0994	1.0599	1.1287
401	ET2(0-40)	0.0938	1.7316×10^{-15}	8.39	1.7230×10^{-15}	2.0145×10^{-15}	1.9497×10^{-15}	1.0050	0.8596	0.8881
402	ET2(40-50)	0.0237	1.8172×10^{-15}	10.74	1.6200×10^{-15}	2.7848×10^{-15}	1.8728×10^{-15}	1.1217	0.6525	0.9703
403	ET2(50-55)	0.0119	1.2123×10^{-15}	11.93	1.8900×10^{-15}	2.1088×10^{-15}	1.5580×10^{-15}	0.6414	0.5749	0.7781
404	ET2(55-65)	0.0238	1.9703×10^{-15}	7.49	2.2270×10^{-15}	2.3323×10^{-15}	1.8300×10^{-15}	0.8848	0.8448	1.0767
405	ET2(65-Surface)	2.14	1.8260×10^{-15}	2.10	1.8410×10^{-15}	1.8679×10^{-15}	1.8276×10^{-15}	0.9918	0.9776	0.9991
500	Oral mucosa tongue	0.0081	2.3939×10^{-15}	13.88	2.7650×10^{-15}	3.2434×10^{-15}	3.6988×10^{-15}	0.8658	0.7381	0.6472
501	Oral mucosa moth floor	0.0098	2.2924×10^{-15}	14.83	2.7850×10^{-15}	3.5499×10^{-15}	2.0891×10^{-15}	0.8231	0.6458	1.0973
600	Oral mucosa lips and cheeks	0.00649	2.2578×10^{-15}	22.56	2.1450×10^{-15}	1.9647×10^{-15}	2.4187×10^{-15}	1.0526	1.1492	0.9335
700	Trachea	0.542	5.1050×10^{-15}	4.75	5.6430×10^{-15}	5.3096×10^{-15}	5.5234×10^{-15}	0.9047	0.9615	0.9243
800	BB(-11–6)	0.00159	9.2013×10^{-15}	13.55	8.6470×10^{-15}	1.1102×10^{-14}	6.1184×10^{-15}	1.0641	0.8288	1.5039
801	BB(-6-0)	0.00204	1.0134×10^{-14}	13.53	8.0690×10^{-15}	1.2917×10^{-14}	5.4600×10^{-15}	1.2559	0.7846	1.8560
802	BB(0-10)	0.00342	1.5322×10^{-14}	14.35	1.0430×10^{-14}	1.0357×10^{-14}	7.5151×10^{-15}	1.4691	1.4794	2.0389
803	BB(10-35)	0.00863	1.0004×10^{-14}	15.16	8.3260×10^{-15}	1.0889×10^{-14}	7.9184×10^{-15}	1.2015	0.9187	1.2634
804	BB(35-40)	0.00174	1.0112×10^{-14}	27.25	8.2300×10^{-15}	1.3175×10^{-14}	8.5682×10^{-15}	1.2287	0.7675	1.1802
805	BB(40-50)	0.0035	1.2082×10^{-14}	20.89	7.5160×10^{-15}	1.1610×10^{-14}	8.5536×10^{-15}	1.6075	1.0407	1.4125
806	BB(50-60)	0.00352	9.7823×10^{-15}	23.23	6.6980×10^{-15}	8.6215×10^{-15}	7.5048×10^{-15}	1.4605	1.1346	1.3035
807	BB(60-70)	0.00353	1.0225×10^{-14}	17.84	7.2200×10^{-15}	9.7958×10^{-15}	9.5953×10^{-15}	1.4162	1.0438	1.0656
808	BB(70-surface)	0.352	9.5128×10^{-15}	5.55	1.0180×10^{-14}	9.0043×10^{-15}	9.6514×10^{-15}	0.9345	1.0565	0.9856
900	Blood in large arteries head	0.0799	2.5771×10^{-15}	12.33	2.3720×10^{-15}	1.5431×10^{-15}	2.5898×10^{-15}	1.0865*	1.6701*	0.9951*
910	Blood in large veins head	0.286	1.8791×10^{-15}	10.39	2.1020×10^{-15}	2.3965×10^{-15}	2.4225×10^{-15}	0.8939*	0.7841*	0.7757*
1000	Blood in large arteries trunk	9.46	8.9381×10^{-15}	0.70	9.0410×10^{-15}	9.0381×10^{-15}	9.2127×10^{-15}	0.9886*	0.9889*	0.9702*
1010	Blood in large veins trunk	23.75	1.0089×10^{-14}	0.46	1.0280×10^{-14}	1.0313×10^{-14}	1.0298×10^{-14}	0.9814*	0.9783*	0.9797*
1100	Blood in large arteries arms	1.76	6.9119×10^{-15}	2.43	6.9370×10^{-15}	6.9617×10^{-15}	7.0735×10^{-15}	0.9964*	0.9929*	0.9772*
1110	Blood in large veins arms	6.79	6.1292×10^{-15}	0.54	6.0620×10^{-15}	6.0317×10^{-15}	6.1209×10^{-15}	1.0111*	1.0162*	1.0013*
1200	Blood in large arteries legs	6.10	7.9720×10^{-16}	2.17	7.9020×10^{-16}	7.7873×10^{-16}	7.7631×10^{-16}	1.0089*	1.0237*	1.0269*
1210	Blood in large veins legs	21.37	8.1175×10^{-16}	1.34	8.5840×10^{-16}	8.4240×10^{-16}	8.5122×10^{-16}	0.9457*	0.9636*	0.9536*
1300	Humeri upper cortical	1.92	4.3723×10^{-15}	3.08	4.2240×10^{-15}	4.3396×10^{-15}	4.3043×10^{-15}	1.0351	1.0075	1.0158
1400	Humeri upper spogiosa	3.55	4.0028×10^{-15}	1.60	3.9870×10^{-15}	3.7970×10^{-15}	3.7855×10^{-15}	1.0040	1.0542	1.0574

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
1500	Humeri upper medullary cavity	0.322	4.3403×10^{-15}	9.16	4.9390×10^{-15}	4.9389×10^{-15}	4.3645×10^{-15}	0.8788	0.8788	0.9945
1600	Humeri lower cortical	1.85	5.1806×10^{-15}	3.20	5.2360×10^{-15}	5.2946×10^{-15}	5.2752×10^{-15}	0.9894	0.9785	0.9821
1700	Humeri lower spongiosa	2.77	5.9941×10^{-15}	2.54	5.8820×10^{-15}	6.0310×10^{-15}	5.7199×10^{-15}	1.0191	0.9939	1.0479
1800	Humeri lower medullary cavity	0.322	5.2225×10^{-15}	3.33	5.4580×10^{-15}	5.9505×10^{-15}	5.4959×10^{-15}	0.9569	0.8777	0.9503
1900	Radii cortical	1.21	6.0241×10^{-15}	3.60	6.0120×10^{-15}	5.7725×10^{-15}	5.9077×10^{-15}	1.0020	1.0436	1.0197
1910	Ulnae cortical	1.55	5.5869×10^{-15}	2.96	5.3350×10^{-15}	5.2704×10^{-15}	5.5996×10^{-15}	1.0472	1.0600	0.9977
2000	Radii spongiosa	1.70	5.4366×10^{-15}	1.95	5.5080×10^{-15}	5.3118×10^{-15}	5.6920×10^{-15}	0.9870	1.0235	0.9551
2010	Ulnae spongiosa	2.16	5.7814×10^{-15}	1.80	5.7160×10^{-15}	5.9840×10^{-15}	5.5060×10^{-15}	1.0114	0.9661	1.0500
2100	Radii medullary cavity	0.122	6.4086×10^{-15}	4.72	6.1540×10^{-15}	6.8034×10^{-15}	6.2068×10^{-15}	1.0414	0.9420	1.0325
2110	Ulnae medullary cavity	0.143	5.3918×10^{-15}	11.08	5.2130×10^{-15}	5.0738×10^{-15}	5.1329×10^{-15}	1.0343	1.0627	1.0504
2200	Wrists and hand bones cortical	0.841	2.5518×10^{-15}	4.57	2.7050×10^{-15}	2.7700×10^{-15}	2.6595×10^{-15}	0.9434	0.9212	0.9595
2300	Wrists and hand bones spongiosa	3.17	2.8224×10^{-15}	4.09	2.9180×10^{-15}	2.8520×10^{-15}	2.8616×10^{-15}	0.9672	0.9896	0.9863
2400	Clavicles cortical	0.915	3.8302×10^{-15}	2.38	3.6040×10^{-15}	3.5996×10^{-15}	3.5044×10^{-15}	1.0628	1.0640	1.0930
2500	Clavicles spongiosa	1.67	3.7938×10^{-15}	3.53	3.8670×10^{-15}	4.0274×10^{-15}	4.0076×10^{-15}	0.9811	0.9420	0.9466
2600	Cranium cortical	21.45	7.0829×10^{-16}	1.11	7.0960×10^{-16}	7.1072×10^{-16}	7.0414×10^{-16}	0.9982	0.9966	1.0059
2700	Cranium spongiosa	78.77	1.1955×10^{-15}	0.61	1.2230×10^{-15}	1.2170×10^{-15}	1.2382×10^{-15}	0.9775	0.9823	0.9655
2800	Femora upper cortical	3.76	1.5247×10^{-15}	2.99	1.5510×10^{-15}	1.4676×10^{-15}	1.5626×10^{-15}	0.9830	1.0389	0.9757
2900	Femora upper spongiosa	5.77	1.8258×10^{-15}	3.20	1.8560×10^{-15}	1.9474×10^{-15}	1.8934×10^{-15}	0.9837	0.9375	0.9643
3000	Femora upper medullary cavity	0.737	1.5465×10^{-15}	6.58	1.3790×10^{-15}	1.4891×10^{-15}	1.5967×10^{-15}	1.1215	1.0386	0.9686
3100	Femora lower cortical	5.51	9.9729×10^{-16}	3.10	9.3410×10^{-16}	9.2049×10^{-16}	9.2487×10^{-16}	1.0676	1.0834	1.0783
3200	Femora lower spongiosa	4.78	7.3704×10^{-16}	3.73	7.6350×10^{-16}	7.7530×10^{-16}	7.5009×10^{-16}	0.9653	0.9507	0.9826
3300	Femora lower medullary cavity	1.17	1.1073×10^{-15}	7.71	1.0630×10^{-15}	9.4158×10^{-16}	1.0551×10^{-15}	1.0417	1.1760	1.0495
3400	Tibiae cortical	4.50	3.9507×10^{-16}	7.25	3.3810×10^{-16}	3.2922×10^{-16}	3.5507×10^{-16}	1.1685	1.2000	1.1126
3410	Fibulae cortical	1.16	2.9303×10^{-16}	10.98	2.8410×10^{-16}	3.0181×10^{-16}	3.2545×10^{-16}	1.0314	0.9709	0.9004
3420	Patellae cortical	0.0374	8.0222×10^{-16}	27.22	6.9990×10^{-16}	4.6609×10^{-16}	9.3503×10^{-16}	1.1462	1.7212	0.8580
3500	Tibiae spongiosa	6.62	4.0246×10^{-16}	5.93	4.1830×10^{-16}	4.1686×10^{-16}	3.6953×10^{-16}	0.9621	0.9655	1.0891
3510	Fibulae spongiosa	1.39	3.2029×10^{-16}	12.77	2.5390×10^{-16}	3.0174×10^{-16}	2.8230×10^{-16}	1.2615	1.0615	1.1346
3520	Patellae spongiosa	0.171	7.0380×10^{-16}	27.42	6.9340×10^{-16}	3.5138×10^{-16}	6.4378×10^{-16}	1.0150	2.0030	1.0932
3600	Tibiae medullary cavity	0.673	4.1436×10^{-16}	18.10	3.1190×10^{-16}	3.4267×10^{-16}	3.3958×10^{-16}	1.3285	1.2092	1.2202
3610	Fibulae medullary cavity	0.107	2.4359×10^{-16}	45.70	3.3370×10^{-16}	3.0057×10^{-16}	5.2152×10^{-16}	0.7300	0.8104	0.4671
3700	Ankles and foot cortical	1.29	2.8190×10^{-16}	8.26	3.1960×10^{-16}	2.1680×10^{-16}	2.4588×10^{-16}	0.8820	1.3002	1.1465
3800	Ankles and foot spongiosa	4.93	2.7261×10^{-16}	10.86	2.5030×10^{-16}	2.6799×10^{-16}	2.2738×10^{-16}	1.0891	1.0172	1.1989
3900	Mandible cortical	1.70	2.3866×10^{-15}	2.36	2.4730×10^{-15}	2.3897×10^{-15}	2.4568×10^{-15}	0.9651	0.9987	0.9714
4000	Mandible spongiosa	5.69	2.4041×10^{-15}	2.68	2.5690×10^{-15}	2.5081×10^{-15}	2.4639×10^{-15}	0.9358	0.9585	0.9757
4100	Pelvis cortical	6.15	3.0348×10^{-15}	2.06	3.0320×10^{-15}	2.9705×10^{-15}	3.0040×10^{-15}	1.0009	1.0217	1.0103
4200	Pelvis spongiosa	8.69	3.1391×10^{-15}	2.19	3.0000×10^{-15}	3.1097×10^{-15}	3.0488×10^{-15}	1.0464	1.0095	1.0296
4300	Ribs cortical	6.39	1.2215×10^{-14}	0.66	1.2150×10^{-14}	1.2294×10^{-14}	1.2215×10^{-14}	1.0053	0.9935	1.0000
4400	Ribs spongiosa	21.36	1.2494×10^{-14}	0.50	1.2500×10^{-14}	1.2633×10^{-14}	1.2471×10^{-14}	0.9995	0.9890	1.0019

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
4500	Scapulae cortical	2.45	4.1381×10^{-15}	2.22	4.1730×10^{-15}	4.1183×10^{-15}	4.1809×10^{-15}	0.9916	1.0048	0.9898
4600	Scapulae spongiosa	4.46	4.0715×10^{-15}	1.62	4.2520×10^{-15}	4.0727×10^{-15}	4.3706×10^{-15}	0.9575	0.9997	0.9316
4700	Cervical spine cortical	3.77	2.7686×10^{-15}	2.23	2.8200×10^{-15}	2.7967×10^{-15}	2.7598×10^{-15}	0.9818	0.9900	1.0032
4800	Cervical spine spongiosa	7.14	2.7661×10^{-15}	1.71	2.7770×10^{-15}	2.8187×10^{-15}	2.8087×10^{-15}	0.9961	0.9813	0.9848
4900	Thoracic spine cortical	8.53	1.0579×10^{-14}	0.78	1.0530×10^{-14}	1.0539×10^{-14}	1.0601×10^{-14}	1.0047	1.0038	0.9979
5000	Thoracic spine spongiosa	10.50	1.1310×10^{-14}	0.99	1.1470×10^{-14}	1.1218×10^{-14}	1.1427×10^{-14}	0.9861	1.0082	0.9898
5100	Lumbar spine cortical	2.73	8.3769×10^{-15}	1.23	8.4030×10^{-15}	8.3007×10^{-15}	8.5916×10^{-15}	0.9969	1.0092	0.9750
5200	Lumbar spine spongiosa	7.85	9.1450×10^{-15}	0.73	9.2720×10^{-15}	9.2095×10^{-15}	9.0740×10^{-15}	0.9863	0.9930	1.0078
5300	Sacrum cortical	1.08	3.4276×10^{-15}	2.66	3.2220×10^{-15}	3.2849×10^{-15}	3.0917×10^{-15}	1.0638	1.0434	1.1086
5400	Sacrum spongiosa	3.14	3.2889×10^{-15}	2.23	3.3220×10^{-15}	3.3269×10^{-15}	3.3537×10^{-15}	0.9900	0.9886	0.9807
5500	Sternum cortical	0.168	1.0438×10^{-14}	3.48	1.1060×10^{-14}	1.0072×10^{-14}	1.0130×10^{-14}	0.9438	1.0363	1.0304
5600	Sternum spongiosa	0.677	1.0489×10^{-14}	3.44	1.0700×10^{-14}	9.9004×10^{-15}	1.0476×10^{-14}	0.9803	1.0595	1.0013
5700	Cartilage head	53.07	9.2002×10^{-16}	1.01	9.1520×10^{-16}	9.2829×10^{-16}	9.0783×10^{-16}	1.0053	0.9911	1.0134
5800	Cartilage trunk	48.20	1.1458×10^{-14}	0.33	1.1460×10^{-14}	1.1455×10^{-14}	1.1437×10^{-14}	0.9999	1.0003	1.0019
5900	Cartilage arms	11.54	4.2487×10^{-15}	1.18	4.3840×10^{-15}	4.3187×10^{-15}	4.2945×10^{-15}	0.9691	0.9838	0.9893
6000	Cartilage legs	15.38	6.8796×10^{-16}	2.39	7.0740×10^{-16}	6.8841×10^{-16}	7.0094×10^{-16}	0.9725	0.9994	0.9815
6100	Brain	395.66	7.3495×10^{-16}	0.50	7.3840×10^{-16}	7.3997×10^{-16}	7.3975×10^{-16}	0.9953	0.9932	0.9935
6200	Breast left adipose tissue	0.087	8.2027×10^{-15}	9.21	1.0610×10^{-14}	1.0741×10^{-14}	1.0537×10^{-14}	0.7731	0.7637	0.7785
6300	Breast left glandular tissue	0.058	8.8024×10^{-15}	11.92	1.1560×10^{-14}	1.1197×10^{-14}	9.3243×10^{-15}	0.7615	0.7861	0.9440
6400	Breast right adipose tissue	0.087	1.3663×10^{-14}	6.27	1.5000×10^{-14}	1.6852×10^{-14}	1.6126×10^{-14}	0.9108	0.8107	0.8472
6500	Breast right glandular tissue	0.058	1.3605×10^{-14}	7.47	1.5540×10^{-14}	1.6200×10^{-14}	1.5012×10^{-14}	0.8755	0.8398	0.9063
6600	Eye lens sensitive left	0.0162	1.0587×10^{-15}	49.01	6.9100×10^{-16}	1.7416×10^{-15}	6.3416×10^{-16}	1.5322	0.6079	1.6695
6601	Eye lens insensitive left	0.0673	1.6367×10^{-15}	26.92	9.6070×10^{-16}	1.4961×10^{-15}	1.4815×10^{-15}	1.7036	1.0939	1.1047
6700	Cornea left	0.582	1.2475×10^{-15}	8.94	1.1200×10^{-15}	1.1987×10^{-15}	1.1689×10^{-15}	1.1138	1.0407	1.0672
6701	Aqueous left	0.174	1.0498×10^{-15}	24.22	1.1130×10^{-15}	1.3753×10^{-15}	1.3550×10^{-15}	0.9432	0.7633	0.7748
6702	Vitreous left	2.41	1.3173×10^{-15}	7.19	1.2090×10^{-15}	1.2183×10^{-15}	1.1963×10^{-15}	1.0896	1.0812	1.1012
6800	Eye lens sensitive right	0.0162	2.1798×10^{-15}	41.73	4.8780×10^{-16}	1.4814×10^{-15}	1.6526×10^{-15}	4.4686	1.4715	1.3190
6801	Eye lens insensitive right	0.0673	2.2289×10^{-15}	33.92	1.0970×10^{-15}	8.6305×10^{-16}	1.0919×10^{-15}	2.0318	2.5826	2.0413
6900	Cornea right	0.582	1.2780×10^{-15}	8.75	1.2660×10^{-15}	1.2442×10^{-15}	1.3394×10^{-15}	1.0095	1.0272	0.9541
6901	Aqueous right	0.174	1.1782×10^{-15}	18.25	1.2580×10^{-15}	1.2305×10^{-15}	1.1787×10^{-15}	0.9365	0.9575	0.9996
6902	Vitreous right	2.41	1.2485×10^{-15}	5.61	1.2840×10^{-15}	1.4151×10^{-15}	1.3467×10^{-15}	0.9724	0.8823	0.9271
7000	Gall bladder wall	0.542	4.0569×10^{-14}	1.25	4.0490×10^{-14}	3.9849×10^{-14}	4.0115×10^{-14}	1.0020	1.0181	1.0113
7100	Gall bladder contents	2.80	4.0532×10^{-14}	1.11	4.0420×10^{-14}	4.1280×10^{-14}	4.1213×10^{-14}	1.0028	0.9819	0.9835
7200	Stomach wall(0-60)	0.408	2.1603×10^{-14}	1.32	2.1810×10^{-14}	2.2023×10^{-14}	2.1646×10^{-14}	0.9905	0.9809	0.9980
7201	Stomach wall(60-100)	0.273	2.2348×10^{-14}	0.81	2.1810×10^{-14}	2.1926×10^{-14}	2.1281×10^{-14}	1.0246	1.0192	1.0501
7202	Stomach wall(100-300)	1.38	2.1789×10^{-14}	0.87	2.1290×10^{-14}	2.1789×10^{-14}	2.1738×10^{-14}	1.0234	1.0000	1.0023
7203	Stomach wall(300-surface)	6.66	2.2069×10^{-14}	0.62	2.1910×10^{-14}	2.2104×10^{-14}	2.2092×10^{-14}	1.0073	0.9984	0.9990
7300	Stomach contents	40.00	2.0088×10^{-14}	0.48	2.0180×10^{-14}	2.0088×10^{-14}	2.0159×10^{-14}	0.9955	1.0000	0.9965

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
7400	Small intestine wall(0-130)	3.22	8.5135×10^{-15}	0.90	8.5400×10^{-15}	8.6371×10^{-15}	8.7092×10^{-15}	0.9969	0.9857	0.9775
7401	Small intestine wall(130-150)	0.503	8.6285×10^{-15}	0.94	8.4930×10^{-15}	8.9372×10^{-15}	8.7923×10^{-15}	1.0160	0.9655	0.9814
7402	Small intestine wall(150-200)	1.27	8.6270×10^{-15}	1.57	8.4460×10^{-15}	8.6351×10^{-15}	8.6846×10^{-15}	1.0214	0.9991	0.9934
7403	Small intestine wall(200-surface)	33.23	8.4123×10^{-15}	0.46	8.4460×10^{-15}	8.4273×10^{-15}	8.5357×10^{-15}	0.9960	0.9982	0.9855
7500	Small intestine contents(-400-0)	9.28	8.6686×10^{-15}	0.79	8.5080×10^{-15}	8.4503×10^{-15}	8.6946×10^{-15}	1.0189	1.0258	0.9970
7501	Small intestine contents(centre–400)	46.72	8.4227×10^{-15}	0.41	8.4050×10^{-15}	8.3967×10^{-15}	8.5254×10^{-15}	1.0021	1.0031	0.9880
7600	Ascending colon wall(0-280)	1.14	6.5325×10^{-15}	3.02	6.3080×10^{-15}	6.1664×10^{-15}	6.5258×10^{-15}	1.0356	1.0594	1.0010
7601	Ascending colon wall(280-300)	0.0832	7.0314×10^{-15}	3.29	6.1790×10^{-15}	6.5524×10^{-15}	6.4379×10^{-15}	1.1379	1.0731	1.0922
7602	Ascending colon wall(300-surface)	2.80	6.4598×10^{-15}	1.64	6.5200×10^{-15}	6.5316×10^{-15}	6.4960×10^{-15}	0.9908	0.9890	0.9944
7700	Ascending colon content	15.59	6.0580×10^{-15}	0.69	5.9520×10^{-15}	5.8976×10^{-15}	5.9769×10^{-15}	1.0178	1.0272	1.0136
7800	Transverse colon wall right(0-280)	0.842	2.3636×10^{-14}	2.24	2.2970×10^{-14}	2.3255×10^{-14}	2.2607×10^{-14}	1.0290	1.0164	1.0455
7801	Transverse colon wall right(280-300)	0.0618	2.4997×10^{-14}	2.02	2.2860×10^{-14}	2.4863×10^{-14}	2.3135×10^{-14}	1.0935	1.0054	1.0805
7802	Transverse colon wall right(300-surface)	3.99	2.2280×10^{-14}	0.77	2.2100×10^{-14}	2.2420×10^{-14}	2.2108×10^{-14}	1.0081	0.9937	1.0078
7900	Transverse colon contents right	8.41	2.3009×10^{-14}	0.84	2.3190×10^{-14}	2.3082×10^{-14}	2.3152×10^{-14}	0.9922	0.9968	0.9938
8000	Transverse colon wall left(0-280)	0.651	9.3896×10^{-15}	1.86	9.4350×10^{-15}	9.0140×10^{-15}	9.1092×10^{-15}	0.9952	1.0417	1.0308
8001	Transverse colon wall left(280-300)	0.0481	9.5073×10^{-15}	5.26	8.9900×10^{-15}	9.5542×10^{-15}	9.3334×10^{-15}	1.0575	0.9951	1.0186
8002	Transverse colon wall left(300-surface)	3.80	9.3231×10^{-15}	1.22	9.1240×10^{-15}	9.1969×10^{-15}	8.8729×10^{-15}	1.0218	1.0137	1.0507
8100	Transverse colon content left	5.42	9.1240×10^{-15}	1.42	9.0750×10^{-15}	9.3435×10^{-15}	9.0507×10^{-15}	1.0054	0.9765	1.0081
8200	Descending colon wall(0-280)	0.78	4.4726×10^{-15}	2.76	4.4620×10^{-15}	4.8744×10^{-15}	4.5379×10^{-15}	1.0024	0.9176	0.9856
8201	Descending colon wall(280-300)	0.0575	4.2375×10^{-15}	4.78	4.7040×10^{-15}	5.1017×10^{-15}	5.0081×10^{-15}	0.9008	0.8306	0.8461
8202	Descending colon wall(300-surface)	3.57	4.6316×10^{-15}	2.73	4.6000×10^{-15}	4.6542×10^{-15}	4.6513×10^{-15}	1.0069	0.9951	0.9958
8300	Descending colon content	6.58	4.5449×10^{-15}	1.76	4.6220×10^{-15}	4.4839×10^{-15}	4.5181×10^{-15}	0.9833	1.0136	1.0059
8400	Sigmoid colon wall(0-280)	0.965	3.6028×10^{-15}	4.20	3.7810×10^{-15}	3.7746×10^{-15}	3.7353×10^{-15}	0.9529	0.9545	0.9645
8401	Sigmoid colon wall(280-300)	0.071	3.4661×10^{-15}	7.89	3.6380×10^{-15}	4.0026×10^{-15}	3.7307×10^{-15}	0.9528	0.8660	0.9291
8402	Sigmoid colon wall(300-surface)	2.22	3.6520×10^{-15}	2.04	3.7540×10^{-15}	3.8236×10^{-15}	3.6631×10^{-15}	0.9728	0.9551	0.9970
8500	Sigmoid colon contents	8.52	3.8599×10^{-15}	1.66	3.9020×10^{-15}	3.7803×10^{-15}	3.8100×10^{-15}	0.9892	1.0211	1.0131
8600	Rectum wall(0-280)	0.418	2.0627×10^{-15}	6.22	1.9740×10^{-15}	2.0459×10^{-15}	1.9964×10^{-15}	1.0449	1.0082	1.0332
8601	Rectum wall(280-300)	0.0309	2.1103×10^{-15}	14.10	1.8210×10^{-15}	2.2790×10^{-15}	2.1721×10^{-15}	1.1588	0.9260	0.9715
8602	Rectum wall(300-surface)	0.111	2.1793×10^{-15}	12.48	1.9780×10^{-15}	1.9752×10^{-15}	2.1460×10^{-15}	1.1018	1.1034	1.0155
8603	Rectum contents	3.48	2.2328×10^{-15}	2.86	1.9650×10^{-15}	1.9824×10^{-15}	2.0409×10^{-15}	1.1363	1.1263	1.0940
8700	Heart wall	23.16	2.2570×10^{-14}	0.41	2.2590×10^{-14}	2.2459×10^{-14}	2.2683×10^{-14}	0.9991	1.0050	0.9950
8800	Blood in heart chamber	26.00	1.8647×10^{-14}	0.40	1.8810×10^{-14}	1.8670×10^{-14}	1.8619×10^{-14}	0.9913	0.9988	1.0015
8900	Kidney left cortex	9.46	7.1830×10^{-15}	1.09	7.2000×10^{-15}	7.0539×10^{-15}	7.0910×10^{-15}	0.9976	1.0183	1.0130
9000	Kidney left medulla	3.38	7.3882×10^{-15}	1.90	7.5580×10^{-15}	7.3205×10^{-15}	7.3577×10^{-15}	0.9775	1.0093	1.0042
9100	Kidney left pelvis	0.676	7.2301×10^{-15}	4.63	6.9220×10^{-15}	7.0421×10^{-15}	6.7786×10^{-15}	1.0445	1.0267	1.0666
9200	Kidney right cortex	9.46	1.5314×10^{-14}	0.91	1.5400×10^{-14}	1.5366×10^{-14}	1.5265×10^{-14}	0.9944	0.9966	1.0032

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
9300	Kidney right medulla	3.38	1.4434×10^{-14}	0.91	1.4020×10^{-14}	1.3967×10^{-14}	1.4186×10^{-14}	1.0295	1.0334	1.0175
9400	Kidney right pelvis	0.676	1.3380×10^{-14}	2.67	1.3760×10^{-14}	1.3532×10^{-14}	1.3744×10^{-14}	0.9724	0.9887	0.9735
9500	Liver	167.41	6.6700×10^{-14}	0.08	6.6730×10^{-14}	6.6656×10^{-14}	6.6897×10^{-14}	0.9996	1.0007	0.9971
9700	Lung(AI) left	28.69	9.8941×10^{-15}	0.62	9.9050×10^{-15}	1.0009×10^{-14}	9.8997×10^{-15}	0.9989	0.9886	0.9994
9900	Lung(AI) right	31.31	1.7890×10^{-14}	0.35	1.7980×10^{-14}	1.7997×10^{-14}	1.8127×10^{-14}	0.9950	0.9940	0.9869
10000	Lymphatic nodes ET	1.24	2.2647×10^{-15}	3.41	2.0730×10^{-15}	2.0411×10^{-15}	2.1438×10^{-15}	1.0925	1.1096	1.0564
10100	Lymphatic nodes thoracic	1.24	1.9530×10^{-14}	1.34	1.9540×10^{-14}	1.9772×10^{-14}	1.9730×10^{-14}	0.9995	0.9878	0.9899
10200	Lymphatic nodes head	0.429	3.4191×10^{-15}	7.80	3.6760×10^{-15}	3.5582×10^{-15}	3.3074×10^{-15}	0.9301	0.9609	1.0338
10300	Lymphatic nodes trunk	10.14	6.4157×10^{-15}	1.38	6.4710×10^{-15}	6.4425×10^{-15}	6.4923×10^{-15}	0.9915	0.9958	0.9882
10400	Lymphatic nodes arms	0.858	8.5933×10^{-15}	4.10	8.2770×10^{-15}	8.3984×10^{-15}	8.5267×10^{-15}	1.0382	1.0232	1.0078
10500	Lymphatic nodes legs	0.858	5.8183×10^{-16}	9.83	6.7200×10^{-16}	6.9916×10^{-16}	6.8798×10^{-16}	0.8658	0.8322	0.8457
10600	Muscle head	125.49	1.6129×10^{-15}	0.67	1.5660×10^{-15}	1.5777×10^{-15}	1.6081×10^{-15}	1.0299	1.0223	1.0030
10700	Muscle trunk	416.87	7.1037×10^{-15}	0.15	7.1950×10^{-15}	7.2130×10^{-15}	7.2368×10^{-15}	0.9873^*	0.9848^*	0.9816^*
10800	Muscle arms	72.88	5.6567×10^{-15}	0.46	5.5250×10^{-15}	5.5224×10^{-15}	5.5478×10^{-15}	1.0238^*	1.0243^*	1.0196^*
10900	Muscle legs	204.51	8.8724×10^{-16}	0.74	9.1360×10^{-16}	9.2461×10^{-16}	9.1959×10^{-16}	0.9711^*	0.9596^*	0.9648^*
11000	Oesophagus wall(0-190)	0.344	1.3470×10^{-14}	2.24	1.2670×10^{-14}	1.3343×10^{-14}	1.2656×10^{-14}	1.0631	1.0095	1.0643
11001	Oesophagus wall(190-200)	0.0188	1.4594×10^{-14}	6.62	1.3020×10^{-14}	1.4365×10^{-14}	1.2702×10^{-14}	1.1209	1.0160	1.1490
11002	Oesophagus wall(200-surface)	2.13	1.4133×10^{-14}	1.74	1.3620×10^{-14}	1.4016×10^{-14}	1.3881×10^{-14}	1.0377	1.0083	1.0182
11003	Oesophagus contents	2.02	1.3364×10^{-14}	2.13	1.2640×10^{-14}	1.2997×10^{-14}	1.2946×10^{-14}	1.0573	1.0282	1.0323
11300	Pancreas	7.25	9.0663×10^{-15}	1.04	9.1970×10^{-15}	9.0353×10^{-15}	9.2857×10^{-15}	0.9858	1.0034	0.9764
11400	Pituitary gland	0.108	7.8023×10^{-16}	31.06	6.9370×10^{-16}	8.4642×10^{-16}	1.2751×10^{-15}	1.1247	0.9218	0.6119
11500	Prostate	0.868	2.0168×10^{-15}	6.58	1.7500×10^{-15}	1.8842×10^{-15}	1.9089×10^{-15}	1.1524	1.0704	1.0565
11600	RST head	170.79	1.6022×10^{-15}	0.62	1.5940×10^{-15}	1.5911×10^{-15}	1.5905×10^{-15}	1.0052^*	1.0070^*	1.0074
11700	RST trunk	645.28	1.0417×10^{-14}	0.08	1.0490×10^{-14}	1.0460×10^{-14}	1.0526×10^{-14}	0.9930^*	0.9958^*	0.9896^*
11800	RST arms	53.00	6.6577×10^{-15}	0.46	6.3900×10^{-15}	6.4344×10^{-15}	6.4382×10^{-15}	1.0419^*	1.0347^*	1.0341^*
11900	RST legs	135.01	9.5994×10^{-16}	0.91	9.7660×10^{-16}	9.7566×10^{-16}	9.8912×10^{-16}	0.9829^*	0.9839^*	0.9705^*
12000	Salivary glands left	3.25	2.3288×10^{-15}	3.16	2.4530×10^{-15}	2.4663×10^{-15}	2.3659×10^{-15}	0.9494	0.9442	0.9843
12100	Salivary glandss right	3.25	2.5840×10^{-15}	3.17	2.6200×10^{-15}	2.7270×10^{-15}	2.6215×10^{-15}	0.9863	0.9476	0.9857
12200	Skin head insensitive	39.28	1.0853×10^{-15}	1.09	1.0730×10^{-15}	1.0637×10^{-15}	1.0555×10^{-15}	1.0115	1.0203	1.0283
12201	Skin head sensitive(40-100)	2.98	9.0071×10^{-16}	2.20	9.1530×10^{-16}	9.4460×10^{-16}	9.0069×10^{-16}	0.9841	0.9535	1.0000
12300	Skin trunk insensitive	65.51	7.1229×10^{-15}	0.32	7.2000×10^{-15}	7.2483×10^{-15}	7.1919×10^{-15}	0.9893^*	0.9827^*	0.9904^*
12301	Skin trunk sensitive(40-100)	4.97	6.2247×10^{-15}	0.75	6.2170×10^{-15}	6.3174×10^{-15}	6.2562×10^{-15}	1.0012^*	0.9853^*	0.9950^*
12400	Skin arms insensitive	24.91	5.2356×10^{-15}	0.45	5.1110×10^{-15}	5.0682×10^{-15}	5.0602×10^{-15}	1.0244^*	1.0330^*	1.0347^*
12401	Skin arms sensitive(40-100)	1.95	4.7233×10^{-15}	1.02	4.6280×10^{-15}	4.6522×10^{-15}	4.6658×10^{-15}	1.0206^*	1.0153^*	1.0123^*
12500	Skin legs insensitive	41.10	6.5938×10^{-16}	1.05	6.5420×10^{-16}	6.5729×10^{-16}	6.6378×10^{-16}	1.0079^*	1.0032^*	0.9934^*
12501	Skin legs sensitive(40-100)	3.17	5.4551×10^{-16}	3.02	5.2600×10^{-16}	5.5674×10^{-16}	5.7409×10^{-16}	1.0371^*	0.9798^*	0.9502^*
12600	Spinal cord	6.11	8.5017×10^{-15}	0.98	8.3040×10^{-15}	8.5717×10^{-15}	8.3817×10^{-15}	1.0238	0.9918	1.0143
12700	Spleen	13.86	9.0634×10^{-15}	0.76	8.9560×10^{-15}	9.0018×10^{-15}	9.0303×10^{-15}	1.0120	1.0068	1.0037

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
12849	Unerupted deciduous enamel	0.19	2.3347×10^{-15}	13.89	2.1550×10^{-15}	2.2778×10^{-15}	2.1129×10^{-15}	1.0834	1.0250	1.1050
12851	Unerupted deciduous dentin	0.425	2.0427×10^{-15}	5.94	2.3470×10^{-15}	2.3225×10^{-15}	2.0648×10^{-15}	0.8703	0.8795	0.9893
12853	Deciduous cementum	0.0337	2.7400×10^{-15}	24.11	2.2060×10^{-15}	2.4524×10^{-15}	2.2385×10^{-15}	1.2421	1.1172	1.2240
12855	Deciduous pulp	0.0607	2.2858×10^{-15}	21.73	3.1030×10^{-15}	2.5715×10^{-15}	2.5063×10^{-15}	0.7367	0.8889	0.9120
12900	Testis left	0.443	1.2759×10^{-15}	9.81	1.1080×10^{-15}	1.3591×10^{-15}	1.4681×10^{-15}	1.1515	0.9388	0.8691
13000	Testis right	0.443	1.3902×10^{-15}	11.29	1.0670×10^{-15}	1.2623×10^{-15}	1.2132×10^{-15}	1.3029	1.1013	1.1459
13100	Thymus	14.10	7.1613×10^{-15}	0.81	7.1430×10^{-15}	7.0993×10^{-15}	7.0154×10^{-15}	1.0026	1.0087	1.0208
13200	Thyroid	1.49	3.9347×10^{-15}	5.20	3.7620×10^{-15}	3.7449×10^{-15}	3.6955×10^{-15}	1.0459	1.0507	1.0647
13300	Tongue upper(food)	3.05	2.4818×10^{-15}	3.99	2.5710×10^{-15}	2.5425×10^{-15}	2.4185×10^{-15}	0.9653	0.9761	1.0262
13301	Tongue lower	0.736	2.8868×10^{-15}	5.21	2.6550×10^{-15}	2.7692×10^{-15}	2.7188×10^{-15}	1.0873	1.0425	1.0618
13400	Tonsils	0.108	2.4460×10^{-15}	15.24	2.4190×10^{-15}	2.2318×10^{-15}	1.9485×10^{-15}	1.0112	1.0960	1.2553
13500	Ureter left	0.418	5.5620×10^{-15}	4.82	5.3680×10^{-15}	6.1971×10^{-15}	5.7010×10^{-15}	1.0361	0.8975	0.9756
13600	Ureter right	0.418	7.7282×10^{-15}	2.32	7.5200×10^{-15}	7.3063×10^{-15}	7.2050×10^{-15}	1.0277	1.0577	1.0726
13700	Urinary bladder wall insensitive	3.57	2.6394×10^{-15}	2.63	2.5890×10^{-15}	2.6970×10^{-15}	2.6058×10^{-15}	1.0195	0.9787	1.0129
13701	Urinary bladder wall sensitive(54-232)	0.51	2.6359×10^{-15}	5.32	2.6120×10^{-15}	2.7021×10^{-15}	2.7163×10^{-15}	1.0091	0.9755	0.9704
13800	Urinary bladder content	12.40	2.7841×10^{-15}	1.14	2.6600×10^{-15}	2.7557×10^{-15}	2.7720×10^{-15}	1.0466	1.0103	1.0044
14000	Air inside body	0.00261	2.4208×10^{-15}	29.90	2.1110×10^{-15}	6.1014×10^{-15}	1.7759×10^{-15}	1.1467	0.3968	1.3631

* For tissues bookkept as head, trunk, arm and leg pieces (muscle, residual soft tissue, skin, large vessels), the volumes tabulated in the distributed cell tables partition the tissue at different boundaries from the transported mesh, and the reference output is normalised by the tabulated volume. The marked ratio therefore compares two different partitions of the same tissue, and is left out of the summary statistics of Table S2.

Table S8. 00M_External. Absorbed dose per primary history, Gy. σ is the FLUKA statistical uncertainty. Ratios are FLUKA / reference.

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
100	Adrenal left	3.60	2.1847×10^{-17}	28.73	3.9760×10^{-17}	4.0106×10^{-17}	3.7940×10^{-17}	0.5495	0.5447	0.5758
200	Adrenal right	3.60	3.4657×10^{-17}	24.55	3.7680×10^{-17}	2.7587×10^{-17}	1.7387×10^{-17}	0.9198	1.2563	1.9933
300	ET1(0-8)	0.000741	0.0000×10^{0}	0.00	–	–	–	–	–	–
301	ET1(8-40)	0.00304	0.0000×10^{0}	0.00	–	–	–	–	–	–
302	ET1(40-50)	0.000976	0.0000×10^{0}	0.00	–	–	–	–	–	–
303	ET1(50-Surface)	0.0968	0.0000×10^{0}	0.00	–	–	–	–	–	–
400	ET2(-15-0)	0.0327	3.1925×10^{-17}	40.78	1.7690×10^{-17}	2.4328×10^{-17}	7.4300×10^{-17}	1.8047	1.3123	0.4297
401	ET2(0-40)	0.0938	6.9149×10^{-17}	81.70	1.5470×10^{-17}	2.0771×10^{-17}	5.4764×10^{-17}	4.4699	3.3291	1.2627
402	ET2(40-50)	0.0237	2.7986×10^{-17}	67.86	7.9450×10^{-18}	1.3769×10^{-17}	3.2855×10^{-17}	3.5224	2.0325	0.8518
403	ET2(50-55)	0.0119	2.6487×10^{-17}	64.04	1.2840×10^{-17}	4.2108×10^{-17}	3.4246×10^{-17}	2.0628	0.6290	0.7734
404	ET2(55-65)	0.0238	2.4053×10^{-17}	62.51	1.0490×10^{-17}	2.1968×10^{-17}	9.8306×10^{-17}	2.2930	1.0949	0.2447
405	ET2(65-Surface)	2.14	3.7925×10^{-17}	39.72	3.2280×10^{-17}	3.4543×10^{-17}	3.9802×10^{-17}	1.1749	1.0979	0.9528
500	Oral mucosa tongue	0.0081	2.4179×10^{-17}	52.58	4.3420×10^{-17}	–	6.8127×10^{-17}	0.5569	–	0.3549
501	Oral mucosa moth floor	0.0098	2.0099×10^{-17}	100.00	4.8510×10^{-17}	1.6209×10^{-18}	8.7484×10^{-17}	0.4143	12.4001	0.2297
600	Oral mucosa lips and cheeks	0.00649	0.0000×10^{0}	0.00	4.4180×10^{-17}	8.0443×10^{-18}	1.1151×10^{-16}	–	–	–
700	Trachea	0.542	6.7385×10^{-17}	50.85	3.0050×10^{-17}	3.7531×10^{-17}	9.6433×10^{-19}	2.2424	1.7955	69.8779
800	BB(-11–6)	0.00159	6.6050×10^{-17}	100.00	–	–	–	–	–	–
801	BB(-6-0)	0.00204	6.7682×10^{-17}	100.00	–	–	–	–	–	–
802	BB(0-10)	0.00342	7.8102×10^{-17}	100.00	–	–	–	–	–	–
803	BB(10-35)	0.00863	9.9674×10^{-17}	100.00	1.3440×10^{-17}	–	–	7.4162	–	–
804	BB(35-40)	0.00174	1.5062×10^{-16}	100.00	–	–	–	–	–	–
805	BB(40-50)	0.0035	2.5790×10^{-16}	100.00	–	–	–	–	–	–
806	BB(50-60)	0.00352	8.6374×10^{-17}	100.00	–	–	–	–	–	–
807	BB(60-70)	0.00353	9.9829×10^{-17}	100.00	–	7.8855×10^{-17}	–	–	1.2660	–
808	BB(70-surface)	0.352	5.2025×10^{-17}	100.00	7.3250×10^{-19}	1.1082×10^{-17}	2.2945×10^{-17}	71.0239	4.6945	2.2674
900	Blood in large arteries head	0.0799	0.0000×10^{0}	0.00	–	–	3.1464×10^{-17}	–	–	–
910	Blood in large veins head	0.286	2.4304×10^{-17}	66.85	8.6250×10^{-18}	5.8277×10^{-17}	3.9075×10^{-17}	2.8179*	0.4170*	0.6220*
1000	Blood in large arteries trunk	9.46	3.0982×10^{-17}	14.32	2.6450×10^{-17}	3.5995×10^{-17}	3.1782×10^{-17}	1.1713*	0.8607*	0.9748*
1010	Blood in large veins trunk	23.75	3.7628×10^{-17}	12.81	3.3330×10^{-17}	3.4284×10^{-17}	3.3312×10^{-17}	1.1290*	1.0975*	1.1296*
1100	Blood in large arteries arms	1.76	2.3287×10^{-17}	33.89	4.0410×10^{-17}	5.3403×10^{-17}	5.2545×10^{-17}	0.5763*	0.4361*	0.4432*
1110	Blood in large veins arms	6.79	4.4227×10^{-17}	8.83	3.1650×10^{-17}	3.1194×10^{-17}	3.2382×10^{-17}	1.3974*	1.4178*	1.3658*
1200	Blood in large arteries legs	6.10	3.5483×10^{-17}	15.14	3.6310×10^{-17}	3.4783×10^{-17}	3.6071×10^{-17}	0.9772*	1.0201*	0.9837*
1210	Blood in large veins legs	21.37	3.2972×10^{-17}	11.29	4.1670×10^{-17}	3.9564×10^{-17}	3.6585×10^{-17}	0.7913*	0.8334*	0.9013*
1300	Humeri upper cortical	1.92	5.0069×10^{-17}	28.07	3.0490×10^{-17}	3.7002×10^{-17}	3.3686×10^{-17}	1.6421	1.3531	1.4863
1400	Humeri upper spogiosa	3.55	2.4095×10^{-17}	33.30	3.7640×10^{-17}	3.6939×10^{-17}	4.2307×10^{-17}	0.6402	0.6523	0.5695

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
1500	Humeri upper medullary cavity	0.322	4.7913×10^{-17}	56.34	2.7420×10^{-17}	8.0524×10^{-17}	1.8865×10^{-17}	1.7474	0.5950	2.5398
1600	Humeri lower cortical	1.85	3.2963×10^{-17}	41.61	2.9780×10^{-17}	5.6848×10^{-17}	5.1468×10^{-17}	1.1069	0.5798	0.6405
1700	Humeri lower spongiosa	2.77	3.5458×10^{-17}	31.64	3.2220×10^{-17}	4.4317×10^{-17}	3.2604×10^{-17}	1.1005	0.8001	1.0875
1800	Humeri lower medullary cavity	0.322	3.3803×10^{-17}	58.70	1.8780×10^{-17}	7.1956×10^{-17}	5.5309×10^{-17}	1.8000	0.4698	0.6112
1900	Radii cortical	1.21	2.4878×10^{-17}	47.95	1.6670×10^{-17}	2.7318×10^{-17}	3.9525×10^{-17}	1.4924	0.9107	0.6294
1910	Ulnae cortical	1.55	3.0874×10^{-17}	41.27	3.7490×10^{-17}	3.3177×10^{-17}	4.8708×10^{-17}	0.8235	0.9306	0.6339
2000	Radii spongiosa	1.70	2.6012×10^{-17}	35.11	4.1000×10^{-17}	3.3324×10^{-17}	3.1241×10^{-17}	0.6344	0.7806	0.8326
2010	Ulnae spongiosa	2.16	4.2478×10^{-17}	29.94	3.1140×10^{-17}	4.8428×10^{-17}	3.6427×10^{-17}	1.3641	0.8771	1.1661
2100	Radii medullary cavity	0.122	1.8896×10^{-17}	100.00	–	–	7.8756×10^{-17}	–	–	0.2399
2110	Ulnae medullary cavity	0.143	3.0722×10^{-17}	87.68	2.5270×10^{-17}	4.9668×10^{-17}	1.9415×10^{-17}	1.2157	0.6185	1.5824
2200	Wrists and hand bones cortical	0.841	2.4954×10^{-17}	28.85	3.0810×10^{-17}	3.7849×10^{-17}	2.8695×10^{-17}	0.8099	0.6593	0.8696
2300	Wrists and hand bones spongiosa	3.17	4.7736×10^{-17}	25.32	2.3950×10^{-17}	3.7245×10^{-17}	3.2944×10^{-17}	1.9931	1.2817	1.4490
2400	Clavicles cortical	0.915	3.6426×10^{-17}	26.57	3.2320×10^{-17}	4.9046×10^{-17}	2.8942×10^{-17}	1.1270	0.7427	1.2586
2500	Clavicles spongiosa	1.67	6.9289×10^{-17}	27.99	3.5950×10^{-17}	4.6585×10^{-17}	3.0806×10^{-17}	1.9274	1.4874	2.2492
2600	Cranium cortical	21.45	2.9998×10^{-17}	12.35	2.9520×10^{-17}	3.1336×10^{-17}	2.6348×10^{-17}	1.0162	0.9573	1.1385
2700	Cranium spongiosa	78.77	3.1917×10^{-17}	3.14	3.3870×10^{-17}	3.0670×10^{-17}	3.2697×10^{-17}	0.9423	1.0407	0.9761
2800	Femora upper cortical	3.76	3.0863×10^{-17}	24.34	3.1540×10^{-17}	2.4113×10^{-17}	2.3086×10^{-17}	0.9785	1.2800	1.3369
2900	Femora upper spongiosa	5.77	3.1901×10^{-17}	22.29	2.4380×10^{-17}	3.2425×10^{-17}	1.1913×10^{-17}	1.3085	0.9838	2.6779
3000	Femora upper medullary cavity	0.737	5.2199×10^{-17}	35.03	8.4690×10^{-17}	3.5609×10^{-17}	6.0767×10^{-17}	0.6164	1.4659	0.8590
3100	Femora lower cortical	5.51	3.0774×10^{-17}	11.48	3.5360×10^{-17}	2.2389×10^{-17}	3.1355×10^{-17}	0.8703	1.3745	0.9815
3200	Femora lower spongiosa	4.78	3.0900×10^{-17}	25.05	4.8440×10^{-17}	2.9158×10^{-17}	4.6880×10^{-17}	0.6379	1.0597	0.6591
3300	Femora lower medullary cavity	1.17	7.1069×10^{-17}	24.97	2.4390×10^{-17}	5.3475×10^{-17}	8.2503×10^{-17}	2.9139	1.3290	0.8614
3400	Tibiae cortical	4.50	4.2857×10^{-17}	23.05	2.7060×10^{-17}	3.8217×10^{-17}	4.3870×10^{-17}	1.5838	1.1214	0.9769
3410	Fibulae cortical	1.16	4.5159×10^{-17}	25.70	4.6300×10^{-17}	3.5490×10^{-17}	2.0291×10^{-17}	0.9753	1.2724	2.2255
3420	Patellae cortical	0.0374	0.0000×10^{0}	0.00	7.0600×10^{-18}	3.8647×10^{-17}	–	–	–	–
3500	Tibiae spongiosa	6.62	3.9524×10^{-17}	11.74	2.8570×10^{-17}	3.9901×10^{-17}	4.4098×10^{-17}	1.3834	0.9905	0.8963
3510	Fibulae spongiosa	1.39	5.2580×10^{-17}	27.49	4.2990×10^{-17}	1.4825×10^{-17}	2.1789×10^{-17}	1.2231	3.5468	2.4132
3520	Patellae spongiosa	0.171	4.9684×10^{-17}	100.00	1.9640×10^{-17}	–	4.4773×10^{-17}	2.5297	–	1.1097
3600	Tibiae medullary cavity	0.673	3.9463×10^{-17}	39.01	5.5840×10^{-17}	2.5722×10^{-17}	1.4491×10^{-17}	0.7067	1.5342	2.7233
3610	Fibulae medullary cavity	0.107	0.0000×10^{0}	0.00	9.3410×10^{-17}	–	–	–	–	–
3700	Ankles and foot cortical	1.29	3.4062×10^{-17}	30.74	3.0530×10^{-17}	3.3043×10^{-17}	5.1809×10^{-17}	1.1157	1.0308	0.6575
3800	Ankles and foot spongiosa	4.93	3.1881×10^{-17}	16.14	3.0150×10^{-17}	4.4063×10^{-17}	4.0005×10^{-17}	1.0574	0.7235	0.7969
3900	Mandible cortical	1.70	3.0447×10^{-17}	32.21	3.6840×10^{-17}	1.6717×10^{-17}	3.6548×10^{-17}	0.8265	1.8213	0.8331
4000	Mandible spongiosa	5.69	3.2699×10^{-17}	16.78	4.4460×10^{-17}	2.8334×10^{-17}	3.7881×10^{-17}	0.7355	1.1540	0.8632
4100	Pelvis cortical	6.15	3.6975×10^{-17}	16.88	2.4360×10^{-17}	3.5027×10^{-17}	3.2041×10^{-17}	1.5179	1.0556	1.1540
4200	Pelvis spongiosa	8.69	3.6176×10^{-17}	12.13	2.2770×10^{-17}	3.3984×10^{-17}	3.0018×10^{-17}	1.5888	1.0645	1.2052
4300	Ribs cortical	6.39	3.0179×10^{-17}	14.97	3.2340×10^{-17}	3.2414×10^{-17}	3.1427×10^{-17}	0.9332	0.9311	0.9603
4400	Ribs spongiosa	21.36	2.9530×10^{-17}	12.51	4.1310×10^{-17}	3.6066×10^{-17}	3.3511×10^{-17}	0.7148	0.8188	0.8812

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
4500	Scapulae cortical	2.45	4.5372×10^{-17}	16.12	3.8580×10^{-17}	3.1245×10^{-17}	3.2086×10^{-17}	1.1761	1.4521	1.4141
4600	Scapulae spongiosa	4.46	4.1793×10^{-17}	18.93	3.1950×10^{-17}	2.4534×10^{-17}	2.6560×10^{-17}	1.3081	1.7034	1.5735
4700	Cervical spine cortical	3.77	2.0083×10^{-17}	26.73	2.0650×10^{-17}	2.3443×10^{-17}	2.8486×10^{-17}	0.9725	0.8567	0.7050
4800	Cervical spine spongiosa	7.14	2.7666×10^{-17}	12.57	2.3530×10^{-17}	2.4599×10^{-17}	3.0321×10^{-17}	1.1758	1.1247	0.9124
4900	Thoracic spine cortical	8.53	2.1764×10^{-17}	13.63	2.4180×10^{-17}	2.5219×10^{-17}	2.6942×10^{-17}	0.9001	0.8630	0.8078
5000	Thoracic spine spongiosa	10.50	2.7907×10^{-17}	17.06	3.5210×10^{-17}	2.7511×10^{-17}	2.4504×10^{-17}	0.7926	1.0144	1.1389
5100	Lumbar spine cortical	2.73	2.8257×10^{-17}	18.59	2.1890×10^{-17}	3.8341×10^{-17}	2.1585×10^{-17}	1.2909	0.7370	1.3091
5200	Lumbar spine spongiosa	7.85	3.5797×10^{-17}	15.45	4.3660×10^{-17}	3.9943×10^{-17}	2.6438×10^{-17}	0.8199	0.8962	1.3540
5300	Sacrum cortical	1.08	3.2940×10^{-17}	56.08	1.7360×10^{-17}	1.7794×10^{-17}	1.6119×10^{-17}	1.8975	1.8512	2.0436
5400	Sacrum spongiosa	3.14	2.7933×10^{-17}	34.88	2.9110×10^{-17}	2.2851×10^{-17}	1.9518×10^{-17}	0.9596	1.2224	1.4311
5500	Sternum cortical	0.168	2.5783×10^{-18}	100.00	4.3040×10^{-18}	2.0520×10^{-17}	3.3843×10^{-17}	0.5991	0.1256	0.0762
5600	Sternum spongiosa	0.677	5.4782×10^{-17}	42.71	1.4630×10^{-17}	5.2612×10^{-17}	6.5422×10^{-17}	3.7445	1.0412	0.8374
5700	Cartilage head	53.07	3.3582×10^{-17}	7.15	3.2290×10^{-17}	3.2085×10^{-17}	3.2603×10^{-17}	1.0400	1.0466	1.0300
5800	Cartilage trunk	48.20	2.6617×10^{-17}	7.26	3.2050×10^{-17}	3.3344×10^{-17}	3.6913×10^{-17}	0.8305	0.7983	0.7211
5900	Cartilage arms	11.54	3.3139×10^{-17}	13.85	2.6560×10^{-17}	3.8522×10^{-17}	3.4348×10^{-17}	1.2477	0.8603	0.9648
6000	Cartilage legs	15.38	4.1693×10^{-17}	12.40	3.5870×10^{-17}	3.7001×10^{-17}	4.4248×10^{-17}	1.1623	1.1268	0.9423
6100	Brain	395.66	3.1978×10^{-17}	2.97	3.1530×10^{-17}	3.3110×10^{-17}	3.0472×10^{-17}	1.0142	0.9658	1.0494
6200	Breast left adipose tissue	0.087	8.6002×10^{-17}	100.00	5.2390×10^{-17}	3.8177×10^{-17}	–	1.6416	2.2527	–
6300	Breast left glandular tissue	0.058	3.5445×10^{-17}	100.00	1.1020×10^{-16}	5.3768×10^{-19}	–	0.3216	65.9214	–
6400	Breast right adipose tissue	0.087	1.8657×10^{-17}	100.00	1.1750×10^{-17}	6.9518×10^{-17}	7.7079×10^{-17}	1.5878	0.2684	0.2420
6500	Breast right glandular tissue	0.058	0.0000×10^{0}	0.00	–	1.0533×10^{-16}	2.9804×10^{-17}	–	–	–
6600	Eye lens sensitive left	0.0162	4.5152×10^{-16}	100.00	–	7.7038×10^{-17}	–	–	5.8610	–
6601	Eye lens insensitive left	0.0673	2.8829×10^{-17}	100.00	2.1160×10^{-17}	1.5448×10^{-16}	–	1.3624	0.1866	–
6700	Cornea left	0.582	1.3679×10^{-17}	67.25	6.3790×10^{-17}	2.8494×10^{-17}	1.6696×10^{-17}	0.2144	0.4801	0.8193
6701	Aqueous left	0.174	7.2595×10^{-17}	57.36	7.5730×10^{-17}	1.3997×10^{-17}	1.9016×10^{-17}	0.9586	5.1866	3.8176
6702	Vitreous left	2.41	2.4274×10^{-17}	22.80	5.2870×10^{-17}	3.7264×10^{-17}	3.3056×10^{-17}	0.4591	0.6514	0.7343
6800	Eye lens sensitive right	0.0162	0.0000×10^{0}	0.00	3.6000×10^{-18}	–	–	–	–	–
6801	Eye lens insensitive right	0.0673	1.4716×10^{-16}	100.00	1.3020×10^{-17}	–	–	11.3027	–	–
6900	Cornea right	0.582	5.5339×10^{-17}	29.05	2.5080×10^{-17}	5.7032×10^{-17}	3.7941×10^{-17}	2.2065	0.9703	1.4586
6901	Aqueous right	0.174	7.5064×10^{-17}	84.27	2.0680×10^{-17}	3.9225×10^{-17}	2.1738×10^{-17}	3.6298	1.9137	3.4531
6902	Vitreous right	2.41	4.4666×10^{-17}	20.22	3.2650×10^{-17}	2.8770×10^{-17}	4.1427×10^{-17}	1.3680	1.5525	1.0782
7000	Gall bladder wall	0.542	2.8304×10^{-17}	47.78	4.0990×10^{-17}	2.2411×10^{-17}	3.9181×10^{-17}	0.6905	1.2630	0.7224
7100	Gall bladder contents	2.80	3.5856×10^{-17}	23.32	2.7480×10^{-17}	2.6462×10^{-17}	2.6299×10^{-17}	1.3048	1.3550	1.3634
7200	Stomach wall(0-60)	0.408	4.4975×10^{-17}	21.97	2.3120×10^{-17}	1.8987×10^{-17}	2.4561×10^{-17}	1.9453	2.3687	1.8312
7201	Stomach wall(60-100)	0.273	4.1661×10^{-17}	24.52	2.1970×10^{-17}	3.1701×10^{-17}	2.2681×10^{-17}	1.8963	1.3142	1.8368
7202	Stomach wall(100-300)	1.38	5.0383×10^{-17}	25.39	2.3620×10^{-17}	3.3017×10^{-17}	3.4542×10^{-17}	2.1331	1.5260	1.4586
7203	Stomach wall(300-surface)	6.66	4.1300×10^{-17}	13.32	3.1630×10^{-17}	3.1841×10^{-17}	3.4364×10^{-17}	1.3057	1.2971	1.2018
7300	Stomach contents	40.00	3.4842×10^{-17}	5.26	3.1970×10^{-17}	3.4471×10^{-17}	3.8637×10^{-17}	1.0898	1.0108	0.9018

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
7400	Small intestine wall(0-130)	3.22	4.7534×10^{-17}	12.48	4.6030×10^{-17}	2.9842×10^{-17}	3.4867×10^{-17}	1.0327	1.5929	1.3633
7401	Small intestine wall(130-150)	0.503	3.9244×10^{-17}	23.57	4.1940×10^{-17}	2.6336×10^{-17}	3.3741×10^{-17}	0.9357	1.4901	1.1631
7402	Small intestine wall(150-200)	1.27	4.2030×10^{-17}	20.65	3.6260×10^{-17}	3.3488×10^{-17}	3.5107×10^{-17}	1.1591	1.2551	1.1972
7403	Small intestine wall(200-surface)	33.23	3.7817×10^{-17}	8.66	4.1700×10^{-17}	3.6359×10^{-17}	3.6164×10^{-17}	0.9069	1.0401	1.0457
7500	Small intestine contents(-400-0)	9.28	3.6023×10^{-17}	5.74	3.6470×10^{-17}	3.6101×10^{-17}	4.0583×10^{-17}	0.9878	0.9979	0.8876
7501	Small intestine contents(centre–400)	46.72	3.9825×10^{-17}	5.22	3.7880×10^{-17}	3.7387×10^{-17}	4.1368×10^{-17}	1.0513	1.0652	0.9627
7600	Ascending colon wall(0-280)	1.14	5.4867×10^{-17}	30.66	4.7890×10^{-17}	4.5266×10^{-17}	3.6577×10^{-17}	1.1457	1.2121	1.5000
7601	Ascending colon wall(280-300)	0.0832	6.5915×10^{-17}	34.20	3.5220×10^{-17}	3.8710×10^{-17}	1.2915×10^{-17}	1.8715	1.7028	5.1038
7602	Ascending colon wall(300-surface)	2.80	3.4196×10^{-17}	18.55	4.3160×10^{-17}	3.7374×10^{-17}	2.9626×10^{-17}	0.7923	0.9150	1.1543
7700	Ascending colon content	15.59	4.2467×10^{-17}	10.73	3.8690×10^{-17}	3.9999×10^{-17}	2.9266×10^{-17}	1.0976	1.0617	1.4511
7800	Transverse colon wall right(0-280)	0.842	2.6712×10^{-17}	36.60	3.3740×10^{-17}	6.3496×10^{-17}	7.1616×10^{-17}	0.7917	0.4207	0.3730
7801	Transverse colon wall right(280-300)	0.0618	1.1404×10^{-17}	55.10	4.1330×10^{-17}	5.1469×10^{-17}	3.2362×10^{-17}	0.2759	0.2216	0.3524
7802	Transverse colon wall right(300-surface)	3.99	4.3025×10^{-17}	25.37	3.8430×10^{-17}	3.7551×10^{-17}	6.3530×10^{-17}	1.1196	1.1458	0.6772
7900	Transverse colon contents right	8.41	3.9063×10^{-17}	22.06	5.2710×10^{-17}	3.7340×10^{-17}	3.3425×10^{-17}	0.7411	1.0461	1.1687
8000	Transverse colon wall left(0-280)	0.651	3.2754×10^{-17}	46.15	3.8950×10^{-17}	3.6510×10^{-17}	4.8340×10^{-17}	0.8409	0.8971	0.6776
8001	Transverse colon wall left(280-300)	0.0481	7.6613×10^{-17}	67.52	2.1490×10^{-17}	2.2365×10^{-17}	1.0905×10^{-16}	3.5651	3.4255	0.7026
8002	Transverse colon wall left(300-surface)	3.80	3.7200×10^{-17}	25.44	3.8230×10^{-17}	3.6460×10^{-17}	4.4702×10^{-17}	0.9731	1.0203	0.8322
8100	Transverse colon content left	5.42	4.6589×10^{-17}	12.94	5.5450×10^{-17}	4.9596×10^{-17}	4.8663×10^{-17}	0.8402	0.9394	0.9574
8200	Descending colon wall(0-280)	0.78	3.4987×10^{-17}	32.83	3.1940×10^{-17}	4.0996×10^{-17}	4.8961×10^{-17}	1.0954	0.8534	0.7146
8201	Descending colon wall(280-300)	0.0575	3.9011×10^{-17}	70.79	1.1110×10^{-17}	1.5038×10^{-17}	3.5322×10^{-17}	3.5113	2.5941	1.1044
8202	Descending colon wall(300-surface)	3.57	3.6717×10^{-17}	16.49	3.3660×10^{-17}	4.7116×10^{-17}	4.9579×10^{-17}	1.0908	0.7793	0.7406
8300	Descending colon content	6.58	3.0339×10^{-17}	20.85	3.5690×10^{-17}	3.3269×10^{-17}	3.4414×10^{-17}	0.8501	0.9119	0.8816
8400	Sigmoid colon wall(0-280)	0.965	2.5878×10^{-17}	37.32	3.1080×10^{-17}	3.6537×10^{-17}	2.0372×10^{-17}	0.8326	0.7083	1.2703
8401	Sigmoid colon wall(280-300)	0.071	1.2677×10^{-17}	58.62	3.3870×10^{-17}	2.4199×10^{-17}	3.5410×10^{-17}	0.3743	0.5239	0.3580
8402	Sigmoid colon wall(300-surface)	2.22	3.5269×10^{-17}	25.27	4.6310×10^{-17}	2.6533×10^{-17}	3.5969×10^{-17}	0.7616	1.3292	0.9805
8500	Sigmoid colon contents	8.52	2.6425×10^{-17}	16.06	2.6130×10^{-17}	3.4713×10^{-17}	2.9175×10^{-17}	1.0113	0.7612	0.9057
8600	Rectum wall(0-280)	0.418	3.5835×10^{-17}	32.51	2.2650×10^{-17}	3.3917×10^{-17}	1.6789×10^{-17}	1.5821	1.0566	2.1345
8601	Rectum wall(280-300)	0.0309	6.4917×10^{-17}	69.55	7.1640×10^{-17}	9.3863×10^{-18}	9.2802×10^{-18}	0.9061	6.9161	6.9952
8602	Rectum wall(300-surface)	0.111	1.8843×10^{-17}	45.12	4.2980×10^{-17}	1.7300×10^{-17}	6.9488×10^{-18}	0.4384	1.0892	2.7117
8603	Rectum contents	3.48	3.8781×10^{-17}	27.28	3.4930×10^{-17}	2.8028×10^{-17}	3.1428×10^{-17}	1.1102	1.3837	1.2340
8700	Heart wall	23.16	3.5263×10^{-17}	9.71	3.3680×10^{-17}	3.2152×10^{-17}	3.9365×10^{-17}	1.0470	1.0968	0.8958
8800	Blood in heart chamber	26.00	3.3871×10^{-17}	6.79	3.7850×10^{-17}	3.2486×10^{-17}	3.6329×10^{-17}	0.8949	1.0426	0.9323
8900	Kidney left cortex	9.46	3.0484×10^{-17}	19.00	3.6920×10^{-17}	3.4865×10^{-17}	3.8222×10^{-17}	0.8257	0.8743	0.7975
9000	Kidney left medulla	3.38	2.3765×10^{-17}	24.41	1.7490×10^{-17}	1.3723×10^{-17}	2.4391×10^{-17}	1.3588	1.7318	0.9743
9100	Kidney left pelvis	0.676	2.9684×10^{-17}	57.52	2.9530×10^{-17}	1.4203×10^{-17}	4.6012×10^{-17}	1.0052	2.0900	0.6451
9200	Kidney right cortex	9.46	2.8441×10^{-17}	16.78	3.6020×10^{-17}	3.7772×10^{-17}	2.6763×10^{-17}	0.7896	0.7529	1.0627

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
9300	Kidney right medulla	3.38	3.2103×10^{-17}	26.28	2.6190×10^{-17}	2.1221×10^{-17}	2.7772×10^{-17}	1.2258	1.5128	1.1559
9400	Kidney right pelvis	0.676	4.3670×10^{-18}	93.88	1.1890×10^{-17}	3.3891×10^{-17}	5.6869×10^{-17}	0.3673	0.1289	0.0768
9500	Liver	167.41	3.2548×10^{-17}	3.09	3.3840×10^{-17}	3.7272×10^{-17}	3.6943×10^{-17}	0.9618	0.8732	0.8810
9700	Lung(AI) left	28.69	3.1850×10^{-17}	9.72	3.3190×10^{-17}	2.6903×10^{-17}	3.3449×10^{-17}	0.9596	1.1839	0.9522
9900	Lung(AI) right	31.31	3.1038×10^{-17}	8.15	3.1410×10^{-17}	3.9497×10^{-17}	3.2304×10^{-17}	0.9881	0.7858	0.9608
10000	Lymphatic nodes ET	1.24	2.4421×10^{-17}	39.74	1.6270×10^{-17}	1.6022×10^{-17}	2.4731×10^{-17}	1.5010	1.5242	0.9875
10100	Lymphatic nodes thoracic	1.24	3.0668×10^{-17}	52.00	2.2620×10^{-17}	4.4691×10^{-17}	1.4797×10^{-17}	1.3558	0.6862	2.0726
10200	Lymphatic nodes head	0.429	9.3524×10^{-18}	96.70	3.9440×10^{-17}	3.6982×10^{-17}	1.9353×10^{-17}	0.2371	0.2529	0.4833
10300	Lymphatic nodes trunk	10.14	2.8180×10^{-17}	13.83	3.5410×10^{-17}	3.4519×10^{-17}	3.0210×10^{-17}	0.7958	0.8164	0.9328
10400	Lymphatic nodes arms	0.858	6.4766×10^{-17}	32.23	1.6040×10^{-17}	4.7765×10^{-17}	2.1413×10^{-17}	4.0378	1.3559	3.0246
10500	Lymphatic nodes legs	0.858	3.7422×10^{-17}	34.05	3.8730×10^{-17}	2.7089×10^{-17}	4.3826×10^{-17}	0.9662	1.3814	0.8539
10600	Muscle head	125.49	3.3430×10^{-17}	4.18	3.0760×10^{-17}	3.0316×10^{-17}	3.0959×10^{-17}	1.0868	1.1027	1.0798
10700	Muscle trunk	416.87	3.5447×10^{-17}	1.81	3.5100×10^{-17}	3.4739×10^{-17}	3.4754×10^{-17}	1.0099*	1.0204*	1.0199*
10800	Muscle arms	72.88	3.6777×10^{-17}	6.38	3.7420×10^{-17}	3.8583×10^{-17}	3.7446×10^{-17}	0.9828*	0.9532*	0.9821*
10900	Muscle legs	204.51	3.7187×10^{-17}	4.18	3.7200×10^{-17}	4.1158×10^{-17}	3.7135×10^{-17}	0.9997*	0.9035*	1.0014*
11000	Oesophagus wall(0-190)	0.344	5.3105×10^{-17}	35.10	2.4160×10^{-17}	2.5881×10^{-17}	6.0955×10^{-17}	2.1981	2.0519	0.8712
11001	Oesophagus wall(190-200)	0.0188	7.3054×10^{-18}	43.34	1.7320×10^{-17}	4.2416×10^{-17}	3.0743×10^{-17}	0.4218	0.1722	0.2376
11002	Oesophagus wall(200-surface)	2.13	2.0292×10^{-17}	31.07	3.5450×10^{-17}	2.8713×10^{-17}	2.5765×10^{-17}	0.5724	0.7067	0.7876
11003	Oesophagus contents	2.02	2.6405×10^{-17}	24.54	3.4410×10^{-17}	1.8085×10^{-17}	3.6527×10^{-17}	0.7674	1.4600	0.7229
11300	Pancreas	7.25	3.5525×10^{-17}	18.74	3.2740×10^{-17}	3.5804×10^{-17}	2.3913×10^{-17}	1.0851	0.9922	1.4856
11400	Pituitary gland	0.108	5.0425×10^{-17}	100.00	1.9680×10^{-17}	–	–	2.5623	–	–
11500	Prostate	0.868	4.6778×10^{-17}	57.77	8.9320×10^{-17}	1.7925×10^{-17}	4.1605×10^{-17}	0.5237	2.6097	1.1243
11600	RST head	170.79	3.3321×10^{-17}	3.29	3.3180×10^{-17}	3.4604×10^{-17}	3.4233×10^{-17}	1.0043*	0.9629*	0.9734
11700	RST trunk	645.28	3.5874×10^{-17}	1.87	3.5230×10^{-17}	3.5316×10^{-17}	3.4983×10^{-17}	1.0183*	1.0158*	1.0255*
11800	RST arms	53.00	3.5919×10^{-17}	6.38	3.9040×10^{-17}	3.7947×10^{-17}	3.8315×10^{-17}	0.9201*	0.9466*	0.9375*
11900	RST legs	135.01	3.4024×10^{-17}	3.07	3.6410×10^{-17}	3.5076×10^{-17}	3.6841×10^{-17}	0.9345*	0.9700*	0.9235*
12000	Salivary glands left	3.25	4.2649×10^{-17}	25.98	2.7060×10^{-17}	4.2374×10^{-17}	2.4588×10^{-17}	1.5761	1.0065	1.7345
12100	Salivary glandss right	3.25	2.9491×10^{-17}	25.01	3.6440×10^{-17}	4.0405×10^{-17}	4.3693×10^{-17}	0.8093	0.7299	0.6750
12200	Skin head insensitive	39.28	2.6905×10^{-17}	6.60	2.4960×10^{-17}	2.1995×10^{-17}	2.2361×10^{-17}	1.0779	1.2233	1.2032
12201	Skin head sensitive(40-100)	2.98	1.7059×10^{-17}	17.72	2.2500×10^{-17}	1.5757×10^{-17}	1.5135×10^{-17}	0.7582	1.0826	1.1271
12300	Skin trunk insensitive	65.51	2.4873×10^{-17}	4.11	2.5580×10^{-17}	2.4347×10^{-17}	2.5219×10^{-17}	0.9724*	1.0216*	0.9863*
12301	Skin trunk sensitive(40-100)	4.97	1.8280×10^{-17}	16.61	1.7150×10^{-17}	1.9312×10^{-17}	1.8667×10^{-17}	1.0659*	0.9466*	0.9793*
12400	Skin arms insensitive	24.91	3.0960×10^{-17}	8.06	2.6680×10^{-17}	3.2179×10^{-17}	3.1243×10^{-17}	1.1604*	0.9621*	0.9909*
12401	Skin arms sensitive(40-100)	1.95	2.1671×10^{-17}	8.95	1.7420×10^{-17}	2.3336×10^{-17}	2.5446×10^{-17}	1.2440*	0.9286*	0.8516*
12500	Skin legs insensitive	41.10	2.6640×10^{-17}	6.13	2.9800×10^{-17}	2.6979×10^{-17}	2.7986×10^{-17}	0.8940*	0.9874*	0.9519*
12501	Skin legs sensitive(40-100)	3.17	2.1542×10^{-17}	14.55	2.4040×10^{-17}	2.0022×10^{-17}	2.2833×10^{-17}	0.8961*	1.0759*	0.9435*
12600	Spinal cord	6.11	2.8578×10^{-17}	15.57	3.1790×10^{-17}	2.2658×10^{-17}	2.8014×10^{-17}	0.8990	1.2613	1.0202
12700	Spleen	13.86	2.7810×10^{-17}	16.77	2.8500×10^{-17}	3.8256×10^{-17}	3.1605×10^{-17}	0.9758	0.7269	0.8799

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
12849	Unerupted deciduous enamel	0.19	4.7926×10^{-17}	62.80	6.2350×10^{-17}	4.7141×10^{-17}	9.6970×10^{-19}	0.7687	1.0166	49.4235
12851	Unerupted deciduous dentin	0.425	3.4717×10^{-17}	65.86	4.3450×10^{-17}	4.4960×10^{-17}	1.4854×10^{-17}	0.7990	0.7722	2.3372
12853	Deciduous cementum	0.0337	9.5844×10^{-18}	66.75	3.7550×10^{-17}	5.0256×10^{-17}	–	0.2552	0.1907	–
12855	Deciduous pulp	0.0607	3.5197×10^{-17}	100.00	–	–	1.0282×10^{-16}	–	–	0.3423
12900	Testis left	0.443	1.7277×10^{-18}	96.08	–	1.4171×10^{-16}	7.0489×10^{-18}	–	0.0122	0.2451
13000	Testis right	0.443	8.5236×10^{-17}	39.09	6.5730×10^{-17}	5.4118×10^{-17}	5.4240×10^{-17}	1.2968	1.5750	1.5715
13100	Thymus	14.10	3.6217×10^{-17}	10.82	3.9280×10^{-17}	3.4993×10^{-17}	3.2122×10^{-17}	0.9220	1.0350	1.1275
13200	Thyroid	1.49	3.6488×10^{-17}	39.07	2.7070×10^{-17}	2.6379×10^{-17}	2.9988×10^{-17}	1.3479	1.3833	1.2168
13300	Tongue upper(food)	3.05	2.5457×10^{-17}	39.82	3.8660×10^{-17}	3.5812×10^{-17}	4.0584×10^{-17}	0.6585	0.7109	0.6273
13301	Tongue lower	0.736	4.9076×10^{-17}	38.34	4.3180×10^{-17}	1.8540×10^{-17}	4.7628×10^{-17}	1.1365	2.6470	1.0304
13400	Tonsils	0.108	0.0000×10^{0}	0.00	–	–	–	–	–	–
13500	Ureter left	0.418	8.2768×10^{-18}	80.33	2.8260×10^{-17}	1.2606×10^{-18}	–	0.2929	6.5656	–
13600	Ureter right	0.418	8.4646×10^{-18}	78.77	4.4690×10^{-17}	2.4664×10^{-18}	9.5193×10^{-18}	0.1894	3.4319	0.8892
13700	Urinary bladder wall insensitive	3.57	2.1180×10^{-17}	26.42	3.2000×10^{-17}	5.6275×10^{-17}	4.0054×10^{-17}	0.6619	0.3764	0.5288
13701	Urinary bladder wall sensitive(54-232)	0.51	2.3989×10^{-17}	35.74	5.6180×10^{-17}	4.5297×10^{-17}	3.2528×10^{-17}	0.4270	0.5296	0.7375
13800	Urinary bladder content	12.40	3.8137×10^{-17}	9.17	3.8840×10^{-17}	4.5127×10^{-17}	3.7083×10^{-17}	0.9819	0.8451	1.0284
14000	Air inside body	0.00261	3.1367×10^{-17}	56.57	2.1360×10^{-18}	6.3831×10^{-18}	2.6488×10^{-17}	14.6851	4.9141	1.1842

* For tissues bookkept as head, trunk, arm and leg pieces (muscle, residual soft tissue, skin, large vessels), the volumes tabulated in the distributed cell tables partition the tissue at different boundaries from the transported mesh, and the reference output is normalised by the tabulated volume. The marked ratio therefore compares two different partitions of the same tissue, and is left out of the summary statistics of Table S2.

Table S9. 00F_Internal. Absorbed dose per primary history, Gy. σ is the FLUKA statistical uncertainty. Ratios are FLUKA / reference.

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
100	Adrenal left	3.60	1.3057×10^{-14}	1.52	–	1.3174×10^{-14}	1.3174×10^{-14}	–	0.9911	0.9911
200	Adrenal right	3.60	2.6409×10^{-14}	1.19	–	2.6982×10^{-14}	2.6917×10^{-14}	–	0.9788	0.9811
300	ET1(0-8)	0.000738	2.1952×10^{-15}	77.99	–	1.8992×10^{-15}	7.4145×10^{-16}	–	1.1559	2.9607
301	ET1(8-40)	0.00303	3.8700×10^{-15}	51.83	–	9.5276×10^{-16}	6.4815×10^{-16}	–	4.0618	5.9708
302	ET1(40-50)	0.000972	3.4255×10^{-15}	59.38	–	4.1245×10^{-16}	5.9387×10^{-16}	–	8.3053	5.7681
303	ET1(50-Surface)	0.0954	1.5497×10^{-15}	24.45	–	1.4663×10^{-15}	1.9158×10^{-15}	–	1.0568	0.8089
400	ET2(-15-0)	0.033	1.2576×10^{-15}	12.76	–	1.8274×10^{-15}	1.9132×10^{-15}	–	0.6882	0.6573
401	ET2(0-40)	0.0945	1.4700×10^{-15}	10.43	–	1.8176×10^{-15}	1.6842×10^{-15}	–	0.8088	0.8728
402	ET2(40-50)	0.0239	1.4655×10^{-15}	11.16	–	2.2906×10^{-15}	1.8974×10^{-15}	–	0.6398	0.7724
403	ET2(50-55)	0.012	1.9384×10^{-15}	22.39	–	2.3059×10^{-15}	2.0665×10^{-15}	–	0.8406	0.9380
404	ET2(55-65)	0.024	1.5995×10^{-15}	17.42	–	2.0136×10^{-15}	1.5917×10^{-15}	–	0.7943	1.0049
405	ET2(65-Surface)	1.68	1.6706×10^{-15}	3.36	–	1.6356×10^{-15}	1.6891×10^{-15}	–	1.0214	0.9891
500	Oral mucosa tongue	0.00811	3.0007×10^{-15}	25.04	–	2.6719×10^{-15}	2.9820×10^{-15}	–	1.1230	1.0063
501	Oral mucosa moth floor	0.0098	1.8835×10^{-15}	25.07	–	3.1522×10^{-15}	1.9718×10^{-15}	–	0.5975	0.9552
600	Oral mucosa lips and cheeks	0.00641	3.1163×10^{-15}	22.99	–	1.7979×10^{-15}	3.0329×10^{-15}	–	1.7333	1.0275
700	Trachea	0.542	5.2120×10^{-15}	5.05	–	5.1104×10^{-15}	5.4805×10^{-15}	–	1.0199	0.9510
800	BB(-11–6)	0.00159	9.0897×10^{-15}	24.54	–	1.0678×10^{-14}	7.2627×10^{-15}	–	0.8512	1.2516
801	BB(-6-0)	0.00203	8.4182×10^{-15}	14.60	–	7.8811×10^{-15}	7.8845×10^{-15}	–	1.0681	1.0677
802	BB(0-10)	0.00341	1.0103×10^{-14}	11.76	–	9.8522×10^{-15}	7.7341×10^{-15}	–	1.0254	1.3062
803	BB(10-35)	0.0086	7.9861×10^{-15}	13.93	–	1.0046×10^{-14}	9.3925×10^{-15}	–	0.7949	0.8503
804	BB(35-40)	0.00173	8.8915×10^{-15}	24.25	–	1.4005×10^{-14}	6.2853×10^{-15}	–	0.6349	1.4147
805	BB(40-50)	0.00348	1.2326×10^{-14}	14.66	–	1.3498×10^{-14}	8.3558×10^{-15}	–	0.9132	1.4752
806	BB(50-60)	0.0035	1.2143×10^{-14}	15.90	–	1.0576×10^{-14}	9.9885×10^{-15}	–	1.1481	1.2157
807	BB(60-70)	0.00352	8.6650×10^{-15}	15.60	–	1.0130×10^{-14}	7.6806×10^{-15}	–	0.8553	1.1282
808	BB(70-surface)	0.35	9.2672×10^{-15}	7.41	–	8.9965×10^{-15}	9.5453×10^{-15}	–	1.0301	0.9709
900	Blood in large arteries head	0.0766	2.5351×10^{-15}	13.02	–	2.0640×10^{-15}	2.4135×10^{-15}	–	1.2283	1.0504
910	Blood in large veins head	0.282	2.2730×10^{-15}	6.46	–	2.3949×10^{-15}	2.3152×10^{-15}	–	0.9491*	0.9818*
1000	Blood in large arteries trunk	9.46	8.8301×10^{-15}	1.11	–	8.9272×10^{-15}	9.2616×10^{-15}	–	0.9891*	0.9534*
1010	Blood in large veins trunk	23.77	1.0031×10^{-14}	0.62	–	1.0358×10^{-14}	1.0348×10^{-14}	–	0.9684*	0.9694*
1100	Blood in large arteries arms	1.76	7.0330×10^{-15}	2.71	–	7.0809×10^{-15}	6.9932×10^{-15}	–	0.9932*	1.0057*
1110	Blood in large veins arms	6.80	6.0857×10^{-15}	1.25	–	6.0192×10^{-15}	6.0734×10^{-15}	–	1.0110*	1.0020*
1200	Blood in large arteries legs	6.10	7.4960×10^{-16}	3.15	–	7.6640×10^{-16}	7.7983×10^{-16}	–	0.9781*	0.9612*
1210	Blood in large veins legs	21.35	8.2181×10^{-16}	2.01	–	8.4987×10^{-16}	8.7009×10^{-16}	–	0.9670*	0.9445*
1300	Humeri upper cortical	1.92	4.3574×10^{-15}	3.14	–	4.3298×10^{-15}	4.2446×10^{-15}	–	1.0064	1.0266
1400	Humeri upper spogiosa	3.55	3.7760×10^{-15}	1.58	–	3.7821×10^{-15}	3.8131×10^{-15}	–	0.9984	0.9903

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
1500	Humeri upper medullary cavity	0.322	4.5978×10^{-15}	10.35	–	4.8762×10^{-15}	4.5943×10^{-15}	–	0.9429	1.0008
1600	Humeri lower cortical	1.85	5.3108×10^{-15}	2.91	–	5.4011×10^{-15}	5.3305×10^{-15}	–	0.9833	0.9963
1700	Humeri lower spongiosa	2.77	5.6121×10^{-15}	1.68	–	5.8665×10^{-15}	5.6610×10^{-15}	–	0.9566	0.9914
1800	Humeri lower medullary cavity	0.322	5.1234×10^{-15}	8.92	–	6.5366×10^{-15}	5.4389×10^{-15}	–	0.7838	0.9420
1900	Radii cortical	1.21	5.8164×10^{-15}	3.22	–	5.8765×10^{-15}	6.1244×10^{-15}	–	0.9898	0.9497
1910	Ulnae cortical	1.55	5.3030×10^{-15}	2.45	–	5.1964×10^{-15}	5.5218×10^{-15}	–	1.0205	0.9604
2000	Radii spongiosa	1.70	5.2732×10^{-15}	3.91	–	5.3371×10^{-15}	5.7221×10^{-15}	–	0.9880	0.9215
2010	Ulnae spongiosa	2.16	5.6519×10^{-15}	2.86	–	6.0063×10^{-15}	5.4528×10^{-15}	–	0.9410	1.0365
2100	Radii medullary cavity	0.122	6.6508×10^{-15}	10.53	–	6.4233×10^{-15}	5.8140×10^{-15}	–	1.0354	1.1439
2110	Ulnae medullary cavity	0.143	5.3046×10^{-15}	13.18	–	5.2563×10^{-15}	5.5046×10^{-15}	–	1.0092	0.9637
2200	Wrists and hand bones cortical	0.841	2.5983×10^{-15}	3.93	–	2.7752×10^{-15}	2.6703×10^{-15}	–	0.9362	0.9730
2300	Wrists and hand bones spongiosa	3.17	2.8178×10^{-15}	2.95	–	2.8973×10^{-15}	2.8522×10^{-15}	–	0.9726	0.9879
2400	Clavicles cortical	0.915	3.6989×10^{-15}	3.17	–	3.6363×10^{-15}	3.5601×10^{-15}	–	1.0172	1.0390
2500	Clavicles spongiosa	1.67	3.7566×10^{-15}	3.11	–	4.0038×10^{-15}	3.9807×10^{-15}	–	0.9383	0.9437
2600	Cranium cortical	21.46	7.2137×10^{-16}	1.47	–	7.3404×10^{-16}	6.9095×10^{-16}	–	0.9828	1.0440
2700	Cranium spongiosa	78.79	1.2212×10^{-15}	1.20	–	1.2186×10^{-15}	1.2450×10^{-15}	–	1.0022	0.9809
2800	Femora upper cortical	3.76	1.5515×10^{-15}	2.04	–	1.5166×10^{-15}	1.5889×10^{-15}	–	1.0230	0.9765
2900	Femora upper spongiosa	5.77	1.9193×10^{-15}	3.19	–	1.9788×10^{-15}	1.8907×10^{-15}	–	0.9699	1.0151
3000	Femora upper medullary cavity	0.737	1.6038×10^{-15}	9.07	–	1.5280×10^{-15}	1.5969×10^{-15}	–	1.0496	1.0043
3100	Femora lower cortical	5.51	9.2150×10^{-16}	5.46	–	8.8218×10^{-16}	9.3940×10^{-16}	–	1.0446	0.9809
3200	Femora lower spongiosa	4.78	7.3608×10^{-16}	4.56	–	7.5662×10^{-16}	7.6812×10^{-16}	–	0.9729	0.9583
3300	Femora lower medullary cavity	1.17	1.0760×10^{-15}	5.29	–	1.0587×10^{-15}	1.0522×10^{-15}	–	1.0163	1.0226
3400	Tibiae cortical	4.50	3.4600×10^{-16}	6.18	–	3.8003×10^{-16}	3.3536×10^{-16}	–	0.9105	1.0317
3410	Fibulae cortical	1.16	2.1296×10^{-16}	13.75	–	3.1819×10^{-16}	3.6339×10^{-16}	–	0.6693	0.5860
3420	Patellae cortical	0.0374	1.7369×10^{-16}	38.64	–	4.9087×10^{-16}	9.1486×10^{-16}	–	0.3538	0.1899
3500	Tibiae spongiosa	6.62	3.8275×10^{-16}	8.51	–	4.2036×10^{-16}	3.6868×10^{-16}	–	0.9105	1.0382
3510	Fibulae spongiosa	1.39	3.9481×10^{-16}	14.83	–	2.5605×10^{-16}	3.1045×10^{-16}	–	1.5420	1.2717
3520	Patellae spongiosa	0.171	3.7949×10^{-16}	18.75	–	1.0659×10^{-15}	5.3882×10^{-16}	–	0.3560	0.7043
3600	Tibiae medullary cavity	0.673	4.6230×10^{-16}	13.78	–	4.3222×10^{-16}	3.5364×10^{-16}	–	1.0696	1.3073
3610	Fibulae medullary cavity	0.107	2.2426×10^{-16}	36.98	–	2.0888×10^{-16}	5.0608×10^{-16}	–	1.0736	0.4431
3700	Ankles and foot cortical	1.29	2.3924×10^{-16}	8.71	–	2.1989×10^{-16}	2.3541×10^{-16}	–	1.0880	1.0163
3800	Ankles and foot spongiosa	4.93	2.3987×10^{-16}	7.72	–	2.5576×10^{-16}	2.1350×10^{-16}	–	0.9379	1.1235
3900	Mandible cortical	1.70	2.4250×10^{-15}	4.42	–	2.3107×10^{-15}	2.4257×10^{-15}	–	1.0495	0.9997
4000	Mandible spongiosa	5.69	2.6356×10^{-15}	2.27	–	2.4927×10^{-15}	2.4668×10^{-15}	–	1.0574	1.0684
4100	Pelvis cortical	6.15	2.9603×10^{-15}	1.99	–	3.0189×10^{-15}	3.0296×10^{-15}	–	0.9806	0.9771
4200	Pelvis spongiosa	8.69	2.9373×10^{-15}	1.46	–	3.0522×10^{-15}	3.0375×10^{-15}	–	0.9624	0.9670
4300	Ribs cortical	6.39	1.2275×10^{-14}	0.80	–	1.2318×10^{-14}	1.2242×10^{-14}	–	0.9965	1.0027
4400	Ribs spongiosa	21.36	1.2556×10^{-14}	0.50	–	1.2658×10^{-14}	1.2483×10^{-14}	–	0.9919	1.0058

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
4500	Scapulae cortical	2.45	4.2087×10^{-15}	2.11	–	4.0616×10^{-15}	4.1944×10^{-15}	–	1.0362	1.0034
4600	Scapulae spongiosa	4.46	4.1867×10^{-15}	2.34	–	4.0779×10^{-15}	4.3734×10^{-15}	–	1.0267	0.9573
4700	Cervical spine cortical	3.77	2.7162×10^{-15}	2.60	–	2.6803×10^{-15}	2.7878×10^{-15}	–	1.0134	0.9743
4800	Cervical spine spongiosa	7.14	2.7115×10^{-15}	2.20	–	2.7736×10^{-15}	2.7860×10^{-15}	–	0.9776	0.9733
4900	Thoracic spine cortical	8.54	1.0587×10^{-14}	0.81	–	1.0555×10^{-14}	1.0592×10^{-14}	–	1.0030	0.9995
5000	Thoracic spine spongiosa	10.50	1.1424×10^{-14}	0.88	–	1.1209×10^{-14}	1.1429×10^{-14}	–	1.0191	0.9995
5100	Lumbar spine cortical	2.73	8.5054×10^{-15}	0.96	–	8.2982×10^{-15}	8.6311×10^{-15}	–	1.0250	0.9854
5200	Lumbar spine spongiosa	7.85	9.1315×10^{-15}	0.58	–	9.1773×10^{-15}	9.0994×10^{-15}	–	0.9950	1.0035
5300	Sacrum cortical	1.08	3.1416×10^{-15}	3.35	–	3.3661×10^{-15}	3.2257×10^{-15}	–	0.9333	0.9739
5400	Sacrum spongiosa	3.14	3.2572×10^{-15}	2.23	–	3.3280×10^{-15}	3.3289×10^{-15}	–	0.9787	0.9785
5500	Sternum cortical	0.168	1.0531×10^{-14}	4.26	–	1.0206×10^{-14}	1.0226×10^{-14}	–	1.0318	1.0298
5600	Sternum spongiosa	0.677	1.0902×10^{-14}	2.25	–	9.8677×10^{-15}	1.0339×10^{-14}	–	1.1048	1.0544
5700	Cartilage head	52.99	9.1847×10^{-16}	1.03	–	9.1715×10^{-16}	9.0927×10^{-16}	–	1.0014	1.0101
5800	Cartilage trunk	48.20	1.1412×10^{-14}	0.43	–	1.1454×10^{-14}	1.1449×10^{-14}	–	0.9963	0.9968
5900	Cartilage arms	11.54	4.2811×10^{-15}	1.20	–	4.3301×10^{-15}	4.3141×10^{-15}	–	0.9887	0.9924
6000	Cartilage legs	15.38	6.9979×10^{-16}	1.97	–	6.7424×10^{-16}	6.8466×10^{-16}	–	1.0379	1.0221
6100	Brain	395.66	7.3491×10^{-16}	0.34	–	7.3714×10^{-16}	7.4067×10^{-16}	–	0.9970	0.9922
6200	Breast left adipose tissue	0.087	1.0512×10^{-14}	6.61	–	1.2257×10^{-14}	9.7990×10^{-15}	–	0.8576	1.0727
6300	Breast left glandular tissue	0.058	1.0153×10^{-14}	5.99	–	1.1496×10^{-14}	9.9020×10^{-15}	–	0.8832	1.0254
6400	Breast right adipose tissue	0.087	1.5664×10^{-14}	6.88	–	1.7864×10^{-14}	1.6031×10^{-14}	–	0.8768	0.9771
6500	Breast right glandular tissue	0.058	1.5963×10^{-14}	5.43	–	1.5206×10^{-14}	1.5765×10^{-14}	–	1.0498	1.0126
6600	Eye lens sensitive left	0.0162	1.0659×10^{-15}	29.58	–	5.2327×10^{-16}	1.7052×10^{-15}	–	2.0370	0.6251
6601	Eye lens insensitive left	0.0673	1.4837×10^{-15}	29.22	–	1.1807×10^{-15}	1.2343×10^{-15}	–	1.2566	1.2020
6700	Cornea left	0.582	1.2608×10^{-15}	6.13	–	1.2874×10^{-15}	1.2095×10^{-15}	–	0.9794	1.0424
6701	Aqueous left	0.174	1.0868×10^{-15}	24.42	–	1.4658×10^{-15}	1.2945×10^{-15}	–	0.7414	0.8395
6702	Vitreous left	2.41	1.2328×10^{-15}	5.97	–	1.2394×10^{-15}	1.2039×10^{-15}	–	0.9947	1.0240
6800	Eye lens sensitive right	0.0162	5.1551×10^{-16}	52.94	–	1.8997×10^{-15}	1.9102×10^{-15}	–	0.2714	0.2699
6801	Eye lens insensitive right	0.0673	8.6855×10^{-16}	28.87	–	6.1056×10^{-16}	1.1816×10^{-15}	–	1.4226	0.7351
6900	Cornea right	0.582	1.5636×10^{-15}	12.18	–	1.3786×10^{-15}	1.2598×10^{-15}	–	1.1341	1.2411
6901	Aqueous right	0.174	8.1574×10^{-16}	18.73	–	9.9282×10^{-16}	1.2874×10^{-15}	–	0.8216	0.6336
6902	Vitreous right	2.41	1.3795×10^{-15}	8.07	–	1.4589×10^{-15}	1.3260×10^{-15}	–	0.9456	1.0404
7000	Gall bladder wall	0.542	4.0168×10^{-14}	0.89	–	3.9769×10^{-14}	4.0319×10^{-14}	–	1.0100	0.9963
7100	Gall bladder contents	2.80	4.0480×10^{-14}	0.50	–	4.1099×10^{-14}	4.1245×10^{-14}	–	0.9849	0.9814
7200	Stomach wall(0-60)	0.408	2.2222×10^{-14}	1.30	–	2.1914×10^{-14}	2.1614×10^{-14}	–	1.0140	1.0281
7201	Stomach wall(60-100)	0.273	2.2404×10^{-14}	1.00	–	2.2349×10^{-14}	2.1633×10^{-14}	–	1.0024	1.0356
7202	Stomach wall(100-300)	1.38	2.2021×10^{-14}	0.98	–	2.1872×10^{-14}	2.1745×10^{-14}	–	1.0068	1.0127
7203	Stomach wall(300-surface)	6.66	2.1964×10^{-14}	0.68	–	2.2275×10^{-14}	2.2059×10^{-14}	–	0.9860	0.9957
7300	Stomach contents	40.00	2.0293×10^{-14}	0.36	–	2.0075×10^{-14}	2.0157×10^{-14}	–	1.0109	1.0067

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
7400	Small intestine wall(0-130)	3.22	8.8067×10^{-15}	0.81	–	8.6646×10^{-15}	8.7031×10^{-15}	–	1.0164	1.0119
7401	Small intestine wall(130-150)	0.503	8.7906×10^{-15}	1.13	–	8.9524×10^{-15}	8.7368×10^{-15}	–	0.9819	1.0062
7402	Small intestine wall(150-200)	1.27	8.7732×10^{-15}	1.05	–	8.8424×10^{-15}	8.6057×10^{-15}	–	0.9922	1.0195
7403	Small intestine wall(200-surface)	33.22	8.4852×10^{-15}	0.62	–	8.4435×10^{-15}	8.5355×10^{-15}	–	1.0049	0.9941
7500	Small intestine contents(-400-0)	9.27	8.5585×10^{-15}	0.83	–	8.4938×10^{-15}	8.6629×10^{-15}	–	1.0076	0.9880
7501	Small intestine contents(centre–400)	46.73	8.3860×10^{-15}	0.62	–	8.4030×10^{-15}	8.5216×10^{-15}	–	0.9980	0.9841
7600	Ascending colon wall(0-280)	1.14	6.1973×10^{-15}	1.64	–	6.0662×10^{-15}	6.5775×10^{-15}	–	1.0216	0.9422
7601	Ascending colon wall(280-300)	0.0832	6.3852×10^{-15}	3.38	–	6.3676×10^{-15}	6.3547×10^{-15}	–	1.0028	1.0048
7602	Ascending colon wall(300-surface)	2.80	6.4961×10^{-15}	2.10	–	6.5780×10^{-15}	6.5088×10^{-15}	–	0.9876	0.9980
7700	Ascending colon content	15.59	5.9818×10^{-15}	0.91	–	5.9050×10^{-15}	5.9808×10^{-15}	–	1.0130	1.0002
7800	Transverse colon wall right(0-280)	0.842	2.2265×10^{-14}	1.55	–	2.3231×10^{-14}	2.2691×10^{-14}	–	0.9584	0.9812
7801	Transverse colon wall right(280-300)	0.0618	2.3148×10^{-14}	2.22	–	2.4983×10^{-14}	2.2744×10^{-14}	–	0.9265	1.0177
7802	Transverse colon wall right(300-surface)	3.99	2.2038×10^{-14}	0.73	–	2.2353×10^{-14}	2.2132×10^{-14}	–	0.9859	0.9957
7900	Transverse colon contents right	8.41	2.2778×10^{-14}	0.63	–	2.3058×10^{-14}	2.3125×10^{-14}	–	0.9879	0.9850
8000	Transverse colon wall left(0-280)	0.651	9.3962×10^{-15}	1.35	–	9.0112×10^{-15}	9.1225×10^{-15}	–	1.0427	1.0300
8001	Transverse colon wall left(280-300)	0.0481	9.1788×10^{-15}	3.87	–	9.1548×10^{-15}	9.0880×10^{-15}	–	1.0026	1.0100
8002	Transverse colon wall left(300-surface)	3.80	8.9610×10^{-15}	1.55	–	9.1375×10^{-15}	8.8978×10^{-15}	–	0.9807	1.0071
8100	Transverse colon content left	5.42	9.1109×10^{-15}	1.18	–	9.3470×10^{-15}	9.0426×10^{-15}	–	0.9747	1.0076
8200	Descending colon wall(0-280)	0.78	4.5512×10^{-15}	3.97	–	4.6849×10^{-15}	4.6866×10^{-15}	–	0.9715	0.9711
8201	Descending colon wall(280-300)	0.0575	4.4855×10^{-15}	6.73	–	5.5629×10^{-15}	4.7373×10^{-15}	–	0.8063	0.9468
8202	Descending colon wall(300-surface)	3.57	4.5767×10^{-15}	1.82	–	4.6478×10^{-15}	4.6322×10^{-15}	–	0.9847	0.9880
8300	Descending colon content	6.58	4.5140×10^{-15}	1.91	–	4.4481×10^{-15}	4.5153×10^{-15}	–	1.0148	0.9997
8400	Sigmoid colon wall(0-280)	0.965	3.7849×10^{-15}	2.81	–	3.7227×10^{-15}	3.7726×10^{-15}	–	1.0167	1.0033
8401	Sigmoid colon wall(280-300)	0.071	4.0928×10^{-15}	6.92	–	3.7902×10^{-15}	3.8370×10^{-15}	–	1.0798	1.0667
8402	Sigmoid colon wall(300-surface)	2.22	3.7893×10^{-15}	3.62	–	3.8645×10^{-15}	3.6845×10^{-15}	–	0.9806	1.0284
8500	Sigmoid colon contents	8.52	3.8812×10^{-15}	0.90	–	3.9147×10^{-15}	3.8019×10^{-15}	–	0.9914	1.0209
8600	Rectum wall(0-280)	0.418	2.0024×10^{-15}	4.15	–	2.0534×10^{-15}	2.1894×10^{-15}	–	0.9752	0.9146
8601	Rectum wall(280-300)	0.0309	2.1993×10^{-15}	17.58	–	1.8934×10^{-15}	1.7943×10^{-15}	–	1.1615	1.2257
8602	Rectum wall(300-surface)	0.111	2.0651×10^{-15}	7.71	–	2.0780×10^{-15}	2.0985×10^{-15}	–	0.9938	0.9841
8603	Rectum contents	3.48	1.9771×10^{-15}	3.26	–	1.8893×10^{-15}	2.0428×10^{-15}	–	1.0465	0.9679
8700	Heart wall	23.16	2.2676×10^{-14}	0.64	–	2.2416×10^{-14}	2.2669×10^{-14}	–	1.0116	1.0003
8800	Blood in heart chamber	26.00	1.8633×10^{-14}	0.29	–	1.8639×10^{-14}	1.8611×10^{-14}	–	0.9997	1.0012
8900	Kidney left cortex	9.46	7.0416×10^{-15}	0.75	–	7.0592×10^{-15}	7.0875×10^{-15}	–	0.9975	0.9935
9000	Kidney left medulla	3.38	7.4578×10^{-15}	1.22	–	7.4144×10^{-15}	7.3633×10^{-15}	–	1.0058	1.0128
9100	Kidney left pelvis	0.676	7.1985×10^{-15}	5.96	–	6.9641×10^{-15}	6.8047×10^{-15}	–	1.0337	1.0579
9200	Kidney right cortex	9.46	1.5178×10^{-14}	0.73	–	1.5365×10^{-14}	1.5279×10^{-14}	–	0.9878	0.9934

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
9300	Kidney right medulla	3.38	1.4231×10^{-14}	1.43	–	1.4101×10^{-14}	1.4176×10^{-14}	–	1.0092	1.0039
9400	Kidney right pelvis	0.676	1.3614×10^{-14}	2.47	–	1.3560×10^{-14}	1.3806×10^{-14}	–	1.0040	0.9861
9500	Liver	167.41	6.6650×10^{-14}	0.08	–	6.6652×10^{-14}	6.6893×10^{-14}	–	1.0000	0.9964
9700	Lung(AI) left	28.69	9.9054×10^{-15}	0.67	–	1.0001×10^{-14}	9.8976×10^{-15}	–	0.9904	1.0008
9900	Lung(AI) right	31.31	1.8033×10^{-14}	0.46	–	1.7971×10^{-14}	1.8129×10^{-14}	–	1.0035	0.9947
10000	Lymphatic nodes ET	1.24	2.0753×10^{-15}	5.20	–	2.3568×10^{-15}	2.2908×10^{-15}	–	0.8805	0.9059
10100	Lymphatic nodes thoracic	1.24	1.9578×10^{-14}	1.24	–	2.0268×10^{-14}	2.0292×10^{-14}	–	0.9660	0.9648
10200	Lymphatic nodes head	0.429	3.0911×10^{-15}	5.54	–	3.2306×10^{-15}	3.3713×10^{-15}	–	0.9568	0.9169
10300	Lymphatic nodes trunk	10.14	6.4560×10^{-15}	0.56	–	6.5791×10^{-15}	6.6484×10^{-15}	–	0.9813	0.9711
10400	Lymphatic nodes arms	0.859	7.6641×10^{-15}	2.66	–	7.9385×10^{-15}	7.8843×10^{-15}	–	0.9654	0.9721
10500	Lymphatic nodes legs	0.859	5.2856×10^{-16}	9.50	–	7.7323×10^{-16}	6.8620×10^{-16}	–	0.6836	0.7703
10600	Muscle head	111.28	1.6106×10^{-15}	0.59	–	1.6149×10^{-15}	1.6309×10^{-15}	–	0.9973	0.9875
10700	Muscle trunk	415.77	6.9809×10^{-15}	0.12	–	7.1168×10^{-15}	7.1421×10^{-15}	–	0.9809*	0.9774*
10800	Muscle arms	76.51	5.6285×10^{-15}	0.59	–	5.4964×10^{-15}	5.5284×10^{-15}	–	1.0240*	1.0181*
10900	Muscle legs	216.20	8.6439×10^{-16}	0.58	–	8.9532×10^{-16}	8.9868×10^{-16}	–	0.9655*	0.9618*
11000	Oesophagus wall(0-190)	0.344	1.3245×10^{-14}	2.53	–	1.3253×10^{-14}	1.2611×10^{-14}	–	0.9994	1.0503
11001	Oesophagus wall(190-200)	0.0188	1.3869×10^{-14}	5.57	–	1.5599×10^{-14}	1.3155×10^{-14}	–	0.8891	1.0542
11002	Oesophagus wall(200-surface)	2.13	1.3812×10^{-14}	1.31	–	1.3943×10^{-14}	1.3871×10^{-14}	–	0.9906	0.9958
11003	Oesophagus contents	2.02	1.2986×10^{-14}	2.07	–	1.2827×10^{-14}	1.2943×10^{-14}	–	1.0124	1.0033
11100	Ovary left	0.168	3.4151×10^{-15}	11.98	–	3.9984×10^{-15}	3.9946×10^{-15}	–	0.8541	0.8549
11200	Ovary right	0.168	3.9996×10^{-15}	11.54	–	4.0032×10^{-15}	4.6675×10^{-15}	–	0.9991	0.8569
11300	Pancreas	7.25	9.2435×10^{-15}	0.85	–	9.0168×10^{-15}	9.2531×10^{-15}	–	1.0251	0.9990
11400	Pituitary gland	0.108	1.0375×10^{-15}	22.73	–	1.0510×10^{-15}	9.6535×10^{-16}	–	0.9871	1.0748
11600	RST head	182.99	1.5789×10^{-15}	0.42	–	1.5636×10^{-15}	1.5627×10^{-15}	–	1.0098	1.0104
11700	RST trunk	633.87	1.0569×10^{-14}	0.11	–	1.0634×10^{-14}	1.0692×10^{-14}	–	0.9939*	0.9885*
11800	RST arms	49.90	6.6291×10^{-15}	0.44	–	6.4289×10^{-15}	6.4245×10^{-15}	–	1.0311*	1.0318*
11900	RST legs	134.95	9.7699×10^{-16}	0.63	–	9.8407×10^{-16}	9.9789×10^{-16}	–	0.9928*	0.9791*
12000	Salivary glands left	3.25	2.4378×10^{-15}	2.91	–	2.5338×10^{-15}	2.3740×10^{-15}	–	0.9621	1.0269
12100	Salivary glandss right	3.25	2.6110×10^{-15}	2.15	–	2.6900×10^{-15}	2.6522×10^{-15}	–	0.9706	0.9845
12200	Skin head insensitive	38.85	1.0757×10^{-15}	0.90	–	1.0589×10^{-15}	1.0633×10^{-15}	–	1.0159	1.0117
12201	Skin head sensitive(40-100)	2.98	9.0296×10^{-16}	1.84	–	9.1516×10^{-16}	9.3974×10^{-16}	–	0.9867	0.9609
12300	Skin trunk insensitive	64.91	7.2213×10^{-15}	0.35	–	7.3814×10^{-15}	7.3237×10^{-15}	–	0.9783*	0.9860*
12301	Skin trunk sensitive(40-100)	4.92	6.3187×10^{-15}	0.65	–	6.4279×10^{-15}	6.4191×10^{-15}	–	0.9830*	0.9844*
12400	Skin arms insensitive	24.67	5.2116×10^{-15}	0.62	–	5.0855×10^{-15}	5.0940×10^{-15}	–	1.0248*	1.0231*
12401	Skin arms sensitive(40-100)	1.95	4.6654×10^{-15}	1.66	–	4.5626×10^{-15}	4.5787×10^{-15}	–	1.0226*	1.0189*
12500	Skin legs insensitive	42.40	6.3410×10^{-16}	1.25	–	6.6706×10^{-16}	6.7024×10^{-16}	–	0.9506*	0.9461*
12501	Skin legs sensitive(40-100)	3.22	5.3341×10^{-16}	4.07	–	5.4956×10^{-16}	5.4743×10^{-16}	–	0.9706*	0.9744*
12600	Spinal cord	6.06	8.4988×10^{-15}	0.69	–	8.5065×10^{-15}	8.3562×10^{-15}	–	0.9991	1.0171

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
12700	Spleen	13.86	9.0405×10^{-15}	0.48	–	9.0097×10^{-15}	9.0332×10^{-15}	–	1.0034	1.0008
12849	Unerupted deciduous enamel	0.19	2.1598×10^{-15}	13.05	–	2.2038×10^{-15}	2.2069×10^{-15}	–	0.9800	0.9786
12851	Unerupted deciduous dentin	0.426	2.3521×10^{-15}	5.27	–	1.9766×10^{-15}	2.0575×10^{-15}	–	1.1900	1.1432
12853	Deciduous cementum	0.0323	2.5352×10^{-15}	11.30	–	1.9460×10^{-15}	1.9776×10^{-15}	–	1.3028	1.2820
12855	Deciduous pulp	0.0607	2.8411×10^{-15}	13.24	–	2.5234×10^{-15}	2.2270×10^{-15}	–	1.1259	1.2758
13100	Thymus	14.10	7.0177×10^{-15}	0.94	–	7.0996×10^{-15}	7.0135×10^{-15}	–	0.9885	1.0006
13200	Thyroid	1.49	3.8568×10^{-15}	2.13	–	3.8766×10^{-15}	3.7900×10^{-15}	–	0.9949	1.0176
13300	Tongue upper(food)	3.05	2.3081×10^{-15}	3.39	–	2.5612×10^{-15}	2.4249×10^{-15}	–	0.9012	0.9519
13301	Tongue lower	0.736	3.0056×10^{-15}	4.04	–	2.8891×10^{-15}	2.6896×10^{-15}	–	1.0403	1.1175
13400	Tonsils	0.108	2.2457×10^{-15}	9.28	–	2.8122×10^{-15}	1.8223×10^{-15}	–	0.7986	1.2324
13500	Ureter left	0.418	5.3831×10^{-15}	6.53	–	5.5298×10^{-15}	5.4346×10^{-15}	–	0.9735	0.9905*
13600	Ureter right	0.418	7.8760×10^{-15}	4.36	–	7.1300×10^{-15}	7.2358×10^{-15}	–	1.1046	1.0885*
13700	Urinary bladder wall insensitive	3.58	2.4066×10^{-15}	2.57	–	2.3016×10^{-15}	2.4538×10^{-15}	–	1.0456	0.9808
13701	Urinary bladder wall sensitive(54-232)	0.505	2.2947×10^{-15}	4.12	–	2.3328×10^{-15}	2.4418×10^{-15}	–	0.9836	0.9397
13800	Urinary bladder content	12.40	2.3378×10^{-15}	1.83	–	2.3719×10^{-15}	2.3273×10^{-15}	–	0.9856	1.0045
13900	Uterus	4.34	2.7795×10^{-15}	2.76	–	2.7482×10^{-15}	2.9114×10^{-15}	–	1.0114	0.9547
14000	Air inside body	0.00192	3.0083×10^{-15}	41.26	–	7.1478×10^{-15}	3.0647×10^{-15}	–	0.4209	0.9816

* For tissues bookkept as head, trunk, arm and leg pieces (muscle, residual soft tissue, skin, large vessels), the volumes tabulated in the distributed cell tables partition the tissue at different boundaries from the transported mesh, and the reference output is normalised by the tabulated volume. The marked ratio therefore compares two different partitions of the same tissue, and is left out of the summary statistics of Table S2.

Table S10. 00F_External. Absorbed dose per primary history, Gy. σ is the FLUKA statistical uncertainty. Ratios are FLUKA / reference.

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
100	Adrenal left	3.60	2.8595×10^{-17}	25.43	–	3.9633×10^{-17}	3.8383×10^{-17}	–	0.7215	0.7450
200	Adrenal right	3.60	2.0717×10^{-17}	36.33	–	2.9413×10^{-17}	1.6230×10^{-17}	–	0.7044	1.2765
300	ET1(0-8)	0.000738	0.0000×10^{0}	0.00	–	–	–	–	–	–
301	ET1(8-40)	0.00303	0.0000×10^{0}	0.00	–	–	–	–	–	–
302	ET1(40-50)	0.000972	0.0000×10^{0}	0.00	–	–	–	–	–	–
303	ET1(50-Surface)	0.0954	0.0000×10^{0}	0.00	–	–	–	–	–	–
400	ET2(-15-0)	0.033	4.4482×10^{-17}	31.07	–	2.8168×10^{-17}	8.8551×10^{-18}	–	1.5792	5.0234
401	ET2(0-40)	0.0945	1.0837×10^{-16}	36.49	–	4.1841×10^{-17}	1.2259×10^{-17}	–	2.5900	8.8398
402	ET2(40-50)	0.0239	7.0837×10^{-17}	35.36	–	5.6186×10^{-17}	8.8585×10^{-17}	–	1.2608	0.7996
403	ET2(50-55)	0.012	1.0929×10^{-16}	40.31	–	6.6060×10^{-17}	2.3141×10^{-17}	–	1.6544	4.7228
404	ET2(55-65)	0.024	1.0695×10^{-16}	38.69	–	6.2328×10^{-17}	5.5819×10^{-17}	–	1.7159	1.9160
405	ET2(65-Surface)	1.68	5.2119×10^{-17}	22.55	–	3.8559×10^{-17}	4.4195×10^{-17}	–	1.3517	1.1793
500	Oral mucosa tongue	0.00811	1.5774×10^{-17}	56.01	–	–	3.6580×10^{-17}	–	–	0.4312
501	Oral mucosa moth floor	0.0098	2.6939×10^{-17}	85.00	–	1.6207×10^{-18}	4.5143×10^{-17}	–	16.6212	0.5967
600	Oral mucosa lips and cheeks	0.00641	6.5585×10^{-17}	100.00	–	2.6871×10^{-17}	1.3215×10^{-16}	–	2.4407	0.4963
700	Trachea	0.542	2.2382×10^{-17}	44.40	–	4.5604×10^{-17}	–	–	0.4908	–
800	BB(-11–6)	0.00159	2.0662×10^{-17}	100.00	–	–	–	–	–	–
801	BB(-6-0)	0.00203	1.3047×10^{-17}	100.00	–	–	–	–	–	–
802	BB(0-10)	0.00341	6.7888×10^{-17}	100.00	–	–	–	–	–	–
803	BB(10-35)	0.0086	2.3491×10^{-16}	100.00	–	–	–	–	–	–
804	BB(35-40)	0.00173	5.5876×10^{-17}	100.00	–	–	–	–	–	–
805	BB(40-50)	0.00348	1.3077×10^{-16}	100.00	–	–	–	–	–	–
806	BB(50-60)	0.0035	9.6577×10^{-17}	100.00	–	–	–	–	–	–
807	BB(60-70)	0.00352	6.0864×10^{-17}	100.00	–	7.9152×10^{-17}	–	–	0.7690	–
808	BB(70-surface)	0.35	7.8626×10^{-17}	49.62	–	2.0473×10^{-17}	1.3208×10^{-17}	–	3.8405	5.9529
900	Blood in large arteries head	0.0766	1.1670×10^{-17}	80.37	–	–	–	–	–	–
910	Blood in large veins head	0.282	1.0183×10^{-18}	100.00	–	5.1726×10^{-17}	3.3927×10^{-17}	–	0.0197*	0.0300*
1000	Blood in large arteries trunk	9.46	3.4477×10^{-17}	13.32	–	3.3221×10^{-17}	3.2862×10^{-17}	–	1.0378*	1.0491*
1010	Blood in large veins trunk	23.77	3.9456×10^{-17}	13.59	–	3.5051×10^{-17}	3.0551×10^{-17}	–	1.1257*	1.2915*
1100	Blood in large arteries arms	1.76	2.3660×10^{-17}	28.73	–	5.0215×10^{-17}	4.8652×10^{-17}	–	0.4712*	0.4863*
1110	Blood in large veins arms	6.80	3.3724×10^{-17}	12.67	–	3.5603×10^{-17}	3.4678×10^{-17}	–	0.9472*	0.9725*
1200	Blood in large arteries legs	6.10	3.9910×10^{-17}	8.66	–	3.8027×10^{-17}	3.5052×10^{-17}	–	1.0495*	1.1386*
1210	Blood in large veins legs	21.35	3.3576×10^{-17}	13.03	–	3.9709×10^{-17}	3.4788×10^{-17}	–	0.8455*	0.9652*
1300	Humeri upper cortical	1.92	3.6137×10^{-17}	39.57	–	4.4092×10^{-17}	4.7248×10^{-17}	–	0.8196	0.7648
1400	Humeri upper spogiosa	3.55	3.9266×10^{-17}	23.58	–	4.2968×10^{-17}	4.3006×10^{-17}	–	0.9138	0.9130

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
1500	Humeri upper medullary cavity	0.322	7.0149×10^{-18}	77.29	–	9.8644×10^{-17}	1.7857×10^{-17}	–	0.0711	0.3928
1600	Humeri lower cortical	1.85	3.7863×10^{-17}	43.13	–	4.8811×10^{-17}	4.0037×10^{-17}	–	0.7757	0.9457
1700	Humeri lower spongiosa	2.77	2.3743×10^{-17}	28.73	–	4.8787×10^{-17}	3.2025×10^{-17}	–	0.4867	0.7414
1800	Humeri lower medullary cavity	0.322	3.7300×10^{-17}	61.95	–	7.1951×10^{-17}	5.2765×10^{-17}	–	0.5184	0.7069
1900	Radii cortical	1.21	2.6158×10^{-17}	47.68	–	2.7950×10^{-17}	2.9342×10^{-17}	–	0.9359	0.8915
1910	Ulnae cortical	1.55	2.7031×10^{-17}	35.32	–	3.5994×10^{-17}	4.2056×10^{-17}	–	0.7510	0.6427
2000	Radii spongiosa	1.70	1.7664×10^{-17}	48.51	–	3.8200×10^{-17}	3.2368×10^{-17}	–	0.4624	0.5457
2010	Ulnae spongiosa	2.16	4.2137×10^{-17}	29.15	–	6.4282×10^{-17}	4.5343×10^{-17}	–	0.6555	0.9293
2100	Radii medullary cavity	0.122	4.6047×10^{-17}	95.13	–	–	7.8758×10^{-17}	–	–	0.5847
2110	Ulnae medullary cavity	0.143	7.1274×10^{-17}	97.88	–	4.9652×10^{-17}	2.9502×10^{-17}	–	1.4355	2.4159
2200	Wrists and hand bones cortical	0.841	2.8808×10^{-17}	31.54	–	2.7324×10^{-17}	2.6186×10^{-17}	–	1.0543	1.1001
2300	Wrists and hand bones spongiosa	3.17	4.5845×10^{-17}	19.11	–	3.5981×10^{-17}	3.5750×10^{-17}	–	1.2741	1.2824
2400	Clavicles cortical	0.915	2.3300×10^{-17}	34.76	–	4.1351×10^{-17}	3.9651×10^{-17}	–	0.5635	0.5876
2500	Clavicles spongiosa	1.67	4.0318×10^{-17}	31.49	–	5.3482×10^{-17}	3.3181×10^{-17}	–	0.7539	1.2151
2600	Cranium cortical	21.46	2.9145×10^{-17}	7.87	–	3.1820×10^{-17}	2.7072×10^{-17}	–	0.9160	1.0766
2700	Cranium spongiosa	78.79	3.1702×10^{-17}	6.04	–	3.0875×10^{-17}	3.2564×10^{-17}	–	1.0268	0.9735
2800	Femora upper cortical	3.76	3.1232×10^{-17}	24.95	–	2.2753×10^{-17}	2.5565×10^{-17}	–	1.3726	1.2217
2900	Femora upper spongiosa	5.77	2.9636×10^{-17}	20.00	–	3.0939×10^{-17}	1.2367×10^{-17}	–	0.9579	2.3964
3000	Femora upper medullary cavity	0.737	3.3756×10^{-17}	47.68	–	3.7276×10^{-17}	5.6089×10^{-17}	–	0.9056	0.6018
3100	Femora lower cortical	5.51	4.0048×10^{-17}	16.98	–	2.0243×10^{-17}	3.5670×10^{-17}	–	1.9784	1.1227
3200	Femora lower spongiosa	4.78	3.2906×10^{-17}	30.14	–	3.0439×10^{-17}	4.8293×10^{-17}	–	1.0810	0.6814
3300	Femora lower medullary cavity	1.17	2.9271×10^{-17}	42.27	–	5.5918×10^{-17}	6.9264×10^{-17}	–	0.5235	0.4226
3400	Tibiae cortical	4.50	4.6274×10^{-17}	21.13	–	4.0780×10^{-17}	4.5508×10^{-17}	–	1.1347	1.0168
3410	Fibulae cortical	1.16	4.2790×10^{-17}	35.37	–	2.2504×10^{-17}	8.4362×10^{-18}	–	1.9014	5.0722
3420	Patellae cortical	0.0374	5.4000×10^{-17}	73.08	–	3.8656×10^{-17}	7.9752×10^{-17}	–	1.3970	0.6771
3500	Tibiae spongiosa	6.62	3.0669×10^{-17}	10.47	–	3.2362×10^{-17}	4.6494×10^{-17}	–	0.9477	0.6596
3510	Fibulae spongiosa	1.39	4.7344×10^{-17}	35.34	–	1.3059×10^{-17}	2.4837×10^{-17}	–	3.6253	1.9062
3520	Patellae spongiosa	0.171	1.2417×10^{-16}	58.75	–	1.8898×10^{-17}	8.2590×10^{-18}	–	6.5707	15.0346
3600	Tibiae medullary cavity	0.673	3.5756×10^{-17}	57.49	–	4.6003×10^{-17}	2.3756×10^{-18}	–	0.7772	15.0512
3610	Fibulae medullary cavity	0.107	3.2118×10^{-17}	100.00	–	–	–	–	–	–
3700	Ankles and foot cortical	1.29	3.2864×10^{-17}	22.13	–	3.8682×10^{-17}	5.5518×10^{-17}	–	0.8496	0.5920
3800	Ankles and foot spongiosa	4.93	3.3065×10^{-17}	16.42	–	4.3254×10^{-17}	3.7587×10^{-17}	–	0.7644	0.8797
3900	Mandible cortical	1.70	3.7468×10^{-17}	24.18	–	1.9872×10^{-17}	3.2621×10^{-17}	–	1.8855	1.1486
4000	Mandible spongiosa	5.69	3.1246×10^{-17}	21.56	–	2.7280×10^{-17}	3.6184×10^{-17}	–	1.1454	0.8635
4100	Pelvis cortical	6.15	3.3590×10^{-17}	16.66	–	3.6178×10^{-17}	3.1640×10^{-17}	–	0.9285	1.0616
4200	Pelvis spongiosa	8.69	3.2735×10^{-17}	7.30	–	3.9203×10^{-17}	2.9807×10^{-17}	–	0.8350	1.0982
4300	Ribs cortical	6.39	3.4209×10^{-17}	16.54	–	3.1841×10^{-17}	3.1857×10^{-17}	–	1.0744	1.0738
4400	Ribs spongiosa	21.36	3.7284×10^{-17}	8.33	–	3.4479×10^{-17}	3.3987×10^{-17}	–	1.0813	1.0970

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
4500	Scapulae cortical	2.45	3.8616×10^{-17}	28.07	–	2.7760×10^{-17}	5.0354×10^{-17}	–	1.3911	0.7669
4600	Scapulae spongiosa	4.46	3.1118×10^{-17}	27.97	–	2.8491×10^{-17}	2.1839×10^{-17}	–	1.0922	1.4249
4700	Cervical spine cortical	3.77	2.3643×10^{-17}	24.22	–	3.0888×10^{-17}	2.9629×10^{-17}	–	0.7654	0.7980
4800	Cervical spine spongiosa	7.14	2.9325×10^{-17}	14.91	–	3.3854×10^{-17}	3.2873×10^{-17}	–	0.8662	0.8921
4900	Thoracic spine cortical	8.54	2.4692×10^{-17}	20.45	–	2.3065×10^{-17}	2.6999×10^{-17}	–	1.0705	0.9145
5000	Thoracic spine spongiosa	10.50	2.3372×10^{-17}	20.94	–	2.5013×10^{-17}	2.2745×10^{-17}	–	0.9344	1.0276
5100	Lumbar spine cortical	2.73	3.4937×10^{-17}	20.17	–	2.7803×10^{-17}	1.8420×10^{-17}	–	1.2566	1.8967
5200	Lumbar spine spongiosa	7.85	2.8262×10^{-17}	12.15	–	3.3213×10^{-17}	2.6478×10^{-17}	–	0.8509	1.0674
5300	Sacrum cortical	1.08	2.8766×10^{-17}	35.54	–	2.3505×10^{-17}	2.8963×10^{-17}	–	1.2238	0.9932
5400	Sacrum spongiosa	3.14	2.1099×10^{-17}	37.75	–	3.1099×10^{-17}	1.9576×10^{-17}	–	0.6785	1.0778
5500	Sternum cortical	0.168	2.4358×10^{-17}	45.05	–	3.9503×10^{-17}	1.5679×10^{-17}	–	0.6166	1.5535
5600	Sternum spongiosa	0.677	5.3446×10^{-17}	38.96	–	4.2529×10^{-17}	7.0002×10^{-17}	–	1.2567	0.7635
5700	Cartilage head	52.99	3.5285×10^{-17}	7.18	–	3.2333×10^{-17}	3.1866×10^{-17}	–	1.0913	1.1073
5800	Cartilage trunk	48.20	3.1444×10^{-17}	6.40	–	3.2129×10^{-17}	3.3949×10^{-17}	–	0.9787	0.9262
5900	Cartilage arms	11.54	3.2196×10^{-17}	13.30	–	4.2715×10^{-17}	3.2783×10^{-17}	–	0.7537	0.9821
6000	Cartilage legs	15.38	4.2812×10^{-17}	6.36	–	3.8366×10^{-17}	4.3573×10^{-17}	–	1.1159	0.9825
6100	Brain	395.66	3.1408×10^{-17}	2.70	–	3.3116×10^{-17}	3.0580×10^{-17}	–	0.9484	1.0271
6200	Breast left adipose tissue	0.087	5.6133×10^{-17}	100.00	–	–	–	–	–	–
6300	Breast left glandular tissue	0.058	4.1383×10^{-17}	100.00	–	–	–	–	–	–
6400	Breast right adipose tissue	0.087	2.0356×10^{-17}	100.00	–	4.6573×10^{-17}	7.1146×10^{-17}	–	0.4371	0.2861
6500	Breast right glandular tissue	0.058	3.1016×10^{-17}	100.00	–	1.7554×10^{-16}	–	–	0.1767	–
6600	Eye lens sensitive left	0.0162	0.0000×10^{0}	0.00	–	7.7038×10^{-17}	–	–	–	–
6601	Eye lens insensitive left	0.0673	0.0000×10^{0}	0.00	–	1.5448×10^{-16}	–	–	–	–
6700	Cornea left	0.582	3.4664×10^{-17}	61.82	–	2.7755×10^{-17}	1.7311×10^{-17}	–	1.2489	2.0024
6701	Aqueous left	0.174	3.9928×10^{-17}	93.37	–	1.3997×10^{-17}	1.9016×10^{-17}	–	2.8527	2.0997
6702	Vitreous left	2.41	2.6948×10^{-17}	38.75	–	3.8247×10^{-17}	3.3240×10^{-17}	–	0.7046	0.8107
6800	Eye lens sensitive right	0.0162	0.0000×10^{0}	0.00	–	–	–	–	–	–
6801	Eye lens insensitive right	0.0673	0.0000×10^{0}	0.00	–	–	–	–	–	–
6900	Cornea right	0.582	4.4688×10^{-17}	44.17	–	5.5537×10^{-17}	3.8015×10^{-17}	–	0.8046	1.1755
6901	Aqueous right	0.174	7.2356×10^{-17}	91.53	–	3.9225×10^{-17}	2.1738×10^{-17}	–	1.8446	3.3286
6902	Vitreous right	2.41	4.0451×10^{-17}	26.75	–	3.0396×10^{-17}	4.1620×10^{-17}	–	1.3308	0.9719
7000	Gall bladder wall	0.542	2.0589×10^{-17}	57.89	–	3.5913×10^{-17}	3.0371×10^{-17}	–	0.5733	0.6779
7100	Gall bladder contents	2.80	3.2159×10^{-17}	29.22	–	2.2883×10^{-17}	2.6548×10^{-17}	–	1.4053	1.2114
7200	Stomach wall(0-60)	0.408	2.9419×10^{-17}	31.31	–	3.2328×10^{-17}	3.0146×10^{-17}	–	0.9100	0.9759
7201	Stomach wall(60-100)	0.273	3.9095×10^{-17}	27.24	–	3.8754×10^{-17}	2.6082×10^{-17}	–	1.0088	1.4989
7202	Stomach wall(100-300)	1.38	2.7614×10^{-17}	19.21	–	3.4984×10^{-17}	4.0425×10^{-17}	–	0.7893	0.6831
7203	Stomach wall(300-surface)	6.66	3.4343×10^{-17}	15.15	–	3.2849×10^{-17}	3.6113×10^{-17}	–	1.0455	0.9510
7300	Stomach contents	40.00	3.4854×10^{-17}	7.24	–	3.4686×10^{-17}	3.8341×10^{-17}	–	1.0048	0.9090

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
7400	Small intestine wall(0-130)	3.22	4.6755×10^{-17}	17.26	–	3.5114×10^{-17}	3.6580×10^{-17}	–	1.3315	1.2782
7401	Small intestine wall(130-150)	0.503	4.1537×10^{-17}	23.01	–	3.2300×10^{-17}	2.5953×10^{-17}	–	1.2860	1.6005
7402	Small intestine wall(150-200)	1.27	4.5624×10^{-17}	19.01	–	3.6439×10^{-17}	3.7404×10^{-17}	–	1.2521	1.2198
7403	Small intestine wall(200-surface)	33.22	4.1400×10^{-17}	9.21	–	3.6777×10^{-17}	3.6162×10^{-17}	–	1.1257	1.1449
7500	Small intestine contents(-400-0)	9.27	4.2770×10^{-17}	10.97	–	3.6368×10^{-17}	4.1501×10^{-17}	–	1.1760	1.0306
7501	Small intestine contents(centre–400)	46.73	4.3412×10^{-17}	5.50	–	3.6864×10^{-17}	4.1382×10^{-17}	–	1.1776	1.0490
7600	Ascending colon wall(0-280)	1.14	3.6828×10^{-17}	27.85	–	3.8539×10^{-17}	2.9407×10^{-17}	–	0.9556	1.2523
7601	Ascending colon wall(280-300)	0.0832	1.8538×10^{-17}	56.72	–	4.1671×10^{-17}	4.4402×10^{-17}	–	0.4449	0.4175
7602	Ascending colon wall(300-surface)	2.80	2.4892×10^{-17}	26.18	–	3.5225×10^{-17}	2.9631×10^{-17}	–	0.7067	0.8401
7700	Ascending colon content	15.59	3.4815×10^{-17}	11.36	–	3.9783×10^{-17}	2.9697×10^{-17}	–	0.8751	1.1723
7800	Transverse colon wall right(0-280)	0.842	4.0014×10^{-17}	36.67	–	5.3104×10^{-17}	6.4488×10^{-17}	–	0.7535	0.6205
7801	Transverse colon wall right(280-300)	0.0618	2.5101×10^{-17}	49.48	–	5.0014×10^{-17}	5.8645×10^{-17}	–	0.5019	0.4280
7802	Transverse colon wall right(300-surface)	3.99	5.0796×10^{-17}	15.86	–	4.8838×10^{-17}	6.1851×10^{-17}	–	1.0401	0.8213
7900	Transverse colon contents right	8.41	3.8409×10^{-17}	13.99	–	3.5802×10^{-17}	3.3790×10^{-17}	–	1.0728	1.1367
8000	Transverse colon wall left(0-280)	0.651	4.4493×10^{-17}	34.70	–	3.3162×10^{-17}	4.2562×10^{-17}	–	1.3417	1.0454
8001	Transverse colon wall left(280-300)	0.0481	1.7897×10^{-17}	35.24	–	2.4469×10^{-17}	2.0383×10^{-17}	–	0.7314	0.8780
8002	Transverse colon wall left(300-surface)	3.80	3.6254×10^{-17}	26.03	–	3.9478×10^{-17}	4.5290×10^{-17}	–	0.9183	0.8005
8100	Transverse colon content left	5.42	5.1583×10^{-17}	20.75	–	5.1630×10^{-17}	4.8063×10^{-17}	–	0.9991	1.0732
8200	Descending colon wall(0-280)	0.78	3.1178×10^{-17}	35.09	–	4.2913×10^{-17}	4.0170×10^{-17}	–	0.7265	0.7761
8201	Descending colon wall(280-300)	0.0575	6.4596×10^{-17}	55.55	–	2.4155×10^{-17}	3.2208×10^{-17}	–	2.6743	2.0056
8202	Descending colon wall(300-surface)	3.57	4.2187×10^{-17}	19.16	–	4.7615×10^{-17}	4.8120×10^{-17}	–	0.8860	0.8767
8300	Descending colon content	6.58	2.4402×10^{-17}	25.44	–	3.7499×10^{-17}	3.5295×10^{-17}	–	0.6507	0.6914
8400	Sigmoid colon wall(0-280)	0.965	6.2762×10^{-17}	29.37	–	4.5917×10^{-17}	2.5609×10^{-17}	–	1.3669	2.4508
8401	Sigmoid colon wall(280-300)	0.071	3.8029×10^{-17}	36.08	–	6.0426×10^{-17}	1.3515×10^{-17}	–	0.6293	2.8138
8402	Sigmoid colon wall(300-surface)	2.22	3.3129×10^{-17}	22.25	–	3.3331×10^{-17}	3.4574×10^{-17}	–	0.9939	0.9582
8500	Sigmoid colon contents	8.52	3.5647×10^{-17}	21.01	–	3.1513×10^{-17}	2.9506×10^{-17}	–	1.1312	1.2081
8600	Rectum wall(0-280)	0.418	2.0043×10^{-17}	48.09	–	1.9573×10^{-17}	4.1489×10^{-17}	–	1.0241	0.4831
8601	Rectum wall(280-300)	0.0309	5.9935×10^{-18}	72.74	–	9.6995×10^{-18}	1.8122×10^{-17}	–	0.6179	0.3307
8602	Rectum wall(300-surface)	0.111	1.3723×10^{-17}	61.81	–	5.4616×10^{-17}	1.5491×10^{-17}	–	0.2513	0.8859
8603	Rectum contents	3.48	3.2760×10^{-17}	32.68	–	2.1214×10^{-17}	3.2164×10^{-17}	–	1.5443	1.0185
8700	Heart wall	23.16	3.3989×10^{-17}	7.14	–	3.2210×10^{-17}	4.0116×10^{-17}	–	1.0552	0.8473
8800	Blood in heart chamber	26.00	3.5944×10^{-17}	9.88	–	3.5950×10^{-17}	3.6942×10^{-17}	–	0.9998	0.9730
8900	Kidney left cortex	9.46	3.4026×10^{-17}	18.75	–	3.3312×10^{-17}	3.7416×10^{-17}	–	1.0214	0.9094
9000	Kidney left medulla	3.38	3.5975×10^{-17}	22.16	–	1.3596×10^{-17}	2.4836×10^{-17}	–	2.6460	1.4485
9100	Kidney left pelvis	0.676	3.2775×10^{-17}	59.36	–	5.6042×10^{-18}	4.6012×10^{-17}	–	5.8484	0.7123
9200	Kidney right cortex	9.46	3.8995×10^{-17}	14.00	–	3.3728×10^{-17}	2.7339×10^{-17}	–	1.1561	1.4263

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
9300	Kidney right medulla	3.38	4.7100×10^{-17}	20.40	–	2.0449×10^{-17}	2.7729×10^{-17}	–	2.3033	1.6986
9400	Kidney right pelvis	0.676	1.7281×10^{-17}	77.23	–	3.9795×10^{-17}	5.7195×10^{-17}	–	0.4342	0.3021
9500	Liver	167.41	3.5674×10^{-17}	2.12	–	3.7073×10^{-17}	3.7053×10^{-17}	–	0.9623	0.9628
9700	Lung(AI) left	28.69	2.7431×10^{-17}	12.54	–	2.9193×10^{-17}	3.3062×10^{-17}	–	0.9396	0.8297
9900	Lung(AI) right	31.31	3.2436×10^{-17}	5.52	–	4.0858×10^{-17}	3.1303×10^{-17}	–	0.7939	1.0362
10000	Lymphatic nodes ET	1.24	1.7695×10^{-17}	32.73	–	1.3580×10^{-17}	2.0269×10^{-17}	–	1.3030	0.8730
10100	Lymphatic nodes thoracic	1.24	5.4884×10^{-17}	30.73	–	2.2719×10^{-17}	3.9091×10^{-17}	–	2.4158	1.4040
10200	Lymphatic nodes head	0.429	2.3279×10^{-17}	58.05	–	2.4990×10^{-17}	–	–	0.9315	–
10300	Lymphatic nodes trunk	10.14	3.3288×10^{-17}	15.83	–	3.4982×10^{-17}	2.5720×10^{-17}	–	0.9516	1.2943
10400	Lymphatic nodes arms	0.859	3.5674×10^{-17}	67.83	–	3.5008×10^{-17}	4.7941×10^{-17}	–	1.0190	0.7441
10500	Lymphatic nodes legs	0.859	4.4371×10^{-17}	30.15	–	4.2339×10^{-17}	2.3764×10^{-17}	–	1.0480	1.8671
10600	Muscle head	111.28	3.4387×10^{-17}	4.89	–	3.1571×10^{-17}	3.2217×10^{-17}	–	1.0892	1.0674
10700	Muscle trunk	415.77	3.6018×10^{-17}	2.48	–	3.4870×10^{-17}	3.4398×10^{-17}	–	1.0329*	1.0471*
10800	Muscle arms	76.51	3.8292×10^{-17}	7.45	–	3.7824×10^{-17}	3.7359×10^{-17}	–	1.0124*	1.0250*
10900	Muscle legs	216.20	3.6475×10^{-17}	3.15	–	4.1118×10^{-17}	3.7603×10^{-17}	–	0.8871*	0.9700*
11000	Oesophagus wall(0-190)	0.344	3.5722×10^{-17}	55.07	–	1.8040×10^{-17}	3.5537×10^{-17}	–	1.9802	1.0052
11001	Oesophagus wall(190-200)	0.0188	2.0818×10^{-17}	47.20	–	1.6689×10^{-17}	2.7309×10^{-17}	–	1.2474	0.7623
11002	Oesophagus wall(200-surface)	2.13	2.1230×10^{-17}	28.98	–	2.4665×10^{-17}	2.9387×10^{-17}	–	0.8607	0.7224
11003	Oesophagus contents	2.02	3.3733×10^{-17}	31.44	–	2.1867×10^{-17}	4.8570×10^{-17}	–	1.5427	0.6945
11100	Ovary left	0.168	1.7322×10^{-17}	100.00	–	3.0903×10^{-17}	1.4676×10^{-17}	–	0.5605	1.1803
11200	Ovary right	0.168	0.0000×10^{0}	0.00	–	6.1394×10^{-18}	1.2478×10^{-17}	–	–	–
11300	Pancreas	7.25	3.2504×10^{-17}	15.52	–	3.7826×10^{-17}	2.3752×10^{-17}	–	0.8593	1.3685
11400	Pituitary gland	0.108	0.0000×10^{0}	0.00	–	–	–	–	–	–
11600	RST head	182.99	3.7977×10^{-17}	4.33	–	3.3686×10^{-17}	3.3443×10^{-17}	–	1.1274	1.1356
11700	RST trunk	633.87	3.6308×10^{-17}	1.63	–	3.4850×10^{-17}	3.5194×10^{-17}	–	1.0418*	1.0316*
11800	RST arms	49.90	3.7410×10^{-17}	4.66	–	3.7914×10^{-17}	3.7711×10^{-17}	–	0.9867*	0.9920*
11900	RST legs	134.95	3.5026×10^{-17}	4.46	–	3.6240×10^{-17}	3.6898×10^{-17}	–	0.9665*	0.9493*
12000	Salivary glands left	3.25	4.6803×10^{-17}	23.37	–	4.1625×10^{-17}	2.4348×10^{-17}	–	1.1244	1.9222
12100	Salivary glandss right	3.25	3.4826×10^{-17}	20.69	–	3.6901×10^{-17}	4.4404×10^{-17}	–	0.9438	0.7843
12200	Skin head insensitive	38.85	2.6325×10^{-17}	6.32	–	2.2580×10^{-17}	2.1703×10^{-17}	–	1.1658	1.2130
12201	Skin head sensitive(40-100)	2.98	2.4398×10^{-17}	15.32	–	1.4890×10^{-17}	1.2374×10^{-17}	–	1.6386	1.9717
12300	Skin trunk insensitive	64.91	2.5136×10^{-17}	6.37	–	2.3157×10^{-17}	2.5573×10^{-17}	–	1.0855*	0.9829*
12301	Skin trunk sensitive(40-100)	4.92	1.6041×10^{-17}	13.81	–	1.5451×10^{-17}	2.2421×10^{-17}	–	1.0382*	0.7155*
12400	Skin arms insensitive	24.67	3.1993×10^{-17}	6.90	–	3.2933×10^{-17}	2.9600×10^{-17}	–	0.9714*	1.0808*
12401	Skin arms sensitive(40-100)	1.95	2.4988×10^{-17}	21.38	–	2.1686×10^{-17}	2.8097×10^{-17}	–	1.1523*	0.8894*
12500	Skin legs insensitive	42.40	2.2556×10^{-17}	8.74	–	2.7053×10^{-17}	2.6478×10^{-17}	–	0.8338*	0.8519*
12501	Skin legs sensitive(40-100)	3.22	1.3726×10^{-17}	19.76	–	2.2819×10^{-17}	1.8338×10^{-17}	–	0.6015*	0.7485*
12600	Spinal cord	6.06	3.3074×10^{-17}	20.44	–	2.3185×10^{-17}	3.0678×10^{-17}	–	1.4266	1.0781

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
12700	Spleen	13.86	3.0021×10^{-17}	13.87	–	3.6100×10^{-17}	3.1788×10^{-17}	–	0.8316	0.9444
12849	Unerupted deciduous enamel	0.19	6.8086×10^{-17}	59.60	–	4.7652×10^{-17}	9.6949×10^{-19}	–	1.4288	70.2289
12851	Unerupted deciduous dentin	0.426	1.8910×10^{-17}	54.66	–	5.2191×10^{-17}	1.4169×10^{-17}	–	0.3623	1.3346
12853	Deciduous cementum	0.0323	1.5000×10^{-17}	100.00	–	1.0620×10^{-16}	1.2008×10^{-17}	–	0.1413	1.2492
12855	Deciduous pulp	0.0607	1.3361×10^{-17}	100.00	–	–	1.0732×10^{-16}	–	–	0.1245
13100	Thymus	14.10	3.4214×10^{-17}	10.53	–	3.5351×10^{-17}	3.5452×10^{-17}	–	0.9679	0.9651
13200	Thyroid	1.49	3.8205×10^{-17}	34.66	–	3.2347×10^{-17}	3.8285×10^{-17}	–	1.1811	0.9979
13300	Tongue upper(food)	3.05	3.3295×10^{-17}	25.08	–	4.1476×10^{-17}	4.1210×10^{-17}	–	0.8028	0.8079
13301	Tongue lower	0.736	3.6627×10^{-17}	40.73	–	1.7713×10^{-17}	4.7748×10^{-17}	–	2.0678	0.7671
13400	Tonsils	0.108	5.2218×10^{-18}	100.00	–	3.1007×10^{-17}	–	–	0.1684	–
13500	Ureter left	0.418	1.8360×10^{-17}	76.89	–	6.8276×10^{-17}	6.0341×10^{-17}	–	0.2689	0.3043*
13600	Ureter right	0.418	2.6965×10^{-18}	71.55	–	1.9447×10^{-17}	1.8784×10^{-18}	–	0.1387	1.4355*
13700	Urinary bladder wall insensitive	3.58	4.6101×10^{-17}	14.91	–	4.0200×10^{-17}	3.0731×10^{-17}	–	1.1468	1.5001
13701	Urinary bladder wall sensitive(54-232)	0.505	2.4428×10^{-17}	28.26	–	4.3428×10^{-17}	4.2169×10^{-17}	–	0.5625	0.5793
13800	Urinary bladder content	12.40	4.1380×10^{-17}	10.65	–	4.4358×10^{-17}	3.9748×10^{-17}	–	0.9329	1.0411
13900	Uterus	4.34	2.9172×10^{-17}	24.60	–	3.0123×10^{-17}	4.5670×10^{-17}	–	0.9684	0.6388
14000	Air inside body	0.00192	1.7688×10^{-17}	42.73	–	3.6464×10^{-18}	5.1919×10^{-18}	–	4.8508	3.4069

* For tissues bookkept as head, trunk, arm and leg pieces (muscle, residual soft tissue, skin, large vessels), the volumes tabulated in the distributed cell tables partition the tissue at different boundaries from the transported mesh, and the reference output is normalised by the tabulated volume. The marked ratio therefore compares two different partitions of the same tissue, and is left out of the summary statistics of Table S2.

Table S11. 01M_Internal. Absorbed dose per primary history, Gy. σ is the FLUKA statistical uncertainty. Ratios are FLUKA / reference.

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
100	Adrenal left	2.26	6.8716×10^{-15}	1.78	–	6.9788×10^{-15}	6.9984×10^{-15}	–	0.9846	0.9819
200	Adrenal right	2.26	1.7918×10^{-14}	1.79	–	1.8360×10^{-14}	1.8032×10^{-14}	–	0.9759	0.9937
300	ET1(0-8)	0.00124	1.2076×10^{-16}	100.00	–	3.9888×10^{-16}	1.9483×10^{-16}	–	0.3027	0.6198
301	ET1(8-40)	0.00504	4.6387×10^{-16}	49.63	–	5.2475×10^{-16}	3.2503×10^{-16}	–	0.8840	1.4272
302	ET1(40-50)	0.0016	3.1398×10^{-15}	52.85	–	4.6635×10^{-16}	1.2662×10^{-16}	–	6.7328	24.7971
303	ET1(50-Surface)	0.218	6.4678×10^{-16}	20.83	–	7.3621×10^{-16}	4.7916×10^{-16}	–	0.8785	1.3498
400	ET2(-15-0)	0.0692	8.2985×10^{-16}	11.51	–	6.3627×10^{-16}	6.0307×10^{-16}	–	1.3042	1.3760
401	ET2(0-40)	0.198	8.0192×10^{-16}	12.27	–	6.5398×10^{-16}	5.3527×10^{-16}	–	1.2262	1.4982
402	ET2(40-50)	0.0498	7.5819×10^{-16}	10.20	–	7.3319×10^{-16}	5.7331×10^{-16}	–	1.0341	1.3225
403	ET2(50-55)	0.0249	7.3854×10^{-16}	19.43	–	7.7001×10^{-16}	4.9313×10^{-16}	–	0.9591	1.4977
404	ET2(55-65)	0.05	5.9913×10^{-16}	18.66	–	8.4531×10^{-16}	6.2331×10^{-16}	–	0.7088	0.9612
405	ET2(65-Surface)	6.93	6.7115×10^{-16}	2.69	–	6.2932×10^{-16}	6.2726×10^{-16}	–	1.0665	1.0700
500	Oral mucosa tongue	0.0187	1.4237×10^{-15}	21.32	–	1.0814×10^{-15}	8.6280×10^{-16}	–	1.3165	1.6500
501	Oral mucosa moth floor	0.0143	7.0794×10^{-16}	19.59	–	9.3959×10^{-16}	6.5849×10^{-16}	–	0.7535	1.0751
600	Oral mucosa lips and cheeks	0.0045	8.9949×10^{-16}	35.29	–	7.3363×10^{-16}	8.5412×10^{-16}	–	1.2261	1.0531
700	Trachea	1.56	1.7261×10^{-15}	5.65	–	1.8705×10^{-15}	1.7393×10^{-15}	–	0.9228	0.9924
800	BB(-11–6)	0.00302	1.9062×10^{-15}	21.79	–	3.2592×10^{-15}	3.3418×10^{-15}	–	0.5849	0.5704
801	BB(-6-0)	0.00387	4.5612×10^{-15}	23.86	–	3.3672×10^{-15}	3.2963×10^{-15}	–	1.3546	1.3837
802	BB(0-10)	0.00648	3.2544×10^{-15}	19.07	–	3.6958×10^{-15}	3.5150×10^{-15}	–	0.8806	0.9259
803	BB(10-35)	0.0163	3.3970×10^{-15}	16.71	–	2.6252×10^{-15}	3.1647×10^{-15}	–	1.2940	1.0734
804	BB(35-40)	0.00329	4.0013×10^{-15}	22.69	–	3.0957×10^{-15}	3.0113×10^{-15}	–	1.2926	1.3288
805	BB(40-50)	0.0066	5.1365×10^{-15}	11.80	–	3.4228×10^{-15}	3.1614×10^{-15}	–	1.5007	1.6248
806	BB(50-60)	0.00663	3.7442×10^{-15}	19.09	–	2.7753×10^{-15}	3.8068×10^{-15}	–	1.3491	0.9836
807	BB(60-70)	0.00667	3.4148×10^{-15}	12.06	–	2.7016×10^{-15}	2.7692×10^{-15}	–	1.2640	1.2331
808	BB(70-surface)	1.92	3.3389×10^{-15}	3.21	–	3.2732×10^{-15}	3.4701×10^{-15}	–	1.0201	0.9622
900	Blood in large arteries head	0.167	1.1014×10^{-15}	14.64	–	1.0661×10^{-15}	1.1058×10^{-15}	–	1.0331*	0.9960*
910	Blood in large veins head	0.56	1.1387×10^{-15}	6.61	–	9.9001×10^{-16}	7.7317×10^{-16}	–	1.1502*	1.4728*
1000	Blood in large arteries trunk	15.42	5.5000×10^{-15}	1.21	–	5.6997×10^{-15}	5.6703×10^{-15}	–	0.9650*	0.9700*
1010	Blood in large veins trunk	42.08	4.4951×10^{-15}	0.52	–	4.7463×10^{-15}	4.7072×10^{-15}	–	0.9471*	0.9549*
1100	Blood in large arteries arms	3.14	3.2833×10^{-15}	1.58	–	3.3371×10^{-15}	3.5009×10^{-15}	–	0.9839*	0.9378*
1110	Blood in large veins arms	13.41	2.9575×10^{-15}	1.41	–	2.9509×10^{-15}	2.8833×10^{-15}	–	1.0022*	1.0257*
1200	Blood in large arteries legs	13.09	3.0591×10^{-16}	6.38	–	3.2144×10^{-16}	3.4304×10^{-16}	–	0.9517*	0.8918*
1210	Blood in large veins legs	39.39	3.2697×10^{-16}	2.19	–	3.7351×10^{-16}	3.6788×10^{-16}	–	0.8754*	0.8888*
1300	Humeri upper cortical	12.67	2.0304×10^{-15}	1.45	–	1.9744×10^{-15}	2.0338×10^{-15}	–	1.0283	0.9983
1400	Humeri upper spogiosa	8.73	1.6065×10^{-15}	2.72	–	1.5425×10^{-15}	1.6741×10^{-15}	–	1.0415	0.9596

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
1500	Humeri upper medullary cavity	0.863	2.6742×10^{-15}	6.93	–	2.3603×10^{-15}	2.6045×10^{-15}	–	1.1330	1.0268
1600	Humeri lower cortical	9.04	3.6897×10^{-15}	1.41	–	3.7082×10^{-15}	3.7775×10^{-15}	–	0.9950	0.9768
1700	Humeri lower spongiosa	4.52	3.7132×10^{-15}	1.82	–	3.9189×10^{-15}	3.6937×10^{-15}	–	0.9475	1.0053
1800	Humeri lower medullary cavity	0.809	4.0769×10^{-15}	4.48	–	4.1741×10^{-15}	3.6368×10^{-15}	–	0.9767	1.1210
1900	Radii cortical	5.62	1.8832×10^{-15}	2.88	–	1.9423×10^{-15}	1.9182×10^{-15}	–	0.9695	0.9817
1910	Ulnae cortical	7.42	2.2034×10^{-15}	2.08	–	2.1537×10^{-15}	2.2114×10^{-15}	–	1.0231	0.9964
2000	Radii spongiosa	1.12	1.7711×10^{-15}	5.62	–	1.8319×10^{-15}	1.6479×10^{-15}	–	0.9668	1.0748
2010	Ulnae spongiosa	1.81	3.1381×10^{-15}	2.88	–	3.0375×10^{-15}	3.0257×10^{-15}	–	1.0331	1.0371
2100	Radii medullary cavity	0.359	1.7813×10^{-15}	12.68	–	2.1585×10^{-15}	1.9398×10^{-15}	–	0.8252	0.9183
2110	Ulnae medullary cavity	0.589	2.2039×10^{-15}	5.11	–	2.1818×10^{-15}	2.2438×10^{-15}	–	1.0101	0.9822
2200	Wrists and hand bones cortical	5.40	6.8729×10^{-16}	3.69	–	6.6835×10^{-16}	6.8609×10^{-16}	–	1.0283	1.0017
2300	Wrists and hand bones spongiosa	14.07	7.0875×10^{-16}	2.48	–	7.3427×10^{-16}	7.3601×10^{-16}	–	0.9652	0.9630
2400	Clavicles cortical	1.37	1.7464×10^{-15}	2.35	–	1.8130×10^{-15}	1.8769×10^{-15}	–	0.9633	0.9305
2500	Clavicles spongiosa	1.96	1.8346×10^{-15}	2.98	–	1.9192×10^{-15}	1.6627×10^{-15}	–	0.9559	1.1034
2600	Cranium cortical	91.49	3.2537×10^{-16}	1.53	–	3.2330×10^{-16}	3.2999×10^{-16}	–	1.0064	0.9860
2700	Cranium spongiosa	289.87	3.2340×10^{-16}	0.66	–	3.2596×10^{-16}	3.2754×10^{-16}	–	0.9921	0.9873
2800	Femora upper cortical	14.11	7.7040×10^{-16}	3.08	–	7.7562×10^{-16}	7.8509×10^{-16}	–	0.9933	0.9813
2900	Femora upper spongiosa	8.87	1.0931×10^{-15}	2.72	–	1.0693×10^{-15}	1.0962×10^{-15}	–	1.0222	0.9972
3000	Femora upper medullary cavity	0.994	5.9262×10^{-16}	10.25	–	6.3560×10^{-16}	5.8060×10^{-16}	–	0.9324	1.0207
3100	Femora lower cortical	20.54	1.9225×10^{-16}	3.47	–	1.9446×10^{-16}	1.8526×10^{-16}	–	0.9886	1.0377
3200	Femora lower spongiosa	8.76	1.4853×10^{-16}	6.64	–	1.4084×10^{-16}	1.6296×10^{-16}	–	1.0546	0.9115
3300	Femora lower medullary cavity	1.57	2.4806×10^{-16}	12.50	–	3.0781×10^{-16}	2.4764×10^{-16}	–	0.8059	1.0017
3400	Tibiae cortical	26.33	7.1433×10^{-17}	6.72	–	8.2070×10^{-17}	7.8320×10^{-17}	–	0.8704	0.9121
3410	Fibulae cortical	3.02	6.2536×10^{-17}	12.42	–	5.2484×10^{-17}	6.9465×10^{-17}	–	1.1915	0.9003
3420	Patellae cortical	0.162	1.1171×10^{-16}	51.60	–	8.8800×10^{-17}	6.0580×10^{-17}	–	1.2580	1.8440
3500	Tibiae spongiosa	7.27	7.8502×10^{-17}	10.36	–	1.0026×10^{-16}	1.2055×10^{-16}	–	0.7830	0.6512
3510	Fibulae spongiosa	0.565	6.4259×10^{-17}	39.09	–	9.1227×10^{-17}	5.7105×10^{-17}	–	0.7044	1.1253
3520	Patellae spongiosa	0.575	6.4110×10^{-17}	28.67	–	7.3369×10^{-17}	9.3341×10^{-17}	–	0.8738	0.6868
3600	Tibiae medullary cavity	2.20	7.0838×10^{-17}	19.08	–	8.9534×10^{-17}	7.5969×10^{-17}	–	0.7912	0.9325
3610	Fibulae medullary cavity	0.204	6.1054×10^{-17}	39.06	–	3.1662×10^{-17}	7.7551×10^{-17}	–	1.9283	0.7873
3700	Ankles and foot cortical	7.24	5.8389×10^{-17}	5.47	–	5.6199×10^{-17}	6.0606×10^{-17}	–	1.0390	0.9634
3800	Ankles and foot spongiosa	20.33	4.5217×10^{-17}	7.76	–	4.8451×10^{-17}	4.6165×10^{-17}	–	0.9332	0.9795
3900	Mandible cortical	7.16	8.6176×10^{-16}	1.91	–	8.6761×10^{-16}	8.9123×10^{-16}	–	0.9933	0.9669
4000	Mandible spongiosa	13.09	9.1196×10^{-16}	2.21	–	9.0311×10^{-16}	9.1537×10^{-16}	–	1.0098	0.9963
4100	Pelvis cortical	21.20	1.9833×10^{-15}	0.75	–	2.0156×10^{-15}	1.9909×10^{-15}	–	0.9840	0.9962
4200	Pelvis spongiosa	34.40	1.9621×10^{-15}	1.06	–	1.9867×10^{-15}	1.9454×10^{-15}	–	0.9876	1.0086
4300	Ribs cortical	20.41	6.2378×10^{-15}	0.50	–	6.2495×10^{-15}	6.3241×10^{-15}	–	0.9981	0.9863
4400	Ribs spongiosa	58.12	6.1698×10^{-15}	0.36	–	6.1880×10^{-15}	6.2462×10^{-15}	–	0.9971	0.9878

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
4500	Scapulae cortical	9.65	1.9987×10^{-15}	1.88	–	1.9725×10^{-15}	1.9939×10^{-15}	–	1.0133	1.0024
4600	Scapulae spongiosa	14.86	2.0056×10^{-15}	1.51	–	1.9168×10^{-15}	2.0121×10^{-15}	–	1.0464	0.9968
4700	Cervical spine cortical	3.28	1.0302×10^{-15}	2.59	–	1.0353×10^{-15}	1.0534×10^{-15}	–	0.9951	0.9780
4800	Cervical spine spongiosa	9.10	1.1099×10^{-15}	1.99	–	1.1058×10^{-15}	1.1043×10^{-15}	–	1.0037	1.0051
4900	Thoracic spine cortical	11.50	5.4804×10^{-15}	0.77	–	5.4414×10^{-15}	5.3729×10^{-15}	–	1.0072	1.0200
5000	Thoracic spine spongiosa	28.11	6.3637×10^{-15}	0.70	–	6.3877×10^{-15}	6.4148×10^{-15}	–	0.9962	0.9920
5100	Lumbar spine cortical	7.07	5.6933×10^{-15}	1.05	–	5.7488×10^{-15}	5.7055×10^{-15}	–	0.9904	0.9979
5200	Lumbar spine spongiosa	21.95	6.2045×10^{-15}	0.61	–	6.1833×10^{-15}	6.2183×10^{-15}	–	1.0034	0.9978
5300	Sacrum cortical	5.45	1.9301×10^{-15}	2.83	–	1.8625×10^{-15}	1.9176×10^{-15}	–	1.0363	1.0065
5400	Sacrum spongiosa	15.12	2.0328×10^{-15}	1.51	–	2.0593×10^{-15}	1.9615×10^{-15}	–	0.9871	1.0363
5500	Sternum cortical	0.63	4.4605×10^{-15}	3.68	–	4.8520×10^{-15}	4.5328×10^{-15}	–	0.9193	0.9840
5600	Sternum spongiosa	2.10	4.3719×10^{-15}	2.47	–	4.4193×10^{-15}	4.6445×10^{-15}	–	0.9893	0.9413
5700	Cartilage head	5.03	1.7775×10^{-16}	11.00	–	1.7675×10^{-16}	1.6727×10^{-16}	–	1.0057	1.0627
5800	Cartilage trunk	36.85	9.4720×10^{-15}	0.43	–	9.4303×10^{-15}	9.5605×10^{-15}	–	1.0044	0.9907
6100	Brain	978.10	2.9499×10^{-16}	0.60	–	2.9523×10^{-16}	2.9282×10^{-16}	–	0.9992	1.0074
6200	Breast left adipose tissue	0.571	3.6417×10^{-15}	3.41	–	3.6100×10^{-15}	4.0213×10^{-15}	–	1.0088	0.9056
6300	Breast left glandular tissue	0.381	4.3100×10^{-15}	7.80	–	3.9967×10^{-15}	4.1176×10^{-15}	–	1.0784	1.0467
6400	Breast right adipose tissue	0.571	6.9072×10^{-15}	3.50	–	6.3511×10^{-15}	6.7871×10^{-15}	–	1.0875	1.0177
6500	Breast right glandular tissue	0.381	6.6813×10^{-15}	3.66	–	5.8231×10^{-15}	6.5240×10^{-15}	–	1.1474	1.0241
6600	Eye lens sensitive left	0.0169	3.1049×10^{-17}	100.00	–	4.2364×10^{-16}	2.3975×10^{-17}	–	0.0733	1.2951
6601	Eye lens insensitive left	0.0978	7.3781×10^{-16}	21.54	–	3.7469×10^{-16}	3.9065×10^{-16}	–	1.9691	1.8887
6700	Cornea left	0.629	3.9308×10^{-16}	11.69	–	4.5975×10^{-16}	3.6556×10^{-16}	–	0.8550	1.0753
6701	Aqueous left	0.275	5.1420×10^{-16}	17.63	–	2.2802×10^{-16}	3.5075×10^{-16}	–	2.2551	1.4660
6702	Vitreous left	2.62	3.5230×10^{-16}	5.71	–	4.1992×10^{-16}	4.1905×10^{-16}	–	0.8390	0.8407
6800	Eye lens sensitive right	0.0169	4.7844×10^{-16}	49.27	–	1.5068×10^{-16}	3.9060×10^{-16}	–	3.1753	1.2249
6801	Eye lens insensitive right	0.0978	8.6962×10^{-16}	28.69	–	1.1228×10^{-16}	2.5912×10^{-16}	–	7.7449	3.3561
6900	Cornea right	0.629	4.5324×10^{-16}	14.81	–	4.3824×10^{-16}	3.6948×10^{-16}	–	1.0342	1.2267
6901	Aqueous right	0.275	3.1871×10^{-16}	19.50	–	4.1619×10^{-16}	5.2425×10^{-16}	–	0.7658	0.6079
6902	Vitreous right	2.62	4.8413×10^{-16}	9.59	–	4.4961×10^{-16}	3.8580×10^{-16}	–	1.0768	1.2549
7000	Gall bladder wall	1.46	2.8645×10^{-14}	0.71	–	2.7987×10^{-14}	2.7335×10^{-14}	–	1.0235	1.0479
7100	Gall bladder contents	8.00	2.7136×10^{-14}	0.67	–	2.7167×10^{-14}	2.7283×10^{-14}	–	0.9989	0.9946
7200	Stomach wall(0-60)	0.814	1.2407×10^{-14}	1.72	–	1.2339×10^{-14}	1.2302×10^{-14}	–	1.0055	1.0085
7201	Stomach wall(60-100)	0.407	1.1699×10^{-14}	1.85	–	1.1800×10^{-14}	1.1088×10^{-14}	–	0.9914	1.0551
7202	Stomach wall(100-300)	2.06	1.1507×10^{-14}	1.39	–	1.1535×10^{-14}	1.1627×10^{-14}	–	0.9975	0.9897
7203	Stomach wall(300-surface)	19.88	1.1545×10^{-14}	0.85	–	1.1694×10^{-14}	1.1692×10^{-14}	–	0.9873	0.9875
7300	Stomach contents	67.00	1.0641×10^{-14}	0.17	–	1.0651×10^{-14}	1.0678×10^{-14}	–	0.9991	0.9965
7400	Small intestine wall(0-130)	4.93	5.2838×10^{-15}	1.51	–	5.1633×10^{-15}	5.1225×10^{-15}	–	1.0234	1.0315
7401	Small intestine wall(130-150)	0.77	5.4707×10^{-15}	1.89	–	5.4571×10^{-15}	5.3017×10^{-15}	–	1.0025	1.0319

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
7402	Small intestine wall(150-200)	1.94	5.3405×10^{-15}	0.90	–	5.2982×10^{-15}	5.0209×10^{-15}	–	1.0080	1.0637
7403	Small intestine wall(200-surface)	92.19	5.2091×10^{-15}	0.40	–	5.1931×10^{-15}	5.2118×10^{-15}	–	1.0031	0.9995
7500	Small intestine contents(-400-0)	14.32	5.1195×10^{-15}	0.80	–	5.1292×10^{-15}	5.1314×10^{-15}	–	0.9981	0.9977
7501	Small intestine contents(centre–400)	78.68	5.0503×10^{-15}	0.39	–	5.1321×10^{-15}	5.0924×10^{-15}	–	0.9841	0.9917
7600	Ascending colon wall(0-280)	1.57	2.9606×10^{-15}	2.57	–	2.9578×10^{-15}	3.2012×10^{-15}	–	1.0009	0.9248
7601	Ascending colon wall(280-300)	0.114	2.8922×10^{-15}	4.15	–	3.0697×10^{-15}	3.0768×10^{-15}	–	0.9422	0.9400
7602	Ascending colon wall(300-surface)	9.36	3.4713×10^{-15}	2.21	–	3.3001×10^{-15}	3.3954×10^{-15}	–	1.0519	1.0224
7700	Ascending colon content	26.02	2.8709×10^{-15}	1.18	–	2.9406×10^{-15}	2.9512×10^{-15}	–	0.9763	0.9728
7800	Transverse colon wall right(0-280)	1.20	1.0297×10^{-14}	1.30	–	9.9999×10^{-15}	9.9868×10^{-15}	–	1.0297	1.0310
7801	Transverse colon wall right(280-300)	0.0878	1.0525×10^{-14}	3.79	–	1.0620×10^{-14}	9.9552×10^{-15}	–	0.9911	1.0572
7802	Transverse colon wall right(300-surface)	11.14	1.0318×10^{-14}	1.16	–	1.0354×10^{-14}	1.0354×10^{-14}	–	0.9965	0.9965
7900	Transverse colon contents right	13.98	9.9431×10^{-15}	0.83	–	1.0020×10^{-14}	1.0104×10^{-14}	–	0.9923	0.9841
8000	Transverse colon wall left(0-280)	1.03	5.9814×10^{-15}	1.92	–	5.9490×10^{-15}	5.6862×10^{-15}	–	1.0054	1.0519
8001	Transverse colon wall left(280-300)	0.0757	5.9478×10^{-15}	4.29	–	6.2165×10^{-15}	5.8586×10^{-15}	–	0.9568	1.0152
8002	Transverse colon wall left(300-surface)	10.30	6.1615×10^{-15}	0.65	–	6.1222×10^{-15}	6.1103×10^{-15}	–	1.0064	1.0084
8100	Transverse colon content left	8.93	5.9788×10^{-15}	1.06	–	5.9536×10^{-15}	5.9188×10^{-15}	–	1.0042	1.0101
8200	Descending colon wall(0-280)	1.23	2.4362×10^{-15}	2.99	–	2.5095×10^{-15}	2.3617×10^{-15}	–	0.9708	1.0315
8201	Descending colon wall(280-300)	0.0906	2.7052×10^{-15}	6.52	–	2.5169×10^{-15}	2.2906×10^{-15}	–	1.0748	1.1810
8202	Descending colon wall(300-surface)	10.75	2.2916×10^{-15}	1.43	–	2.2767×10^{-15}	2.2830×10^{-15}	–	1.0066	1.0038
8300	Descending colon content	11.07	2.3396×10^{-15}	1.43	–	2.4015×10^{-15}	2.2734×10^{-15}	–	0.9742	1.0291
8400	Sigmoid colon wall(0-280)	1.56	1.5079×10^{-15}	3.07	–	1.4696×10^{-15}	1.5165×10^{-15}	–	1.0261	0.9943
8401	Sigmoid colon wall(280-300)	0.115	1.4642×10^{-15}	6.86	–	1.4727×10^{-15}	1.4939×10^{-15}	–	0.9942	0.9801
8402	Sigmoid colon wall(300-surface)	8.00	1.5090×10^{-15}	1.63	–	1.5971×10^{-15}	1.5335×10^{-15}	–	0.9449	0.9840
8500	Sigmoid colon contents	13.98	1.4520×10^{-15}	1.14	–	1.4632×10^{-15}	1.4481×10^{-15}	–	0.9923	1.0027
8600	Rectum wall(0-280)	0.657	8.4661×10^{-16}	6.60	–	9.8294×10^{-16}	8.8152×10^{-16}	–	0.8613	0.9604
8601	Rectum wall(280-300)	0.0484	1.0066×10^{-15}	16.78	–	1.1747×10^{-15}	1.1206×10^{-15}	–	0.8568	0.8982
8602	Rectum wall(300-surface)	1.36	9.1722×10^{-16}	6.00	–	1.0376×10^{-15}	9.3018×10^{-16}	–	0.8840	0.9861
8603	Rectum contents	6.02	9.4900×10^{-16}	3.58	–	9.1941×10^{-16}	8.8926×10^{-16}	–	1.0322	1.0672
8700	Heart wall	55.04	9.2001×10^{-15}	0.27	–	9.1491×10^{-15}	9.1603×10^{-15}	–	1.0056	1.0043
8800	Blood in heart chamber	48.00	9.1585×10^{-15}	0.37	–	9.0656×10^{-15}	9.1709×10^{-15}	–	1.0102	0.9986
8900	Kidney left cortex	27.76	5.1706×10^{-15}	0.70	–	5.0693×10^{-15}	5.1046×10^{-15}	–	1.0200	1.0129
9000	Kidney left medulla	9.92	5.7724×10^{-15}	1.13	–	5.8206×10^{-15}	5.8491×10^{-15}	–	0.9917	0.9869
9100	Kidney left pelvis	1.98	6.2721×10^{-15}	2.53	–	6.2522×10^{-15}	5.9874×10^{-15}	–	1.0032	1.0475
9200	Kidney right cortex	27.76	1.5466×10^{-14}	0.33	–	1.5416×10^{-14}	1.5464×10^{-14}	–	1.0033	1.0001
9300	Kidney right medulla	9.92	1.5052×10^{-14}	0.81	–	1.4951×10^{-14}	1.4988×10^{-14}	–	1.0068	1.0043
9400	Kidney right pelvis	1.98	1.4891×10^{-14}	1.60	–	1.4654×10^{-14}	1.4887×10^{-14}	–	1.0161	1.0003

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
9500	Liver	390.44	3.8011×10^{-14}	0.09	–	3.8036×10^{-14}	3.8216×10^{-14}	–	0.9993	0.9946
9700	Lung(AI) left	68.84	3.8246×10^{-15}	0.45	–	3.7648×10^{-15}	3.8350×10^{-15}	–	1.0159	0.9973
9900	Lung(AI) right	81.16	9.0299×10^{-15}	0.26	–	9.0480×10^{-15}	9.0839×10^{-15}	–	0.9980	0.9941
10000	Lymphatic nodes ET	2.18	6.5824×10^{-16}	7.85	–	7.1090×10^{-16}	6.9333×10^{-16}	–	0.9259	0.9494
10100	Lymphatic nodes thoracic	2.18	1.2411×10^{-14}	1.73	–	1.2571×10^{-14}	1.2559×10^{-14}	–	0.9873	0.9882
10200	Lymphatic nodes head	0.752	1.4223×10^{-15}	8.25	–	1.5240×10^{-15}	1.3550×10^{-15}	–	0.9332	1.0497
10300	Lymphatic nodes trunk	17.76	3.4337×10^{-15}	0.70	–	3.4366×10^{-15}	3.4798×10^{-15}	–	0.9991	0.9867
10400	Lymphatic nodes arms	1.50	4.0159×10^{-15}	2.15	–	3.9494×10^{-15}	3.9176×10^{-15}	–	1.0168	1.0251
10500	Lymphatic nodes legs	1.50	1.3575×10^{-16}	17.71	–	1.5271×10^{-16}	1.6622×10^{-16}	–	0.8889	0.8167
10600	Muscle head	267.77	7.2570×10^{-16}	0.66	–	7.1964×10^{-16}	7.2185×10^{-16}	–	1.0084*	1.0053*
10700	Muscle trunk	972.02	3.4683×10^{-15}	0.15	–	3.5359×10^{-15}	3.5568×10^{-15}	–	0.9809*	0.9751*
10800	Muscle arms	190.86	2.4037×10^{-15}	0.50	–	2.3692×10^{-15}	2.3703×10^{-15}	–	1.0145*	1.0141*
10900	Muscle legs	498.56	3.0111×10^{-16}	0.85	–	3.3120×10^{-16}	3.2996×10^{-16}	–	0.9092*	0.9126*
11000	Oesophagus wall(0-190)	0.545	4.3978×10^{-15}	5.19	–	4.0146×10^{-15}	4.6608×10^{-15}	–	1.0955	0.9436
11001	Oesophagus wall(190-200)	0.0296	4.8013×10^{-15}	4.72	–	4.7291×10^{-15}	4.8500×10^{-15}	–	1.0153	0.9900
11002	Oesophagus wall(200-surface)	5.20	4.4604×10^{-15}	1.43	–	4.4432×10^{-15}	4.5884×10^{-15}	–	1.0039	0.9721
11003	Oesophagus contents	3.79	4.3343×10^{-15}	1.49	–	4.2901×10^{-15}	4.4646×10^{-15}	–	1.0103	0.9708
11300	Pancreas	22.65	6.2213×10^{-15}	0.60	–	6.1446×10^{-15}	6.1996×10^{-15}	–	1.0125	1.0035
11400	Pituitary gland	0.156	1.8964×10^{-16}	27.49	–	3.8670×10^{-16}	3.7561×10^{-16}	–	0.4904	0.5049
11500	Prostate	1.04	8.2996×10^{-16}	9.80	–	7.4343×10^{-16}	1.0011×10^{-15}	–	1.1164	0.8291
11600	RST head	551.24	5.5660×10^{-16}	0.55	–	5.4957×10^{-16}	5.5429×10^{-16}	–	1.0128*	1.0042*
11700	RST trunk	2719.28	4.8975×10^{-15}	0.06	–	5.0120×10^{-15}	5.0370×10^{-15}	–	0.9771*	0.9723*
11800	RST arms	185.29	2.6579×10^{-15}	0.31	–	2.5487×10^{-15}	2.5615×10^{-15}	–	1.0429*	1.0377*
11900	RST legs	837.15	3.4450×10^{-16}	0.42	–	3.9477×10^{-16}	3.9322×10^{-16}	–	0.8727*	0.8761*
12000	Salivary glands left	12.48	8.6238×10^{-16}	2.33	–	8.6330×10^{-16}	8.1874×10^{-16}	–	0.9989	1.0533
12100	Salivary glandss right	12.49	1.0316×10^{-15}	1.96	–	9.8811×10^{-16}	1.0319×10^{-15}	–	1.0440	0.9997
12200	Skin head insensitive	67.32	4.0556×10^{-16}	1.45	–	3.9987×10^{-16}	4.0884×10^{-16}	–	1.0142	0.9920
12201	Skin head sensitive(40-100)	5.47	3.5000×10^{-16}	3.38	–	3.4655×10^{-16}	3.3573×10^{-16}	–	1.0099	1.0425
12300	Skin trunk insensitive	127.43	3.0346×10^{-15}	0.21	–	3.2000×10^{-15}	3.1763×10^{-15}	–	0.9483*	0.9554*
12301	Skin trunk sensitive(40-100)	11.20	2.5129×10^{-15}	0.76	–	2.7931×10^{-15}	2.7748×10^{-15}	–	0.8997*	0.9056*
12400	Skin arms insensitive	48.55	2.1644×10^{-15}	0.75	–	2.1201×10^{-15}	2.1284×10^{-15}	–	1.0209*	1.0169*
12401	Skin arms sensitive(40-100)	4.05	1.9687×10^{-15}	0.95	–	1.9781×10^{-15}	1.9424×10^{-15}	–	0.9952*	1.0135*
12500	Skin legs insensitive	90.30	2.2725×10^{-16}	1.32	–	2.6172×10^{-16}	2.5838×10^{-16}	–	0.8683*	0.8795*
12501	Skin legs sensitive(40-100)	6.63	1.8896×10^{-16}	4.31	–	2.2397×10^{-16}	2.2090×10^{-16}	–	0.8437*	0.8554*
12600	Spinal cord	10.64	4.1497×10^{-15}	0.65	–	4.3076×10^{-15}	4.2320×10^{-15}	–	0.9633	0.9806
12700	Spleen	37.35	3.9920×10^{-15}	0.48	–	3.9516×10^{-15}	3.9551×10^{-15}	–	1.0102	1.0093
12802	Erupted upper front buccal deciduous ena	0.0709	5.1063×10^{-16}	35.03	–	9.9270×10^{-16}	7.4584×10^{-16}	–	0.5144	0.6846

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
12803	Erupted upper front lingual deciduous en	0.0598	3.3263×10^{-16}	40.17	–	7.8306×10^{-16}	9.0214×10^{-16}	–	0.4248	0.3687
12818	Erupted lower front buccal deciduous ena	0.0453	1.0612×10^{-15}	37.88	–	5.5923×10^{-16}	6.7878×10^{-16}	–	1.8975	1.5633
12819	Erupted lower front lingual deciduous en	0.0425	6.6805×10^{-16}	45.75	–	9.6148×10^{-16}	9.8600×10^{-16}	–	0.6948	0.6775
12833	Erupted upper front deciduous dentin	0.3	5.7031×10^{-16}	12.38	–	5.6629×10^{-16}	7.2428×10^{-16}	–	1.0071	0.7874
12841	Erupted lower front deciduous dentin	0.199	7.2577×10^{-16}	15.53	–	7.7664×10^{-16}	9.8070×10^{-16}	–	0.9345	0.7401
12848	Unerupted permanent enamel	0.146	9.6480×10^{-16}	23.32	–	7.0228×10^{-16}	6.2546×10^{-16}	–	1.3738	1.5425
12849	Unerupted deciduous enamel	0.952	6.8932×10^{-16}	9.57	–	7.6019×10^{-16}	7.3589×10^{-16}	–	0.9068	0.9367
12850	Unerupted permanent dentin	0.486	8.2772×10^{-16}	11.34	–	4.8393×10^{-16}	6.1691×10^{-16}	–	1.7104	1.3417
12851	Unerupted deciduous dentin	2.20	7.7467×10^{-16}	6.53	–	8.2666×10^{-16}	7.6695×10^{-16}	–	0.9371	1.0101
12852	Permanent cementum	0.0239	9.0428×10^{-16}	23.75	–	7.3008×10^{-16}	1.2067×10^{-15}	–	1.2386	0.7494
12853	Deciduous cementum	0.125	5.8583×10^{-16}	14.86	–	8.1322×10^{-16}	8.6841×10^{-16}	–	0.7204	0.6746
12854	Permanent pulp	0.0146	1.1296×10^{-15}	42.42	–	5.2566×10^{-16}	8.4230×10^{-16}	–	2.1490	1.3411
12855	Deciduous pulp	0.374	8.4870×10^{-16}	17.34	–	7.3313×10^{-16}	1.0087×10^{-15}	–	1.1576	0.8414
12856	Teeth retention region	0.00345	8.4743×10^{-16}	35.17	–	3.7529×10^{-16}	3.7926×10^{-16}	–	2.2581	2.2344
12900	Testis left	0.773	6.5442×10^{-16}	12.80	–	7.6667×10^{-16}	8.5564×10^{-16}	–	0.8536	0.7648
13000	Testis right	0.773	7.5538×10^{-16}	8.26	–	7.7094×10^{-16}	7.2316×10^{-16}	–	0.9798	1.0446
13100	Thymus	31.21	3.0002×10^{-15}	0.87	–	3.0409×10^{-15}	3.0204×10^{-15}	–	0.9866	0.9933
13200	Thyroid	1.97	1.3925×10^{-15}	5.77	–	1.4048×10^{-15}	1.1925×10^{-15}	–	0.9913	1.1677
13300	Tongue upper(food)	5.97	7.9063×10^{-16}	3.10	–	7.5295×10^{-16}	7.7290×10^{-16}	–	1.0500	1.0229
13301	Tongue lower	4.42	9.0690×10^{-16}	4.14	–	9.1748×10^{-16}	9.7103×10^{-16}	–	0.9885	0.9340
13400	Tonsils	0.521	8.9683×10^{-16}	14.25	–	8.0852×10^{-16}	7.4883×10^{-16}	–	1.1092	1.1976
13500	Ureter left	1.15	3.7923×10^{-15}	3.42	–	3.9251×10^{-15}	3.8093×10^{-15}	–	0.9662	0.9955
13600	Ureter right	1.15	5.5071×10^{-15}	3.08	–	5.6565×10^{-15}	5.3272×10^{-15}	–	0.9736	1.0338
13700	Urinary bladder wall insensitive	8.16	1.4394×10^{-15}	2.23	–	1.4737×10^{-15}	1.4152×10^{-15}	–	0.9767	1.0171
13701	Urinary bladder wall sensitive(71-238)	0.948	1.4134×10^{-15}	4.06	–	1.4653×10^{-15}	1.5051×10^{-15}	–	0.9646	0.9391
13800	Urinary bladder content	32.90	1.4412×10^{-15}	1.21	–	1.4431×10^{-15}	1.4307×10^{-15}	–	0.9987	1.0074
14000	Air inside body	0.00925	2.4675×10^{-15}	13.96	–	3.1262×10^{-15}	2.2743×10^{-15}	–	0.7893	1.0849

* For tissues bookkept as head, trunk, arm and leg pieces (muscle, residual soft tissue, skin, large vessels), the volumes tabulated in the distributed cell tables partition the tissue at different boundaries from the transported mesh, and the reference output is normalised by the tabulated volume. The marked ratio therefore compares two different partitions of the same tissue, and is left out of the summary statistics of Table S2.

Table S12. 01M_External. Absorbed dose per primary history, Gy. σ is the FLUKA statistical uncertainty. Ratios are FLUKA / reference.

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
100	Adrenal left	2.26	1.0159×10^{-17}	58.15	–	2.1712×10^{-17}	2.9179×10^{-17}	–	0.4679	0.3482
200	Adrenal right	2.26	2.4808×10^{-17}	37.31	–	2.1536×10^{-17}	3.0615×10^{-17}	–	1.1519	0.8103
300	ET1(0-8)	0.00124	4.0857×10^{-17}	100.00	–	5.5960×10^{-17}	–	–	0.7301	–
301	ET1(8-40)	0.00504	5.1327×10^{-17}	100.00	–	4.3884×10^{-17}	–	–	1.1696	–
302	ET1(40-50)	0.0016	6.7447×10^{-17}	100.00	–	5.2466×10^{-17}	–	–	1.2856	–
303	ET1(50-Surface)	0.218	5.9640×10^{-17}	59.05	–	5.8523×10^{-17}	7.6410×10^{-17}	–	1.0191	0.7805
400	ET2(-15-0)	0.0692	7.6057×10^{-17}	47.01	–	7.2551×10^{-17}	1.9629×10^{-17}	–	1.0483	3.8747
401	ET2(0-40)	0.198	9.8774×10^{-17}	34.24	–	4.8691×10^{-17}	5.2820×10^{-17}	–	2.0286	1.8700
402	ET2(40-50)	0.0498	9.3531×10^{-17}	38.62	–	1.8803×10^{-17}	6.2459×10^{-17}	–	4.9742	1.4975
403	ET2(50-55)	0.0249	7.3313×10^{-17}	44.01	–	1.5235×10^{-17}	1.8318×10^{-17}	–	4.8123	4.0023
404	ET2(55-65)	0.05	5.8554×10^{-17}	39.99	–	2.7892×10^{-17}	1.4818×10^{-17}	–	2.0993	3.9515
405	ET2(65-Surface)	6.93	4.8911×10^{-17}	12.11	–	4.5042×10^{-17}	3.4420×10^{-17}	–	1.0859	1.4210
500	Oral mucosa tongue	0.0187	1.3243×10^{-16}	95.39	–	4.7493×10^{-17}	4.3326×10^{-17}	–	2.7885	3.0567
501	Oral mucosa moth floor	0.0143	0.0000×10^{0}	0.00	–	8.4025×10^{-17}	1.4619×10^{-16}	–	–	–
600	Oral mucosa lips and cheeks	0.0045	0.0000×10^{0}	0.00	–	4.4819×10^{-17}	2.8266×10^{-17}	–	–	–
700	Trachea	1.56	5.1149×10^{-17}	37.04	–	3.7825×10^{-17}	5.4623×10^{-17}	–	1.3523	0.9364
800	BB(-11–6)	0.00302	6.0016×10^{-17}	68.19	–	–	–	–	–	–
801	BB(-6-0)	0.00387	1.0833×10^{-16}	67.99	–	5.3419×10^{-17}	–	–	2.0279	–
802	BB(0-10)	0.00648	1.8198×10^{-16}	74.73	–	2.5423×10^{-17}	–	–	7.1579	–
803	BB(10-35)	0.0163	2.1804×10^{-16}	75.52	–	2.0309×10^{-17}	–	–	10.7365	–
804	BB(35-40)	0.00329	9.6666×10^{-17}	74.90	–	3.6299×10^{-17}	–	–	2.6630	–
805	BB(40-50)	0.0066	1.4500×10^{-16}	82.19	–	4.3818×10^{-17}	–	–	3.3090	–
806	BB(50-60)	0.00663	3.2273×10^{-16}	68.07	–	2.2520×10^{-17}	4.6570×10^{-17}	–	14.3307	6.9299
807	BB(60-70)	0.00667	5.2064×10^{-16}	80.53	–	3.1493×10^{-17}	4.9568×10^{-17}	–	16.5316	10.5035
808	BB(70-surface)	1.92	4.0000×10^{-17}	28.20	–	2.2165×10^{-17}	3.1784×10^{-17}	–	1.8046	1.2585
900	Blood in large arteries head	0.167	5.3484×10^{-17}	67.28	–	2.1902×10^{-17}	4.5180×10^{-17}	–	2.4419*	1.1838*
910	Blood in large veins head	0.56	5.3656×10^{-18}	48.79	–	3.3534×10^{-17}	3.5772×10^{-17}	–	0.1600*	0.1500*
1000	Blood in large arteries trunk	15.42	2.7158×10^{-17}	11.77	–	3.7387×10^{-17}	3.4783×10^{-17}	–	0.7264*	0.7808*
1010	Blood in large veins trunk	42.08	3.4318×10^{-17}	5.71	–	3.5763×10^{-17}	3.2740×10^{-17}	–	0.9596*	1.0482*
1100	Blood in large arteries arms	3.14	2.1266×10^{-17}	22.11	–	4.4520×10^{-17}	3.1637×10^{-17}	–	0.4777*	0.6722*
1110	Blood in large veins arms	13.41	4.1882×10^{-17}	10.29	–	4.0617×10^{-17}	3.0620×10^{-17}	–	1.0311*	1.3678*
1200	Blood in large arteries legs	13.09	3.5735×10^{-17}	12.21	–	3.6472×10^{-17}	3.6608×10^{-17}	–	0.9798*	0.9762*
1210	Blood in large veins legs	39.39	3.7493×10^{-17}	6.45	–	3.1188×10^{-17}	3.6188×10^{-17}	–	1.2021*	1.0360*
1300	Humeri upper cortical	12.67	2.9533×10^{-17}	16.90	–	4.3270×10^{-17}	2.6570×10^{-17}	–	0.6825	1.1115
1400	Humeri upper spogiosa	8.73	2.8318×10^{-17}	21.52	–	3.3173×10^{-17}	4.4434×10^{-17}	–	0.8537	0.6373

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
1500	Humeri upper medullary cavity	0.863	3.1886×10^{-17}	64.48	–	2.8971×10^{-17}	3.3252×10^{-17}	–	1.1006	0.9589
1600	Humeri lower cortical	9.04	3.3762×10^{-17}	19.89	–	3.3891×10^{-17}	2.7190×10^{-17}	–	0.9962	1.2417
1700	Humeri lower spongiosa	4.52	3.6739×10^{-17}	15.65	–	2.7131×10^{-17}	4.7635×10^{-17}	–	1.3541	0.7713
1800	Humeri lower medullary cavity	0.809	4.9880×10^{-17}	50.02	–	5.3248×10^{-17}	2.7742×10^{-17}	–	0.9367	1.7980
1900	Radii cortical	5.62	4.6545×10^{-17}	20.52	–	3.7276×10^{-17}	4.5694×10^{-17}	–	1.2487	1.0186
1910	Ulnae cortical	7.42	3.6155×10^{-17}	13.77	–	3.3814×10^{-17}	4.1183×10^{-17}	–	1.0692	0.8779
2000	Radii spongiosa	1.12	3.9777×10^{-17}	51.24	–	4.6262×10^{-17}	3.4738×10^{-17}	–	0.8598	1.1451
2010	Ulnae spongiosa	1.81	5.5266×10^{-17}	21.64	–	3.5560×10^{-17}	3.8246×10^{-17}	–	1.5542	1.4450
2100	Radii medullary cavity	0.359	4.3301×10^{-17}	44.46	–	1.0718×10^{-16}	5.3857×10^{-17}	–	0.4040	0.8040
2110	Ulnae medullary cavity	0.589	4.2348×10^{-17}	47.73	–	2.6559×10^{-17}	5.2469×10^{-17}	–	1.5945	0.8071
2200	Wrists and hand bones cortical	5.40	3.0645×10^{-17}	19.30	–	3.9250×10^{-17}	4.3311×10^{-17}	–	0.7808	0.7076
2300	Wrists and hand bones spongiosa	14.07	3.1751×10^{-17}	13.04	–	3.4950×10^{-17}	4.4776×10^{-17}	–	0.9085	0.7091
2400	Clavicles cortical	1.37	2.4871×10^{-17}	38.08	–	2.2693×10^{-17}	3.0912×10^{-17}	–	1.0960	0.8046
2500	Clavicles spongiosa	1.96	3.5460×10^{-17}	37.41	–	5.0517×10^{-17}	5.3153×10^{-17}	–	0.7019	0.6671
2600	Cranium cortical	91.49	2.8293×10^{-17}	2.33	–	2.8676×10^{-17}	2.8657×10^{-17}	–	0.9866	0.9873
2700	Cranium spongiosa	289.87	2.7263×10^{-17}	2.55	–	2.8109×10^{-17}	2.7927×10^{-17}	–	0.9699	0.9762
2800	Femora upper cortical	14.11	3.6174×10^{-17}	8.51	–	3.3062×10^{-17}	2.9848×10^{-17}	–	1.0941	1.2119
2900	Femora upper spongiosa	8.87	3.9522×10^{-17}	17.39	–	3.3762×10^{-17}	3.4415×10^{-17}	–	1.1706	1.1484
3000	Femora upper medullary cavity	0.994	1.3571×10^{-17}	32.22	–	2.6154×10^{-17}	6.8610×10^{-17}	–	0.5189	0.1978
3100	Femora lower cortical	20.54	2.6888×10^{-17}	9.73	–	2.9673×10^{-17}	3.3731×10^{-17}	–	0.9061	0.7971
3200	Femora lower spongiosa	8.76	3.9083×10^{-17}	14.47	–	2.5437×10^{-17}	3.1304×10^{-17}	–	1.5364	1.2485
3300	Femora lower medullary cavity	1.57	3.1141×10^{-17}	57.28	–	2.6879×10^{-17}	4.4821×10^{-17}	–	1.1586	0.6948
3400	Tibiae cortical	26.33	3.3485×10^{-17}	7.18	–	3.1282×10^{-17}	3.3896×10^{-17}	–	1.0704	0.9879
3410	Fibulae cortical	3.02	1.8794×10^{-17}	32.98	–	1.6322×10^{-17}	2.7675×10^{-17}	–	1.1515	0.6791
3420	Patellae cortical	0.162	1.2971×10^{-16}	53.91	–	1.2177×10^{-17}	3.5854×10^{-17}	–	10.6517	3.6176
3500	Tibiae spongiosa	7.27	2.8225×10^{-17}	18.50	–	3.4288×10^{-17}	3.4799×10^{-17}	–	0.8232	0.8111
3510	Fibulae spongiosa	0.565	2.1879×10^{-17}	65.80	–	2.2521×10^{-17}	1.1466×10^{-17}	–	0.9715	1.9081
3520	Patellae spongiosa	0.575	3.2714×10^{-17}	79.25	–	2.0802×10^{-17}	2.5804×10^{-17}	–	1.5726	1.2678
3600	Tibiae medullary cavity	2.20	4.1061×10^{-17}	27.56	–	4.2803×10^{-17}	3.6850×10^{-17}	–	0.9593	1.1143
3610	Fibulae medullary cavity	0.204	9.2132×10^{-18}	100.00	–	5.4799×10^{-17}	–	–	0.1681	–
3700	Ankles and foot cortical	7.24	2.9747×10^{-17}	15.18	–	2.8913×10^{-17}	3.4554×10^{-17}	–	1.0289	0.8609
3800	Ankles and foot spongiosa	20.33	2.9102×10^{-17}	11.04	–	3.3325×10^{-17}	3.3087×10^{-17}	–	0.8733	0.8796
3900	Mandible cortical	7.16	3.6618×10^{-17}	12.54	–	2.9833×10^{-17}	3.5137×10^{-17}	–	1.2274	1.0421
4000	Mandible spongiosa	13.09	4.3729×10^{-17}	7.28	–	3.9248×10^{-17}	3.0404×10^{-17}	–	1.1142	1.4383
4100	Pelvis cortical	21.20	3.6460×10^{-17}	7.91	–	2.9572×10^{-17}	2.7802×10^{-17}	–	1.2329	1.3114
4200	Pelvis spongiosa	34.40	3.2909×10^{-17}	7.79	–	3.3129×10^{-17}	3.2249×10^{-17}	–	0.9934	1.0205
4300	Ribs cortical	20.41	2.9432×10^{-17}	7.98	–	2.7972×10^{-17}	2.6766×10^{-17}	–	1.0522	1.0996
4400	Ribs spongiosa	58.12	2.8279×10^{-17}	6.29	–	3.1959×10^{-17}	3.2466×10^{-17}	–	0.8848	0.8710

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
4500	Scapulae cortical	9.65	2.1917×10^{-17}	12.11	–	2.0939×10^{-17}	2.7065×10^{-17}	–	1.0467	0.8098
4600	Scapulae spongiosa	14.86	1.9793×10^{-17}	19.92	–	2.6170×10^{-17}	2.5178×10^{-17}	–	0.7563	0.7861
4700	Cervical spine cortical	3.28	2.6758×10^{-17}	18.92	–	2.6170×10^{-17}	2.4179×10^{-17}	–	1.0225	1.1067
4800	Cervical spine spongiosa	9.10	4.0933×10^{-17}	9.08	–	2.4698×10^{-17}	3.3438×10^{-17}	–	1.6574	1.2242
4900	Thoracic spine cortical	11.50	2.2223×10^{-17}	14.36	–	2.6067×10^{-17}	2.1484×10^{-17}	–	0.8525	1.0344
5000	Thoracic spine spongiosa	28.11	2.8981×10^{-17}	8.54	–	2.2272×10^{-17}	2.2087×10^{-17}	–	1.3012	1.3121
5100	Lumbar spine cortical	7.07	2.1491×10^{-17}	13.27	–	2.4281×10^{-17}	2.5965×10^{-17}	–	0.8851	0.8277
5200	Lumbar spine spongiosa	21.95	2.6834×10^{-17}	6.66	–	2.8397×10^{-17}	2.4225×10^{-17}	–	0.9450	1.1077
5300	Sacrum cortical	5.45	2.3702×10^{-17}	14.84	–	2.3944×10^{-17}	2.8079×10^{-17}	–	0.9899	0.8441
5400	Sacrum spongiosa	15.12	2.8770×10^{-17}	11.18	–	2.8052×10^{-17}	3.0903×10^{-17}	–	1.0256	0.9310
5500	Sternum cortical	0.63	4.4887×10^{-17}	37.57	–	2.2356×10^{-17}	4.2906×10^{-17}	–	2.0078	1.0462
5600	Sternum spongiosa	2.10	3.6137×10^{-17}	20.55	–	4.2802×10^{-17}	4.2700×10^{-17}	–	0.8443	0.8463
5700	Cartilage head	5.03	2.8404×10^{-17}	15.81	–	2.4684×10^{-17}	2.3037×10^{-17}	–	1.1507	1.2330
5800	Cartilage trunk	36.85	3.3019×10^{-17}	8.43	–	3.6140×10^{-17}	3.3471×10^{-17}	–	0.9136	0.9865
6100	Brain	978.10	2.6938×10^{-17}	1.66	–	2.7098×10^{-17}	2.6824×10^{-17}	–	0.9941	1.0043
6200	Breast left adipose tissue	0.571	3.4159×10^{-17}	63.37	–	7.9375×10^{-18}	8.5340×10^{-17}	–	4.3035	0.4003
6300	Breast left glandular tissue	0.381	6.7608×10^{-17}	52.88	–	1.3779×10^{-19}	4.7738×10^{-17}	–	490.6541	1.4162
6400	Breast right adipose tissue	0.571	2.9553×10^{-17}	57.50	–	8.4204×10^{-17}	3.7109×10^{-17}	–	0.3510	0.7964
6500	Breast right glandular tissue	0.381	2.8561×10^{-17}	58.62	–	7.3573×10^{-17}	3.2071×10^{-17}	–	0.3882	0.8905
6600	Eye lens sensitive left	0.0169	0.0000×10^{0}	0.00	–	–	1.1160×10^{-16}	–	–	–
6601	Eye lens insensitive left	0.0978	0.0000×10^{0}	0.00	–	–	4.0823×10^{-17}	–	–	–
6700	Cornea left	0.629	3.6352×10^{-17}	44.22	–	3.1802×10^{-17}	5.1214×10^{-17}	–	1.1431	0.7098
6701	Aqueous left	0.275	7.9789×10^{-17}	64.43	–	7.5140×10^{-18}	7.2162×10^{-17}	–	10.6188	1.1057
6702	Vitreous left	2.62	1.9824×10^{-17}	41.00	–	3.3552×10^{-17}	2.2097×10^{-17}	–	0.5908	0.8971
6800	Eye lens sensitive right	0.0169	0.0000×10^{0}	0.00	–	–	–	–	–	–
6801	Eye lens insensitive right	0.0978	0.0000×10^{0}	0.00	–	3.9090×10^{-19}	–	–	–	–
6900	Cornea right	0.629	1.6366×10^{-18}	96.94	–	2.5418×10^{-17}	3.2937×10^{-17}	–	0.0644	0.0497
6901	Aqueous right	0.275	5.8635×10^{-17}	52.45	–	4.8426×10^{-17}	2.9525×10^{-17}	–	1.2108	1.9859
6902	Vitreous right	2.62	3.1952×10^{-17}	30.84	–	5.6407×10^{-17}	3.1745×10^{-17}	–	0.5665	1.0065
7000	Gall bladder wall	1.46	2.6380×10^{-17}	36.20	–	2.9129×10^{-17}	1.7963×10^{-17}	–	0.9056	1.4686
7100	Gall bladder contents	8.00	2.7660×10^{-17}	20.57	–	4.3712×10^{-17}	2.9581×10^{-17}	–	0.6328	0.9351
7200	Stomach wall(0-60)	0.814	1.7464×10^{-17}	28.81	–	2.9477×10^{-17}	2.4185×10^{-17}	–	0.5925	0.7221
7201	Stomach wall(60-100)	0.407	2.9346×10^{-17}	46.96	–	2.6715×10^{-17}	1.8290×10^{-17}	–	1.0985	1.6045
7202	Stomach wall(100-300)	2.06	3.8541×10^{-17}	23.20	–	3.8441×10^{-17}	3.6157×10^{-17}	–	1.0026	1.0659
7203	Stomach wall(300-surface)	19.88	3.3841×10^{-17}	6.35	–	3.8427×10^{-17}	2.8704×10^{-17}	–	0.8806	1.1789
7300	Stomach contents	67.00	3.3135×10^{-17}	6.27	–	3.3585×10^{-17}	3.1930×10^{-17}	–	0.9866	1.0378
7400	Small intestine wall(0-130)	4.93	3.1876×10^{-17}	8.60	–	3.5161×10^{-17}	3.2218×10^{-17}	–	0.9066	0.9894
7401	Small intestine wall(130-150)	0.77	4.0336×10^{-17}	20.42	–	3.1505×10^{-17}	3.9033×10^{-17}	–	1.2803	1.0334

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
7402	Small intestine wall(150-200)	1.94	3.6958×10^{-17}	13.40	–	3.7949×10^{-17}	2.9248×10^{-17}	–	0.9739	1.2636
7403	Small intestine wall(200-surface)	92.19	3.5500×10^{-17}	5.49	–	3.8650×10^{-17}	3.6799×10^{-17}	–	0.9185	0.9647
7500	Small intestine contents(-400-0)	14.32	3.6863×10^{-17}	10.20	–	3.3555×10^{-17}	3.6196×10^{-17}	–	1.0986	1.0184
7501	Small intestine contents(centre–400)	78.68	3.5872×10^{-17}	5.93	–	3.5841×10^{-17}	3.2332×10^{-17}	–	1.0009	1.1095
7600	Ascending colon wall(0-280)	1.57	3.6866×10^{-17}	13.15	–	3.3228×10^{-17}	3.3417×10^{-17}	–	1.1095	1.1032
7601	Ascending colon wall(280-300)	0.114	4.2329×10^{-17}	51.27	–	2.4291×10^{-17}	2.6680×10^{-17}	–	1.7426	1.5866
7602	Ascending colon wall(300-surface)	9.36	4.9226×10^{-17}	10.64	–	4.9336×10^{-17}	3.3206×10^{-17}	–	0.9978	1.4825
7700	Ascending colon content	26.02	4.1557×10^{-17}	6.07	–	3.7664×10^{-17}	3.5661×10^{-17}	–	1.1034	1.1653
7800	Transverse colon wall right(0-280)	1.20	5.6664×10^{-17}	21.68	–	4.0939×10^{-17}	5.1734×10^{-17}	–	1.3841	1.0953
7801	Transverse colon wall right(280-300)	0.0878	5.0747×10^{-17}	44.72	–	3.3454×10^{-17}	8.6989×10^{-17}	–	1.5169	0.5834
7802	Transverse colon wall right(300-surface)	11.14	3.8474×10^{-17}	10.72	–	4.4590×10^{-17}	4.7414×10^{-17}	–	0.8628	0.8114
7900	Transverse colon contents right	13.98	3.4208×10^{-17}	13.21	–	5.0496×10^{-17}	4.1077×10^{-17}	–	0.6774	0.8328
8000	Transverse colon wall left(0-280)	1.03	5.2463×10^{-17}	25.62	–	3.8947×10^{-17}	3.8100×10^{-17}	–	1.3470	1.3770
8001	Transverse colon wall left(280-300)	0.0757	5.9489×10^{-17}	65.30	–	8.0530×10^{-17}	1.5814×10^{-17}	–	0.7387	3.7618
8002	Transverse colon wall left(300-surface)	10.30	4.2123×10^{-17}	15.68	–	3.8755×10^{-17}	4.4429×10^{-17}	–	1.0869	0.9481
8100	Transverse colon content left	8.93	3.2633×10^{-17}	14.43	–	3.3025×10^{-17}	3.3189×10^{-17}	–	0.9881	0.9832
8200	Descending colon wall(0-280)	1.23	2.9761×10^{-17}	21.57	–	3.4806×10^{-17}	2.5087×10^{-17}	–	0.8551	1.1863
8201	Descending colon wall(280-300)	0.0906	3.4706×10^{-17}	34.94	–	2.0418×10^{-17}	2.9724×10^{-17}	–	1.6998	1.1676
8202	Descending colon wall(300-surface)	10.75	3.2512×10^{-17}	13.98	–	3.3465×10^{-17}	3.3937×10^{-17}	–	0.9715	0.9580
8300	Descending colon content	11.07	2.7979×10^{-17}	20.20	–	2.8666×10^{-17}	2.9806×10^{-17}	–	0.9760	0.9387
8400	Sigmoid colon wall(0-280)	1.56	4.6016×10^{-17}	19.91	–	3.3156×10^{-17}	2.0980×10^{-17}	–	1.3878	2.1933
8401	Sigmoid colon wall(280-300)	0.115	2.0004×10^{-17}	39.90	–	2.5743×10^{-17}	6.2161×10^{-17}	–	0.7771	0.3218
8402	Sigmoid colon wall(300-surface)	8.00	3.3475×10^{-17}	9.94	–	2.6599×10^{-17}	2.7052×10^{-17}	–	1.2585	1.2374
8500	Sigmoid colon contents	13.98	3.6385×10^{-17}	11.97	–	2.3033×10^{-17}	3.5621×10^{-17}	–	1.5797	1.0214
8600	Rectum wall(0-280)	0.657	1.3424×10^{-17}	45.44	–	2.7136×10^{-17}	3.2853×10^{-17}	–	0.4947	0.4086
8601	Rectum wall(280-300)	0.0484	7.3113×10^{-18}	56.27	–	1.8259×10^{-17}	2.1669×10^{-17}	–	0.4004	0.3374
8602	Rectum wall(300-surface)	1.36	3.0164×10^{-17}	23.40	–	2.1935×10^{-17}	2.3991×10^{-17}	–	1.3752	1.2573
8603	Rectum contents	6.02	2.3653×10^{-17}	14.28	–	3.0740×10^{-17}	2.9945×10^{-17}	–	0.7695	0.7899
8700	Heart wall	55.04	3.2068×10^{-17}	7.41	–	3.2723×10^{-17}	3.6407×10^{-17}	–	0.9800	0.8808
8800	Blood in heart chamber	48.00	3.4803×10^{-17}	4.71	–	3.0697×10^{-17}	3.4701×10^{-17}	–	1.1337	1.0029
8900	Kidney left cortex	27.76	2.6710×10^{-17}	15.97	–	2.4036×10^{-17}	2.5969×10^{-17}	–	1.1113	1.0285
9000	Kidney left medulla	9.92	2.6215×10^{-17}	15.94	–	3.0387×10^{-17}	2.3259×10^{-17}	–	0.8627	1.1271
9100	Kidney left pelvis	1.98	4.3311×10^{-17}	28.62	–	4.1371×10^{-17}	4.6894×10^{-17}	–	1.0469	0.9236
9200	Kidney right cortex	27.76	3.1644×10^{-17}	7.89	–	2.3823×10^{-17}	2.5921×10^{-17}	–	1.3283	1.2208
9300	Kidney right medulla	9.92	2.3159×10^{-17}	19.14	–	3.3164×10^{-17}	3.3090×10^{-17}	–	0.6983	0.6999
9400	Kidney right pelvis	1.98	3.8679×10^{-18}	50.27	–	2.7383×10^{-17}	2.4052×10^{-17}	–	0.1413	0.1608

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
9500	Liver	390.44	3.3590×10^{-17}	1.78	–	3.4959×10^{-17}	3.4092×10^{-17}	–	0.9608	0.9853
9700	Lung(AI) left	68.84	3.4927×10^{-17}	6.20	–	3.2187×10^{-17}	3.2581×10^{-17}	–	1.0851	1.0720
9900	Lung(AI) right	81.16	3.3906×10^{-17}	6.55	–	3.3332×10^{-17}	3.4037×10^{-17}	–	1.0172	0.9961
10000	Lymphatic nodes ET	2.18	2.5584×10^{-17}	24.54	–	5.1473×10^{-17}	3.9971×10^{-17}	–	0.4970	0.6401
10100	Lymphatic nodes thoracic	2.18	4.2030×10^{-17}	32.78	–	4.0665×10^{-17}	2.8403×10^{-17}	–	1.0336	1.4798
10200	Lymphatic nodes head	0.752	7.6294×10^{-17}	24.68	–	1.5341×10^{-17}	4.8560×10^{-17}	–	4.9732	1.5711
10300	Lymphatic nodes trunk	17.76	3.3234×10^{-17}	11.88	–	3.7009×10^{-17}	3.8106×10^{-17}	–	0.8980	0.8721
10400	Lymphatic nodes arms	1.50	4.2642×10^{-17}	35.18	–	2.7113×10^{-17}	3.5055×10^{-17}	–	1.5728	1.2164
10500	Lymphatic nodes legs	1.50	4.2148×10^{-17}	21.87	–	6.2254×10^{-17}	2.9886×10^{-17}	–	0.6770	1.4103
10600	Muscle head	267.77	3.3405×10^{-17}	2.01	–	3.1583×10^{-17}	3.2646×10^{-17}	–	1.0577*	1.0232*
10700	Muscle trunk	972.02	3.2172×10^{-17}	1.95	–	3.2761×10^{-17}	3.2039×10^{-17}	–	0.9820*	1.0042*
10800	Muscle arms	190.86	3.7798×10^{-17}	2.75	–	3.7767×10^{-17}	3.6609×10^{-17}	–	1.0008*	1.0325*
10900	Muscle legs	498.56	3.2263×10^{-17}	2.40	–	3.6330×10^{-17}	3.4112×10^{-17}	–	0.8880*	0.9458*
11000	Oesophagus wall(0-190)	0.545	3.8975×10^{-17}	32.14	–	1.7673×10^{-17}	3.1881×10^{-17}	–	2.2053	1.2225
11001	Oesophagus wall(190-200)	0.0296	2.8902×10^{-17}	57.05	–	3.6563×10^{-17}	1.1933×10^{-17}	–	0.7905	2.4220
11002	Oesophagus wall(200-surface)	5.20	3.4270×10^{-17}	20.02	–	3.1172×10^{-17}	2.4846×10^{-17}	–	1.0994	1.3793
11003	Oesophagus contents	3.79	1.9492×10^{-17}	31.07	–	2.9074×10^{-17}	2.8094×10^{-17}	–	0.6704	0.6938
11300	Pancreas	22.65	3.6463×10^{-17}	10.84	–	2.7741×10^{-17}	3.2674×10^{-17}	–	1.3144	1.1160
11400	Pituitary gland	0.156	0.0000×10^{0}	0.00	–	1.0170×10^{-16}	6.4923×10^{-17}	–	–	–
11500	Prostate	1.04	3.2688×10^{-17}	44.51	–	4.3476×10^{-17}	2.7905×10^{-17}	–	0.7519	1.1714
11600	RST head	551.24	3.1714×10^{-17}	1.76	–	3.0883×10^{-17}	3.2740×10^{-17}	–	1.0269*	0.9687*
11700	RST trunk	2719.28	3.4417×10^{-17}	1.09	–	3.4338×10^{-17}	3.4402×10^{-17}	–	1.0023*	1.0004*
11800	RST arms	185.29	3.7625×10^{-17}	4.28	–	3.9088×10^{-17}	3.8352×10^{-17}	–	0.9626*	0.9810*
11900	RST legs	837.15	3.5125×10^{-17}	1.32	–	3.5012×10^{-17}	3.4884×10^{-17}	–	1.0032*	1.0069*
12000	Salivary glands left	12.48	3.7052×10^{-17}	12.78	–	3.6509×10^{-17}	3.4424×10^{-17}	–	1.0149	1.0763
12100	Salivary glandss right	12.49	3.1734×10^{-17}	5.40	–	3.3812×10^{-17}	2.1605×10^{-17}	–	0.9385	1.4688
12200	Skin head insensitive	67.32	2.2896×10^{-17}	7.72	–	2.3119×10^{-17}	2.4164×10^{-17}	–	0.9904	0.9475
12201	Skin head sensitive(40-100)	5.47	1.4666×10^{-17}	17.02	–	1.5368×10^{-17}	2.0259×10^{-17}	–	0.9543	0.7239
12300	Skin trunk insensitive	127.43	2.4082×10^{-17}	3.34	–	2.4015×10^{-17}	2.5014×10^{-17}	–	1.0028*	0.9628*
12301	Skin trunk sensitive(40-100)	11.20	1.8366×10^{-17}	9.85	–	1.6965×10^{-17}	1.7610×10^{-17}	–	1.0826*	1.0429*
12400	Skin arms insensitive	48.55	2.9883×10^{-17}	6.31	–	2.9328×10^{-17}	3.2302×10^{-17}	–	1.0189*	0.9251*
12401	Skin arms sensitive(40-100)	4.05	2.5637×10^{-17}	12.41	–	2.5458×10^{-17}	2.5199×10^{-17}	–	1.0070*	1.0174*
12500	Skin legs insensitive	90.30	2.4287×10^{-17}	3.10	–	2.5306×10^{-17}	2.7766×10^{-17}	–	0.9598*	0.8747*
12501	Skin legs sensitive(40-100)	6.63	1.7290×10^{-17}	13.93	–	1.9131×10^{-17}	1.7683×10^{-17}	–	0.9038*	0.9778*
12600	Spinal cord	10.64	2.1399×10^{-17}	16.70	–	2.0792×10^{-17}	2.8201×10^{-17}	–	1.0292	0.7588
12700	Spleen	37.35	3.4698×10^{-17}	8.05	–	3.5106×10^{-17}	2.5344×10^{-17}	–	0.9884	1.3691
12802	Erupted upper front buccal deciduous ena	0.0709	0.0000×10^{0}	0.00	–	1.0832×10^{-16}	1.5929×10^{-16}	–	–	–

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
12803	Erupted upper front lingual deciduous en	0.0598	6.2757×10^{-18}	100.00	–	1.6622×10^{-16}	9.9273×10^{-18}	–	0.0378	0.6322
12818	Erupted lower front buccal deciduous ena	0.0453	3.3049×10^{-17}	100.00	–	1.0846×10^{-16}	2.3010×10^{-16}	–	0.3047	0.1436
12819	Erupted lower front lingual deciduous en	0.0425	1.2310×10^{-17}	100.00	–	2.5286×10^{-16}	1.1600×10^{-16}	–	0.0487	0.1061
12833	Erupted upper front deciduous dentin	0.3	1.1641×10^{-17}	100.00	–	1.3811×10^{-17}	6.0558×10^{-17}	–	0.8428	0.1922
12841	Erupted lower front deciduous dentin	0.199	2.5668×10^{-17}	100.00	–	5.0953×10^{-17}	1.1240×10^{-16}	–	0.5038	0.2284
12848	Unerupted permanent enamel	0.146	1.9218×10^{-17}	100.00	–	3.9699×10^{-18}	5.5237×10^{-18}	–	4.8410	3.4792
12849	Unerupted deciduous enamel	0.952	3.5009×10^{-17}	40.38	–	4.5615×10^{-17}	4.5443×10^{-17}	–	0.7675	0.7704
12850	Unerupted permanent dentin	0.486	2.5820×10^{-17}	51.70	–	7.0046×10^{-17}	4.9368×10^{-17}	–	0.3686	0.5230
12851	Unerupted deciduous dentin	2.20	4.6975×10^{-17}	17.25	–	2.5410×10^{-17}	3.5887×10^{-17}	–	1.8487	1.3090
12852	Permanent cementum	0.0239	2.0948×10^{-17}	100.00	–	6.6870×10^{-17}	1.5121×10^{-17}	–	0.3133	1.3854
12853	Deciduous cementum	0.125	5.1844×10^{-17}	38.76	–	3.3830×10^{-18}	3.9351×10^{-17}	–	15.3248	1.3175
12854	Permanent pulp	0.0146	0.0000×10^{0}	0.00	–	–	–	–	–	–
12855	Deciduous pulp	0.374	2.1738×10^{-17}	73.69	–	6.1588×10^{-17}	2.1646×10^{-17}	–	0.3529	1.0042
12856	Teeth retention region	0.00345	0.0000×10^{0}	0.00	–	6.5039×10^{-17}	6.0547×10^{-17}	–	–	–
12900	Testis left	0.773	2.1525×10^{-17}	75.44	–	4.1464×10^{-17}	6.1448×10^{-17}	–	0.5191	0.3503
13000	Testis right	0.773	4.9355×10^{-17}	40.71	–	2.5585×10^{-17}	4.3380×10^{-17}	–	1.9290	1.1377
13100	Thymus	31.21	3.9588×10^{-17}	8.45	–	3.7814×10^{-17}	3.8227×10^{-17}	–	1.0469	1.0356
13200	Thyroid	1.97	3.9500×10^{-17}	31.84	–	3.2396×10^{-17}	3.9913×10^{-17}	–	1.2193	0.9896
13300	Tongue upper(food)	5.97	4.0427×10^{-17}	15.96	–	3.8644×10^{-17}	4.4930×10^{-17}	–	1.0461	0.8998
13301	Tongue lower	4.42	3.0079×10^{-17}	26.35	–	5.1158×10^{-17}	4.3239×10^{-17}	–	0.5880	0.6956
13400	Tonsils	0.521	8.3726×10^{-17}	37.30	–	2.0336×10^{-17}	4.4602×10^{-17}	–	4.1171	1.8772
13500	Ureter left	1.15	4.2300×10^{-17}	48.26	–	3.2646×10^{-17}	5.1967×10^{-18}	–	1.2957	8.1397
13600	Ureter right	1.15	1.7980×10^{-17}	45.02	–	1.5687×10^{-17}	1.3555×10^{-17}	–	1.1462	1.3264
13700	Urinary bladder wall insensitive	8.16	3.9536×10^{-17}	11.05	–	2.7824×10^{-17}	3.1814×10^{-17}	–	1.4210	1.2427
13701	Urinary bladder wall sensitive(71-238)	0.948	3.9239×10^{-17}	27.98	–	5.7834×10^{-17}	3.5660×10^{-17}	–	0.6785	1.1004
13800	Urinary bladder content	32.90	3.4883×10^{-17}	7.25	–	3.7599×10^{-17}	3.2473×10^{-17}	–	0.9278	1.0742
14000	Air inside body	0.00925	5.2374×10^{-17}	39.81	–	4.7518×10^{-17}	1.5411×10^{-17}	–	1.1022	3.3985

* For tissues bookkept as head, trunk, arm and leg pieces (muscle, residual soft tissue, skin, large vessels), the volumes tabulated in the distributed cell tables partition the tissue at different boundaries from the transported mesh, and the reference output is normalised by the tabulated volume. The marked ratio therefore compares two different partitions of the same tissue, and is left out of the summary statistics of Table S2.

Table S13. 01F_Internal. Absorbed dose per primary history, Gy. σ is the FLUKA statistical uncertainty. Ratios are FLUKA / reference.

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
100	Adrenal left	2.26	6.9269×10^{-15}	2.12	–	6.9068×10^{-15}	6.9486×10^{-15}	–	1.0029	0.9969
200	Adrenal right	2.26	1.8365×10^{-14}	2.15	–	1.8070×10^{-14}	1.8096×10^{-14}	–	1.0163	1.0148
300	ET1(0-8)	0.00125	1.8602×10^{-15}	83.42	–	8.5777×10^{-17}	7.4976×10^{-17}	–	21.6859	24.8100
301	ET1(8-40)	0.0051	6.6799×10^{-16}	71.50	–	9.5022×10^{-17}	4.6529×10^{-16}	–	7.0298	1.4356
302	ET1(40-50)	0.00163	2.6563×10^{-15}	62.21	–	8.3924×10^{-16}	1.2674×10^{-16}	–	3.1651	20.9586
303	ET1(50-Surface)	0.212	8.4260×10^{-16}	19.75	–	4.9376×10^{-16}	4.1377×10^{-16}	–	1.7065	2.0364
400	ET2(-15-0)	0.0693	7.1015×10^{-16}	12.69	–	6.4771×10^{-16}	6.9386×10^{-16}	–	1.0964	1.0235
401	ET2(0-40)	0.197	6.6215×10^{-16}	11.40	–	6.6354×10^{-16}	7.2431×10^{-16}	–	0.9979	0.9142
402	ET2(40-50)	0.0496	6.9490×10^{-16}	9.73	–	7.7456×10^{-16}	5.5001×10^{-16}	–	0.8971	1.2634
403	ET2(50-55)	0.0249	7.5079×10^{-16}	19.17	–	9.8920×10^{-16}	6.1511×10^{-16}	–	0.7590	1.2206
404	ET2(55-65)	0.0498	7.6171×10^{-16}	15.18	–	5.9977×10^{-16}	5.6049×10^{-16}	–	1.2700	1.3590
405	ET2(65-Surface)	6.03	6.1498×10^{-16}	4.12	–	5.9968×10^{-16}	6.0387×10^{-16}	–	1.0255	1.0184
500	Oral mucosa tongue	0.0186	6.3862×10^{-16}	20.05	–	9.9805×10^{-16}	8.6761×10^{-16}	–	0.6399	0.7361
501	Oral mucosa mouth floor	0.0143	5.0233×10^{-16}	37.35	–	7.1196×10^{-16}	9.8115×10^{-16}	–	0.7056	0.5120
600	Oral mucosa lips and cheeks	0.00425	1.3704×10^{-15}	31.26	–	1.1437×10^{-15}	6.7034×10^{-16}	–	1.1982	2.0444
700	Trachea	1.56	1.7616×10^{-15}	3.91	–	1.7460×10^{-15}	1.7138×10^{-15}	–	1.0090	1.0279
800	BB(-11–6)	0.00302	4.5705×10^{-15}	17.40	–	2.9994×10^{-15}	4.5314×10^{-15}	–	1.5238	1.0086
801	BB(-6-0)	0.00385	3.6656×10^{-15}	25.38	–	4.0802×10^{-15}	2.8459×10^{-15}	–	0.8984	1.2880
802	BB(0-10)	0.00645	3.6440×10^{-15}	13.74	–	2.5620×10^{-15}	3.3161×10^{-15}	–	1.4223	1.0989
803	BB(10-35)	0.0163	2.8584×10^{-15}	12.54	–	2.9983×10^{-15}	3.7474×10^{-15}	–	0.9533	0.7628
804	BB(35-40)	0.00327	2.3698×10^{-15}	22.91	–	2.4866×10^{-15}	4.0766×10^{-15}	–	0.9530	0.5813
805	BB(40-50)	0.00657	2.0702×10^{-15}	16.77	–	3.2622×10^{-15}	3.2926×10^{-15}	–	0.6346	0.6287
806	BB(50-60)	0.0066	3.2383×10^{-15}	15.31	–	3.1109×10^{-15}	3.7763×10^{-15}	–	1.0410	0.8575
807	BB(60-70)	0.00663	4.2455×10^{-15}	17.91	–	3.3388×10^{-15}	4.7722×10^{-15}	–	1.2716	0.8896
808	BB(70-surface)	1.91	3.3708×10^{-15}	3.06	–	3.2768×10^{-15}	3.4225×10^{-15}	–	1.0287	0.9849
900	Blood in large arteries head	0.167	9.7719×10^{-16}	14.60	–	8.1886×10^{-16}	9.3259×10^{-16}	–	1.1934*	1.0478*
910	Blood in large veins head	0.557	9.4496×10^{-16}	8.95	–	1.0300×10^{-15}	8.6271×10^{-16}	–	0.9174*	1.0953*
1000	Blood in large arteries trunk	15.47	5.5115×10^{-15}	0.60	–	5.6594×10^{-15}	5.7115×10^{-15}	–	0.9739*	0.9650*
1010	Blood in large veins trunk	42.36	4.5761×10^{-15}	0.61	–	4.7688×10^{-15}	4.8086×10^{-15}	–	0.9596*	0.9516*
1100	Blood in large arteries arms	3.09	3.2017×10^{-15}	1.31	–	3.2279×10^{-15}	3.3561×10^{-15}	–	0.9919*	0.9540*
1110	Blood in large veins arms	13.39	2.9320×10^{-15}	1.24	–	2.8970×10^{-15}	2.8512×10^{-15}	–	1.0121*	1.0283*
1200	Blood in large arteries legs	13.08	2.9342×10^{-16}	3.89	–	3.2856×10^{-16}	3.4608×10^{-16}	–	0.8930*	0.8478*
1210	Blood in large veins legs	39.13	3.4881×10^{-16}	2.32	–	3.6963×10^{-16}	3.7285×10^{-16}	–	0.9436*	0.9355*
1300	Humeri upper cortical	12.67	2.0441×10^{-15}	1.79	–	1.9341×10^{-15}	2.0018×10^{-15}	–	1.0569	1.0212
1400	Humeri upper spogiosa	8.73	1.6356×10^{-15}	1.11	–	1.5527×10^{-15}	1.6823×10^{-15}	–	1.0534	0.9723

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
1500	Humeri upper medullary cavity	0.863	2.7103×10^{-15}	5.63	–	2.3432×10^{-15}	2.7339×10^{-15}	–	1.1567	0.9913
1600	Humeri lower cortical	9.04	3.6823×10^{-15}	1.30	–	3.6985×10^{-15}	3.7919×10^{-15}	–	0.9956	0.9711
1700	Humeri lower spongiosa	4.52	3.8582×10^{-15}	1.85	–	3.8264×10^{-15}	3.7092×10^{-15}	–	1.0083	1.0402
1800	Humeri lower medullary cavity	0.809	3.9247×10^{-15}	4.39	–	3.9971×10^{-15}	3.7451×10^{-15}	–	0.9819	1.0480
1900	Radii cortical	5.62	1.7959×10^{-15}	3.08	–	1.9594×10^{-15}	1.9474×10^{-15}	–	0.9165	0.9222
1910	Ulnae cortical	7.42	2.2827×10^{-15}	1.36	–	2.1735×10^{-15}	2.2438×10^{-15}	–	1.0502	1.0173
2000	Radii spongiosa	1.12	1.6578×10^{-15}	4.26	–	1.7468×10^{-15}	1.6249×10^{-15}	–	0.9491	1.0203
2010	Ulnae spongiosa	1.81	3.0441×10^{-15}	3.95	–	3.0716×10^{-15}	3.0290×10^{-15}	–	0.9911	1.0050
2100	Radii medullary cavity	0.359	2.2376×10^{-15}	11.94	–	2.0966×10^{-15}	1.9668×10^{-15}	–	1.0673	1.1377
2110	Ulnae medullary cavity	0.589	2.1671×10^{-15}	6.32	–	2.2025×10^{-15}	2.1177×10^{-15}	–	0.9839	1.0233
2200	Wrists and hand bones cortical	5.40	6.8243×10^{-16}	3.97	–	7.3160×10^{-16}	7.1724×10^{-16}	–	0.9328	0.9515
2300	Wrists and hand bones spongiosa	14.07	7.5065×10^{-16}	1.92	–	7.4181×10^{-16}	7.1481×10^{-16}	–	1.0119	1.0501
2400	Clavicles cortical	1.37	1.6752×10^{-15}	3.69	–	1.8971×10^{-15}	1.8908×10^{-15}	–	0.8830	0.8860
2500	Clavicles spongiosa	1.96	1.8934×10^{-15}	5.94	–	1.9592×10^{-15}	1.6671×10^{-15}	–	0.9664	1.1357
2600	Cranium cortical	91.49	3.3480×10^{-16}	1.51	–	3.2941×10^{-16}	3.2828×10^{-16}	–	1.0164	1.0199
2700	Cranium spongiosa	289.96	3.2363×10^{-16}	1.00	–	3.2670×10^{-16}	3.2587×10^{-16}	–	0.9906	0.9931
2800	Femora upper cortical	14.11	7.6963×10^{-16}	2.56	–	7.9438×10^{-16}	7.8190×10^{-16}	–	0.9688	0.9843
2900	Femora upper spongiosa	8.86	1.0592×10^{-15}	2.31	–	1.0541×10^{-15}	1.1019×10^{-15}	–	1.0048	0.9612
3000	Femora upper medullary cavity	0.994	5.4369×10^{-16}	10.12	–	4.9538×10^{-16}	5.9773×10^{-16}	–	1.0975	0.9096
3100	Femora lower cortical	20.53	1.8214×10^{-16}	5.32	–	1.8141×10^{-16}	1.8761×10^{-16}	–	1.0041	0.9709
3200	Femora lower spongiosa	8.77	1.4901×10^{-16}	8.86	–	1.4560×10^{-16}	1.5653×10^{-16}	–	1.0235	0.9520
3300	Femora lower medullary cavity	1.57	2.1520×10^{-16}	10.79	–	2.0335×10^{-16}	2.3594×10^{-16}	–	1.0583	0.9121
3400	Tibiae cortical	26.33	8.1053×10^{-17}	3.35	–	7.7775×10^{-17}	8.5062×10^{-17}	–	1.0422	0.9529
3410	Fibulae cortical	3.02	5.9731×10^{-17}	17.30	–	4.8842×10^{-17}	6.0921×10^{-17}	–	1.2229	0.9805
3420	Patellae cortical	0.162	4.7977×10^{-17}	53.85	–	1.1034×10^{-16}	7.6279×10^{-17}	–	0.4348	0.6290
3500	Tibiae spongiosa	7.27	1.0360×10^{-16}	7.63	–	1.0614×10^{-16}	1.1246×10^{-16}	–	0.9760	0.9212
3510	Fibulae spongiosa	0.565	8.0597×10^{-17}	30.66	–	7.5189×10^{-17}	4.1418×10^{-17}	–	1.0719	1.9459
3520	Patellae spongiosa	0.575	1.1246×10^{-16}	24.92	–	9.3798×10^{-17}	8.1973×10^{-17}	–	1.1990	1.3719
3600	Tibiae medullary cavity	2.20	1.0985×10^{-16}	23.22	–	7.4093×10^{-17}	8.0476×10^{-17}	–	1.4826	1.3650
3610	Fibulae medullary cavity	0.204	3.3511×10^{-17}	60.98	–	8.0251×10^{-17}	4.8745×10^{-17}	–	0.4176	0.6875
3700	Ankles and foot cortical	7.24	5.2477×10^{-17}	8.66	–	5.3234×10^{-17}	5.8806×10^{-17}	–	0.9858	0.8924
3800	Ankles and foot spongiosa	20.33	5.6160×10^{-17}	5.15	–	4.9730×10^{-17}	5.3064×10^{-17}	–	1.1293	1.0584
3900	Mandible cortical	7.16	9.0959×10^{-16}	3.48	–	8.6130×10^{-16}	9.0897×10^{-16}	–	1.0561	1.0007
4000	Mandible spongiosa	13.09	8.9050×10^{-16}	1.77	–	8.6940×10^{-16}	8.9504×10^{-16}	–	1.0243	0.9949
4100	Pelvis cortical	21.20	1.9617×10^{-15}	1.05	–	2.0093×10^{-15}	1.9866×10^{-15}	–	0.9763	0.9875
4200	Pelvis spongiosa	34.40	1.9753×10^{-15}	0.89	–	2.0015×10^{-15}	1.9598×10^{-15}	–	0.9869	1.0079
4300	Ribs cortical	20.41	6.2670×10^{-15}	0.47	–	6.2746×10^{-15}	6.3345×10^{-15}	–	0.9988	0.9893
4400	Ribs spongiosa	58.12	6.1712×10^{-15}	0.34	–	6.2021×10^{-15}	6.2507×10^{-15}	–	0.9950	0.9873

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
4500	Scapulae cortical	9.65	1.9858×10^{-15}	1.63	–	1.9587×10^{-15}	1.9994×10^{-15}	–	1.0138	0.9932
4600	Scapulae spongiosa	14.86	1.9720×10^{-15}	1.57	–	1.9437×10^{-15}	2.0132×10^{-15}	–	1.0146	0.9795
4700	Cervical spine cortical	3.28	9.5798×10^{-16}	3.31	–	1.0773×10^{-15}	9.8945×10^{-16}	–	0.8893	0.9682
4800	Cervical spine spongiosa	9.10	1.0503×10^{-15}	1.80	–	1.0937×10^{-15}	1.0722×10^{-15}	–	0.9603	0.9796
4900	Thoracic spine cortical	11.50	5.4145×10^{-15}	0.88	–	5.3902×10^{-15}	5.3768×10^{-15}	–	1.0045	1.0070
5000	Thoracic spine spongiosa	28.11	6.3585×10^{-15}	0.79	–	6.3600×10^{-15}	6.3571×10^{-15}	–	0.9998	1.0002
5100	Lumbar spine cortical	7.07	5.7475×10^{-15}	0.99	–	5.8949×10^{-15}	5.7828×10^{-15}	–	0.9750	0.9939
5200	Lumbar spine spongiosa	21.94	6.3516×10^{-15}	0.78	–	6.3475×10^{-15}	6.3013×10^{-15}	–	1.0006	1.0080
5300	Sacrum cortical	5.45	1.9443×10^{-15}	2.07	–	1.9733×10^{-15}	1.9253×10^{-15}	–	0.9853	1.0099
5400	Sacrum spongiosa	15.12	2.1789×10^{-15}	1.33	–	2.1294×10^{-15}	2.0817×10^{-15}	–	1.0232	1.0467
5500	Sternum cortical	0.63	4.4678×10^{-15}	3.66	–	4.6141×10^{-15}	4.6036×10^{-15}	–	0.9683	0.9705
5600	Sternum spongiosa	2.10	4.4206×10^{-15}	2.72	–	4.2660×10^{-15}	4.6440×10^{-15}	–	1.0362	0.9519
5700	Cartilage head	5.03	1.8112×10^{-16}	11.13	–	2.0290×10^{-16}	1.8072×10^{-16}	–	0.8927	1.0022
5800	Cartilage trunk	36.85	9.8268×10^{-15}	0.36	–	9.8579×10^{-15}	9.9281×10^{-15}	–	0.9969	0.9898
6100	Brain	978.10	2.9142×10^{-16}	0.54	–	2.9453×10^{-16}	2.9201×10^{-16}	–	0.9894	0.9980
6200	Breast left adipose tissue	0.572	3.8855×10^{-15}	4.05	–	3.9938×10^{-15}	4.1306×10^{-15}	–	0.9729	0.9407
6300	Breast left glandular tissue	0.381	3.9316×10^{-15}	6.91	–	3.8587×10^{-15}	4.1029×10^{-15}	–	1.0189	0.9582
6400	Breast right adipose tissue	0.572	6.1595×10^{-15}	3.60	–	6.2654×10^{-15}	6.7462×10^{-15}	–	0.9831	0.9130
6500	Breast right glandular tissue	0.381	6.7337×10^{-15}	5.78	–	6.1506×10^{-15}	6.5859×10^{-15}	–	1.0948	1.0224
6600	Eye lens sensitive left	0.0169	2.2173×10^{-16}	92.79	–	6.5115×10^{-16}	9.4148×10^{-18}	–	0.3405	23.5509
6601	Eye lens insensitive left	0.0978	4.6253×10^{-16}	37.04	–	4.8751×10^{-16}	5.8715×10^{-16}	–	0.9488	0.7878
6700	Cornea left	0.629	5.0433×10^{-16}	9.50	–	4.5613×10^{-16}	4.0040×10^{-16}	–	1.1057	1.2596
6701	Aqueous left	0.275	4.7239×10^{-16}	21.95	–	2.6681×10^{-16}	3.9941×10^{-16}	–	1.7705	1.1827
6702	Vitreous left	2.62	4.5249×10^{-16}	9.35	–	4.1958×10^{-16}	3.9436×10^{-16}	–	1.0785	1.1474
6800	Eye lens sensitive right	0.0169	3.8235×10^{-16}	59.68	–	4.6130×10^{-16}	2.8621×10^{-16}	–	0.8289	1.3359
6801	Eye lens insensitive right	0.0978	2.6737×10^{-16}	32.87	–	2.8754×10^{-16}	2.8112×10^{-16}	–	0.9298	0.9511
6900	Cornea right	0.629	4.2579×10^{-16}	10.11	–	4.4992×10^{-16}	4.1046×10^{-16}	–	0.9464	1.0373
6901	Aqueous right	0.275	4.9207×10^{-16}	18.73	–	4.4857×10^{-16}	5.5076×10^{-16}	–	1.0970	0.8934
6902	Vitreous right	2.62	4.3020×10^{-16}	9.46	–	3.8557×10^{-16}	3.7859×10^{-16}	–	1.1157	1.1363
7000	Gall bladder wall	1.46	2.7400×10^{-14}	1.30	–	2.7648×10^{-14}	2.7106×10^{-14}	–	0.9910	1.0109
7100	Gall bladder contents	8.00	2.7432×10^{-14}	0.71	–	2.7265×10^{-14}	2.7258×10^{-14}	–	1.0062	1.0064
7200	Stomach wall(0-60)	0.814	1.2535×10^{-14}	1.40	–	1.2547×10^{-14}	1.2288×10^{-14}	–	0.9991	1.0201
7201	Stomach wall(60-100)	0.407	1.1910×10^{-14}	1.10	–	1.1648×10^{-14}	1.1353×10^{-14}	–	1.0225	1.0490
7202	Stomach wall(100-300)	2.06	1.1623×10^{-14}	0.80	–	1.1273×10^{-14}	1.1538×10^{-14}	–	1.0310	1.0073
7203	Stomach wall(300-surface)	19.88	1.1675×10^{-14}	0.39	–	1.1648×10^{-14}	1.1682×10^{-14}	–	1.0024	0.9994
7300	Stomach contents	67.00	1.0564×10^{-14}	0.36	–	1.0651×10^{-14}	1.0681×10^{-14}	–	0.9918	0.9891
7400	Small intestine wall(0-130)	4.93	5.2852×10^{-15}	0.90	–	5.1888×10^{-15}	5.1537×10^{-15}	–	1.0186	1.0255
7401	Small intestine wall(130-150)	0.77	5.3055×10^{-15}	2.33	–	5.4968×10^{-15}	5.1915×10^{-15}	–	0.9652	1.0220

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
7402	Small intestine wall(150-200)	1.94	5.3387×10^{-15}	1.50	–	5.3383×10^{-15}	5.1118×10^{-15}	–	1.0001	1.0444
7403	Small intestine wall(200-surface)	92.27	5.1881×10^{-15}	0.47	–	5.2041×10^{-15}	5.2032×10^{-15}	–	0.9969	0.9971
7500	Small intestine contents(-400-0)	14.32	5.1767×10^{-15}	0.52	–	5.1147×10^{-15}	5.1322×10^{-15}	–	1.0121	1.0087
7501	Small intestine contents(centre–400)	78.68	5.0793×10^{-15}	0.39	–	5.1046×10^{-15}	5.0895×10^{-15}	–	0.9950	0.9980
7600	Ascending colon wall(0-280)	1.57	3.1227×10^{-15}	2.38	–	3.0210×10^{-15}	3.1323×10^{-15}	–	1.0337	0.9969
7601	Ascending colon wall(280-300)	0.114	3.3782×10^{-15}	4.23	–	3.1987×10^{-15}	2.9611×10^{-15}	–	1.0561	1.1409
7602	Ascending colon wall(300-surface)	9.37	3.3491×10^{-15}	2.02	–	3.3765×10^{-15}	3.3844×10^{-15}	–	0.9919	0.9896
7700	Ascending colon content	26.02	2.9189×10^{-15}	1.12	–	2.9807×10^{-15}	2.9606×10^{-15}	–	0.9792	0.9859
7800	Transverse colon wall right(0-280)	1.20	1.0413×10^{-14}	1.63	–	9.9156×10^{-15}	9.9314×10^{-15}	–	1.0501	1.0485
7801	Transverse colon wall right(280-300)	0.0878	1.0925×10^{-14}	2.95	–	1.0744×10^{-14}	9.2913×10^{-15}	–	1.0169	1.1759
7802	Transverse colon wall right(300-surface)	11.14	1.0299×10^{-14}	0.66	–	1.0401×10^{-14}	1.0322×10^{-14}	–	0.9902	0.9978
7900	Transverse colon contents right	13.98	1.0084×10^{-14}	0.86	–	1.0066×10^{-14}	1.0118×10^{-14}	–	1.0019	0.9967
8000	Transverse colon wall left(0-280)	1.03	5.8116×10^{-15}	2.98	–	6.1510×10^{-15}	5.7574×10^{-15}	–	0.9448	1.0094
8001	Transverse colon wall left(280-300)	0.0757	5.9255×10^{-15}	5.15	–	6.2991×10^{-15}	5.9294×10^{-15}	–	0.9407	0.9993
8002	Transverse colon wall left(300-surface)	10.30	5.9900×10^{-15}	1.04	–	6.0995×10^{-15}	6.1123×10^{-15}	–	0.9820	0.9800
8100	Transverse colon content left	8.93	6.0093×10^{-15}	1.45	–	5.8946×10^{-15}	5.9117×10^{-15}	–	1.0195	1.0165
8200	Descending colon wall(0-280)	1.23	2.1553×10^{-15}	1.83	–	2.3333×10^{-15}	2.2879×10^{-15}	–	0.9237	0.9420
8201	Descending colon wall(280-300)	0.0906	2.3742×10^{-15}	5.83	–	2.1610×10^{-15}	2.3883×10^{-15}	–	1.0987	0.9941
8202	Descending colon wall(300-surface)	10.75	2.1693×10^{-15}	1.76	–	2.2384×10^{-15}	2.2899×10^{-15}	–	0.9691	0.9473
8300	Descending colon content	11.07	2.3231×10^{-15}	1.65	–	2.2664×10^{-15}	2.2663×10^{-15}	–	1.0250	1.0251
8400	Sigmoid colon wall(0-280)	1.57	1.4491×10^{-15}	2.37	–	1.5495×10^{-15}	1.5038×10^{-15}	–	0.9352	0.9636
8401	Sigmoid colon wall(280-300)	0.115	1.4230×10^{-15}	8.56	–	1.6198×10^{-15}	1.4885×10^{-15}	–	0.8785	0.9560
8402	Sigmoid colon wall(300-surface)	7.99	1.4165×10^{-15}	1.95	–	1.5280×10^{-15}	1.4773×10^{-15}	–	0.9270	0.9589
8500	Sigmoid colon contents	13.98	1.4548×10^{-15}	1.68	–	1.5114×10^{-15}	1.4640×10^{-15}	–	0.9625	0.9937
8600	Rectum wall(0-280)	0.657	9.4307×10^{-16}	9.66	–	8.9115×10^{-16}	7.8470×10^{-16}	–	1.0583	1.2018
8601	Rectum wall(280-300)	0.0484	9.4291×10^{-16}	17.50	–	9.7053×10^{-16}	1.0010×10^{-15}	–	0.9715	0.9420
8602	Rectum wall(300-surface)	1.36	8.6497×10^{-16}	3.91	–	1.0308×10^{-15}	8.7674×10^{-16}	–	0.8391	0.9866
8603	Rectum contents	6.02	9.9453×10^{-16}	4.43	–	9.1402×10^{-16}	9.2336×10^{-16}	–	1.0881	1.0771
8700	Heart wall	55.04	9.1571×10^{-15}	0.49	–	9.1676×10^{-15}	9.1740×10^{-15}	–	0.9988	0.9982
8800	Blood in heart chamber	48.00	9.1369×10^{-15}	0.29	–	9.1225×10^{-15}	9.1851×10^{-15}	–	1.0016	0.9948
8900	Kidney left cortex	27.76	5.0938×10^{-15}	0.87	–	5.0697×10^{-15}	5.1163×10^{-15}	–	1.0048	0.9956
9000	Kidney left medulla	9.92	5.7066×10^{-15}	1.09	–	5.8123×10^{-15}	5.8214×10^{-15}	–	0.9818	0.9803
9100	Kidney left pelvis	1.98	5.9486×10^{-15}	2.00	–	5.9829×10^{-15}	6.0414×10^{-15}	–	0.9943	0.9846
9200	Kidney right cortex	27.76	1.5399×10^{-14}	0.47	–	1.5424×10^{-14}	1.5507×10^{-14}	–	0.9984	0.9931
9300	Kidney right medulla	9.92	1.4865×10^{-14}	0.85	–	1.4917×10^{-14}	1.4980×10^{-14}	–	0.9965	0.9923
9400	Kidney right pelvis	1.98	1.4623×10^{-14}	1.78	–	1.4654×10^{-14}	1.4839×10^{-14}	–	0.9979	0.9854

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
9500	Liver	390.44	3.8066×10^{-14}	0.07	–	3.7985×10^{-14}	3.8125×10^{-14}	–	1.0021	0.9985
9700	Lung(AI) left	68.84	3.7813×10^{-15}	0.61	–	3.7734×10^{-15}	3.8356×10^{-15}	–	1.0021	0.9859
9900	Lung(AI) right	81.16	9.0000×10^{-15}	0.33	–	9.0657×10^{-15}	9.0947×10^{-15}	–	0.9927	0.9896
10000	Lymphatic nodes ET	2.18	7.0272×10^{-16}	5.20	–	6.9015×10^{-16}	6.8107×10^{-16}	–	1.0182	1.0318
10100	Lymphatic nodes thoracic	2.18	1.2613×10^{-14}	1.05	–	1.2675×10^{-14}	1.2505×10^{-14}	–	0.9951	1.0086
10200	Lymphatic nodes head	0.752	1.4207×10^{-15}	7.43	–	1.3249×10^{-15}	1.6444×10^{-15}	–	1.0723	0.8640
10300	Lymphatic nodes trunk	17.76	3.4504×10^{-15}	0.79	–	3.4591×10^{-15}	3.4596×10^{-15}	–	0.9975	0.9973
10400	Lymphatic nodes arms	1.50	3.1634×10^{-15}	3.09	–	3.2381×10^{-15}	3.2942×10^{-15}	–	0.9769	0.9603
10500	Lymphatic nodes legs	1.50	1.7853×10^{-16}	12.90	–	8.1387×10^{-17}	1.3840×10^{-16}	–	2.1936	1.2899
10600	Muscle head	240.46	7.5487×10^{-16}	0.53	–	7.5198×10^{-16}	7.5364×10^{-16}	–	1.0039*	1.0016*
10700	Muscle trunk	885.17	3.5473×10^{-15}	0.18	–	3.6454×10^{-15}	3.6604×10^{-15}	–	0.9731*	0.9691*
10800	Muscle arms	204.07	2.4665×10^{-15}	0.38	–	2.4633×10^{-15}	2.4634×10^{-15}	–	1.0013*	1.0013*
10900	Muscle legs	599.50	3.0795×10^{-16}	0.49	–	3.3793×10^{-16}	3.3643×10^{-16}	–	0.9113*	0.9153*
11000	Oesophagus wall(0-190)	0.567	4.5340×10^{-15}	3.83	–	4.0641×10^{-15}	4.4424×10^{-15}	–	1.1156	1.0206
11001	Oesophagus wall(190-200)	0.0308	5.2534×10^{-15}	6.57	–	4.6177×10^{-15}	4.1786×10^{-15}	–	1.1377	1.2572
11002	Oesophagus wall(200-surface)	5.19	4.5259×10^{-15}	2.05	–	4.2978×10^{-15}	4.5208×10^{-15}	–	1.0531	1.0011
11003	Oesophagus contents	3.79	4.3798×10^{-15}	1.56	–	4.3456×10^{-15}	4.5627×10^{-15}	–	1.0079	0.9599
11100	Ovary left	0.423	1.7956×10^{-15}	11.41	–	1.3923×10^{-15}	1.3722×10^{-15}	–	1.2897	1.3086
11200	Ovary right	0.423	1.8694×10^{-15}	9.09	–	2.1033×10^{-15}	2.1465×10^{-15}	–	0.8888	0.8709
11300	Pancreas	22.65	6.1775×10^{-15}	0.99	–	6.1739×10^{-15}	6.2043×10^{-15}	–	1.0006	0.9957
11400	Pituitary gland	0.156	3.8187×10^{-16}	32.81	–	2.2359×10^{-16}	3.3714×10^{-16}	–	1.7079	1.1327
11600	RST head	572.11	5.5846×10^{-16}	0.34	–	5.5223×10^{-16}	5.5591×10^{-16}	–	1.0113*	1.0046*
11700	RST trunk	2832.52	4.8212×10^{-15}	0.05	–	4.9422×10^{-15}	4.9754×10^{-15}	–	0.9755*	0.9690*
11800	RST arms	193.18	2.5591×10^{-15}	0.31	–	2.4687×10^{-15}	2.4910×10^{-15}	–	1.0366*	1.0273*
11900	RST legs	696.24	3.5416×10^{-16}	0.50	–	4.2204×10^{-16}	4.2078×10^{-16}	–	0.8392*	0.8417*
12000	Salivary glands left	12.49	8.6945×10^{-16}	2.04	–	8.2978×10^{-16}	8.2645×10^{-16}	–	1.0478	1.0520
12100	Salivary glandss right	12.49	1.0182×10^{-15}	1.96	–	9.8354×10^{-16}	1.0292×10^{-15}	–	1.0352	0.9893
12200	Skin head insensitive	68.00	4.1446×10^{-16}	1.42	–	4.0712×10^{-16}	4.0793×10^{-16}	–	1.0180	1.0160
12201	Skin head sensitive(40-100)	5.45	3.4878×10^{-16}	3.15	–	3.6098×10^{-16}	3.5912×10^{-16}	–	0.9662*	0.9712*
12300	Skin trunk insensitive	126.04	3.0656×10^{-15}	0.41	–	3.2159×10^{-15}	3.2003×10^{-15}	–	0.9533*	0.9579*
12301	Skin trunk sensitive(40-100)	10.41	2.6532×10^{-15}	0.70	–	2.8016×10^{-15}	2.7881×10^{-15}	–	0.9470*	0.9516*
12400	Skin arms insensitive	49.86	2.1222×10^{-15}	0.58	–	2.0832×10^{-15}	2.0836×10^{-15}	–	1.0187*	1.0185*
12401	Skin arms sensitive(40-100)	4.11	1.9725×10^{-15}	0.95	–	1.9147×10^{-15}	1.8618×10^{-15}	–	1.0302*	1.0594*
12500	Skin legs insensitive	89.78	2.2394×10^{-16}	1.60	–	2.5710×10^{-16}	2.6306×10^{-16}	–	0.8710*	0.8513*
12501	Skin legs sensitive(40-100)	7.28	1.9945×10^{-16}	3.96	–	2.1807×10^{-16}	2.3181×10^{-16}	–	0.9146*	0.8604*
12600	Spinal cord	10.67	4.1162×10^{-15}	1.26	–	4.2037×10^{-15}	4.1134×10^{-15}	–	0.9792	1.0007
12700	Spleen	37.35	3.9242×10^{-15}	0.40	–	3.9384×10^{-15}	3.9597×10^{-15}	–	0.9964	0.9910

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
12802	Erupted upper front buccal deciduous ena	0.0689	1.3525×10^{-15}	12.84	–	6.9986×10^{-16}	1.1957×10^{-15}	–	1.9325	1.1311
12803	Erupted upper front lingual deciduous en	0.0617	4.4469×10^{-16}	43.65	–	6.4997×10^{-16}	6.6991×10^{-16}	–	0.6842	0.6638
12818	Erupted lower front buccal deciduous ena	0.0506	4.3154×10^{-16}	36.19	–	6.7362×10^{-16}	8.2233×10^{-16}	–	0.6406	0.5248
12819	Erupted lower front lingual deciduous en	0.0373	7.3700×10^{-16}	44.50	–	8.2897×10^{-16}	8.7121×10^{-16}	–	0.8890	0.8459
12833	Erupted upper front deciduous dentin	0.3	7.1327×10^{-16}	16.87	–	6.4321×10^{-16}	8.1046×10^{-16}	–	1.1089	0.8801
12841	Erupted lower front deciduous dentin	0.2	5.4380×10^{-16}	24.38	–	7.1926×10^{-16}	9.9895×10^{-16}	–	0.7561	0.5444
12848	Unerupted permanent enamel	0.146	8.8846×10^{-16}	16.99	–	5.9276×10^{-16}	4.3087×10^{-16}	–	1.4989	2.0620
12849	Unerupted deciduous enamel	0.952	6.6563×10^{-16}	8.39	–	7.3221×10^{-16}	8.4752×10^{-16}	–	0.9091	0.7854
12850	Unerupted permanent dentin	0.489	8.5405×10^{-16}	6.30	–	6.4019×10^{-16}	8.0427×10^{-16}	–	1.3340	1.0619
12851	Unerupted deciduous dentin	2.21	7.7668×10^{-16}	4.55	–	7.5697×10^{-16}	7.3694×10^{-16}	–	1.0260	1.0539
12852	Permanent cementum	0.0203	1.2620×10^{-15}	24.49	–	9.5199×10^{-16}	3.4573×10^{-16}	–	1.3257	3.6504
12853	Deciduous cementum	0.122	6.2121×10^{-16}	17.25	–	5.3848×10^{-16}	6.5211×10^{-16}	–	1.1536	0.9526
12854	Permanent pulp	0.0146	5.5660×10^{-16}	52.59	–	1.1364×10^{-15}	5.4694×10^{-16}	–	0.4898	1.0177
12855	Deciduous pulp	0.374	7.3355×10^{-16}	12.99	–	8.9118×10^{-16}	7.1106×10^{-16}	–	0.8231	1.0316
12856	Teeth retention region	0.00345	1.4300×10^{-15}	53.85	–	8.5978×10^{-16}	8.2832×10^{-16}	–	1.6632	1.7263
13100	Thymus	31.19	3.0195×10^{-15}	1.11	–	3.0350×10^{-15}	3.0171×10^{-15}	–	0.9949	1.0008
13200	Thyroid	1.97	1.3005×10^{-15}	5.06	–	1.4465×10^{-15}	1.2073×10^{-15}	–	0.8990	1.0772
13300	Tongue upper(food)	5.97	7.6560×10^{-16}	4.22	–	7.2534×10^{-16}	7.6320×10^{-16}	–	1.0555	1.0031
13301	Tongue lower	4.42	8.6915×10^{-16}	6.11	–	8.7600×10^{-16}	8.7647×10^{-16}	–	0.9922	0.9917
13400	Tonsils	0.521	8.9001×10^{-16}	15.39	–	7.6539×10^{-16}	7.6756×10^{-16}	–	1.1628	1.1595
13500	Ureter left	1.15	4.1147×10^{-15}	3.76	–	4.0437×10^{-15}	4.2529×10^{-15}	–	1.0175	0.9675
13600	Ureter right	1.15	5.3321×10^{-15}	2.56	–	5.6734×10^{-15}	4.9728×10^{-15}	–	0.9398	1.0722
13700	Urinary bladder wall insensitive	8.17	1.4478×10^{-15}	2.23	–	1.4569×10^{-15}	1.4254×10^{-15}	–	0.9938	1.0157
13701	Urinary bladder wall sensitive(71-238)	0.947	1.4766×10^{-15}	6.04	–	1.4200×10^{-15}	1.4363×10^{-15}	–	1.0399	1.0281
13800	Urinary bladder content	32.90	1.4276×10^{-15}	0.88	–	1.4247×10^{-15}	1.4236×10^{-15}	–	1.0020	1.0028
13900	Uterus	1.56	1.5332×10^{-15}	4.30	–	1.5682×10^{-15}	1.4993×10^{-15}	–	0.9777	1.0226
14000	Air inside body	0.00736	2.5953×10^{-15}	11.19	–	3.7669×10^{-15}	2.9893×10^{-15}	–	0.6890	0.8682

* For tissues bookkept as head, trunk, arm and leg pieces (muscle, residual soft tissue, skin, large vessels), the volumes tabulated in the distributed cell tables partition the tissue at different boundaries from the transported mesh, and the reference output is normalised by the tabulated volume. The marked ratio therefore compares two different partitions of the same tissue, and is left out of the summary statistics of Table S2.

Table S14. 01F_External. Absorbed dose per primary history, Gy. σ is the FLUKA statistical uncertainty. Ratios are FLUKA / reference.

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
100	Adrenal left	2.26	1.5561×10^{-17}	38.47	–	1.9911×10^{-17}	2.5389×10^{-17}	–	0.7815	0.6129
200	Adrenal right	2.26	2.7345×10^{-17}	34.16	–	1.8808×10^{-17}	2.7201×10^{-17}	–	1.4539	1.0053
300	ET1(0-8)	0.00125	6.3879×10^{-17}	100.00	–	–	9.3662×10^{-17}	–	–	0.6820
301	ET1(8-40)	0.0051	3.9193×10^{-17}	100.00	–	–	8.2910×10^{-17}	–	–	0.4727
302	ET1(40-50)	0.00163	4.4882×10^{-17}	100.00	–	–	1.1711×10^{-16}	–	–	0.3832
303	ET1(50-Surface)	0.212	6.8843×10^{-17}	72.82	–	4.9825×10^{-17}	1.2308×10^{-16}	–	1.3817	0.5593
400	ET2(-15-0)	0.0693	1.6672×10^{-17}	55.28	–	8.0398×10^{-17}	4.6672×10^{-17}	–	0.2074	0.3572
401	ET2(0-40)	0.197	3.2029×10^{-17}	46.07	–	5.6587×10^{-17}	5.7290×10^{-17}	–	0.5660	0.5591
402	ET2(40-50)	0.0496	1.6812×10^{-17}	36.20	–	2.0545×10^{-17}	1.1225×10^{-16}	–	0.8183	0.1498
403	ET2(50-55)	0.0249	7.8638×10^{-17}	82.61	–	8.2852×10^{-17}	1.1248×10^{-16}	–	0.9491	0.6991
404	ET2(55-65)	0.0498	1.4592×10^{-17}	38.48	–	3.0947×10^{-17}	7.0616×10^{-17}	–	0.4715	0.2066
405	ET2(65-Surface)	6.03	3.6033×10^{-17}	18.96	–	4.0241×10^{-17}	3.4615×10^{-17}	–	0.8954	1.0410
500	Oral mucosa tongue	0.0186	2.1389×10^{-17}	56.96	–	6.6097×10^{-17}	3.2669×10^{-17}	–	0.3236	0.6547
501	Oral mucosa mouth floor	0.0143	3.5254×10^{-18}	68.98	–	6.7400×10^{-17}	4.1638×10^{-17}	–	0.0523	0.0847
600	Oral mucosa lips and cheeks	0.00425	8.5558×10^{-18}	100.00	–	7.5150×10^{-17}	9.0252×10^{-18}	–	0.1138	0.9480
700	Trachea	1.56	3.1421×10^{-17}	44.41	–	3.6814×10^{-17}	3.3214×10^{-17}	–	0.8535	0.9460
800	BB(-11–6)	0.00302	2.1837×10^{-17}	75.43	–	4.0823×10^{-17}	4.6896×10^{-17}	–	0.5349	0.4656
801	BB(-6-0)	0.00385	2.9052×10^{-16}	97.48	–	7.7161×10^{-17}	4.5755×10^{-17}	–	3.7651	6.3495
802	BB(0-10)	0.00645	3.8046×10^{-17}	70.20	–	5.9922×10^{-17}	1.0603×10^{-16}	–	0.6349	0.3588
803	BB(10-35)	0.0163	2.1647×10^{-17}	87.61	–	4.8331×10^{-17}	9.2650×10^{-17}	–	0.4479	0.2336
804	BB(35-40)	0.00327	2.8935×10^{-17}	82.78	–	5.0457×10^{-17}	6.8557×10^{-17}	–	0.5735	0.4221
805	BB(40-50)	0.00657	2.5086×10^{-17}	75.78	–	5.0519×10^{-17}	2.4815×10^{-16}	–	0.4966	0.1011
806	BB(50-60)	0.0066	2.5118×10^{-17}	85.12	–	7.8136×10^{-17}	6.4632×10^{-17}	–	0.3215	0.3886
807	BB(60-70)	0.00663	1.3068×10^{-16}	97.72	–	3.3939×10^{-16}	6.3645×10^{-17}	–	0.3850	2.0532
808	BB(70-surface)	1.91	4.2399×10^{-17}	29.00	–	1.8923×10^{-17}	2.2599×10^{-17}	–	2.2406	1.8762
900	Blood in large arteries head	0.167	2.7904×10^{-17}	57.11	–	2.8409×10^{-17}	7.3603×10^{-17}	–	0.9822*	0.3791*
910	Blood in large veins head	0.557	4.8057×10^{-17}	61.93	–	3.4849×10^{-17}	1.0909×10^{-17}	–	1.3790*	4.4053*
1000	Blood in large arteries trunk	15.47	2.7737×10^{-17}	8.67	–	3.6916×10^{-17}	3.7478×10^{-17}	–	0.7514*	0.7401*
1010	Blood in large veins trunk	42.36	3.2882×10^{-17}	6.20	–	3.3090×10^{-17}	3.2738×10^{-17}	–	0.9937*	1.0044*
1100	Blood in large arteries arms	3.09	4.3101×10^{-17}	28.36	–	3.3968×10^{-17}	3.7248×10^{-17}	–	1.2689*	1.1571*
1110	Blood in large veins arms	13.39	3.7232×10^{-17}	12.86	–	4.4041×10^{-17}	3.3555×10^{-17}	–	0.8454*	1.1096*
1200	Blood in large arteries legs	13.08	3.5437×10^{-17}	8.08	–	3.3930×10^{-17}	3.4901×10^{-17}	–	1.0444*	1.0154*
1210	Blood in large veins legs	39.13	3.6561×10^{-17}	6.39	–	3.1738×10^{-17}	3.5983×10^{-17}	–	1.1519*	1.0161*
1300	Humeri upper cortical	12.67	2.6468×10^{-17}	15.49	–	4.1141×10^{-17}	3.0182×10^{-17}	–	0.6433	0.8769
1400	Humeri upper spogiosa	8.73	3.3961×10^{-17}	16.19	–	3.0431×10^{-17}	3.9423×10^{-17}	–	1.1160	0.8615

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
1500	Humeri upper medullary cavity	0.863	5.5121×10^{-17}	44.44	–	5.3065×10^{-17}	2.1338×10^{-17}	–	1.0387	2.5832
1600	Humeri lower cortical	9.04	3.6393×10^{-17}	13.64	–	3.3882×10^{-17}	3.1248×10^{-17}	–	1.0741	1.1646
1700	Humeri lower spongiosa	4.52	3.9689×10^{-17}	11.08	–	3.9441×10^{-17}	5.0938×10^{-17}	–	1.0063	0.7792
1800	Humeri lower medullary cavity	0.809	1.6636×10^{-17}	48.12	–	6.5602×10^{-17}	3.1991×10^{-17}	–	0.2536	0.5200
1900	Radii cortical	5.62	5.1102×10^{-17}	17.35	–	4.7961×10^{-17}	3.9820×10^{-17}	–	1.0655	1.2833
1910	Ulnae cortical	7.42	4.0583×10^{-17}	13.08	–	3.4605×10^{-17}	3.8630×10^{-17}	–	1.1728	1.0506
2000	Radii spongiosa	1.12	2.8523×10^{-17}	47.42	–	4.5702×10^{-17}	3.1544×10^{-17}	–	0.6241	0.9042
2010	Ulnae spongiosa	1.81	4.8580×10^{-17}	34.79	–	4.2656×10^{-17}	4.0020×10^{-17}	–	1.1389	1.2139
2100	Radii medullary cavity	0.359	5.5746×10^{-17}	51.25	–	7.0602×10^{-17}	5.0744×10^{-17}	–	0.7896	1.0986
2110	Ulnae medullary cavity	0.589	4.8573×10^{-17}	51.97	–	2.1355×10^{-17}	6.2866×10^{-17}	–	2.2745	0.7726
2200	Wrists and hand bones cortical	5.40	2.3110×10^{-17}	18.98	–	3.7034×10^{-17}	4.7768×10^{-17}	–	0.6240	0.4838
2300	Wrists and hand bones spongiosa	14.07	2.8811×10^{-17}	10.53	–	3.9658×10^{-17}	3.9185×10^{-17}	–	0.7265	0.7353
2400	Clavicles cortical	1.37	4.6821×10^{-17}	25.11	–	3.4763×10^{-17}	2.7997×10^{-17}	–	1.3469	1.6723
2500	Clavicles spongiosa	1.96	4.8520×10^{-17}	27.26	–	5.4429×10^{-17}	5.0569×10^{-17}	–	0.8914	0.9595
2600	Cranium cortical	91.49	3.0756×10^{-17}	4.78	–	2.8685×10^{-17}	2.9253×10^{-17}	–	1.0722	1.0514
2700	Cranium spongiosa	289.96	2.7487×10^{-17}	1.70	–	2.7456×10^{-17}	2.7948×10^{-17}	–	1.0011	0.9835
2800	Femora upper cortical	14.11	3.5537×10^{-17}	15.18	–	3.0975×10^{-17}	3.1386×10^{-17}	–	1.1473	1.1323
2900	Femora upper spongiosa	8.86	3.6290×10^{-17}	11.19	–	3.3689×10^{-17}	3.4667×10^{-17}	–	1.0772	1.0468
3000	Femora upper medullary cavity	0.994	1.1050×10^{-17}	57.17	–	2.6611×10^{-17}	6.9035×10^{-17}	–	0.4152	0.1601
3100	Femora lower cortical	20.53	2.3830×10^{-17}	12.40	–	2.9202×10^{-17}	3.6137×10^{-17}	–	0.8160	0.6594
3200	Femora lower spongiosa	8.77	4.1330×10^{-17}	8.80	–	2.6568×10^{-17}	3.3172×10^{-17}	–	1.5556	1.2459
3300	Femora lower medullary cavity	1.57	1.4359×10^{-17}	43.01	–	2.6490×10^{-17}	3.7558×10^{-17}	–	0.5420	0.3823
3400	Tibiae cortical	26.33	3.2380×10^{-17}	9.41	–	3.2636×10^{-17}	3.4616×10^{-17}	–	0.9922	0.9354
3410	Fibulae cortical	3.02	1.8366×10^{-17}	28.99	–	1.8314×10^{-17}	2.1108×10^{-17}	–	1.0028	0.8701
3420	Patellae cortical	0.162	4.7892×10^{-17}	49.29	–	1.3882×10^{-17}	3.9755×10^{-17}	–	3.4499	1.2047
3500	Tibiae spongiosa	7.27	2.7718×10^{-17}	14.47	–	3.5231×10^{-17}	3.5741×10^{-17}	–	0.7867	0.7755
3510	Fibulae spongiosa	0.565	3.6316×10^{-17}	51.18	–	4.4111×10^{-17}	2.2859×10^{-17}	–	0.8233	1.5887
3520	Patellae spongiosa	0.575	3.3918×10^{-17}	57.66	–	4.2248×10^{-17}	3.0901×10^{-17}	–	0.8028	1.0976
3600	Tibiae medullary cavity	2.20	4.7710×10^{-17}	26.23	–	3.4953×10^{-17}	3.6883×10^{-17}	–	1.3649	1.2935
3610	Fibulae medullary cavity	0.204	8.7781×10^{-19}	100.00	–	7.4515×10^{-17}	2.3521×10^{-17}	–	0.0118	0.0373
3700	Ankles and foot cortical	7.24	3.2566×10^{-17}	17.99	–	3.1667×10^{-17}	3.3177×10^{-17}	–	1.0284	0.9816
3800	Ankles and foot spongiosa	20.33	2.7528×10^{-17}	10.45	–	2.9560×10^{-17}	3.1270×10^{-17}	–	0.9313	0.8803
3900	Mandible cortical	7.16	3.2636×10^{-17}	13.15	–	3.1851×10^{-17}	4.0708×10^{-17}	–	1.0246	0.8017
4000	Mandible spongiosa	13.09	4.2129×10^{-17}	9.69	–	3.9686×10^{-17}	3.0298×10^{-17}	–	1.0616	1.3905
4100	Pelvis cortical	21.20	3.4965×10^{-17}	6.74	–	2.7390×10^{-17}	2.8287×10^{-17}	–	1.2766	1.2361
4200	Pelvis spongiosa	34.40	3.2923×10^{-17}	7.05	–	3.2495×10^{-17}	3.1509×10^{-17}	–	1.0132	1.0449
4300	Ribs cortical	20.41	2.6174×10^{-17}	9.98	–	3.0004×10^{-17}	2.6419×10^{-17}	–	0.8723	0.9907
4400	Ribs spongiosa	58.12	2.8684×10^{-17}	7.04	–	3.2033×10^{-17}	3.4446×10^{-17}	–	0.8955	0.8327

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
4500	Scapulae cortical	9.65	1.9085×10^{-17}	13.13	–	2.5536×10^{-17}	2.1769×10^{-17}	–	0.7473	0.8767
4600	Scapulae spongiosa	14.86	2.3317×10^{-17}	21.54	–	2.6416×10^{-17}	2.6620×10^{-17}	–	0.8827	0.8759
4700	Cervical spine cortical	3.28	4.1581×10^{-17}	15.78	–	3.1837×10^{-17}	2.9739×10^{-17}	–	1.3061	1.3982
4800	Cervical spine spongiosa	9.10	3.2300×10^{-17}	10.22	–	3.5267×10^{-17}	3.3962×10^{-17}	–	0.9159	0.9511
4900	Thoracic spine cortical	11.50	2.4695×10^{-17}	14.47	–	2.5275×10^{-17}	2.1036×10^{-17}	–	0.9771	1.1739
5000	Thoracic spine spongiosa	28.11	2.5758×10^{-17}	11.19	–	2.4846×10^{-17}	2.3217×10^{-17}	–	1.0367	1.1095
5100	Lumbar spine cortical	7.07	2.7728×10^{-17}	16.19	–	2.7470×10^{-17}	2.4152×10^{-17}	–	1.0094	1.1481
5200	Lumbar spine spongiosa	21.94	2.4536×10^{-17}	9.47	–	3.1618×10^{-17}	2.5428×10^{-17}	–	0.7760	0.9649
5300	Sacrum cortical	5.45	2.8563×10^{-17}	17.45	–	2.0286×10^{-17}	2.5213×10^{-17}	–	1.4080	1.1329
5400	Sacrum spongiosa	15.12	2.8671×10^{-17}	11.73	–	2.5923×10^{-17}	3.0681×10^{-17}	–	1.1060	0.9345
5500	Sternum cortical	0.63	4.8175×10^{-17}	31.12	–	3.8046×10^{-17}	4.3621×10^{-17}	–	1.2662	1.1044
5600	Sternum spongiosa	2.10	4.6051×10^{-17}	25.14	–	3.3920×10^{-17}	4.5823×10^{-17}	–	1.3577	1.0050
5700	Cartilage head	5.03	2.9215×10^{-17}	27.11	–	2.5626×10^{-17}	2.3724×10^{-17}	–	1.1401	1.2315
5800	Cartilage trunk	36.85	3.5233×10^{-17}	8.47	–	3.6323×10^{-17}	3.2199×10^{-17}	–	0.9700	1.0942
6100	Brain	978.10	2.6489×10^{-17}	1.48	–	2.7081×10^{-17}	2.6763×10^{-17}	–	0.9781	0.9898
6200	Breast left adipose tissue	0.572	3.7021×10^{-17}	62.23	–	7.6181×10^{-18}	7.8904×10^{-17}	–	4.8596	0.4692
6300	Breast left glandular tissue	0.381	6.6636×10^{-17}	51.52	–	5.3369×10^{-19}	3.7102×10^{-17}	–	124.8591	1.7960
6400	Breast right adipose tissue	0.572	4.1978×10^{-17}	46.68	–	1.0538×10^{-16}	3.5113×10^{-17}	–	0.3983	1.1955
6500	Breast right glandular tissue	0.381	3.1690×10^{-17}	54.40	–	7.3291×10^{-17}	4.2321×10^{-17}	–	0.4324	0.7488
6600	Eye lens sensitive left	0.0169	0.0000×10^{0}	0.00	–	–	1.1160×10^{-16}	–	–	–
6601	Eye lens insensitive left	0.0978	0.0000×10^{0}	0.00	–	3.4665×10^{-18}	4.0823×10^{-17}	–	–	–
6700	Cornea left	0.629	3.3941×10^{-17}	49.83	–	2.6666×10^{-17}	5.6497×10^{-17}	–	1.2728	0.6008
6701	Aqueous left	0.275	7.9789×10^{-17}	64.43	–	7.5140×10^{-18}	5.9432×10^{-17}	–	10.6188	1.3425
6702	Vitreous left	2.62	1.7988×10^{-17}	43.52	–	3.3007×10^{-17}	2.2960×10^{-17}	–	0.5450	0.7835
6800	Eye lens sensitive right	0.0169	0.0000×10^{0}	0.00	–	–	–	–	–	–
6801	Eye lens insensitive right	0.0978	0.0000×10^{0}	0.00	–	–	–	–	–	–
6900	Cornea right	0.629	1.9865×10^{-18}	79.79	–	2.5734×10^{-17}	3.0026×10^{-17}	–	0.0772	0.0662
6901	Aqueous right	0.275	5.8635×10^{-17}	52.45	–	4.8426×10^{-17}	2.9525×10^{-17}	–	1.2108	1.9859
6902	Vitreous right	2.62	3.0511×10^{-17}	32.25	–	5.6277×10^{-17}	3.1709×10^{-17}	–	0.5422	0.9622
7000	Gall bladder wall	1.46	3.0724×10^{-17}	32.31	–	2.3272×10^{-17}	1.2918×10^{-17}	–	1.3202	2.3784
7100	Gall bladder contents	8.00	2.6722×10^{-17}	19.81	–	4.4795×10^{-17}	3.0737×10^{-17}	–	0.5965	0.8694
7200	Stomach wall(0-60)	0.814	4.4115×10^{-17}	24.37	–	3.2352×10^{-17}	2.8419×10^{-17}	–	1.3636	1.5523
7201	Stomach wall(60-100)	0.407	2.9638×10^{-17}	28.49	–	2.4638×10^{-17}	3.9286×10^{-17}	–	1.2030	0.7544
7202	Stomach wall(100-300)	2.06	3.9702×10^{-17}	19.03	–	4.2604×10^{-17}	3.1390×10^{-17}	–	0.9319	1.2648
7203	Stomach wall(300-surface)	19.88	3.2509×10^{-17}	9.65	–	3.6344×10^{-17}	3.0428×10^{-17}	–	0.8945	1.0684
7300	Stomach contents	67.00	3.3197×10^{-17}	5.85	–	3.2075×10^{-17}	3.1515×10^{-17}	–	1.0350	1.0534
7400	Small intestine wall(0-130)	4.93	3.4731×10^{-17}	15.62	–	3.1649×10^{-17}	3.3294×10^{-17}	–	1.0974	1.0432
7401	Small intestine wall(130-150)	0.77	3.8094×10^{-17}	20.93	–	3.3576×10^{-17}	3.5774×10^{-17}	–	1.1346	1.0649

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
7402	Small intestine wall(150-200)	1.94	3.6763×10^{-17}	9.57	–	3.3423×10^{-17}	3.2377×10^{-17}	–	1.0999	1.1355
7403	Small intestine wall(200-surface)	92.27	3.6753×10^{-17}	5.23	–	3.7181×10^{-17}	3.6347×10^{-17}	–	0.9885	1.0112
7500	Small intestine contents(-400-0)	14.32	3.4562×10^{-17}	9.16	–	3.4105×10^{-17}	3.3320×10^{-17}	–	1.0134	1.0373
7501	Small intestine contents(centre–400)	78.68	3.4282×10^{-17}	5.64	–	3.4194×10^{-17}	3.2092×10^{-17}	–	1.0026	1.0682
7600	Ascending colon wall(0-280)	1.57	3.5102×10^{-17}	23.55	–	2.4586×10^{-17}	4.8374×10^{-17}	–	1.4277	0.7256
7601	Ascending colon wall(280-300)	0.114	5.0541×10^{-17}	43.19	–	3.7978×10^{-17}	6.9613×10^{-17}	–	1.3308	0.7260
7602	Ascending colon wall(300-surface)	9.37	4.4563×10^{-17}	9.80	–	5.2420×10^{-17}	3.4178×10^{-17}	–	0.8501	1.3038
7700	Ascending colon content	26.02	4.1283×10^{-17}	8.27	–	3.7650×10^{-17}	3.5124×10^{-17}	–	1.0965	1.1753
7800	Transverse colon wall right(0-280)	1.20	4.4050×10^{-17}	29.67	–	4.9037×10^{-17}	3.1381×10^{-17}	–	0.8983	1.4037
7801	Transverse colon wall right(280-300)	0.0878	1.7625×10^{-17}	26.47	–	3.6032×10^{-17}	1.0239×10^{-16}	–	0.4892	0.1721
7802	Transverse colon wall right(300-surface)	11.14	3.4662×10^{-17}	10.89	–	4.6701×10^{-17}	4.8444×10^{-17}	–	0.7422	0.7155
7900	Transverse colon contents right	13.98	3.4885×10^{-17}	10.43	–	4.9345×10^{-17}	4.2607×10^{-17}	–	0.7070	0.8188
8000	Transverse colon wall left(0-280)	1.03	4.2220×10^{-17}	25.01	–	3.2554×10^{-17}	4.4441×10^{-17}	–	1.2969	0.9500
8001	Transverse colon wall left(280-300)	0.0757	6.0947×10^{-17}	47.34	–	7.9635×10^{-17}	4.4467×10^{-17}	–	0.7653	1.3706
8002	Transverse colon wall left(300-surface)	10.30	3.9432×10^{-17}	15.06	–	3.6128×10^{-17}	4.3432×10^{-17}	–	1.0914	0.9079
8100	Transverse colon content left	8.93	3.3981×10^{-17}	14.86	–	3.2522×10^{-17}	3.3748×10^{-17}	–	1.0449	1.0069
8200	Descending colon wall(0-280)	1.23	2.9759×10^{-17}	31.84	–	2.5192×10^{-17}	2.2707×10^{-17}	–	1.1813	1.3106
8201	Descending colon wall(280-300)	0.0906	3.4399×10^{-17}	34.45	–	4.0753×10^{-17}	6.0556×10^{-17}	–	0.8441	0.5681
8202	Descending colon wall(300-surface)	10.75	3.2157×10^{-17}	9.05	–	2.6673×10^{-17}	3.4940×10^{-17}	–	1.2056	0.9204
8300	Descending colon content	11.07	2.7594×10^{-17}	20.05	–	3.2699×10^{-17}	3.1713×10^{-17}	–	0.8439	0.8701
8400	Sigmoid colon wall(0-280)	1.57	4.1833×10^{-17}	27.48	–	3.2769×10^{-17}	2.9030×10^{-17}	–	1.2766	1.4410
8401	Sigmoid colon wall(280-300)	0.115	2.9105×10^{-17}	52.49	–	4.3857×10^{-17}	1.8614×10^{-17}	–	0.6636	1.5636
8402	Sigmoid colon wall(300-surface)	7.99	2.5472×10^{-17}	23.86	–	3.6834×10^{-17}	2.5676×10^{-17}	–	0.6915	0.9921
8500	Sigmoid colon contents	13.98	4.2556×10^{-17}	12.31	–	2.6115×10^{-17}	2.9586×10^{-17}	–	1.6295	1.4384
8600	Rectum wall(0-280)	0.657	1.6622×10^{-17}	54.39	–	3.0153×10^{-17}	2.9717×10^{-17}	–	0.5513	0.5593
8601	Rectum wall(280-300)	0.0484	7.8134×10^{-18}	50.90	–	2.6507×10^{-17}	1.3390×10^{-17}	–	0.2948	0.5835
8602	Rectum wall(300-surface)	1.36	1.6102×10^{-17}	29.73	–	1.6616×10^{-17}	3.1633×10^{-17}	–	0.9691	0.5090
8603	Rectum contents	6.02	2.1116×10^{-17}	19.85	–	3.2067×10^{-17}	2.6070×10^{-17}	–	0.6585	0.8100
8700	Heart wall	55.04	3.4198×10^{-17}	7.05	–	3.2963×10^{-17}	3.5273×10^{-17}	–	1.0375	0.9695
8800	Blood in heart chamber	48.00	3.6779×10^{-17}	4.54	–	3.2394×10^{-17}	3.5307×10^{-17}	–	1.1354	1.0417
8900	Kidney left cortex	27.76	2.5087×10^{-17}	15.99	–	2.1676×10^{-17}	2.7229×10^{-17}	–	1.1573	0.9213
9000	Kidney left medulla	9.92	2.8163×10^{-17}	11.33	–	2.9156×10^{-17}	2.2596×10^{-17}	–	0.9659	1.2464
9100	Kidney left pelvis	1.98	5.1834×10^{-17}	27.61	–	3.6268×10^{-17}	3.9900×10^{-17}	–	1.4292	1.2991
9200	Kidney right cortex	27.76	2.9577×10^{-17}	8.35	–	2.5191×10^{-17}	2.5380×10^{-17}	–	1.1741	1.1654
9300	Kidney right medulla	9.92	1.9620×10^{-17}	21.48	–	3.4632×10^{-17}	3.3445×10^{-17}	–	0.5665	0.5866
9400	Kidney right pelvis	1.98	9.2023×10^{-18}	47.25	–	3.0092×10^{-17}	2.5924×10^{-17}	–	0.3058	0.3550

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
9500	Liver	390.44	3.3197×10^{-17}	2.59	–	3.4706×10^{-17}	3.4088×10^{-17}	–	0.9565	0.9738
9700	Lung(AI) left	68.84	3.4404×10^{-17}	5.68	–	3.2133×10^{-17}	3.2447×10^{-17}	–	1.0707	1.0603
9900	Lung(AI) right	81.16	3.5100×10^{-17}	5.92	–	3.3304×10^{-17}	3.4196×10^{-17}	–	1.0539	1.0264
10000	Lymphatic nodes ET	2.18	4.8830×10^{-17}	28.00	–	3.7569×10^{-17}	3.0824×10^{-17}	–	1.2997	1.5842
10100	Lymphatic nodes thoracic	2.18	2.3930×10^{-17}	25.84	–	3.0451×10^{-17}	2.7142×10^{-17}	–	0.7859	0.8817
10200	Lymphatic nodes head	0.752	7.6236×10^{-17}	30.34	–	6.2839×10^{-17}	2.6886×10^{-17}	–	1.2132	2.8355
10300	Lymphatic nodes trunk	17.76	3.5665×10^{-17}	11.24	–	3.3541×10^{-17}	4.4149×10^{-17}	–	1.0633	0.8078
10400	Lymphatic nodes arms	1.50	1.7696×10^{-17}	40.99	–	3.0215×10^{-17}	3.9134×10^{-17}	–	0.5857	0.4522
10500	Lymphatic nodes legs	1.50	4.3960×10^{-17}	28.95	–	5.2476×10^{-17}	4.8829×10^{-17}	–	0.8377	0.9003
10600	Muscle head	240.46	3.2734×10^{-17}	2.63	–	3.2809×10^{-17}	3.2224×10^{-17}	–	0.9977*	1.0158*
10700	Muscle trunk	885.17	3.1619×10^{-17}	1.50	–	3.2550×10^{-17}	3.1650×10^{-17}	–	0.9714*	0.9990*
10800	Muscle arms	204.07	3.6765×10^{-17}	3.90	–	3.7264×10^{-17}	3.7787×10^{-17}	–	0.9866*	0.9730*
10900	Muscle legs	599.50	3.2041×10^{-17}	1.81	–	3.5981×10^{-17}	3.3723×10^{-17}	–	0.8905*	0.9501*
11000	Oesophagus wall(0-190)	0.567	2.7839×10^{-17}	29.28	–	2.2582×10^{-17}	2.3900×10^{-17}	–	1.2328	1.1648
11001	Oesophagus wall(190-200)	0.0308	8.0806×10^{-17}	68.35	–	1.2673×10^{-17}	1.9407×10^{-17}	–	6.3760	4.1638
11002	Oesophagus wall(200-surface)	5.19	3.4709×10^{-17}	22.07	–	3.4766×10^{-17}	3.1502×10^{-17}	–	0.9984	1.1018
11003	Oesophagus contents	3.79	2.0433×10^{-17}	21.20	–	3.0195×10^{-17}	2.3340×10^{-17}	–	0.6767	0.8755
11100	Ovary left	0.423	3.9326×10^{-17}	61.55	–	3.6703×10^{-17}	5.8799×10^{-18}	–	1.0715	6.6881
11200	Ovary right	0.423	3.3809×10^{-17}	64.21	–	9.3038×10^{-17}	4.3573×10^{-17}	–	0.3634	0.7759
11300	Pancreas	22.65	3.6055×10^{-17}	10.48	–	2.8848×10^{-17}	3.2105×10^{-17}	–	1.2498	1.1230
11400	Pituitary gland	0.156	1.6910×10^{-18}	100.00	–	5.6032×10^{-17}	4.1588×10^{-17}	–	0.0302	0.0407
11600	RST head	572.11	3.2031×10^{-17}	2.59	–	3.1607×10^{-17}	3.2874×10^{-17}	–	1.0134*	0.9744*
11700	RST trunk	2832.52	3.4156×10^{-17}	1.36	–	3.4297×10^{-17}	3.4360×10^{-17}	–	0.9959*	0.9941*
11800	RST arms	193.18	3.8920×10^{-17}	4.26	–	3.8100×10^{-17}	3.6200×10^{-17}	–	1.0215*	1.0751*
11900	RST legs	696.24	3.4824×10^{-17}	2.11	–	3.5808×10^{-17}	3.5272×10^{-17}	–	0.9725*	0.9873*
12000	Salivary glands left	12.49	3.5452×10^{-17}	14.50	–	3.7588×10^{-17}	3.2926×10^{-17}	–	0.9432	1.0767
12100	Salivary glandss right	12.49	3.1400×10^{-17}	9.63	–	3.5160×10^{-17}	2.1933×10^{-17}	–	0.8930	1.4316
12200	Skin head insensitive	68.00	2.3297×10^{-17}	6.81	–	2.4439×10^{-17}	2.5727×10^{-17}	–	0.9533	0.9055
12201	Skin head sensitive(40-100)	5.45	1.8013×10^{-17}	14.29	–	1.6451×10^{-17}	2.1316×10^{-17}	–	1.0950*	0.8451*
12300	Skin trunk insensitive	126.04	2.3908×10^{-17}	4.02	–	2.3808×10^{-17}	2.6919×10^{-17}	–	1.0042*	0.8881*
12301	Skin trunk sensitive(40-100)	10.41	1.7373×10^{-17}	9.30	–	1.9333×10^{-17}	1.9745×10^{-17}	–	0.8986*	0.8799*
12400	Skin arms insensitive	49.86	2.9863×10^{-17}	8.87	–	2.9643×10^{-17}	3.2509×10^{-17}	–	1.0074*	0.9186*
12401	Skin arms sensitive(40-100)	4.11	2.3279×10^{-17}	11.12	–	2.3316×10^{-17}	2.8594×10^{-17}	–	0.9984*	0.8141*
12500	Skin legs insensitive	89.78	2.5303×10^{-17}	2.79	–	2.6252×10^{-17}	2.5032×10^{-17}	–	0.9638*	1.0108*
12501	Skin legs sensitive(40-100)	7.28	1.7896×10^{-17}	6.03	–	1.7616×10^{-17}	1.5259×10^{-17}	–	1.0159*	1.1728*
12600	Spinal cord	10.67	2.3885×10^{-17}	16.32	–	3.0711×10^{-17}	2.4611×10^{-17}	–	0.7777	0.9705
12700	Spleen	37.35	3.3554×10^{-17}	5.88	–	3.5324×10^{-17}	2.5117×10^{-17}	–	0.9499	1.3359

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
12802	Erupted upper front buccal deciduous ena	0.0689	0.0000×10^{0}	0.00	–	1.0829×10^{-16}	5.5147×10^{-17}	–	–	–
12803	Erupted upper front lingual deciduous en	0.0617	0.0000×10^{0}	0.00	–	1.5603×10^{-16}	9.0705×10^{-17}	–	–	–
12818	Erupted lower front buccal deciduous ena	0.0506	2.9600×10^{-17}	100.00	–	–	3.0490×10^{-16}	–	–	0.0971
12819	Erupted lower front lingual deciduous en	0.0373	0.0000×10^{0}	0.00	–	–	–	–	–	–
12833	Erupted upper front deciduous dentin	0.3	3.3936×10^{-18}	100.00	–	3.9007×10^{-18}	4.6536×10^{-17}	–	0.8700	0.0729
12841	Erupted lower front deciduous dentin	0.2	2.8153×10^{-17}	100.00	–	2.8612×10^{-17}	1.0917×10^{-16}	–	0.9839	0.2579
12848	Unerupted permanent enamel	0.146	1.9636×10^{-17}	100.00	–	1.4435×10^{-17}	1.7941×10^{-17}	–	1.3603	1.0945
12849	Unerupted deciduous enamel	0.952	5.7082×10^{-17}	29.74	–	3.7578×10^{-17}	4.0161×10^{-17}	–	1.5190	1.4213
12850	Unerupted permanent dentin	0.489	2.4865×10^{-17}	49.03	–	9.9892×10^{-17}	2.8460×10^{-17}	–	0.2489	0.8737
12851	Unerupted deciduous dentin	2.21	3.4026×10^{-17}	25.89	–	2.3443×10^{-17}	3.5383×10^{-17}	–	1.4514	0.9616
12852	Permanent cementum	0.0203	1.0842×10^{-16}	100.00	–	6.7235×10^{-18}	5.2431×10^{-17}	–	16.1258	2.0679
12853	Deciduous cementum	0.122	2.6409×10^{-17}	54.59	–	1.5634×10^{-17}	4.7329×10^{-17}	–	1.6892	0.5580
12854	Permanent pulp	0.0146	0.0000×10^{0}	0.00	–	–	–	–	–	–
12855	Deciduous pulp	0.374	2.5087×10^{-17}	49.47	–	2.5546×10^{-17}	4.4542×10^{-17}	–	0.9820	0.5632
12856	Teeth retention region	0.00345	4.5388×10^{-17}	100.00	–	–	4.7806×10^{-17}	–	–	0.9494
13100	Thymus	31.19	4.0477×10^{-17}	6.97	–	3.5378×10^{-17}	3.8018×10^{-17}	–	1.1441	1.0647
13200	Thyroid	1.97	5.5624×10^{-17}	30.54	–	3.0368×10^{-17}	1.9069×10^{-17}	–	1.8317	2.9170
13300	Tongue upper(food)	5.97	4.2324×10^{-17}	11.46	–	4.5009×10^{-17}	4.1601×10^{-17}	–	0.9404	1.0174
13301	Tongue lower	4.42	3.3236×10^{-17}	29.26	–	4.0809×10^{-17}	4.0238×10^{-17}	–	0.8144	0.8260
13400	Tonsils	0.521	4.0204×10^{-17}	46.96	–	8.8871×10^{-18}	4.3331×10^{-17}	–	4.5239	0.9278
13500	Ureter left	1.15	2.5366×10^{-17}	69.92	–	2.6023×10^{-17}	6.7747×10^{-18}	–	0.9748	3.7442
13600	Ureter right	1.15	3.0220×10^{-17}	43.17	–	2.6516×10^{-17}	1.4415×10^{-17}	–	1.1397	2.0964
13700	Urinary bladder wall insensitive	8.17	3.6181×10^{-17}	13.31	–	3.6179×10^{-17}	3.1005×10^{-17}	–	1.0001	1.1670
13701	Urinary bladder wall sensitive(71-238)	0.947	3.8260×10^{-17}	21.57	–	4.0804×10^{-17}	3.6342×10^{-17}	–	0.9376	1.0528
13800	Urinary bladder content	32.90	3.3940×10^{-17}	7.25	–	3.4553×10^{-17}	3.2717×10^{-17}	–	0.9823	1.0374
13900	Uterus	1.56	5.2039×10^{-17}	31.61	–	5.9171×10^{-17}	2.1625×10^{-17}	–	0.8795	2.4064
14000	Air inside body	0.00736	1.0485×10^{-17}	50.47	–	3.2203×10^{-17}	2.4958×10^{-16}	–	0.3256	0.0420

* For tissues bookkept as head, trunk, arm and leg pieces (muscle, residual soft tissue, skin, large vessels), the volumes tabulated in the distributed cell tables partition the tissue at different boundaries from the transported mesh, and the reference output is normalised by the tabulated volume. The marked ratio therefore compares two different partitions of the same tissue, and is left out of the summary statistics of Table S2.

Table S15. 05M_Internal. Absorbed dose per primary history, Gy. σ is the FLUKA statistical uncertainty. Ratios are FLUKA / reference.

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
100	Adrenal left	2.97	5.0012×10^{-15}	2.42	–	4.9227×10^{-15}	4.7618×10^{-15}	–	1.0160	1.0503
200	Adrenal right	2.97	1.2492×10^{-14}	1.27	–	1.2964×10^{-14}	1.2778×10^{-14}	–	0.9636	0.9776
300	ET1(0-8)	0.00268	2.4890×10^{-16}	77.14	–	1.0010×10^{-16}	1.2526×10^{-16}	–	2.4866	1.9871
301	ET1(8-40)	0.0109	2.4650×10^{-16}	64.21	–	1.5370×10^{-16}	1.8274×10^{-16}	–	1.6038	1.3489
302	ET1(40-50)	0.00345	7.6901×10^{-16}	63.28	–	5.1639×10^{-16}	3.9668×10^{-16}	–	1.4892	1.9386
303	ET1(50-Surface)	0.316	3.4079×10^{-16}	19.08	–	2.3320×10^{-16}	2.6827×10^{-16}	–	1.4613	1.2703
400	ET2(-15-0)	0.127	3.5693×10^{-16}	15.13	–	3.6673×10^{-16}	3.0194×10^{-16}	–	0.9733	1.1821
401	ET2(0-40)	0.364	3.5964×10^{-16}	6.75	–	4.2619×10^{-16}	3.9218×10^{-16}	–	0.8438	0.9170
402	ET2(40-50)	0.0917	3.6775×10^{-16}	17.10	–	4.2565×10^{-16}	3.1812×10^{-16}	–	0.8640	1.1560
403	ET2(50-55)	0.046	3.3816×10^{-16}	21.88	–	4.3641×10^{-16}	3.8549×10^{-16}	–	0.7749	0.8772
404	ET2(55-65)	0.0921	4.4703×10^{-16}	9.34	–	3.7594×10^{-16}	3.7876×10^{-16}	–	1.1891	1.1802
405	ET2(65-Surface)	16.63	3.3908×10^{-16}	3.20	–	3.4534×10^{-16}	3.0803×10^{-16}	–	0.9819	1.1008
500	Oral mucosa tongue	0.0311	4.6746×10^{-16}	18.80	–	4.4637×10^{-16}	4.9176×10^{-16}	–	1.0472	0.9506
501	Oral mucosa moth floor	0.019	5.7295×10^{-16}	30.94	–	5.1150×10^{-16}	3.0612×10^{-16}	–	1.1201	1.8717
600	Oral mucosa lips and cheeks	0.0133	6.5872×10^{-16}	37.20	–	5.0464×10^{-16}	5.1530×10^{-16}	–	1.3053	1.2783
700	Trachea	2.60	1.2821×10^{-15}	4.33	–	1.1781×10^{-15}	1.2621×10^{-15}	–	1.0883	1.0159
800	BB(-11–6)	0.0022	1.6347×10^{-15}	44.04	–	1.8764×10^{-15}	1.3207×10^{-15}	–	0.8712	1.2378
801	BB(-6-0)	0.00282	9.3954×10^{-16}	39.60	–	2.4350×10^{-15}	1.1205×10^{-15}	–	0.3859	0.8385
802	BB(0-10)	0.00473	1.6458×10^{-15}	35.87	–	2.6051×10^{-15}	2.8097×10^{-15}	–	0.6317	0.5857
803	BB(10-35)	0.012	1.7412×10^{-15}	30.76	–	2.1837×10^{-15}	2.4659×10^{-15}	–	0.7974	0.7061
804	BB(35-40)	0.00243	1.8851×10^{-15}	28.26	–	3.3280×10^{-15}	1.5369×10^{-15}	–	0.5664	1.2265
805	BB(40-50)	0.00489	2.4682×10^{-15}	33.86	–	2.1594×10^{-15}	1.8696×10^{-15}	–	1.1430	1.3202
806	BB(50-60)	0.00493	1.5956×10^{-15}	25.26	–	2.8927×10^{-15}	2.0525×10^{-15}	–	0.5516	0.7774
807	BB(60-70)	0.00497	1.8857×10^{-15}	31.25	–	2.1785×10^{-15}	3.1345×10^{-15}	–	0.8656	0.6016
808	BB(70-surface)	3.38	2.1726×10^{-15}	2.82	–	2.0433×10^{-15}	2.1959×10^{-15}	–	1.0633	0.9894
900	Blood in large arteries head	0.271	7.6513×10^{-16}	14.29	–	5.5107×10^{-16}	5.5828×10^{-16}	–	1.3885*	1.3705*
910	Blood in large veins head	1.77	5.2066×10^{-16}	8.38	–	6.6341×10^{-16}	5.9591×10^{-16}	–	0.7848*	0.8737*
1000	Blood in large arteries trunk	38.26	3.3221×10^{-15}	0.75	–	3.3749×10^{-15}	3.4152×10^{-15}	–	0.9844*	0.9727*
1010	Blood in large veins trunk	107.57	3.2139×10^{-15}	0.40	–	3.3095×10^{-15}	3.3299×10^{-15}	–	0.9711*	0.9652*
1100	Blood in large arteries arms	6.61	1.9041×10^{-15}	1.68	–	1.7893×10^{-15}	1.8154×10^{-15}	–	1.0642*	1.0489*
1110	Blood in large veins arms	42.98	1.3188×10^{-15}	0.94	–	1.3353×10^{-15}	1.3127×10^{-15}	–	0.9877*	1.0046*
1200	Blood in large arteries legs	44.81	8.1998×10^{-17}	3.37	–	8.7267×10^{-17}	8.5197×10^{-17}	–	0.9396*	0.9625*
1210	Blood in large veins legs	117.52	9.7886×10^{-17}	1.94	–	1.0109×10^{-16}	1.0243×10^{-16}	–	0.9683*	0.9556*
1300	Humeri upper cortical	28.83	1.1811×10^{-15}	1.62	–	1.1408×10^{-15}	1.1292×10^{-15}	–	1.0353	1.0460
1400	Humeri upper spogiosa	20.16	8.8131×10^{-16}	2.51	–	8.1706×10^{-16}	8.3197×10^{-16}	–	1.0786	1.0593

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
1500	Humeri upper medullary cavity	3.76	1.4671×10^{-15}	3.07	–	1.4539×10^{-15}	1.5536×10^{-15}	–	1.0091	0.9443
1600	Humeri lower cortical	22.27	1.9143×10^{-15}	1.25	–	1.9226×10^{-15}	1.8996×10^{-15}	–	0.9957	1.0077
1700	Humeri lower spongiosa	11.03	1.6819×10^{-15}	2.12	–	1.6598×10^{-15}	1.6977×10^{-15}	–	1.0133	0.9907
1800	Humeri lower medullary cavity	3.56	2.1364×10^{-15}	2.96	–	2.1376×10^{-15}	2.1198×10^{-15}	–	0.9994	1.0078
1900	Radii cortical	14.85	6.3428×10^{-16}	2.11	–	6.4536×10^{-16}	6.2437×10^{-16}	–	0.9828	1.0159
1910	Ulnae cortical	18.78	8.6449×10^{-16}	1.54	–	8.9349×10^{-16}	8.4352×10^{-16}	–	0.9675	1.0249
2000	Radii spongiosa	3.31	6.0367×10^{-16}	5.21	–	6.2397×10^{-16}	5.7079×10^{-16}	–	0.9675	1.0576
2010	Ulnae spongiosa	5.49	1.3286×10^{-15}	3.90	–	1.2756×10^{-15}	1.2193×10^{-15}	–	1.0416	1.0897
2100	Radii medullary cavity	2.30	7.0114×10^{-16}	6.37	–	6.7636×10^{-16}	7.6198×10^{-16}	–	1.0366	0.9202
2110	Ulnae medullary cavity	2.63	6.5742×10^{-16}	4.85	–	6.3986×10^{-16}	7.1442×10^{-16}	–	1.0274	0.9202
2200	Wrists and hand bones cortical	9.33	2.1747×10^{-16}	3.86	–	2.2321×10^{-16}	2.2274×10^{-16}	–	0.9743	0.9763
2300	Wrists and hand bones spongiosa	16.48	2.3355×10^{-16}	3.72	–	2.2791×10^{-16}	2.5002×10^{-16}	–	1.0247	0.9341
2400	Clavicles cortical	6.18	1.0097×10^{-15}	2.12	–	1.1016×10^{-15}	9.8581×10^{-16}	–	0.9166	1.0243
2500	Clavicles spongiosa	7.96	1.0936×10^{-15}	2.19	–	1.1193×10^{-15}	1.1434×10^{-15}	–	0.9770	0.9565
2600	Cranium cortical	229.09	1.9399×10^{-16}	1.23	–	1.9095×10^{-16}	1.8853×10^{-16}	–	1.0159	1.0290
2700	Cranium spongiosa	520.37	1.7338×10^{-16}	0.45	–	1.7563×10^{-16}	1.7213×10^{-16}	–	0.9871	1.0072
2800	Femora upper cortical	49.47	2.2616×10^{-16}	2.71	–	2.2921×10^{-16}	2.1351×10^{-16}	–	0.9867	1.0592
2900	Femora upper spongiosa	26.10	3.5667×10^{-16}	2.84	–	3.5110×10^{-16}	3.5725×10^{-16}	–	1.0159	0.9984
3000	Femora upper medullary cavity	8.22	1.4907×10^{-16}	7.74	–	1.5103×10^{-16}	1.2266×10^{-16}	–	0.9870	1.2153
3100	Femora lower cortical	48.06	2.9850×10^{-17}	9.02	–	2.9655×10^{-17}	3.4680×10^{-17}	–	1.0066	0.8607
3200	Femora lower spongiosa	28.74	2.7042×10^{-17}	10.23	–	2.7274×10^{-17}	2.2049×10^{-17}	–	0.9915	1.2265
3300	Femora lower medullary cavity	5.79	3.5201×10^{-17}	12.26	–	2.9284×10^{-17}	3.4985×10^{-17}	–	1.2020	1.0062
3400	Tibiae cortical	73.04	9.9339×10^{-18}	6.78	–	1.1575×10^{-17}	1.1551×10^{-17}	–	0.8582	0.8600
3410	Fibulae cortical	10.25	6.1092×10^{-18}	25.75	–	9.7879×10^{-18}	7.6756×10^{-18}	–	0.6242	0.7959
3420	Patellae cortical	1.19	1.3986×10^{-17}	47.98	–	1.8916×10^{-17}	8.9505×10^{-18}	–	0.7394	1.5625
3500	Tibiae spongiosa	23.03	1.6440×10^{-17}	14.97	–	1.6394×10^{-17}	1.7362×10^{-17}	–	1.0028	0.9469
3510	Fibulae spongiosa	2.79	8.0858×10^{-18}	36.95	–	1.0169×10^{-17}	2.3358×10^{-17}	–	0.7951	0.3462
3520	Patellae spongiosa	5.57	2.6283×10^{-17}	24.91	–	2.9092×10^{-17}	2.1767×10^{-17}	–	0.9034	1.2075
3600	Tibiae medullary cavity	8.26	9.7276×10^{-18}	17.91	–	1.1586×10^{-17}	8.1130×10^{-18}	–	0.8396	1.1990
3610	Fibulae medullary cavity	1.87	7.2802×10^{-18}	62.80	–	1.7119×10^{-17}	1.3990×10^{-17}	–	0.4253	0.5204
3700	Ankles and foot cortical	33.25	1.3916×10^{-17}	12.81	–	1.4232×10^{-17}	1.3670×10^{-17}	–	0.9778	1.0180
3800	Ankles and foot spongiosa	70.79	9.3330×10^{-18}	5.91	–	9.0711×10^{-18}	1.0585×10^{-17}	–	1.0289	0.8817
3900	Mandible cortical	19.29	4.5670×10^{-16}	2.21	–	4.7243×10^{-16}	4.7182×10^{-16}	–	0.9667	0.9680
4000	Mandible spongiosa	29.49	4.7982×10^{-16}	2.10	–	4.8192×10^{-16}	5.0400×10^{-16}	–	0.9956	0.9520
4100	Pelvis cortical	58.23	7.3827×10^{-16}	1.15	–	7.3695×10^{-16}	7.3929×10^{-16}	–	1.0018	0.9986
4200	Pelvis spongiosa	99.47	7.3182×10^{-16}	0.79	–	7.3484×10^{-16}	7.2398×10^{-16}	–	0.9959	1.0108
4300	Ribs cortical	38.51	4.0294×10^{-15}	0.46	–	3.9553×10^{-15}	4.0183×10^{-15}	–	1.0188	1.0028
4400	Ribs spongiosa	79.11	4.1146×10^{-15}	0.47	–	4.1278×10^{-15}	4.1269×10^{-15}	–	0.9968	0.9970

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
4500	Scapulae cortical	29.30	1.4459×10^{-15}	1.31	–	1.3945×10^{-15}	1.4435×10^{-15}	–	1.0369	1.0017
4600	Scapulae spongiosa	42.72	1.3354×10^{-15}	1.42	–	1.3380×10^{-15}	1.3542×10^{-15}	–	0.9981	0.9862
4700	Cervical spine cortical	6.99	5.7896×10^{-16}	4.22	–	5.8387×10^{-16}	5.5920×10^{-16}	–	0.9916	1.0353
4800	Cervical spine spongiosa	14.08	6.0694×10^{-16}	2.89	–	5.6765×10^{-16}	5.8282×10^{-16}	–	1.0692	1.0414
4900	Thoracic spine cortical	33.10	4.2244×10^{-15}	0.77	–	4.1677×10^{-15}	4.1682×10^{-15}	–	1.0136	1.0135
5000	Thoracic spine spongiosa	74.60	4.8863×10^{-15}	0.58	–	4.8801×10^{-15}	4.8844×10^{-15}	–	1.0013	1.0004
5100	Lumbar spine cortical	16.80	3.7064×10^{-15}	0.88	–	3.7641×10^{-15}	3.7330×10^{-15}	–	0.9847	0.9929
5200	Lumbar spine spongiosa	58.99	4.0138×10^{-15}	0.51	–	4.0553×10^{-15}	4.0562×10^{-15}	–	0.9898	0.9896
5300	Sacrum cortical	11.70	9.7734×10^{-16}	1.43	–	1.0120×10^{-15}	9.9903×10^{-16}	–	0.9657	0.9783
5400	Sacrum spongiosa	34.52	1.1156×10^{-15}	1.04	–	1.1330×10^{-15}	1.1255×10^{-15}	–	0.9846	0.9912
5500	Sternum cortical	2.47	3.6806×10^{-15}	2.87	–	3.6556×10^{-15}	3.9445×10^{-15}	–	1.0068	0.9331
5600	Sternum spongiosa	8.50	2.7506×10^{-15}	1.05	–	2.8030×10^{-15}	2.8322×10^{-15}	–	0.9813	0.9712
5800	Cartilage trunk	55.14	5.6723×10^{-15}	0.58	–	5.7008×10^{-15}	5.7246×10^{-15}	–	0.9950	0.9909
6100	Brain	1374.20	1.6423×10^{-16}	0.78	–	1.6467×10^{-16}	1.6534×10^{-16}	–	0.9973	0.9933
6200	Breast left adipose tissue	1.18	2.4561×10^{-15}	3.97	–	2.5590×10^{-15}	2.4067×10^{-15}	–	0.9598	1.0205
6300	Breast left glandular tissue	0.785	2.4790×10^{-15}	5.34	–	2.3234×10^{-15}	2.3882×10^{-15}	–	1.0669	1.0380
6400	Breast right adipose tissue	1.18	5.2954×10^{-15}	3.85	–	4.9321×10^{-15}	5.2956×10^{-15}	–	1.0737	1.0000
6500	Breast right glandular tissue	0.785	5.0671×10^{-15}	5.24	–	4.8269×10^{-15}	4.9619×10^{-15}	–	1.0498	1.0212
6600	Eye lens sensitive left	0.0202	1.4795×10^{-16}	80.19	–	–	2.8752×10^{-16}	–	–	0.5146
6601	Eye lens insensitive left	0.114	2.1472×10^{-16}	53.97	–	1.1989×10^{-16}	2.4340×10^{-16}	–	1.7910	0.8822
6700	Cornea left	0.87	2.0973×10^{-16}	14.11	–	1.7162×10^{-16}	1.8017×10^{-16}	–	1.2221	1.1641
6701	Aqueous left	0.294	1.5736×10^{-16}	31.81	–	2.2571×10^{-16}	2.2221×10^{-16}	–	0.6972	0.7082
6702	Vitreous left	4.40	1.8840×10^{-16}	9.98	–	2.0390×10^{-16}	2.2568×10^{-16}	–	0.9240	0.8348
6800	Eye lens sensitive right	0.0202	0.0000×10^{0}	0.00	–	2.7576×10^{-16}	6.1961×10^{-16}	–	–	–
6801	Eye lens insensitive right	0.114	2.0632×10^{-16}	36.91	–	2.4554×10^{-16}	4.5915×10^{-16}	–	0.8403	0.4493
6900	Cornea right	0.87	2.0741×10^{-16}	16.14	–	2.0815×10^{-16}	2.3458×10^{-16}	–	0.9964	0.8842
6901	Aqueous right	0.294	2.6429×10^{-16}	18.34	–	2.2899×10^{-16}	2.9527×10^{-16}	–	1.1541	0.8951
6902	Vitreous right	4.40	2.3905×10^{-16}	5.35	–	2.6109×10^{-16}	2.5421×10^{-16}	–	0.9156	0.9403
7000	Gall bladder wall	2.70	1.8634×10^{-14}	1.22	–	1.8372×10^{-14}	1.8364×10^{-14}	–	1.0143	1.0147
7100	Gall bladder contents	15.00	1.7552×10^{-14}	0.51	–	1.7547×10^{-14}	1.7642×10^{-14}	–	1.0003	0.9949
7200	Stomach wall(0-60)	0.745	5.9554×10^{-15}	1.62	–	5.9816×10^{-15}	5.9546×10^{-15}	–	0.9956	1.0001
7201	Stomach wall(60-100)	0.489	5.9308×10^{-15}	1.42	–	6.0143×10^{-15}	5.9725×10^{-15}	–	0.9861	0.9930
7202	Stomach wall(100-300)	2.47	5.8508×10^{-15}	1.59	–	5.9267×10^{-15}	5.8564×10^{-15}	–	0.9872	0.9990
7203	Stomach wall(300-surface)	57.96	6.0089×10^{-15}	0.46	–	5.9428×10^{-15}	6.0537×10^{-15}	–	1.0111	0.9926
7300	Stomach contents	83.00	5.5351×10^{-15}	0.47	–	5.5082×10^{-15}	5.5495×10^{-15}	–	1.0049	0.9974
7400	Small intestine wall(0-130)	6.59	2.4923×10^{-15}	0.97	–	2.4736×10^{-15}	2.4341×10^{-15}	–	1.0075	1.0239
7401	Small intestine wall(130-150)	1.03	2.5503×10^{-15}	2.20	–	2.5419×10^{-15}	2.5109×10^{-15}	–	1.0033	1.0157
7402	Small intestine wall(150-200)	2.60	2.5553×10^{-15}	1.31	–	2.5138×10^{-15}	2.4643×10^{-15}	–	1.0165	1.0369

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
7403	Small intestine wall(200-surface)	266.74	2.5179×10^{-15}	0.20	–	2.5133×10^{-15}	2.5193×10^{-15}	–	1.0019	0.9995
7500	Small intestine contents(-400-0)	19.02	2.4594×10^{-15}	0.64	–	2.4569×10^{-15}	2.4487×10^{-15}	–	1.0010	1.0044
7501	Small intestine contents(centre–400)	97.98	2.3820×10^{-15}	0.54	–	2.3730×10^{-15}	2.3683×10^{-15}	–	1.0038	1.0058
7600	Ascending colon wall(0-280)	1.91	1.6141×10^{-15}	4.37	–	1.6562×10^{-15}	1.7117×10^{-15}	–	0.9746	0.9430
7601	Ascending colon wall(280-300)	0.139	1.7654×10^{-15}	8.13	–	1.6820×10^{-15}	1.7873×10^{-15}	–	1.0496	0.9877
7602	Ascending colon wall(300-surface)	26.67	1.6927×10^{-15}	1.18	–	1.7225×10^{-15}	1.6974×10^{-15}	–	0.9827	0.9972
7700	Ascending colon content	31.35	1.5350×10^{-15}	1.27	–	1.5729×10^{-15}	1.5628×10^{-15}	–	0.9758	0.9822
7800	Transverse colon wall right(0-280)	1.58	7.2990×10^{-15}	0.93	–	6.8988×10^{-15}	7.1187×10^{-15}	–	1.0580	1.0253
7801	Transverse colon wall right(280-300)	0.116	7.3585×10^{-15}	2.54	–	7.0145×10^{-15}	7.1100×10^{-15}	–	1.0491	1.0350
7802	Transverse colon wall right(300-surface)	31.21	7.5065×10^{-15}	0.75	–	7.3834×10^{-15}	7.4226×10^{-15}	–	1.0167	1.0113
7900	Transverse colon contents right	18.65	7.1936×10^{-15}	0.86	–	7.2283×10^{-15}	7.2598×10^{-15}	–	0.9952	0.9909
8000	Transverse colon wall left(0-280)	1.10	3.0953×10^{-15}	2.27	–	3.0772×10^{-15}	2.9896×10^{-15}	–	1.0059	1.0354
8001	Transverse colon wall left(280-300)	0.0808	2.8963×10^{-15}	3.54	–	3.3552×10^{-15}	3.1255×10^{-15}	–	0.8632	0.9267
8002	Transverse colon wall left(300-surface)	27.09	3.0645×10^{-15}	1.22	–	3.0156×10^{-15}	2.9937×10^{-15}	–	1.0162	1.0236
8100	Transverse colon content left	10.92	3.0310×10^{-15}	0.77	–	3.0828×10^{-15}	3.0515×10^{-15}	–	0.9832	0.9933
8200	Descending colon wall(0-280)	1.50	1.0203×10^{-15}	2.65	–	1.0684×10^{-15}	1.1198×10^{-15}	–	0.9550	0.9111
8201	Descending colon wall(280-300)	0.11	1.2155×10^{-15}	4.71	–	1.1787×10^{-15}	9.9796×10^{-16}	–	1.0313	1.2180
8202	Descending colon wall(300-surface)	31.74	1.0367×10^{-15}	1.70	–	1.0692×10^{-15}	1.0554×10^{-15}	–	0.9695	0.9823
8300	Descending colon content	14.08	1.0548×10^{-15}	2.88	–	1.1252×10^{-15}	1.0326×10^{-15}	–	0.9374	1.0215
8400	Sigmoid colon wall(0-280)	1.81	7.0578×10^{-16}	3.79	–	6.7667×10^{-16}	7.3820×10^{-16}	–	1.0430	0.9561
8401	Sigmoid colon wall(280-300)	0.133	8.5255×10^{-16}	7.97	–	6.1713×10^{-16}	7.1919×10^{-16}	–	1.3815	1.1854
8402	Sigmoid colon wall(300-surface)	22.78	7.5648×10^{-16}	1.58	–	7.5180×10^{-16}	7.7592×10^{-16}	–	1.0062	0.9749
8500	Sigmoid colon contents	17.99	7.0175×10^{-16}	2.17	–	7.2762×10^{-16}	6.9913×10^{-16}	–	0.9645	1.0038
8600	Rectum wall(0-280)	0.717	3.9987×10^{-16}	8.71	–	4.0057×10^{-16}	3.4737×10^{-16}	–	0.9982	1.1511
8601	Rectum wall(280-300)	0.0527	3.0229×10^{-16}	20.31	–	2.6993×10^{-16}	3.4657×10^{-16}	–	1.1199	0.8722
8602	Rectum wall(300-surface)	2.18	3.4136×10^{-16}	8.51	–	3.7936×10^{-16}	3.5642×10^{-16}	–	0.8998	0.9578
8603	Rectum contents	7.01	3.6534×10^{-16}	3.33	–	3.5804×10^{-16}	4.0107×10^{-16}	–	1.0204	0.9109
8700	Heart wall	97.70	6.7001×10^{-15}	0.31	–	6.6554×10^{-15}	6.7437×10^{-15}	–	1.0067	0.9935
8800	Blood in heart chamber	135.00	5.9741×10^{-15}	0.46	–	5.9642×10^{-15}	5.9937×10^{-15}	–	1.0017	0.9967
8900	Kidney left cortex	49.84	3.3245×10^{-15}	0.50	–	3.2958×10^{-15}	3.3489×10^{-15}	–	1.0087	0.9927
9000	Kidney left medulla	17.80	3.5056×10^{-15}	1.36	–	3.5498×10^{-15}	3.5675×10^{-15}	–	0.9876	0.9826
9100	Kidney left pelvis	3.56	3.5193×10^{-15}	2.02	–	3.7737×10^{-15}	3.7041×10^{-15}	–	0.9326	0.9501
9200	Kidney right cortex	49.85	8.4600×10^{-15}	0.31	–	8.5345×10^{-15}	8.5121×10^{-15}	–	0.9913	0.9939
9300	Kidney right medulla	17.78	8.1360×10^{-15}	0.70	–	8.0957×10^{-15}	8.0382×10^{-15}	–	1.0050	1.0122
9400	Kidney right pelvis	3.56	7.6965×10^{-15}	1.69	–	7.7766×10^{-15}	7.6080×10^{-15}	–	0.9897	1.0116
9500	Liver	724.41	2.5669×10^{-14}	0.07	–	2.5636×10^{-14}	2.5732×10^{-14}	–	1.0013	0.9975

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
9700	Lung(AI) left	139.44	1.9722×10^{-15}	0.36	–	1.9823×10^{-15}	1.9801×10^{-15}	–	0.9949	0.9960
9900	Lung(AI) right	160.56	4.9153×10^{-15}	0.28	–	4.9283×10^{-15}	4.9487×10^{-15}	–	0.9974	0.9933
10000	Lymphatic nodes ET	4.31	3.7923×10^{-16}	7.27	–	3.9131×10^{-16}	4.0165×10^{-16}	–	0.9691	0.9442
10100	Lymphatic nodes thoracic	4.31	6.9700×10^{-15}	1.67	–	7.1025×10^{-15}	7.0485×10^{-15}	–	0.9813	0.9889
10200	Lymphatic nodes head	1.49	7.1202×10^{-16}	6.25	–	6.7363×10^{-16}	7.0080×10^{-16}	–	1.0570	1.0160
10300	Lymphatic nodes trunk	35.19	2.0094×10^{-15}	0.59	–	2.0535×10^{-15}	2.0220×10^{-15}	–	0.9785	0.9938
10400	Lymphatic nodes arms	2.98	1.4118×10^{-15}	4.36	–	1.5168×10^{-15}	1.5820×10^{-15}	–	0.9307	0.8924
10500	Lymphatic nodes legs	2.98	3.6575×10^{-17}	22.85	–	1.6930×10^{-17}	3.0638×10^{-17}	–	2.1604	1.1938
10600	Muscle head	407.78	4.0139×10^{-16}	0.70	–	4.1138×10^{-16}	4.1230×10^{-16}	–	0.9757*	0.9735*
10700	Muscle trunk	2521.09	2.1480×10^{-15}	0.08	–	2.1991×10^{-15}	2.2011×10^{-15}	–	0.9767*	0.9759*
10800	Muscle arms	464.26	1.1859×10^{-15}	0.44	–	1.2016×10^{-15}	1.2101×10^{-15}	–	0.9869*	0.9800*
10900	Muscle legs	2334.63	1.0348×10^{-16}	0.48	–	1.0808×10^{-16}	1.0687×10^{-16}	–	0.9574*	0.9682*
11000	Oesophagus wall(0-190)	0.918	4.6974×10^{-15}	2.24	–	4.4589×10^{-15}	4.6404×10^{-15}	–	1.0535	1.0123
11001	Oesophagus wall(190-200)	0.0496	4.8549×10^{-15}	4.92	–	5.2745×10^{-15}	4.7085×10^{-15}	–	0.9205	1.0311
11002	Oesophagus wall(200-surface)	11.37	4.5642×10^{-15}	1.26	–	4.5683×10^{-15}	4.7140×10^{-15}	–	0.9991	0.9682
11003	Oesophagus contents	7.14	4.7997×10^{-15}	1.34	–	4.7862×10^{-15}	5.0255×10^{-15}	–	1.0028	0.9551
11300	Pancreas	41.89	4.5116×10^{-15}	0.71	–	4.5057×10^{-15}	4.5747×10^{-15}	–	1.0013	0.9862
11400	Pituitary gland	0.26	1.7136×10^{-16}	27.46	–	1.1279×10^{-16}	2.3546×10^{-16}	–	1.5193	0.7278
11500	Prostate	1.25	3.3380×10^{-16}	17.24	–	3.9716×10^{-16}	3.8794×10^{-16}	–	0.8405	0.8604
11600	RST head	468.28	3.1687×10^{-16}	0.71	–	3.1509×10^{-16}	3.1883×10^{-16}	–	1.0056	0.9938
11700	RST trunk	3350.84	3.4089×10^{-15}	0.07	–	3.4455×10^{-15}	3.4689×10^{-15}	–	0.9894*	0.9827*
11800	RST arms	613.67	1.3429×10^{-15}	0.21	–	1.3840×10^{-15}	1.3899×10^{-15}	–	0.9703*	0.9662*
11900	RST legs	1817.51	7.6749×10^{-17}	0.68	–	8.0104×10^{-17}	8.0890×10^{-17}	–	0.9581*	0.9488*
12000	Salivary glands left	17.67	3.7245×10^{-16}	3.43	–	4.2499×10^{-16}	3.9931×10^{-16}	–	0.8764	0.9327
12100	Salivary glandss right	17.67	5.1015×10^{-16}	3.20	–	5.1030×10^{-16}	5.2917×10^{-16}	–	0.9997	0.9641
12200	Skin head insensitive	75.38	2.4168×10^{-16}	1.15	–	2.3333×10^{-16}	2.3948×10^{-16}	–	1.0357*	1.0092*
12201	Skin head sensitive(40-100)	5.81	2.1772×10^{-16}	1.80	–	2.0913×10^{-16}	1.9252×10^{-16}	–	1.0411*	1.1309*
12300	Skin trunk insensitive	213.00	1.8977×10^{-15}	0.26	–	2.0798×10^{-15}	2.0816×10^{-15}	–	0.9125*	0.9117*
12301	Skin trunk sensitive(40-100)	19.59	1.6098×10^{-15}	0.63	–	1.8077×10^{-15}	1.8253×10^{-15}	–	0.8905*	0.8819*
12400	Skin arms insensitive	66.16	8.7320×10^{-16}	0.63	–	1.0376×10^{-15}	1.0424×10^{-15}	–	0.8415*	0.8377*
12401	Skin arms sensitive(40-100)	5.09	7.7708×10^{-16}	1.72	–	9.4192×10^{-16}	9.3543×10^{-16}	–	0.8250*	0.8307*
12500	Skin legs insensitive	194.72	6.4904×10^{-17}	2.30	–	6.5083×10^{-17}	6.8037×10^{-17}	–	0.9972*	0.9539*
12501	Skin legs sensitive(40-100)	16.66	5.0184×10^{-17}	3.85	–	5.7736×10^{-17}	5.8535×10^{-17}	–	0.8692*	0.8573*
12600	Spinal cord	23.82	3.1931×10^{-15}	0.93	–	3.2156×10^{-15}	3.2668×10^{-15}	–	0.9930	0.9774
12700	Spleen	71.31	2.4998×10^{-15}	0.44	–	2.5153×10^{-15}	2.5490×10^{-15}	–	0.9939	0.9807
12802	Erupted upper front buccal deciduous ena	0.193	4.4360×10^{-16}	33.06	–	3.5460×10^{-16}	5.2001×10^{-16}	–	1.2510	0.8531

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
12803	Erupted upper front lingual deciduous en	0.165	3.2417×10^{-16}	25.48	–	3.6081×10^{-16}	4.8594×10^{-16}	–	0.8985	0.6671
12810	Erupted upper left buccal deciduous enam	0.149	3.4382×10^{-16}	29.41	–	2.1927×10^{-16}	2.6371×10^{-16}	–	1.5680	1.3038
12811	Erupted upper left lingual deciduous ena	0.159	6.1639×10^{-16}	27.50	–	4.6125×10^{-16}	4.2432×10^{-16}	–	1.3364	1.4527
12814	Erupted upper right buccal deciduous ena	0.149	5.9471×10^{-16}	40.35	–	4.2330×10^{-16}	4.5305×10^{-16}	–	1.4049	1.3127
12815	Erupted upper right lingual deciduous en	0.158	4.1480×10^{-16}	39.69	–	6.0072×10^{-16}	5.9225×10^{-16}	–	0.6905	0.7004
12818	Erupted lower front buccal deciduous ena	0.135	6.1807×10^{-16}	26.41	–	2.6941×10^{-16}	4.7836×10^{-16}	–	2.2942	1.2921
12819	Erupted lower front lingual deciduous en	0.138	3.6085×10^{-16}	33.95	–	3.4628×10^{-16}	3.4276×10^{-16}	–	1.0421	1.0528
12826	Erupted lower left buccal deciduous enam	0.175	4.4218×10^{-16}	20.42	–	3.5065×10^{-16}	4.8806×10^{-16}	–	1.2610	0.9060
12827	Erupted lower left lingual deciduous ena	0.234	5.5753×10^{-16}	18.40	–	3.1922×10^{-16}	6.2617×10^{-16}	–	1.7465	0.8904
12830	Erupted lower right buccal deciduous ena	0.197	4.7157×10^{-16}	23.07	–	3.3410×10^{-16}	2.5967×10^{-16}	–	1.4115	1.8160
12831	Erupted lower right lingual deciduous en	0.213	3.9075×10^{-16}	24.32	–	4.5668×10^{-16}	5.2335×10^{-16}	–	0.8556	0.7466
12833	Erupted upper front deciduous dentin	0.821	3.1549×10^{-16}	14.68	–	3.9623×10^{-16}	5.0099×10^{-16}	–	0.7962	0.6297
12837	Erupted upper left deciduous dentin	0.708	3.9200×10^{-16}	12.71	–	2.6992×10^{-16}	3.4311×10^{-16}	–	1.4523	1.1425
12839	Erupted upper right deciduous dentin	0.708	4.1405×10^{-16}	14.14	–	3.9562×10^{-16}	3.7106×10^{-16}	–	1.0466	1.1159
12841	Erupted lower front deciduous dentin	0.617	4.0829×10^{-16}	15.77	–	4.5041×10^{-16}	4.6174×10^{-16}	–	0.9065	0.8843
12845	Erupted lower left deciduous dentin	0.948	4.7975×10^{-16}	10.84	–	4.8962×10^{-16}	4.3491×10^{-16}	–	0.9799	1.1031
12847	Erupted lower right deciduous dentin	0.948	5.9948×10^{-16}	13.21	–	3.9418×10^{-16}	6.1114×10^{-16}	–	1.5208	0.9809
12848	Unerupted permanent enamel	1.63	3.8877×10^{-16}	8.92	–	3.0291×10^{-16}	3.9419×10^{-16}	–	1.2834	0.9862
12850	Unerupted permanent dentin	5.40	4.5536×10^{-16}	5.50	–	4.1661×10^{-16}	4.2260×10^{-16}	–	1.0930	1.0775
12852	Permanent cementum	0.284	4.6886×10^{-16}	13.22	–	4.0110×10^{-16}	3.6314×10^{-16}	–	1.1689	1.2911
12853	Deciduous cementum	0.242	4.2403×10^{-16}	10.16	–	3.9584×10^{-16}	4.4818×10^{-16}	–	1.0712	0.9461
12854	Permanent pulp	0.163	6.8748×10^{-16}	26.14	–	5.6198×10^{-16}	3.6377×10^{-16}	–	1.2233	1.8899
12855	Deciduous pulp	0.661	5.9678×10^{-16}	9.92	–	4.7291×10^{-16}	4.7514×10^{-16}	–	1.2619	1.2560
12856	Teeth retention region	0.0211	5.8459×10^{-16}	27.51	–	5.2321×10^{-16}	5.7291×10^{-16}	–	1.1173	1.0204
12900	Testis left	0.907	2.9485×10^{-16}	15.20	–	2.2486×10^{-16}	3.3148×10^{-16}	–	1.3113	0.8895
13000	Testis right	0.907	3.2380×10^{-16}	14.79	–	2.5210×10^{-16}	3.2742×10^{-16}	–	1.2844	0.9889
13100	Thymus	31.20	1.9404×10^{-15}	1.38	–	1.9464×10^{-15}	1.9594×10^{-15}	–	0.9969	0.9903
13200	Thyroid	3.87	9.7200×10^{-16}	2.67	–	9.7955×10^{-16}	9.4858×10^{-16}	–	0.9923	1.0247

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
13300	Tongue upper(food)	9.15	4.2002×10^{-16}	3.14	–	4.4647×10^{-16}	4.2913×10^{-16}	–	0.9408	0.9788
13301	Tongue lower	10.58	4.9031×10^{-16}	2.86	–	5.2053×10^{-16}	4.9033×10^{-16}	–	0.9419	1.0000
13400	Tonsils	2.08	4.3957×10^{-16}	6.17	–	5.0302×10^{-16}	5.0877×10^{-16}	–	0.8739	0.8640
13500	Ureter left	2.18	2.0091×10^{-15}	3.19	–	2.1420×10^{-15}	1.9789×10^{-15}	–	0.9379	1.0153
13600	Ureter right	2.18	3.1470×10^{-15}	3.37	–	3.2855×10^{-15}	3.0183×10^{-15}	–	0.9579	1.0426
13700	Urinary bladder wall insensitive	15.42	6.0769×10^{-16}	1.66	–	6.1547×10^{-16}	6.0271×10^{-16}	–	0.9874	1.0083
13701	Urinary bladder wall sensitive(86-193)	0.882	7.1752×10^{-16}	8.22	–	6.0613×10^{-16}	6.3613×10^{-16}	–	1.1838	1.1279
13800	Urinary bladder content	64.70	5.9453×10^{-16}	1.38	–	5.9405×10^{-16}	5.9618×10^{-16}	–	1.0008	0.9972
14000	Air inside body	0.024	1.7696×10^{-15}	8.02	–	2.3732×10^{-15}	1.8344×10^{-15}	–	0.7456	0.9647

* For tissues bookkept as head, trunk, arm and leg pieces (muscle, residual soft tissue, skin, large vessels), the volumes tabulated in the distributed cell tables partition the tissue at different boundaries from the transported mesh, and the reference output is normalised by the tabulated volume. The marked ratio therefore compares two different partitions of the same tissue, and is left out of the summary statistics of Table S2.

Table S16. 05M_External. Absorbed dose per primary history, Gy. σ is the FLUKA statistical uncertainty. Ratios are FLUKA / reference.

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
100	Adrenal left	2.97	8.5439×10^{-18}	31.92	–	3.3557×10^{-17}	3.0078×10^{-17}	–	0.2546	0.2841
200	Adrenal right	2.97	1.8865×10^{-17}	44.17	–	2.3778×10^{-17}	3.3259×10^{-17}	–	0.7934	0.5672
300	ET1(0-8)	0.00268	1.5683×10^{-17}	100.00	–	4.1365×10^{-17}	3.5617×10^{-16}	–	0.3791	0.0440
301	ET1(8-40)	0.0109	5.9462×10^{-17}	100.00	–	1.9198×10^{-17}	8.4402×10^{-17}	–	3.0973	0.7045
302	ET1(40-50)	0.00345	2.0646×10^{-17}	100.00	–	1.4382×10^{-17}	7.4090×10^{-17}	–	1.4356	0.2787
303	ET1(50-Surface)	0.316	7.8882×10^{-17}	52.22	–	4.9638×10^{-17}	4.1388×10^{-17}	–	1.5891	1.9059
400	ET2(-15-0)	0.127	3.5388×10^{-17}	47.09	–	3.6877×10^{-17}	5.5399×10^{-17}	–	0.9596	0.6388
401	ET2(0-40)	0.364	1.9901×10^{-17}	34.50	–	4.6217×10^{-17}	4.4861×10^{-17}	–	0.4306	0.4436
402	ET2(40-50)	0.0917	3.9204×10^{-17}	64.74	–	4.4962×10^{-17}	9.1202×10^{-17}	–	0.8719	0.4299
403	ET2(50-55)	0.046	2.0749×10^{-17}	30.01	–	3.7934×10^{-17}	7.8558×10^{-17}	–	0.5470	0.2641
404	ET2(55-65)	0.0921	5.4814×10^{-17}	43.53	–	5.4153×10^{-17}	4.8133×10^{-17}	–	1.0122	1.1388
405	ET2(65-Surface)	16.63	2.9225×10^{-17}	14.96	–	3.1230×10^{-17}	4.3014×10^{-17}	–	0.9358	0.6794
500	Oral mucosa tongue	0.0311	9.5726×10^{-18}	33.72	–	1.7195×10^{-17}	3.0882×10^{-17}	–	0.5567	0.3100
501	Oral mucosa moth floor	0.019	2.1194×10^{-17}	50.96	–	4.6838×10^{-17}	1.0817×10^{-16}	–	0.4525	0.1959
600	Oral mucosa lips and cheeks	0.0133	0.0000×10^{0}	0.00	–	3.7259×10^{-17}	2.2916×10^{-17}	–	–	–
700	Trachea	2.60	3.0168×10^{-17}	30.31	–	3.4522×10^{-17}	5.3209×10^{-17}	–	0.8739	0.5670
800	BB(-11–6)	0.0022	0.0000×10^{0}	0.00	–	1.5158×10^{-17}	3.1076×10^{-17}	–	–	–
801	BB(-6-0)	0.00282	0.0000×10^{0}	0.00	–	6.0739×10^{-17}	5.9593×10^{-16}	–	–	–
802	BB(0-10)	0.00473	0.0000×10^{0}	0.00	–	9.7965×10^{-18}	1.0805×10^{-16}	–	–	–
803	BB(10-35)	0.012	0.0000×10^{0}	0.00	–	2.9717×10^{-17}	6.1223×10^{-17}	–	–	–
804	BB(35-40)	0.00243	0.0000×10^{0}	0.00	–	1.4207×10^{-17}	2.3018×10^{-17}	–	–	–
805	BB(40-50)	0.00489	0.0000×10^{0}	0.00	–	1.3878×10^{-17}	2.2780×10^{-17}	–	–	–
806	BB(50-60)	0.00493	5.9546×10^{-17}	100.00	–	2.2950×10^{-17}	2.8251×10^{-17}	–	2.5946	2.1078
807	BB(60-70)	0.00497	2.0613×10^{-17}	100.00	–	2.3510×10^{-17}	2.8161×10^{-17}	–	0.8768	0.7320
808	BB(70-surface)	3.38	2.3182×10^{-17}	31.07	–	3.8968×10^{-17}	4.1953×10^{-17}	–	0.5949	0.5526
900	Blood in large arteries head	0.271	5.8382×10^{-17}	73.49	–	4.4992×10^{-17}	1.6013×10^{-17}	–	1.2976*	3.6459*
910	Blood in large veins head	1.77	3.8735×10^{-17}	34.63	–	4.3345×10^{-17}	3.1596×10^{-17}	–	0.8936*	1.2259*
1000	Blood in large arteries trunk	38.26	3.3695×10^{-17}	4.38	–	3.5646×10^{-17}	3.4021×10^{-17}	–	0.9453*	0.9904*
1010	Blood in large veins trunk	107.57	3.5716×10^{-17}	5.09	–	3.5555×10^{-17}	3.3372×10^{-17}	–	1.0045*	1.0703*
1100	Blood in large arteries arms	6.61	2.8542×10^{-17}	21.64	–	3.4925×10^{-17}	3.2948×10^{-17}	–	0.8173*	0.8663*
1110	Blood in large veins arms	42.98	3.6889×10^{-17}	8.32	–	3.3822×10^{-17}	3.8485×10^{-17}	–	1.0907*	0.9585*
1200	Blood in large arteries legs	44.81	3.5964×10^{-17}	8.07	–	3.3678×10^{-17}	3.1651×10^{-17}	–	1.0679*	1.1363*
1210	Blood in large veins legs	117.52	3.1947×10^{-17}	3.91	–	3.2013×10^{-17}	3.2510×10^{-17}	–	0.9979*	0.9827*
1300	Humeri upper cortical	28.83	3.3849×10^{-17}	9.52	–	2.8954×10^{-17}	3.0252×10^{-17}	–	1.1690	1.1189
1400	Humeri upper spogiosa	20.16	3.4770×10^{-17}	11.08	–	3.2804×10^{-17}	2.8856×10^{-17}	–	1.0600	1.2050

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
1500	Humeri upper medullary cavity	3.76	3.5042×10^{-17}	18.59	–	1.7388×10^{-17}	4.8104×10^{-17}	–	2.0153	0.7285
1600	Humeri lower cortical	22.27	3.0007×10^{-17}	8.68	–	3.3162×10^{-17}	3.5794×10^{-17}	–	0.9049	0.8383
1700	Humeri lower spongiosa	11.03	4.1535×10^{-17}	12.93	–	3.5260×10^{-17}	3.3282×10^{-17}	–	1.1780	1.2480
1800	Humeri lower medullary cavity	3.56	4.9484×10^{-17}	32.33	–	3.3045×10^{-17}	2.7815×10^{-17}	–	1.4975	1.7790
1900	Radii cortical	14.85	3.4288×10^{-17}	17.94	–	3.7955×10^{-17}	3.3517×10^{-17}	–	0.9034	1.0230
1910	Ulnae cortical	18.78	3.7668×10^{-17}	11.16	–	4.1612×10^{-17}	2.9612×10^{-17}	–	0.9052	1.2721
2000	Radii spongiosa	3.31	3.7627×10^{-17}	21.40	–	4.4328×10^{-17}	5.4612×10^{-17}	–	0.8488	0.6890
2010	Ulnae spongiosa	5.49	3.5556×10^{-17}	18.82	–	3.0809×10^{-17}	2.8528×10^{-17}	–	1.1541	1.2464
2100	Radii medullary cavity	2.30	5.0638×10^{-17}	18.63	–	6.2289×10^{-17}	5.6503×10^{-17}	–	0.8129	0.8962
2110	Ulnae medullary cavity	2.63	5.5328×10^{-17}	21.92	–	4.7271×10^{-17}	3.7628×10^{-17}	–	1.1704	1.4704
2200	Wrists and hand bones cortical	9.33	3.1471×10^{-17}	11.70	–	2.8489×10^{-17}	2.6781×10^{-17}	–	1.1046	1.1751
2300	Wrists and hand bones spongiosa	16.48	3.3277×10^{-17}	9.10	–	3.8970×10^{-17}	3.0385×10^{-17}	–	0.8539	1.0952
2400	Clavicles cortical	6.18	3.4491×10^{-17}	21.05	–	4.2126×10^{-17}	3.7565×10^{-17}	–	0.8188	0.9182
2500	Clavicles spongiosa	7.96	4.0558×10^{-17}	18.47	–	3.4073×10^{-17}	3.7184×10^{-17}	–	1.1903	1.0907
2600	Cranium cortical	229.09	2.3364×10^{-17}	4.55	–	2.4322×10^{-17}	2.5373×10^{-17}	–	0.9606	0.9208
2700	Cranium spongiosa	520.37	2.3200×10^{-17}	2.66	–	2.3535×10^{-17}	2.2775×10^{-17}	–	0.9858	1.0186
2800	Femora upper cortical	49.47	3.2150×10^{-17}	5.55	–	3.0414×10^{-17}	3.2632×10^{-17}	–	1.0571	0.9852
2900	Femora upper spongiosa	26.10	2.9291×10^{-17}	7.58	–	3.2227×10^{-17}	2.8496×10^{-17}	–	0.9089	1.0279
3000	Femora upper medullary cavity	8.22	2.9582×10^{-17}	14.62	–	4.0093×10^{-17}	3.5006×10^{-17}	–	0.7378	0.8451
3100	Femora lower cortical	48.06	2.7914×10^{-17}	6.47	–	2.6356×10^{-17}	2.7004×10^{-17}	–	1.0591	1.0337
3200	Femora lower spongiosa	28.74	2.8655×10^{-17}	7.15	–	2.9304×10^{-17}	2.6691×10^{-17}	–	0.9778	1.0736
3300	Femora lower medullary cavity	5.79	3.4650×10^{-17}	16.52	–	2.9061×10^{-17}	3.6761×10^{-17}	–	1.1923	0.9426
3400	Tibiae cortical	73.04	2.9915×10^{-17}	4.71	–	2.7314×10^{-17}	3.0721×10^{-17}	–	1.0953	0.9738
3410	Fibulae cortical	10.25	2.4648×10^{-17}	16.66	–	2.2375×10^{-17}	2.1150×10^{-17}	–	1.1016	1.1654
3420	Patellae cortical	1.19	5.5416×10^{-17}	36.34	–	8.3934×10^{-17}	3.8966×10^{-17}	–	0.6602	1.4222
3500	Tibiae spongiosa	23.03	3.2738×10^{-17}	8.53	–	2.7468×10^{-17}	2.7064×10^{-17}	–	1.1918	1.2096
3510	Fibulae spongiosa	2.79	1.4978×10^{-17}	37.80	–	3.3880×10^{-17}	3.3090×10^{-17}	–	0.4421	0.4527
3520	Patellae spongiosa	5.57	4.0266×10^{-17}	11.89	–	4.9859×10^{-17}	2.7474×10^{-17}	–	0.8076	1.4656
3600	Tibiae medullary cavity	8.26	2.7812×10^{-17}	18.76	–	2.8016×10^{-17}	2.3037×10^{-17}	–	0.9927	1.2073
3610	Fibulae medullary cavity	1.87	2.5068×10^{-17}	41.69	–	1.4459×10^{-17}	2.5571×10^{-17}	–	1.7337	0.9803
3700	Ankles and foot cortical	33.25	2.6154×10^{-17}	8.31	–	2.8716×10^{-17}	2.6933×10^{-17}	–	0.9108	0.9711
3800	Ankles and foot spongiosa	70.79	2.3877×10^{-17}	9.81	–	3.0358×10^{-17}	2.6349×10^{-17}	–	0.7865	0.9062
3900	Mandible cortical	19.29	3.3507×10^{-17}	13.82	–	3.0958×10^{-17}	3.6559×10^{-17}	–	1.0823	0.9165
4000	Mandible spongiosa	29.49	4.4375×10^{-17}	9.59	–	3.3792×10^{-17}	3.8131×10^{-17}	–	1.3132	1.1638
4100	Pelvis cortical	58.23	3.6751×10^{-17}	5.50	–	3.2071×10^{-17}	2.7156×10^{-17}	–	1.1460	1.3533
4200	Pelvis spongiosa	99.47	3.0039×10^{-17}	3.14	–	3.1720×10^{-17}	3.2048×10^{-17}	–	0.9470	0.9373
4300	Ribs cortical	38.51	2.5793×10^{-17}	5.92	–	2.6177×10^{-17}	2.8834×10^{-17}	–	0.9853	0.8945
4400	Ribs spongiosa	79.11	2.9230×10^{-17}	5.72	–	3.2825×10^{-17}	2.9343×10^{-17}	–	0.8905	0.9961

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
4500	Scapulae cortical	29.30	2.7182×10^{-17}	7.99	–	2.5454×10^{-17}	2.6411×10^{-17}	–	1.0679	1.0292
4600	Scapulae spongiosa	42.72	2.5918×10^{-17}	4.84	–	2.7693×10^{-17}	2.4679×10^{-17}	–	0.9359	1.0502
4700	Cervical spine cortical	6.99	2.5209×10^{-17}	16.67	–	3.1166×10^{-17}	2.8784×10^{-17}	–	0.8089	0.8758
4800	Cervical spine spongiosa	14.08	2.6548×10^{-17}	18.46	–	3.0720×10^{-17}	2.5425×10^{-17}	–	0.8642	1.0442
4900	Thoracic spine cortical	33.10	2.1674×10^{-17}	7.99	–	2.3180×10^{-17}	2.2766×10^{-17}	–	0.9350	0.9520
5000	Thoracic spine spongiosa	74.60	2.3803×10^{-17}	6.22	–	2.5671×10^{-17}	2.6013×10^{-17}	–	0.9272	0.9150
5100	Lumbar spine cortical	16.80	2.5950×10^{-17}	7.64	–	2.5393×10^{-17}	2.8928×10^{-17}	–	1.0219	0.8970
5200	Lumbar spine spongiosa	58.99	3.0368×10^{-17}	5.66	–	2.8104×10^{-17}	2.4494×10^{-17}	–	1.0805	1.2398
5300	Sacrum cortical	11.70	2.5316×10^{-17}	12.76	–	3.1218×10^{-17}	2.1625×10^{-17}	–	0.8109	1.1707
5400	Sacrum spongiosa	34.52	2.8972×10^{-17}	6.68	–	2.5692×10^{-17}	2.9505×10^{-17}	–	1.1277	0.9820
5500	Sternum cortical	2.47	4.9116×10^{-17}	20.52	–	3.7211×10^{-17}	3.6485×10^{-17}	–	1.3199	1.3462
5600	Sternum spongiosa	8.50	3.9883×10^{-17}	16.02	–	3.7382×10^{-17}	4.6882×10^{-17}	–	1.0669	0.8507
5800	Cartilage trunk	55.14	3.3525×10^{-17}	8.53	–	3.1109×10^{-17}	2.8097×10^{-17}	–	1.0777	1.1932
6100	Brain	1374.20	2.2066×10^{-17}	1.67	–	2.2747×10^{-17}	2.2043×10^{-17}	–	0.9701	1.0011
6200	Breast left adipose tissue	1.18	3.1333×10^{-17}	44.83	–	3.6336×10^{-17}	3.4587×10^{-17}	–	0.8623	0.9059
6300	Breast left glandular tissue	0.785	2.2777×10^{-17}	49.16	–	4.7218×10^{-17}	3.1599×10^{-17}	–	0.4824	0.7208
6400	Breast right adipose tissue	1.18	5.2667×10^{-17}	41.66	–	4.6505×10^{-17}	6.1490×10^{-17}	–	1.1325	0.8565
6500	Breast right glandular tissue	0.785	5.5472×10^{-17}	51.71	–	4.8401×10^{-17}	4.8243×10^{-17}	–	1.1461	1.1499
6600	Eye lens sensitive left	0.0202	0.0000×10^{0}	0.00	–	–	5.0013×10^{-17}	–	–	–
6601	Eye lens insensitive left	0.114	1.7054×10^{-17}	100.00	–	1.1535×10^{-17}	7.7190×10^{-17}	–	1.4784	0.2209
6700	Cornea left	0.87	2.7850×10^{-17}	33.92	–	2.1577×10^{-17}	4.9277×10^{-17}	–	1.2907	0.5652
6701	Aqueous left	0.294	5.7031×10^{-17}	72.25	–	2.3123×10^{-17}	2.0387×10^{-17}	–	2.4664	2.7974
6702	Vitreous left	4.40	2.6706×10^{-17}	20.26	–	4.1604×10^{-17}	4.8046×10^{-17}	–	0.6419	0.5558
6800	Eye lens sensitive right	0.0202	0.0000×10^{0}	0.00	–	2.1171×10^{-17}	–	–	–	–
6801	Eye lens insensitive right	0.114	0.0000×10^{0}	0.00	–	1.0450×10^{-16}	–	–	–	–
6900	Cornea right	0.87	7.4539×10^{-17}	30.71	–	3.4798×10^{-17}	2.0927×10^{-17}	–	2.1420	3.5619
6901	Aqueous right	0.294	1.8636×10^{-17}	68.17	–	4.8077×10^{-17}	9.9208×10^{-18}	–	0.3876	1.8785
6902	Vitreous right	4.40	4.4092×10^{-17}	20.34	–	4.4053×10^{-17}	3.4191×10^{-17}	–	1.0009	1.2896
7000	Gall bladder wall	2.70	4.0035×10^{-17}	19.93	–	3.1920×10^{-17}	4.4876×10^{-17}	–	1.2542	0.8921
7100	Gall bladder contents	15.00	3.1668×10^{-17}	14.94	–	4.3775×10^{-17}	3.9524×10^{-17}	–	0.7234	0.8012
7200	Stomach wall(0-60)	0.745	3.1870×10^{-17}	21.24	–	4.1052×10^{-17}	2.8552×10^{-17}	–	0.7763	1.1162
7201	Stomach wall(60-100)	0.489	4.2010×10^{-17}	25.18	–	3.6227×10^{-17}	3.7623×10^{-17}	–	1.1596	1.1166
7202	Stomach wall(100-300)	2.47	3.7770×10^{-17}	16.51	–	4.0936×10^{-17}	4.2653×10^{-17}	–	0.9227	0.8855
7203	Stomach wall(300-surface)	57.96	4.0891×10^{-17}	5.34	–	3.6408×10^{-17}	3.3795×10^{-17}	–	1.1231	1.2100
7300	Stomach contents	83.00	3.7554×10^{-17}	4.35	–	3.5108×10^{-17}	3.6016×10^{-17}	–	1.0697	1.0427
7400	Small intestine wall(0-130)	6.59	3.4440×10^{-17}	13.85	–	3.6872×10^{-17}	3.6324×10^{-17}	–	0.9341	0.9481
7401	Small intestine wall(130-150)	1.03	3.2580×10^{-17}	14.40	–	3.0177×10^{-17}	3.0715×10^{-17}	–	1.0796	1.0607
7402	Small intestine wall(150-200)	2.60	3.7625×10^{-17}	11.56	–	3.9126×10^{-17}	3.7668×10^{-17}	–	0.9616	0.9989

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
7403	Small intestine wall(200-surface)	266.74	3.8566×10^{-17}	2.65	–	3.7477×10^{-17}	3.6853×10^{-17}	–	1.0290	1.0465
7500	Small intestine contents(-400-0)	19.02	3.6588×10^{-17}	8.85	–	3.7969×10^{-17}	3.7135×10^{-17}	–	0.9636	0.9853
7501	Small intestine contents(centre–400)	97.98	3.9425×10^{-17}	4.54	–	3.7699×10^{-17}	3.8645×10^{-17}	–	1.0458	1.0202
7600	Ascending colon wall(0-280)	1.91	2.7075×10^{-17}	25.03	–	4.3514×10^{-17}	3.0146×10^{-17}	–	0.6222	0.8981
7601	Ascending colon wall(280-300)	0.139	3.3860×10^{-17}	46.71	–	4.0375×10^{-17}	4.2157×10^{-17}	–	0.8386	0.8032
7602	Ascending colon wall(300-surface)	26.67	3.3984×10^{-17}	9.38	–	3.8627×10^{-17}	3.7739×10^{-17}	–	0.8798	0.9005
7700	Ascending colon content	31.35	3.2670×10^{-17}	7.49	–	3.3459×10^{-17}	3.2711×10^{-17}	–	0.9764	0.9988
7800	Transverse colon wall right(0-280)	1.58	2.5318×10^{-17}	33.24	–	3.9295×10^{-17}	2.8760×10^{-17}	–	0.6443	0.8803
7801	Transverse colon wall right(280-300)	0.116	6.1159×10^{-17}	49.13	–	2.9964×10^{-17}	2.0259×10^{-17}	–	2.0411	3.0189
7802	Transverse colon wall right(300-surface)	31.21	4.0393×10^{-17}	7.86	–	3.3878×10^{-17}	3.8693×10^{-17}	–	1.1923	1.0439
7900	Transverse colon contents right	18.65	3.6613×10^{-17}	11.11	–	3.6601×10^{-17}	3.5085×10^{-17}	–	1.0003	1.0436
8000	Transverse colon wall left(0-280)	1.10	5.2362×10^{-17}	35.26	–	1.9144×10^{-17}	2.6688×10^{-17}	–	2.7352	1.9620
8001	Transverse colon wall left(280-300)	0.0808	4.4020×10^{-17}	43.46	–	8.3985×10^{-18}	2.5171×10^{-17}	–	5.2415	1.7489
8002	Transverse colon wall left(300-surface)	27.09	4.0726×10^{-17}	7.13	–	4.0338×10^{-17}	3.9380×10^{-17}	–	1.0096	1.0342
8100	Transverse colon content left	10.92	3.9892×10^{-17}	15.08	–	3.8937×10^{-17}	3.5966×10^{-17}	–	1.0245	1.1092
8200	Descending colon wall(0-280)	1.50	3.2592×10^{-17}	22.96	–	2.5577×10^{-17}	1.3426×10^{-17}	–	1.2743	2.4276
8201	Descending colon wall(280-300)	0.11	5.3866×10^{-17}	45.98	–	3.5705×10^{-17}	5.9162×10^{-18}	–	1.5086	9.1048
8202	Descending colon wall(300-surface)	31.74	3.2868×10^{-17}	7.29	–	3.4780×10^{-17}	3.0904×10^{-17}	–	0.9450	1.0635
8300	Descending colon content	14.08	3.9177×10^{-17}	11.73	–	3.2451×10^{-17}	3.5039×10^{-17}	–	1.2073	1.1181
8400	Sigmoid colon wall(0-280)	1.81	2.5158×10^{-17}	27.64	–	2.7359×10^{-17}	1.9159×10^{-17}	–	0.9196	1.3131
8401	Sigmoid colon wall(280-300)	0.133	1.5248×10^{-17}	30.67	–	2.7774×10^{-17}	4.0414×10^{-17}	–	0.5490	0.3773
8402	Sigmoid colon wall(300-surface)	22.78	3.0149×10^{-17}	7.96	–	3.1373×10^{-17}	2.9511×10^{-17}	–	0.9610	1.0216
8500	Sigmoid colon contents	17.99	3.0901×10^{-17}	12.38	–	3.1899×10^{-17}	2.0564×10^{-17}	–	0.9687	1.5027
8600	Rectum wall(0-280)	0.717	2.5730×10^{-17}	30.97	–	4.3535×10^{-17}	4.3339×10^{-17}	–	0.5910	0.5937
8601	Rectum wall(280-300)	0.0527	2.1664×10^{-17}	33.72	–	3.3721×10^{-17}	3.3115×10^{-17}	–	0.6424	0.6542
8602	Rectum wall(300-surface)	2.18	3.9898×10^{-17}	21.14	–	3.1221×10^{-17}	3.6001×10^{-17}	–	1.2779	1.1082
8603	Rectum contents	7.01	3.7127×10^{-17}	26.69	–	1.9463×10^{-17}	2.2176×10^{-17}	–	1.9076	1.6742
8700	Heart wall	97.70	3.1550×10^{-17}	5.20	–	3.4517×10^{-17}	3.2985×10^{-17}	–	0.9140	0.9565
8800	Blood in heart chamber	135.00	3.4167×10^{-17}	2.45	–	3.4290×10^{-17}	3.2616×10^{-17}	–	0.9964	1.0476
8900	Kidney left cortex	49.84	2.5331×10^{-17}	6.89	–	2.7084×10^{-17}	2.5741×10^{-17}	–	0.9353	0.9841
9000	Kidney left medulla	17.80	2.6053×10^{-17}	15.01	–	3.4991×10^{-17}	2.7546×10^{-17}	–	0.7446	0.9458
9100	Kidney left pelvis	3.56	1.7244×10^{-17}	44.03	–	1.8258×10^{-17}	2.3729×10^{-17}	–	0.9445	0.7267
9200	Kidney right cortex	49.85	2.6070×10^{-17}	8.34	–	2.8410×10^{-17}	2.7728×10^{-17}	–	0.9177	0.9402
9300	Kidney right medulla	17.78	3.0386×10^{-17}	11.13	–	2.8267×10^{-17}	2.9413×10^{-17}	–	1.0749	1.0331
9400	Kidney right pelvis	3.56	2.9152×10^{-17}	22.41	–	2.3076×10^{-17}	2.8409×10^{-17}	–	1.2633	1.0262
9500	Liver	724.41	3.5515×10^{-17}	1.53	–	3.3909×10^{-17}	3.4049×10^{-17}	–	1.0474	1.0431

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
9700	Lung(AI) left	139.44	2.8409×10^{-17}	3.18	–	2.9571×10^{-17}	3.1065×10^{-17}	–	0.9607	0.9145
9900	Lung(AI) right	160.56	3.2955×10^{-17}	4.23	–	3.2390×10^{-17}	3.4778×10^{-17}	–	1.0175	0.9476
10000	Lymphatic nodes ET	4.31	2.7877×10^{-17}	30.10	–	2.4476×10^{-17}	2.6257×10^{-17}	–	1.1390	1.0617
10100	Lymphatic nodes thoracic	4.31	3.9153×10^{-17}	13.19	–	3.6235×10^{-17}	3.4068×10^{-17}	–	1.0805	1.1493
10200	Lymphatic nodes head	1.49	3.6686×10^{-17}	32.37	–	1.9926×10^{-17}	3.1951×10^{-17}	–	1.8411	1.1482
10300	Lymphatic nodes trunk	35.19	3.7281×10^{-17}	9.89	–	3.0770×10^{-17}	3.4532×10^{-17}	–	1.2116	1.0796
10400	Lymphatic nodes arms	2.98	3.6587×10^{-17}	27.93	–	4.3785×10^{-17}	4.0066×10^{-17}	–	0.8356	0.9132
10500	Lymphatic nodes legs	2.98	4.2181×10^{-17}	27.13	–	4.7209×10^{-17}	2.5537×10^{-17}	–	0.8935	1.6518
10600	Muscle head	407.78	2.9590×10^{-17}	1.98	–	3.0874×10^{-17}	3.1320×10^{-17}	–	0.9584*	0.9448*
10700	Muscle trunk	2521.09	3.1493×10^{-17}	1.03	–	3.1680×10^{-17}	3.1101×10^{-17}	–	0.9941*	1.0126*
10800	Muscle arms	464.26	3.6153×10^{-17}	2.06	–	3.4862×10^{-17}	3.3650×10^{-17}	–	1.0370*	1.0744*
10900	Muscle legs	2334.63	3.1401×10^{-17}	0.81	–	3.2433×10^{-17}	3.1511×10^{-17}	–	0.9682*	0.9965*
11000	Oesophagus wall(0-190)	0.918	2.3718×10^{-17}	42.12	–	2.5223×10^{-17}	2.3800×10^{-17}	–	0.9403	0.9965
11001	Oesophagus wall(190-200)	0.0496	6.1002×10^{-17}	82.21	–	2.5565×10^{-17}	2.4196×10^{-17}	–	2.3862	2.5212
11002	Oesophagus wall(200-surface)	11.37	3.1562×10^{-17}	11.09	–	3.2474×10^{-17}	3.5337×10^{-17}	–	0.9719	0.8932
11003	Oesophagus contents	7.14	3.0886×10^{-17}	19.08	–	3.1661×10^{-17}	2.5254×10^{-17}	–	0.9755	1.2230
11300	Pancreas	41.89	3.0274×10^{-17}	12.30	–	3.5729×10^{-17}	3.6464×10^{-17}	–	0.8473	0.8302
11400	Pituitary gland	0.26	1.7468×10^{-17}	100.00	–	7.0432×10^{-17}	6.9709×10^{-17}	–	0.2480	0.2506
11500	Prostate	1.25	5.4963×10^{-17}	31.46	–	4.4485×10^{-17}	3.8575×10^{-17}	–	1.2355	1.4248
11600	RST head	468.28	3.1848×10^{-17}	1.91	–	2.8891×10^{-17}	3.2463×10^{-17}	–	1.1023	0.9811
11700	RST trunk	3350.84	3.2568×10^{-17}	1.35	–	3.3182×10^{-17}	3.3141×10^{-17}	–	0.9815*	0.9827*
11800	RST arms	613.67	3.7110×10^{-17}	1.99	–	3.7418×10^{-17}	3.7076×10^{-17}	–	0.9918*	1.0009*
11900	RST legs	1817.51	3.3425×10^{-17}	0.89	–	3.2969×10^{-17}	3.3482×10^{-17}	–	1.0138*	0.9983*
12000	Salivary glands left	17.67	2.4798×10^{-17}	12.47	–	2.3898×10^{-17}	3.1003×10^{-17}	–	1.0376	0.7999
12100	Salivary glandss right	17.67	2.7524×10^{-17}	10.80	–	3.0137×10^{-17}	3.5803×10^{-17}	–	0.9133	0.7688
12200	Skin head insensitive	75.38	2.3327×10^{-17}	6.50	–	2.0652×10^{-17}	2.4870×10^{-17}	–	1.1295*	0.9379*
12201	Skin head sensitive(40-100)	5.81	1.4257×10^{-17}	18.04	–	1.4702×10^{-17}	1.7899×10^{-17}	–	0.9697*	0.7965*
12300	Skin trunk insensitive	213.00	2.4092×10^{-17}	3.15	–	2.1524×10^{-17}	2.2388×10^{-17}	–	1.1193*	1.0761*
12301	Skin trunk sensitive(40-100)	19.59	1.6891×10^{-17}	6.22	–	1.4342×10^{-17}	1.7429×10^{-17}	–	1.1777*	0.9691*
12400	Skin arms insensitive	66.16	2.6256×10^{-17}	4.47	–	2.7105×10^{-17}	2.7516×10^{-17}	–	0.9687*	0.9542*
12401	Skin arms sensitive(40-100)	5.09	1.9217×10^{-17}	6.94	–	2.3223×10^{-17}	2.1555×10^{-17}	–	0.8275*	0.8915*
12500	Skin legs insensitive	194.72	2.3084×10^{-17}	2.05	–	2.4994×10^{-17}	2.4511×10^{-17}	–	0.9236*	0.9418*
12501	Skin legs sensitive(40-100)	16.66	1.8303×10^{-17}	5.78	–	2.0263×10^{-17}	1.7359×10^{-17}	–	0.9033*	1.0544*
12600	Spinal cord	23.82	2.6066×10^{-17}	10.81	–	2.5871×10^{-17}	2.6464×10^{-17}	–	1.0075	0.9849
12700	Spleen	71.31	2.7406×10^{-17}	3.09	–	2.6362×10^{-17}	2.5609×10^{-17}	–	1.0396	1.0702
12802	Erupted upper front buccal deciduous ena	0.193	4.3052×10^{-17}	80.05	–	1.0424×10^{-17}	5.6399×10^{-17}	–	4.1300	0.7633

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
12803	Erupted upper front lingual deciduous en	0.165	7.0481×10^{-17}	68.85	–	2.4206×10^{-17}	1.5505×10^{-17}	–	2.9118	4.5457
12810	Erupted upper left buccal deciduous enam	0.149	1.8232×10^{-16}	48.30	–	1.2851×10^{-17}	6.4473×10^{-17}	–	14.1870	2.8278
12811	Erupted upper left lingual deciduous ena	0.159	1.7375×10^{-17}	100.00	–	7.1268×10^{-17}	2.6186×10^{-17}	–	0.2438	0.6635
12814	Erupted upper right buccal deciduous ena	0.149	0.0000×10^{0}	0.00	–	–	3.8023×10^{-17}	–	–	–
12815	Erupted upper right lingual deciduous en	0.158	6.5190×10^{-18}	100.00	–	6.3040×10^{-17}	5.9783×10^{-17}	–	0.1034	0.1090
12818	Erupted lower front buccal deciduous ena	0.135	1.0782×10^{-17}	100.00	–	9.2214×10^{-18}	1.4988×10^{-17}	–	1.1692	0.7193
12819	Erupted lower front lingual deciduous en	0.138	4.9159×10^{-17}	90.96	–	–	–	–	–	–
12826	Erupted lower left buccal deciduous enam	0.175	2.3662×10^{-17}	66.85	–	6.9445×10^{-17}	2.4657×10^{-17}	–	0.3407	0.9596
12827	Erupted lower left lingual deciduous ena	0.234	2.8839×10^{-18}	100.00	–	5.9616×10^{-18}	8.4834×10^{-17}	–	0.4838	0.0340
12830	Erupted lower right buccal deciduous ena	0.197	6.6200×10^{-17}	78.22	–	1.2753×10^{-17}	5.4200×10^{-17}	–	5.1909	1.2214
12831	Erupted lower right lingual deciduous en	0.213	0.0000×10^{0}	0.00	–	4.4489×10^{-18}	–	–	–	–
12833	Erupted upper front deciduous dentin	0.821	2.4757×10^{-17}	48.93	–	4.2002×10^{-17}	5.2313×10^{-17}	–	0.5894	0.4732
12837	Erupted upper left deciduous dentin	0.708	3.1563×10^{-17}	40.64	–	1.8887×10^{-17}	4.6180×10^{-17}	–	1.6712	0.6835
12839	Erupted upper right deciduous dentin	0.708	4.3978×10^{-17}	47.14	–	2.5916×10^{-19}	8.6588×10^{-18}	–	169.6925	5.0790
12841	Erupted lower front deciduous dentin	0.617	7.2512×10^{-17}	52.61	–	5.2323×10^{-17}	4.5541×10^{-17}	–	1.3859	1.5922
12845	Erupted lower left deciduous dentin	0.948	3.1491×10^{-17}	37.31	–	5.1696×10^{-17}	2.7058×10^{-17}	–	0.6092	1.1638
12847	Erupted lower right deciduous dentin	0.948	4.4222×10^{-17}	38.28	–	3.1072×10^{-17}	3.0330×10^{-17}	–	1.4232	1.4580
12848	Unerupted permanent enamel	1.63	1.8232×10^{-17}	45.40	–	2.9881×10^{-17}	4.7469×10^{-17}	–	0.6102	0.3841
12850	Unerupted permanent dentin	5.40	2.8020×10^{-17}	21.23	–	3.9146×10^{-17}	4.8050×10^{-17}	–	0.7158	0.5831
12852	Permanent cementum	0.284	5.7064×10^{-17}	35.75	–	4.8799×10^{-17}	3.1509×10^{-17}	–	1.1694	1.8110
12853	Deciduous cementum	0.242	1.8682×10^{-17}	32.27	–	4.7957×10^{-17}	6.7948×10^{-17}	–	0.3896	0.2749
12854	Permanent pulp	0.163	2.7918×10^{-17}	73.89	–	1.4494×10^{-17}	1.1917×10^{-16}	–	1.9262	0.2343
12855	Deciduous pulp	0.661	4.3695×10^{-17}	34.32	–	5.5184×10^{-17}	3.1589×10^{-17}	–	0.7918	1.3832
12856	Teeth retention region	0.0211	2.0710×10^{-17}	42.78	–	1.0369×10^{-17}	9.8716×10^{-18}	–	1.9972	2.0979
12900	Testis left	0.907	3.8129×10^{-17}	45.64	–	2.6020×10^{-17}	8.3535×10^{-17}	–	1.4654	0.4564
13000	Testis right	0.907	5.1849×10^{-17}	36.78	–	2.1090×10^{-17}	3.6806×10^{-17}	–	2.4585	1.4087
13100	Thymus	31.20	3.6371×10^{-17}	5.66	–	4.0350×10^{-17}	3.5070×10^{-17}	–	0.9014	1.0371
13200	Thyroid	3.87	4.9744×10^{-17}	19.60	–	3.9616×10^{-17}	4.2785×10^{-17}	–	1.2557	1.1627

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
13300	Tongue upper(food)	9.15	3.6469×10^{-17}	21.05	–	3.5067×10^{-17}	3.8046×10^{-17}	–	1.0400	0.9585
13301	Tongue lower	10.58	2.4819×10^{-17}	19.89	–	3.7583×10^{-17}	2.7532×10^{-17}	–	0.6604	0.9015
13400	Tonsils	2.08	1.0802×10^{-17}	29.09	–	1.3818×10^{-17}	2.5024×10^{-17}	–	0.7817	0.4317
13500	Ureter left	2.18	3.1391×10^{-17}	38.65	–	5.0846×10^{-17}	5.4629×10^{-17}	–	0.6174	0.5746
13600	Ureter right	2.18	3.9034×10^{-17}	35.85	–	3.3105×10^{-17}	1.6802×10^{-17}	–	1.1791	2.3232
13700	Urinary bladder wall insensitive	15.42	4.2843×10^{-17}	7.36	–	4.0733×10^{-17}	4.2404×10^{-17}	–	1.0518	1.0104
13701	Urinary bladder wall sensitive(86-193)	0.882	6.4283×10^{-17}	16.56	–	6.3739×10^{-17}	3.7178×10^{-17}	–	1.0085	1.7291
13800	Urinary bladder content	64.70	3.2812×10^{-17}	6.17	–	3.6920×10^{-17}	3.6451×10^{-17}	–	0.8887	0.9002
14000	Air inside body	0.024	2.3143×10^{-17}	34.97	–	1.0664×10^{-16}	3.6214×10^{-17}	–	0.2170	0.6391

* For tissues bookkept as head, trunk, arm and leg pieces (muscle, residual soft tissue, skin, large vessels), the volumes tabulated in the distributed cell tables partition the tissue at different boundaries from the transported mesh, and the reference output is normalised by the tabulated volume. The marked ratio therefore compares two different partitions of the same tissue, and is left out of the summary statistics of Table S2.

Table S17. 05F_Internal. Absorbed dose per primary history, Gy. σ is the FLUKA statistical uncertainty. Ratios are FLUKA / reference.

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
100	Adrenal left	2.97	5.0225×10^{-15}	1.68	–	5.1536×10^{-15}	4.7578×10^{-15}	–	0.9746	1.0556
200	Adrenal right	2.97	1.2620×10^{-14}	1.55	–	1.2445×10^{-14}	1.2798×10^{-14}	–	1.0141	0.9861
300	ET1(0-8)	0.00259	2.7438×10^{-16}	77.89	–	3.7465×10^{-16}	2.1458×10^{-17}	–	0.7324	12.7870
301	ET1(8-40)	0.0105	4.3106×10^{-16}	61.07	–	4.0599×10^{-16}	2.1177×10^{-17}	–	1.0618	20.3553
302	ET1(40-50)	0.00333	7.3208×10^{-16}	75.81	–	5.3826×10^{-16}	2.9835×10^{-17}	–	1.3601	24.5375
303	ET1(50-Surface)	0.33	2.4704×10^{-16}	25.53	–	4.2938×10^{-16}	2.3250×10^{-16}	–	0.5753	1.0625
400	ET2(-15-0)	0.129	4.1118×10^{-16}	18.92	–	3.8051×10^{-16}	2.3636×10^{-16}	–	1.0806	1.7396
401	ET2(0-40)	0.367	3.1721×10^{-16}	11.12	–	3.6905×10^{-16}	2.7383×10^{-16}	–	0.8595	1.1584
402	ET2(40-50)	0.0925	3.8709×10^{-16}	15.98	–	4.0801×10^{-16}	2.1605×10^{-16}	–	0.9487	1.7917
403	ET2(50-55)	0.0463	3.0561×10^{-16}	16.83	–	4.9133×10^{-16}	3.4044×10^{-16}	–	0.6220	0.8977
404	ET2(55-65)	0.0928	3.0371×10^{-16}	16.21	–	3.9856×10^{-16}	3.1976×10^{-16}	–	0.7620	0.9498
405	ET2(65-Surface)	19.23	3.1944×10^{-16}	2.18	–	3.2973×10^{-16}	3.1770×10^{-16}	–	0.9688	1.0055
500	Oral mucosa tongue	0.0313	6.3321×10^{-16}	19.70	–	5.3807×10^{-16}	4.0484×10^{-16}	–	1.1768	1.5641
501	Oral mucosa moth floor	0.0187	2.4743×10^{-16}	20.75	–	4.8217×10^{-16}	3.4252×10^{-16}	–	0.5132	0.7224
600	Oral mucosa lips and cheeks	0.0144	5.5438×10^{-16}	52.03	–	5.7042×10^{-16}	1.8678×10^{-16}	–	0.9719	2.9681
700	Trachea	2.58	1.3503×10^{-15}	4.83	–	1.2050×10^{-15}	1.2706×10^{-15}	–	1.1206	1.0627
800	BB(-11–6)	0.0022	2.1042×10^{-15}	39.83	–	2.4027×10^{-15}	1.4746×10^{-15}	–	0.8758	1.4270
801	BB(-6-0)	0.00281	1.7788×10^{-15}	36.18	–	1.8116×10^{-15}	1.7523×10^{-15}	–	0.9819	1.0151
802	BB(0-10)	0.00471	1.5521×10^{-15}	30.71	–	1.7039×10^{-15}	1.6165×10^{-15}	–	0.9109	0.9602
803	BB(10-35)	0.0119	2.4535×10^{-15}	17.87	–	2.4259×10^{-15}	2.8185×10^{-15}	–	1.0114	0.8705
804	BB(35-40)	0.00242	2.2521×10^{-15}	28.31	–	2.0682×10^{-15}	1.0996×10^{-15}	–	1.0889	2.0481
805	BB(40-50)	0.00487	1.1437×10^{-15}	27.82	–	1.4747×10^{-15}	1.3614×10^{-15}	–	0.7755	0.8401
806	BB(40-50)	0.0049	1.4804×10^{-15}	31.26	–	1.3876×10^{-15}	2.1509×10^{-15}	–	1.0669	0.6883
807	BB(60-70)	0.00494	1.1600×10^{-15}	20.22	–	1.7997×10^{-15}	1.3787×10^{-15}	–	0.6446	0.8414
808	BB(70-surface)	3.37	2.1130×10^{-15}	1.79	–	2.1851×10^{-15}	2.1892×10^{-15}	–	0.9670	0.9652
900	Blood in large arteries head	0.275	5.8029×10^{-16}	19.50	–	6.1901×10^{-16}	5.8317×10^{-16}	–	0.9375*	0.9951*
910	Blood in large veins head	1.77	5.8830×10^{-16}	8.08	–	6.3968×10^{-16}	5.9298×10^{-16}	–	0.9197*	0.9921*
1000	Blood in large arteries trunk	38.12	3.3602×10^{-15}	0.43	–	3.4237×10^{-15}	3.4354×10^{-15}	–	0.9815*	0.9781*
1010	Blood in large veins trunk	107.85	3.2187×10^{-15}	0.30	–	3.3618×10^{-15}	3.3570×10^{-15}	–	0.9574*	0.9588*
1100	Blood in large arteries arms	6.62	1.8012×10^{-15}	3.86	–	1.8303×10^{-15}	1.7943×10^{-15}	–	0.9841*	1.0038*
1110	Blood in large veins arms	43.12	1.3435×10^{-15}	1.38	–	1.3529×10^{-15}	1.3215×10^{-15}	–	0.9931*	1.0167*
1200	Blood in large arteries legs	44.93	8.0211×10^{-17}	4.74	–	8.4425×10^{-17}	8.8571×10^{-17}	–	0.9501*	0.9056*
1210	Blood in large veins legs	117.10	9.9952×10^{-17}	2.25	–	1.0059×10^{-16}	1.0456×10^{-16}	–	0.9936*	0.9559*
1300	Humeri upper cortical	28.83	1.1786×10^{-15}	1.02	–	1.1661×10^{-15}	1.1250×10^{-15}	–	1.0107	1.0477
1400	Humeri upper spogiosa	20.16	8.0922×10^{-16}	2.47	–	8.2049×10^{-16}	8.2772×10^{-16}	–	0.9863	0.9776

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
1500	Humeri upper medullary cavity	3.76	1.4564×10^{-15}	3.20	–	1.4673×10^{-15}	1.5546×10^{-15}	–	0.9926	0.9368
1600	Humeri lower cortical	22.27	1.8914×10^{-15}	0.90	–	1.9448×10^{-15}	1.8942×10^{-15}	–	0.9725	0.9985
1700	Humeri lower spongiosa	11.03	1.7166×10^{-15}	2.51	–	1.7165×10^{-15}	1.6985×10^{-15}	–	1.0001	1.0107
1800	Humeri lower medullary cavity	3.56	2.1363×10^{-15}	3.98	–	2.1807×10^{-15}	2.0890×10^{-15}	–	0.9796	1.0226
1900	Radii cortical	14.85	6.2656×10^{-16}	2.78	–	6.4148×10^{-16}	6.1094×10^{-16}	–	0.9767	1.0256
1910	Ulnae cortical	18.78	8.1924×10^{-16}	1.89	–	8.5373×10^{-16}	8.4833×10^{-16}	–	0.9596	0.9657
2000	Radii spongiosa	3.31	6.2749×10^{-16}	4.69	–	6.2881×10^{-16}	5.8651×10^{-16}	–	0.9979	1.0699
2010	Ulnae spongiosa	5.49	1.2433×10^{-15}	2.87	–	1.3202×10^{-15}	1.2215×10^{-15}	–	0.9417	1.0178
2100	Radii medullary cavity	2.30	7.5337×10^{-16}	6.49	–	7.6190×10^{-16}	7.6384×10^{-16}	–	0.9888	0.9863
2110	Ulnae medullary cavity	2.63	6.5649×10^{-16}	7.20	–	7.1112×10^{-16}	6.7302×10^{-16}	–	0.9232	0.9754
2200	Wrists and hand bones cortical	9.33	2.2868×10^{-16}	4.55	–	2.1774×10^{-16}	2.3426×10^{-16}	–	1.0503	0.9762
2300	Wrists and hand bones spongiosa	16.48	2.2950×10^{-16}	3.17	–	2.3838×10^{-16}	2.4199×10^{-16}	–	0.9628	0.9484
2400	Clavicles cortical	6.18	1.0525×10^{-15}	2.83	–	1.0613×10^{-15}	9.8233×10^{-16}	–	0.9918	1.0715
2500	Clavicles spongiosa	7.96	1.1151×10^{-15}	2.40	–	1.1025×10^{-15}	1.1402×10^{-15}	–	1.0114	0.9780
2600	Cranium cortical	229.12	1.8814×10^{-16}	1.65	–	1.9380×10^{-16}	1.9158×10^{-16}	–	0.9708	0.9820
2700	Cranium spongiosa	520.56	1.7364×10^{-16}	1.16	–	1.7017×10^{-16}	1.7122×10^{-16}	–	1.0204	1.0141
2800	Femora upper cortical	49.47	2.1876×10^{-16}	2.42	–	2.2552×10^{-16}	2.1369×10^{-16}	–	0.9700	1.0237
2900	Femora upper spongiosa	26.10	3.6059×10^{-16}	4.03	–	3.6171×10^{-16}	3.6386×10^{-16}	–	0.9969	0.9910
3000	Femora upper medullary cavity	8.22	1.3072×10^{-16}	5.23	–	1.4494×10^{-16}	1.1942×10^{-16}	–	0.9019	1.0946
3100	Femora lower cortical	48.06	3.1100×10^{-17}	9.41	–	3.5488×10^{-17}	3.3044×10^{-17}	–	0.8764	0.9412
3200	Femora lower spongiosa	28.74	2.7837×10^{-17}	8.22	–	2.5079×10^{-17}	2.2363×10^{-17}	–	1.1100	1.2448
3300	Femora lower medullary cavity	5.79	4.7262×10^{-17}	22.75	–	3.8474×10^{-17}	3.5908×10^{-17}	–	1.2284	1.3162
3400	Tibiae cortical	73.04	1.0245×10^{-17}	11.87	–	9.8975×10^{-18}	1.2630×10^{-17}	–	1.0351	0.8112
3410	Fibulae cortical	10.25	8.7216×10^{-18}	23.61	–	1.2746×10^{-17}	6.5579×10^{-18}	–	0.6843	1.3299
3420	Patellae cortical	1.19	2.9526×10^{-17}	34.75	–	2.4585×10^{-17}	9.0763×10^{-18}	–	1.2010	3.2531
3500	Tibiae spongiosa	23.03	1.9707×10^{-17}	14.58	–	1.4911×10^{-17}	1.5969×10^{-17}	–	1.3216	1.2341
3510	Fibulae spongiosa	2.79	1.8391×10^{-17}	29.70	–	1.7824×10^{-17}	1.7526×10^{-17}	–	1.0318	1.0494
3520	Patellae spongiosa	5.57	1.8904×10^{-17}	24.95	–	2.8220×10^{-17}	1.7964×10^{-17}	–	0.6699	1.0523
3600	Tibiae medullary cavity	8.26	1.0108×10^{-17}	17.65	–	1.1556×10^{-17}	8.0465×10^{-18}	–	0.8747	1.2562
3610	Fibulae medullary cavity	1.87	1.1106×10^{-17}	51.55	–	1.1630×10^{-17}	1.7540×10^{-17}	–	0.9549	0.6332
3700	Ankles and foot cortical	33.25	1.3284×10^{-17}	14.01	–	1.2201×10^{-17}	1.0715×10^{-17}	–	1.0887	1.2397
3800	Ankles and foot spongiosa	70.79	1.0552×10^{-17}	9.04	–	1.0361×10^{-17}	1.0033×10^{-17}	–	1.0184	1.0517
3900	Mandible cortical	19.28	4.5106×10^{-16}	2.12	–	4.4540×10^{-16}	4.5292×10^{-16}	–	1.0127	0.9959
4000	Mandible spongiosa	29.48	5.0546×10^{-16}	1.37	–	4.8883×10^{-16}	5.0739×10^{-16}	–	1.0340	0.9962
4100	Pelvis cortical	58.23	7.4031×10^{-16}	1.25	–	7.5275×10^{-16}	7.4140×10^{-16}	–	0.9835	0.9985
4200	Pelvis spongiosa	99.47	7.0502×10^{-16}	1.44	–	7.2652×10^{-16}	7.2319×10^{-16}	–	0.9704	0.9749
4300	Ribs cortical	38.51	3.9916×10^{-15}	0.49	–	3.9873×10^{-15}	4.0172×10^{-15}	–	1.0011	0.9936
4400	Ribs spongiosa	79.11	4.1156×10^{-15}	0.57	–	4.1149×10^{-15}	4.1256×10^{-15}	–	1.0002	0.9976

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
4500	Scapulae cortical	29.30	1.4021×10^{-15}	1.96	–	1.4332×10^{-15}	1.4538×10^{-15}	–	0.9783	0.9645
4600	Scapulae spongiosa	42.72	1.3476×10^{-15}	1.04	–	1.3247×10^{-15}	1.3488×10^{-15}	–	1.0173	0.9991
4700	Cervical spine cortical	6.99	5.4391×10^{-16}	3.34	–	5.8563×10^{-16}	5.4921×10^{-16}	–	0.9288	0.9903
4800	Cervical spine spongiosa	14.08	5.4843×10^{-16}	3.10	–	6.0504×10^{-16}	5.7916×10^{-16}	–	0.9064	0.9469
4900	Thoracic spine cortical	33.10	4.1661×10^{-15}	0.54	–	4.1566×10^{-15}	4.1266×10^{-15}	–	1.0023	1.0096
5000	Thoracic spine spongiosa	74.60	4.8346×10^{-15}	0.37	–	4.8291×10^{-15}	4.8740×10^{-15}	–	1.0011	0.9919
5100	Lumbar spine cortical	16.80	3.7439×10^{-15}	0.85	–	3.7353×10^{-15}	3.7116×10^{-15}	–	1.0023	1.0087
5200	Lumbar spine spongiosa	58.99	3.9821×10^{-15}	0.72	–	4.0299×10^{-15}	4.0674×10^{-15}	–	0.9881	0.9790
5300	Sacrum cortical	11.70	9.8619×10^{-16}	2.08	–	1.0336×10^{-15}	1.0016×10^{-15}	–	0.9542	0.9846
5400	Sacrum spongiosa	34.52	1.1438×10^{-15}	1.21	–	1.1324×10^{-15}	1.1288×10^{-15}	–	1.0101	1.0133
5500	Sternum cortical	2.47	3.7440×10^{-15}	2.09	–	3.7748×10^{-15}	3.8959×10^{-15}	–	0.9918	0.9610
5600	Sternum spongiosa	8.50	2.7894×10^{-15}	2.00	–	2.8311×10^{-15}	2.8207×10^{-15}	–	0.9853	0.9889
5800	Cartilage trunk	55.14	5.6815×10^{-15}	0.67	–	5.7025×10^{-15}	5.7183×10^{-15}	–	0.9963	0.9936
6100	Brain	1244.20	1.6952×10^{-16}	0.46	–	1.7021×10^{-16}	1.7031×10^{-16}	–	0.9960	0.9954
6200	Breast left adipose tissue	1.17	2.3092×10^{-15}	4.69	–	2.5085×10^{-15}	2.4635×10^{-15}	–	0.9206	0.9374
6300	Breast left glandular tissue	0.781	2.4000×10^{-15}	5.49	–	2.4344×10^{-15}	2.3556×10^{-15}	–	0.9859	1.0189
6400	Breast right adipose tissue	1.17	5.0845×10^{-15}	2.96	–	5.1992×10^{-15}	5.3118×10^{-15}	–	0.9779	0.9572
6500	Breast right glandular tissue	0.781	5.3450×10^{-15}	2.66	–	5.3512×10^{-15}	5.0062×10^{-15}	–	0.9988	1.0677
6600	Eye lens sensitive left	0.0202	1.8734×10^{-16}	64.49	–	1.6815×10^{-16}	1.9464×10^{-16}	–	1.1141	0.9625
6601	Eye lens insensitive left	0.114	1.1861×10^{-16}	35.87	–	2.3505×10^{-16}	1.9331×10^{-16}	–	0.5046	0.6136
6700	Cornea left	0.87	2.3981×10^{-16}	14.38	–	1.5763×10^{-16}	1.9346×10^{-16}	–	1.5213	1.2396
6701	Aqueous left	0.294	2.3854×10^{-16}	24.40	–	1.9795×10^{-16}	1.4388×10^{-16}	–	1.2050	1.6579
6702	Vitreous left	4.40	2.0320×10^{-16}	10.22	–	2.0889×10^{-16}	2.1379×10^{-16}	–	0.9728	0.9504
6800	Eye lens sensitive right	0.0202	1.4991×10^{-16}	100.00	–	4.0392×10^{-17}	3.7489×10^{-16}	–	3.7114	0.3999
6801	Eye lens insensitive right	0.114	2.1097×10^{-16}	45.20	–	9.2833×10^{-17}	6.2776×10^{-16}	–	2.2726	0.3361
6900	Cornea right	0.87	2.1547×10^{-16}	13.24	–	2.1919×10^{-16}	1.8596×10^{-16}	–	0.9830	1.1587
6901	Aqueous right	0.294	2.0225×10^{-16}	24.38	–	2.1117×10^{-16}	2.6188×10^{-16}	–	0.9577	0.7723
6902	Vitreous right	4.40	2.1708×10^{-16}	11.36	–	2.4431×10^{-16}	2.4035×10^{-16}	–	0.8886	0.9032
7000	Gall bladder wall	2.69	1.8375×10^{-14}	1.02	–	1.8489×10^{-14}	1.8304×10^{-14}	–	0.9939	1.0039
7100	Gall bladder contents	15.00	1.7574×10^{-14}	0.49	–	1.7733×10^{-14}	1.7683×10^{-14}	–	0.9910	0.9939
7200	Stomach wall(0-60)	0.745	6.1643×10^{-15}	2.33	–	6.0925×10^{-15}	5.9312×10^{-15}	–	1.0118	1.0393
7201	Stomach wall(60-100)	0.489	5.9009×10^{-15}	3.38	–	6.1527×10^{-15}	6.0093×10^{-15}	–	0.9591	0.9820
7202	Stomach wall(100-300)	2.47	5.7863×10^{-15}	1.05	–	5.9827×10^{-15}	5.8821×10^{-15}	–	0.9672	0.9837
7203	Stomach wall(300-surface)	57.96	5.9897×10^{-15}	0.37	–	6.0203×10^{-15}	6.0513×10^{-15}	–	0.9949	0.9898
7300	Stomach contents	83.00	5.5196×10^{-15}	0.31	–	5.4983×10^{-15}	5.5439×10^{-15}	–	1.0039	0.9956
7400	Small intestine wall(0-130)	6.59	2.4551×10^{-15}	0.61	–	2.4953×10^{-15}	2.4250×10^{-15}	–	0.9839	1.0124
7401	Small intestine wall(130-150)	1.03	2.4959×10^{-15}	2.10	–	2.5588×10^{-15}	2.5081×10^{-15}	–	0.9754	0.9951
7402	Small intestine wall(150-200)	2.60	2.5430×10^{-15}	1.42	–	2.5210×10^{-15}	2.4674×10^{-15}	–	1.0087	1.0306

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
7403	Small intestine wall(200-surface)	266.74	2.5085×10^{-15}	0.26	–	2.5224×10^{-15}	2.5260×10^{-15}	–	0.9945	0.9931
7500	Small intestine contents(-400-0)	19.02	2.4827×10^{-15}	0.74	–	2.4702×10^{-15}	2.4655×10^{-15}	–	1.0051	1.0070
7501	Small intestine contents(centre–400)	97.98	2.4123×10^{-15}	0.53	–	2.4020×10^{-15}	2.3738×10^{-15}	–	1.0043	1.0162
7600	Ascending colon wall(0-280)	1.91	1.8008×10^{-15}	3.15	–	1.6821×10^{-15}	1.7418×10^{-15}	–	1.0706	1.0339
7601	Ascending colon wall(280-300)	0.139	1.8240×10^{-15}	5.67	–	1.8125×10^{-15}	1.6512×10^{-15}	–	1.0063	1.1047
7602	Ascending colon wall(300-surface)	26.67	1.6661×10^{-15}	1.20	–	1.7388×10^{-15}	1.7026×10^{-15}	–	0.9582	0.9785
7700	Ascending colon content	31.35	1.5571×10^{-15}	1.25	–	1.5555×10^{-15}	1.5659×10^{-15}	–	1.0010	0.9944
7800	Transverse colon wall right(0-280)	1.58	7.3648×10^{-15}	1.36	–	7.1043×10^{-15}	7.1365×10^{-15}	–	1.0367	1.0320
7801	Transverse colon wall right(280-300)	0.116	7.4279×10^{-15}	2.56	–	7.1336×10^{-15}	7.1515×10^{-15}	–	1.0413	1.0387
7802	Transverse colon wall right(300-surface)	31.23	7.4627×10^{-15}	0.60	–	7.3912×10^{-15}	7.4391×10^{-15}	–	1.0097	1.0032
7900	Transverse colon contents right	18.65	7.1419×10^{-15}	0.69	–	7.1675×10^{-15}	7.2731×10^{-15}	–	0.9964	0.9820
8000	Transverse colon wall left(0-280)	1.10	3.1910×10^{-15}	2.77	–	3.1263×10^{-15}	3.0488×10^{-15}	–	1.0207	1.0466
8001	Transverse colon wall left(280-300)	0.0808	3.2542×10^{-15}	6.62	–	3.1535×10^{-15}	2.9567×10^{-15}	–	1.0319	1.1006
8002	Transverse colon wall left(300-surface)	27.08	2.9956×10^{-15}	0.77	–	3.0251×10^{-15}	2.9992×10^{-15}	–	0.9902	0.9988
8100	Transverse colon content left	10.92	3.1459×10^{-15}	1.31	–	3.1287×10^{-15}	3.0518×10^{-15}	–	1.0055	1.0308
8200	Descending colon wall(0-280)	1.50	1.0783×10^{-15}	5.45	–	1.1081×10^{-15}	1.1340×10^{-15}	–	0.9731	0.9509
8201	Descending colon wall(280-300)	0.11	9.4029×10^{-16}	7.80	–	1.1266×10^{-15}	1.0612×10^{-15}	–	0.8347	0.8861
8202	Descending colon wall(300-surface)	31.75	1.0339×10^{-15}	1.20	–	1.0654×10^{-15}	1.0612×10^{-15}	–	0.9704	0.9743
8300	Descending colon content	14.08	1.0428×10^{-15}	1.58	–	1.0642×10^{-15}	1.0285×10^{-15}	–	0.9799	1.0139
8400	Sigmoid colon wall(0-280)	1.81	7.1605×10^{-16}	3.72	–	6.8702×10^{-16}	7.0802×10^{-16}	–	1.0422	1.0113
8401	Sigmoid colon wall(280-300)	0.133	7.1072×10^{-16}	11.23	–	7.5637×10^{-16}	6.9342×10^{-16}	–	0.9396	1.0249
8402	Sigmoid colon wall(300-surface)	22.78	7.4400×10^{-16}	2.11	–	7.7476×10^{-16}	7.7570×10^{-16}	–	0.9603	0.9591
8500	Sigmoid colon contents	17.99	7.1207×10^{-16}	2.60	–	7.1525×10^{-16}	7.0434×10^{-16}	–	0.9956	1.0110
8600	Rectum wall(0-280)	0.717	4.3553×10^{-16}	9.49	–	3.8010×10^{-16}	3.3456×10^{-16}	–	1.1458	1.3018
8601	Rectum wall(280-300)	0.0527	3.9683×10^{-16}	22.43	–	4.2403×10^{-16}	2.6406×10^{-16}	–	0.9358	1.5028
8602	Rectum wall(300-surface)	2.18	3.2306×10^{-16}	6.47	–	3.5715×10^{-16}	3.4980×10^{-16}	–	0.9045	0.9235
8603	Rectum contents	7.01	3.4605×10^{-16}	5.16	–	3.5941×10^{-16}	4.1108×10^{-16}	–	0.9628	0.8418
8700	Heart wall	97.62	6.6050×10^{-15}	0.39	–	6.6196×10^{-15}	6.6747×10^{-15}	–	0.9978	0.9896
8800	Blood in heart chamber	135.00	5.9090×10^{-15}	0.43	–	5.9044×10^{-15}	5.9319×10^{-15}	–	1.0008	0.9961
8900	Kidney left cortex	49.85	3.3120×10^{-15}	0.57	–	3.3074×10^{-15}	3.3582×10^{-15}	–	1.0014	0.9863
9000	Kidney left medulla	17.81	3.5124×10^{-15}	1.05	–	3.5217×10^{-15}	3.5633×10^{-15}	–	0.9974	0.9857
9100	Kidney left pelvis	3.56	3.7464×10^{-15}	2.44	–	3.6894×10^{-15}	3.7099×10^{-15}	–	1.0155	1.0098
9200	Kidney right cortex	49.85	8.5268×10^{-15}	0.24	–	8.5979×10^{-15}	8.5358×10^{-15}	–	0.9917	0.9989
9300	Kidney right medulla	17.81	8.0766×10^{-15}	0.61	–	8.1185×10^{-15}	8.0489×10^{-15}	–	0.9948	1.0034
9400	Kidney right pelvis	3.56	8.2978×10^{-15}	1.40	–	7.7583×10^{-15}	7.5919×10^{-15}	–	1.0695	1.0930
9500	Liver	723.34	2.5681×10^{-14}	0.08	–	2.5667×10^{-14}	2.5761×10^{-14}	–	1.0005	0.9969

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
9700	Lung(AI) left	139.44	1.9746×10^{-15}	0.55	–	1.9799×10^{-15}	1.9718×10^{-15}	–	0.9973	1.0014
9900	Lung(AI) right	160.56	4.9090×10^{-15}	0.29	–	4.9109×10^{-15}	4.9460×10^{-15}	–	0.9996	0.9925
10000	Lymphatic nodes ET	4.31	4.3345×10^{-16}	5.29	–	4.4068×10^{-16}	4.3662×10^{-16}	–	0.9836	0.9927
10100	Lymphatic nodes thoracic	4.31	7.5608×10^{-15}	0.93	–	7.5232×10^{-15}	7.4300×10^{-15}	–	1.0050	1.0176
10200	Lymphatic nodes head	1.49	7.1405×10^{-16}	12.62	–	6.6269×10^{-16}	6.8181×10^{-16}	–	1.0775	1.0473
10300	Lymphatic nodes trunk	35.19	2.0647×10^{-15}	0.92	–	2.0716×10^{-15}	2.0732×10^{-15}	–	0.9967	0.9959
10400	Lymphatic nodes arms	2.98	1.5046×10^{-15}	4.05	–	1.4159×10^{-15}	1.5345×10^{-15}	–	1.0626	0.9805
10500	Lymphatic nodes legs	2.98	2.3521×10^{-17}	29.62	–	2.6289×10^{-17}	3.9294×10^{-17}	–	0.8947	0.5986
10600	Muscle head	448.65	3.9630×10^{-16}	0.62	–	3.9587×10^{-16}	3.9717×10^{-16}	–	1.0011*	0.9978*
10700	Muscle trunk	2449.49	2.1559×10^{-15}	0.16	–	2.2092×10^{-15}	2.2145×10^{-15}	–	0.9759*	0.9735*
10800	Muscle arms	463.65	1.1996×10^{-15}	0.19	–	1.2009×10^{-15}	1.2093×10^{-15}	–	0.9989*	0.9920*
10900	Muscle legs	2364.96	1.0308×10^{-16}	0.71	–	1.0933×10^{-16}	1.0816×10^{-16}	–	0.9428*	0.9530*
11000	Oesophagus wall(0-190)	0.895	4.7478×10^{-15}	1.88	–	4.6474×10^{-15}	4.6812×10^{-15}	–	1.0216	1.0142
11001	Oesophagus wall(190-200)	0.0483	5.1403×10^{-15}	4.47	–	4.6514×10^{-15}	4.3481×10^{-15}	–	1.1051	1.1822
11002	Oesophagus wall(200-surface)	11.39	4.7320×10^{-15}	1.19	–	4.6182×10^{-15}	4.7996×10^{-15}	–	1.0246	0.9859
11003	Oesophagus contents	7.14	4.9188×10^{-15}	0.98	–	4.8744×10^{-15}	5.1054×10^{-15}	–	1.0091	0.9634
11100	Ovary left	1.06	8.7017×10^{-16}	4.91	–	7.4047×10^{-16}	7.2138×10^{-16}	–	1.1752	1.2063
11200	Ovary right	1.06	1.0671×10^{-15}	2.98	–	9.5777×10^{-16}	1.0052×10^{-15}	–	1.1141	1.0616
11300	Pancreas	41.88	4.4937×10^{-15}	0.82	–	4.5086×10^{-15}	4.5712×10^{-15}	–	0.9967	0.9830
11400	Pituitary gland	0.259	1.5251×10^{-16}	36.62	–	2.4997×10^{-16}	2.7075×10^{-16}	–	0.6101	0.5633
11600	RST head	529.93	2.8777×10^{-16}	0.56	–	2.8691×10^{-16}	2.8861×10^{-16}	–	1.0030	0.9971
11700	RST trunk	3518.43	3.3398×10^{-15}	0.05	–	3.3817×10^{-15}	3.4011×10^{-15}	–	0.9876*	0.9820*
11800	RST arms	550.67	1.3035×10^{-15}	0.27	–	1.3406×10^{-15}	1.3493×10^{-15}	–	0.9723*	0.9661*
11900	RST legs	1779.66	7.7258×10^{-17}	0.46	–	8.1783×10^{-17}	8.3478×10^{-17}	–	0.9447*	0.9255*
12000	Salivary glands left	17.57	4.0260×10^{-16}	2.91	–	4.0625×10^{-16}	3.9576×10^{-16}	–	0.9910	1.0173
12100	Salivary glandss right	17.57	5.2110×10^{-16}	2.32	–	5.3146×10^{-16}	5.2771×10^{-16}	–	0.9805	0.9875
12200	Skin head insensitive	79.46	2.4526×10^{-16}	1.67	–	2.4272×10^{-16}	2.4307×10^{-16}	–	1.0105	1.0090
12201	Skin head sensitive(40-100)	6.78	2.0196×10^{-16}	4.03	–	2.1328×10^{-16}	1.9708×10^{-16}	–	0.9469	1.0247
12300	Skin trunk insensitive	199.43	1.9695×10^{-15}	0.32	–	2.0924×10^{-15}	2.0862×10^{-15}	–	0.9413*	0.9441*
12301	Skin trunk sensitive(40-100)	17.99	1.6866×10^{-15}	0.39	–	1.8388×10^{-15}	1.8373×10^{-15}	–	0.9172*	0.9180*
12400	Skin arms insensitive	72.70	9.4958×10^{-16}	1.42	–	1.0191×10^{-15}	1.0285×10^{-15}	–	0.9318*	0.9233*
12401	Skin arms sensitive(40-100)	5.85	8.5299×10^{-16}	1.94	–	9.2714×10^{-16}	9.0794×10^{-16}	–	0.9200*	0.9395*
12500	Skin legs insensitive	197.85	6.4899×10^{-17}	1.88	–	6.6255×10^{-17}	6.7682×10^{-17}	–	0.9795*	0.9589*
12501	Skin legs sensitive(40-100)	16.35	5.3158×10^{-17}	4.81	–	5.8134×10^{-17}	5.3682×10^{-17}	–	0.9144*	0.9902*
12600	Spinal cord	23.12	3.1570×10^{-15}	0.82	–	3.1625×10^{-15}	3.2044×10^{-15}	–	0.9983	0.9852
12700	Spleen	71.26	2.5326×10^{-15}	0.57	–	2.5221×10^{-15}	2.5458×10^{-15}	–	1.0042	0.9948
12802	Erupted upper front buccal deciduous ena	0.193	4.2066×10^{-16}	17.06	–	3.2306×10^{-16}	5.0080×10^{-16}	–	1.3021	0.8400

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
12803	Erupted upper front lingual deciduous en	0.165	3.4144×10^{-16}	36.76	–	3.8541×10^{-16}	4.1955×10^{-16}	–	0.8859	0.8138
12810	Erupted upper left buccal deciduous enam	0.149	3.8465×10^{-16}	23.95	–	2.4908×10^{-16}	1.7311×10^{-16}	–	1.5443	2.2220
12811	Erupted upper left lingual deciduous ena	0.159	3.4355×10^{-16}	26.55	–	4.1854×10^{-16}	5.3115×10^{-16}	–	0.8208	0.6468
12814	Erupted upper right buccal deciduous ena	0.149	3.3337×10^{-16}	26.76	–	3.8801×10^{-16}	3.5775×10^{-16}	–	0.8592	0.9319
12815	Erupted upper right lingual deciduous en	0.158	5.0504×10^{-16}	24.54	–	3.8850×10^{-16}	3.8858×10^{-16}	–	1.3000	1.2997
12818	Erupted lower front buccal deciduous ena	0.135	3.3829×10^{-16}	27.46	–	4.8116×10^{-16}	3.8417×10^{-16}	–	0.7031	0.8806
12819	Erupted lower front lingual deciduous en	0.138	2.4034×10^{-16}	24.48	–	2.9536×10^{-16}	4.1066×10^{-16}	–	0.8137	0.5853
12826	Erupted lower left buccal deciduous enam	0.175	3.3696×10^{-16}	34.66	–	4.3277×10^{-16}	3.4561×10^{-16}	–	0.7786	0.9750
12827	Erupted lower left lingual deciduous ena	0.234	3.4010×10^{-16}	21.81	–	3.6967×10^{-16}	5.9289×10^{-16}	–	0.9200	0.5736
12830	Erupted lower right buccal deciduous ena	0.197	6.1385×10^{-16}	20.00	–	5.5846×10^{-16}	4.9495×10^{-16}	–	1.0992	1.2402
12831	Erupted lower right lingual deciduous en	0.213	4.8877×10^{-16}	29.47	–	4.9036×10^{-16}	5.9420×10^{-16}	–	0.9968	0.8226
12833	Erupted upper front deciduous dentin	0.821	3.3295×10^{-16}	11.13	–	3.7471×10^{-16}	5.0488×10^{-16}	–	0.8886	0.6595
12837	Erupted upper left deciduous dentin	0.708	4.1005×10^{-16}	14.30	–	3.6108×10^{-16}	3.8490×10^{-16}	–	1.1356	1.0653
12839	Erupted upper right deciduous dentin	0.708	3.7249×10^{-16}	7.34	–	4.3190×10^{-16}	3.6214×10^{-16}	–	0.8625	1.0286
12841	Erupted lower front deciduous dentin	0.617	5.5687×10^{-16}	11.13	–	4.0852×10^{-16}	4.5164×10^{-16}	–	1.3631	1.2330
12845	Erupted lower left deciduous dentin	0.948	3.7251×10^{-16}	12.97	–	4.6574×10^{-16}	4.9027×10^{-16}	–	0.7998	0.7598
12847	Erupted lower right deciduous dentin	0.948	5.1262×10^{-16}	13.56	–	4.7505×10^{-16}	5.5290×10^{-16}	–	1.0791	0.9271
12848	Unerupted permanent enamel	1.63	5.3650×10^{-16}	8.54	–	3.8621×10^{-16}	4.2800×10^{-16}	–	1.3892	1.2535
12850	Unerupted permanent dentin	5.43	4.0127×10^{-16}	5.44	–	4.0151×10^{-16}	3.9707×10^{-16}	–	0.9994	1.0106
12852	Permanent cementum	0.256	4.5344×10^{-16}	14.29	–	4.1723×10^{-16}	3.7241×10^{-16}	–	1.0868	1.2176
12853	Deciduous cementum	0.242	4.9757×10^{-16}	14.57	–	4.1365×10^{-16}	3.6060×10^{-16}	–	1.2029	1.3798
12854	Permanent pulp	0.163	5.2063×10^{-16}	24.48	–	4.9081×10^{-16}	4.7395×10^{-16}	–	1.0607	1.0985
12855	Deciduous pulp	0.661	5.0839×10^{-16}	8.07	–	4.1022×10^{-16}	4.8764×10^{-16}	–	1.2393	1.0425
12856	Teeth retention region	0.0212	5.7276×10^{-16}	33.27	–	5.6201×10^{-16}	5.0857×10^{-16}	–	1.0191	1.1262
13100	Thymus	31.01	1.9682×10^{-15}	1.35	–	1.9078×10^{-15}	1.9531×10^{-15}	–	1.0317	1.0077
13200	Thyroid	3.87	9.7072×10^{-16}	5.18	–	1.0181×10^{-15}	9.4146×10^{-16}	–	0.9535	1.0311
13300	Tongue upper(food)	9.12	4.1295×10^{-16}	3.29	–	4.2857×10^{-16}	4.1461×10^{-16}	–	0.9635	0.9960
13301	Tongue lower	10.50	5.0408×10^{-16}	2.45	–	5.1330×10^{-16}	5.0329×10^{-16}	–	0.9820	1.0016

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
13400	Tonsils	2.07	4.9475×10^{-16}	9.71	–	4.0739×10^{-16}	4.6211×10^{-16}	–	1.2145	1.0706
13500	Ureter left	2.17	2.1748×10^{-15}	3.96	–	2.2596×10^{-15}	2.0551×10^{-15}	–	0.9625	1.0582
13600	Ureter right	2.17	2.8834×10^{-15}	4.66	–	3.3314×10^{-15}	2.8676×10^{-15}	–	0.8655	1.0055
13700	Urinary bladder wall insensitive	15.42	6.4010×10^{-16}	1.56	–	6.2158×10^{-16}	6.2944×10^{-16}	–	1.0298	1.0169
13701	Urinary bladder wall sensitive(86-193)	0.884	6.5314×10^{-16}	8.05	–	6.7939×10^{-16}	6.0861×10^{-16}	–	0.9614	1.0732
13800	Urinary bladder content	64.70	6.3020×10^{-16}	1.19	–	6.4395×10^{-16}	6.4420×10^{-16}	–	0.9786	0.9783
13900	Uterus	3.10	8.6096×10^{-16}	4.44	–	8.2105×10^{-16}	7.9239×10^{-16}	–	1.0486	1.0865
14000	Air inside body	0.0229	1.8623×10^{-15}	9.87	–	2.1392×10^{-15}	1.8874×10^{-15}	–	0.8705	0.9867

* For tissues bookkept as head, trunk, arm and leg pieces (muscle, residual soft tissue, skin, large vessels), the volumes tabulated in the distributed cell tables partition the tissue at different boundaries from the transported mesh, and the reference output is normalised by the tabulated volume. The marked ratio therefore compares two different partitions of the same tissue, and is left out of the summary statistics of Table S2.

Table S18. 05F_External. Absorbed dose per primary history, Gy. σ is the FLUKA statistical uncertainty. Ratios are FLUKA / reference.

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
100	Adrenal left	2.97	8.5817×10^{-18}	42.24	–	3.1690×10^{-17}	2.6907×10^{-17}	–	0.2708	0.3189
200	Adrenal right	2.97	2.4549×10^{-17}	27.65	–	2.7786×10^{-17}	3.1599×10^{-17}	–	0.8835	0.7769
300	ET1(0-8)	0.00259	0.0000×10^{0}	0.00	–	5.7008×10^{-17}	1.1970×10^{-16}	–	–	–
301	ET1(8-40)	0.0105	0.0000×10^{0}	0.00	–	4.2121×10^{-17}	1.0403×10^{-16}	–	–	–
302	ET1(40-50)	0.00333	0.0000×10^{0}	0.00	–	2.7269×10^{-17}	4.0681×10^{-17}	–	–	–
303	ET1(50-Surface)	0.33	4.8057×10^{-17}	63.95	–	4.5569×10^{-17}	1.8543×10^{-17}	–	1.0546	2.5917
400	ET2(-15-0)	0.129	2.4206×10^{-17}	73.14	–	5.0120×10^{-17}	2.7275×10^{-17}	–	0.4830	0.8875
401	ET2(0-40)	0.367	2.0831×10^{-17}	43.83	–	3.7686×10^{-17}	3.8968×10^{-17}	–	0.5527	0.5346
402	ET2(40-50)	0.0925	4.4249×10^{-17}	57.75	–	2.3862×10^{-17}	2.0083×10^{-17}	–	1.8543	2.2033
403	ET2(50-55)	0.0463	1.3460×10^{-17}	44.60	–	2.5347×10^{-17}	4.7393×10^{-17}	–	0.5310	0.2840
404	ET2(55-65)	0.0928	2.1781×10^{-17}	59.19	–	5.3502×10^{-17}	6.3650×10^{-17}	–	0.4071	0.3422
405	ET2(65-Surface)	19.23	3.0387×10^{-17}	14.02	–	3.2693×10^{-17}	3.9042×10^{-17}	–	0.9295	0.7783
500	Oral mucosa tongue	0.0313	5.2541×10^{-17}	46.69	–	-3.3189×10^{-17}	2.3148×10^{-17}	–	-1.5831	2.2698
501	Oral mucosa moth floor	0.0187	1.8875×10^{-17}	85.58	–	8.7325×10^{-17}	5.0676×10^{-18}	–	0.2162	3.7247
600	Oral mucosa lips and cheeks	0.0144	2.9922×10^{-17}	75.26	–	9.2044×10^{-17}	4.4195×10^{-17}	–	0.3251	0.6770
700	Trachea	2.58	3.3229×10^{-17}	28.27	–	4.0398×10^{-17}	4.9605×10^{-17}	–	0.8225	0.6699
800	BB(-11–6)	0.0022	0.0000×10^{0}	0.00	–	4.8633×10^{-18}	8.5066×10^{-16}	–	–	–
801	BB(-6-0)	0.00281	0.0000×10^{0}	0.00	–	1.0626×10^{-17}	2.2509×10^{-17}	–	–	–
802	BB(0-10)	0.00471	0.0000×10^{0}	0.00	–	6.4813×10^{-18}	3.6850×10^{-16}	–	–	–
803	BB(10-35)	0.0119	0.0000×10^{0}	0.00	–	1.6066×10^{-17}	4.0211×10^{-17}	–	–	–
804	BB(35-40)	0.00242	0.0000×10^{0}	0.00	–	7.1539×10^{-18}	2.7408×10^{-17}	–	–	–
805	BB(40-50)	0.00487	0.0000×10^{0}	0.00	–	1.0500×10^{-17}	2.7334×10^{-17}	–	–	–
806	BB(40-50)	0.0049	0.0000×10^{0}	0.00	–	1.3658×10^{-17}	2.6837×10^{-17}	–	–	–
807	BB(60-70)	0.00494	0.0000×10^{0}	0.00	–	1.6508×10^{-17}	2.6827×10^{-17}	–	–	–
808	BB(70-surface)	3.37	2.4931×10^{-17}	28.50	–	3.1507×10^{-17}	3.5265×10^{-17}	–	0.7913	0.7070
900	Blood in large arteries head	0.275	3.7261×10^{-17}	60.63	–	5.7194×10^{-17}	1.3654×10^{-17}	–	0.6515*	2.7289*
910	Blood in large veins head	1.77	2.7213×10^{-17}	37.16	–	2.7007×10^{-17}	2.5479×10^{-17}	–	1.0076*	1.0681*
1000	Blood in large arteries trunk	38.12	3.2813×10^{-17}	3.51	–	3.7856×10^{-17}	3.2949×10^{-17}	–	0.8668*	0.9959*
1010	Blood in large veins trunk	107.85	3.6240×10^{-17}	4.16	–	3.4045×10^{-17}	3.1222×10^{-17}	–	1.0645*	1.1607*
1100	Blood in large arteries arms	6.62	2.8157×10^{-17}	13.51	–	3.6066×10^{-17}	3.7290×10^{-17}	–	0.7807*	0.7551*
1110	Blood in large veins arms	43.12	3.7116×10^{-17}	7.02	–	3.2844×10^{-17}	3.9469×10^{-17}	–	1.1301*	0.9404*
1200	Blood in large arteries legs	44.93	3.5685×10^{-17}	4.86	–	3.1984×10^{-17}	3.3693×10^{-17}	–	1.1157*	1.0591*
1210	Blood in large veins legs	117.10	3.1536×10^{-17}	3.61	–	3.2222×10^{-17}	3.2215×10^{-17}	–	0.9787*	0.9789*
1300	Humeri upper cortical	28.83	3.7662×10^{-17}	9.87	–	2.9245×10^{-17}	3.2566×10^{-17}	–	1.2878	1.1565
1400	Humeri upper spogiosa	20.16	3.1150×10^{-17}	11.89	–	3.3784×10^{-17}	2.7028×10^{-17}	–	0.9220	1.1525

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
1500	Humeri upper medullary cavity	3.76	3.1513×10^{-17}	22.13	–	1.3998×10^{-17}	4.7074×10^{-17}	–	2.2513	0.6694
1600	Humeri lower cortical	22.27	3.4927×10^{-17}	10.18	–	3.4556×10^{-17}	3.3353×10^{-17}	–	1.0108	1.0472
1700	Humeri lower spongiosa	11.03	3.8474×10^{-17}	15.15	–	3.5117×10^{-17}	3.2281×10^{-17}	–	1.0956	1.1918
1800	Humeri lower medullary cavity	3.56	4.2215×10^{-17}	30.74	–	3.1287×10^{-17}	3.6551×10^{-17}	–	1.3493	1.1550
1900	Radii cortical	14.85	3.1829×10^{-17}	14.37	–	3.3136×10^{-17}	3.3523×10^{-17}	–	0.9606	0.9495
1910	Ulnae cortical	18.78	3.7147×10^{-17}	9.73	–	4.4153×10^{-17}	3.0665×10^{-17}	–	0.8413	1.2114
2000	Radii spongiosa	3.31	4.5300×10^{-17}	24.64	–	4.1882×10^{-17}	5.5819×10^{-17}	–	1.0816	0.8115
2010	Ulnae spongiosa	5.49	3.2421×10^{-17}	17.76	–	3.3213×10^{-17}	2.8051×10^{-17}	–	0.9761	1.1558
2100	Radii medullary cavity	2.30	5.4425×10^{-17}	19.99	–	7.0211×10^{-17}	5.4268×10^{-17}	–	0.7752	1.0029
2110	Ulnae medullary cavity	2.63	5.4542×10^{-17}	25.07	–	4.5031×10^{-17}	3.7184×10^{-17}	–	1.2112	1.4668
2200	Wrists and hand bones cortical	9.33	3.7493×10^{-17}	12.63	–	2.7213×10^{-17}	2.5997×10^{-17}	–	1.3778	1.4422
2300	Wrists and hand bones spongiosa	16.48	2.9533×10^{-17}	9.35	–	4.0508×10^{-17}	3.0007×10^{-17}	–	0.7291	0.9842
2400	Clavicles cortical	6.18	3.8948×10^{-17}	15.21	–	3.7413×10^{-17}	3.3934×10^{-17}	–	1.0410	1.1478
2500	Clavicles spongiosa	7.96	3.0319×10^{-17}	18.45	–	3.3778×10^{-17}	3.4469×10^{-17}	–	0.8976	0.8796
2600	Cranium cortical	229.12	2.3737×10^{-17}	1.68	–	2.5040×10^{-17}	2.6495×10^{-17}	–	0.9479	0.8959
2700	Cranium spongiosa	520.56	2.3132×10^{-17}	2.09	–	2.3603×10^{-17}	2.2861×10^{-17}	–	0.9801	1.0119
2800	Femora upper cortical	49.47	3.0202×10^{-17}	7.74	–	2.9863×10^{-17}	3.3682×10^{-17}	–	1.0114	0.8967
2900	Femora upper spongiosa	26.10	3.0762×10^{-17}	8.74	–	3.1279×10^{-17}	2.7815×10^{-17}	–	0.9835	1.1059
3000	Femora upper medullary cavity	8.22	3.3227×10^{-17}	16.19	–	3.8984×10^{-17}	3.4653×10^{-17}	–	0.8523	0.9589
3100	Femora lower cortical	48.06	3.0906×10^{-17}	9.17	–	2.7431×10^{-17}	2.8383×10^{-17}	–	1.1267	1.0889
3200	Femora lower spongiosa	28.74	2.8772×10^{-17}	6.08	–	2.8831×10^{-17}	2.8043×10^{-17}	–	0.9979	1.0260
3300	Femora lower medullary cavity	5.79	2.8888×10^{-17}	14.97	–	3.0074×10^{-17}	3.7476×10^{-17}	–	0.9605	0.7708
3400	Tibiae cortical	73.04	3.1577×10^{-17}	4.54	–	2.6994×10^{-17}	3.1456×10^{-17}	–	1.1698	1.0039
3410	Fibulae cortical	10.25	2.2012×10^{-17}	12.60	–	2.5130×10^{-17}	1.7934×10^{-17}	–	0.8759	1.2274
3420	Patellae cortical	1.19	4.2623×10^{-17}	31.04	–	6.8438×10^{-17}	2.2466×10^{-17}	–	0.6228	1.8972
3500	Tibiae spongiosa	23.03	3.1568×10^{-17}	7.47	–	2.7981×10^{-17}	2.7219×10^{-17}	–	1.1282	1.1598
3510	Fibulae spongiosa	2.79	2.2409×10^{-17}	25.66	–	2.9560×10^{-17}	3.8706×10^{-17}	–	0.7581	0.5789
3520	Patellae spongiosa	5.57	3.6610×10^{-17}	12.77	–	4.1000×10^{-17}	2.9826×10^{-17}	–	0.8929	1.2275
3600	Tibiae medullary cavity	8.26	3.2036×10^{-17}	13.58	–	2.6812×10^{-17}	2.1240×10^{-17}	–	1.1948	1.5083
3610	Fibulae medullary cavity	1.87	4.4881×10^{-17}	25.12	–	2.0024×10^{-17}	2.6011×10^{-17}	–	2.2414	1.7255
3700	Ankles and foot cortical	33.25	2.5188×10^{-17}	8.80	–	2.8343×10^{-17}	2.5916×10^{-17}	–	0.8887	0.9719
3800	Ankles and foot spongiosa	70.79	2.8675×10^{-17}	7.17	–	2.7909×10^{-17}	2.6840×10^{-17}	–	1.0275	1.0684
3900	Mandible cortical	19.28	3.3373×10^{-17}	11.72	–	3.1849×10^{-17}	3.6759×10^{-17}	–	1.0478	0.9079
4000	Mandible spongiosa	29.48	4.5716×10^{-17}	8.71	–	3.3910×10^{-17}	3.5360×10^{-17}	–	1.3481	1.2929
4100	Pelvis cortical	58.23	3.4841×10^{-17}	3.92	–	3.0824×10^{-17}	2.8480×10^{-17}	–	1.1303	1.2233
4200	Pelvis spongiosa	99.47	2.9344×10^{-17}	3.76	–	3.0620×10^{-17}	3.2745×10^{-17}	–	0.9583	0.8961
4300	Ribs cortical	38.51	2.7180×10^{-17}	3.84	–	2.5232×10^{-17}	2.7052×10^{-17}	–	1.0772	1.0047
4400	Ribs spongiosa	79.11	2.9342×10^{-17}	6.89	–	3.2753×10^{-17}	3.0622×10^{-17}	–	0.8959	0.9582

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
4500	Scapulae cortical	29.30	2.6195×10^{-17}	8.69	–	2.5685×10^{-17}	2.7606×10^{-17}	–	1.0199	0.9489
4600	Scapulae spongiosa	42.72	2.4059×10^{-17}	4.17	–	2.6175×10^{-17}	2.4380×10^{-17}	–	0.9191	0.9868
4700	Cervical spine cortical	6.99	3.1593×10^{-17}	10.18	–	2.3505×10^{-17}	2.5250×10^{-17}	–	1.3441	1.2512
4800	Cervical spine spongiosa	14.08	3.0040×10^{-17}	17.15	–	3.2928×10^{-17}	2.6686×10^{-17}	–	0.9123	1.1257
4900	Thoracic spine cortical	33.10	2.5696×10^{-17}	8.76	–	2.7756×10^{-17}	2.4211×10^{-17}	–	0.9258	1.0613
5000	Thoracic spine spongiosa	74.60	2.4381×10^{-17}	6.07	–	2.4910×10^{-17}	2.6434×10^{-17}	–	0.9787	0.9223
5100	Lumbar spine cortical	16.80	2.3449×10^{-17}	8.79	–	2.6275×10^{-17}	2.5653×10^{-17}	–	0.8925	0.9141
5200	Lumbar spine spongiosa	58.99	3.0139×10^{-17}	6.90	–	2.8981×10^{-17}	2.4338×10^{-17}	–	1.0400	1.2384
5300	Sacrum cortical	11.70	3.0170×10^{-17}	13.37	–	2.3105×10^{-17}	2.2856×10^{-17}	–	1.3058	1.3200
5400	Sacrum spongiosa	34.52	3.0717×10^{-17}	7.73	–	2.5086×10^{-17}	3.1181×10^{-17}	–	1.2244	0.9851
5500	Sternum cortical	2.47	4.4294×10^{-17}	25.52	–	3.1098×10^{-17}	5.5013×10^{-17}	–	1.4243	0.8052
5600	Sternum spongiosa	8.50	4.4811×10^{-17}	15.16	–	3.7608×10^{-17}	4.0178×10^{-17}	–	1.1915	1.1153
5800	Cartilage trunk	55.14	3.4528×10^{-17}	9.66	–	3.1602×10^{-17}	2.9988×10^{-17}	–	1.0926	1.1514
6100	Brain	1244.20	2.2549×10^{-17}	1.28	–	2.2594×10^{-17}	2.2249×10^{-17}	–	0.9980	1.0135
6200	Breast left adipose tissue	1.17	2.3487×10^{-17}	51.99	–	3.0726×10^{-17}	3.5508×10^{-17}	–	0.7644	0.6615
6300	Breast left glandular tissue	0.781	2.4482×10^{-17}	45.38	–	4.6801×10^{-17}	3.0373×10^{-17}	–	0.5231	0.8061
6400	Breast right adipose tissue	1.17	4.7687×10^{-17}	38.10	–	4.1023×10^{-17}	5.8358×10^{-17}	–	1.1624	0.8172
6500	Breast right glandular tissue	0.781	5.1426×10^{-17}	51.15	–	5.4605×10^{-17}	5.3808×10^{-17}	–	0.9418	0.9557
6600	Eye lens sensitive left	0.0202	0.0000×10^{0}	0.00	–	–	5.0013×10^{-17}	–	–	–
6601	Eye lens insensitive left	0.114	1.7054×10^{-17}	100.00	–	1.1535×10^{-17}	7.7190×10^{-17}	–	1.4784	0.2209
6700	Cornea left	0.87	3.1396×10^{-17}	31.17	–	2.1568×10^{-17}	4.7272×10^{-17}	–	1.4557	0.6642
6701	Aqueous left	0.294	7.7866×10^{-17}	55.04	–	2.4051×10^{-17}	1.9857×10^{-17}	–	3.2376	3.9213
6702	Vitreous left	4.40	2.5657×10^{-17}	24.37	–	3.6872×10^{-17}	4.9023×10^{-17}	–	0.6958	0.5234
6800	Eye lens sensitive right	0.0202	0.0000×10^{0}	0.00	–	2.1172×10^{-17}	–	–	–	–
6801	Eye lens insensitive right	0.114	0.0000×10^{0}	0.00	–	1.0572×10^{-16}	–	–	–	–
6900	Cornea right	0.87	6.1464×10^{-17}	29.69	–	2.8223×10^{-17}	2.1659×10^{-17}	–	2.1778	2.8378
6901	Aqueous right	0.294	1.1098×10^{-17}	100.00	–	4.8077×10^{-17}	9.9208×10^{-18}	–	0.2308	1.1186
6902	Vitreous right	4.40	4.1475×10^{-17}	26.86	–	4.3179×10^{-17}	3.5826×10^{-17}	–	0.9605	1.1577
7000	Gall bladder wall	2.69	3.6069×10^{-17}	22.00	–	3.7638×10^{-17}	4.5922×10^{-17}	–	0.9583	0.7854
7100	Gall bladder contents	15.00	3.6238×10^{-17}	8.72	–	4.2618×10^{-17}	3.9891×10^{-17}	–	0.8503	0.9084
7200	Stomach wall(0-60)	0.745	2.5541×10^{-17}	19.56	–	4.9716×10^{-17}	2.8534×10^{-17}	–	0.5137	0.8951
7201	Stomach wall(60-100)	0.489	3.4755×10^{-17}	25.11	–	3.9065×10^{-17}	2.9188×10^{-17}	–	0.8897	1.1907
7202	Stomach wall(100-300)	2.47	4.5539×10^{-17}	13.38	–	3.5559×10^{-17}	2.9167×10^{-17}	–	1.2807	1.5613
7203	Stomach wall(300-surface)	57.96	4.0885×10^{-17}	4.46	–	3.6567×10^{-17}	3.4253×10^{-17}	–	1.1181	1.1936
7300	Stomach contents	83.00	3.8228×10^{-17}	4.82	–	3.5589×10^{-17}	3.6340×10^{-17}	–	1.0741	1.0520
7400	Small intestine wall(0-130)	6.59	3.1161×10^{-17}	7.92	–	3.4374×10^{-17}	3.1130×10^{-17}	–	0.9065	1.0010
7401	Small intestine wall(130-150)	1.03	2.7671×10^{-17}	21.37	–	3.4756×10^{-17}	3.7194×10^{-17}	–	0.7962	0.7440
7402	Small intestine wall(150-200)	2.60	3.6245×10^{-17}	7.18	–	3.7123×10^{-17}	3.1164×10^{-17}	–	0.9763	1.1630

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
7403	Small intestine wall(200-surface)	266.74	3.8928×10^{-17}	3.03	–	3.6854×10^{-17}	3.7417×10^{-17}	–	1.0563	1.0404
7500	Small intestine contents(-400-0)	19.02	3.7175×10^{-17}	7.45	–	3.6879×10^{-17}	3.1417×10^{-17}	–	1.0080	1.1833
7501	Small intestine contents(centre–400)	97.98	3.8221×10^{-17}	5.14	–	3.7092×10^{-17}	3.9004×10^{-17}	–	1.0304	0.9799
7600	Ascending colon wall(0-280)	1.91	3.0697×10^{-17}	25.38	–	4.0266×10^{-17}	3.9738×10^{-17}	–	0.7624	0.7725
7601	Ascending colon wall(280-300)	0.139	2.3032×10^{-17}	54.45	–	8.1013×10^{-17}	3.6458×10^{-17}	–	0.2843	0.6317
7602	Ascending colon wall(300-surface)	26.67	3.4576×10^{-17}	8.34	–	3.9535×10^{-17}	3.6588×10^{-17}	–	0.8746	0.9450
7700	Ascending colon content	31.35	3.1210×10^{-17}	7.82	–	3.1440×10^{-17}	3.2281×10^{-17}	–	0.9927	0.9668
7800	Transverse colon wall right(0-280)	1.58	2.3134×10^{-17}	36.54	–	3.3008×10^{-17}	1.9596×10^{-17}	–	0.7009	1.1805
7801	Transverse colon wall right(280-300)	0.116	6.7193×10^{-17}	36.99	–	3.5123×10^{-17}	3.3235×10^{-17}	–	1.9131	2.0218
7802	Transverse colon wall right(300-surface)	31.23	3.7024×10^{-17}	8.59	–	3.2476×10^{-17}	3.8682×10^{-17}	–	1.1400	0.9571
7900	Transverse colon contents right	18.65	3.7429×10^{-17}	12.07	–	3.8654×10^{-17}	3.4914×10^{-17}	–	0.9683	1.0720
8000	Transverse colon wall left(0-280)	1.10	3.2210×10^{-17}	23.88	–	2.9213×10^{-17}	4.4770×10^{-17}	–	1.1026	0.7195
8001	Transverse colon wall left(280-300)	0.0808	1.5130×10^{-17}	36.80	–	1.8973×10^{-17}	6.7323×10^{-17}	–	0.7974	0.2247
8002	Transverse colon wall left(300-surface)	27.08	3.8490×10^{-17}	7.80	–	3.8363×10^{-17}	3.9389×10^{-17}	–	1.0033	0.9772
8100	Transverse colon content left	10.92	4.0814×10^{-17}	12.03	–	3.8996×10^{-17}	3.1172×10^{-17}	–	1.0466	1.3093
8200	Descending colon wall(0-280)	1.50	3.6023×10^{-17}	25.96	–	3.5602×10^{-17}	2.6271×10^{-17}	–	1.0118	1.3712
8201	Descending colon wall(280-300)	0.11	3.8478×10^{-17}	60.09	–	3.9423×10^{-17}	2.9160×10^{-17}	–	0.9760	1.3195
8202	Descending colon wall(300-surface)	31.75	3.1311×10^{-17}	7.93	–	3.8192×10^{-17}	2.9767×10^{-17}	–	0.8198	1.0519
8300	Descending colon content	14.08	3.5765×10^{-17}	11.14	–	2.8815×10^{-17}	3.4188×10^{-17}	–	1.2412	1.0461
8400	Sigmoid colon wall(0-280)	1.81	3.4890×10^{-17}	27.62	–	3.3948×10^{-17}	3.4734×10^{-17}	–	1.0278	1.0045
8401	Sigmoid colon wall(280-300)	0.133	1.1839×10^{-17}	44.58	–	2.7102×10^{-17}	2.4072×10^{-17}	–	0.4368	0.4918
8402	Sigmoid colon wall(300-surface)	22.78	3.2593×10^{-17}	11.30	–	2.9813×10^{-17}	2.9028×10^{-17}	–	1.0932	1.1228
8500	Sigmoid colon contents	17.99	3.6092×10^{-17}	10.49	–	3.3092×10^{-17}	1.9876×10^{-17}	–	1.0907	1.8158
8600	Rectum wall(0-280)	0.717	2.3563×10^{-17}	57.34	–	4.7296×10^{-17}	2.6784×10^{-17}	–	0.4982	0.8798
8601	Rectum wall(280-300)	0.0527	4.8929×10^{-17}	72.14	–	4.0663×10^{-17}	8.2589×10^{-18}	–	1.2033	5.9244
8602	Rectum wall(300-surface)	2.18	2.2611×10^{-17}	29.13	–	3.8542×10^{-17}	2.2268×10^{-17}	–	0.5867	1.0154
8603	Rectum contents	7.01	3.0972×10^{-17}	23.63	–	3.1315×10^{-17}	3.1550×10^{-17}	–	0.9890	0.9817
8700	Heart wall	97.62	3.1789×10^{-17}	4.29	–	3.4810×10^{-17}	3.2945×10^{-17}	–	0.9132	0.9649
8800	Blood in heart chamber	135.00	3.3935×10^{-17}	2.96	–	3.3082×10^{-17}	3.2540×10^{-17}	–	1.0258	1.0429
8900	Kidney left cortex	49.85	2.5491×10^{-17}	6.04	–	2.4624×10^{-17}	2.6693×10^{-17}	–	1.0352	0.9550
9000	Kidney left medulla	17.81	2.4437×10^{-17}	14.73	–	3.2916×10^{-17}	2.5828×10^{-17}	–	0.7424	0.9461
9100	Kidney left pelvis	3.56	3.0599×10^{-17}	34.03	–	1.6760×10^{-17}	2.2560×10^{-17}	–	1.8257	1.3563
9200	Kidney right cortex	49.85	2.7311×10^{-17}	7.83	–	2.6997×10^{-17}	2.6989×10^{-17}	–	1.0117	1.0119
9300	Kidney right medulla	17.81	2.8292×10^{-17}	10.86	–	3.1422×10^{-17}	3.0377×10^{-17}	–	0.9004	0.9314
9400	Kidney right pelvis	3.56	3.1296×10^{-17}	30.04	–	2.2811×10^{-17}	2.5930×10^{-17}	–	1.3719	1.2069
9500	Liver	723.34	3.5681×10^{-17}	1.54	–	3.3892×10^{-17}	3.3930×10^{-17}	–	1.0528	1.0516

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
9700	Lung(AI) left	139.44	2.8046×10^{-17}	3.63	–	2.9393×10^{-17}	3.1180×10^{-17}	–	0.9542	0.8995
9900	Lung(AI) right	160.56	3.2386×10^{-17}	2.94	–	3.2424×10^{-17}	3.4702×10^{-17}	–	0.9988	0.9333
10000	Lymphatic nodes ET	4.31	2.1075×10^{-17}	27.73	–	3.5566×10^{-17}	2.3582×10^{-17}	–	0.5925	0.8937
10100	Lymphatic nodes thoracic	4.31	2.8673×10^{-17}	23.87	–	3.2799×10^{-17}	3.6418×10^{-17}	–	0.8742	0.7873
10200	Lymphatic nodes head	1.49	4.8953×10^{-17}	30.36	–	1.4725×10^{-17}	5.6086×10^{-17}	–	3.3245	0.8728
10300	Lymphatic nodes trunk	35.19	3.4443×10^{-17}	4.85	–	3.8689×10^{-17}	3.5303×10^{-17}	–	0.8902	0.9756
10400	Lymphatic nodes arms	2.98	3.4481×10^{-17}	28.74	–	3.0932×10^{-17}	3.2028×10^{-17}	–	1.1147	1.0766
10500	Lymphatic nodes legs	2.98	3.1409×10^{-17}	25.36	–	2.9414×10^{-17}	4.0324×10^{-17}	–	1.0678	0.7789
10600	Muscle head	448.65	2.9272×10^{-17}	1.19	–	3.0411×10^{-17}	3.0782×10^{-17}	–	0.9625*	0.9509*
10700	Muscle trunk	2449.49	3.1449×10^{-17}	0.69	–	3.1182×10^{-17}	3.0703×10^{-17}	–	1.0086*	1.0243*
10800	Muscle arms	463.65	3.6389×10^{-17}	1.86	–	3.5212×10^{-17}	3.3464×10^{-17}	–	1.0334*	1.0874*
10900	Muscle legs	2364.96	3.1358×10^{-17}	0.88	–	3.2091×10^{-17}	3.1360×10^{-17}	–	0.9771*	0.9999*
11000	Oesophagus wall(0-190)	0.895	4.3254×10^{-17}	32.33	–	3.6417×10^{-17}	2.7319×10^{-17}	–	1.1878	1.5833
11001	Oesophagus wall(190-200)	0.0483	2.2978×10^{-17}	41.16	–	4.0732×10^{-17}	1.0029×10^{-17}	–	0.5641	2.2912
11002	Oesophagus wall(200-surface)	11.39	3.0273×10^{-17}	13.17	–	3.6550×10^{-17}	2.3989×10^{-17}	–	0.8282	1.2619
11003	Oesophagus contents	7.14	2.6686×10^{-17}	16.02	–	3.0546×10^{-17}	3.0903×10^{-17}	–	0.8736	0.8635
11100	Ovary left	1.06	4.7449×10^{-17}	35.05	–	4.1463×10^{-17}	2.2547×10^{-17}	–	1.1444	2.1045
11200	Ovary right	1.06	5.0325×10^{-17}	33.62	–	7.7456×10^{-17}	6.0522×10^{-17}	–	0.6497	0.8315
11300	Pancreas	41.88	2.9334×10^{-17}	11.70	–	3.3259×10^{-17}	3.5304×10^{-17}	–	0.8820	0.8309
11400	Pituitary gland	0.259	2.5802×10^{-17}	71.88	–	8.3172×10^{-17}	8.8456×10^{-17}	–	0.3102	0.2917
11600	RST head	529.93	2.9415×10^{-17}	1.04	–	2.6959×10^{-17}	3.0128×10^{-17}	–	1.0911	0.9763
11700	RST trunk	3518.43	3.2792×10^{-17}	1.44	–	3.3215×10^{-17}	3.3290×10^{-17}	–	0.9872*	0.9850*
11800	RST arms	550.67	3.7012×10^{-17}	2.53	–	3.7793×10^{-17}	3.7105×10^{-17}	–	0.9793*	0.9975*
11900	RST legs	1779.66	3.3607×10^{-17}	1.20	–	3.3149×10^{-17}	3.3699×10^{-17}	–	1.0138*	0.9973*
12000	Salivary glands left	17.57	2.3325×10^{-17}	16.27	–	2.5663×10^{-17}	2.9851×10^{-17}	–	0.9089	0.7814
12100	Salivary glandss right	17.57	2.6597×10^{-17}	11.59	–	3.1951×10^{-17}	3.4756×10^{-17}	–	0.8325	0.7653
12200	Skin head insensitive	79.46	2.2744×10^{-17}	5.91	–	2.0691×10^{-17}	2.2972×10^{-17}	–	1.0992	0.9901
12201	Skin head sensitive(40-100)	6.78	1.5084×10^{-17}	16.04	–	1.5037×10^{-17}	1.7804×10^{-17}	–	1.0031	0.8472
12300	Skin trunk insensitive	199.43	2.4808×10^{-17}	3.86	–	2.3213×10^{-17}	2.3436×10^{-17}	–	1.0687*	1.0585*
12301	Skin trunk sensitive(40-100)	17.99	1.7252×10^{-17}	7.36	–	1.6158×10^{-17}	1.8130×10^{-17}	–	1.0677*	0.9515*
12400	Skin arms insensitive	72.70	2.7777×10^{-17}	5.06	–	2.8538×10^{-17}	2.7073×10^{-17}	–	0.9733*	1.0260*
12401	Skin arms sensitive(40-100)	5.85	2.2334×10^{-17}	11.01	–	2.2042×10^{-17}	2.0605×10^{-17}	–	1.0133*	1.0839*
12500	Skin legs insensitive	197.85	2.4557×10^{-17}	4.19	–	2.4238×10^{-17}	2.4819×10^{-17}	–	1.0132*	0.9894*
12501	Skin legs sensitive(40-100)	16.35	1.9378×10^{-17}	5.20	–	1.9294×10^{-17}	1.7983×10^{-17}	–	1.0044*	1.0776*
12600	Spinal cord	23.12	2.0870×10^{-17}	10.31	–	3.2857×10^{-17}	2.3120×10^{-17}	–	0.6352	0.9027
12700	Spleen	71.26	2.7962×10^{-17}	2.78	–	2.5561×10^{-17}	2.5380×10^{-17}	–	1.0939	1.1017
12802	Erupted upper front buccal deciduous ena	0.193	5.5430×10^{-17}	90.21	–	8.2420×10^{-18}	6.5000×10^{-17}	–	6.7253	0.8528

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
12803	Erupted upper front lingual deciduous en	0.165	5.1223×10^{-17}	66.75	–	1.4632×10^{-17}	1.5606×10^{-17}	–	3.5006	3.2822
12810	Erupted upper left buccal deciduous enam	0.149	1.1922×10^{-16}	56.85	–	5.6604×10^{-18}	6.4473×10^{-17}	–	21.0620	1.8491
12811	Erupted upper left lingual deciduous ena	0.159	0.0000×10^{0}	0.00	–	2.0699×10^{-16}	7.9965×10^{-18}	–	–	–
12814	Erupted upper right buccal deciduous ena	0.149	0.0000×10^{0}	0.00	–	–	3.8023×10^{-17}	–	–	–
12815	Erupted upper right lingual deciduous en	0.158	4.2075×10^{-18}	100.00	–	7.5193×10^{-17}	6.1310×10^{-17}	–	0.0560	0.0686
12818	Erupted lower front buccal deciduous ena	0.135	8.0731×10^{-18}	100.00	–	3.3826×10^{-17}	–	–	0.2387	–
12819	Erupted lower front lingual deciduous en	0.138	5.1799×10^{-17}	91.40	–	–	2.2459×10^{-17}	–	–	2.3064
12826	Erupted lower left buccal deciduous enam	0.175	6.5300×10^{-17}	85.31	–	8.0066×10^{-17}	4.8554×10^{-17}	–	0.8156	1.3449
12827	Erupted lower left lingual deciduous ena	0.234	3.4187×10^{-18}	84.07	–	1.0038×10^{-18}	8.2126×10^{-17}	–	3.4058	0.0416
12830	Erupted lower right buccal deciduous ena	0.197	1.4931×10^{-17}	100.00	–	8.1941×10^{-18}	5.1163×10^{-17}	–	1.8222	0.2918
12831	Erupted lower right lingual deciduous en	0.213	0.0000×10^{0}	0.00	–	8.4971×10^{-17}	–	–	–	–
12833	Erupted upper front deciduous dentin	0.821	3.4159×10^{-17}	42.89	–	2.5567×10^{-17}	4.9446×10^{-17}	–	1.3361	0.6908
12837	Erupted upper left deciduous dentin	0.708	4.3188×10^{-17}	48.60	–	3.2329×10^{-17}	5.4150×10^{-17}	–	1.3359	0.7976
12839	Erupted upper right deciduous dentin	0.708	4.8212×10^{-17}	45.68	–	1.9667×10^{-17}	7.4802×10^{-18}	–	2.4515	6.4453
12841	Erupted lower front deciduous dentin	0.617	2.8651×10^{-17}	57.40	–	2.5919×10^{-17}	3.3098×10^{-17}	–	1.1054	0.8656
12845	Erupted lower left deciduous dentin	0.948	1.6795×10^{-17}	42.91	–	3.3384×10^{-17}	2.7636×10^{-17}	–	0.5031	0.6077
12847	Erupted lower right deciduous dentin	0.948	3.8123×10^{-17}	55.43	–	3.4812×10^{-17}	3.1878×10^{-17}	–	1.0951	1.1959
12848	Unerupted permanent enamel	1.63	3.3874×10^{-17}	32.76	–	3.3839×10^{-17}	4.4040×10^{-17}	–	1.0010	0.7692
12850	Unerupted permanent dentin	5.43	3.5977×10^{-17}	28.20	–	4.4792×10^{-17}	4.5076×10^{-17}	–	0.8032	0.7981
12852	Permanent cementum	0.256	4.7361×10^{-17}	35.08	–	3.9820×10^{-17}	3.0825×10^{-17}	–	1.1894	1.5364
12853	Deciduous cementum	0.242	4.5764×10^{-17}	40.25	–	3.0939×10^{-17}	4.1652×10^{-17}	–	1.4792	1.0987
12854	Permanent pulp	0.163	1.7367×10^{-17}	73.30	–	–	1.5320×10^{-16}	–	–	0.1134
12855	Deciduous pulp	0.661	3.4786×10^{-17}	56.55	–	7.0050×10^{-17}	4.4256×10^{-17}	–	0.4966	0.7860
12856	Teeth retention region	0.0212	1.6757×10^{-17}	40.25	–	3.7011×10^{-17}	1.0678×10^{-17}	–	0.4528	1.5693
13100	Thymus	31.01	3.5421×10^{-17}	5.64	–	3.7823×10^{-17}	3.5984×10^{-17}	–	0.9365	0.9844
13200	Thyroid	3.87	3.7943×10^{-17}	14.26	–	4.4151×10^{-17}	3.9739×10^{-17}	–	0.8594	0.9548
13300	Tongue upper(food)	9.12	3.4745×10^{-17}	17.41	–	3.6328×10^{-17}	3.5905×10^{-17}	–	0.9564	0.9677
13301	Tongue lower	10.50	3.0550×10^{-17}	20.05	–	2.0707×10^{-17}	3.0473×10^{-17}	–	1.4753	1.0025

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
13400	Tonsils	2.07	1.0876×10^{-17}	34.55	–	2.0103×10^{-17}	2.3272×10^{-17}	–	0.5410	0.4673
13500	Ureter left	2.17	2.1649×10^{-17}	43.02	–	3.4036×10^{-17}	3.4157×10^{-17}	–	0.6360	0.6338
13600	Ureter right	2.17	4.6784×10^{-17}	25.34	–	3.4645×10^{-17}	3.2372×10^{-17}	–	1.3504	1.4452
13700	Urinary bladder wall insensitive	15.42	3.9672×10^{-17}	11.13	–	4.1759×10^{-17}	3.2505×10^{-17}	–	0.9500	1.2205
13701	Urinary bladder wall sensitive(86-193)	0.884	3.7661×10^{-17}	18.30	–	3.5693×10^{-17}	3.6022×10^{-17}	–	1.0551	1.0455
13800	Urinary bladder content	64.70	3.8304×10^{-17}	4.13	–	3.8747×10^{-17}	4.2119×10^{-17}	–	0.9886	0.9094
13900	Uterus	3.10	3.0169×10^{-17}	31.32	–	2.7637×10^{-17}	2.4287×10^{-17}	–	1.0916	1.2422
14000	Air inside body	0.0229	1.2486×10^{-17}	36.65	–	1.2407×10^{-16}	2.0086×10^{-17}	–	0.1006	0.6216

* For tissues bookkept as head, trunk, arm and leg pieces (muscle, residual soft tissue, skin, large vessels), the volumes tabulated in the distributed cell tables partition the tissue at different boundaries from the transported mesh, and the reference output is normalised by the tabulated volume. The marked ratio therefore compares two different partitions of the same tissue, and is left out of the summary statistics of Table S2.

Table S19. 10M_Internal. Absorbed dose per primary history, Gy. σ is the FLUKA statistical uncertainty. Ratios are FLUKA / reference.

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
100	Adrenal left	4.18	3.1836×10^{-15}	2.67	–	3.2109×10^{-15}	3.2436×10^{-15}	–	0.9915	0.9815
200	Adrenal right	4.18	7.9878×10^{-15}	1.91	–	7.5718×10^{-15}	7.7492×10^{-15}	–	1.0549	1.0308
300	ET1(0-8)	0.0026	4.2181×10^{-17}	69.48	–	2.0906×10^{-17}	1.2308×10^{-16}	–	2.0176	0.3427
301	ET1(8-40)	0.0105	3.9683×10^{-16}	53.26	–	3.3459×10^{-17}	1.6761×10^{-16}	–	11.8604	2.3676
302	ET1(40-50)	0.00334	9.1520×10^{-17}	54.63	–	5.4078×10^{-17}	5.8647×10^{-16}	–	1.6924	0.1561
303	ET1(50-Surface)	0.667	1.9993×10^{-16}	18.90	–	1.6457×10^{-16}	1.4628×10^{-16}	–	1.2149	1.3668
400	ET2(-15-0)	0.138	1.8519×10^{-16}	14.60	–	2.0150×10^{-16}	2.1839×10^{-16}	–	0.9190	0.8480
401	ET2(0-40)	0.395	1.8614×10^{-16}	12.14	–	1.9850×10^{-16}	2.3237×10^{-16}	–	0.9377	0.8010
402	ET2(40-50)	0.0995	2.5708×10^{-16}	24.59	–	2.5832×10^{-16}	2.4890×10^{-16}	–	0.9952	1.0329
403	ET2(50-55)	0.0498	2.2328×10^{-16}	32.29	–	1.9023×10^{-16}	2.9103×10^{-16}	–	1.1737	0.7672
404	ET2(55-65)	0.0999	2.0174×10^{-16}	25.03	–	1.9408×10^{-16}	2.4704×10^{-16}	–	1.0395	0.8166
405	ET2(65-Surface)	19.65	2.1561×10^{-16}	3.04	–	2.0396×10^{-16}	2.0091×10^{-16}	–	1.0571	1.0731
500	Oral mucosa tongue	0.0509	3.5689×10^{-16}	11.41	–	3.8978×10^{-16}	2.3683×10^{-16}	–	0.9156	1.5069
501	Oral mucosa moth floor	0.0245	2.8595×10^{-16}	33.25	–	3.5391×10^{-16}	1.8677×10^{-16}	–	0.8080	1.5310
600	Oral mucosa lips and cheeks	0.0211	3.1964×10^{-16}	16.52	–	2.0103×10^{-16}	1.5551×10^{-16}	–	1.5900	2.0554
700	Trachea	4.70	8.7545×10^{-16}	4.74	–	7.8398×10^{-16}	8.5085×10^{-16}	–	1.1167	1.0289
800	BB(-11–6)	0.00499	8.9028×10^{-16}	23.61	–	1.4863×10^{-15}	1.4111×10^{-15}	–	0.5990	0.6309
801	BB(-6-0)	0.00639	1.7445×10^{-15}	46.85	–	1.4433×10^{-15}	1.5325×10^{-15}	–	1.2086	1.1383
802	BB(0-10)	0.0107	1.5865×10^{-15}	31.12	–	1.4993×10^{-15}	9.8560×10^{-16}	–	1.0582	1.6097
803	BB(10-35)	0.0268	1.6522×10^{-15}	24.11	–	1.6888×10^{-15}	1.3544×10^{-15}	–	0.9783	1.2199
804	BB(35-40)	0.00539	2.2231×10^{-15}	41.68	–	1.4524×10^{-15}	9.0087×10^{-16}	–	1.5307	2.4677
805	BB(40-50)	0.0108	9.2433×10^{-16}	31.65	–	1.3316×10^{-15}	1.5836×10^{-15}	–	0.6941	0.5837
806	BB(50-60)	0.0109	1.3754×10^{-15}	28.82	–	1.4624×10^{-15}	1.2807×10^{-15}	–	0.9405	1.0739
807	BB(60-70)	0.0109	1.1332×10^{-15}	24.95	–	3.0887×10^{-15}	1.0115×10^{-15}	–	0.3669	1.1203
808	BB(70-surface)	2.82	1.5228×10^{-15}	3.43	–	1.5109×10^{-15}	1.6179×10^{-15}	–	1.0079	0.9412
900	Blood in large arteries head	0.631	3.2054×10^{-16}	12.38	–	3.1587×10^{-16}	3.8067×10^{-16}	–	1.0148*	0.8420*
910	Blood in large veins head	1.91	2.8140×10^{-16}	6.49	–	2.7990×10^{-16}	2.6000×10^{-16}	–	1.0054	1.0823
1000	Blood in large arteries trunk	58.29	2.0550×10^{-15}	0.81	–	2.0747×10^{-15}	2.0715×10^{-15}	–	0.9905*	0.9920*
1010	Blood in large veins trunk	135.44	1.7810×10^{-15}	0.46	–	1.8135×10^{-15}	1.8466×10^{-15}	–	0.9821*	0.9645*
1100	Blood in large arteries arms	12.25	1.3698×10^{-15}	2.82	–	1.3850×10^{-15}	1.3504×10^{-15}	–	0.9890*	1.0144*
1110	Blood in large veins arms	90.31	8.8544×10^{-16}	1.20	–	8.7115×10^{-16}	8.9183×10^{-16}	–	1.0164*	0.9928*
1200	Blood in large arteries legs	78.73	2.7481×10^{-17}	6.37	–	2.8105×10^{-17}	3.1032×10^{-17}	–	0.9778	0.8856
1210	Blood in large veins legs	222.01	2.9074×10^{-17}	3.31	–	3.0012×10^{-17}	2.9735×10^{-17}	–	0.9687	0.9778
1300	Humeri upper cortical	88.31	7.8077×10^{-16}	1.16	–	7.7443×10^{-16}	8.0162×10^{-16}	–	1.0082	0.9740
1400	Humeri upper spogiosa	52.65	5.3021×10^{-16}	1.09	–	5.4087×10^{-16}	5.4129×10^{-16}	–	0.9803	0.9795

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
1500	Humeri upper medullary cavity	9.79	9.8631×10^{-16}	2.45	–	9.9813×10^{-16}	1.0845×10^{-15}	–	0.9882	0.9095
1600	Humeri lower cortical	67.36	1.3822×10^{-15}	1.18	–	1.3754×10^{-15}	1.3914×10^{-15}	–	1.0050	0.9934
1700	Humeri lower spongiosa	26.26	1.2026×10^{-15}	1.53	–	1.1667×10^{-15}	1.1865×10^{-15}	–	1.0308	1.0136
1800	Humeri lower medullary cavity	9.77	1.5536×10^{-15}	2.36	–	1.5946×10^{-15}	1.5763×10^{-15}	–	0.9743	0.9856
1900	Radii cortical	44.01	3.6426×10^{-16}	1.91	–	3.6071×10^{-16}	3.5262×10^{-16}	–	1.0098	1.0330
1910	Ulnae cortical	56.67	5.4437×10^{-16}	1.15	–	5.3477×10^{-16}	5.4054×10^{-16}	–	1.0180	1.0071
2000	Radii spongiosa	7.24	3.4033×10^{-16}	5.19	–	2.8105×10^{-16}	3.4876×10^{-16}	–	1.2109	0.9758
2010	Ulnae spongiosa	12.63	7.9380×10^{-16}	2.04	–	7.5794×10^{-16}	7.8958×10^{-16}	–	1.0473	1.0053
2100	Radii medullary cavity	6.29	3.7394×10^{-16}	2.98	–	3.9871×10^{-16}	3.6949×10^{-16}	–	0.9379	1.0120
2110	Ulnae medullary cavity	6.91	4.0346×10^{-16}	4.44	–	3.9878×10^{-16}	4.0359×10^{-16}	–	1.0117	0.9997
2200	Wrists and hand bones cortical	32.59	1.0955×10^{-16}	4.17	–	1.0121×10^{-16}	1.1111×10^{-16}	–	1.0824	0.9859
2300	Wrists and hand bones spongiosa	37.06	1.1989×10^{-16}	3.49	–	1.2167×10^{-16}	1.2973×10^{-16}	–	0.9854	0.9242
2400	Clavicles cortical	14.38	6.7805×10^{-16}	3.17	–	6.8844×10^{-16}	6.5117×10^{-16}	–	0.9849	1.0413
2500	Clavicles spongiosa	17.02	7.2504×10^{-16}	2.11	–	6.7890×10^{-16}	7.2143×10^{-16}	–	1.0680	1.0050
2600	Cranium cortical	328.09	1.0767×10^{-16}	0.95	–	1.0951×10^{-16}	1.1060×10^{-16}	–	0.9832	0.9735
2700	Cranium spongiosa	497.53	1.1828×10^{-16}	1.07	–	1.1958×10^{-16}	1.1821×10^{-16}	–	0.9892	1.0006
2800	Femora upper cortical	164.62	7.4724×10^{-17}	2.01	–	7.4291×10^{-17}	7.4002×10^{-17}	–	1.0058	1.0098
2900	Femora upper spongiosa	75.47	1.1993×10^{-16}	2.49	–	1.2542×10^{-16}	1.2426×10^{-16}	–	0.9562	0.9652
3000	Femora upper medullary cavity	25.19	4.1018×10^{-17}	9.06	–	3.8960×10^{-17}	3.9262×10^{-17}	–	1.0528	1.0447
3100	Femora lower cortical	152.10	8.3697×10^{-18}	8.28	–	9.0634×10^{-18}	8.5927×10^{-18}	–	0.9235	0.9741
3200	Femora lower spongiosa	90.51	4.9130×10^{-18}	18.03	–	6.5731×10^{-18}	6.7103×10^{-18}	–	0.7474	0.7322
3300	Femora lower medullary cavity	16.91	9.5354×10^{-18}	17.78	–	8.7808×10^{-18}	8.5250×10^{-18}	–	1.0859	1.1185
3400	Tibiae cortical	240.89	3.5961×10^{-18}	9.73	–	3.7479×10^{-18}	3.0058×10^{-18}	–	0.9595	1.1964
3410	Fibulae cortical	32.68	2.9771×10^{-18}	23.06	–	1.1092×10^{-18}	2.2789×10^{-18}	–	2.6839	1.3064
3420	Patellae cortical	2.60	1.2120×10^{-17}	38.40	–	6.0845×10^{-18}	6.1980×10^{-18}	–	1.9920	1.9555
3500	Tibiae spongiosa	62.57	3.7833×10^{-18}	21.82	–	7.5469×10^{-18}	5.2518×10^{-18}	–	0.5013	0.7204
3510	Fibulae spongiosa	7.38	9.7115×10^{-19}	61.43	–	9.3040×10^{-19}	1.8503×10^{-18}	–	1.0438	0.5249
3520	Patellae spongiosa	12.63	4.7212×10^{-18}	24.47	–	8.5487×10^{-18}	9.6235×10^{-18}	–	0.5523	0.4906
3600	Tibiae medullary cavity	26.27	2.6342×10^{-18}	25.75	–	1.8917×10^{-18}	2.8509×10^{-18}	–	1.3925	0.9240
3610	Fibulae medullary cavity	5.58	3.1084×10^{-18}	50.06	–	1.8417×10^{-18}	1.5352×10^{-18}	–	1.6878	2.0248
3700	Ankles and foot cortical	93.86	5.7841×10^{-18}	9.42	–	5.0951×10^{-18}	5.2443×10^{-18}	–	1.1352	1.1029
3800	Ankles and foot spongiosa	159.84	4.7665×10^{-18}	9.83	–	4.3393×10^{-18}	4.3748×10^{-18}	–	1.0984	1.0895
3900	Mandible cortical	24.91	3.2756×10^{-16}	2.37	–	3.0749×10^{-16}	2.9777×10^{-16}	–	1.0653	1.1000
4000	Mandible spongiosa	27.51	3.3242×10^{-16}	3.15	–	3.0447×10^{-16}	3.1451×10^{-16}	–	1.0918	1.0570
4100	Pelvis cortical	128.93	3.4122×10^{-16}	1.10	–	3.4990×10^{-16}	3.3937×10^{-16}	–	0.9752	1.0054
4200	Pelvis spongiosa	285.90	3.1561×10^{-16}	0.83	–	3.1789×10^{-16}	3.1288×10^{-16}	–	0.9928	1.0087
4300	Ribs cortical	70.42	2.8146×10^{-15}	0.52	–	2.8478×10^{-15}	2.8048×10^{-15}	–	0.9883	1.0035
4400	Ribs spongiosa	138.70	2.8126×10^{-15}	0.28	–	2.8102×10^{-15}	2.8348×10^{-15}	–	1.0009	0.9922

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
4500	Scapulae cortical	59.27	7.9018×10^{-16}	0.86	–	7.8164×10^{-16}	7.9606×10^{-16}	–	1.0109	0.9926
4600	Scapulae spongiosa	90.28	7.3782×10^{-16}	0.82	–	7.3851×10^{-16}	7.3352×10^{-16}	–	0.9991	1.0059
4700	Cervical spine cortical	14.13	3.9474×10^{-16}	2.81	–	4.0357×10^{-16}	3.6291×10^{-16}	–	0.9781	1.0877
4800	Cervical spine spongiosa	30.75	3.7589×10^{-16}	2.27	–	3.9318×10^{-16}	3.7967×10^{-16}	–	0.9560	0.9901
4900	Thoracic spine cortical	68.39	2.5787×10^{-15}	0.33	–	2.5685×10^{-15}	2.5931×10^{-15}	–	1.0040	0.9945
5000	Thoracic spine spongiosa	211.32	2.8906×10^{-15}	0.40	–	2.9115×10^{-15}	2.9302×10^{-15}	–	0.9928	0.9865
5100	Lumbar spine cortical	36.75	1.3073×10^{-15}	0.84	–	1.3375×10^{-15}	1.2954×10^{-15}	–	0.9774	1.0092
5200	Lumbar spine spongiosa	161.65	1.3681×10^{-15}	0.58	–	1.3702×10^{-15}	1.3603×10^{-15}	–	0.9985	1.0057
5300	Sacrum cortical	16.67	3.0761×10^{-16}	3.58	–	3.0276×10^{-16}	2.9003×10^{-16}	–	1.0160	1.0606
5400	Sacrum spongiosa	55.24	3.5142×10^{-16}	2.08	–	3.5967×10^{-16}	3.5072×10^{-16}	–	0.9771	1.0020
5500	Sternum cortical	5.28	2.0616×10^{-15}	2.50	–	2.0249×10^{-15}	2.0778×10^{-15}	–	1.0181	0.9922
5600	Sternum spongiosa	20.82	1.9859×10^{-15}	1.60	–	1.9779×10^{-15}	1.9166×10^{-15}	–	1.0040	1.0361
5800	Cartilage trunk	129.15	3.3446×10^{-15}	0.54	–	3.3874×10^{-15}	3.3893×10^{-15}	–	0.9873	0.9868
6100	Brain	1467.08	1.0545×10^{-16}	0.76	–	1.0514×10^{-16}	1.0560×10^{-16}	–	1.0030	0.9986
6200	Breast left adipose tissue	2.04	1.5209×10^{-15}	3.94	–	1.5259×10^{-15}	1.4279×10^{-15}	–	0.9968	1.0652
6300	Breast left glandular tissue	1.36	1.5393×10^{-15}	4.26	–	1.5047×10^{-15}	1.5198×10^{-15}	–	1.0230	1.0128
6400	Breast right adipose tissue	2.04	2.7313×10^{-15}	3.69	–	2.5932×10^{-15}	2.5599×10^{-15}	–	1.0532	1.0669
6500	Breast right glandular tissue	1.36	2.5807×10^{-15}	2.74	–	2.5974×10^{-15}	2.4883×10^{-15}	–	0.9935	1.0371
6600	Eye lens sensitive left	0.0184	1.0027×10^{-16}	100.00	–	7.0563×10^{-17}	–	–	1.4209	–
6601	Eye lens insensitive left	0.113	3.8465×10^{-17}	68.83	–	1.4548×10^{-16}	2.6963×10^{-16}	–	0.2644	0.1427
6700	Cornea left	0.941	8.5910×10^{-17}	22.73	–	1.2927×10^{-16}	1.5736×10^{-16}	–	0.6646	0.5459
6701	Aqueous left	0.327	2.0664×10^{-16}	39.73	–	1.0355×10^{-16}	5.9052×10^{-17}	–	1.9954	3.4992
6702	Vitreous left	4.85	1.3323×10^{-16}	10.15	–	1.2336×10^{-16}	1.4257×10^{-16}	–	1.0800	0.9345
6800	Eye lens sensitive right	0.0184	1.2889×10^{-16}	55.36	–	–	6.0955×10^{-17}	–	–	2.1145
6801	Eye lens insensitive right	0.113	1.6225×10^{-16}	49.84	–	5.0585×10^{-17}	2.0340×10^{-16}	–	3.2075	0.7977
6900	Cornea right	0.942	1.3888×10^{-16}	18.96	–	1.6323×10^{-16}	1.6558×10^{-16}	–	0.8508	0.8387
6901	Aqueous right	0.327	1.6158×10^{-16}	17.25	–	1.1211×10^{-16}	2.2771×10^{-16}	–	1.4413	0.7096
6902	Vitreous right	4.85	1.4856×10^{-16}	9.33	–	1.6376×10^{-16}	1.5925×10^{-16}	–	0.9072	0.9329
7000	Gall bladder wall	4.60	1.4612×10^{-14}	1.08	–	1.4763×10^{-14}	1.4791×10^{-14}	–	0.9898	0.9879
7100	Gall bladder contents	26.00	1.3978×10^{-14}	0.58	–	1.4019×10^{-14}	1.4154×10^{-14}	–	0.9971	0.9876
7200	Stomach wall(0-60)	0.998	5.1949×10^{-15}	1.52	–	5.1787×10^{-15}	5.0948×10^{-15}	–	1.0031	1.0196
7201	Stomach wall(60-100)	0.667	5.1075×10^{-15}	2.02	–	5.1092×10^{-15}	5.0226×10^{-15}	–	0.9997	1.0169
7202	Stomach wall(100-300)	3.36	5.0678×10^{-15}	1.26	–	5.1416×10^{-15}	5.0334×10^{-15}	–	0.9856	1.0068
7203	Stomach wall(300-surface)	100.42	5.2361×10^{-15}	0.33	–	5.2459×10^{-15}	5.2913×10^{-15}	–	0.9981	0.9896
7300	Stomach contents	117.00	4.5083×10^{-15}	0.37	–	4.5512×10^{-15}	4.5638×10^{-15}	–	0.9906	0.9878
7400	Small intestine wall(0-130)	8.74	1.6924×10^{-15}	1.61	–	1.6193×10^{-15}	1.6503×10^{-15}	–	1.0451	1.0255
7401	Small intestine wall(130-150)	1.36	1.6778×10^{-15}	1.79	–	1.6581×10^{-15}	1.6490×10^{-15}	–	1.0118	1.0174
7402	Small intestine wall(150-200)	3.44	1.6620×10^{-15}	1.08	–	1.6337×10^{-15}	1.6648×10^{-15}	–	1.0173	0.9983

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
7403	Small intestine wall(200-surface)	454.79	1.6758×10^{-15}	0.32	–	1.6770×10^{-15}	1.6796×10^{-15}	–	0.9992	0.9977
7500	Small intestine contents(-500-0)	31.33	1.6301×10^{-15}	1.34	–	1.6161×10^{-15}	1.6529×10^{-15}	–	1.0087	0.9862
7501	Small intestine contents(centre–500)	131.67	1.5981×10^{-15}	0.70	–	1.6044×10^{-15}	1.6069×10^{-15}	–	0.9961	0.9945
7600	Ascending colon wall(0-280)	2.59	1.1993×10^{-15}	2.76	–	1.2777×10^{-15}	1.1997×10^{-15}	–	0.9387	0.9997
7601	Ascending colon wall(280-300)	0.188	1.1226×10^{-15}	6.52	–	1.2657×10^{-15}	1.2044×10^{-15}	–	0.8869	0.9321
7602	Ascending colon wall(300-surface)	50.90	1.2686×10^{-15}	1.68	–	1.2847×10^{-15}	1.2508×10^{-15}	–	0.9874	1.0142
7700	Ascending colon content	45.28	1.1374×10^{-15}	1.44	–	1.1429×10^{-15}	1.1382×10^{-15}	–	0.9952	0.9993
7800	Transverse colon wall right(0-280)	1.92	5.0899×10^{-15}	2.80	–	4.9951×10^{-15}	5.0554×10^{-15}	–	1.0190	1.0068
7801	Transverse colon wall right(280-300)	0.14	5.1986×10^{-15}	2.85	–	5.3704×10^{-15}	4.8537×10^{-15}	–	0.9680	1.0711
7802	Transverse colon wall right(300-surface)	51.69	5.0974×10^{-15}	0.42	–	5.1042×10^{-15}	5.1349×10^{-15}	–	0.9987	0.9927
7900	Transverse colon contents right	24.72	4.9910×10^{-15}	1.01	–	4.9982×10^{-15}	5.0386×10^{-15}	–	0.9986	0.9906
8000	Transverse colon wall left(0-280)	1.49	1.9981×10^{-15}	2.72	–	1.9755×10^{-15}	2.0057×10^{-15}	–	1.0114	0.9962
8001	Transverse colon wall left(280-300)	0.109	2.0349×10^{-15}	5.88	–	2.1688×10^{-15}	1.9038×10^{-15}	–	0.9383	1.0689
8002	Transverse colon wall left(300-surface)	45.61	2.0723×10^{-15}	0.93	–	2.0533×10^{-15}	2.0850×10^{-15}	–	1.0093	0.9939
8100	Transverse colon content left	15.05	1.9677×10^{-15}	1.82	–	2.0426×10^{-15}	1.9962×10^{-15}	–	0.9633	0.9857
8200	Descending colon wall(0-280)	2.01	7.0401×10^{-16}	2.49	–	7.1964×10^{-16}	7.7907×10^{-16}	–	0.9783	0.9036
8201	Descending colon wall(280-300)	0.148	7.0017×10^{-16}	6.21	–	8.5389×10^{-16}	7.5230×10^{-16}	–	0.8200	0.9307
8202	Descending colon wall(300-surface)	58.01	7.3478×10^{-16}	1.64	–	7.2889×10^{-16}	7.1415×10^{-16}	–	1.0081	1.0289
8300	Descending colon content	19.95	7.1529×10^{-16}	1.70	–	7.3842×10^{-16}	7.2517×10^{-16}	–	0.9687	0.9864
8400	Sigmoid colon wall(0-280)	2.45	3.5698×10^{-16}	3.04	–	3.3577×10^{-16}	3.3450×10^{-16}	–	1.0632	1.0672
8401	Sigmoid colon wall(280-300)	0.18	3.9764×10^{-16}	9.06	–	3.7309×10^{-16}	3.4728×10^{-16}	–	1.0658	1.1450
8402	Sigmoid colon wall(300-surface)	42.08	3.7666×10^{-16}	2.24	–	3.6966×10^{-16}	3.6247×10^{-16}	–	1.0190	1.0392
8500	Sigmoid colon contents	24.52	3.5465×10^{-16}	2.52	–	3.4491×10^{-16}	3.3973×10^{-16}	–	1.0283	1.0439
8600	Rectum wall(0-280)	1.03	1.4772×10^{-16}	8.76	–	1.5055×10^{-16}	1.4315×10^{-16}	–	0.9811	1.0319
8601	Rectum wall(280-300)	0.0758	1.7968×10^{-16}	26.30	–	1.3026×10^{-16}	2.1486×10^{-16}	–	1.3794	0.8363
8602	Rectum wall(300-surface)	4.71	1.5337×10^{-16}	6.40	–	1.4706×10^{-16}	1.6318×10^{-16}	–	1.0429	0.9399
8603	Rectum contents	10.48	1.4065×10^{-16}	5.47	–	1.7126×10^{-16}	1.6485×10^{-16}	–	0.8213	0.8532
8700	Heart wall	161.50	4.7543×10^{-15}	0.31	–	4.7483×10^{-15}	4.7331×10^{-15}	–	1.0013	1.0045
8800	Blood in heart chamber	230.00	4.1338×10^{-15}	0.22	–	4.1291×10^{-15}	4.1420×10^{-15}	–	1.0011	0.9980
8900	Kidney left cortex	82.03	1.7673×10^{-15}	0.42	–	1.7760×10^{-15}	1.7498×10^{-15}	–	0.9951	1.0100
9000	Kidney left medulla	29.30	1.8277×10^{-15}	1.27	–	1.8317×10^{-15}	1.7969×10^{-15}	–	0.9978	1.0172
9100	Kidney left pelvis	5.86	1.9224×10^{-15}	2.05	–	1.9280×10^{-15}	1.8547×10^{-15}	–	0.9971	1.0365
9200	Kidney right cortex	82.03	5.1109×10^{-15}	0.45	–	5.1134×10^{-15}	5.1542×10^{-15}	–	0.9995	0.9916
9300	Kidney right medulla	29.30	4.5320×10^{-15}	1.01	–	4.5488×10^{-15}	4.5798×10^{-15}	–	0.9963	0.9896
9400	Kidney right pelvis	5.86	4.1828×10^{-15}	1.46	–	4.4255×10^{-15}	4.2058×10^{-15}	–	0.9452	0.9945
9500	Liver	1059.76	1.8979×10^{-14}	0.07	–	1.8960×10^{-14}	1.9023×10^{-14}	–	1.0010	0.9977

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
9700	Lung(AI) left	222.22	1.3463×10^{-15}	0.34	–	1.3346×10^{-15}	1.3515×10^{-15}	–	1.0088	0.9962
9900	Lung(AI) right	277.78	3.6091×10^{-15}	0.28	–	3.6244×10^{-15}	3.6167×10^{-15}	–	0.9958	0.9979
10000	Lymphatic nodes ET	7.16	2.7234×10^{-16}	5.03	–	2.7669×10^{-16}	3.0063×10^{-16}	–	0.9842	0.9059
10100	Lymphatic nodes thoracic	7.16	5.5070×10^{-15}	0.98	–	5.3590×10^{-15}	5.4552×10^{-15}	–	1.0276	1.0095
10200	Lymphatic nodes head	2.47	5.0865×10^{-16}	6.97	–	5.5435×10^{-16}	4.9227×10^{-16}	–	0.9176	1.0333
10300	Lymphatic nodes trunk	58.43	8.5404×10^{-16}	1.40	–	8.5268×10^{-16}	8.4988×10^{-16}	–	1.0016	1.0049
10400	Lymphatic nodes arms	4.94	1.0979×10^{-15}	2.72	–	1.0505×10^{-15}	1.0665×10^{-15}	–	1.0451	1.0294
10500	Lymphatic nodes legs	4.95	8.8034×10^{-18}	36.58	–	8.7768×10^{-18}	5.4036×10^{-18}	–	1.0030	1.6292
10600	Muscle head	402.29	2.3307×10^{-16}	0.90	–	2.3407×10^{-16}	2.3320×10^{-16}	–	0.9957	0.9994
10700	Muscle trunk	5431.32	1.1502×10^{-15}	0.08	–	1.1575×10^{-15}	1.1580×10^{-15}	–	0.9937*	0.9932*
10800	Muscle arms	1166.19	7.8004×10^{-16}	0.39	–	7.7186×10^{-16}	7.7153×10^{-16}	–	1.0106*	1.0110*
10900	Muscle legs	4258.32	3.5779×10^{-17}	0.47	–	3.6999×10^{-17}	3.7006×10^{-17}	–	0.9670	0.9668
11000	Oesophagus wall(0-190)	1.60	2.8268×10^{-15}	2.80	–	2.6549×10^{-15}	2.6829×10^{-15}	–	1.0647	1.0536
11001	Oesophagus wall(190-200)	0.0858	3.1286×10^{-15}	4.94	–	3.1543×10^{-15}	2.8738×10^{-15}	–	0.9919	1.0887
11002	Oesophagus wall(200-surface)	20.60	2.9233×10^{-15}	1.34	–	2.9593×10^{-15}	3.0341×10^{-15}	–	0.9878	0.9635
11003	Oesophagus contents	11.91	3.1019×10^{-15}	1.25	–	3.1040×10^{-15}	3.2488×10^{-15}	–	0.9993	0.9548
11300	Pancreas	72.13	4.0335×10^{-15}	0.57	–	4.0411×10^{-15}	4.0231×10^{-15}	–	0.9981	1.0026
11400	Pituitary gland	0.366	1.0006×10^{-16}	33.22	–	1.0173×10^{-16}	1.5805×10^{-16}	–	0.9837	0.6331
11500	Prostate	1.67	1.3631×10^{-16}	18.77	–	1.5843×10^{-16}	1.4351×10^{-16}	–	0.8604	0.9498
11600	RST head	476.06	1.9265×10^{-16}	1.18	–	1.8899×10^{-16}	1.8885×10^{-16}	–	1.0194	1.0201
11700	RST trunk	6679.34	2.1561×10^{-15}	0.09	–	2.1724×10^{-15}	2.1871×10^{-15}	–	0.9925*	0.9858*
11800	RST arms	629.22	8.0027×10^{-16}	0.26	–	8.0744×10^{-16}	8.1388×10^{-16}	–	0.9911*	0.9833*
11900	RST legs	1784.08	3.8036×10^{-17}	0.98	–	4.0399×10^{-17}	3.9616×10^{-17}	–	0.9415*	0.9601*
12000	Salivary glands left	22.98	2.7263×10^{-16}	4.64	–	2.4677×10^{-16}	2.6645×10^{-16}	–	1.1048	1.0232
12100	Salivary glandss right	22.98	3.0476×10^{-16}	2.10	–	3.1446×10^{-16}	3.0260×10^{-16}	–	0.9691	1.0071
12200	Skin head insensitive	80.29	1.4346×10^{-16}	2.18	–	1.4263×10^{-16}	1.4866×10^{-16}	–	1.0058	0.9650
12201	Skin head sensitive(40-100)	7.01	1.2906×10^{-16}	4.82	–	1.2807×10^{-16}	1.2369×10^{-16}	–	1.0077*	1.0434*
12300	Skin trunk insensitive	345.73	1.1199×10^{-15}	0.40	–	1.2965×10^{-15}	1.2911×10^{-15}	–	0.8638*	0.8674*
12301	Skin trunk sensitive(40-100)	40.69	7.4987×10^{-16}	0.41	–	1.1459×10^{-15}	1.1370×10^{-15}	–	0.6544*	0.6595*
12400	Skin arms insensitive	120.70	5.5647×10^{-16}	0.96	–	6.4663×10^{-16}	6.3876×10^{-16}	–	0.8606*	0.8712*
12401	Skin arms sensitive(40-100)	9.94	4.8829×10^{-16}	1.67	–	5.9517×10^{-16}	5.7997×10^{-16}	–	0.8204*	0.8419*
12500	Skin legs insensitive	243.30	2.2743×10^{-17}	2.85	–	2.3302×10^{-17}	2.3517×10^{-17}	–	0.9760*	0.9671*
12501	Skin legs sensitive(40-100)	11.26	2.1379×10^{-17}	10.35	–	2.1703×10^{-17}	2.0743×10^{-17}	–	0.9851*	1.0307*
12600	Spinal cord	45.70	1.8051×10^{-15}	0.95	–	1.8081×10^{-15}	1.8029×10^{-15}	–	0.9984	1.0012
12700	Spleen	114.98	1.6241×10^{-15}	0.71	–	1.6249×10^{-15}	1.6229×10^{-15}	–	0.9995	1.0007
12800	Erupted upper front buccal permanent ena	0.352	3.1807×10^{-16}	25.05	–	3.2497×10^{-16}	3.4195×10^{-16}	–	0.9788	0.9302

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
12801	Erupted upper front lingual permanent en	0.354	1.7445×10^{-16}	20.09	–	3.1323×10^{-16}	2.9107×10^{-16}	–	0.5569	0.5993
12802	Erupted upper front buccal deciduous ena	0.068	2.7865×10^{-16}	43.95	–	9.4486×10^{-17}	5.2197×10^{-16}	–	2.9491	0.5338
12803	Erupted upper front lingual deciduous en	0.0873	7.8672×10^{-17}	58.76	–	3.3833×10^{-16}	2.0984×10^{-16}	–	0.2325	0.3749
12804	Erupted upper front-left buccal permanen	0.117	1.3246×10^{-16}	45.44	–	2.3922×10^{-16}	3.6048×10^{-16}	–	0.5537	0.3674
12805	Erupted upper front-left lingual permane	0.0766	2.9580×10^{-16}	32.12	–	1.4812×10^{-16}	2.6067×10^{-16}	–	1.9970	1.1348
12806	Erupted upper front-right buccal permane	0.0755	1.7523×10^{-16}	38.41	–	1.5610×10^{-16}	5.9380×10^{-16}	–	1.1225	0.2951
12807	Erupted upper front-right lingual perman	0.118	6.4479×10^{-16}	27.14	–	4.7407×10^{-16}	1.4916×10^{-16}	–	1.3601	4.3228
12808	Erupted upper left buccal permanent enam	0.303	2.9814×10^{-16}	24.83	–	2.3393×10^{-16}	3.4577×10^{-16}	–	1.2745	0.8622
12809	Erupted upper left lingual permanent ena	0.145	2.1614×10^{-16}	33.08	–	1.3510×10^{-16}	2.8029×10^{-16}	–	1.5999	0.7711
12810	Erupted upper left buccal deciduous enam	0.0983	1.8932×10^{-16}	41.84	–	2.8460×10^{-16}	3.2218×10^{-16}	–	0.6652	0.5876
12811	Erupted upper left lingual deciduous ena	0.0569	1.8450×10^{-16}	41.75	–	1.5024×10^{-16}	2.0399×10^{-16}	–	1.2280	0.9045
12812	Erupted upper right buccal permanent ena	0.216	1.2895×10^{-16}	41.86	–	1.8657×10^{-16}	2.1875×10^{-16}	–	0.6911	0.5895
12813	Erupted upper right lingual permanent en	0.232	1.4445×10^{-16}	21.19	–	2.7525×10^{-16}	2.3162×10^{-16}	–	0.5248	0.6236
12814	Erupted upper right buccal deciduous ena	0.076	3.4402×10^{-16}	49.99	–	2.9793×10^{-16}	4.9083×10^{-16}	–	1.1547	0.7009
12815	Erupted upper right lingual deciduous en	0.0792	3.4108×10^{-16}	31.27	–	6.6820×10^{-16}	2.0304×10^{-16}	–	0.5104	1.6799
12816	Erupted lower front buccal permanent ena	0.532	3.0560×10^{-16}	20.29	–	3.6288×10^{-16}	2.3931×10^{-16}	–	0.8421	1.2770
12817	Erupted lower front lingual permanent en	0.412	3.0858×10^{-16}	21.00	–	4.1572×10^{-16}	2.6056×10^{-16}	–	0.7423	1.1843
12824	Erupted lower left buccal permanent enam	0.176	1.6892×10^{-16}	26.90	–	3.0083×10^{-16}	1.7195×10^{-16}	–	0.5615	0.9824
12825	Erupted lower left lingual permanent ena	0.302	2.8611×10^{-16}	28.68	–	2.4340×10^{-16}	3.6347×10^{-16}	–	1.1754	0.7872
12826	Erupted lower left buccal deciduous enam	0.146	1.3625×10^{-16}	27.94	–	2.1260×10^{-16}	4.2923×10^{-16}	–	0.6409	0.3174

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
12827	Erupted lower left lingual deciduous ena	0.111	4.3153×10^{-16}	30.89	–	2.9378×10^{-16}	1.6381×10^{-16}	–	1.4689	2.6343
12828	Erupted lower right buccal permanent ena	0.232	3.8611×10^{-16}	27.80	–	4.0972×10^{-16}	3.7520×10^{-16}	–	0.9424	1.0291
12829	Erupted lower right lingual permanent en	0.246	2.1755×10^{-16}	27.11	–	2.1343×10^{-16}	3.5137×10^{-16}	–	1.0193	0.6192
12830	Erupted lower right buccal deciduous ena	0.14	5.4116×10^{-16}	30.37	–	1.4406×10^{-16}	2.3520×10^{-16}	–	3.7565	2.3009
12831	Erupted lower right lingual deciduous en	0.117	3.6922×10^{-16}	32.32	–	2.0148×10^{-16}	3.1652×10^{-16}	–	1.8325	1.1665
12832	Erupted upper front permanent dentin	2.39	2.2632×10^{-16}	15.46	–	2.5295×10^{-16}	2.0927×10^{-16}	–	0.8947	1.0815
12833	Erupted upper front deciduous dentin	0.35	2.8867×10^{-16}	28.45	–	2.4085×10^{-16}	1.5089×10^{-16}	–	1.1986	1.9131
12834	Erupted upper front-left permanent denti	0.655	3.1873×10^{-16}	16.59	–	1.9488×10^{-16}	1.4604×10^{-16}	–	1.6355	2.1825
12835	Erupted upper front-right permanent dent	0.653	3.0151×10^{-16}	22.07	–	3.0272×10^{-16}	2.1690×10^{-16}	–	0.9960	1.3901
12836	Erupted upper left permanent dentin	1.52	2.9458×10^{-16}	9.82	–	2.6554×10^{-16}	2.1351×10^{-16}	–	1.1094	1.3797
12837	Erupted upper left deciduous dentin	0.35	2.6438×10^{-16}	24.45	–	1.6714×10^{-16}	4.0552×10^{-16}	–	1.5818	0.6520
12838	Erupted upper right permanent dentin	1.51	2.3345×10^{-16}	11.98	–	2.2548×10^{-16}	2.7975×10^{-16}	–	1.0354	0.8345
12839	Erupted upper right deciduous dentin	0.355	2.2265×10^{-16}	22.15	–	2.0188×10^{-16}	3.3403×10^{-16}	–	1.1029	0.6665
12840	Erupted lower front permanent dentin	3.14	2.9571×10^{-16}	6.36	–	2.9310×10^{-16}	2.9667×10^{-16}	–	1.0089	0.9968
12844	Erupted lower left permanent dentin	1.63	3.1476×10^{-16}	10.45	–	3.0153×10^{-16}	3.8549×10^{-16}	–	1.0439	0.8165
12845	Erupted lower left deciduous dentin	0.591	3.2799×10^{-16}	21.19	–	2.4594×10^{-16}	3.0446×10^{-16}	–	1.3336	1.0773
12846	Erupted lower right permanent dentin	1.62	3.6195×10^{-16}	12.07	–	3.0569×10^{-16}	3.2445×10^{-16}	–	1.1840	1.1156
12847	Erupted lower right deciduous dentin	0.595	3.1911×10^{-16}	12.18	–	3.3115×10^{-16}	3.4265×10^{-16}	–	0.9636	0.9313
12848	Unerupted permanent enamel	1.93	2.7600×10^{-16}	10.46	–	2.6934×10^{-16}	2.4551×10^{-16}	–	1.0247	1.1242
12850	Unerupted permanent dentin	6.44	2.6831×10^{-16}	8.41	–	2.7508×10^{-16}	3.0797×10^{-16}	–	0.9754	0.8712
12852	Permanent cementum	0.786	3.1625×10^{-16}	6.41	–	2.5506×10^{-16}	3.1152×10^{-16}	–	1.2399	1.0152
12853	Deciduous cementum	0.125	2.3054×10^{-16}	18.29	–	2.8779×10^{-16}	2.6705×10^{-16}	–	0.8011	0.8633
12854	Permanent pulp	0.582	3.2701×10^{-16}	9.68	–	2.8095×10^{-16}	2.6478×10^{-16}	–	1.1639	1.2350
12855	Deciduous pulp	0.313	2.9350×10^{-16}	38.06	–	1.8018×10^{-16}	2.4808×10^{-16}	–	1.6289	1.1831
12856	Teeth retention region	0.0382	2.5600×10^{-16}	29.56	–	3.3992×10^{-16}	4.1703×10^{-16}	–	0.7531	0.6139
12900	Testis left	1.09	9.7153×10^{-17}	13.64	–	9.6086×10^{-17}	6.6731×10^{-17}	–	1.0111	1.4559
13000	Testis right	1.09	1.3640×10^{-16}	12.43	–	1.2115×10^{-16}	8.5179×10^{-17}	–	1.1259	1.6014
13100	Thymus	41.79	1.1264×10^{-15}	1.18	–	1.1275×10^{-15}	1.1070×10^{-15}	–	0.9990	1.0175
13200	Thyroid	9.02	5.7168×10^{-16}	2.74	–	5.5580×10^{-16}	6.2186×10^{-16}	–	1.0286	0.9193
13300	Tongue upper(food)	12.99	3.0080×10^{-16}	3.74	–	2.8274×10^{-16}	3.0265×10^{-16}	–	1.0639	0.9939

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
13301	Tongue lower	20.41	3.2922×10^{-16}	1.67	–	3.2669×10^{-16}	3.5615×10^{-16}	–	1.0077	0.9244
13400	Tonsils	3.13	2.7987×10^{-16}	7.60	–	2.5362×10^{-16}	3.0045×10^{-16}	–	1.1035	0.9315
13500	Ureter left	3.66	9.8908×10^{-16}	3.27	–	1.0037×10^{-15}	9.4408×10^{-16}	–	0.9855	1.0477
13600	Ureter right	3.66	1.3175×10^{-15}	3.14	–	1.3605×10^{-15}	1.3336×10^{-15}	–	0.9684	0.9879
13700	Urinary bladder wall insensitive	24.21	2.4468×10^{-16}	1.80	–	2.5651×10^{-16}	2.4536×10^{-16}	–	0.9539	0.9972
13701	Urinary bladder wall sensitive(99-212)	1.29	2.3457×10^{-16}	5.56	–	2.3892×10^{-16}	2.1438×10^{-16}	–	0.9818	1.0942
13800	Urinary bladder content	103.00	2.3370×10^{-16}	1.47	–	2.4626×10^{-16}	2.4203×10^{-16}	–	0.9490	0.9656
14000	Air inside body	0.0529	3.0353×10^{-15}	6.20	–	3.0547×10^{-15}	3.0287×10^{-15}	–	0.9937	1.0022

* For tissues bookkept as head, trunk, arm and leg pieces (muscle, residual soft tissue, skin, large vessels), the volumes tabulated in the distributed cell tables partition the tissue at different boundaries from the transported mesh, and the reference output is normalised by the tabulated volume. The marked ratio therefore compares two different partitions of the same tissue, and is left out of the summary statistics of Table S2.

Table S20. 10M_External. Absorbed dose per primary history, Gy. σ is the FLUKA statistical uncertainty. Ratios are FLUKA / reference.

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
100	Adrenal left	4.18	1.7213×10^{-17}	39.30	–	2.1224×10^{-17}	2.9432×10^{-17}	–	0.8110	0.5848
200	Adrenal right	4.18	1.6115×10^{-17}	41.80	–	1.2462×10^{-17}	3.1782×10^{-17}	–	1.2932	0.5071
300	ET1(0-8)	0.0026	0.0000×10^{0}	0.00	–	–	–	–	–	–
301	ET1(8-40)	0.0105	1.7103×10^{-17}	100.00	–	–	–	–	–	–
302	ET1(40-50)	0.00334	0.0000×10^{0}	0.00	–	–	–	–	–	–
303	ET1(50-Surface)	0.667	1.4329×10^{-17}	42.55	–	5.8192×10^{-17}	4.0828×10^{-17}	–	0.2462	0.3510
400	ET2(-15-0)	0.138	1.0194×10^{-17}	35.38	–	3.1303×10^{-17}	4.5042×10^{-17}	–	0.3257	0.2263
401	ET2(0-40)	0.395	1.4792×10^{-17}	44.86	–	2.4006×10^{-17}	2.3025×10^{-17}	–	0.6162	0.6424
402	ET2(40-50)	0.0995	2.8205×10^{-17}	59.54	–	1.7709×10^{-17}	3.3080×10^{-17}	–	1.5927	0.8526
403	ET2(50-55)	0.0498	1.3511×10^{-17}	76.22	–	2.0178×10^{-17}	1.5809×10^{-17}	–	0.6696	0.8546
404	ET2(55-65)	0.0999	4.3311×10^{-17}	59.16	–	7.7566×10^{-17}	2.5911×10^{-17}	–	0.5584	1.6715
405	ET2(65-Surface)	19.65	3.1065×10^{-17}	7.57	–	3.4939×10^{-17}	2.7167×10^{-17}	–	0.8891	1.1435
500	Oral mucosa tongue	0.0509	1.1896×10^{-17}	40.91	–	7.9660×10^{-17}	1.7934×10^{-17}	–	0.1493	0.6633
501	Oral mucosa moth floor	0.0245	7.3295×10^{-18}	42.69	–	3.3872×10^{-17}	5.6302×10^{-18}	–	0.2164	1.3018
600	Oral mucosa lips and cheeks	0.0211	1.5654×10^{-17}	75.02	–	1.1477×10^{-16}	3.1513×10^{-17}	–	0.1364	0.4968
700	Trachea	4.70	3.0031×10^{-17}	22.18	–	1.9509×10^{-17}	3.4955×10^{-17}	–	1.5393	0.8591
800	BB(-11–6)	0.00499	0.0000×10^{0}	0.00	–	–	3.2464×10^{-17}	–	–	–
801	BB(-6-0)	0.00639	0.0000×10^{0}	0.00	–	–	3.4359×10^{-17}	–	–	–
802	BB(0-10)	0.0107	1.0359×10^{-17}	100.00	–	–	3.2787×10^{-17}	–	–	0.3159
803	BB(10-35)	0.0268	0.0000×10^{0}	0.00	–	–	2.8994×10^{-17}	–	–	–
804	BB(35-40)	0.00539	0.0000×10^{0}	0.00	–	–	2.8697×10^{-17}	–	–	–
805	BB(40-50)	0.0108	0.0000×10^{0}	0.00	–	–	2.8327×10^{-17}	–	–	–
806	BB(50-60)	0.0109	0.0000×10^{0}	0.00	–	4.7417×10^{-19}	2.5979×10^{-17}	–	–	–
807	BB(60-70)	0.0109	0.0000×10^{0}	0.00	–	2.7692×10^{-17}	2.5129×10^{-17}	–	–	–
808	BB(70-surface)	2.82	1.2518×10^{-17}	36.39	–	2.1903×10^{-17}	2.6706×10^{-17}	–	0.5715	0.4687
900	Blood in large arteries head	0.631	3.1903×10^{-17}	60.84	–	3.7084×10^{-17}	7.7878×10^{-17}	–	0.8603*	0.4097*
910	Blood in large veins head	1.91	1.6619×10^{-17}	38.66	–	1.4185×10^{-17}	3.5460×10^{-17}	–	1.1716	0.4687
1000	Blood in large arteries trunk	58.29	3.0078×10^{-17}	7.99	–	3.1296×10^{-17}	3.0338×10^{-17}	–	0.9611*	0.9914*
1010	Blood in large veins trunk	135.44	2.9869×10^{-17}	2.80	–	3.3164×10^{-17}	3.5440×10^{-17}	–	0.9007*	0.8428*
1100	Blood in large arteries arms	12.25	5.1047×10^{-17}	10.78	–	3.9250×10^{-17}	3.5906×10^{-17}	–	1.3006*	1.4217*
1110	Blood in large veins arms	90.31	3.6510×10^{-17}	2.81	–	3.4939×10^{-17}	3.8423×10^{-17}	–	1.0450*	0.9502*
1200	Blood in large arteries legs	78.73	3.0197×10^{-17}	4.82	–	2.9657×10^{-17}	3.0794×10^{-17}	–	1.0182	0.9806
1210	Blood in large veins legs	222.01	3.2253×10^{-17}	4.60	–	3.5027×10^{-17}	3.4625×10^{-17}	–	0.9208	0.9315
1300	Humeri upper cortical	88.31	3.0988×10^{-17}	5.73	–	2.8792×10^{-17}	2.7912×10^{-17}	–	1.0763	1.1102
1400	Humeri upper spogiosa	52.65	3.1936×10^{-17}	5.47	–	3.2507×10^{-17}	2.9271×10^{-17}	–	0.9824	1.0910

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
1500	Humeri upper medullary cavity	9.79	2.0287×10^{-17}	16.04	–	3.4012×10^{-17}	3.2307×10^{-17}	–	0.5965	0.6279
1600	Humeri lower cortical	67.36	3.5488×10^{-17}	4.20	–	3.4941×10^{-17}	3.6133×10^{-17}	–	1.0156	0.9821
1700	Humeri lower spongiosa	26.26	4.0777×10^{-17}	9.23	–	3.6607×10^{-17}	3.4938×10^{-17}	–	1.1139	1.1671
1800	Humeri lower medullary cavity	9.77	3.7014×10^{-17}	18.57	–	3.3029×10^{-17}	3.7092×10^{-17}	–	1.1206	0.9979
1900	Radii cortical	44.01	3.4556×10^{-17}	6.70	–	4.3117×10^{-17}	3.8621×10^{-17}	–	0.8015	0.8947
1910	Ulnae cortical	56.67	3.2883×10^{-17}	7.38	–	3.4108×10^{-17}	3.3549×10^{-17}	–	0.9641	0.9802
2000	Radii spongiosa	7.24	3.7764×10^{-17}	13.97	–	4.1725×10^{-17}	4.5790×10^{-17}	–	0.9051	0.8247
2010	Ulnae spongiosa	12.63	3.7261×10^{-17}	13.00	–	2.7682×10^{-17}	3.8463×10^{-17}	–	1.3460	0.9688
2100	Radii medullary cavity	6.29	5.0822×10^{-17}	12.75	–	4.5946×10^{-17}	5.5825×10^{-17}	–	1.1061	0.9104
2110	Ulnae medullary cavity	6.91	3.1685×10^{-17}	21.91	–	4.0869×10^{-17}	4.1271×10^{-17}	–	0.7753	0.7677
2200	Wrists and hand bones cortical	32.59	3.8736×10^{-17}	9.88	–	3.6836×10^{-17}	3.3860×10^{-17}	–	1.0516	1.1440
2300	Wrists and hand bones spongiosa	37.06	3.8549×10^{-17}	6.47	–	3.4723×10^{-17}	3.8261×10^{-17}	–	1.1102	1.0075
2400	Clavicles cortical	14.38	3.0228×10^{-17}	14.33	–	3.8870×10^{-17}	3.0574×10^{-17}	–	0.7777	0.9887
2500	Clavicles spongiosa	17.02	3.0986×10^{-17}	14.40	–	3.6186×10^{-17}	3.2420×10^{-17}	–	0.8563	0.9558
2600	Cranium cortical	328.09	1.9285×10^{-17}	3.61	–	1.9516×10^{-17}	1.9400×10^{-17}	–	0.9882	0.9941
2700	Cranium spongiosa	497.53	2.1662×10^{-17}	3.84	–	2.1848×10^{-17}	2.1940×10^{-17}	–	0.9915	0.9873
2800	Femora upper cortical	164.62	3.0434×10^{-17}	3.73	–	3.0658×10^{-17}	3.0079×10^{-17}	–	0.9927	1.0118
2900	Femora upper spongiosa	75.47	3.1479×10^{-17}	6.29	–	3.1466×10^{-17}	3.0684×10^{-17}	–	1.0004	1.0259
3000	Femora upper medullary cavity	25.19	3.6974×10^{-17}	6.44	–	2.9794×10^{-17}	2.8285×10^{-17}	–	1.2410	1.3072
3100	Femora lower cortical	152.10	2.9542×10^{-17}	4.57	–	2.9689×10^{-17}	2.6285×10^{-17}	–	0.9951	1.1239
3200	Femora lower spongiosa	90.51	3.0485×10^{-17}	5.84	–	3.1805×10^{-17}	3.1039×10^{-17}	–	0.9585	0.9822
3300	Femora lower medullary cavity	16.91	2.8880×10^{-17}	10.33	–	3.2422×10^{-17}	2.8554×10^{-17}	–	0.8908	1.0114
3400	Tibiae cortical	240.89	2.6413×10^{-17}	1.13	–	2.5350×10^{-17}	2.6242×10^{-17}	–	1.0419	1.0065
3410	Fibulae cortical	32.68	2.3570×10^{-17}	9.05	–	2.3157×10^{-17}	2.4905×10^{-17}	–	1.0178	0.9464
3420	Patellae cortical	2.60	3.5433×10^{-17}	15.45	–	3.6732×10^{-17}	1.4319×10^{-17}	–	0.9646	2.4745
3500	Tibiae spongiosa	62.57	2.8987×10^{-17}	6.53	–	3.1545×10^{-17}	3.1840×10^{-17}	–	0.9189	0.9104
3510	Fibulae spongiosa	7.38	2.6240×10^{-17}	15.19	–	1.9173×10^{-17}	3.2700×10^{-17}	–	1.3686	0.8025
3520	Patellae spongiosa	12.63	3.9763×10^{-17}	15.50	–	2.9436×10^{-17}	2.7746×10^{-17}	–	1.3508	1.4331
3600	Tibiae medullary cavity	26.27	2.8586×10^{-17}	10.17	–	3.1088×10^{-17}	3.1671×10^{-17}	–	0.9195	0.9026
3610	Fibulae medullary cavity	5.58	2.5956×10^{-17}	26.70	–	3.2111×10^{-17}	2.0666×10^{-17}	–	0.8083	1.2560
3700	Ankles and foot cortical	93.86	2.2249×10^{-17}	3.59	–	2.1674×10^{-17}	2.3327×10^{-17}	–	1.0265	0.9538
3800	Ankles and foot spongiosa	159.84	2.3527×10^{-17}	4.85	–	2.3591×10^{-17}	2.5274×10^{-17}	–	0.9973	0.9309
3900	Mandible cortical	24.91	3.6358×10^{-17}	10.15	–	3.3661×10^{-17}	3.1077×10^{-17}	–	1.0801	1.1699
4000	Mandible spongiosa	27.51	3.5147×10^{-17}	6.23	–	3.2252×10^{-17}	3.2179×10^{-17}	–	1.0897	1.0922
4100	Pelvis cortical	128.93	2.9349×10^{-17}	4.11	–	3.0442×10^{-17}	2.9721×10^{-17}	–	0.9641	0.9875
4200	Pelvis spongiosa	285.90	3.0878×10^{-17}	2.14	–	3.1097×10^{-17}	3.0375×10^{-17}	–	0.9930	1.0166
4300	Ribs cortical	70.42	2.2632×10^{-17}	5.79	–	2.4541×10^{-17}	2.5117×10^{-17}	–	0.9222	0.9011
4400	Ribs spongiosa	138.70	2.7787×10^{-17}	3.46	–	2.5700×10^{-17}	2.5408×10^{-17}	–	1.0812	1.0936

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
4500	Scapulae cortical	59.27	1.7676×10^{-17}	4.48	–	2.0657×10^{-17}	2.0345×10^{-17}	–	0.8557	0.8688
4600	Scapulae spongiosa	90.28	1.9770×10^{-17}	4.25	–	1.9781×10^{-17}	2.2401×10^{-17}	–	0.9994	0.8825
4700	Cervical spine cortical	14.13	2.3372×10^{-17}	12.57	–	3.2352×10^{-17}	2.7939×10^{-17}	–	0.7224	0.8365
4800	Cervical spine spongiosa	30.75	2.5682×10^{-17}	7.78	–	2.9380×10^{-17}	2.9587×10^{-17}	–	0.8741	0.8680
4900	Thoracic spine cortical	68.39	1.9418×10^{-17}	5.03	–	2.0383×10^{-17}	1.9648×10^{-17}	–	0.9527	0.9883
5000	Thoracic spine spongiosa	211.32	1.8432×10^{-17}	3.99	–	1.9817×10^{-17}	2.1142×10^{-17}	–	0.9301	0.8718
5100	Lumbar spine cortical	36.75	2.2770×10^{-17}	6.71	–	1.9593×10^{-17}	2.2221×10^{-17}	–	1.1621	1.0247
5200	Lumbar spine spongiosa	161.65	2.3334×10^{-17}	3.05	–	2.6021×10^{-17}	2.4949×10^{-17}	–	0.8967	0.9353
5300	Sacrum cortical	16.67	2.3053×10^{-17}	5.97	–	2.3335×10^{-17}	2.1399×10^{-17}	–	0.9879	1.0773
5400	Sacrum spongiosa	55.24	2.3879×10^{-17}	7.76	–	2.2577×10^{-17}	2.4912×10^{-17}	–	1.0577	0.9585
5500	Sternum cortical	5.28	3.2991×10^{-17}	9.21	–	3.3054×10^{-17}	4.0816×10^{-17}	–	0.9981	0.8083
5600	Sternum spongiosa	20.82	4.0446×10^{-17}	7.36	–	3.2459×10^{-17}	3.6956×10^{-17}	–	1.2460	1.0944
5800	Cartilage trunk	129.15	2.9844×10^{-17}	4.21	–	3.2336×10^{-17}	3.0717×10^{-17}	–	0.9229	0.9716
6100	Brain	1467.08	1.9073×10^{-17}	1.29	–	1.9712×10^{-17}	1.9181×10^{-17}	–	0.9676	0.9944
6200	Breast left adipose tissue	2.04	2.1936×10^{-17}	29.50	–	2.8134×10^{-17}	5.3512×10^{-17}	–	0.7797	0.4099
6300	Breast left glandular tissue	1.36	3.3578×10^{-17}	32.83	–	3.1724×10^{-17}	3.2184×10^{-17}	–	1.0584	1.0433
6400	Breast right adipose tissue	2.04	3.8336×10^{-17}	27.39	–	3.9979×10^{-17}	3.8334×10^{-17}	–	0.9589	1.0000
6500	Breast right glandular tissue	1.36	4.8076×10^{-17}	36.06	–	5.0670×10^{-17}	2.6277×10^{-17}	–	0.9488	1.8296
6600	Eye lens sensitive left	0.0184	0.0000×10^{0}	0.00	–	–	–	–	–	–
6601	Eye lens insensitive left	0.113	0.0000×10^{0}	0.00	–	–	–	–	–	–
6700	Cornea left	0.941	1.5617×10^{-17}	24.91	–	2.3821×10^{-17}	1.6324×10^{-17}	–	0.6556	0.9567
6701	Aqueous left	0.327	3.7159×10^{-18}	72.48	–	9.7626×10^{-18}	2.2333×10^{-17}	–	0.3806	0.1664
6702	Vitreous left	4.85	3.8244×10^{-17}	18.60	–	5.2269×10^{-17}	4.7206×10^{-17}	–	0.7317	0.8101
6800	Eye lens sensitive right	0.0184	0.0000×10^{0}	0.00	–	3.0439×10^{-17}	2.3365×10^{-17}	–	–	–
6801	Eye lens insensitive right	0.113	1.0092×10^{-17}	100.00	–	6.4121×10^{-17}	1.8519×10^{-18}	–	0.1574	5.4495
6900	Cornea right	0.942	1.3894×10^{-17}	56.04	–	2.9394×10^{-17}	3.1570×10^{-17}	–	0.4727	0.4401
6901	Aqueous right	0.327	2.8262×10^{-17}	55.84	–	5.5244×10^{-17}	–	–	0.5116	–
6902	Vitreous right	4.85	2.4164×10^{-17}	20.54	–	3.8396×10^{-17}	3.4690×10^{-17}	–	0.6293	0.6966
7000	Gall bladder wall	4.60	2.5133×10^{-17}	20.69	–	2.8821×10^{-17}	2.3762×10^{-17}	–	0.8720	1.0577
7100	Gall bladder contents	26.00	3.3521×10^{-17}	7.62	–	3.4092×10^{-17}	3.5984×10^{-17}	–	0.9832	0.9315
7200	Stomach wall(0-60)	0.998	4.4181×10^{-17}	18.03	–	2.8403×10^{-17}	4.2944×10^{-17}	–	1.5555	1.0288
7201	Stomach wall(60-100)	0.667	3.3714×10^{-17}	14.15	–	2.6968×10^{-17}	5.0287×10^{-17}	–	1.2502	0.6704
7202	Stomach wall(100-300)	3.36	3.2794×10^{-17}	16.46	–	3.6077×10^{-17}	3.2710×10^{-17}	–	0.9090	1.0026
7203	Stomach wall(300-surface)	100.42	3.1873×10^{-17}	4.43	–	3.0622×10^{-17}	3.0440×10^{-17}	–	1.0409	1.0471
7300	Stomach contents	117.00	3.5252×10^{-17}	5.06	–	3.2661×10^{-17}	3.1922×10^{-17}	–	1.0793	1.1043
7400	Small intestine wall(0-130)	8.74	4.0858×10^{-17}	4.63	–	3.7331×10^{-17}	3.6590×10^{-17}	–	1.0945	1.1166
7401	Small intestine wall(130-150)	1.36	3.7248×10^{-17}	6.30	–	4.4800×10^{-17}	3.3158×10^{-17}	–	0.8314	1.1233
7402	Small intestine wall(150-200)	3.44	4.2153×10^{-17}	6.83	–	3.6288×10^{-17}	3.0976×10^{-17}	–	1.1616	1.3608

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
7403	Small intestine wall(200-surface)	454.79	3.7395×10^{-17}	1.68	–	3.6610×10^{-17}	3.5814×10^{-17}	–	1.0214	1.0441
7500	Small intestine contents(-500-0)	31.33	3.8328×10^{-17}	8.28	–	3.6929×10^{-17}	3.7802×10^{-17}	–	1.0379	1.0139
7501	Small intestine contents(centre–500)	131.67	3.5137×10^{-17}	3.50	–	3.6101×10^{-17}	3.4851×10^{-17}	–	0.9733	1.0082
7600	Ascending colon wall(0-280)	2.59	3.0025×10^{-17}	24.00	–	3.1797×10^{-17}	4.0735×10^{-17}	–	0.9442	0.7371
7601	Ascending colon wall(280-300)	0.188	2.5138×10^{-17}	29.16	–	3.9109×10^{-17}	2.9200×10^{-17}	–	0.6428	0.8609
7602	Ascending colon wall(300-surface)	50.90	3.2771×10^{-17}	6.73	–	3.5935×10^{-17}	3.4866×10^{-17}	–	0.9120	0.9399
7700	Ascending colon content	45.28	3.3839×10^{-17}	7.51	–	3.5648×10^{-17}	3.4142×10^{-17}	–	0.9493	0.9911
7800	Transverse colon wall right(0-280)	1.92	3.5066×10^{-17}	23.11	–	3.7892×10^{-17}	3.5585×10^{-17}	–	0.9254	0.9854
7801	Transverse colon wall right(280-300)	0.14	1.9781×10^{-17}	18.59	–	4.2673×10^{-17}	2.9603×10^{-17}	–	0.4636	0.6682
7802	Transverse colon wall right(300-surface)	51.69	3.2570×10^{-17}	9.38	–	3.6421×10^{-17}	4.2091×10^{-17}	–	0.8943	0.7738
7900	Transverse colon contents right	24.72	3.5026×10^{-17}	11.40	–	3.5497×10^{-17}	3.7171×10^{-17}	–	0.9867	0.9423
8000	Transverse colon wall left(0-280)	1.49	3.7013×10^{-17}	21.24	–	3.3835×10^{-17}	4.8779×10^{-17}	–	1.0939	0.7588
8001	Transverse colon wall left(280-300)	0.109	4.7239×10^{-17}	42.15	–	4.3496×10^{-17}	3.3733×10^{-17}	–	1.0861	1.4004
8002	Transverse colon wall left(300-surface)	45.61	3.9789×10^{-17}	5.17	–	3.9768×10^{-17}	3.9980×10^{-17}	–	1.0005	0.9952
8100	Transverse colon content left	15.05	3.9514×10^{-17}	10.69	–	3.8547×10^{-17}	4.0547×10^{-17}	–	1.0251	0.9745
8200	Descending colon wall(0-280)	2.01	3.4547×10^{-17}	21.18	–	3.9107×10^{-17}	3.5719×10^{-17}	–	0.8834	0.9672
8201	Descending colon wall(280-300)	0.148	3.3823×10^{-17}	45.54	–	2.2773×10^{-17}	4.8170×10^{-17}	–	1.4852	0.7022
8202	Descending colon wall(300-surface)	58.01	3.5975×10^{-17}	7.67	–	3.7604×10^{-17}	3.5643×10^{-17}	–	0.9567	1.0093
8300	Descending colon content	19.95	4.0725×10^{-17}	11.22	–	3.6214×10^{-17}	3.7006×10^{-17}	–	1.1246	1.1005
8400	Sigmoid colon wall(0-280)	2.45	3.9854×10^{-17}	19.57	–	2.8029×10^{-17}	3.2053×10^{-17}	–	1.4219	1.2434
8401	Sigmoid colon wall(280-300)	0.18	1.2188×10^{-17}	23.49	–	1.6907×10^{-17}	5.6148×10^{-17}	–	0.7209	0.2171
8402	Sigmoid colon wall(300-surface)	42.08	3.1677×10^{-17}	5.84	–	3.0968×10^{-17}	2.8806×10^{-17}	–	1.0229	1.0997
8500	Sigmoid colon contents	24.52	3.1770×10^{-17}	10.32	–	2.7173×10^{-17}	3.4714×10^{-17}	–	1.1692	0.9152
8600	Rectum wall(0-280)	1.03	2.0405×10^{-17}	27.10	–	1.1102×10^{-17}	2.6414×10^{-17}	–	1.8379	0.7725
8601	Rectum wall(280-300)	0.0758	1.3345×10^{-17}	35.77	–	4.4660×10^{-17}	2.1635×10^{-17}	–	0.2988	0.6168
8602	Rectum wall(300-surface)	4.71	2.4664×10^{-17}	26.36	–	2.2739×10^{-17}	2.4770×10^{-17}	–	1.0846	0.9957
8603	Rectum contents	10.48	2.7838×10^{-17}	21.52	–	2.5124×10^{-17}	1.9707×10^{-17}	–	1.1080	1.4126
8700	Heart wall	161.50	3.0669×10^{-17}	2.85	–	3.3466×10^{-17}	3.1084×10^{-17}	–	0.9164	0.9866
8800	Blood in heart chamber	230.00	3.0713×10^{-17}	3.18	–	2.8967×10^{-17}	3.0265×10^{-17}	–	1.0603	1.0148
8900	Kidney left cortex	82.03	1.9382×10^{-17}	8.33	–	2.2240×10^{-17}	2.4569×10^{-17}	–	0.8715	0.7889
9000	Kidney left medulla	29.30	1.8898×10^{-17}	6.90	–	1.9352×10^{-17}	2.2668×10^{-17}	–	0.9766	0.8337
9100	Kidney left pelvis	5.86	2.3225×10^{-17}	10.09	–	1.9488×10^{-17}	2.2764×10^{-17}	–	1.1918	1.0203
9200	Kidney right cortex	82.03	2.0958×10^{-17}	3.83	–	2.4873×10^{-17}	2.2692×10^{-17}	–	0.8426	0.9236
9300	Kidney right medulla	29.30	2.0543×10^{-17}	10.71	–	2.2246×10^{-17}	2.3692×10^{-17}	–	0.9234	0.8671
9400	Kidney right pelvis	5.86	1.7829×10^{-17}	27.94	–	2.3645×10^{-17}	2.7031×10^{-17}	–	0.7540	0.6596
9500	Liver	1059.76	3.2206×10^{-17}	1.84	–	3.1196×10^{-17}	3.1397×10^{-17}	–	1.0324	1.0258

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
9700	Lung(AI) left	222.22	2.6763×10^{-17}	3.26	–	2.6160×10^{-17}	2.7604×10^{-17}	–	1.0231	0.9696
9900	Lung(AI) right	277.78	2.8124×10^{-17}	3.34	–	2.6746×10^{-17}	2.5907×10^{-17}	–	1.0515	1.0856
10000	Lymphatic nodes ET	7.16	2.5159×10^{-17}	26.88	–	2.5039×10^{-17}	3.3091×10^{-17}	–	1.0048	0.7603
10100	Lymphatic nodes thoracic	7.16	2.8245×10^{-17}	19.68	–	1.9236×10^{-17}	2.7547×10^{-17}	–	1.4683	1.0253
10200	Lymphatic nodes head	2.47	2.7086×10^{-17}	30.02	–	2.8766×10^{-17}	3.6145×10^{-17}	–	0.9416	0.7494
10300	Lymphatic nodes trunk	58.43	3.4917×10^{-17}	7.84	–	3.2860×10^{-17}	3.1785×10^{-17}	–	1.0626	1.0985
10400	Lymphatic nodes arms	4.94	3.4510×10^{-17}	26.21	–	2.7078×10^{-17}	4.0122×10^{-17}	–	1.2745	0.8601
10500	Lymphatic nodes legs	4.95	3.3094×10^{-17}	28.30	–	3.3154×10^{-17}	4.1840×10^{-17}	–	0.9982	0.7910
10600	Muscle head	402.29	2.8640×10^{-17}	3.02	–	2.9077×10^{-17}	2.8236×10^{-17}	–	0.9850	1.0143
10700	Muscle trunk	5431.32	2.9450×10^{-17}	0.54	–	2.9094×10^{-17}	2.9121×10^{-17}	–	1.0123*	1.0113*
10800	Muscle arms	1166.19	3.4633×10^{-17}	2.06	–	3.4454×10^{-17}	3.3756×10^{-17}	–	1.0052*	1.0260*
10900	Muscle legs	4258.32	3.0982×10^{-17}	0.92	–	3.1269×10^{-17}	3.1314×10^{-17}	–	0.9908	0.9894
11000	Oesophagus wall(0-190)	1.60	2.8108×10^{-17}	21.26	–	2.4954×10^{-17}	3.1101×10^{-17}	–	1.1264	0.9038
11001	Oesophagus wall(190-200)	0.0858	3.6168×10^{-17}	66.81	–	1.8275×10^{-17}	4.0027×10^{-17}	–	1.9791	0.9036
11002	Oesophagus wall(200-surface)	20.60	2.4481×10^{-17}	11.03	–	2.6050×10^{-17}	2.7648×10^{-17}	–	0.9398	0.8855
11003	Oesophagus contents	11.91	2.5912×10^{-17}	19.50	–	3.2738×10^{-17}	2.3163×10^{-17}	–	0.7915	1.1187
11300	Pancreas	72.13	2.7701×10^{-17}	5.71	–	2.9847×10^{-17}	3.0822×10^{-17}	–	0.9281	0.8987
11400	Pituitary gland	0.366	1.9384×10^{-17}	97.71	–	7.0808×10^{-17}	–	–	0.2738	–
11500	Prostate	1.67	3.1208×10^{-17}	33.46	–	4.8342×10^{-17}	3.7657×10^{-17}	–	0.6456	0.8287
11600	RST head	476.06	2.7282×10^{-17}	2.61	–	2.7051×10^{-17}	2.7198×10^{-17}	–	1.0085	1.0031
11700	RST trunk	6679.34	3.1247×10^{-17}	0.70	–	3.0943×10^{-17}	3.1134×10^{-17}	–	1.0098*	1.0036*
11800	RST arms	629.22	3.7031×10^{-17}	1.58	–	3.6006×10^{-17}	3.6335×10^{-17}	–	1.0285*	1.0192*
11900	RST legs	1784.08	3.2163×10^{-17}	1.13	–	3.1689×10^{-17}	3.1835×10^{-17}	–	1.0150*	1.0103*
12000	Salivary glands left	22.98	2.8040×10^{-17}	11.03	–	3.3810×10^{-17}	3.0971×10^{-17}	–	0.8293	0.9054
12100	Salivary glandss right	22.98	3.4686×10^{-17}	9.80	–	3.0261×10^{-17}	2.8304×10^{-17}	–	1.1462	1.2255
12200	Skin head insensitive	80.29	2.0051×10^{-17}	5.18	–	1.8923×10^{-17}	1.9115×10^{-17}	–	1.0596	1.0490
12201	Skin head sensitive(40-100)	7.01	1.4135×10^{-17}	12.17	–	1.3353×10^{-17}	1.4790×10^{-17}	–	1.0586*	0.9557*
12300	Skin trunk insensitive	345.73	2.0835×10^{-17}	3.56	–	2.1165×10^{-17}	2.2373×10^{-17}	–	0.9844*	0.9312*
12301	Skin trunk sensitive(40-100)	40.69	1.5716×10^{-17}	4.66	–	1.5096×10^{-17}	1.6126×10^{-17}	–	1.0411*	0.9746*
12400	Skin arms insensitive	120.70	2.7213×10^{-17}	3.08	–	2.6309×10^{-17}	2.6624×10^{-17}	–	1.0344*	1.0221*
12401	Skin arms sensitive(40-100)	9.94	2.5612×10^{-17}	5.37	–	2.1345×10^{-17}	2.0999×10^{-17}	–	1.1999*	1.2197*
12500	Skin legs insensitive	243.30	2.2808×10^{-17}	2.30	–	2.3311×10^{-17}	2.3744×10^{-17}	–	0.9784*	0.9606*
12501	Skin legs sensitive(40-100)	11.26	1.4200×10^{-17}	9.67	–	1.8521×10^{-17}	1.6517×10^{-17}	–	0.7667*	0.8597*
12600	Spinal cord	45.70	2.1427×10^{-17}	6.02	–	2.3544×10^{-17}	2.0356×10^{-17}	–	0.9101	1.0526
12700	Spleen	114.98	2.6164×10^{-17}	5.30	–	2.4243×10^{-17}	2.7114×10^{-17}	–	1.0792	0.9650
12800	Erupted upper front buccal permanent ena	0.352	1.2591×10^{-17}	71.75	–	7.3037×10^{-17}	–	–	0.1724	–

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
12801	Erupted upper front lingual permanent en	0.354	5.3034×10^{-17}	44.91	–	2.7379×10^{-17}	5.7836×10^{-18}	–	1.9370	9.1697
12802	Erupted upper front buccal deciduous ena	0.068	0.0000×10^{0}	0.00	–	–	1.0649×10^{-17}	–	–	–
12803	Erupted upper front lingual deciduous en	0.0873	1.1102×10^{-18}	100.00	–	–	1.3993×10^{-16}	–	–	0.0079
12804	Erupted upper front-left buccal permanen	0.117	5.1530×10^{-17}	84.60	–	5.8770×10^{-17}	1.1752×10^{-16}	–	0.8768	0.4385
12805	Erupted upper front-left lingual permane	0.0766	0.0000×10^{0}	0.00	–	–	1.3598×10^{-17}	–	–	–
12806	Erupted upper front-right buccal permane	0.0755	0.0000×10^{0}	0.00	–	6.1644×10^{-19}	–	–	–	–
12807	Erupted upper front-right lingual perman	0.118	3.8075×10^{-17}	100.00	–	–	–	–	–	–
12808	Erupted upper left buccal permanent enam	0.303	8.5658×10^{-17}	53.28	–	3.2615×10^{-17}	9.2160×10^{-17}	–	2.6263	0.9294
12809	Erupted upper left lingual permanent ena	0.145	2.1498×10^{-18}	100.00	–	9.8605×10^{-17}	7.5891×10^{-17}	–	0.0218	0.0283
12810	Erupted upper left buccal deciduous enam	0.0983	0.0000×10^{0}	0.00	–	1.2464×10^{-17}	5.6968×10^{-17}	–	–	–
12811	Erupted upper left lingual deciduous ena	0.0569	9.7523×10^{-17}	100.00	–	–	1.8293×10^{-17}	–	–	5.3312
12812	Erupted upper right buccal permanent ena	0.216	1.4779×10^{-17}	100.00	–	2.3998×10^{-18}	3.0343×10^{-17}	–	6.1584	0.4871
12813	Erupted upper right lingual permanent en	0.232	8.3288×10^{-18}	87.50	–	7.1849×10^{-17}	1.7462×10^{-17}	–	0.1159	0.4770
12814	Erupted upper right buccal deciduous ena	0.076	6.3959×10^{-17}	96.39	–	1.3959×10^{-17}	–	–	4.5821	–
12815	Erupted upper right lingual deciduous en	0.0792	0.0000×10^{0}	0.00	–	–	1.5413×10^{-16}	–	–	–
12816	Erupted lower front buccal permanent ena	0.532	2.9901×10^{-17}	78.00	–	3.2615×10^{-17}	4.3496×10^{-17}	–	0.9168	0.6874
12817	Erupted lower front lingual permanent en	0.412	3.9200×10^{-17}	77.45	–	1.0503×10^{-16}	1.2253×10^{-17}	–	0.3732	3.1992
12824	Erupted lower left buccal permanent enam	0.176	6.1727×10^{-17}	100.00	–	3.6450×10^{-17}	8.4894×10^{-17}	–	1.6935	0.7271
12825	Erupted lower left lingual permanent ena	0.302	2.2090×10^{-17}	57.38	–	1.4329×10^{-17}	3.5819×10^{-17}	–	1.5416	0.6167
12826	Erupted lower left buccal deciduous enam	0.146	2.3102×10^{-17}	66.67	–	1.6053×10^{-17}	4.6815×10^{-17}	–	1.4391	0.4935

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
12827	Erupted lower left lingual deciduous ena	0.111	0.0000×10^{0}	0.00	–	4.6853×10^{-18}	–	–	–	–
12828	Erupted lower right buccal permanent ena	0.232	6.0296×10^{-18}	69.56	–	5.9161×10^{-19}	–	–	10.1919	–
12829	Erupted lower right lingual permanent en	0.246	2.6437×10^{-17}	100.00	–	–	5.3270×10^{-17}	–	–	0.4963
12830	Erupted lower right buccal deciduous ena	0.14	0.0000×10^{0}	0.00	–	2.1138×10^{-17}	3.7313×10^{-17}	–	–	–
12831	Erupted lower right lingual deciduous en	0.117	9.2126×10^{-19}	100.00	–	–	–	–	–	–
12832	Erupted upper front permanent dentin	2.39	5.1777×10^{-17}	24.74	–	3.0279×10^{-17}	1.2545×10^{-17}	–	1.7100	4.1273
12833	Erupted upper front deciduous dentin	0.35	3.6370×10^{-17}	61.51	–	6.4626×10^{-17}	5.6377×10^{-17}	–	0.5628	0.6451
12834	Erupted upper front-left permanent denti	0.655	4.2005×10^{-17}	63.51	–	1.0460×10^{-17}	4.3594×10^{-17}	–	4.0156	0.9636
12835	Erupted upper front-right permanent dent	0.653	3.2139×10^{-17}	57.20	–	4.5903×10^{-17}	1.9856×10^{-17}	–	0.7001	1.6186
12836	Erupted upper left permanent dentin	1.52	4.4972×10^{-17}	41.81	–	2.4365×10^{-17}	5.0546×10^{-17}	–	1.8458	0.8897
12837	Erupted upper left deciduous dentin	0.35	1.7253×10^{-17}	73.15	–	7.5134×10^{-17}	5.5662×10^{-17}	–	0.2296	0.3100
12838	Erupted upper right permanent dentin	1.51	2.9698×10^{-17}	37.95	–	1.2481×10^{-17}	4.0022×10^{-17}	–	2.3796	0.7421
12839	Erupted upper right deciduous dentin	0.355	1.0312×10^{-17}	50.18	–	1.6201×10^{-17}	2.6915×10^{-17}	–	0.6365	0.3831
12840	Erupted lower front permanent dentin	3.14	3.9011×10^{-17}	32.31	–	5.3732×10^{-17}	3.4106×10^{-17}	–	0.7260	1.1438
12844	Erupted lower left permanent dentin	1.63	4.5024×10^{-17}	29.35	–	1.7879×10^{-17}	1.5821×10^{-17}	–	2.5182	2.8459
12845	Erupted lower left deciduous dentin	0.591	1.4431×10^{-17}	60.36	–	2.3502×10^{-17}	2.8522×10^{-17}	–	0.6140	0.5060
12846	Erupted lower right permanent dentin	1.62	1.5215×10^{-17}	40.32	–	1.2224×10^{-17}	2.4697×10^{-17}	–	1.2447	0.6161
12847	Erupted lower right deciduous dentin	0.595	4.2170×10^{-17}	49.48	–	5.4157×10^{-18}	7.5554×10^{-17}	–	7.7867	0.5582
12848	Unerupted permanent enamel	1.93	1.5044×10^{-17}	45.48	–	2.8436×10^{-17}	4.3113×10^{-17}	–	0.5290	0.3489
12850	Unerupted permanent dentin	6.44	3.5351×10^{-17}	17.05	–	2.2045×10^{-17}	3.7128×10^{-17}	–	1.6036	0.9521
12852	Permanent cementum	0.786	3.5773×10^{-17}	30.99	–	2.4571×10^{-17}	2.7723×10^{-17}	–	1.4559	1.2904
12853	Deciduous cementum	0.125	0.0000×10^{0}	0.00	–	2.1112×10^{-17}	2.6903×10^{-17}	–	–	–
12854	Permanent pulp	0.582	3.2386×10^{-17}	56.30	–	5.2308×10^{-17}	2.2667×10^{-17}	–	0.6191	1.4288
12855	Deciduous pulp	0.313	4.0930×10^{-17}	67.16	–	1.6576×10^{-17}	4.5612×10^{-17}	–	2.4692	0.8974
12856	Teeth retention region	0.0382	2.1236×10^{-17}	37.44	–	5.4978×10^{-17}	1.9153×10^{-17}	–	0.3863	1.1087
12900	Testis left	1.09	6.7734×10^{-17}	23.32	–	4.1338×10^{-17}	3.2307×10^{-17}	–	1.6386	2.0966
13000	Testis right	1.09	2.1195×10^{-17}	79.66	–	4.5453×10^{-17}	4.4776×10^{-17}	–	0.4663	0.4734
13100	Thymus	41.79	3.2463×10^{-17}	6.38	–	3.1473×10^{-17}	3.1221×10^{-17}	–	1.0315	1.0398
13200	Thyroid	9.02	3.6934×10^{-17}	10.90	–	3.8131×10^{-17}	2.6576×10^{-17}	–	0.9686	1.3898
13300	Tongue upper(food)	12.99	3.3563×10^{-17}	15.58	–	2.9818×10^{-17}	2.7966×10^{-17}	–	1.1256	1.2001

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
13301	Tongue lower	20.41	3.2986×10^{-17}	12.97	–	3.1398×10^{-17}	2.9491×10^{-17}	–	1.0506	1.1185
13400	Tonsils	3.13	2.7627×10^{-17}	19.10	–	5.5064×10^{-17}	2.4978×10^{-17}	–	0.5017	1.1061
13500	Ureter left	3.66	3.4422×10^{-17}	31.72	–	2.4716×10^{-17}	3.2674×10^{-17}	–	1.3927	1.0535
13600	Ureter right	3.66	3.6453×10^{-17}	25.20	–	1.6948×10^{-17}	2.4043×10^{-17}	–	2.1509	1.5162
13700	Urinary bladder wall insensitive	24.21	3.6863×10^{-17}	11.03	–	3.2961×10^{-17}	2.7940×10^{-17}	–	1.1184	1.3194
13701	Urinary bladder wall sensitive(99-212)	1.29	3.5117×10^{-17}	22.05	–	3.6924×10^{-17}	3.9505×10^{-17}	–	0.9511	0.8889
13800	Urinary bladder content	103.00	3.2560×10^{-17}	2.60	–	3.6136×10^{-17}	3.5557×10^{-17}	–	0.9010	0.9157
14000	Air inside body	0.0529	1.6103×10^{-17}	26.35	–	2.7551×10^{-17}	2.0422×10^{-17}	–	0.5845	0.7885

* For tissues bookkept as head, trunk, arm and leg pieces (muscle, residual soft tissue, skin, large vessels), the volumes tabulated in the distributed cell tables partition the tissue at different boundaries from the transported mesh, and the reference output is normalised by the tabulated volume. The marked ratio therefore compares two different partitions of the same tissue, and is left out of the summary statistics of Table S2.

Table S21. 10F_Internal. Absorbed dose per primary history, Gy. σ is the FLUKA statistical uncertainty. Ratios are FLUKA / reference.

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
100	Adrenal left	4.18	3.2577×10^{-15}	1.88	–	3.2039×10^{-15}	3.2305×10^{-15}	–	1.0168	1.0084
200	Adrenal right	4.18	8.0323×10^{-15}	1.43	–	7.5837×10^{-15}	7.7064×10^{-15}	–	1.0591	1.0423
300	ET1(0-8)	0.00259	2.9445×10^{-16}	78.85	–	2.1444×10^{-16}	6.8678×10^{-17}	–	1.3731	4.2874
301	ET1(8-40)	0.0105	3.6974×10^{-16}	66.58	–	1.6531×10^{-16}	1.1772×10^{-16}	–	2.2367	3.1409
302	ET1(40-50)	0.00332	2.1117×10^{-16}	59.74	–	2.6377×10^{-16}	9.1920×10^{-17}	–	0.8006	2.2973
303	ET1(50-Surface)	0.664	2.1217×10^{-16}	26.35	–	2.2990×10^{-16}	1.7248×10^{-16}	–	0.9229	1.2301
400	ET2(-15-0)	0.136	2.9198×10^{-16}	20.87	–	2.0414×10^{-16}	1.9960×10^{-16}	–	1.4303	1.4628
401	ET2(0-40)	0.387	2.2292×10^{-16}	8.21	–	2.0163×10^{-16}	1.9066×10^{-16}	–	1.1056	1.1692
402	ET2(40-50)	0.0973	1.7170×10^{-16}	21.02	–	2.1846×10^{-16}	2.4390×10^{-16}	–	0.7860	0.7040
403	ET2(50-55)	0.0488	2.5661×10^{-16}	30.53	–	1.8238×10^{-16}	3.6235×10^{-16}	–	1.4071	0.7082
404	ET2(55-65)	0.0977	1.8873×10^{-16}	16.43	–	2.2011×10^{-16}	2.0770×10^{-16}	–	0.8574	0.9086
405	ET2(65-Surface)	22.19	1.9588×10^{-16}	4.95	–	2.0221×10^{-16}	1.9450×10^{-16}	–	0.9687	1.0071
500	Oral mucosa tongue	0.0502	4.5830×10^{-16}	16.25	–	3.7641×10^{-16}	3.4529×10^{-16}	–	1.2176	1.3273
501	Oral mucosa moth floor	0.0244	3.8617×10^{-16}	28.41	–	6.1096×10^{-16}	3.0708×10^{-16}	–	0.6321	1.2576
600	Oral mucosa lips and cheeks	0.0212	2.0239×10^{-16}	40.60	–	2.9009×10^{-16}	2.5029×10^{-16}	–	0.6977	0.8086
700	Trachea	4.68	7.8843×10^{-16}	2.59	–	7.5727×10^{-16}	8.4447×10^{-16}	–	1.0411	0.9336
800	BB(-11–6)	0.00499	2.5738×10^{-15}	25.77	–	1.4408×10^{-15}	1.7976×10^{-15}	–	1.7863	1.4318
801	BB(-6-0)	0.00636	1.9290×10^{-15}	26.84	–	1.4941×10^{-15}	1.2626×10^{-15}	–	1.2911	1.5278
802	BB(0-10)	0.0106	1.6905×10^{-15}	17.86	–	1.1890×10^{-15}	1.0873×10^{-15}	–	1.4217	1.5547
803	BB(10-35)	0.0267	1.5412×10^{-15}	20.95	–	1.4454×10^{-15}	1.4205×10^{-15}	–	1.0663	1.0850
804	BB(35-40)	0.00537	1.8066×10^{-15}	29.02	–	1.3247×10^{-15}	1.8678×10^{-15}	–	1.3638	0.9673
805	BB(40-50)	0.0108	1.3761×10^{-15}	17.75	–	1.5652×10^{-15}	1.8936×10^{-15}	–	0.8792	0.7267
806	BB(50-60)	0.0108	2.1061×10^{-15}	20.74	–	1.1435×10^{-15}	1.6711×10^{-15}	–	1.8418	1.2603
807	BB(60-70)	0.0108	1.2697×10^{-15}	34.15	–	2.5969×10^{-15}	1.2820×10^{-15}	–	0.4889	0.9904
808	BB(70-surface)	2.81	1.5720×10^{-15}	3.30	–	1.5117×10^{-15}	1.6111×10^{-15}	–	1.0399	0.9757
900	Blood in large arteries head	0.62	3.4923×10^{-16}	14.66	–	3.1724×10^{-16}	4.3968×10^{-16}	–	1.1008*	0.7943*
910	Blood in large veins head	1.89	2.8067×10^{-16}	6.12	–	3.0582×10^{-16}	2.6369×10^{-16}	–	0.9178*	1.0644*
1000	Blood in large arteries trunk	58.82	2.0592×10^{-15}	0.76	–	2.0732×10^{-15}	2.0830×10^{-15}	–	0.9932*	0.9886*
1010	Blood in large veins trunk	137.53	1.7985×10^{-15}	0.51	–	1.8451×10^{-15}	1.8667×10^{-15}	–	0.9747*	0.9635*
1100	Blood in large arteries arms	11.71	1.3645×10^{-15}	1.52	–	1.3741×10^{-15}	1.3408×10^{-15}	–	0.9930*	1.0177*
1110	Blood in large veins arms	88.06	8.9378×10^{-16}	0.71	–	8.6776×10^{-16}	8.9226×10^{-16}	–	1.0300*	1.0017*
1200	Blood in large arteries legs	78.74	2.9450×10^{-17}	6.77	–	2.5193×10^{-17}	3.1126×10^{-17}	–	1.1690	0.9462
1210	Blood in large veins legs	222.21	2.9419×10^{-17}	3.08	–	2.9918×10^{-17}	2.9764×10^{-17}	–	0.9833	0.9884
1300	Humeri upper cortical	88.31	7.7950×10^{-16}	1.06	–	7.7861×10^{-16}	8.0036×10^{-16}	–	1.0011	0.9739
1400	Humeri upper spogiosa	52.65	5.2465×10^{-16}	1.25	–	5.3791×10^{-16}	5.4269×10^{-16}	–	0.9753	0.9668

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
1500	Humeri upper medullary cavity	9.79	1.0042×10^{-15}	2.13	–	1.0048×10^{-15}	1.0892×10^{-15}	–	0.9995	0.9220
1600	Humeri lower cortical	67.36	1.3799×10^{-15}	0.87	–	1.3904×10^{-15}	1.3952×10^{-15}	–	0.9925	0.9890
1700	Humeri lower spongiosa	26.26	1.1354×10^{-15}	1.63	–	1.1859×10^{-15}	1.1916×10^{-15}	–	0.9575	0.9529
1800	Humeri lower medullary cavity	9.77	1.5195×10^{-15}	1.89	–	1.5366×10^{-15}	1.5867×10^{-15}	–	0.9888	0.9576
1900	Radii cortical	44.01	3.7123×10^{-16}	2.88	–	3.5988×10^{-16}	3.6219×10^{-16}	–	1.0315	1.0250
1910	Ulnae cortical	56.67	5.3980×10^{-16}	1.66	–	5.4370×10^{-16}	5.4442×10^{-16}	–	0.9928	0.9915
2000	Radii spongiosa	7.24	3.2741×10^{-16}	3.01	–	2.8796×10^{-16}	3.6210×10^{-16}	–	1.1370	0.9042
2010	Ulnae spongiosa	12.63	7.7372×10^{-16}	3.63	–	7.6824×10^{-16}	8.0347×10^{-16}	–	1.0071	0.9630
2100	Radii medullary cavity	6.29	3.4760×10^{-16}	5.18	–	4.0718×10^{-16}	3.7241×10^{-16}	–	0.8537	0.9334
2110	Ulnae medullary cavity	6.91	4.0783×10^{-16}	3.95	–	3.9694×10^{-16}	4.0053×10^{-16}	–	1.0274	1.0182
2200	Wrists and hand bones cortical	32.59	1.1398×10^{-16}	3.48	–	1.1335×10^{-16}	1.1410×10^{-16}	–	1.0056	0.9989
2300	Wrists and hand bones spongiosa	37.06	1.1693×10^{-16}	3.36	–	1.1925×10^{-16}	1.2845×10^{-16}	–	0.9806	0.9103
2400	Clavicles cortical	14.38	6.6333×10^{-16}	2.01	–	6.8869×10^{-16}	6.4815×10^{-16}	–	0.9632	1.0234
2500	Clavicles spongiosa	17.02	7.2573×10^{-16}	2.07	–	6.9435×10^{-16}	7.2426×10^{-16}	–	1.0452	1.0020
2600	Cranium cortical	328.08	1.1290×10^{-16}	1.05	–	1.1341×10^{-16}	1.1407×10^{-16}	–	0.9955	0.9897
2700	Cranium spongiosa	497.18	1.1593×10^{-16}	1.30	–	1.1662×10^{-16}	1.1656×10^{-16}	–	0.9941	0.9946
2800	Femora upper cortical	164.70	7.4404×10^{-17}	2.44	–	7.6790×10^{-17}	7.3485×10^{-17}	–	0.9689	1.0125
2900	Femora upper spongiosa	75.47	1.2099×10^{-16}	1.82	–	1.2118×10^{-16}	1.2391×10^{-16}	–	0.9985	0.9764
3000	Femora upper medullary cavity	25.19	4.1061×10^{-17}	4.04	–	3.9599×10^{-17}	3.8884×10^{-17}	–	1.0369	1.0560
3100	Femora lower cortical	152.14	7.8465×10^{-18}	6.09	–	6.7939×10^{-18}	8.2025×10^{-18}	–	1.1549	0.9566
3200	Femora lower spongiosa	90.51	6.0330×10^{-18}	9.57	–	6.0558×10^{-18}	7.0861×10^{-18}	–	0.9962	0.8514
3300	Femora lower medullary cavity	16.91	1.0319×10^{-17}	17.77	–	8.3176×10^{-18}	9.4044×10^{-18}	–	1.2406	1.0972
3400	Tibiae cortical	240.89	3.1633×10^{-18}	12.16	–	3.1474×10^{-18}	2.9244×10^{-18}	–	1.0050	1.0817
3410	Fibulae cortical	32.68	1.4132×10^{-18}	32.72	–	1.5780×10^{-18}	2.6112×10^{-18}	–	0.8956	0.5412
3420	Patellae cortical	2.60	6.3069×10^{-18}	48.54	–	8.2391×10^{-18}	3.8307×10^{-18}	–	0.7655	1.6464
3500	Tibiae spongiosa	62.57	4.4770×10^{-18}	15.71	–	5.2899×10^{-18}	4.7710×10^{-18}	–	0.8463	0.9384
3510	Fibulae spongiosa	7.38	3.9166×10^{-18}	59.11	–	1.3451×10^{-18}	3.3037×10^{-19}	–	2.9117	11.8551
3520	Patellae spongiosa	12.63	6.1363×10^{-18}	33.31	–	5.1233×10^{-18}	8.5040×10^{-18}	–	1.1977	0.7216
3600	Tibiae medullary cavity	26.27	2.5246×10^{-18}	63.49	–	1.5857×10^{-18}	2.7939×10^{-18}	–	1.5921	0.9036
3610	Fibulae medullary cavity	5.58	2.0291×10^{-19}	100.00	–	4.5076×10^{-19}	2.9905×10^{-18}	–	0.4502	0.0679
3700	Ankles and foot cortical	93.86	5.7428×10^{-18}	13.11	–	5.4505×10^{-18}	5.6188×10^{-18}	–	1.0536	1.0221
3800	Ankles and foot spongiosa	159.84	3.7591×10^{-18}	16.04	–	4.4823×10^{-18}	4.0207×10^{-18}	–	0.8387	0.9349
3900	Mandible cortical	24.91	3.1551×10^{-16}	2.08	–	2.9808×10^{-16}	3.1719×10^{-16}	–	1.0584	0.9947
4000	Mandible spongiosa	27.51	3.0703×10^{-16}	2.49	–	3.0856×10^{-16}	3.1098×10^{-16}	–	0.9950	0.9873
4100	Pelvis cortical	128.93	3.4293×10^{-16}	1.11	–	3.3905×10^{-16}	3.3623×10^{-16}	–	1.0115	1.0199
4200	Pelvis spongiosa	285.90	3.1755×10^{-16}	0.49	–	3.1955×10^{-16}	3.1345×10^{-16}	–	0.9938	1.0131
4300	Ribs cortical	70.42	2.8101×10^{-15}	0.60	–	2.8440×10^{-15}	2.8077×10^{-15}	–	0.9881	1.0009
4400	Ribs spongiosa	138.70	2.8082×10^{-15}	0.22	–	2.7985×10^{-15}	2.8305×10^{-15}	–	1.0035	0.9921

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
4500	Scapulae cortical	59.27	8.0255×10^{-16}	1.33	–	7.9242×10^{-16}	7.9923×10^{-16}	–	1.0128	1.0041
4600	Scapulae spongiosa	90.28	7.3201×10^{-16}	0.83	–	7.3077×10^{-16}	7.3052×10^{-16}	–	1.0017	1.0020
4700	Cervical spine cortical	14.13	3.8409×10^{-16}	2.74	–	3.7205×10^{-16}	3.6328×10^{-16}	–	1.0324	1.0573
4800	Cervical spine spongiosa	30.75	4.0871×10^{-16}	2.62	–	4.0007×10^{-16}	3.8395×10^{-16}	–	1.0216	1.0645
4900	Thoracic spine cortical	68.39	2.5803×10^{-15}	0.51	–	2.5604×10^{-15}	2.5833×10^{-15}	–	1.0077	0.9988
5000	Thoracic spine spongiosa	211.32	2.8893×10^{-15}	0.46	–	2.9167×10^{-15}	2.9218×10^{-15}	–	0.9906	0.9889
5100	Lumbar spine cortical	36.75	1.3053×10^{-15}	1.12	–	1.3236×10^{-15}	1.2890×10^{-15}	–	0.9862	1.0126
5200	Lumbar spine spongiosa	161.65	1.3627×10^{-15}	0.69	–	1.3697×10^{-15}	1.3604×10^{-15}	–	0.9949	1.0017
5300	Sacrum cortical	16.67	3.0590×10^{-16}	1.52	–	3.0669×10^{-16}	2.9688×10^{-16}	–	0.9974	1.0304
5400	Sacrum spongiosa	55.24	3.3626×10^{-16}	1.80	–	3.5000×10^{-16}	3.5025×10^{-16}	–	0.9607	0.9601
5500	Sternum cortical	5.28	2.0473×10^{-15}	2.17	–	2.0496×10^{-15}	2.0619×10^{-15}	–	0.9989	0.9929
5600	Sternum spongiosa	20.82	1.9602×10^{-15}	0.74	–	1.9995×10^{-15}	1.9185×10^{-15}	–	0.9803	1.0217
5800	Cartilage trunk	129.15	3.3741×10^{-15}	0.31	–	3.3929×10^{-15}	3.3893×10^{-15}	–	0.9945	0.9955
6100	Brain	1287.08	1.0707×10^{-16}	0.54	–	1.0711×10^{-16}	1.0835×10^{-16}	–	0.9996	0.9882
6200	Breast left adipose tissue	2.03	1.5945×10^{-15}	5.05	–	1.4498×10^{-15}	1.4276×10^{-15}	–	1.0998	1.1169
6300	Breast left glandular tissue	1.35	1.7696×10^{-15}	5.66	–	1.5782×10^{-15}	1.5235×10^{-15}	–	1.1212	1.1615
6400	Breast right adipose tissue	2.03	2.6112×10^{-15}	4.02	–	2.5791×10^{-15}	2.5374×10^{-15}	–	1.0125	1.0291
6500	Breast right glandular tissue	1.35	2.4162×10^{-15}	2.63	–	2.5860×10^{-15}	2.4899×10^{-15}	–	0.9343	0.9704
6600	Eye lens sensitive left	0.0184	1.7481×10^{-16}	68.71	–	–	3.6228×10^{-16}	–	–	0.4825
6601	Eye lens insensitive left	0.113	1.0014×10^{-16}	59.41	–	9.0459×10^{-17}	1.7741×10^{-16}	–	1.1071	0.5645
6700	Cornea left	0.941	1.1989×10^{-16}	13.98	–	1.2209×10^{-16}	1.6831×10^{-16}	–	0.9820	0.7123
6701	Aqueous left	0.327	6.4783×10^{-17}	41.46	–	1.0568×10^{-16}	1.8347×10^{-16}	–	0.6130	0.3531
6702	Vitreous left	4.85	1.4603×10^{-16}	9.62	–	1.3425×10^{-16}	1.2737×10^{-16}	–	1.0878	1.1465
6800	Eye lens sensitive right	0.0184	5.0299×10^{-18}	100.00	–	5.1676×10^{-17}	6.3759×10^{-16}	–	0.0973	0.0079
6801	Eye lens insensitive right	0.113	8.7655×10^{-17}	50.56	–	5.9028×10^{-17}	2.1847×10^{-16}	–	1.4850	0.4012
6900	Cornea right	0.941	1.4307×10^{-16}	15.88	–	1.3659×10^{-16}	1.2868×10^{-16}	–	1.0475	1.1118
6901	Aqueous right	0.327	2.0666×10^{-16}	33.22	–	1.4129×10^{-16}	2.6998×10^{-16}	–	1.4627	0.7655
6902	Vitreous right	4.85	1.4438×10^{-16}	7.08	–	1.4457×10^{-16}	1.6547×10^{-16}	–	0.9987	0.8726
7000	Gall bladder wall	4.58	1.4622×10^{-14}	0.74	–	1.4690×10^{-14}	1.4725×10^{-14}	–	0.9954	0.9930
7100	Gall bladder contents	26.00	1.3999×10^{-14}	0.30	–	1.4082×10^{-14}	1.4142×10^{-14}	–	0.9941	0.9899
7200	Stomach wall(0-60)	0.998	5.1503×10^{-15}	1.78	–	5.0844×10^{-15}	5.1312×10^{-15}	–	1.0130	1.0037
7201	Stomach wall(60-100)	0.667	5.2185×10^{-15}	1.71	–	5.1175×10^{-15}	4.9428×10^{-15}	–	1.0197	1.0558
7202	Stomach wall(100-300)	3.36	5.1854×10^{-15}	0.94	–	5.1739×10^{-15}	5.0050×10^{-15}	–	1.0022	1.0360
7203	Stomach wall(300-surface)	100.41	5.2260×10^{-15}	0.51	–	5.2373×10^{-15}	5.2901×10^{-15}	–	0.9978	0.9879
7300	Stomach contents	117.00	4.5363×10^{-15}	0.57	–	4.5526×10^{-15}	4.5613×10^{-15}	–	0.9964	0.9945
7400	Small intestine wall(0-130)	8.70	1.6607×10^{-15}	1.38	–	1.6144×10^{-15}	1.6444×10^{-15}	–	1.0287	1.0099
7401	Small intestine wall(130-150)	1.36	1.6733×10^{-15}	2.50	–	1.6585×10^{-15}	1.5932×10^{-15}	–	1.0089	1.0502
7402	Small intestine wall(150-200)	3.44	1.7046×10^{-15}	1.38	–	1.6876×10^{-15}	1.6674×10^{-15}	–	1.0101	1.0223

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
7403	Small intestine wall(200-surface)	454.52	1.6724×10^{-15}	0.42	–	1.6745×10^{-15}	1.6770×10^{-15}	–	0.9988	0.9973
7500	Small intestine contents(-500-0)	31.37	1.6400×10^{-15}	0.90	–	1.6024×10^{-15}	1.6542×10^{-15}	–	1.0235	0.9914
7501	Small intestine contents(centre–500)	131.67	1.5975×10^{-15}	0.49	–	1.6065×10^{-15}	1.6048×10^{-15}	–	0.9944	0.9954
7600	Ascending colon wall(0-280)	2.59	1.2379×10^{-15}	3.49	–	1.2534×10^{-15}	1.1996×10^{-15}	–	0.9877	1.0320
7601	Ascending colon wall(280-300)	0.188	1.4291×10^{-15}	5.12	–	1.2226×10^{-15}	1.3061×10^{-15}	–	1.1689	1.0942
7602	Ascending colon wall(300-surface)	50.88	1.2669×10^{-15}	0.76	–	1.2843×10^{-15}	1.2469×10^{-15}	–	0.9865	1.0161
7700	Ascending colon content	45.28	1.1293×10^{-15}	0.99	–	1.1491×10^{-15}	1.1388×10^{-15}	–	0.9828	0.9917
7800	Transverse colon wall right(0-280)	1.92	4.9194×10^{-15}	2.13	–	4.8585×10^{-15}	5.0463×10^{-15}	–	1.0125	0.9749
7801	Transverse colon wall right(280-300)	0.14	5.0357×10^{-15}	1.53	–	5.4963×10^{-15}	4.9318×10^{-15}	–	0.9162	1.0211
7802	Transverse colon wall right(300-surface)	51.71	5.0964×10^{-15}	0.52	–	5.0748×10^{-15}	5.1256×10^{-15}	–	1.0043	0.9943
7900	Transverse colon contents right	24.72	4.8775×10^{-15}	0.82	–	4.9487×10^{-15}	5.0322×10^{-15}	–	0.9856	0.9693
8000	Transverse colon wall left(0-280)	1.49	2.1398×10^{-15}	3.03	–	1.9536×10^{-15}	2.0119×10^{-15}	–	1.0953	1.0636
8001	Transverse colon wall left(280-300)	0.109	2.1281×10^{-15}	6.69	–	2.1718×10^{-15}	1.8978×10^{-15}	–	0.9799	1.1214
8002	Transverse colon wall left(300-surface)	45.61	2.0601×10^{-15}	1.15	–	2.0639×10^{-15}	2.0841×10^{-15}	–	0.9982	0.9885
8100	Transverse colon content left	15.05	2.0626×10^{-15}	2.03	–	2.0495×10^{-15}	1.9910×10^{-15}	–	1.0064	1.0360
8200	Descending colon wall(0-280)	2.01	7.0859×10^{-16}	5.11	–	7.2745×10^{-16}	7.7498×10^{-16}	–	0.9741	0.9143
8201	Descending colon wall(280-300)	0.148	8.6191×10^{-16}	6.34	–	7.9636×10^{-16}	7.2628×10^{-16}	–	1.0823	1.1867
8202	Descending colon wall(300-surface)	58.06	7.1888×10^{-16}	1.43	–	7.3036×10^{-16}	7.1294×10^{-16}	–	0.9843	1.0083
8300	Descending colon content	19.95	7.1967×10^{-16}	2.57	–	7.3563×10^{-16}	7.2661×10^{-16}	–	0.9783	0.9905
8400	Sigmoid colon wall(0-280)	2.45	3.4608×10^{-16}	4.84	–	3.5577×10^{-16}	2.9990×10^{-16}	–	0.9728	1.1540
8401	Sigmoid colon wall(280-300)	0.18	2.9420×10^{-16}	10.92	–	3.4263×10^{-16}	2.7429×10^{-16}	–	0.8586	1.0726
8402	Sigmoid colon wall(300-surface)	42.10	3.6704×10^{-16}	1.94	–	3.7406×10^{-16}	3.6409×10^{-16}	–	0.9812	1.0081
8500	Sigmoid colon contents	24.52	3.6990×10^{-16}	2.40	–	3.5417×10^{-16}	3.3821×10^{-16}	–	1.0444	1.0937
8600	Rectum wall(0-280)	1.03	1.0887×10^{-16}	16.07	–	1.6008×10^{-16}	1.4289×10^{-16}	–	0.6801	0.7619
8601	Rectum wall(280-300)	0.0758	1.4213×10^{-16}	37.32	–	1.8176×10^{-16}	1.1527×10^{-16}	–	0.7819	1.2330
8602	Rectum wall(300-surface)	4.71	1.3724×10^{-16}	6.48	–	1.5177×10^{-16}	1.6639×10^{-16}	–	0.9043	0.8248
8603	Rectum contents	10.48	1.4930×10^{-16}	4.86	–	1.7072×10^{-16}	1.6639×10^{-16}	–	0.8745	0.8973
8700	Heart wall	161.50	4.7398×10^{-15}	0.39	–	4.7510×10^{-15}	4.7307×10^{-15}	–	0.9976	1.0019
8800	Blood in heart chamber	230.00	4.1044×10^{-15}	0.37	–	4.1261×10^{-15}	4.1416×10^{-15}	–	0.9948	0.9910
8900	Kidney left cortex	82.03	1.7683×10^{-15}	0.70	–	1.7684×10^{-15}	1.7507×10^{-15}	–	0.9999	1.0100
9000	Kidney left medulla	29.30	1.8430×10^{-15}	0.88	–	1.8213×10^{-15}	1.7919×10^{-15}	–	1.0119	1.0285
9100	Kidney left pelvis	5.86	1.8670×10^{-15}	1.67	–	1.8697×10^{-15}	1.8568×10^{-15}	–	0.9985	1.0055
9200	Kidney right cortex	82.03	5.1238×10^{-15}	0.54	–	5.1257×10^{-15}	5.1489×10^{-15}	–	0.9996	0.9951
9300	Kidney right medulla	29.30	4.5736×10^{-15}	0.66	–	4.5599×10^{-15}	4.5787×10^{-15}	–	1.0030	0.9989
9400	Kidney right pelvis	5.86	4.2050×10^{-15}	1.67	–	4.3752×10^{-15}	4.2042×10^{-15}	–	0.9611	1.0002
9500	Liver	1059.77	1.8969×10^{-14}	0.06	–	1.8941×10^{-14}	1.9020×10^{-14}	–	1.0015	0.9973

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
9700	Lung(AI) left	222.24	1.3459×10^{-15}	0.45	–	1.3483×10^{-15}	1.3494×10^{-15}	–	0.9982	0.9974
9900	Lung(AI) right	277.76	3.6108×10^{-15}	0.29	–	3.6172×10^{-15}	3.6161×10^{-15}	–	0.9983	0.9985
10000	Lymphatic nodes ET	7.16	2.7436×10^{-16}	5.02	–	2.9430×10^{-16}	2.9196×10^{-16}	–	0.9322	0.9397
10100	Lymphatic nodes thoracic	7.16	5.2564×10^{-15}	0.73	–	5.3429×10^{-15}	5.2809×10^{-15}	–	0.9838	0.9954
10200	Lymphatic nodes head	2.47	4.5775×10^{-16}	6.15	–	5.3500×10^{-16}	4.8242×10^{-16}	–	0.8556	0.9489
10300	Lymphatic nodes trunk	58.43	8.5852×10^{-16}	1.36	–	8.5446×10^{-16}	8.3772×10^{-16}	–	1.0047	1.0248
10400	Lymphatic nodes arms	4.94	9.8376×10^{-16}	2.63	–	9.5469×10^{-16}	1.0010×10^{-15}	–	1.0304	0.9828
10500	Lymphatic nodes legs	4.94	5.2119×10^{-18}	39.39	–	1.6773×10^{-17}	1.2508×10^{-17}	–	0.3107	0.4167
10600	Muscle head	457.24	2.2156×10^{-16}	0.86	–	2.2203×10^{-16}	2.2237×10^{-16}	–	0.9979	0.9964
10700	Muscle trunk	5450.50	1.1359×10^{-15}	0.10	–	1.1501×10^{-15}	1.1522×10^{-15}	–	0.9877*	0.9859*
10800	Muscle arms	1076.44	7.7969×10^{-16}	0.40	–	7.7364×10^{-16}	7.7759×10^{-16}	–	1.0078*	1.0027*
10900	Muscle legs	4272.92	3.5772×10^{-17}	0.66	–	3.6821×10^{-17}	3.6668×10^{-17}	–	0.9715*	0.9756*
11000	Oesophagus wall(0-190)	1.57	2.9282×10^{-15}	1.97	–	2.7249×10^{-15}	2.8005×10^{-15}	–	1.0746	1.0456
11001	Oesophagus wall(190-200)	0.0843	3.0346×10^{-15}	5.66	–	2.9884×10^{-15}	2.8257×10^{-15}	–	1.0155	1.0739
11002	Oesophagus wall(200-surface)	20.66	2.9349×10^{-15}	1.14	–	2.9234×10^{-15}	2.9899×10^{-15}	–	1.0039	0.9816
11003	Oesophagus contents	11.91	3.1626×10^{-15}	1.27	–	3.1478×10^{-15}	3.2989×10^{-15}	–	1.0047	0.9587
11100	Ovary left	1.84	3.4522×10^{-16}	7.81	–	4.1524×10^{-16}	3.7137×10^{-16}	–	0.8314	0.9296
11200	Ovary right	1.84	4.4387×10^{-16}	5.09	–	3.4426×10^{-16}	3.6047×10^{-16}	–	1.2893	1.2314
11300	Pancreas	72.13	4.0636×10^{-15}	0.56	–	4.0395×10^{-15}	4.0238×10^{-15}	–	1.0060	1.0099
11400	Pituitary gland	0.364	1.2387×10^{-16}	32.17	–	2.6947×10^{-16}	5.0165×10^{-17}	–	0.4597	2.4693
11600	RST head	571.92	1.7054×10^{-16}	0.69	–	1.6752×10^{-16}	1.6760×10^{-16}	–	1.0181	1.0176
11700	RST trunk	6580.66	2.1529×10^{-15}	0.08	–	2.1629×10^{-15}	2.1761×10^{-15}	–	0.9954*	0.9893*
11800	RST arms	730.89	7.6233×10^{-16}	0.32	–	7.7853×10^{-16}	7.8365×10^{-16}	–	0.9792*	0.9728*
11900	RST legs	1867.01	3.8380×10^{-17}	0.98	–	3.9208×10^{-17}	3.9532×10^{-17}	–	0.9789*	0.9709*
12000	Salivary glands left	22.86	2.5073×10^{-16}	3.50	–	2.6040×10^{-16}	2.7009×10^{-16}	–	0.9629	0.9283
12100	Salivary glandss right	22.86	2.9900×10^{-16}	2.84	–	3.0537×10^{-16}	3.1343×10^{-16}	–	0.9791	0.9540
12200	Skin head insensitive	82.16	1.4225×10^{-16}	1.27	–	1.4334×10^{-16}	1.4506×10^{-16}	–	0.9924*	0.9806*
12201	Skin head sensitive(40-100)	6.89	1.1937×10^{-16}	4.77	–	1.3621×10^{-16}	1.2407×10^{-16}	–	0.8764*	0.9622*
12300	Skin trunk insensitive	327.47	1.2438×10^{-15}	0.33	–	1.3101×10^{-15}	1.3106×10^{-15}	–	0.9494*	0.9491*
12301	Skin trunk sensitive(40-100)	35.11	9.4204×10^{-16}	0.47	–	1.1552×10^{-15}	1.1418×10^{-15}	–	0.8155*	0.8250*
12400	Skin arms insensitive	108.23	4.5261×10^{-16}	0.86	–	6.4135×10^{-16}	6.4374×10^{-16}	–	0.7057*	0.7031*
12401	Skin arms sensitive(40-100)	8.58	3.7411×10^{-16}	1.99	–	6.0018×10^{-16}	5.8633×10^{-16}	–	0.6233*	0.6381*
12500	Skin legs insensitive	271.96	2.2968×10^{-17}	2.78	–	2.3393×10^{-17}	2.4235×10^{-17}	–	0.9818*	0.9477*
12501	Skin legs sensitive(40-100)	18.51	1.2548×10^{-17}	6.31	–	2.1949×10^{-17}	2.0717×10^{-17}	–	0.5717*	0.6057*
12600	Spinal cord	45.34	1.7874×10^{-15}	0.87	–	1.8097×10^{-15}	1.7985×10^{-15}	–	0.9877	0.9938
12700	Spleen	114.98	1.6124×10^{-15}	0.52	–	1.6301×10^{-15}	1.6199×10^{-15}	–	0.9892	0.9954
12800	Erupted upper front buccal permanent ena	0.35	2.4563×10^{-16}	18.58	–	2.9705×10^{-16}	3.3321×10^{-16}	–	0.8269	0.7372

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
12801	Erupted upper front lingual permanent en	0.356	2.1351×10^{-16}	23.58	–	2.6043×10^{-16}	2.6182×10^{-16}	–	0.8198	0.8155
12802	Erupted upper front buccal deciduous ena	0.0649	4.0421×10^{-16}	44.34	–	1.1836×10^{-16}	2.5840×10^{-16}	–	3.4151	1.5643
12803	Erupted upper front lingual deciduous en	0.0903	4.1674×10^{-16}	29.59	–	4.3445×10^{-16}	2.9529×10^{-16}	–	0.9592	1.4113
12804	Erupted upper front-left buccal permanen	0.108	1.2474×10^{-16}	66.67	–	3.9526×10^{-16}	2.8895×10^{-16}	–	0.3156	0.4317
12805	Erupted upper front-left lingual permane	0.0859	2.8535×10^{-16}	38.66	–	2.4511×10^{-16}	2.6740×10^{-16}	–	1.1642	1.0671
12806	Erupted upper front-right buccal permane	0.125	2.4209×10^{-16}	27.66	–	1.1234×10^{-16}	3.0401×10^{-16}	–	2.1549	0.7963
12807	Erupted upper front-right lingual perman	0.0693	2.0518×10^{-16}	39.52	–	2.0592×10^{-16}	1.1662×10^{-16}	–	0.9964	1.7594
12808	Erupted upper left buccal permanent enam	0.25	3.4253×10^{-16}	15.08	–	2.4105×10^{-16}	3.2334×10^{-16}	–	1.4210	1.0594
12809	Erupted upper left lingual permanent ena	0.198	1.3054×10^{-16}	40.66	–	1.1616×10^{-16}	3.2732×10^{-16}	–	1.1238	0.3988
12810	Erupted upper left buccal deciduous enam	0.0804	1.7499×10^{-16}	47.83	–	2.2127×10^{-16}	3.1293×10^{-16}	–	0.7908	0.5592
12811	Erupted upper left lingual deciduous ena	0.0749	2.6093×10^{-16}	90.34	–	5.7081×10^{-17}	3.4036×10^{-16}	–	4.5711	0.7666
12812	Erupted upper right buccal permanent ena	0.218	3.0892×10^{-16}	28.59	–	5.0638×10^{-16}	3.3369×10^{-16}	–	0.6101	0.9258
12813	Erupted upper right lingual permanent en	0.229	4.4872×10^{-16}	30.42	–	3.1296×10^{-16}	3.2919×10^{-16}	–	1.4338	1.3631
12814	Erupted upper right buccal deciduous ena	0.0705	6.1335×10^{-17}	54.06	–	1.8577×10^{-16}	7.1317×10^{-16}	–	0.3302	0.0860
12815	Erupted upper right lingual deciduous en	0.0848	2.5632×10^{-16}	45.57	–	2.4438×10^{-16}	2.4854×10^{-16}	–	1.0489	1.0313
12816	Erupted lower front buccal permanent ena	0.516	2.7412×10^{-16}	14.35	–	1.8027×10^{-16}	1.9688×10^{-16}	–	1.5206	1.3923
12817	Erupted lower front lingual permanent en	0.427	2.7961×10^{-16}	21.95	–	2.0909×10^{-16}	2.2343×10^{-16}	–	1.3373	1.2514
12824	Erupted lower left buccal permanent enam	0.27	3.2169×10^{-16}	26.87	–	1.9975×10^{-16}	2.9742×10^{-16}	–	1.6105	1.0816
12825	Erupted lower left lingual permanent ena	0.208	2.4890×10^{-16}	30.28	–	1.7678×10^{-16}	3.6764×10^{-16}	–	1.4079	0.6770
12826	Erupted lower left buccal deciduous enam	0.141	2.6186×10^{-16}	27.04	–	1.8106×10^{-16}	3.1043×10^{-16}	–	1.4463	0.8435

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
12827	Erupted lower left lingual deciduous ena	0.116	2.6727×10^{-16}	39.62	–	4.6782×10^{-16}	2.0687×10^{-16}	–	0.5713	1.2920
12828	Erupted lower right buccal permanent ena	0.237	3.2565×10^{-16}	22.15	–	2.9473×10^{-16}	2.8335×10^{-16}	–	1.1049	1.1493
12829	Erupted lower right lingual permanent en	0.241	2.4192×10^{-16}	31.53	–	3.1642×10^{-16}	2.6388×10^{-16}	–	0.7646	0.9168
12830	Erupted lower right buccal deciduous ena	0.121	2.6566×10^{-16}	48.39	–	1.4563×10^{-16}	1.6711×10^{-16}	–	1.8242	1.5897
12831	Erupted lower right lingual deciduous en	0.136	2.1479×10^{-16}	41.89	–	3.5753×10^{-16}	4.2392×10^{-16}	–	0.6008	0.5067
12832	Erupted upper front permanent dentin	2.39	2.0175×10^{-16}	11.94	–	2.2862×10^{-16}	2.4357×10^{-16}	–	0.8825	0.8283
12833	Erupted upper front deciduous dentin	0.35	2.8728×10^{-16}	17.80	–	2.8648×10^{-16}	2.5529×10^{-16}	–	1.0028	1.1253
12834	Erupted upper front-left permanent denti	0.654	3.1873×10^{-16}	26.75	–	1.9531×10^{-16}	1.3026×10^{-16}	–	1.6319	2.4469
12835	Erupted upper front-right permanent dent	0.653	2.3877×10^{-16}	24.65	–	2.7888×10^{-16}	2.2390×10^{-16}	–	0.8562	1.0664
12836	Erupted upper left permanent dentin	1.51	2.5111×10^{-16}	14.27	–	2.4940×10^{-16}	2.1635×10^{-16}	–	1.0069	1.1607
12837	Erupted upper left deciduous dentin	0.354	2.3103×10^{-16}	23.60	–	2.5815×10^{-16}	3.3657×10^{-16}	–	0.8949	0.6864
12838	Erupted upper right permanent dentin	1.51	2.9157×10^{-16}	14.36	–	2.0516×10^{-16}	2.2095×10^{-16}	–	1.4212	1.3196
12839	Erupted upper right deciduous dentin	0.352	1.2679×10^{-16}	28.84	–	3.0421×10^{-16}	2.7624×10^{-16}	–	0.4168	0.4590
12840	Erupted lower front permanent dentin	3.16	3.2489×10^{-16}	8.38	–	2.6924×10^{-16}	3.1135×10^{-16}	–	1.2067	1.0435
12844	Erupted lower left permanent dentin	1.63	2.6022×10^{-16}	12.45	–	2.6775×10^{-16}	3.6629×10^{-16}	–	0.9719	0.7104
12845	Erupted lower left deciduous dentin	0.59	2.5658×10^{-16}	15.95	–	2.9131×10^{-16}	3.2690×10^{-16}	–	0.8808	0.7849
12846	Erupted lower right permanent dentin	1.63	2.8447×10^{-16}	15.46	–	2.8586×10^{-16}	3.1516×10^{-16}	–	0.9951	0.9026
12847	Erupted lower right deciduous dentin	0.593	2.6447×10^{-16}	24.18	–	3.4468×10^{-16}	2.8199×10^{-16}	–	0.7673	0.9379
12848	Unerupted permanent enamel	1.93	2.6033×10^{-16}	7.38	–	2.9721×10^{-16}	2.6920×10^{-16}	–	0.8759	0.9670
12850	Unerupted permanent dentin	6.44	2.7582×10^{-16}	6.56	–	2.6173×10^{-16}	2.9724×10^{-16}	–	1.0538	0.9279
12852	Permanent cementum	0.779	3.2210×10^{-16}	11.63	–	2.9339×10^{-16}	2.4509×10^{-16}	–	1.0979	1.3142
12853	Deciduous cementum	0.128	3.1743×10^{-16}	15.32	–	2.5706×10^{-16}	2.9641×10^{-16}	–	1.2349	1.0709
12854	Permanent pulp	0.582	2.4366×10^{-16}	12.32	–	3.2879×10^{-16}	1.8474×10^{-16}	–	0.7411	1.3189
12855	Deciduous pulp	0.313	4.0721×10^{-16}	21.08	–	3.5475×10^{-16}	2.3377×10^{-16}	–	1.1479	1.7419
12856	Teeth retention region	0.0386	2.2821×10^{-16}	28.60	–	3.4303×10^{-16}	4.1424×10^{-16}	–	0.6653	0.5509
13100	Thymus	36.39	1.1140×10^{-15}	0.72	–	1.1108×10^{-15}	1.1042×10^{-15}	–	1.0028	1.0088
13200	Thyroid	9.02	6.4506×10^{-16}	3.45	–	5.7997×10^{-16}	6.2756×10^{-16}	–	1.1122	1.0279
13300	Tongue upper(food)	12.95	2.8774×10^{-16}	5.21	–	2.7821×10^{-16}	2.8938×10^{-16}	–	1.0343	0.9943
13301	Tongue lower	20.29	3.4345×10^{-16}	3.35	–	3.2055×10^{-16}	3.5733×10^{-16}	–	1.0715	0.9612
13400	Tonsils	3.12	2.7040×10^{-16}	5.42	–	2.4535×10^{-16}	2.5730×10^{-16}	–	1.1021	1.0509

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
13500	Ureter left	3.64	9.0457×10^{-16}	3.49	–	1.0872×10^{-15}	1.0177×10^{-15}	–	0.8320	0.8888
13600	Ureter right	3.64	1.3593×10^{-15}	4.17	–	1.4772×10^{-15}	1.4031×10^{-15}	–	0.9201	0.9688
13700	Urinary bladder wall insensitive	24.21	2.6163×10^{-16}	2.91	–	2.7594×10^{-16}	2.5883×10^{-16}	–	0.9481	1.0108
13701	Urinary bladder wall sensitive(99-212)	1.29	2.2620×10^{-16}	9.80	–	2.8392×10^{-16}	2.6746×10^{-16}	–	0.7967	0.8457
13800	Urinary bladder content	103.00	2.6540×10^{-16}	1.97	–	2.6302×10^{-16}	2.6338×10^{-16}	–	1.0091	1.0077
13900	Uterus	4.16	3.3995×10^{-16}	8.44	–	3.2530×10^{-16}	3.4880×10^{-16}	–	1.0450	0.9746
14000	Air inside body	0.0513	3.1047×10^{-15}	5.77	–	3.3825×10^{-15}	3.0557×10^{-15}	–	0.9179	1.0160

* For tissues bookkept as head, trunk, arm and leg pieces (muscle, residual soft tissue, skin, large vessels), the volumes tabulated in the distributed cell tables partition the tissue at different boundaries from the transported mesh, and the reference output is normalised by the tabulated volume. The marked ratio therefore compares two different partitions of the same tissue, and is left out of the summary statistics of Table S2.

Table S22. 10F_External. Absorbed dose per primary history, Gy. σ is the FLUKA statistical uncertainty. Ratios are FLUKA / reference.

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
100	Adrenal left	4.18	1.4522×10^{-17}	35.21	–	1.8115×10^{-17}	2.5317×10^{-17}	–	0.8017	0.5736
200	Adrenal right	4.18	1.9523×10^{-17}	22.50	–	1.3599×10^{-17}	3.2250×10^{-17}	–	1.4356	0.6054
300	ET1(0-8)	0.00259	0.0000×10^{0}	0.00	–	–	–	–	–	–
301	ET1(8-40)	0.0105	0.0000×10^{0}	0.00	–	–	6.6732×10^{-17}	–	–	–
302	ET1(40-50)	0.00332	0.0000×10^{0}	0.00	–	–	8.1610×10^{-17}	–	–	–
303	ET1(50-Surface)	0.664	2.8033×10^{-17}	47.41	–	6.0090×10^{-17}	4.0332×10^{-17}	–	0.4665	0.6950
400	ET2(-15-0)	0.136	2.5724×10^{-17}	30.86	–	4.7383×10^{-17}	3.1414×10^{-17}	–	0.5429	0.8189
401	ET2(0-40)	0.387	3.6015×10^{-17}	37.48	–	2.1345×10^{-17}	3.7204×10^{-17}	–	1.6872	0.9680
402	ET2(40-50)	0.0973	2.1178×10^{-17}	33.91	–	1.4721×10^{-17}	5.2711×10^{-17}	–	1.4386	0.4018
403	ET2(50-55)	0.0488	1.6465×10^{-17}	33.61	–	4.2095×10^{-17}	4.1726×10^{-17}	–	0.3911	0.3946
404	ET2(55-65)	0.0977	1.4801×10^{-17}	27.76	–	2.4288×10^{-17}	6.7896×10^{-17}	–	0.6094	0.2180
405	ET2(65-Surface)	22.19	2.8581×10^{-17}	10.13	–	2.9195×10^{-17}	2.5083×10^{-17}	–	0.9790	1.1395
500	Oral mucosa tongue	0.0502	8.3028×10^{-18}	32.66	–	2.2650×10^{-17}	2.4915×10^{-17}	–	0.3666	0.3332
501	Oral mucosa moth floor	0.0244	1.3231×10^{-17}	82.62	–	2.1939×10^{-17}	5.1309×10^{-18}	–	0.6031	2.5786
600	Oral mucosa lips and cheeks	0.0212	3.7631×10^{-17}	58.82	–	1.0241×10^{-17}	1.4739×10^{-17}	–	3.6743	2.5531
700	Trachea	4.68	3.5196×10^{-17}	24.93	–	1.9155×10^{-17}	3.3319×10^{-17}	–	1.8374	1.0563
800	BB(-11–6)	0.00499	0.0000×10^{0}	0.00	–	–	3.1799×10^{-17}	–	–	–
801	BB(-6-0)	0.00636	0.0000×10^{0}	0.00	–	–	4.3313×10^{-17}	–	–	–
802	BB(0-10)	0.0106	0.0000×10^{0}	0.00	–	–	3.4940×10^{-17}	–	–	–
803	BB(10-35)	0.0267	0.0000×10^{0}	0.00	–	–	2.9861×10^{-17}	–	–	–
804	BB(35-40)	0.00537	2.0595×10^{-17}	100.00	–	–	2.3979×10^{-17}	–	–	0.8589
805	BB(40-50)	0.0108	0.0000×10^{0}	0.00	–	–	2.4223×10^{-17}	–	–	–
806	BB(50-60)	0.0108	0.0000×10^{0}	0.00	–	4.5010×10^{-17}	2.3861×10^{-17}	–	–	–
807	BB(60-70)	0.0108	0.0000×10^{0}	0.00	–	3.3059×10^{-17}	2.3712×10^{-17}	–	–	–
808	BB(70-surface)	2.81	1.4572×10^{-17}	42.35	–	1.9088×10^{-17}	3.3744×10^{-17}	–	0.7634	0.4318
900	Blood in large arteries head	0.62	1.3764×10^{-17}	77.68	–	2.0132×10^{-17}	6.7585×10^{-17}	–	0.6837*	0.2037*
910	Blood in large veins head	1.89	2.4451×10^{-17}	30.03	–	3.5063×10^{-17}	2.7265×10^{-17}	–	0.6973*	0.8968*
1000	Blood in large arteries trunk	58.82	2.9016×10^{-17}	5.76	–	3.2779×10^{-17}	2.9789×10^{-17}	–	0.8852*	0.9741*
1010	Blood in large veins trunk	137.53	3.0854×10^{-17}	2.62	–	3.2835×10^{-17}	3.4683×10^{-17}	–	0.9397*	0.8896*
1100	Blood in large arteries arms	11.71	4.6748×10^{-17}	10.15	–	3.5016×10^{-17}	3.8173×10^{-17}	–	1.3351*	1.2246*
1110	Blood in large veins arms	88.06	3.8143×10^{-17}	3.22	–	3.4478×10^{-17}	3.7913×10^{-17}	–	1.1063*	1.0061*
1200	Blood in large arteries legs	78.74	3.1614×10^{-17}	7.89	–	2.9972×10^{-17}	3.0941×10^{-17}	–	1.0548	1.0217
1210	Blood in large veins legs	222.21	3.2743×10^{-17}	3.22	–	3.5265×10^{-17}	3.3899×10^{-17}	–	0.9285	0.9659
1300	Humeri upper cortical	88.31	3.1501×10^{-17}	6.16	–	2.7041×10^{-17}	2.7601×10^{-17}	–	1.1649	1.1413
1400	Humeri upper spogiosa	52.65	2.7144×10^{-17}	8.64	–	3.1586×10^{-17}	2.9870×10^{-17}	–	0.8594	0.9087

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
1500	Humeri upper medullary cavity	9.79	1.9813×10^{-17}	15.60	–	3.3470×10^{-17}	3.2044×10^{-17}	–	0.5919	0.6183
1600	Humeri lower cortical	67.36	3.6322×10^{-17}	3.56	–	3.3984×10^{-17}	3.5950×10^{-17}	–	1.0688	1.0104
1700	Humeri lower spongiosa	26.26	3.9381×10^{-17}	11.17	–	3.6402×10^{-17}	3.3958×10^{-17}	–	1.0818	1.1597
1800	Humeri lower medullary cavity	9.77	3.0677×10^{-17}	15.97	–	3.2121×10^{-17}	4.1102×10^{-17}	–	0.9551	0.7464
1900	Radii cortical	44.01	3.6001×10^{-17}	4.84	–	4.3173×10^{-17}	3.8451×10^{-17}	–	0.8339	0.9363
1910	Ulnae cortical	56.67	3.4355×10^{-17}	6.12	–	3.2617×10^{-17}	3.3705×10^{-17}	–	1.0533	1.0193
2000	Radii spongiosa	7.24	4.4834×10^{-17}	16.43	–	4.3445×10^{-17}	4.6022×10^{-17}	–	1.0320	0.9742
2010	Ulnae spongiosa	12.63	3.6968×10^{-17}	14.00	–	3.1338×10^{-17}	3.7447×10^{-17}	–	1.1797	0.9872
2100	Radii medullary cavity	6.29	5.1438×10^{-17}	14.28	–	4.4666×10^{-17}	5.4378×10^{-17}	–	1.1516	0.9459
2110	Ulnae medullary cavity	6.91	2.7868×10^{-17}	18.64	–	4.0795×10^{-17}	4.0866×10^{-17}	–	0.6831	0.6819
2200	Wrists and hand bones cortical	32.59	3.6013×10^{-17}	9.09	–	3.7893×10^{-17}	3.2053×10^{-17}	–	0.9504	1.1235
2300	Wrists and hand bones spongiosa	37.06	3.8196×10^{-17}	7.06	–	3.5866×10^{-17}	3.8358×10^{-17}	–	1.0650	0.9958
2400	Clavicles cortical	14.38	2.3255×10^{-17}	9.53	–	3.6759×10^{-17}	2.3477×10^{-17}	–	0.6326	0.9906
2500	Clavicles spongiosa	17.02	3.2469×10^{-17}	12.48	–	3.4563×10^{-17}	3.3342×10^{-17}	–	0.9394	0.9738
2600	Cranium cortical	328.08	1.9457×10^{-17}	2.50	–	2.0355×10^{-17}	1.9548×10^{-17}	–	0.9559	0.9953
2700	Cranium spongiosa	497.18	2.1546×10^{-17}	2.31	–	2.1538×10^{-17}	2.1874×10^{-17}	–	1.0004	0.9850
2800	Femora upper cortical	164.70	3.1505×10^{-17}	4.28	–	2.9851×10^{-17}	3.0206×10^{-17}	–	1.0554	1.0430
2900	Femora upper spongiosa	75.47	3.4451×10^{-17}	6.04	–	3.1181×10^{-17}	3.2103×10^{-17}	–	1.1049	1.0731
3000	Femora upper medullary cavity	25.19	3.5770×10^{-17}	4.91	–	2.6823×10^{-17}	2.8498×10^{-17}	–	1.3335	1.2552
3100	Femora lower cortical	152.14	2.8637×10^{-17}	5.33	–	2.8837×10^{-17}	2.5987×10^{-17}	–	0.9931	1.1020
3200	Femora lower spongiosa	90.51	3.1039×10^{-17}	4.61	–	3.1346×10^{-17}	3.0664×10^{-17}	–	0.9902	1.0122
3300	Femora lower medullary cavity	16.91	3.3335×10^{-17}	10.37	–	3.0197×10^{-17}	3.2063×10^{-17}	–	1.1039	1.0397
3400	Tibiae cortical	240.89	2.6925×10^{-17}	3.89	–	2.5148×10^{-17}	2.6243×10^{-17}	–	1.0707	1.0260
3410	Fibulae cortical	32.68	2.3410×10^{-17}	10.99	–	2.3701×10^{-17}	2.5449×10^{-17}	–	0.9877	0.9199
3420	Patellae cortical	2.60	3.4093×10^{-17}	25.66	–	4.1774×10^{-17}	2.0165×10^{-17}	–	0.8161	1.6907
3500	Tibiae spongiosa	62.57	2.8219×10^{-17}	7.40	–	3.0100×10^{-17}	3.1827×10^{-17}	–	0.9375	0.8866
3510	Fibulae spongiosa	7.38	2.4170×10^{-17}	14.50	–	1.7985×10^{-17}	2.8707×10^{-17}	–	1.3439	0.8419
3520	Patellae spongiosa	12.63	3.4196×10^{-17}	13.52	–	2.9128×10^{-17}	2.7209×10^{-17}	–	1.1740	1.2568
3600	Tibiae medullary cavity	26.27	2.9158×10^{-17}	14.90	–	3.3104×10^{-17}	3.2234×10^{-17}	–	0.8808	0.9046
3610	Fibulae medullary cavity	5.58	2.4196×10^{-17}	19.74	–	3.4077×10^{-17}	1.4808×10^{-17}	–	0.7101	1.6340
3700	Ankles and foot cortical	93.86	2.4394×10^{-17}	2.99	–	2.2678×10^{-17}	2.3112×10^{-17}	–	1.0756	1.0555
3800	Ankles and foot spongiosa	159.84	2.2845×10^{-17}	5.80	–	2.3686×10^{-17}	2.5421×10^{-17}	–	0.9645	0.8987
3900	Mandible cortical	24.91	3.3756×10^{-17}	8.63	–	3.4478×10^{-17}	3.5415×10^{-17}	–	0.9791	0.9532
4000	Mandible spongiosa	27.51	3.3275×10^{-17}	7.88	–	3.2730×10^{-17}	2.7198×10^{-17}	–	1.0167	1.2234
4100	Pelvis cortical	128.93	3.0484×10^{-17}	2.62	–	3.1285×10^{-17}	2.9509×10^{-17}	–	0.9744	1.0330
4200	Pelvis spongiosa	285.90	3.1231×10^{-17}	1.31	–	3.1420×10^{-17}	3.0184×10^{-17}	–	0.9940	1.0347
4300	Ribs cortical	70.42	2.2980×10^{-17}	5.08	–	2.3279×10^{-17}	2.6123×10^{-17}	–	0.9871	0.8797
4400	Ribs spongiosa	138.70	2.7235×10^{-17}	4.87	–	2.5235×10^{-17}	2.4436×10^{-17}	–	1.0792	1.1145

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
4500	Scapulae cortical	59.27	1.8733×10^{-17}	9.82	–	1.8886×10^{-17}	2.1145×10^{-17}	–	0.9919	0.8859
4600	Scapulae spongiosa	90.28	2.1298×10^{-17}	6.42	–	1.8799×10^{-17}	2.2773×10^{-17}	–	1.1329	0.9352
4700	Cervical spine cortical	14.13	3.1173×10^{-17}	11.19	–	2.7193×10^{-17}	3.0092×10^{-17}	–	1.1464	1.0359
4800	Cervical spine spongiosa	30.75	2.9561×10^{-17}	6.56	–	3.0007×10^{-17}	3.0128×10^{-17}	–	0.9851	0.9812
4900	Thoracic spine cortical	68.39	2.0399×10^{-17}	5.03	–	2.0056×10^{-17}	1.8758×10^{-17}	–	1.0171	1.0875
5000	Thoracic spine spongiosa	211.32	1.9916×10^{-17}	4.58	–	2.0488×10^{-17}	2.2482×10^{-17}	–	0.9721	0.8859
5100	Lumbar spine cortical	36.75	2.3642×10^{-17}	9.31	–	2.2433×10^{-17}	2.4738×10^{-17}	–	1.0539	0.9557
5200	Lumbar spine spongiosa	161.65	2.3778×10^{-17}	4.51	–	2.5502×10^{-17}	2.4867×10^{-17}	–	0.9324	0.9562
5300	Sacrum cortical	16.67	2.4789×10^{-17}	9.54	–	1.6825×10^{-17}	2.1348×10^{-17}	–	1.4733	1.1612
5400	Sacrum spongiosa	55.24	2.6634×10^{-17}	6.23	–	2.2412×10^{-17}	2.5894×10^{-17}	–	1.1884	1.0286
5500	Sternum cortical	5.28	3.3697×10^{-17}	21.34	–	3.4231×10^{-17}	3.3872×10^{-17}	–	0.9844	0.9948
5600	Sternum spongiosa	20.82	3.9268×10^{-17}	10.69	–	3.3047×10^{-17}	3.8989×10^{-17}	–	1.1882	1.0071
5800	Cartilage trunk	129.15	3.0106×10^{-17}	4.88	–	3.1728×10^{-17}	3.0047×10^{-17}	–	0.9489	1.0020
6100	Brain	1287.08	1.9057×10^{-17}	1.74	–	1.9932×10^{-17}	1.9249×10^{-17}	–	0.9561	0.9900
6200	Breast left adipose tissue	2.03	3.0926×10^{-17}	30.50	–	2.5645×10^{-17}	5.2745×10^{-17}	–	1.2059	0.5863
6300	Breast left glandular tissue	1.35	3.0892×10^{-17}	30.38	–	2.9459×10^{-17}	3.3163×10^{-17}	–	1.0486	0.9315
6400	Breast right adipose tissue	2.03	4.9639×10^{-17}	32.34	–	4.8697×10^{-17}	3.1507×10^{-17}	–	1.0193	1.5755
6500	Breast right glandular tissue	1.35	4.4368×10^{-17}	30.26	–	5.0144×10^{-17}	2.8975×10^{-17}	–	0.8848	1.5312
6600	Eye lens sensitive left	0.0184	0.0000×10^{0}	0.00	–	–	–	–	–	–
6601	Eye lens insensitive left	0.113	0.0000×10^{0}	0.00	–	–	–	–	–	–
6700	Cornea left	0.941	4.3289×10^{-17}	23.55	–	1.8189×10^{-17}	1.5772×10^{-17}	–	2.3799	2.7447
6701	Aqueous left	0.327	1.1222×10^{-17}	67.94	–	7.8515×10^{-18}	3.8226×10^{-17}	–	1.4292	0.2936
6702	Vitreous left	4.85	4.4426×10^{-17}	16.46	–	4.9589×10^{-17}	4.6241×10^{-17}	–	0.8959	0.9608
6800	Eye lens sensitive right	0.0184	0.0000×10^{0}	0.00	–	3.0439×10^{-17}	–	–	–	–
6801	Eye lens insensitive right	0.113	2.2436×10^{-17}	100.00	–	6.5857×10^{-17}	–	–	0.3407	–
6900	Cornea right	0.941	3.9114×10^{-17}	25.36	–	2.7639×10^{-17}	2.9625×10^{-17}	–	1.4152	1.3203
6901	Aqueous right	0.327	4.4955×10^{-17}	81.16	–	5.7447×10^{-17}	–	–	0.7825	–
6902	Vitreous right	4.85	2.6180×10^{-17}	20.06	–	4.1766×10^{-17}	3.3713×10^{-17}	–	0.6268	0.7765
7000	Gall bladder wall	4.58	2.9312×10^{-17}	18.90	–	3.0874×10^{-17}	3.3794×10^{-17}	–	0.9494	0.8674
7100	Gall bladder contents	26.00	3.5527×10^{-17}	9.49	–	3.7036×10^{-17}	3.4409×10^{-17}	–	0.9593	1.0325
7200	Stomach wall(0-60)	0.998	4.5169×10^{-17}	13.86	–	3.0210×10^{-17}	3.1994×10^{-17}	–	1.4952	1.4118
7201	Stomach wall(60-100)	0.667	3.3657×10^{-17}	26.43	–	3.1338×10^{-17}	2.4836×10^{-17}	–	1.0740	1.3552
7202	Stomach wall(100-300)	3.36	3.5885×10^{-17}	9.88	–	3.3428×10^{-17}	2.8411×10^{-17}	–	1.0735	1.2631
7203	Stomach wall(300-surface)	100.41	3.0994×10^{-17}	5.10	–	3.0404×10^{-17}	3.1877×10^{-17}	–	1.0194	0.9723
7300	Stomach contents	117.00	3.4370×10^{-17}	5.47	–	3.2610×10^{-17}	3.1688×10^{-17}	–	1.0540	1.0846
7400	Small intestine wall(0-130)	8.70	4.2586×10^{-17}	9.14	–	3.6247×10^{-17}	3.7066×10^{-17}	–	1.1749	1.1489
7401	Small intestine wall(130-150)	1.36	3.9043×10^{-17}	14.61	–	4.2429×10^{-17}	3.9856×10^{-17}	–	0.9202	0.9796
7402	Small intestine wall(150-200)	3.44	3.9845×10^{-17}	10.40	–	3.3398×10^{-17}	3.7485×10^{-17}	–	1.1930	1.0630

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
7403	Small intestine wall(200-surface)	454.52	3.7673×10^{-17}	2.20	–	3.6127×10^{-17}	3.6216×10^{-17}	–	1.0428	1.0402
7500	Small intestine contents(-500-0)	31.37	4.0710×10^{-17}	4.11	–	3.6650×10^{-17}	3.6432×10^{-17}	–	1.1108	1.1174
7501	Small intestine contents(centre–500)	131.67	3.6619×10^{-17}	5.23	–	3.6867×10^{-17}	3.4750×10^{-17}	–	0.9933	1.0538
7600	Ascending colon wall(0-280)	2.59	2.9994×10^{-17}	28.22	–	3.7937×10^{-17}	4.7666×10^{-17}	–	0.7906	0.6292
7601	Ascending colon wall(280-300)	0.188	4.1570×10^{-17}	44.27	–	5.8421×10^{-17}	4.4598×10^{-17}	–	0.7116	0.9321
7602	Ascending colon wall(300-surface)	50.88	3.2759×10^{-17}	7.54	–	3.7958×10^{-17}	3.5012×10^{-17}	–	0.8630	0.9357
7700	Ascending colon content	45.28	3.3044×10^{-17}	9.69	–	3.4528×10^{-17}	3.3734×10^{-17}	–	0.9570	0.9795
7800	Transverse colon wall right(0-280)	1.92	3.4357×10^{-17}	21.32	–	4.2777×10^{-17}	3.9432×10^{-17}	–	0.8032	0.8713
7801	Transverse colon wall right(280-300)	0.14	4.7836×10^{-17}	56.21	–	2.8921×10^{-17}	4.2528×10^{-17}	–	1.6540	1.1248
7802	Transverse colon wall right(300-surface)	51.71	3.8293×10^{-17}	5.87	–	3.7123×10^{-17}	3.9236×10^{-17}	–	1.0315	0.9760
7900	Transverse colon contents right	24.72	3.6521×10^{-17}	9.95	–	3.5247×10^{-17}	3.8219×10^{-17}	–	1.0362	0.9556
8000	Transverse colon wall left(0-280)	1.49	2.6168×10^{-17}	26.56	–	3.8721×10^{-17}	5.1327×10^{-17}	–	0.6758	0.5098
8001	Transverse colon wall left(280-300)	0.109	5.1172×10^{-17}	51.48	–	4.7243×10^{-17}	6.0717×10^{-17}	–	1.0832	0.8428
8002	Transverse colon wall left(300-surface)	45.61	3.8283×10^{-17}	4.30	–	3.9625×10^{-17}	3.7922×10^{-17}	–	0.9661	1.0095
8100	Transverse colon content left	15.05	3.9538×10^{-17}	10.43	–	3.7617×10^{-17}	3.9273×10^{-17}	–	1.0511	1.0068
8200	Descending colon wall(0-280)	2.01	5.6456×10^{-17}	17.06	–	3.9533×10^{-17}	4.1114×10^{-17}	–	1.4281	1.3732
8201	Descending colon wall(280-300)	0.148	5.3606×10^{-17}	27.72	–	1.5286×10^{-17}	4.7254×10^{-17}	–	3.5069	1.1344
8202	Descending colon wall(300-surface)	58.06	3.7492×10^{-17}	4.07	–	3.7218×10^{-17}	3.5327×10^{-17}	–	1.0074	1.0613
8300	Descending colon content	19.95	4.1853×10^{-17}	11.45	–	3.3564×10^{-17}	3.5472×10^{-17}	–	1.2470	1.1799
8400	Sigmoid colon wall(0-280)	2.45	2.5348×10^{-17}	21.61	–	3.1458×10^{-17}	2.7075×10^{-17}	–	0.8057	0.9362
8401	Sigmoid colon wall(280-300)	0.18	1.4548×10^{-17}	36.62	–	4.2467×10^{-17}	2.0915×10^{-17}	–	0.3426	0.6956
8402	Sigmoid colon wall(300-surface)	42.10	3.0526×10^{-17}	4.17	–	3.1706×10^{-17}	2.8832×10^{-17}	–	0.9628	1.0588
8500	Sigmoid colon contents	24.52	3.2737×10^{-17}	8.67	–	3.2371×10^{-17}	3.3948×10^{-17}	–	1.0113	0.9643
8600	Rectum wall(0-280)	1.03	1.9373×10^{-17}	36.70	–	1.7316×10^{-17}	2.6807×10^{-17}	–	1.1189	0.7227
8601	Rectum wall(280-300)	0.0758	1.3485×10^{-17}	40.65	–	3.7708×10^{-17}	4.4256×10^{-17}	–	0.3576	0.3047
8602	Rectum wall(300-surface)	4.71	2.8106×10^{-17}	22.29	–	2.9112×10^{-17}	2.0427×10^{-17}	–	0.9654	1.3759
8603	Rectum contents	10.48	2.5833×10^{-17}	12.79	–	2.5364×10^{-17}	2.4135×10^{-17}	–	1.0185	1.0704
8700	Heart wall	161.50	3.1955×10^{-17}	4.06	–	3.3724×10^{-17}	3.1504×10^{-17}	–	0.9475	1.0143
8800	Blood in heart chamber	230.00	3.0093×10^{-17}	4.25	–	2.8745×10^{-17}	3.0244×10^{-17}	–	1.0469	0.9950
8900	Kidney left cortex	82.03	2.0993×10^{-17}	7.42	–	2.2774×10^{-17}	2.3641×10^{-17}	–	0.9218	0.8880
9000	Kidney left medulla	29.30	2.1924×10^{-17}	7.27	–	2.0705×10^{-17}	2.3550×10^{-17}	–	1.0589	0.9310
9100	Kidney left pelvis	5.86	1.6273×10^{-17}	22.98	–	1.7999×10^{-17}	2.3134×10^{-17}	–	0.9041	0.7034
9200	Kidney right cortex	82.03	2.0563×10^{-17}	4.65	–	2.4532×10^{-17}	2.4015×10^{-17}	–	0.8382	0.8563
9300	Kidney right medulla	29.30	2.0260×10^{-17}	9.17	–	2.2732×10^{-17}	2.2406×10^{-17}	–	0.8912	0.9042
9400	Kidney right pelvis	5.86	1.6718×10^{-17}	19.81	–	2.3206×10^{-17}	3.2079×10^{-17}	–	0.7204	0.5211
9500	Liver	1059.77	3.2137×10^{-17}	1.91	–	3.1517×10^{-17}	3.2023×10^{-17}	–	1.0197	1.0036

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
9700	Lung(AI) left	222.24	2.6147×10^{-17}	3.91	–	2.5147×10^{-17}	2.7753×10^{-17}	–	1.0398	0.9421
9900	Lung(AI) right	277.76	2.8049×10^{-17}	2.30	–	2.6951×10^{-17}	2.6239×10^{-17}	–	1.0407	1.0690
10000	Lymphatic nodes ET	7.16	3.0786×10^{-17}	19.45	–	3.0921×10^{-17}	2.5325×10^{-17}	–	0.9956	1.2156
10100	Lymphatic nodes thoracic	7.16	2.7678×10^{-17}	13.23	–	2.6264×10^{-17}	1.9707×10^{-17}	–	1.0538	1.4045
10200	Lymphatic nodes head	2.47	2.9420×10^{-17}	33.97	–	2.3708×10^{-17}	3.2885×10^{-17}	–	1.2409	0.8946
10300	Lymphatic nodes trunk	58.43	3.1926×10^{-17}	7.29	–	3.0160×10^{-17}	3.1089×10^{-17}	–	1.0586	1.0269
10400	Lymphatic nodes arms	4.94	3.5467×10^{-17}	19.61	–	3.0891×10^{-17}	3.3191×10^{-17}	–	1.1481	1.0686
10500	Lymphatic nodes legs	4.94	2.6553×10^{-17}	26.90	–	2.7197×10^{-17}	2.9667×10^{-17}	–	0.9763	0.8950
10600	Muscle head	457.24	2.7682×10^{-17}	2.96	–	2.7322×10^{-17}	2.7073×10^{-17}	–	1.0132	1.0225
10700	Muscle trunk	5450.50	2.8915×10^{-17}	0.87	–	2.8764×10^{-17}	2.8757×10^{-17}	–	1.0052*	1.0055*
10800	Muscle arms	1076.44	3.4562×10^{-17}	1.67	–	3.5110×10^{-17}	3.4434×10^{-17}	–	0.9844*	1.0037*
10900	Muscle legs	4272.92	3.0996×10^{-17}	0.98	–	3.1354×10^{-17}	3.1106×10^{-17}	–	0.9886*	0.9965*
11000	Oesophagus wall(0-190)	1.57	1.8578×10^{-17}	28.45	–	2.6676×10^{-17}	2.9804×10^{-17}	–	0.6964	0.6233
11001	Oesophagus wall(190-200)	0.0843	7.7662×10^{-18}	25.62	–	2.8233×10^{-17}	5.8248×10^{-17}	–	0.2751	0.1333
11002	Oesophagus wall(200-surface)	20.66	2.4800×10^{-17}	14.42	–	2.9941×10^{-17}	3.0393×10^{-17}	–	0.8283	0.8160
11003	Oesophagus contents	11.91	2.0698×10^{-17}	14.53	–	2.5875×10^{-17}	2.1475×10^{-17}	–	0.7999	0.9638
11100	Ovary left	1.84	3.1009×10^{-17}	31.19	–	2.7328×10^{-17}	1.2820×10^{-17}	–	1.1347	2.4188
11200	Ovary right	1.84	2.4394×10^{-17}	39.85	–	3.3023×10^{-17}	3.0799×10^{-17}	–	0.7387	0.7921
11300	Pancreas	72.13	3.1964×10^{-17}	6.26	–	3.0700×10^{-17}	3.0207×10^{-17}	–	1.0412	1.0582
11400	Pituitary gland	0.364	1.2582×10^{-17}	89.56	–	9.5313×10^{-17}	–	–	0.1320	–
11600	RST head	571.92	2.4909×10^{-17}	2.60	–	2.4718×10^{-17}	2.5449×10^{-17}	–	1.0078	0.9788
11700	RST trunk	6580.66	3.1503×10^{-17}	0.65	–	3.1480×10^{-17}	3.1392×10^{-17}	–	1.0007*	1.0035*
11800	RST arms	730.89	3.5060×10^{-17}	1.68	–	3.4489×10^{-17}	3.4865×10^{-17}	–	1.0166*	1.0056*
11900	RST legs	1867.01	3.1756×10^{-17}	1.49	–	3.1881×10^{-17}	3.2075×10^{-17}	–	0.9961*	0.9901*
12000	Salivary glands left	22.86	2.9183×10^{-17}	5.25	–	3.0996×10^{-17}	3.1516×10^{-17}	–	0.9415	0.9260
12100	Salivary glandss right	22.86	3.0394×10^{-17}	11.20	–	3.3079×10^{-17}	3.0769×10^{-17}	–	0.9188	0.9878
12200	Skin head insensitive	82.16	1.9570×10^{-17}	4.43	–	1.8625×10^{-17}	2.0424×10^{-17}	–	1.0508*	0.9582*
12201	Skin head sensitive(40-100)	6.89	1.5248×10^{-17}	10.08	–	1.4095×10^{-17}	1.4314×10^{-17}	–	1.0818*	1.0652*
12300	Skin trunk insensitive	327.47	2.0634×10^{-17}	2.34	–	2.0625×10^{-17}	2.1793×10^{-17}	–	1.0004*	0.9468*
12301	Skin trunk sensitive(40-100)	35.11	1.4807×10^{-17}	3.51	–	1.3873×10^{-17}	1.5285×10^{-17}	–	1.0673*	0.9687*
12400	Skin arms insensitive	108.23	2.8607×10^{-17}	3.94	–	2.6515×10^{-17}	2.6971×10^{-17}	–	1.0789*	1.0607*
12401	Skin arms sensitive(40-100)	8.58	2.2969×10^{-17}	9.31	–	2.2551×10^{-17}	1.9164×10^{-17}	–	1.0185*	1.1985*
12500	Skin legs insensitive	271.96	2.3599×10^{-17}	2.89	–	2.2898×10^{-17}	2.3957×10^{-17}	–	1.0306*	0.9851*
12501	Skin legs sensitive(40-100)	18.51	1.8357×10^{-17}	4.48	–	1.6745×10^{-17}	1.7817×10^{-17}	–	1.0963*	1.0303*
12600	Spinal cord	45.34	2.0289×10^{-17}	5.93	–	2.3209×10^{-17}	1.9066×10^{-17}	–	0.8742	1.0641
12700	Spleen	114.98	2.5090×10^{-17}	4.83	–	2.3681×10^{-17}	2.6015×10^{-17}	–	1.0595	0.9644
12800	Erupted upper front buccal permanent ena	0.35	4.6574×10^{-17}	53.89	–	5.6305×10^{-17}	3.8478×10^{-17}	–	0.8272	1.2104

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
12801	Erupted upper front lingual permanent en	0.356	4.8829×10^{-17}	87.92	–	1.2277×10^{-17}	9.5627×10^{-19}	–	3.9772	51.0620
12802	Erupted upper front buccal deciduous ena	0.0649	0.0000×10^{0}	0.00	–	1.4518×10^{-17}	1.4134×10^{-18}	–	–	–
12803	Erupted upper front lingual deciduous en	0.0903	0.0000×10^{0}	0.00	–	1.9810×10^{-18}	2.0748×10^{-16}	–	–	–
12804	Erupted upper front-left buccal permanen	0.108	3.9546×10^{-17}	100.00	–	–	1.1574×10^{-16}	–	–	0.3417
12805	Erupted upper front-left lingual permane	0.0859	1.3605×10^{-16}	66.41	–	1.4094×10^{-16}	–	–	0.9653	–
12806	Erupted upper front-right buccal permane	0.125	1.9998×10^{-17}	78.88	–	–	–	–	–	–
12807	Erupted upper front-right lingual perman	0.0693	1.2258×10^{-16}	100.00	–	–	–	–	–	–
12808	Erupted upper left buccal permanent enam	0.25	6.5192×10^{-17}	77.31	–	2.9486×10^{-17}	1.2229×10^{-16}	–	2.2110	0.5331
12809	Erupted upper left lingual permanent ena	0.198	8.3519×10^{-18}	92.37	–	–	6.3918×10^{-17}	–	–	0.1307
12810	Erupted upper left buccal deciduous enam	0.0804	0.0000×10^{0}	0.00	–	–	–	–	–	–
12811	Erupted upper left lingual deciduous ena	0.0749	0.0000×10^{0}	0.00	–	2.9686×10^{-17}	2.0156×10^{-17}	–	–	–
12812	Erupted upper right buccal permanent ena	0.218	9.2480×10^{-19}	100.00	–	5.1886×10^{-17}	2.5492×10^{-17}	–	0.0178	0.0363
12813	Erupted upper right lingual permanent en	0.229	5.1603×10^{-18}	78.16	–	7.5701×10^{-17}	2.5857×10^{-19}	–	0.0682	19.9569
12814	Erupted upper right buccal deciduous ena	0.0705	6.8007×10^{-19}	100.00	–	–	–	–	–	–
12815	Erupted upper right lingual deciduous en	0.0848	1.2499×10^{-17}	100.00	–	–	9.8326×10^{-17}	–	–	0.1271
12816	Erupted lower front buccal permanent ena	0.516	2.5626×10^{-17}	77.84	–	4.4517×10^{-17}	3.7953×10^{-17}	–	0.5757	0.6752
12817	Erupted lower front lingual permanent en	0.427	7.1040×10^{-17}	38.89	–	6.4781×10^{-17}	1.2694×10^{-17}	–	1.0966	5.5964
12824	Erupted lower left buccal permanent enam	0.27	4.9205×10^{-17}	100.00	–	2.7765×10^{-17}	7.1072×10^{-17}	–	1.7722	0.6923
12825	Erupted lower left lingual permanent ena	0.208	2.6530×10^{-17}	61.14	–	4.0957×10^{-17}	–	–	0.6478	–
12826	Erupted lower left buccal deciduous enam	0.141	0.0000×10^{0}	0.00	–	–	6.4729×10^{-17}	–	–	–

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
12827	Erupted lower left lingual deciduous ena	0.116	0.0000×10^{0}	0.00	–	3.4212×10^{-18}	4.5454×10^{-18}	–	–	–
12828	Erupted lower right buccal permanent ena	0.237	1.2044×10^{-17}	100.00	–	3.3823×10^{-18}	–	–	3.5608	–
12829	Erupted lower right lingual permanent en	0.241	8.4827×10^{-18}	91.40	–	2.5000×10^{-17}	7.1893×10^{-17}	–	0.3393	0.1180
12830	Erupted lower right buccal deciduous ena	0.121	0.0000×10^{0}	0.00	–	7.1578×10^{-17}	1.6960×10^{-17}	–	–	–
12831	Erupted lower right lingual deciduous en	0.136	5.5291×10^{-17}	83.70	–	–	1.3442×10^{-17}	–	–	4.1133
12832	Erupted upper front permanent dentin	2.39	4.2069×10^{-17}	32.85	–	1.8003×10^{-17}	1.0755×10^{-17}	–	2.3367	3.9116
12833	Erupted upper front deciduous dentin	0.35	3.4511×10^{-17}	60.91	–	1.1427×10^{-17}	6.5674×10^{-17}	–	3.0202	0.5255
12834	Erupted upper front-left permanent denti	0.654	6.3521×10^{-17}	43.21	–	2.0321×10^{-17}	3.0560×10^{-17}	–	3.1259	2.0786
12835	Erupted upper front-right permanent dent	0.653	4.3754×10^{-17}	84.26	–	1.6336×10^{-17}	2.0710×10^{-17}	–	2.6784	2.1127
12836	Erupted upper left permanent dentin	1.51	2.7657×10^{-17}	68.15	–	3.0456×10^{-17}	5.3732×10^{-17}	–	0.9081	0.5147
12837	Erupted upper left deciduous dentin	0.354	3.3971×10^{-17}	57.15	–	4.0437×10^{-18}	4.8175×10^{-17}	–	8.4010	0.7052
12838	Erupted upper right permanent dentin	1.51	1.9244×10^{-17}	41.97	–	4.7889×10^{-17}	4.1751×10^{-17}	–	0.4019	0.4609
12839	Erupted upper right deciduous dentin	0.352	3.1935×10^{-18}	100.00	–	2.7078×10^{-17}	2.7868×10^{-17}	–	0.1179	0.1146
12840	Erupted lower front permanent dentin	3.16	3.6713×10^{-17}	22.49	–	3.4311×10^{-17}	3.7024×10^{-17}	–	1.0700	0.9916
12844	Erupted lower left permanent dentin	1.63	5.0114×10^{-17}	27.03	–	2.6026×10^{-17}	2.5473×10^{-17}	–	1.9256	1.9674
12845	Erupted lower left deciduous dentin	0.59	1.4847×10^{-17}	56.98	–	9.8307×10^{-18}	3.4418×10^{-17}	–	1.5103	0.4314
12846	Erupted lower right permanent dentin	1.63	8.3647×10^{-18}	38.19	–	1.7772×10^{-17}	2.1940×10^{-17}	–	0.4707	0.3813
12847	Erupted lower right deciduous dentin	0.593	2.7543×10^{-17}	60.46	–	1.4704×10^{-17}	8.9900×10^{-17}	–	1.8732	0.3064
12848	Unerupted permanent enamel	1.93	2.6028×10^{-17}	26.34	–	5.1823×10^{-17}	3.7564×10^{-17}	–	0.5022	0.6929
12850	Unerupted permanent dentin	6.44	2.1717×10^{-17}	23.74	–	3.4172×10^{-17}	3.6190×10^{-17}	–	0.6355	0.6001
12852	Permanent cementum	0.779	2.3820×10^{-17}	37.68	–	2.6215×10^{-17}	2.7323×10^{-17}	–	0.9086	0.8718
12853	Deciduous cementum	0.128	8.6090×10^{-18}	100.00	–	2.3621×10^{-17}	1.1506×10^{-18}	–	0.3645	7.4822
12854	Permanent pulp	0.582	5.2692×10^{-17}	67.20	–	4.8993×10^{-17}	9.4036×10^{-18}	–	1.0755	5.6034
12855	Deciduous pulp	0.313	3.4923×10^{-17}	67.22	–	4.3700×10^{-17}	3.8159×10^{-17}	–	0.7992	0.9152
12856	Teeth retention region	0.0386	1.2241×10^{-17}	22.03	–	3.1860×10^{-17}	2.2740×10^{-17}	–	0.3842	0.5383
13100	Thymus	36.39	3.1889×10^{-17}	11.23	–	3.2754×10^{-17}	3.0496×10^{-17}	–	0.9736	1.0457
13200	Thyroid	9.02	2.4343×10^{-17}	15.61	–	2.6951×10^{-17}	3.0445×10^{-17}	–	0.9032	0.7996
13300	Tongue upper(food)	12.95	2.9331×10^{-17}	19.88	–	2.9225×10^{-17}	2.8972×10^{-17}	–	1.0036	1.0124
13301	Tongue lower	20.29	3.4198×10^{-17}	14.98	–	3.2804×10^{-17}	2.9928×10^{-17}	–	1.0425	1.1427
13400	Tonsils	3.12	2.2042×10^{-17}	20.66	–	3.9404×10^{-17}	2.5633×10^{-17}	–	0.5594	0.8599

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
13500	Ureter left	3.64	2.5796×10^{-17}	22.33	–	1.8155×10^{-17}	2.5735×10^{-17}	–	1.4209	1.0024
13600	Ureter right	3.64	3.0529×10^{-17}	38.60	–	2.0838×10^{-17}	4.1331×10^{-17}	–	1.4650	0.7386
13700	Urinary bladder wall insensitive	24.21	3.3054×10^{-17}	7.82	–	3.4613×10^{-17}	4.3212×10^{-17}	–	0.9550	0.7649
13701	Urinary bladder wall sensitive(99-212)	1.29	2.6653×10^{-17}	20.34	–	4.2493×10^{-17}	3.7939×10^{-17}	–	0.6272	0.7025
13800	Urinary bladder content	103.00	3.6122×10^{-17}	3.54	–	3.8429×10^{-17}	3.7832×10^{-17}	–	0.9400	0.9548
13900	Uterus	4.16	2.3286×10^{-17}	20.39	–	4.1332×10^{-17}	4.4841×10^{-17}	–	0.5634	0.5193
14000	Air inside body	0.0513	2.2450×10^{-17}	23.30	–	2.4123×10^{-17}	1.3228×10^{-17}	–	0.9306	1.6972

* For tissues bookkept as head, trunk, arm and leg pieces (muscle, residual soft tissue, skin, large vessels), the volumes tabulated in the distributed cell tables partition the tissue at different boundaries from the transported mesh, and the reference output is normalised by the tabulated volume. The marked ratio therefore compares two different partitions of the same tissue, and is left out of the summary statistics of Table S2.

Table S23. 15M_Internal. Absorbed dose per primary history, Gy. σ is the FLUKA statistical uncertainty. Ratios are FLUKA / reference.

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
100	Adrenal left	6.21	2.1292×10^{-15}	2.10	–	2.0886×10^{-15}	2.0990×10^{-15}	–	1.0194	1.0144
200	Adrenal right	6.21	5.0248×10^{-15}	1.03	–	5.1249×10^{-15}	5.1307×10^{-15}	–	0.9805	0.9794
300	ET1(0-8)	0.0106	4.6482×10^{-16}	47.62	–	2.6933×10^{-18}	1.0404×10^{-16}	–	172.5816	4.4677
301	ET1(8-40)	0.0425	1.4288×10^{-16}	43.04	–	8.1327×10^{-18}	2.2519×10^{-16}	–	17.5680	0.6345
302	ET1(40-50)	0.0134	2.0661×10^{-16}	28.43	–	9.2165×10^{-17}	1.1273×10^{-16}	–	2.2418	1.8328
303	ET1(50-Surface)	4.15	1.0345×10^{-16}	10.10	–	8.6609×10^{-17}	1.1126×10^{-16}	–	1.1944	0.9298
400	ET2(-15-0)	0.2	9.9762×10^{-17}	25.43	–	1.2426×10^{-16}	1.0748×10^{-16}	–	0.8029	0.9282
401	ET2(0-40)	0.572	9.6119×10^{-17}	13.14	–	1.3299×10^{-16}	1.2193×10^{-16}	–	0.7228	0.7883
402	ET2(40-50)	0.144	1.2525×10^{-16}	19.32	–	1.4793×10^{-16}	1.6013×10^{-16}	–	0.8467	0.7822
403	ET2(50-55)	0.072	1.2554×10^{-16}	31.14	–	1.1944×10^{-16}	8.4226×10^{-17}	–	1.0511	1.4906
404	ET2(55-65)	0.144	1.2261×10^{-16}	17.65	–	1.0068×10^{-16}	1.1056×10^{-16}	–	1.2179	1.1090
405	ET2(65-Surface)	30.93	9.8862×10^{-17}	3.74	–	9.9957×10^{-17}	1.0658×10^{-16}	–	0.9890	0.9276
500	Oral mucosa tongue	0.0767	1.2429×10^{-16}	29.93	–	2.6989×10^{-16}	1.9233×10^{-16}	–	0.4605	0.6462
501	Oral mucosa moth floor	0.0317	9.6387×10^{-17}	29.76	–	1.1579×10^{-16}	5.2870×10^{-17}	–	0.8324	1.8231
600	Oral mucosa lips and cheeks	0.0212	7.5376×10^{-17}	57.18	–	3.3822×10^{-17}	1.6750×10^{-16}	–	2.2286	0.4500
700	Trachea	7.93	3.9226×10^{-16}	2.83	–	3.6110×10^{-16}	3.8912×10^{-16}	–	1.0863	1.0081
800	BB(-11–6)	0.00916	5.7201×10^{-16}	23.81	–	7.3301×10^{-16}	8.2666×10^{-16}	–	0.7804	0.6920
801	BB(-6-0)	0.0117	8.7851×10^{-16}	41.88	–	9.3268×10^{-16}	1.0822×10^{-15}	–	0.9419	0.8118
802	BB(0-10)	0.0196	1.0485×10^{-15}	16.27	–	7.8402×10^{-16}	6.8524×10^{-16}	–	1.3373	1.5301
803	BB(10-35)	0.0492	8.9149×10^{-16}	22.28	–	9.2473×10^{-16}	9.8118×10^{-16}	–	0.9641	0.9086
804	BB(35-40)	0.00987	7.4057×10^{-16}	31.90	–	8.0804×10^{-16}	6.9724×10^{-16}	–	0.9165	1.0622
805	BB(40-50)	0.0198	7.5235×10^{-16}	22.27	–	8.0553×10^{-16}	6.7593×10^{-16}	–	0.9340	1.1131
806	BB(50-60)	0.0198	8.4728×10^{-16}	22.84	–	8.0230×10^{-16}	9.0517×10^{-16}	–	1.0561	0.9360
807	BB(60-70)	0.0199	5.0843×10^{-16}	22.89	–	7.1369×10^{-16}	8.5266×10^{-16}	–	0.7124	0.5963
808	BB(70-surface)	7.24	7.0945×10^{-16}	4.64	–	6.8409×10^{-16}	6.8815×10^{-16}	–	1.0371	1.0310
900	Blood in large arteries head	1.59	1.2782×10^{-16}	25.35	–	1.3280×10^{-16}	1.3092×10^{-16}	–	0.9625*	0.9763*
910	Blood in large veins head	5.22	1.3208×10^{-16}	10.77	–	1.2011×10^{-16}	1.0678×10^{-16}	–	1.0997*	1.2370*
1000	Blood in large arteries trunk	103.81	9.8397×10^{-16}	0.86	–	9.9389×10^{-16}	1.0050×10^{-15}	–	0.9900*	0.9791*
1010	Blood in large veins trunk	249.10	1.0509×10^{-15}	0.55	–	1.1127×10^{-15}	1.1037×10^{-15}	–	0.9444*	0.9521*
1100	Blood in large arteries arms	31.47	6.7794×10^{-16}	1.90	–	6.8045×10^{-16}	6.7442×10^{-16}	–	0.9963*	1.0052*
1110	Blood in large veins arms	146.40	4.8168×10^{-16}	0.90	–	4.9298×10^{-16}	4.8682×10^{-16}	–	0.9771*	0.9894*
1200	Blood in large arteries legs	150.90	1.9057×10^{-17}	4.80	–	2.1917×10^{-17}	2.0915×10^{-17}	–	0.8695*	0.9112*
1210	Blood in large veins legs	462.58	1.9408×10^{-17}	2.26	–	2.0677×10^{-17}	2.2239×10^{-17}	–	0.9386*	0.8727*
1300	Humeri upper cortical	153.12	4.3774×10^{-16}	1.11	–	4.4372×10^{-16}	4.2865×10^{-16}	–	0.9865	1.0212
1400	Humeri upper spogiosa	172.91	2.7340×10^{-16}	1.11	–	2.7299×10^{-16}	2.7471×10^{-16}	–	1.0015	0.9952

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
1500	Humeri upper medullary cavity	19.32	4.9939×10^{-16}	2.72	–	4.8251×10^{-16}	4.9860×10^{-16}	–	1.0350	1.0016
1600	Humeri lower cortical	167.64	7.4534×10^{-16}	0.69	–	7.7167×10^{-16}	7.5243×10^{-16}	–	0.9659	0.9906
1700	Humeri lower spongiosa	70.77	7.0026×10^{-16}	1.30	–	7.0420×10^{-16}	7.1157×10^{-16}	–	0.9944	0.9841
1800	Humeri lower medullary cavity	16.56	8.7738×10^{-16}	2.63	–	8.7452×10^{-16}	8.9380×10^{-16}	–	1.0033	0.9816
1900	Radii cortical	97.60	2.4493×10^{-16}	1.56	–	2.3573×10^{-16}	2.4008×10^{-16}	–	1.0390	1.0202
1910	Ulnae cortical	124.44	2.6642×10^{-16}	0.94	–	2.6196×10^{-16}	2.6436×10^{-16}	–	1.0170	1.0078
2000	Radii spongiosa	37.90	1.9977×10^{-16}	3.14	–	2.0032×10^{-16}	2.0857×10^{-16}	–	0.9973	0.9578
2010	Ulnae spongiosa	54.22	4.4581×10^{-16}	2.05	–	4.2845×10^{-16}	4.1322×10^{-16}	–	1.0405	1.0789
2100	Radii medullary cavity	6.59	2.6056×10^{-16}	7.07	–	2.3932×10^{-16}	2.1237×10^{-16}	–	1.0887	1.2269
2110	Ulnae medullary cavity	8.39	2.0137×10^{-16}	6.21	–	2.3662×10^{-16}	2.0059×10^{-16}	–	0.8510	1.0039
2200	Wrists and hand bones cortical	130.63	5.4788×10^{-17}	3.68	–	5.4099×10^{-17}	5.4655×10^{-17}	–	1.0127	1.0024
2300	Wrists and hand bones spongiosa	59.94	6.2884×10^{-17}	6.66	–	6.4161×10^{-17}	6.2822×10^{-17}	–	0.9801	1.0010
2400	Clavicles cortical	43.29	3.1302×10^{-16}	2.07	–	3.1525×10^{-16}	3.2032×10^{-16}	–	0.9929	0.9772
2500	Clavicles spongiosa	21.10	3.1366×10^{-16}	3.63	–	3.2138×10^{-16}	3.2212×10^{-16}	–	0.9760	0.9737
2600	Cranium cortical	544.13	5.2403×10^{-17}	1.52	–	5.3057×10^{-17}	5.3448×10^{-17}	–	0.9877	0.9804
2700	Cranium spongiosa	493.24	5.5762×10^{-17}	2.18	–	5.6230×10^{-17}	5.5112×10^{-17}	–	0.9917	1.0118
2800	Femora upper cortical	315.08	4.3049×10^{-17}	1.42	–	4.1233×10^{-17}	4.2663×10^{-17}	–	1.0440	1.0091
2900	Femora upper spongiosa	251.20	1.1874×10^{-16}	1.49	–	1.1556×10^{-16}	1.1818×10^{-16}	–	1.0276	1.0048
3000	Femora upper medullary cavity	49.29	3.0117×10^{-17}	6.96	–	3.0589×10^{-17}	3.5035×10^{-17}	–	0.9846	0.8596
3100	Femora lower cortical	243.61	5.1878×10^{-18}	7.00	–	5.2904×10^{-18}	5.2112×10^{-18}	–	0.9806	0.9955
3200	Femora lower spongiosa	308.42	3.4738×10^{-18}	9.25	–	3.7022×10^{-18}	3.2642×10^{-18}	–	0.9383	1.0642
3300	Femora lower medullary cavity	36.21	4.4939×10^{-18}	16.21	–	6.4110×10^{-18}	7.0411×10^{-18}	–	0.7010	0.6382
3400	Tibiae cortical	381.34	1.3695×10^{-18}	8.95	–	1.7736×10^{-18}	1.4648×10^{-18}	–	0.7721	0.9349
3410	Fibulae cortical	72.32	5.3344×10^{-19}	25.07	–	1.0726×10^{-18}	8.5557×10^{-19}	–	0.4973	0.6235
3420	Patellae cortical	21.42	4.9927×10^{-18}	13.54	–	4.2635×10^{-18}	3.1387×10^{-18}	–	1.1710	1.5907
3500	Tibiae spongiosa	264.54	2.0520×10^{-18}	13.15	–	2.0588×10^{-18}	2.1930×10^{-18}	–	0.9967	0.9357
3510	Fibulae spongiosa	31.69	1.1016×10^{-18}	39.66	–	1.6221×10^{-18}	1.1159×10^{-18}	–	0.6791	0.9872
3520	Patellae spongiosa	22.26	5.4069×10^{-18}	18.67	–	7.1266×10^{-18}	2.3389×10^{-18}	–	0.7587	2.3117
3600	Tibiae medullary cavity	59.42	1.7162×10^{-18}	34.11	–	1.8233×10^{-18}	1.3146×10^{-18}	–	0.9412	1.3055
3610	Fibulae medullary cavity	8.19	2.0043×10^{-19}	67.05	–	9.7531×10^{-21}	–	–	20.5509	–
3700	Ankles and foot cortical	217.34	4.2314×10^{-18}	12.11	–	4.9165×10^{-18}	4.8744×10^{-18}	–	0.8607	0.8681
3800	Ankles and foot spongiosa	387.18	2.7960×10^{-18}	7.14	–	2.6296×10^{-18}	3.2449×10^{-18}	–	1.0633	0.8617
3900	Mandible cortical	59.29	1.4534×10^{-16}	3.23	–	1.4106×10^{-16}	1.4205×10^{-16}	–	1.0303	1.0231
4000	Mandible spongiosa	37.75	1.4673×10^{-16}	3.92	–	1.5309×10^{-16}	1.4981×10^{-16}	–	0.9585	0.9794
4100	Pelvis cortical	188.61	3.8859×10^{-16}	1.23	–	3.8698×10^{-16}	3.8204×10^{-16}	–	1.0042	1.0172
4200	Pelvis spongiosa	512.50	3.5238×10^{-16}	0.50	–	3.5903×10^{-16}	3.5537×10^{-16}	–	0.9815	0.9916
4300	Ribs cortical	169.28	1.2051×10^{-15}	0.51	–	1.2095×10^{-15}	1.1976×10^{-15}	–	0.9963	1.0062
4400	Ribs spongiosa	198.25	1.1486×10^{-15}	0.51	–	1.1408×10^{-15}	1.1481×10^{-15}	–	1.0068	1.0005

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
4500	Scapulae cortical	124.48	4.6352×10^{-16}	1.04	–	4.6396×10^{-16}	4.6494×10^{-16}	–	0.9990	0.9969
4600	Scapulae spongiosa	110.81	4.3575×10^{-16}	0.93	–	4.3542×10^{-16}	4.3327×10^{-16}	–	1.0008	1.0057
4700	Cervical spine cortical	33.76	1.5907×10^{-16}	4.16	–	1.5046×10^{-16}	1.5424×10^{-16}	–	1.0572	1.0313
4800	Cervical spine spongiosa	56.49	1.5787×10^{-16}	3.31	–	1.5708×10^{-16}	1.5336×10^{-16}	–	1.0051	1.0294
4900	Thoracic spine cortical	90.86	8.9755×10^{-16}	1.05	–	9.0104×10^{-16}	9.0940×10^{-16}	–	0.9961	0.9870
5000	Thoracic spine spongiosa	264.35	1.0217×10^{-15}	0.31	–	1.0102×10^{-15}	1.0313×10^{-15}	–	1.0114	0.9906
5100	Lumbar spine cortical	35.53	1.3742×10^{-15}	0.79	–	1.3748×10^{-15}	1.3438×10^{-15}	–	0.9995	1.0226
5200	Lumbar spine spongiosa	255.53	1.4622×10^{-15}	0.44	–	1.4523×10^{-15}	1.4547×10^{-15}	–	1.0068	1.0051
5300	Sacrum cortical	69.68	3.9496×10^{-16}	1.42	–	4.0162×10^{-16}	4.0703×10^{-16}	–	0.9834	0.9703
5400	Sacrum spongiosa	118.65	4.1794×10^{-16}	0.95	–	4.2317×10^{-16}	4.2126×10^{-16}	–	0.9876	0.9921
5500	Sternum cortical	22.16	1.1002×10^{-15}	1.64	–	1.1028×10^{-15}	1.1405×10^{-15}	–	0.9977	0.9647
5600	Sternum spongiosa	42.39	9.8495×10^{-16}	2.00	–	9.8059×10^{-16}	9.5359×10^{-16}	–	1.0044	1.0329
5800	Cartilage trunk	152.19	2.1432×10^{-15}	0.48	–	2.1613×10^{-15}	2.1391×10^{-15}	–	0.9916	1.0019
6100	Brain	1496.74	4.5875×10^{-17}	1.01	–	4.6326×10^{-17}	4.6084×10^{-17}	–	0.9903	0.9955
6200	Breast left adipose tissue	4.76	1.0125×10^{-15}	3.44	–	1.0385×10^{-15}	1.0046×10^{-15}	–	0.9750	1.0079
6300	Breast left glandular tissue	3.17	9.6444×10^{-16}	4.38	–	9.5258×10^{-16}	1.0680×10^{-15}	–	1.0125	0.9030
6400	Breast right adipose tissue	4.76	2.4104×10^{-15}	2.84	–	2.3387×10^{-15}	2.3746×10^{-15}	–	1.0307	1.0151
6500	Breast right glandular tissue	3.17	2.4041×10^{-15}	2.40	–	2.6440×10^{-15}	2.5354×10^{-15}	–	0.9093	0.9482
6600	Eye lens sensitive left	0.0203	0.0000×10^{0}	0.00	–	5.2566×10^{-18}	–	–	–	–
6601	Eye lens insensitive left	0.117	6.8796×10^{-17}	88.00	–	4.1091×10^{-18}	2.0244×10^{-16}	–	16.7422	0.3398
6700	Cornea left	0.996	5.3017×10^{-17}	32.77	–	8.6022×10^{-17}	6.9568×10^{-17}	–	0.6163	0.7621
6701	Aqueous left	0.344	1.2274×10^{-16}	51.82	–	8.3073×10^{-18}	1.4349×10^{-16}	–	14.7749	0.8554
6702	Vitreous left	5.32	6.6843×10^{-17}	17.38	–	5.8423×10^{-17}	5.7740×10^{-17}	–	1.1441	1.1577
6800	Eye lens sensitive right	0.0203	2.6423×10^{-17}	100.00	–	–	1.2364×10^{-16}	–	–	0.2137
6801	Eye lens insensitive right	0.117	1.2076×10^{-16}	72.45	–	7.7343×10^{-17}	9.8624×10^{-17}	–	1.5613	1.2244
6900	Cornea right	0.996	7.5474×10^{-17}	32.45	–	7.7789×10^{-17}	5.5657×10^{-17}	–	0.9702	1.3561
6901	Aqueous right	0.344	1.2925×10^{-17}	41.77	–	1.1820×10^{-16}	8.4560×10^{-17}	–	0.1094	0.1529
6902	Vitreous right	5.32	8.5438×10^{-17}	6.47	–	8.2618×10^{-17}	7.6746×10^{-17}	–	1.0341	1.1133
7000	Gall bladder wall	8.14	1.0593×10^{-14}	0.84	–	1.0486×10^{-14}	1.0303×10^{-14}	–	1.0102	1.0281
7100	Gall bladder contents	45.00	1.0035×10^{-14}	0.49	–	9.9851×10^{-15}	1.0115×10^{-14}	–	1.0050	0.9921
7200	Stomach wall(0-60)	2.41	2.8372×10^{-15}	1.39	–	2.8219×10^{-15}	2.8977×10^{-15}	–	1.0054	0.9791
7201	Stomach wall(60-100)	1.61	2.8651×10^{-15}	1.96	–	2.8432×10^{-15}	2.9005×10^{-15}	–	1.0077	0.9878
7202	Stomach wall(100-300)	8.07	2.8796×10^{-15}	0.85	–	2.8691×10^{-15}	2.9298×10^{-15}	–	1.0036	0.9829
7203	Stomach wall(300-surface)	143.66	2.9122×10^{-15}	0.34	–	2.9425×10^{-15}	2.9526×10^{-15}	–	0.9897	0.9863
7300	Stomach contents	200.00	2.9655×10^{-15}	0.44	–	2.9932×10^{-15}	2.9885×10^{-15}	–	0.9907	0.9923
7400	Small intestine wall(0-130)	12.50	1.4005×10^{-15}	1.11	–	1.3686×10^{-15}	1.3451×10^{-15}	–	1.0234	1.0412
7401	Small intestine wall(130-150)	1.95	1.4594×10^{-15}	1.14	–	1.4102×10^{-15}	1.3139×10^{-15}	–	1.0348	1.1107
7402	Small intestine wall(150-200)	4.90	1.4063×10^{-15}	1.32	–	1.3754×10^{-15}	1.3314×10^{-15}	–	1.0224	1.0562

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
7403	Small intestine wall(200-surface)	673.31	1.3634×10^{-15}	0.14	–	1.3626×10^{-15}	1.3597×10^{-15}	–	1.0006	1.0027
7500	Small intestine contents(-500-0)	45.27	1.3497×10^{-15}	0.47	–	1.3612×10^{-15}	1.3531×10^{-15}	–	0.9916	0.9975
7501	Small intestine contents(centre–500)	234.73	1.3437×10^{-15}	0.43	–	1.3538×10^{-15}	1.3547×10^{-15}	–	0.9925	0.9918
7600	Ascending colon wall(0-280)	3.64	8.9988×10^{-16}	3.01	–	8.7257×10^{-16}	8.2361×10^{-16}	–	1.0313	1.0926
7601	Ascending colon wall(280-300)	0.264	8.4542×10^{-16}	8.11	–	9.1604×10^{-16}	7.9059×10^{-16}	–	0.9229	1.0694
7602	Ascending colon wall(300-surface)	76.36	8.8528×10^{-16}	1.19	–	9.0368×10^{-16}	8.7889×10^{-16}	–	0.9796	1.0073
7700	Ascending colon content	77.81	8.0391×10^{-16}	1.35	–	7.9739×10^{-16}	7.9279×10^{-16}	–	1.0082	1.0140
7800	Transverse colon wall right(0-280)	2.66	3.2404×10^{-15}	1.26	–	3.1408×10^{-15}	3.2012×10^{-15}	–	1.0317	1.0123
7801	Transverse colon wall right(280-300)	0.193	3.2155×10^{-15}	3.46	–	3.2359×10^{-15}	3.2310×10^{-15}	–	0.9937	0.9952
7802	Transverse colon wall right(300-surface)	79.88	3.2696×10^{-15}	0.54	–	3.3000×10^{-15}	3.2847×10^{-15}	–	0.9908	0.9954
7900	Transverse colon contents right	42.19	3.2147×10^{-15}	0.80	–	3.3153×10^{-15}	3.2580×10^{-15}	–	0.9696	0.9867
8000	Transverse colon wall left(0-280)	2.19	1.1345×10^{-15}	3.44	–	1.1647×10^{-15}	1.2405×10^{-15}	–	0.9740	0.9145
8001	Transverse colon wall left(280-300)	0.16	1.1349×10^{-15}	8.19	–	1.2080×10^{-15}	1.1460×10^{-15}	–	0.9394	0.9903
8002	Transverse colon wall left(300-surface)	74.08	1.1960×10^{-15}	1.22	–	1.2103×10^{-15}	1.1983×10^{-15}	–	0.9882	0.9981
8100	Transverse colon content left	27.01	1.2063×10^{-15}	1.35	–	1.1697×10^{-15}	1.1843×10^{-15}	–	1.0313	1.0186
8200	Descending colon wall(0-280)	2.69	5.0280×10^{-16}	4.25	–	5.2235×10^{-16}	5.1213×10^{-16}	–	0.9626	0.9818
8201	Descending colon wall(280-300)	0.197	4.4006×10^{-16}	7.71	–	5.3029×10^{-16}	5.0281×10^{-16}	–	0.8298	0.8752
8202	Descending colon wall(300-surface)	83.74	5.3155×10^{-16}	1.46	–	5.3954×10^{-16}	5.4771×10^{-16}	–	0.9852	0.9705
8300	Descending colon content	32.99	5.1822×10^{-16}	2.52	–	5.1296×10^{-16}	5.0394×10^{-16}	–	1.0103	1.0283
8400	Sigmoid colon wall(0-280)	3.26	2.3222×10^{-16}	8.99	–	2.6541×10^{-16}	2.4688×10^{-16}	–	0.8749	0.9406
8401	Sigmoid colon wall(280-300)	0.238	2.9457×10^{-16}	8.93	–	2.5753×10^{-16}	1.8951×10^{-16}	–	1.1438	1.5544
8402	Sigmoid colon wall(300-surface)	61.66	2.6070×10^{-16}	1.93	–	2.7526×10^{-16}	2.6826×10^{-16}	–	0.9471	0.9718
8500	Sigmoid colon contents	41.77	2.5968×10^{-16}	2.68	–	2.4302×10^{-16}	2.5087×10^{-16}	–	1.0686	1.0351
8600	Rectum wall(0-280)	1.49	1.2479×10^{-16}	9.17	–	1.1919×10^{-16}	1.4612×10^{-16}	–	1.0470	0.8540
8601	Rectum wall(280-300)	0.109	1.2336×10^{-16}	25.30	–	1.9909×10^{-16}	1.4244×10^{-16}	–	0.6196	0.8661
8602	Rectum wall(300-surface)	8.11	1.0837×10^{-16}	9.59	–	1.1005×10^{-16}	1.4074×10^{-16}	–	0.9847	0.7700
8603	Rectum contents	18.23	1.2510×10^{-16}	2.71	–	1.1993×10^{-16}	1.1361×10^{-16}	–	1.0431	1.1011
8700	Heart wall	269.65	1.9832×10^{-15}	0.52	–	1.9824×10^{-15}	1.9815×10^{-15}	–	1.0004	1.0009
8800	Blood in heart chamber	430.00	1.8654×10^{-15}	0.29	–	1.8539×10^{-15}	1.8647×10^{-15}	–	1.0062	1.0004
8900	Kidney left cortex	119.39	1.0886×10^{-15}	0.81	–	1.0912×10^{-15}	1.0903×10^{-15}	–	0.9977	0.9985
9000	Kidney left medulla	42.64	1.1672×10^{-15}	1.32	–	1.1813×10^{-15}	1.1918×10^{-15}	–	0.9880	0.9793
9100	Kidney left pelvis	8.53	1.2061×10^{-15}	2.24	–	1.1776×10^{-15}	1.1623×10^{-15}	–	1.0242	1.0377
9200	Kidney right cortex	119.39	2.2905×10^{-15}	0.57	–	2.2690×10^{-15}	2.3011×10^{-15}	–	1.0095	0.9954
9300	Kidney right medulla	42.64	2.4657×10^{-15}	0.56	–	2.4675×10^{-15}	2.4588×10^{-15}	–	0.9993	1.0028
9400	Kidney right pelvis	8.53	2.4240×10^{-15}	2.16	–	2.5223×10^{-15}	2.3970×10^{-15}	–	0.9610	1.0112
9500	Liver	1707.82	1.4263×10^{-14}	0.05	–	1.4248×10^{-14}	1.4299×10^{-14}	–	1.0010	0.9975

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
9700	Lung(AI) left	389.51	6.6062×10^{-16}	0.47	–	6.6697×10^{-16}	6.6675×10^{-16}	–	0.9905	0.9908
9900	Lung(AI) right	510.49	1.9905×10^{-15}	0.36	–	1.9983×10^{-15}	2.0054×10^{-15}	–	0.9961	0.9926
10000	Lymphatic nodes ET	12.73	1.3407×10^{-16}	6.82	–	1.1682×10^{-16}	1.2847×10^{-16}	–	1.1476	1.0436
10100	Lymphatic nodes thoracic	12.73	2.8502×10^{-15}	1.30	–	2.8319×10^{-15}	2.8835×10^{-15}	–	1.0065	0.9884
10200	Lymphatic nodes head	4.40	2.4172×10^{-16}	4.80	–	2.3897×10^{-16}	2.5025×10^{-16}	–	1.0115	0.9659
10300	Lymphatic nodes trunk	103.94	1.1232×10^{-15}	0.45	–	1.1259×10^{-15}	1.1272×10^{-15}	–	0.9976	0.9965
10400	Lymphatic nodes arms	8.80	6.0347×10^{-16}	3.29	–	6.3506×10^{-16}	6.3899×10^{-16}	–	0.9503	0.9444
10500	Lymphatic nodes legs	8.80	3.4894×10^{-18}	30.22	–	3.3819×10^{-18}	5.4762×10^{-18}	–	1.0318	0.6372
10600	Muscle head	563.16	1.1098×10^{-16}	1.33	–	1.0474×10^{-16}	1.0574×10^{-16}	–	1.0596*	1.0495*
10700	Muscle trunk	11667.40	6.3395×10^{-16}	0.08	–	6.5698×10^{-16}	6.5869×10^{-16}	–	0.9649*	0.9624*
10800	Muscle arms	2810.41	3.9415×10^{-16}	0.26	–	3.9034×10^{-16}	3.9219×10^{-16}	–	1.0098*	1.0050*
10900	Muscle legs	9616.52	2.7603×10^{-17}	0.55	–	3.1911×10^{-17}	3.1856×10^{-17}	–	0.8650*	0.8665*
11000	Oesophagus wall(0-190)	2.17	1.4481×10^{-15}	2.53	–	1.3753×10^{-15}	1.4176×10^{-15}	–	1.0529	1.0215
11001	Oesophagus wall(190-200)	0.116	1.5848×10^{-15}	3.02	–	1.6413×10^{-15}	1.6529×10^{-15}	–	0.9656	0.9588
11002	Oesophagus wall(200-surface)	36.64	1.5198×10^{-15}	1.07	–	1.4908×10^{-15}	1.5377×10^{-15}	–	1.0195	0.9884
11003	Oesophagus contents	21.84	1.4211×10^{-15}	1.47	–	1.4778×10^{-15}	1.4892×10^{-15}	–	0.9617	0.9543
11300	Pancreas	136.66	1.9367×10^{-15}	0.70	–	1.9435×10^{-15}	1.9533×10^{-15}	–	0.9965	0.9915
11400	Pituitary gland	0.529	7.9346×10^{-17}	51.04	–	2.7243×10^{-17}	3.3653×10^{-17}	–	2.9125	2.3578
11500	Prostate	4.55	1.3085×10^{-16}	6.77	–	9.8687×10^{-17}	9.3759×10^{-17}	–	1.3259	1.3956
11600	RST head	459.74	9.7192×10^{-17}	1.03	–	8.9064×10^{-17}	9.0678×10^{-17}	–	1.0913*	1.0718*
11700	RST trunk	8275.44	1.8787×10^{-15}	0.06	–	1.8851×10^{-15}	1.8932×10^{-15}	–	0.9966	0.9924
11800	RST arms	1102.01	4.0163×10^{-16}	0.40	–	3.9658×10^{-16}	3.9769×10^{-16}	–	1.0127*	1.0099*
11900	RST legs	2501.20	7.8650×10^{-18}	1.51	–	9.1789×10^{-18}	9.3420×10^{-18}	–	0.8568*	0.8419*
12000	Salivary glands left	35.95	1.0007×10^{-16}	4.81	–	1.0563×10^{-16}	1.0384×10^{-16}	–	0.9474	0.9637
12100	Salivary glandss right	35.95	1.3529×10^{-16}	4.14	–	1.3599×10^{-16}	1.3598×10^{-16}	–	0.9949	0.9950
12200	Skin head insensitive	153.24	7.6498×10^{-17}	1.62	–	7.1760×10^{-17}	7.2879×10^{-17}	–	1.0660*	1.0497*
12201	Skin head sensitive(50-100)	6.53	6.1346×10^{-17}	5.25	–	5.9891×10^{-17}	6.7054×10^{-17}	–	1.0243*	0.9149*
12300	Skin trunk insensitive	663.99	8.9665×10^{-16}	0.31	–	9.2202×10^{-16}	9.2905×10^{-16}	–	0.9725*	0.9651*
12301	Skin trunk sensitive(50-100)	28.45	7.8956×10^{-16}	1.04	–	7.9188×10^{-16}	7.8519×10^{-16}	–	0.9971*	1.0056*
12400	Skin arms insensitive	412.60	3.3334×10^{-16}	0.59	–	3.3652×10^{-16}	3.3690×10^{-16}	–	0.9906*	0.9894*
12401	Skin arms sensitive(50-100)	17.73	3.0341×10^{-16}	1.69	–	2.9762×10^{-16}	3.0706×10^{-16}	–	1.0195*	0.9881*
12500	Skin legs insensitive	786.97	1.5523×10^{-17}	0.74	–	1.7181×10^{-17}	1.7631×10^{-17}	–	0.9035*	0.8804*
12501	Skin legs sensitive(50-100)	33.57	1.3575×10^{-17}	6.77	–	1.6475×10^{-17}	1.4812×10^{-17}	–	0.8240*	0.9165*
12600	Spinal cord	40.22	7.8531×10^{-16}	1.58	–	7.9032×10^{-16}	7.9174×10^{-16}	–	0.9937	0.9919
12700	Spleen	198.77	9.3677×10^{-16}	0.50	–	9.3666×10^{-16}	9.2909×10^{-16}	–	1.0001	1.0083
12800	Erupted upper front buccal permanent ena	0.65	1.9139×10^{-16}	28.52	–	9.3970×10^{-17}	1.5638×10^{-16}	–	2.0367	1.2239

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
12801	Erupted upper front lingual permanent en	0.823	1.7335×10^{-16}	23.39	–	1.6021×10^{-16}	1.4497×10^{-16}	–	1.0820	1.1958
12804	Erupted upper front-left buccal permanen	0.317	1.3049×10^{-16}	33.06	–	1.0050×10^{-16}	6.6954×10^{-17}	–	1.2984	1.9490
12805	Erupted upper front-left lingual permane	0.224	1.7070×10^{-16}	38.31	–	1.3070×10^{-16}	1.3470×10^{-16}	–	1.3060	1.2673
12806	Erupted upper front-right buccal permane	0.317	8.7846×10^{-17}	29.41	–	9.9344×10^{-17}	1.9246×10^{-16}	–	0.8843	0.4564
12807	Erupted upper front-right lingual perman	0.224	1.0177×10^{-16}	57.41	–	1.1641×10^{-16}	2.0518×10^{-16}	–	0.8743	0.4960
12808	Erupted upper left buccal permanent enam	0.513	1.0931×10^{-16}	32.07	–	5.6316×10^{-17}	1.0035×10^{-16}	–	1.9410	1.0893
12809	Erupted upper left lingual permanent ena	0.494	1.0230×10^{-16}	18.83	–	1.7119×10^{-16}	1.1292×10^{-16}	–	0.5976	0.9059
12812	Erupted upper right buccal permanent ena	0.477	1.5961×10^{-16}	28.72	–	1.2170×10^{-16}	1.4882×10^{-16}	–	1.3115	1.0725
12813	Erupted upper right lingual permanent en	0.531	1.4198×10^{-16}	27.65	–	6.8372×10^{-17}	1.6160×10^{-16}	–	2.0766	0.8786
12816	Erupted lower front buccal permanent ena	0.605	9.9221×10^{-17}	26.16	–	7.1127×10^{-17}	1.1143×10^{-16}	–	1.3950	0.8904
12817	Erupted lower front lingual permanent en	0.54	7.7270×10^{-17}	27.26	–	7.8578×10^{-17}	2.4016×10^{-16}	–	0.9834	0.3217
12820	Erupted lower front-left buccal permanen	0.239	6.0850×10^{-17}	63.02	–	1.2895×10^{-16}	8.7584×10^{-17}	–	0.4719	0.6948
12821	Erupted lower front-left lingual permane	0.228	9.8994×10^{-17}	34.41	–	7.3706×10^{-17}	1.7688×10^{-16}	–	1.3431	0.5597
12822	Erupted lower front-right buccal permane	0.244	6.8376×10^{-17}	42.43	–	6.8165×10^{-17}	5.9038×10^{-17}	–	1.0031	1.1582
12823	Erupted lower front-right lingual perman	0.224	1.3042×10^{-16}	49.23	–	2.9030×10^{-17}	2.0766×10^{-16}	–	4.4925	0.6280
12824	Erupted lower left buccal permanent enam	0.536	1.4146×10^{-16}	20.63	–	7.0679×10^{-17}	6.6301×10^{-17}	–	2.0015	2.1337
12825	Erupted lower left lingual permanent ena	0.494	5.6114×10^{-17}	39.99	–	1.2451×10^{-16}	9.5867×10^{-17}	–	0.4507	0.5853
12828	Erupted lower right buccal permanent ena	0.594	1.3322×10^{-16}	21.00	–	8.6593×10^{-17}	1.8482×10^{-16}	–	1.5385	0.7208
12829	Erupted lower right lingual permanent en	0.437	7.8514×10^{-17}	47.57	–	1.2726×10^{-16}	1.2850×10^{-16}	–	0.6169	0.6110
12832	Erupted upper front permanent dentin	4.94	1.2009×10^{-16}	9.33	–	1.2398×10^{-16}	1.3148×10^{-16}	–	0.9686	0.9134

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
12834	Erupted upper front-left permanent denti	1.81	1.1517×10^{-16}	20.12	–	1.0451×10^{-16}	9.2031×10^{-17}	–	1.1020	1.2514
12835	Erupted upper front-right permanent dent	1.82	1.1475×10^{-16}	10.83	–	1.3014×10^{-16}	1.1351×10^{-16}	–	0.8817	1.0109
12836	Erupted upper left permanent dentin	3.39	8.7419×10^{-17}	11.30	–	9.5663×10^{-17}	1.3072×10^{-16}	–	0.9138	0.6688
12838	Erupted upper right permanent dentin	3.39	1.1887×10^{-16}	13.18	–	1.0264×10^{-16}	1.5156×10^{-16}	–	1.1582	0.7843
12840	Erupted lower front permanent dentin	3.76	1.2584×10^{-16}	11.77	–	1.3703×10^{-16}	1.7166×10^{-16}	–	0.9183	0.7330
12842	Erupted lower front-left permanent denti	1.56	1.2043×10^{-16}	12.52	–	1.1162×10^{-16}	1.2433×10^{-16}	–	1.0789	0.9686
12843	Erupted lower front-right permanent dent	1.56	1.0962×10^{-16}	14.19	–	1.6674×10^{-16}	1.6691×10^{-16}	–	0.6574	0.6568
12844	Erupted lower left permanent dentin	3.47	1.3807×10^{-16}	10.08	–	1.2886×10^{-16}	1.2545×10^{-16}	–	1.0714	1.1006
12846	Erupted lower right permanent dentin	3.46	1.6008×10^{-16}	6.58	–	1.4715×10^{-16}	1.4764×10^{-16}	–	1.0879	1.0843
12848	Unerupted permanent enamel	1.22	1.6377×10^{-16}	14.83	–	1.1639×10^{-16}	1.2336×10^{-16}	–	1.4071	1.3276
12850	Unerupted permanent dentin	4.11	1.4859×10^{-16}	10.30	–	1.0605×10^{-16}	1.2603×10^{-16}	–	1.4012	1.1790
12852	Permanent cementum	1.45	1.2962×10^{-16}	10.66	–	1.1936×10^{-16}	1.2544×10^{-16}	–	1.0860	1.0333
12854	Permanent pulp	0.993	1.4003×10^{-16}	15.71	–	1.1165×10^{-16}	1.4019×10^{-16}	–	1.2542	0.9989
12856	Teeth retention region	0.0505	1.4912×10^{-16}	28.86	–	1.0055×10^{-16}	1.9539×10^{-16}	–	1.4830	0.7632
12900	Testis left	8.52	9.3869×10^{-17}	7.82	–	8.3923×10^{-17}	1.0550×10^{-16}	–	1.1185	0.8898
13000	Testis right	8.52	9.7705×10^{-17}	6.99	–	1.0154×10^{-16}	1.0242×10^{-16}	–	0.9622	0.9540
13100	Thymus	37.00	4.7961×10^{-16}	1.35	–	4.7408×10^{-16}	4.5554×10^{-16}	–	1.0117	1.0528
13200	Thyroid	14.04	2.9123×10^{-16}	3.89	–	3.0975×10^{-16}	3.0471×10^{-16}	–	0.9402	0.9558
13300	Tongue upper(food)	19.03	1.2813×10^{-16}	6.26	–	1.2964×10^{-16}	1.1938×10^{-16}	–	0.9883	1.0733
13301	Tongue lower	40.04	1.5237×10^{-16}	3.02	–	1.4979×10^{-16}	1.5215×10^{-16}	–	1.0172	1.0014
13400	Tonsils	3.17	1.2768×10^{-16}	16.05	–	1.3124×10^{-16}	1.1962×10^{-16}	–	0.9728	1.0673
13500	Ureter left	6.34	5.9112×10^{-16}	3.84	–	7.0222×10^{-16}	6.0148×10^{-16}	–	0.8418	0.9828
13600	Ureter right	6.34	8.1739×10^{-16}	3.83	–	8.9565×10^{-16}	8.2369×10^{-16}	–	0.9126	0.9924
13700	Urinary bladder wall insensitive	39.00	2.0313×10^{-16}	2.79	–	1.8979×10^{-16}	1.9441×10^{-16}	–	1.0703	1.0449
13701	Urinary bladder wall sensitive(116-238)	1.87	2.0518×10^{-16}	8.80	–	1.6778×10^{-16}	1.7748×10^{-16}	–	1.2229	1.1561
13800	Urinary bladder content	160.00	1.8347×10^{-16}	1.07	–	1.9526×10^{-16}	1.9327×10^{-16}	–	0.9396	0.9493
14000	Air inside body	0.384	1.8768×10^{-15}	3.92	–	2.0705×10^{-15}	2.0603×10^{-15}	–	0.9064	0.9109

* For tissues bookkept as head, trunk, arm and leg pieces (muscle, residual soft tissue, skin, large vessels), the volumes tabulated in the distributed cell tables partition the tissue at different boundaries from the transported mesh, and the reference output is normalised by the tabulated volume. The marked ratio therefore compares two different partitions of the same tissue, and is left out of the summary statistics of Table S2.

Table S24. 15M_External. Absorbed dose per primary history, Gy. σ is the FLUKA statistical uncertainty. Ratios are FLUKA / reference.

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
100	Adrenal left	6.21	1.7978×10^{-17}	22.30	–	1.9544×10^{-17}	2.4247×10^{-17}	–	0.9199	0.7415
200	Adrenal right	6.21	1.7551×10^{-17}	21.67	–	2.5366×10^{-17}	1.8728×10^{-17}	–	0.6919	0.9372
300	ET1(0-8)	0.0106	9.5203×10^{-17}	63.89	–	2.2357×10^{-17}	6.4550×10^{-17}	–	4.2583	1.4749
301	ET1(8-40)	0.0425	5.3102×10^{-17}	36.21	–	1.7985×10^{-17}	4.1771×10^{-17}	–	2.9526	1.2713
302	ET1(40-50)	0.0134	3.2122×10^{-17}	35.92	–	–	6.5727×10^{-17}	–	–	0.4887
303	ET1(50-Surface)	4.15	3.4576×10^{-17}	20.20	–	3.3858×10^{-17}	2.4971×10^{-17}	–	1.0212	1.3847
400	ET2(-15-0)	0.2	2.6518×10^{-17}	44.39	–	3.7144×10^{-17}	2.1904×10^{-17}	–	0.7139	1.2106
401	ET2(0-40)	0.572	3.6702×10^{-17}	40.81	–	4.3232×10^{-17}	3.1421×10^{-17}	–	0.8489	1.1681
402	ET2(40-50)	0.144	2.6264×10^{-17}	39.21	–	2.2884×10^{-17}	1.6786×10^{-17}	–	1.1477	1.5646
403	ET2(50-55)	0.072	2.6187×10^{-17}	39.05	–	3.7892×10^{-17}	1.8084×10^{-17}	–	0.6911	1.4481
404	ET2(55-65)	0.144	3.2659×10^{-17}	56.77	–	1.9427×10^{-17}	2.4334×10^{-17}	–	1.6811	1.3421
405	ET2(65-Surface)	30.93	2.8151×10^{-17}	9.09	–	3.1347×10^{-17}	2.3612×10^{-17}	–	0.8980	1.1922
500	Oral mucosa tongue	0.0767	3.7978×10^{-17}	56.84	–	2.3564×10^{-17}	6.3558×10^{-17}	–	1.6117	0.5975
501	Oral mucosa moth floor	0.0317	3.7722×10^{-17}	75.78	–	1.3976×10^{-17}	3.7092×10^{-17}	–	2.6990	1.0170
600	Oral mucosa lips and cheeks	0.0212	9.0739×10^{-17}	89.39	–	7.0970×10^{-17}	2.0039×10^{-17}	–	1.2786	4.5281
700	Trachea	7.93	2.3610×10^{-17}	25.43	–	2.5969×10^{-17}	2.4263×10^{-17}	–	0.9092	0.9731
800	BB(-11–6)	0.00916	4.2783×10^{-16}	97.82	–	1.0992×10^{-16}	6.9566×10^{-18}	–	3.8922	61.4998
801	BB(-6-0)	0.0117	2.6578×10^{-17}	68.56	–	5.5471×10^{-17}	7.0393×10^{-18}	–	0.4791	3.7757
802	BB(0-10)	0.0196	3.0740×10^{-17}	73.18	–	8.9268×10^{-17}	6.9506×10^{-18}	–	0.3444	4.4226
803	BB(10-35)	0.0492	2.5882×10^{-17}	66.96	–	2.0544×10^{-17}	5.7637×10^{-18}	–	1.2598	4.4905
804	BB(35-40)	0.00987	3.1102×10^{-17}	72.52	–	1.7883×10^{-17}	5.7124×10^{-18}	–	1.7392	5.4446
805	BB(40-50)	0.0198	1.9528×10^{-17}	56.47	–	2.2360×10^{-17}	5.6668×10^{-18}	–	0.8733	3.4460
806	BB(50-60)	0.0198	2.5251×10^{-17}	62.26	–	1.7792×10^{-17}	5.4861×10^{-18}	–	1.4192	4.6027
807	BB(60-70)	0.0199	2.1535×10^{-17}	60.20	–	2.4436×10^{-17}	5.3975×10^{-18}	–	0.8813	3.9898
808	BB(70-surface)	7.24	2.8412×10^{-17}	18.06	–	2.7332×10^{-17}	2.3725×10^{-17}	–	1.0395	1.1976
900	Blood in large arteries head	1.59	4.3175×10^{-17}	26.82	–	1.0817×10^{-17}	4.2071×10^{-17}	–	3.9915*	1.0262*
910	Blood in large veins head	5.22	2.8697×10^{-17}	22.15	–	3.9109×10^{-17}	2.0837×10^{-17}	–	0.7338*	1.3772*
1000	Blood in large arteries trunk	103.81	2.8586×10^{-17}	4.78	–	2.5943×10^{-17}	2.6613×10^{-17}	–	1.1019*	1.0741*
1010	Blood in large veins trunk	249.10	3.0005×10^{-17}	2.05	–	2.8772×10^{-17}	2.7147×10^{-17}	–	1.0428*	1.1053*
1100	Blood in large arteries arms	31.47	3.6427×10^{-17}	10.30	–	3.7330×10^{-17}	2.9093×10^{-17}	–	0.9758*	1.2521*
1110	Blood in large veins arms	146.40	3.5042×10^{-17}	2.93	–	3.3658×10^{-17}	3.3109×10^{-17}	–	1.0411*	1.0584*
1200	Blood in large arteries legs	150.90	3.0574×10^{-17}	1.27	–	2.8557×10^{-17}	3.0527×10^{-17}	–	1.0707*	1.0016*
1210	Blood in large veins legs	462.58	3.0074×10^{-17}	1.47	–	2.9810×10^{-17}	2.9715×10^{-17}	–	1.0088*	1.0121*
1300	Humeri upper cortical	153.12	2.4316×10^{-17}	3.96	–	2.4203×10^{-17}	2.5109×10^{-17}	–	1.0047	0.9684
1400	Humeri upper spogiosa	172.91	2.1652×10^{-17}	3.80	–	2.3001×10^{-17}	2.1772×10^{-17}	–	0.9414	0.9945

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
1500	Humeri upper medullary cavity	19.32	3.0660×10^{-17}	14.01	–	2.9085×10^{-17}	3.5772×10^{-17}	–	1.0541	0.8571
1600	Humeri lower cortical	167.64	2.7858×10^{-17}	5.29	–	2.8048×10^{-17}	3.1552×10^{-17}	–	0.9932	0.8829
1700	Humeri lower spongiosa	70.77	3.5177×10^{-17}	6.63	–	3.0884×10^{-17}	3.2584×10^{-17}	–	1.1390	1.0796
1800	Humeri lower medullary cavity	16.56	2.6803×10^{-17}	12.76	–	3.0891×10^{-17}	3.4370×10^{-17}	–	0.8677	0.7798
1900	Radii cortical	97.60	3.3510×10^{-17}	6.43	–	3.5463×10^{-17}	3.8886×10^{-17}	–	0.9449	0.8618
1910	Ulnae cortical	124.44	2.8819×10^{-17}	5.22	–	3.0133×10^{-17}	2.9293×10^{-17}	–	0.9564	0.9838
2000	Radii spongiosa	37.90	3.4027×10^{-17}	8.63	–	4.1305×10^{-17}	3.5711×10^{-17}	–	0.8238	0.9529
2010	Ulnae spongiosa	54.22	3.0787×10^{-17}	6.55	–	2.7761×10^{-17}	3.0739×10^{-17}	–	1.1090	1.0016
2100	Radii medullary cavity	6.59	3.7428×10^{-17}	25.25	–	2.3353×10^{-17}	5.8253×10^{-17}	–	1.6027	0.6425
2110	Ulnae medullary cavity	8.39	3.9971×10^{-17}	21.37	–	3.4518×10^{-17}	3.1111×10^{-17}	–	1.1580	1.2848
2200	Wrists and hand bones cortical	130.63	3.4514×10^{-17}	4.32	–	3.2317×10^{-17}	3.1384×10^{-17}	–	1.0680	1.0997
2300	Wrists and hand bones spongiosa	59.94	3.2860×10^{-17}	4.47	–	2.7171×10^{-17}	3.4270×10^{-17}	–	1.2094	0.9589
2400	Clavicles cortical	43.29	2.4572×10^{-17}	10.27	–	2.0729×10^{-17}	2.3711×10^{-17}	–	1.1854	1.0363
2500	Clavicles spongiosa	21.10	2.5606×10^{-17}	11.79	–	2.7326×10^{-17}	1.9043×10^{-17}	–	0.9371	1.3447
2600	Cranium cortical	544.13	1.6517×10^{-17}	3.26	–	1.7521×10^{-17}	1.7993×10^{-17}	–	0.9427	0.9180
2700	Cranium spongiosa	493.24	1.7956×10^{-17}	2.47	–	1.9037×10^{-17}	1.7635×10^{-17}	–	0.9432	1.0182
2800	Femora upper cortical	315.08	2.7087×10^{-17}	2.28	–	2.7860×10^{-17}	2.7606×10^{-17}	–	0.9723	0.9812
2900	Femora upper spongiosa	251.20	3.0944×10^{-17}	2.68	–	2.9611×10^{-17}	2.8274×10^{-17}	–	1.0450	1.0944
3000	Femora upper medullary cavity	49.29	2.5873×10^{-17}	4.93	–	2.8106×10^{-17}	2.8832×10^{-17}	–	0.9205	0.8974
3100	Femora lower cortical	243.61	2.6711×10^{-17}	3.68	–	2.6725×10^{-17}	2.6279×10^{-17}	–	0.9995	1.0164
3200	Femora lower spongiosa	308.42	2.7966×10^{-17}	3.57	–	2.6058×10^{-17}	2.7285×10^{-17}	–	1.0732	1.0250
3300	Femora lower medullary cavity	36.21	3.4005×10^{-17}	3.72	–	3.1105×10^{-17}	3.2302×10^{-17}	–	1.0932	1.0527
3400	Tibiae cortical	381.34	2.4686×10^{-17}	3.35	–	2.2231×10^{-17}	2.3903×10^{-17}	–	1.1104	1.0328
3410	Fibulae cortical	72.32	2.1122×10^{-17}	7.90	–	2.1008×10^{-17}	2.0000×10^{-17}	–	1.0054	1.0561
3420	Patellae cortical	21.42	3.4831×10^{-17}	11.09	–	2.6489×10^{-17}	3.2885×10^{-17}	–	1.3149	1.0592
3500	Tibiae spongiosa	264.54	2.5628×10^{-17}	2.66	–	2.6145×10^{-17}	2.7229×10^{-17}	–	0.9802	0.9412
3510	Fibulae spongiosa	31.69	2.1802×10^{-17}	14.89	–	2.3144×10^{-17}	1.8296×10^{-17}	–	0.9420	1.1916
3520	Patellae spongiosa	22.26	3.4372×10^{-17}	9.95	–	3.6947×10^{-17}	3.7398×10^{-17}	–	0.9303	0.9191
3600	Tibiae medullary cavity	59.42	2.8780×10^{-17}	7.97	–	2.5046×10^{-17}	2.3851×10^{-17}	–	1.1491	1.2067
3610	Fibulae medullary cavity	8.19	2.9954×10^{-17}	16.98	–	2.1845×10^{-17}	2.7432×10^{-17}	–	1.3713	1.0920
3700	Ankles and foot cortical	217.34	2.0484×10^{-17}	4.13	–	2.0863×10^{-17}	2.0192×10^{-17}	–	0.9818	1.0144
3800	Ankles and foot spongiosa	387.18	2.0884×10^{-17}	5.49	–	2.0535×10^{-17}	2.0240×10^{-17}	–	1.0170	1.0318
3900	Mandible cortical	59.29	2.6773×10^{-17}	6.77	–	2.6118×10^{-17}	2.3520×10^{-17}	–	1.0251	1.1383
4000	Mandible spongiosa	37.75	2.6467×10^{-17}	10.70	–	2.7250×10^{-17}	2.6977×10^{-17}	–	0.9713	0.9811
4100	Pelvis cortical	188.61	2.9050×10^{-17}	3.25	–	2.7293×10^{-17}	2.8886×10^{-17}	–	1.0644	1.0057
4200	Pelvis spongiosa	512.50	2.9660×10^{-17}	1.58	–	2.9749×10^{-17}	2.9010×10^{-17}	–	0.9970	1.0224
4300	Ribs cortical	169.28	2.3925×10^{-17}	3.22	–	2.1908×10^{-17}	2.3991×10^{-17}	–	1.0920	0.9972
4400	Ribs spongiosa	198.25	2.3757×10^{-17}	4.46	–	2.3364×10^{-17}	2.4322×10^{-17}	–	1.0168	0.9768

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
4500	Scapulae cortical	124.48	1.7077×10^{-17}	3.93	–	1.7361×10^{-17}	1.6441×10^{-17}	–	0.9836	1.0387
4600	Scapulae spongiosa	110.81	1.6219×10^{-17}	7.93	–	1.7924×10^{-17}	1.8339×10^{-17}	–	0.9049	0.8844
4700	Cervical spine cortical	33.76	2.1103×10^{-17}	10.29	–	2.2780×10^{-17}	2.2312×10^{-17}	–	0.9264	0.9458
4800	Cervical spine spongiosa	56.49	2.0957×10^{-17}	10.29	–	2.3152×10^{-17}	2.1443×10^{-17}	–	0.9052	0.9773
4900	Thoracic spine cortical	90.86	1.6247×10^{-17}	6.51	–	1.6601×10^{-17}	1.4876×10^{-17}	–	0.9787	1.0922
5000	Thoracic spine spongiosa	264.35	1.5441×10^{-17}	4.82	–	1.6128×10^{-17}	1.6943×10^{-17}	–	0.9574	0.9114
5100	Lumbar spine cortical	35.53	1.8647×10^{-17}	5.57	–	1.8151×10^{-17}	2.1591×10^{-17}	–	1.0273	0.8636
5200	Lumbar spine spongiosa	255.53	2.0987×10^{-17}	3.78	–	2.1649×10^{-17}	2.0139×10^{-17}	–	0.9694	1.0421
5300	Sacrum cortical	69.68	2.3364×10^{-17}	5.01	–	2.2887×10^{-17}	2.2597×10^{-17}	–	1.0209	1.0340
5400	Sacrum spongiosa	118.65	2.1418×10^{-17}	5.60	–	2.1922×10^{-17}	2.2835×10^{-17}	–	0.9770	0.9379
5500	Sternum cortical	22.16	3.3054×10^{-17}	8.56	–	3.6292×10^{-17}	3.0433×10^{-17}	–	0.9108	1.0861
5600	Sternum spongiosa	42.39	3.7624×10^{-17}	9.43	–	3.6487×10^{-17}	3.7019×10^{-17}	–	1.0312	1.0163
5800	Cartilage trunk	152.19	3.1248×10^{-17}	4.39	–	2.8154×10^{-17}	3.1341×10^{-17}	–	1.1099	0.9970
6100	Brain	1496.74	1.6128×10^{-17}	2.14	–	1.6478×10^{-17}	1.6185×10^{-17}	–	0.9788	0.9965
6200	Breast left adipose tissue	4.76	4.8358×10^{-17}	15.09	–	4.6935×10^{-17}	5.3840×10^{-17}	–	1.0303	0.8982
6300	Breast left glandular tissue	3.17	4.5166×10^{-17}	13.17	–	4.5793×10^{-17}	4.1261×10^{-17}	–	0.9863	1.0946
6400	Breast right adipose tissue	4.76	3.9740×10^{-17}	19.01	–	3.6941×10^{-17}	4.8273×10^{-17}	–	1.0758	0.8232
6500	Breast right glandular tissue	3.17	2.5451×10^{-17}	43.07	–	3.4328×10^{-17}	4.1692×10^{-17}	–	0.7414	0.6105
6600	Eye lens sensitive left	0.0203	0.0000×10^{0}	0.00	–	–	–	–	–	–
6601	Eye lens insensitive left	0.117	4.7874×10^{-17}	100.00	–	3.5039×10^{-18}	–	–	13.6630	–
6700	Cornea left	0.996	1.0099×10^{-17}	50.59	–	3.5040×10^{-17}	3.4382×10^{-17}	–	0.2882	0.2937
6701	Aqueous left	0.344	3.5028×10^{-19}	100.00	–	6.6494×10^{-19}	1.9505×10^{-17}	–	0.5268	0.0180
6702	Vitreous left	5.32	2.3764×10^{-17}	22.03	–	3.2173×10^{-17}	3.7410×10^{-17}	–	0.7386	0.6352
6800	Eye lens sensitive right	0.0203	0.0000×10^{0}	0.00	–	–	–	–	–	–
6801	Eye lens insensitive right	0.117	0.0000×10^{0}	0.00	–	7.3662×10^{-17}	1.5089×10^{-17}	–	–	–
6900	Cornea right	0.996	3.2535×10^{-17}	22.39	–	3.8073×10^{-17}	3.2312×10^{-17}	–	0.8545	1.0069
6901	Aqueous right	0.344	2.3157×10^{-17}	98.36	–	1.9949×10^{-17}	8.0677×10^{-18}	–	1.1608	2.8703
6902	Vitreous right	5.32	4.2777×10^{-17}	18.62	–	5.0060×10^{-17}	2.9909×10^{-17}	–	0.8545	1.4302
7000	Gall bladder wall	8.14	3.0445×10^{-17}	14.88	–	3.1911×10^{-17}	2.1380×10^{-17}	–	0.9541	1.4240
7100	Gall bladder contents	45.00	3.2465×10^{-17}	5.44	–	2.7146×10^{-17}	3.2310×10^{-17}	–	1.1960	1.0048
7200	Stomach wall(0-60)	2.41	3.3024×10^{-17}	14.72	–	2.9449×10^{-17}	3.2189×10^{-17}	–	1.1214	1.0259
7201	Stomach wall(60-100)	1.61	3.5317×10^{-17}	12.82	–	3.9916×10^{-17}	3.8036×10^{-17}	–	0.8848	0.9285
7202	Stomach wall(100-300)	8.07	3.4270×10^{-17}	10.52	–	3.1160×10^{-17}	3.2299×10^{-17}	–	1.0998	1.0610
7203	Stomach wall(300-surface)	143.66	3.3265×10^{-17}	3.96	–	3.4200×10^{-17}	3.1230×10^{-17}	–	0.9727	1.0652
7300	Stomach contents	200.00	3.3162×10^{-17}	3.27	–	3.4772×10^{-17}	3.3397×10^{-17}	–	0.9537	0.9930
7400	Small intestine wall(0-130)	12.50	3.3527×10^{-17}	6.27	–	3.3590×10^{-17}	3.3561×10^{-17}	–	0.9981	0.9990
7401	Small intestine wall(130-150)	1.95	3.9213×10^{-17}	12.20	–	3.1026×10^{-17}	2.9321×10^{-17}	–	1.2639	1.3374
7402	Small intestine wall(150-200)	4.90	3.8644×10^{-17}	6.67	–	3.6974×10^{-17}	3.2239×10^{-17}	–	1.0452	1.1987

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
7403	Small intestine wall(200-surface)	673.31	3.3798×10^{-17}	1.41	–	3.4253×10^{-17}	3.2643×10^{-17}	–	0.9867	1.0354
7500	Small intestine contents(-500-0)	45.27	3.2451×10^{-17}	7.18	–	3.1036×10^{-17}	3.1711×10^{-17}	–	1.0456	1.0233
7501	Small intestine contents(centre–500)	234.73	3.2697×10^{-17}	1.29	–	3.3150×10^{-17}	3.2973×10^{-17}	–	0.9863	0.9916
7600	Ascending colon wall(0-280)	3.64	4.1020×10^{-17}	15.68	–	3.6469×10^{-17}	3.5106×10^{-17}	–	1.1248	1.1685
7601	Ascending colon wall(280-300)	0.264	3.0562×10^{-17}	28.53	–	3.2590×10^{-17}	3.2298×10^{-17}	–	0.9378	0.9462
7602	Ascending colon wall(300-surface)	76.36	3.6090×10^{-17}	4.35	–	3.5204×10^{-17}	3.4520×10^{-17}	–	1.0252	1.0455
7700	Ascending colon content	77.81	3.6629×10^{-17}	6.71	–	3.3518×10^{-17}	3.4883×10^{-17}	–	1.0928	1.0501
7800	Transverse colon wall right(0-280)	2.66	3.3189×10^{-17}	17.49	–	4.0959×10^{-17}	3.3259×10^{-17}	–	0.8103	0.9979
7801	Transverse colon wall right(280-300)	0.193	3.4287×10^{-17}	41.97	–	5.9611×10^{-17}	4.7917×10^{-17}	–	0.5752	0.7155
7802	Transverse colon wall right(300-surface)	79.88	4.0719×10^{-17}	6.96	–	3.9954×10^{-17}	4.1980×10^{-17}	–	1.0191	0.9700
7900	Transverse colon contents right	42.19	3.9034×10^{-17}	8.06	–	4.2808×10^{-17}	3.7200×10^{-17}	–	0.9118	1.0493
8000	Transverse colon wall left(0-280)	2.19	2.2029×10^{-17}	24.64	–	2.3361×10^{-17}	4.6725×10^{-17}	–	0.9430	0.4715
8001	Transverse colon wall left(280-300)	0.16	2.2084×10^{-17}	43.70	–	9.0870×10^{-19}	7.0556×10^{-17}	–	24.3029	0.3130
8002	Transverse colon wall left(300-surface)	74.08	4.2162×10^{-17}	4.73	–	3.8096×10^{-17}	4.4930×10^{-17}	–	1.1067	0.9384
8100	Transverse colon content left	27.01	4.2243×10^{-17}	6.98	–	4.1673×10^{-17}	3.6922×10^{-17}	–	1.0137	1.1441
8200	Descending colon wall(0-280)	2.69	2.6621×10^{-17}	22.37	–	3.0182×10^{-17}	4.0200×10^{-17}	–	0.8820	0.6622
8201	Descending colon wall(280-300)	0.197	1.7216×10^{-17}	31.10	–	5.2984×10^{-17}	5.6104×10^{-17}	–	0.3249	0.3068
8202	Descending colon wall(300-surface)	83.74	3.4730×10^{-17}	7.07	–	3.4884×10^{-17}	3.7309×10^{-17}	–	0.9956	0.9309
8300	Descending colon content	32.99	3.3328×10^{-17}	10.50	–	3.0506×10^{-17}	4.1443×10^{-17}	–	1.0925	0.8042
8400	Sigmoid colon wall(0-280)	3.26	3.0891×10^{-17}	19.32	–	1.5416×10^{-17}	2.5275×10^{-17}	–	2.0039	1.2222
8401	Sigmoid colon wall(280-300)	0.238	2.4364×10^{-17}	37.86	–	9.9150×10^{-18}	1.7043×10^{-17}	–	2.4573	1.4296
8402	Sigmoid colon wall(300-surface)	61.66	2.9848×10^{-17}	6.53	–	2.4918×10^{-17}	2.6209×10^{-17}	–	1.1979	1.1388
8500	Sigmoid colon contents	41.77	2.5375×10^{-17}	11.42	–	2.4409×10^{-17}	2.6102×10^{-17}	–	1.0396	0.9722
8600	Rectum wall(0-280)	1.49	2.9371×10^{-17}	26.05	–	2.7421×10^{-17}	6.8867×10^{-18}	–	1.0711	4.2649
8601	Rectum wall(280-300)	0.109	9.0070×10^{-18}	34.09	–	3.6484×10^{-17}	5.7820×10^{-18}	–	0.2469	1.5578
8602	Rectum wall(300-surface)	8.11	2.8844×10^{-17}	14.52	–	3.0111×10^{-17}	1.8205×10^{-17}	–	0.9579	1.5844
8603	Rectum contents	18.23	3.2054×10^{-17}	11.88	–	2.9285×10^{-17}	2.1935×10^{-17}	–	1.0946	1.4613
8700	Heart wall	269.65	2.9396×10^{-17}	3.05	–	2.9833×10^{-17}	2.9035×10^{-17}	–	0.9854	1.0124
8800	Blood in heart chamber	430.00	2.8814×10^{-17}	2.68	–	2.9763×10^{-17}	2.9244×10^{-17}	–	0.9681	0.9853
8900	Kidney left cortex	119.39	2.2351×10^{-17}	4.86	–	2.3003×10^{-17}	2.4264×10^{-17}	–	0.9717	0.9212
9000	Kidney left medulla	42.64	2.1067×10^{-17}	13.70	–	1.9589×10^{-17}	2.5651×10^{-17}	–	1.0754	0.8213
9100	Kidney left pelvis	8.53	2.1039×10^{-17}	12.86	–	2.9100×10^{-17}	2.0839×10^{-17}	–	0.7230	1.0096
9200	Kidney right cortex	119.39	2.1519×10^{-17}	5.15	–	2.2152×10^{-17}	2.0831×10^{-17}	–	0.9714	1.0330
9300	Kidney right medulla	42.64	2.1641×10^{-17}	6.78	–	1.9929×10^{-17}	1.9975×10^{-17}	–	1.0859	1.0834
9400	Kidney right pelvis	8.53	2.0255×10^{-17}	16.76	–	1.9073×10^{-17}	1.7603×10^{-17}	–	1.0620	1.1507
9500	Liver	1707.82	3.5115×10^{-17}	1.18	–	3.4061×10^{-17}	3.4500×10^{-17}	–	1.0310	1.0178

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
9700	Lung(AI) left	389.51	2.4032×10^{-17}	1.58	–	2.2894×10^{-17}	2.2679×10^{-17}	–	1.0497	1.0597
9900	Lung(AI) right	510.49	2.7207×10^{-17}	3.74	–	2.6204×10^{-17}	2.6333×10^{-17}	–	1.0383	1.0332
10000	Lymphatic nodes ET	12.73	2.9111×10^{-17}	16.05	–	2.3363×10^{-17}	2.9044×10^{-17}	–	1.2460	1.0023
10100	Lymphatic nodes thoracic	12.73	2.1005×10^{-17}	20.48	–	2.3797×10^{-17}	1.8757×10^{-17}	–	0.8827	1.1198
10200	Lymphatic nodes head	4.40	3.5230×10^{-17}	17.15	–	1.9324×10^{-17}	2.4283×10^{-17}	–	1.8231	1.4508
10300	Lymphatic nodes trunk	103.94	3.0825×10^{-17}	3.18	–	3.1516×10^{-17}	2.8414×10^{-17}	–	0.9781	1.0848
10400	Lymphatic nodes arms	8.80	2.8118×10^{-17}	20.62	–	3.7123×10^{-17}	2.9628×10^{-17}	–	0.7574	0.9490
10500	Lymphatic nodes legs	8.80	2.5107×10^{-17}	27.13	–	2.1095×10^{-17}	3.9860×10^{-17}	–	1.1902	0.6299
10600	Muscle head	563.16	2.4367×10^{-17}	2.63	–	2.3695×10^{-17}	2.4441×10^{-17}	–	1.0283*	0.9970*
10700	Muscle trunk	11667.40	2.6689×10^{-17}	0.40	–	2.6642×10^{-17}	2.6796×10^{-17}	–	1.0018*	0.9960*
10800	Muscle arms	2810.41	3.1346×10^{-17}	1.18	–	3.1440×10^{-17}	3.1346×10^{-17}	–	0.9970*	1.0000*
10900	Muscle legs	9616.52	2.8634×10^{-17}	0.53	–	2.8834×10^{-17}	2.8483×10^{-17}	–	0.9931*	1.0053*
11000	Oesophagus wall(0-190)	2.17	1.8092×10^{-17}	33.19	–	3.0882×10^{-17}	2.2059×10^{-17}	–	0.5858	0.8201
11001	Oesophagus wall(190-200)	0.116	1.8104×10^{-17}	52.20	–	2.2935×10^{-17}	4.8760×10^{-17}	–	0.7894	0.3713
11002	Oesophagus wall(200-surface)	36.64	1.9611×10^{-17}	11.27	–	2.1299×10^{-17}	2.3083×10^{-17}	–	0.9208	0.8496
11003	Oesophagus contents	21.84	2.3153×10^{-17}	16.57	–	2.2280×10^{-17}	2.3353×10^{-17}	–	1.0392	0.9915
11300	Pancreas	136.66	2.9686×10^{-17}	4.16	–	2.4675×10^{-17}	2.7526×10^{-17}	–	1.2031	1.0785
11400	Pituitary gland	0.529	2.6571×10^{-17}	40.52	–	2.0073×10^{-17}	1.4185×10^{-17}	–	1.3237	1.8732
11500	Prostate	4.55	3.2793×10^{-17}	28.21	–	2.7353×10^{-17}	4.7910×10^{-17}	–	1.1989	0.6845
11600	RST head	459.74	2.4028×10^{-17}	2.30	–	2.3943×10^{-17}	2.3107×10^{-17}	–	1.0036*	1.0399*
11700	RST trunk	8275.44	2.8014×10^{-17}	0.60	–	2.7895×10^{-17}	2.7998×10^{-17}	–	1.0043	1.0006
11800	RST arms	1102.01	3.4604×10^{-17}	1.23	–	3.4354×10^{-17}	3.3919×10^{-17}	–	1.0073*	1.0202*
11900	RST legs	2501.20	2.7709×10^{-17}	1.13	–	2.7511×10^{-17}	2.8086×10^{-17}	–	1.0072*	0.9866*
12000	Salivary glands left	35.95	2.1647×10^{-17}	9.85	–	2.8001×10^{-17}	2.7254×10^{-17}	–	0.7731	0.7943
12100	Salivary glandss right	35.95	2.5024×10^{-17}	5.83	–	2.1290×10^{-17}	2.5182×10^{-17}	–	1.1754	0.9937
12200	Skin head insensitive	153.24	1.7036×10^{-17}	4.81	–	1.7785×10^{-17}	1.6681×10^{-17}	–	0.9579*	1.0213*
12201	Skin head sensitive(50-100)	6.53	1.1459×10^{-17}	14.46	–	1.3532×10^{-17}	1.0499×10^{-17}	–	0.8467*	1.0914*
12300	Skin trunk insensitive	663.99	2.1962×10^{-17}	1.47	–	2.1172×10^{-17}	2.2233×10^{-17}	–	1.0373*	0.9878*
12301	Skin trunk sensitive(50-100)	28.45	1.4524×10^{-17}	5.98	–	1.3648×10^{-17}	1.3154×10^{-17}	–	1.0642*	1.1041*
12400	Skin arms insensitive	412.60	2.7243×10^{-17}	2.80	–	2.7100×10^{-17}	2.8113×10^{-17}	–	1.0053*	0.9691*
12401	Skin arms sensitive(50-100)	17.73	1.9287×10^{-17}	4.79	–	1.7423×10^{-17}	2.0996×10^{-17}	–	1.1070*	0.9186*
12500	Skin legs insensitive	786.97	2.2577×10^{-17}	1.05	–	2.2234×10^{-17}	2.2660×10^{-17}	–	1.0154*	0.9964*
12501	Skin legs sensitive(50-100)	33.57	1.6955×10^{-17}	4.43	–	1.4664×10^{-17}	1.4995×10^{-17}	–	1.1562*	1.1307*
12600	Spinal cord	40.22	1.9943×10^{-17}	7.16	–	2.2483×10^{-17}	1.7182×10^{-17}	–	0.8871	1.1607
12700	Spleen	198.77	2.5702×10^{-17}	4.24	–	2.7284×10^{-17}	2.6647×10^{-17}	–	0.9420	0.9645
12800	Erupted upper front buccal permanent ena	0.65	2.8271×10^{-17}	67.11	–	2.1770×10^{-17}	2.1443×10^{-17}	–	1.2986	1.3184

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
12801	Erupted upper front lingual permanent en	0.823	3.0306×10^{-17}	44.34	–	2.2568×10^{-17}	2.9064×10^{-17}	–	1.3429	1.0427
12804	Erupted upper front-left buccal permanen	0.317	3.9815×10^{-17}	78.21	–	7.6167×10^{-18}	8.6784×10^{-18}	–	5.2272	4.5878
12805	Erupted upper front-left lingual permane	0.224	5.2211×10^{-17}	87.93	–	2.1888×10^{-18}	4.0996×10^{-17}	–	23.8536	1.2736
12806	Erupted upper front-right buccal permane	0.317	2.1503×10^{-17}	87.83	–	7.7267×10^{-18}	3.5378×10^{-17}	–	2.7829	0.6078
12807	Erupted upper front-right lingual perman	0.224	3.1158×10^{-17}	68.18	–	6.5868×10^{-17}	5.9541×10^{-17}	–	0.4730	0.5233
12808	Erupted upper left buccal permanent enam	0.513	2.3825×10^{-18}	100.00	–	1.9951×10^{-17}	2.4016×10^{-17}	–	0.1194	0.0992
12809	Erupted upper left lingual permanent ena	0.494	1.9762×10^{-17}	42.46	–	2.8975×10^{-17}	3.3342×10^{-19}	–	0.6821	59.2718
12812	Erupted upper right buccal permanent ena	0.477	3.3122×10^{-17}	92.80	–	2.9618×10^{-17}	1.1485×10^{-17}	–	1.1183	2.8839
12813	Erupted upper right lingual permanent en	0.531	9.8308×10^{-18}	80.95	–	1.6417×10^{-17}	9.0568×10^{-17}	–	0.5988	0.1085
12816	Erupted lower front buccal permanent ena	0.605	4.4522×10^{-17}	50.13	–	7.2337×10^{-18}	2.9784×10^{-17}	–	6.1547	1.4948
12817	Erupted lower front lingual permanent en	0.54	2.8692×10^{-18}	100.00	–	1.4612×10^{-17}	6.0801×10^{-18}	–	0.1964	0.4719
12820	Erupted lower front-left buccal permanen	0.239	0.0000×10^{0}	0.00	–	6.8055×10^{-17}	5.4790×10^{-17}	–	–	–
12821	Erupted lower front-left lingual permane	0.228	6.7803×10^{-18}	100.00	–	1.1376×10^{-16}	3.5504×10^{-17}	–	0.0596	0.1910
12822	Erupted lower front-right buccal permane	0.244	1.0085×10^{-17}	75.62	–	8.8793×10^{-18}	8.0409×10^{-18}	–	1.1358	1.2542
12823	Erupted lower front-right lingual perman	0.224	3.7175×10^{-18}	100.00	–	7.3960×10^{-17}	4.5045×10^{-18}	–	0.0503	0.8253
12824	Erupted lower left buccal permanent enam	0.536	3.3784×10^{-17}	61.10	–	2.7466×10^{-17}	5.9656×10^{-17}	–	1.2300	0.5663
12825	Erupted lower left lingual permanent ena	0.494	1.8106×10^{-17}	73.75	–	1.4809×10^{-17}	7.8901×10^{-18}	–	1.2226	2.2948
12828	Erupted lower right buccal permanent ena	0.594	2.6198×10^{-17}	42.93	–	3.4605×10^{-17}	5.5915×10^{-17}	–	0.7571	0.4685
12829	Erupted lower right lingual permanent en	0.437	0.0000×10^{0}	0.00	–	3.6931×10^{-17}	4.3034×10^{-17}	–	–	–
12832	Erupted upper front permanent dentin	4.94	2.5246×10^{-17}	22.26	–	2.6930×10^{-17}	2.9861×10^{-17}	–	0.9375	0.8454

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
12834	Erupted upper front-left permanent denti	1.81	2.0795×10^{-17}	44.25	–	3.1752×10^{-17}	2.9965×10^{-17}	–	0.6549	0.6940
12835	Erupted upper front-right permanent dent	1.82	1.3281×10^{-17}	54.48	–	3.0380×10^{-17}	2.9712×10^{-17}	–	0.4371	0.4470
12836	Erupted upper left permanent dentin	3.39	2.5043×10^{-17}	32.45	–	1.8897×10^{-17}	1.3540×10^{-17}	–	1.3252	1.8495
12838	Erupted upper right permanent dentin	3.39	2.0399×10^{-17}	35.08	–	3.8443×10^{-17}	2.5699×10^{-17}	–	0.5306	0.7938
12840	Erupted lower front permanent dentin	3.76	2.8446×10^{-17}	29.45	–	2.7127×10^{-17}	3.5191×10^{-17}	–	1.0486	0.8083
12842	Erupted lower front-left permanent denti	1.56	3.9016×10^{-17}	40.40	–	4.2161×10^{-17}	4.2283×10^{-17}	–	0.9254	0.9227
12843	Erupted lower front-right permanent dent	1.56	2.3044×10^{-17}	49.10	–	3.6672×10^{-17}	6.9892×10^{-17}	–	0.6284	0.3297
12844	Erupted lower left permanent dentin	3.47	1.9920×10^{-17}	37.21	–	2.9511×10^{-17}	4.0199×10^{-17}	–	0.6750	0.4955
12846	Erupted lower right permanent dentin	3.46	2.2546×10^{-17}	28.16	–	2.8725×10^{-17}	3.4248×10^{-17}	–	0.7849	0.6583
12848	Unerupted permanent enamel	1.22	3.1312×10^{-17}	28.68	–	2.0712×10^{-17}	2.5293×10^{-17}	–	1.5118	1.2380
12850	Unerupted permanent dentin	4.11	2.2522×10^{-17}	38.44	–	3.1337×10^{-17}	3.3248×10^{-17}	–	0.7187	0.6774
12852	Permanent cementum	1.45	2.4298×10^{-17}	17.87	–	2.8841×10^{-17}	2.4583×10^{-17}	–	0.8425	0.9884
12854	Permanent pulp	0.993	6.1725×10^{-17}	31.33	–	9.3421×10^{-18}	2.8058×10^{-17}	–	6.6072	2.1999
12856	Teeth retention region	0.0505	2.5431×10^{-17}	36.31	–	9.1245×10^{-18}	3.1235×10^{-17}	–	2.7871	0.8142
12900	Testis left	8.52	3.9761×10^{-17}	8.34	–	4.3507×10^{-17}	3.8043×10^{-17}	–	0.9139	1.0451
13000	Testis right	8.52	3.9894×10^{-17}	16.63	–	3.9542×10^{-17}	4.1438×10^{-17}	–	1.0089	0.9627
13100	Thymus	37.00	2.9581×10^{-17}	5.96	–	2.3972×10^{-17}	2.6509×10^{-17}	–	1.2340	1.1159
13200	Thyroid	14.04	2.7927×10^{-17}	12.12	–	3.1656×10^{-17}	2.8446×10^{-17}	–	0.8822	0.9818
13300	Tongue upper(food)	19.03	2.1523×10^{-17}	9.71	–	2.0624×10^{-17}	3.0390×10^{-17}	–	1.0436	0.7082
13301	Tongue lower	40.04	2.6100×10^{-17}	8.36	–	2.3194×10^{-17}	2.7388×10^{-17}	–	1.1253	0.9530
13400	Tonsils	3.17	2.3945×10^{-17}	38.70	–	2.5802×10^{-17}	1.3493×10^{-17}	–	0.9280	1.7746
13500	Ureter left	6.34	3.3665×10^{-17}	20.33	–	2.9153×10^{-17}	3.4251×10^{-17}	–	1.1548	0.9829
13600	Ureter right	6.34	1.6579×10^{-17}	21.73	–	2.4816×10^{-17}	2.3685×10^{-17}	–	0.6681	0.7000
13700	Urinary bladder wall insensitive	39.00	2.9954×10^{-17}	10.34	–	3.0785×10^{-17}	3.0004×10^{-17}	–	0.9730	0.9983
13701	Urinary bladder wall sensitive(116-238)	1.87	3.9668×10^{-17}	15.54	–	3.6173×10^{-17}	3.0537×10^{-17}	–	1.0966	1.2990
13800	Urinary bladder content	160.00	3.1548×10^{-17}	3.45	–	3.2743×10^{-17}	3.1083×10^{-17}	–	0.9635	1.0150
14000	Air inside body	0.384	2.4337×10^{-17}	9.92	–	4.0055×10^{-17}	2.2956×10^{-17}	–	0.6076	1.0601

* For tissues bookkept as head, trunk, arm and leg pieces (muscle, residual soft tissue, skin, large vessels), the volumes tabulated in the distributed cell tables partition the tissue at different boundaries from the transported mesh, and the reference output is normalised by the tabulated volume. The marked ratio therefore compares two different partitions of the same tissue, and is left out of the summary statistics of Table S2.

Table S25. 15F_Internal. Absorbed dose per primary history, Gy. σ is the FLUKA statistical uncertainty. Ratios are FLUKA / reference.

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
100	Adrenal left	5.24	2.5499×10^{-15}	2.21	–	2.6899×10^{-15}	2.6544×10^{-15}	–	0.9480	0.9606
200	Adrenal right	5.24	5.3613×10^{-15}	1.22	–	5.1104×10^{-15}	5.3589×10^{-15}	–	1.0491	1.0005
300	ET1(0-8)	0.00194	0.0000×10^{0}	0.00	–	1.1758×10^{-16}	1.0458×10^{-16}	–	–	–
301	ET1(8-40)	0.0079	5.3929×10^{-17}	100.00	–	1.2355×10^{-16}	1.6654×10^{-16}	–	0.4365	0.3238
302	ET1(40-50)	0.00251	4.0579×10^{-17}	100.00	–	1.0964×10^{-16}	1.9024×10^{-16}	–	0.3701	0.2133
303	ET1(50-Surface)	0.729	1.0906×10^{-16}	18.00	–	1.3063×10^{-16}	1.2696×10^{-16}	–	0.8349	0.8590
400	ET2(-15-0)	0.15	2.1094×10^{-16}	18.95	–	1.6540×10^{-16}	1.5896×10^{-16}	–	1.2753*	1.3270*
401	ET2(0-40)	0.426	2.1120×10^{-16}	8.13	–	1.6315×10^{-16}	1.5129×10^{-16}	–	1.2945*	1.3960*
402	ET2(40-50)	0.107	1.4925×10^{-16}	13.79	–	2.8703×10^{-16}	1.2249×10^{-16}	–	0.5200*	1.2185*
403	ET2(50-55)	0.0537	1.1466×10^{-16}	29.88	–	2.3785×10^{-16}	1.5793×10^{-16}	–	0.4821	0.7260
404	ET2(55-65)	0.108	1.2468×10^{-16}	24.59	–	1.5682×10^{-16}	1.2519×10^{-16}	–	0.7950*	0.9959*
405	ET2(65-Surface)	24.77	1.4830×10^{-16}	3.23	–	1.4959×10^{-16}	1.5078×10^{-16}	–	0.9914	0.9836
500	Oral mucosa tongue	0.071	1.9227×10^{-16}	20.12	–	2.6195×10^{-16}	2.1865×10^{-16}	–	0.7340	0.8794
501	Oral mucosa moth floor	0.0292	1.8439×10^{-16}	21.42	–	2.2045×10^{-16}	1.8166×10^{-16}	–	0.8364	1.0150
600	Oral mucosa lips and cheeks	0.0178	1.0331×10^{-16}	24.98	–	2.4978×10^{-16}	1.6670×10^{-16}	–	0.4136	0.6197
700	Trachea	6.20	6.2201×10^{-16}	3.56	–	6.4231×10^{-16}	5.7830×10^{-16}	–	0.9684	1.0756
800	BB(-11–6)	0.00858	1.5392×10^{-15}	23.31	–	1.1065×10^{-15}	9.2635×10^{-16}	–	1.3910	1.6615
801	BB(-6-0)	0.0109	1.2491×10^{-15}	13.54	–	1.3220×10^{-15}	7.9815×10^{-16}	–	0.9449	1.5650
802	BB(0-10)	0.0183	1.0646×10^{-15}	13.76	–	1.3385×10^{-15}	1.6274×10^{-15}	–	0.7954	0.6542
803	BB(10-35)	0.0458	1.1889×10^{-15}	12.73	–	1.1007×10^{-15}	1.6338×10^{-15}	–	1.0801	0.7277
804	BB(35-40)	0.0092	1.1059×10^{-15}	16.15	–	1.4268×10^{-15}	1.3961×10^{-15}	–	0.7750	0.7921
805	BB(40-50)	0.0184	1.9905×10^{-15}	24.18	–	1.2898×10^{-15}	1.1731×10^{-15}	–	1.5433	1.6968
806	BB(50-60)	0.0185	1.2668×10^{-15}	18.29	–	1.4156×10^{-15}	1.2363×10^{-15}	–	0.8949	1.0247
807	BB(60-70)	0.0185	1.2544×10^{-15}	17.86	–	1.3101×10^{-15}	1.4410×10^{-15}	–	0.9575	0.8705
808	BB(70-surface)	6.64	1.2377×10^{-15}	2.62	–	1.1917×10^{-15}	1.2059×10^{-15}	–	1.0386	1.0264
900	Blood in large arteries head	0.718	2.2856×10^{-16}	18.07	–	2.4246×10^{-16}	2.8784×10^{-16}	–	0.9427*	0.7941*
910	Blood in large veins head	3.30	1.2472×10^{-16}	15.40	–	1.6467×10^{-16}	2.0808×10^{-16}	–	0.7574*	0.5994*
1000	Blood in large arteries trunk	85.09	1.4251×10^{-15}	0.55	–	1.4379×10^{-15}	1.4402×10^{-15}	–	0.9911*	0.9895*
1010	Blood in large veins trunk	171.13	1.3976×10^{-15}	0.58	–	1.4215×10^{-15}	1.4237×10^{-15}	–	0.9832*	0.9817*
1100	Blood in large arteries arms	25.60	7.2589×10^{-16}	1.52	–	7.3486×10^{-16}	7.2702×10^{-16}	–	0.9878*	0.9985*
1110	Blood in large veins arms	95.90	6.0880×10^{-16}	0.81	–	6.1718×10^{-16}	6.1854×10^{-16}	–	0.9864*	0.9843*
1200	Blood in large arteries legs	98.57	1.8116×10^{-17}	4.51	–	1.6635×10^{-17}	1.8582×10^{-17}	–	1.0890	0.9749
1210	Blood in large veins legs	359.61	1.5434×10^{-17}	1.79	–	1.5754×10^{-17}	1.5845×10^{-17}	–	0.9797	0.9740
1300	Humeri upper cortical	124.84	6.1704×10^{-16}	0.90	–	6.3666×10^{-16}	6.0789×10^{-16}	–	0.9692	1.0150
1400	Humeri upper spogiosa	133.67	3.7290×10^{-16}	1.26	–	3.6420×10^{-16}	3.7268×10^{-16}	–	1.0239	1.0006

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
1500	Humeri upper medullary cavity	15.04	6.2536×10^{-16}	3.60	–	6.7993×10^{-16}	6.5015×10^{-16}	–	0.9198	0.9619
1600	Humeri lower cortical	133.91	8.5684×10^{-16}	0.52	–	8.9325×10^{-16}	8.6689×10^{-16}	–	0.9592	0.9884
1700	Humeri lower spongiosa	55.19	6.8681×10^{-16}	2.10	–	7.0479×10^{-16}	7.0860×10^{-16}	–	0.9745	0.9692
1800	Humeri lower medullary cavity	12.75	9.6047×10^{-16}	2.87	–	9.7375×10^{-16}	9.7059×10^{-16}	–	0.9864	0.9896
1900	Radii cortical	84.25	2.4066×10^{-16}	2.31	–	2.3528×10^{-16}	2.3098×10^{-16}	–	1.0228	1.0419
1910	Ulnae cortical	105.37	2.8010×10^{-16}	1.45	–	2.8780×10^{-16}	2.8069×10^{-16}	–	0.9732	0.9979
2000	Radii spongiosa	28.35	2.1227×10^{-16}	2.52	–	2.0056×10^{-16}	2.1153×10^{-16}	–	1.0584	1.0035
2010	Ulnae spongiosa	41.42	4.2921×10^{-16}	2.24	–	4.2989×10^{-16}	4.3642×10^{-16}	–	0.9984	0.9835
2100	Radii medullary cavity	5.64	2.3381×10^{-16}	8.32	–	2.2818×10^{-16}	2.1749×10^{-16}	–	1.0247	1.0750
2110	Ulnae medullary cavity	6.99	2.3563×10^{-16}	6.64	–	2.3346×10^{-16}	2.1592×10^{-16}	–	1.0093	1.0913
2200	Wrists and hand bones cortical	108.28	5.4402×10^{-17}	3.09	–	5.4267×10^{-17}	5.2734×10^{-17}	–	1.0025	1.0316
2300	Wrists and hand bones spongiosa	48.59	5.8915×10^{-17}	8.05	–	5.5701×10^{-17}	5.9184×10^{-17}	–	1.0577	0.9955
2400	Clavicles cortical	56.94	4.6044×10^{-16}	1.61	–	4.4940×10^{-16}	4.5458×10^{-16}	–	1.0245	1.0129
2500	Clavicles spongiosa	26.60	4.6392×10^{-16}	2.72	–	4.4482×10^{-16}	4.5982×10^{-16}	–	1.0430	1.0089
2600	Cranium cortical	450.60	7.0208×10^{-17}	1.78	–	7.3318×10^{-17}	7.1466×10^{-17}	–	0.9576	0.9824
2700	Cranium spongiosa	391.55	7.1060×10^{-17}	2.07	–	7.3540×10^{-17}	7.2722×10^{-17}	–	0.9663	0.9771
2800	Femora upper cortical	236.16	5.4273×10^{-17}	2.15	–	5.3904×10^{-17}	5.3162×10^{-17}	–	1.0068	1.0209
2900	Femora upper spongiosa	186.87	1.1811×10^{-16}	1.42	–	1.1955×10^{-16}	1.1615×10^{-16}	–	0.9879	1.0168
3000	Femora upper medullary cavity	34.93	4.3409×10^{-17}	5.57	–	4.2450×10^{-17}	4.5466×10^{-17}	–	1.0226	0.9547
3100	Femora lower cortical	255.04	7.7220×10^{-18}	6.46	–	7.5819×10^{-18}	7.7510×10^{-18}	–	1.0185	0.9963
3200	Femora lower spongiosa	271.34	4.2801×10^{-18}	6.71	–	4.3932×10^{-18}	3.9633×10^{-18}	–	0.9743	1.0799
3300	Femora lower medullary cavity	38.34	9.8270×10^{-18}	12.73	–	7.4267×10^{-18}	6.1932×10^{-18}	–	1.3232	1.5867
3400	Tibiae cortical	325.55	1.8477×10^{-18}	14.16	–	1.4889×10^{-18}	1.7207×10^{-18}	–	1.2410	1.0738
3410	Fibulae cortical	54.33	1.1761×10^{-18}	38.30	–	9.2908×10^{-19}	6.9570×10^{-19}	–	1.2659	1.6905
3420	Patellae cortical	17.54	3.2496×10^{-18}	27.57	–	2.8609×10^{-18}	3.0324×10^{-18}	–	1.1359	1.0716
3500	Tibiae spongiosa	227.06	2.1694×10^{-18}	12.62	–	2.4518×10^{-18}	2.2034×10^{-18}	–	0.8848	0.9846
3510	Fibulae spongiosa	24.54	6.1179×10^{-19}	44.43	–	1.1837×10^{-18}	6.0954×10^{-19}	–	0.5168	1.0037
3520	Patellae spongiosa	17.83	3.2704×10^{-18}	33.91	–	4.4763×10^{-18}	3.4895×10^{-18}	–	0.7306	0.9372
3600	Tibiae medullary cavity	48.77	1.4245×10^{-18}	20.77	–	1.2768×10^{-18}	1.2037×10^{-18}	–	1.1157	1.1834
3610	Fibulae medullary cavity	5.81	0.0000×10^{0}	0.00	–	1.9307×10^{-18}	3.7342×10^{-19}	–	–	–
3700	Ankles and foot cortical	176.95	3.5803×10^{-18}	13.89	–	3.5931×10^{-18}	3.1554×10^{-18}	–	0.9965	1.1347
3800	Ankles and foot spongiosa	306.52	2.7592×10^{-18}	12.43	–	2.6637×10^{-18}	2.4383×10^{-18}	–	1.0359	1.1316
3900	Mandible cortical	47.27	1.8486×10^{-16}	1.77	–	1.8441×10^{-16}	1.8033×10^{-16}	–	1.0024	1.0251
4000	Mandible spongiosa	28.85	1.9523×10^{-16}	3.84	–	2.0126×10^{-16}	1.9767×10^{-16}	–	0.9700	0.9876
4100	Pelvis cortical	215.14	3.0155×10^{-16}	1.05	–	3.0834×10^{-16}	3.0428×10^{-16}	–	0.9780	0.9910
4200	Pelvis spongiosa	553.25	3.0997×10^{-16}	0.54	–	3.1001×10^{-16}	3.1098×10^{-16}	–	0.9999	0.9967
4300	Ribs cortical	186.41	1.8236×10^{-15}	0.42	–	1.8269×10^{-15}	1.8307×10^{-15}	–	0.9982	0.9961
4400	Ribs spongiosa	208.05	1.6815×10^{-15}	0.43	–	1.6719×10^{-15}	1.6923×10^{-15}	–	1.0058	0.9936

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
4500	Scapulae cortical	172.60	6.3149×10^{-16}	0.73	–	6.3499×10^{-16}	6.4538×10^{-16}	–	0.9945	0.9785
4600	Scapulae spongiosa	147.47	6.2908×10^{-16}	1.15	–	6.2078×10^{-16}	6.2548×10^{-16}	–	1.0134	1.0058
4700	Cervical spine cortical	42.94	2.3124×10^{-16}	1.97	–	2.2819×10^{-16}	2.2558×10^{-16}	–	1.0134	1.0251
4800	Cervical spine spongiosa	68.33	2.2877×10^{-16}	1.95	–	2.3504×10^{-16}	2.3790×10^{-16}	–	0.9733	0.9616
4900	Thoracic spine cortical	84.06	1.5350×10^{-15}	0.75	–	1.5298×10^{-15}	1.5156×10^{-15}	–	1.0034	1.0128
5000	Thoracic spine spongiosa	230.52	1.6230×10^{-15}	0.47	–	1.6289×10^{-15}	1.6395×10^{-15}	–	0.9963	0.9899
5100	Lumbar spine cortical	41.25	1.5801×10^{-15}	0.73	–	1.6080×10^{-15}	1.6022×10^{-15}	–	0.9827	0.9862
5200	Lumbar spine spongiosa	270.61	1.6660×10^{-15}	0.33	–	1.6652×10^{-15}	1.6691×10^{-15}	–	1.0005	0.9981
5300	Sacrum cortical	86.54	4.4879×10^{-16}	1.50	–	4.5333×10^{-16}	4.4306×10^{-16}	–	0.9900	1.0129
5400	Sacrum spongiosa	140.21	4.2418×10^{-16}	0.69	–	4.2563×10^{-16}	4.2617×10^{-16}	–	0.9966	0.9953
5500	Sternum cortical	15.67	9.5519×10^{-16}	1.75	–	9.6100×10^{-16}	9.6687×10^{-16}	–	0.9940	0.9879
5600	Sternum spongiosa	28.43	9.1680×10^{-16}	1.18	–	8.8416×10^{-16}	9.2788×10^{-16}	–	1.0369	0.9881
5800	Cartilage trunk	161.94	2.9397×10^{-15}	0.51	–	2.9320×10^{-15}	2.9557×10^{-15}	–	1.0026	0.9946
6100	Brain	1348.99	6.5706×10^{-17}	0.84	–	6.7966×10^{-17}	6.6135×10^{-17}	–	0.9667	0.9935
6200	Breast left adipose tissue	77.62	1.0039×10^{-15}	0.76	–	1.0058×10^{-15}	1.0208×10^{-15}	–	0.9981	0.9834
6300	Breast left glandular tissue	51.75	1.0473×10^{-15}	1.09	–	1.0585×10^{-15}	1.0455×10^{-15}	–	0.9895	1.0017
6400	Breast right adipose tissue	77.62	2.1000×10^{-15}	0.56	–	2.0835×10^{-15}	2.1117×10^{-15}	–	1.0079	0.9944
6500	Breast right glandular tissue	51.75	2.2026×10^{-15}	0.72	–	2.1674×10^{-15}	2.1674×10^{-15}	–	1.0162	1.0162
6600	Eye lens sensitive left	0.0203	1.5353×10^{-16}	100.00	–	2.3964×10^{-16}	–	–	0.6407	–
6601	Eye lens insensitive left	0.117	8.1992×10^{-17}	80.08	–	6.7078×10^{-17}	5.5820×10^{-17}	–	1.2223	1.4689
6700	Cornea left	0.996	1.1002×10^{-16}	30.15	–	8.2915×10^{-17}	5.8969×10^{-17}	–	1.3269	1.8658
6701	Aqueous left	0.344	1.3015×10^{-16}	38.27	–	1.4198×10^{-16}	3.4765×10^{-17}	–	0.9167	3.7438
6702	Vitreous left	5.32	8.5063×10^{-17}	9.97	–	8.4503×10^{-17}	6.8821×10^{-17}	–	1.0066	1.2360
6800	Eye lens sensitive right	0.0203	8.0086×10^{-17}	98.00	–	5.8209×10^{-16}	2.0760×10^{-16}	–	0.1376	0.3858
6801	Eye lens insensitive right	0.117	7.7368×10^{-17}	71.81	–	8.3350×10^{-17}	5.2565×10^{-18}	–	0.9282	14.7186
6900	Cornea right	0.996	5.7081×10^{-17}	24.16	–	8.7501×10^{-17}	9.8839×10^{-17}	–	0.6523	0.5775
6901	Aqueous right	0.344	7.4193×10^{-17}	47.93	–	6.5009×10^{-17}	9.5526×10^{-17}	–	1.1413	0.7767
6902	Vitreous right	5.32	8.6612×10^{-17}	14.52	–	1.0712×10^{-16}	1.0630×10^{-16}	–	0.8086	0.8148
7000	Gall bladder wall	7.56	1.1323×10^{-14}	0.76	–	1.1617×10^{-14}	1.1418×10^{-14}	–	0.9747	0.9917
7100	Gall bladder contents	42.00	1.0907×10^{-14}	0.37	–	1.0892×10^{-14}	1.0924×10^{-14}	–	1.0014	0.9985
7200	Stomach wall(0-60)	1.60	3.5753×10^{-15}	1.63	–	3.6115×10^{-15}	3.4716×10^{-15}	–	0.9900	1.0299
7201	Stomach wall(60-100)	1.07	3.5411×10^{-15}	1.58	–	3.6701×10^{-15}	3.5683×10^{-15}	–	0.9648	0.9924
7202	Stomach wall(100-300)	5.39	3.5625×10^{-15}	0.86	–	3.5521×10^{-15}	3.4802×10^{-15}	–	1.0030	1.0237
7203	Stomach wall(300-surface)	136.31	3.7040×10^{-15}	0.29	–	3.6740×10^{-15}	3.6892×10^{-15}	–	1.0081	1.0040
7300	Stomach contents	200.00	3.2610×10^{-15}	0.41	–	3.3004×10^{-15}	3.3107×10^{-15}	–	0.9881	0.9850
7400	Small intestine wall(0-130)	12.46	1.7979×10^{-15}	1.40	–	1.7065×10^{-15}	1.7497×10^{-15}	–	1.0536	1.0275
7401	Small intestine wall(130-150)	1.94	1.8132×10^{-15}	1.97	–	1.7871×10^{-15}	1.7391×10^{-15}	–	1.0146	1.0426
7402	Small intestine wall(150-200)	4.88	1.7938×10^{-15}	1.57	–	1.7566×10^{-15}	1.7626×10^{-15}	–	1.0212	1.0177

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
7403	Small intestine wall(200-surface)	616.55	1.7289×10^{-15}	0.13	–	1.7271×10^{-15}	1.7281×10^{-15}	–	1.0011	1.0005
7500	Small intestine contents(-500-0)	45.19	1.7391×10^{-15}	0.52	–	1.7288×10^{-15}	1.7374×10^{-15}	–	1.0059	1.0010
7501	Small intestine contents(centre–500)	234.81	1.7293×10^{-15}	0.44	–	1.7278×10^{-15}	1.7325×10^{-15}	–	1.0009	0.9981
7600	Ascending colon wall(0-280)	3.66	7.5984×10^{-16}	3.46	–	6.9878×10^{-16}	6.9999×10^{-16}	–	1.0874	1.0855
7601	Ascending colon wall(280-300)	0.265	6.9565×10^{-16}	7.56	–	7.5326×10^{-16}	6.6607×10^{-16}	–	0.9235	1.0444
7602	Ascending colon wall(300-surface)	68.67	7.3377×10^{-16}	1.15	–	7.4262×10^{-16}	7.5216×10^{-16}	–	0.9881	0.9756
7700	Ascending colon content	77.81	6.7889×10^{-16}	1.04	–	6.8148×10^{-16}	6.8914×10^{-16}	–	0.9962	0.9851
7800	Transverse colon wall right(0-280)	2.67	3.2730×10^{-15}	1.51	–	3.2940×10^{-15}	3.3049×10^{-15}	–	0.9936	0.9903
7801	Transverse colon wall right(280-300)	0.194	3.2962×10^{-15}	3.63	–	3.3802×10^{-15}	3.3233×10^{-15}	–	0.9751	0.9918
7802	Transverse colon wall right(300-surface)	72.88	3.2073×10^{-15}	0.50	–	3.2502×10^{-15}	3.2486×10^{-15}	–	0.9868	0.9873
7900	Transverse colon contents right	42.19	3.2261×10^{-15}	0.72	–	3.2577×10^{-15}	3.2279×10^{-15}	–	0.9903	0.9994
8000	Transverse colon wall left(0-280)	2.15	1.5388×10^{-15}	2.75	–	1.5006×10^{-15}	1.5394×10^{-15}	–	1.0255	0.9996
8001	Transverse colon wall left(280-300)	0.157	1.5840×10^{-15}	6.00	–	1.6805×10^{-15}	1.4656×10^{-15}	–	0.9426	1.0808
8002	Transverse colon wall left(300-surface)	67.63	1.4938×10^{-15}	0.64	–	1.5053×10^{-15}	1.5211×10^{-15}	–	0.9923	0.9820
8100	Transverse colon content left	27.01	1.5386×10^{-15}	1.12	–	1.5351×10^{-15}	1.5044×10^{-15}	–	1.0023	1.0227
8200	Descending colon wall(0-280)	2.69	5.5202×10^{-16}	4.43	–	5.6537×10^{-16}	5.7131×10^{-16}	–	0.9764	0.9662
8201	Descending colon wall(280-300)	0.196	6.2231×10^{-16}	10.07	–	6.2721×10^{-16}	5.5685×10^{-16}	–	0.9922	1.1176
8202	Descending colon wall(300-surface)	75.44	5.4814×10^{-16}	0.93	–	5.5737×10^{-16}	5.3369×10^{-16}	–	0.9834	1.0271
8300	Descending colon content	32.99	5.3373×10^{-16}	1.37	–	5.6802×10^{-16}	5.5991×10^{-16}	–	0.9396	0.9532
8400	Sigmoid colon wall(0-280)	3.22	2.3345×10^{-16}	6.00	–	1.9582×10^{-16}	2.4778×10^{-16}	–	1.1921	0.9422
8401	Sigmoid colon wall(280-300)	0.235	1.8256×10^{-16}	14.30	–	1.7992×10^{-16}	1.9236×10^{-16}	–	1.0146	0.9490
8402	Sigmoid colon wall(300-surface)	55.58	2.3436×10^{-16}	2.27	–	2.2691×10^{-16}	2.2415×10^{-16}	–	1.0328	1.0455
8500	Sigmoid colon contents	41.77	2.0896×10^{-16}	2.25	–	2.1343×10^{-16}	2.1867×10^{-16}	–	0.9790	0.9556
8600	Rectum wall(0-280)	1.44	1.0803×10^{-16}	10.64	–	9.4870×10^{-17}	1.1529×10^{-16}	–	1.1387	0.9370
8601	Rectum wall(280-300)	0.105	1.0358×10^{-16}	38.23	–	1.0085×10^{-16}	1.2616×10^{-16}	–	1.0271	0.8210
8602	Rectum wall(300-surface)	7.53	1.0612×10^{-16}	6.76	–	1.1784×10^{-16}	1.0585×10^{-16}	–	0.9006	1.0026
8603	Rectum contents	18.23	1.1913×10^{-16}	3.39	–	1.0842×10^{-16}	1.1347×10^{-16}	–	1.0988	1.0499
8700	Heart wall	251.49	3.0114×10^{-15}	0.25	–	2.9999×10^{-15}	3.0129×10^{-15}	–	1.0038	0.9995
8800	Blood in heart chamber	320.00	2.4060×10^{-15}	0.34	–	2.4034×10^{-15}	2.4094×10^{-15}	–	1.0011	0.9986
8900	Kidney left cortex	105.69	1.3117×10^{-15}	0.55	–	1.3127×10^{-15}	1.3220×10^{-15}	–	0.9993	0.9922
9000	Kidney left medulla	37.75	1.2891×10^{-15}	1.78	–	1.2943×10^{-15}	1.2862×10^{-15}	–	0.9960	1.0023
9100	Kidney left pelvis	7.55	1.3810×10^{-15}	3.24	–	1.3824×10^{-15}	1.3797×10^{-15}	–	0.9990	1.0009
9200	Kidney right cortex	105.69	2.8213×10^{-15}	0.55	–	2.7940×10^{-15}	2.8504×10^{-15}	–	1.0098	0.9898
9300	Kidney right medulla	37.75	2.9934×10^{-15}	1.29	–	2.9800×10^{-15}	3.0131×10^{-15}	–	1.0045	0.9935
9400	Kidney right pelvis	7.55	3.0015×10^{-15}	1.20	–	3.0583×10^{-15}	3.0633×10^{-15}	–	0.9814	0.9798
9500	Liver	1628.97	1.4660×10^{-14}	0.04	–	1.4627×10^{-14}	1.4693×10^{-14}	–	1.0022	0.9977

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
9700	Lung(AI) left	330.58	9.6611×10^{-16}	0.45	–	9.5984×10^{-16}	9.6224×10^{-16}	–	1.0065	1.0040
9900	Lung(AI) right	419.42	2.5192×10^{-15}	0.25	–	2.5132×10^{-15}	2.5446×10^{-15}	–	1.0024	0.9900
10000	Lymphatic nodes ET	11.52	1.8106×10^{-16}	5.26	–	1.7737×10^{-16}	1.7806×10^{-16}	–	1.0208	1.0169
10100	Lymphatic nodes thoracic	11.52	3.7404×10^{-15}	1.33	–	3.8572×10^{-15}	3.8745×10^{-15}	–	0.9697	0.9654
10200	Lymphatic nodes head	4.00	2.9138×10^{-16}	7.61	–	2.9141×10^{-16}	2.9633×10^{-16}	–	0.9999*	0.9833*
10300	Lymphatic nodes trunk	94.06	7.4712×10^{-16}	1.06	–	7.5607×10^{-16}	7.6726×10^{-16}	–	0.9882	0.9737
10400	Lymphatic nodes arms	7.96	6.8077×10^{-16}	4.51	–	7.2325×10^{-16}	7.0805×10^{-16}	–	0.9413	0.9615
10500	Lymphatic nodes legs	7.96	2.6486×10^{-18}	26.08	–	1.8981×10^{-18}	3.6407×10^{-18}	–	1.3954	0.7275
10600	Muscle head	416.65	1.4720×10^{-16}	1.25	–	1.5673×10^{-16}	1.4828×10^{-16}	–	0.9392*	0.9927*
10700	Muscle trunk	7978.95	9.1395×10^{-16}	0.10	–	9.2148×10^{-16}	9.2222×10^{-16}	–	0.9918*	0.9910*
10800	Muscle arms	1519.22	4.7593×10^{-16}	0.35	–	5.0463×10^{-16}	4.9339×10^{-16}	–	0.9431*	0.9646*
10900	Muscle legs	7446.05	2.0808×10^{-17}	0.57	–	2.0672×10^{-17}	2.0826×10^{-17}	–	1.0065	0.9991
11000	Oesophagus wall(0-190)	1.78	2.1804×10^{-15}	2.06	–	2.2810×10^{-15}	2.2564×10^{-15}	–	0.9559	0.9663
11001	Oesophagus wall(190-200)	0.0957	2.2656×10^{-15}	6.04	–	2.3423×10^{-15}	2.2816×10^{-15}	–	0.9673	0.9930
11002	Oesophagus wall(200-surface)	34.20	2.2563×10^{-15}	0.70	–	2.2349×10^{-15}	2.2840×10^{-15}	–	1.0095	0.9879
11003	Oesophagus contents	21.03	2.2826×10^{-15}	1.57	–	2.2492×10^{-15}	2.2762×10^{-15}	–	1.0148	1.0028
11100	Ovary left	3.19	2.7463×10^{-16}	8.76	–	2.6749×10^{-16}	2.6905×10^{-16}	–	1.0267	1.0208
11200	Ovary right	3.19	2.7210×10^{-16}	10.18	–	2.4548×10^{-16}	2.2103×10^{-16}	–	1.1084	1.2310
11300	Pancreas	117.63	2.8308×10^{-15}	0.61	–	2.8437×10^{-15}	2.8778×10^{-15}	–	0.9955	0.9837
11400	Pituitary gland	0.518	1.2801×10^{-16}	36.91	–	1.3442×10^{-16}	5.4578×10^{-17}	–	0.9523	2.3454
11600	RST head	746.19	1.2924×10^{-16}	0.65	–	1.3087×10^{-16}	1.3030×10^{-16}	–	0.9876	0.9919
11700	RST trunk	10421.35	1.4682×10^{-15}	0.08	–	1.4607×10^{-15}	1.5133×10^{-15}	–	1.0051*	0.9702*
11800	RST arms	1774.29	5.7707×10^{-16}	0.31	–	5.9697×10^{-16}	5.2763×10^{-16}	–	0.9667*	1.0937*
11900	RST legs	5479.97	2.4115×10^{-17}	0.43	–	2.4395×10^{-17}	2.4343×10^{-17}	–	0.9885	0.9906
12000	Salivary glands left	33.65	1.4785×10^{-16}	3.49	–	1.4078×10^{-16}	1.5399×10^{-16}	–	1.0502	0.9601
12100	Salivary glandss right	33.65	1.9718×10^{-16}	2.02	–	2.0265×10^{-16}	1.8714×10^{-16}	–	0.9730	1.0536
12200	Skin head insensitive	130.08	1.0230×10^{-16}	2.51	–	1.0594×10^{-16}	1.0508×10^{-16}	–	0.9656	0.9736
12201	Skin head sensitive(50-100)	6.25	9.0498×10^{-17}	4.15	–	9.6000×10^{-17}	9.0576×10^{-17}	–	0.9427	0.9991
12300	Skin trunk insensitive	559.14	8.8598×10^{-16}	0.26	–	9.3130×10^{-16}	9.2899×10^{-16}	–	0.9513*	0.9537*
12301	Skin trunk sensitive(50-100)	26.45	7.8933×10^{-16}	1.11	–	8.0386×10^{-16}	8.1068×10^{-16}	–	0.9819*	0.9737*
12400	Skin arms insensitive	314.14	4.0452×10^{-16}	0.69	–	4.4314×10^{-16}	3.4833×10^{-16}	–	0.9129*	1.1613*
12401	Skin arms sensitive(50-100)	15.16	3.7298×10^{-16}	1.50	–	4.0231×10^{-16}	2.6926×10^{-16}	–	0.9271*	1.3852*
12500	Skin legs insensitive	691.14	1.3901×10^{-17}	2.69	–	1.4125×10^{-17}	1.4061×10^{-17}	–	0.9841*	0.9886*
12501	Skin legs sensitive(50-100)	33.06	1.2183×10^{-17}	5.52	–	1.3019×10^{-17}	1.3098×10^{-17}	–	0.9359	0.9302
12600	Spinal cord	54.22	1.0923×10^{-15}	1.16	–	1.0732×10^{-15}	1.0977×10^{-15}	–	1.0179	0.9951
12700	Spleen	179.32	1.2438×10^{-15}	0.66	–	1.2425×10^{-15}	1.2379×10^{-15}	–	1.0010	1.0047
12800	Erupted upper front buccal permanent ena	0.564	1.4945×10^{-16}	14.46	–	1.7296×10^{-16}	1.1390×10^{-16}	–	0.8641	1.3122

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
12801	Erupted upper front lingual permanent en	0.58	1.9111×10^{-16}	20.03	–	1.3376×10^{-16}	1.7527×10^{-16}	–	1.4287	1.0904
12804	Erupted upper front-left buccal permanen	0.219	6.5134×10^{-17}	35.05	–	1.1085×10^{-16}	5.8249×10^{-17}	–	0.5876	1.1182
12805	Erupted upper front-left lingual permane	0.201	1.0908×10^{-16}	43.41	–	1.8564×10^{-16}	8.5722×10^{-17}	–	0.5876	1.2725
12806	Erupted upper front-right buccal permane	0.201	1.6856×10^{-16}	29.83	–	4.9430×10^{-17}	1.3764×10^{-16}	–	3.4100	1.2246
12807	Erupted upper front-right lingual perman	0.219	1.4230×10^{-16}	44.20	–	1.7518×10^{-16}	1.2509×10^{-16}	–	0.8123	1.1376
12808	Erupted upper left buccal permanent enam	0.426	1.3309×10^{-16}	20.51	–	1.3191×10^{-16}	1.1163×10^{-16}	–	1.0090	1.1923
12809	Erupted upper left lingual permanent ena	0.356	1.2806×10^{-16}	31.04	–	1.2343×10^{-16}	1.6409×10^{-16}	–	1.0375	0.7804
12812	Erupted upper right buccal permanent ena	0.381	1.9718×10^{-16}	25.23	–	1.2164×10^{-16}	2.1160×10^{-16}	–	1.6210	0.9319
12813	Erupted upper right lingual permanent en	0.4	2.2736×10^{-16}	32.19	–	1.2083×10^{-16}	2.1171×10^{-16}	–	1.8817	1.0739
12816	Erupted lower front buccal permanent ena	0.444	1.6085×10^{-16}	32.16	–	1.0312×10^{-16}	2.3170×10^{-16}	–	1.5597	0.6942
12817	Erupted lower front lingual permanent en	0.445	1.5007×10^{-16}	19.96	–	1.5301×10^{-16}	1.5306×10^{-16}	–	0.9808	0.9804
12820	Erupted lower front-left buccal permanen	0.181	3.2891×10^{-17}	48.16	–	1.1434×10^{-16}	8.6765×10^{-17}	–	0.2876	0.3791
12821	Erupted lower front-left lingual permane	0.182	1.0862×10^{-16}	37.89	–	5.5973×10^{-17}	6.1544×10^{-17}	–	1.9405	1.7649
12822	Erupted lower front-right buccal permane	0.2	1.5170×10^{-16}	26.53	–	1.7436×10^{-16}	2.6424×10^{-16}	–	0.8700	0.5741
12823	Erupted lower front-right lingual perman	0.163	2.2478×10^{-16}	38.88	–	1.0438×10^{-16}	1.1355×10^{-16}	–	2.1535	1.9795
12824	Erupted lower left buccal permanent enam	0.334	7.0537×10^{-17}	35.72	–	1.3422×10^{-16}	1.2477×10^{-16}	–	0.5255	0.5653
12825	Erupted lower left lingual permanent ena	0.466	1.3463×10^{-16}	31.25	–	1.7343×10^{-16}	1.3254×10^{-16}	–	0.7763	1.0158
12828	Erupted lower right buccal permanent ena	0.396	2.2872×10^{-16}	26.27	–	1.7529×10^{-16}	1.1259×10^{-16}	–	1.3048	2.0314
12829	Erupted lower right lingual permanent en	0.404	1.1767×10^{-16}	35.50	–	1.2439×10^{-16}	1.5773×10^{-16}	–	0.9459	0.7460
12832	Erupted upper front permanent dentin	3.83	1.4339×10^{-16}	15.73	–	1.4563×10^{-16}	1.4480×10^{-16}	–	0.9847	0.9903

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
12834	Erupted upper front-left permanent denti	1.41	1.1469×10^{-16}	15.57	–	9.3450×10^{-17}	1.0749×10^{-16}	–	1.2273	1.0670
12835	Erupted upper front-right permanent dent	1.41	1.1307×10^{-16}	22.46	–	1.6686×10^{-16}	1.5030×10^{-16}	–	0.6776	0.7523
12836	Erupted upper left permanent dentin	2.62	1.2041×10^{-16}	13.34	–	1.4550×10^{-16}	1.7472×10^{-16}	–	0.8276	0.6892
12838	Erupted upper right permanent dentin	2.63	1.3591×10^{-16}	8.46	–	1.2825×10^{-16}	1.5531×10^{-16}	–	1.0597	0.8751
12840	Erupted lower front permanent dentin	2.93	1.6073×10^{-16}	8.60	–	1.4821×10^{-16}	1.8385×10^{-16}	–	1.0845	0.8742
12842	Erupted lower front-left permanent denti	1.21	1.8206×10^{-16}	25.29	–	1.4009×10^{-16}	1.5955×10^{-16}	–	1.2996	1.1411
12843	Erupted lower front-right permanent dent	1.21	1.8593×10^{-16}	14.28	–	1.4668×10^{-16}	2.0098×10^{-16}	–	1.2676	0.9251
12844	Erupted lower left permanent dentin	2.69	1.4905×10^{-16}	12.59	–	1.4763×10^{-16}	1.6003×10^{-16}	–	1.0096	0.9314
12846	Erupted lower right permanent dentin	2.68	1.9770×10^{-16}	9.88	–	1.9505×10^{-16}	2.0117×10^{-16}	–	1.0136	0.9827
12848	Unerupted permanent enamel	0.945	1.4806×10^{-16}	15.24	–	1.3989×10^{-16}	2.3136×10^{-16}	–	1.0584	0.6399
12850	Unerupted permanent dentin	3.18	1.4872×10^{-16}	14.05	–	1.5116×10^{-16}	1.4095×10^{-16}	–	0.9838	1.0551
12852	Permanent cementum	1.14	1.5228×10^{-16}	8.59	–	1.6846×10^{-16}	1.3445×10^{-16}	–	0.9040	1.1327
12854	Permanent pulp	0.771	2.2992×10^{-16}	10.01	–	1.5554×10^{-16}	1.6067×10^{-16}	–	1.4782	1.4310
12856	Teeth retention region	0.0412	1.4402×10^{-16}	31.65	–	1.5776×10^{-16}	1.7872×10^{-16}	–	0.9129	0.8058
13100	Thymus	31.05	6.1452×10^{-16}	1.98	–	6.2227×10^{-16}	6.2584×10^{-16}	–	0.9875	0.9819
13200	Thyroid	13.40	4.4631×10^{-16}	2.60	–	4.0275×10^{-16}	4.6967×10^{-16}	–	1.1082*	0.9503*
13300	Tongue upper(food)	18.08	1.6123×10^{-16}	4.45	–	1.6052×10^{-16}	1.7085×10^{-16}	–	1.0044	0.9437
13301	Tongue lower	36.65	1.9769×10^{-16}	4.15	–	2.0616×10^{-16}	2.0292×10^{-16}	–	0.9589	0.9742
13400	Tonsils	3.11	1.5043×10^{-16}	7.23	–	1.6658×10^{-16}	1.6005×10^{-16}	–	0.9031	0.9399
13500	Ureter left	6.21	6.8446×10^{-16}	3.80	–	7.2635×10^{-16}	6.7162×10^{-16}	–	0.9423	1.0191
13600	Ureter right	6.21	1.0093×10^{-15}	4.05	–	1.0627×10^{-15}	1.0112×10^{-15}	–	0.9497	0.9981
13700	Urinary bladder wall insensitive	33.93	1.8619×10^{-16}	2.60	–	1.8892×10^{-16}	1.8851×10^{-16}	–	0.9855	0.9877
13701	Urinary bladder wall sensitive(111-227)	1.66	1.9196×10^{-16}	5.09	–	2.0170×10^{-16}	1.8565×10^{-16}	–	0.9517	1.0340
13800	Urinary bladder content	140.00	1.9227×10^{-16}	1.31	–	1.8473×10^{-16}	1.8713×10^{-16}	–	1.0408	1.0274
13900	Uterus	31.06	2.2741×10^{-16}	3.24	–	2.2938×10^{-16}	2.2844×10^{-16}	–	0.9914	0.9955
14000	Air inside body	0.127	2.4415×10^{-15}	3.45	–	2.8129×10^{-15}	2.3194×10^{-15}	–	0.8680	1.0526

* For tissues bookkept as head, trunk, arm and leg pieces (muscle, residual soft tissue, skin, large vessels), the volumes tabulated in the distributed cell tables partition the tissue at different boundaries from the transported mesh, and the reference output is normalised by the tabulated volume. The marked ratio therefore compares two different partitions of the same tissue, and is left out of the summary statistics of Table S2.

Table S26. 15F_External. Absorbed dose per primary history, Gy. σ is the FLUKA statistical uncertainty. Ratios are FLUKA / reference.

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
100	Adrenal left	5.24	1.9140×10^{-17}	24.29	–	2.5198×10^{-17}	1.3417×10^{-17}	–	0.7596	1.4266
200	Adrenal right	5.24	1.3761×10^{-17}	25.36	–	1.8483×10^{-17}	2.1245×10^{-17}	–	0.7445	0.6477
300	ET1(0-8)	0.00194	0.0000×10^{0}	0.00	–	–	–	–	–	–
301	ET1(8-40)	0.0079	0.0000×10^{0}	0.00	–	–	–	–	–	–
302	ET1(40-50)	0.00251	0.0000×10^{0}	0.00	–	–	–	–	–	–
303	ET1(50-Surface)	0.729	1.3345×10^{-17}	54.53	–	2.8255×10^{-17}	2.7218×10^{-17}	–	0.4723	0.4903
400	ET2(-15-0)	0.15	5.6777×10^{-17}	40.43	–	3.6124×10^{-17}	1.6903×10^{-17}	–	1.5717*	3.3590*
401	ET2(0-40)	0.426	4.6070×10^{-17}	38.28	–	1.3117×10^{-17}	1.7285×10^{-17}	–	3.5124*	2.6653*
402	ET2(40-50)	0.107	6.1735×10^{-17}	29.09	–	3.3559×10^{-17}	1.1829×10^{-17}	–	1.8396*	5.2189*
403	ET2(50-55)	0.0537	3.3212×10^{-17}	21.88	–	3.5040×10^{-17}	1.3851×10^{-17}	–	0.9479	2.3978
404	ET2(55-65)	0.108	3.3731×10^{-17}	37.66	–	3.9958×10^{-17}	3.1306×10^{-17}	–	0.8442*	1.0775*
405	ET2(65-Surface)	24.77	2.9447×10^{-17}	11.50	–	3.2518×10^{-17}	2.3415×10^{-17}	–	0.9055	1.2576
500	Oral mucosa tongue	0.071	8.5158×10^{-18}	28.17	–	3.1800×10^{-17}	1.6826×10^{-17}	–	0.2678	0.5061
501	Oral mucosa moth floor	0.0292	1.3647×10^{-17}	70.62	–	2.7693×10^{-17}	3.8514×10^{-17}	–	0.4928	0.3543
600	Oral mucosa lips and cheeks	0.0178	3.1148×10^{-17}	47.00	–	2.1180×10^{-18}	5.2741×10^{-17}	–	14.7065	0.5906
700	Trachea	6.20	2.9555×10^{-17}	16.90	–	2.4508×10^{-17}	2.0592×10^{-17}	–	1.2059	1.4353
800	BB(-11–6)	0.00858	1.0395×10^{-17}	68.23	–	3.4675×10^{-17}	1.3072×10^{-17}	–	0.2998	0.7952
801	BB(-6-0)	0.0109	1.3072×10^{-17}	66.68	–	9.1351×10^{-17}	1.3087×10^{-17}	–	0.1431	0.9988
802	BB(0-10)	0.0183	2.2449×10^{-17}	74.06	–	9.7925×10^{-18}	1.8769×10^{-17}	–	2.2924	1.1960
803	BB(10-35)	0.0458	4.0178×10^{-17}	83.74	–	1.6560×10^{-17}	1.4054×10^{-17}	–	2.4262	2.8588
804	BB(35-40)	0.0092	3.1707×10^{-17}	69.04	–	7.3667×10^{-18}	1.4872×10^{-17}	–	4.3040	2.1320
805	BB(40-50)	0.0184	2.9271×10^{-17}	83.39	–	4.8923×10^{-18}	1.6693×10^{-17}	–	5.9832	1.7535
806	BB(50-60)	0.0185	3.7103×10^{-17}	85.30	–	6.0102×10^{-18}	1.5898×10^{-17}	–	6.1733	2.3338
807	BB(60-70)	0.0185	8.3469×10^{-17}	82.28	–	3.5031×10^{-17}	1.3673×10^{-17}	–	2.3827	6.1047
808	BB(70-surface)	6.64	1.9162×10^{-17}	18.25	–	2.3786×10^{-17}	1.5040×10^{-17}	–	0.8056	1.2740
900	Blood in large arteries head	0.718	7.1179×10^{-18}	66.56	–	2.1852×10^{-17}	2.4740×10^{-17}	–	0.3257*	0.2877*
910	Blood in large veins head	3.30	2.7366×10^{-17}	25.83	–	2.3371×10^{-17}	2.1484×10^{-17}	–	1.1709*	1.2738*
1000	Blood in large arteries trunk	85.09	2.6290×10^{-17}	5.23	–	2.6617×10^{-17}	2.4295×10^{-17}	–	0.9877*	1.0821*
1010	Blood in large veins trunk	171.13	3.0966×10^{-17}	2.62	–	3.2622×10^{-17}	3.0379×10^{-17}	–	0.9492*	1.0193*
1100	Blood in large arteries arms	25.60	3.2482×10^{-17}	7.88	–	3.0097×10^{-17}	3.2961×10^{-17}	–	1.0793*	0.9855*
1110	Blood in large veins arms	95.90	3.4350×10^{-17}	4.24	–	3.3559×10^{-17}	3.1583×10^{-17}	–	1.0236*	1.0876*
1200	Blood in large arteries legs	98.57	2.6264×10^{-17}	3.01	–	2.8728×10^{-17}	2.4993×10^{-17}	–	0.9142	1.0508
1210	Blood in large veins legs	359.61	2.7937×10^{-17}	2.45	–	2.7963×10^{-17}	2.8687×10^{-17}	–	0.9991	0.9739
1300	Humeri upper cortical	124.84	2.7289×10^{-17}	5.32	–	2.4825×10^{-17}	2.6468×10^{-17}	–	1.0992	1.0310
1400	Humeri upper spogiosa	133.67	2.3713×10^{-17}	4.53	–	2.4477×10^{-17}	2.3929×10^{-17}	–	0.9688	0.9910

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
1500	Humeri upper medullary cavity	15.04	2.3892×10^{-17}	10.02	–	2.9490×10^{-17}	2.6981×10^{-17}	–	0.8102	0.8855
1600	Humeri lower cortical	133.91	2.9418×10^{-17}	4.44	–	2.9291×10^{-17}	2.8303×10^{-17}	–	1.0043	1.0394
1700	Humeri lower spongiosa	55.19	3.3166×10^{-17}	10.74	–	2.9040×10^{-17}	3.2971×10^{-17}	–	1.1421	1.0059
1800	Humeri lower medullary cavity	12.75	2.3069×10^{-17}	21.14	–	3.3677×10^{-17}	4.1750×10^{-17}	–	0.6850	0.5526
1900	Radii cortical	84.25	3.7808×10^{-17}	4.42	–	3.8257×10^{-17}	3.5337×10^{-17}	–	0.9883	1.0699
1910	Ulnae cortical	105.37	2.7948×10^{-17}	5.34	–	3.0357×10^{-17}	2.9064×10^{-17}	–	0.9207	0.9616
2000	Radii spongiosa	28.35	3.8835×10^{-17}	11.75	–	3.1196×10^{-17}	4.6257×10^{-17}	–	1.2449	0.8395
2010	Ulnae spongiosa	41.42	3.1983×10^{-17}	5.74	–	2.8602×10^{-17}	3.1138×10^{-17}	–	1.1182	1.0271
2100	Radii medullary cavity	5.64	3.6955×10^{-17}	20.86	–	3.4547×10^{-17}	3.2976×10^{-17}	–	1.0697	1.1207
2110	Ulnae medullary cavity	6.99	4.2238×10^{-17}	14.39	–	3.1518×10^{-17}	3.6479×10^{-17}	–	1.3401	1.1579
2200	Wrists and hand bones cortical	108.28	3.2355×10^{-17}	6.24	–	3.4845×10^{-17}	3.1562×10^{-17}	–	0.9285	1.0251
2300	Wrists and hand bones spongiosa	48.59	3.3660×10^{-17}	3.77	–	3.3640×10^{-17}	3.3741×10^{-17}	–	1.0006	0.9976
2400	Clavicles cortical	56.94	2.1635×10^{-17}	4.78	–	2.2691×10^{-17}	2.0128×10^{-17}	–	0.9534	1.0749
2500	Clavicles spongiosa	26.60	2.3327×10^{-17}	10.18	–	2.2985×10^{-17}	2.6255×10^{-17}	–	1.0149	0.8885
2600	Cranium cortical	450.60	1.7751×10^{-17}	3.29	–	1.7437×10^{-17}	1.7107×10^{-17}	–	1.0180	1.0377
2700	Cranium spongiosa	391.55	1.8160×10^{-17}	2.40	–	1.8589×10^{-17}	1.8639×10^{-17}	–	0.9769	0.9743
2800	Femora upper cortical	236.16	2.9369×10^{-17}	2.64	–	3.1196×10^{-17}	2.8849×10^{-17}	–	0.9415	1.0180
2900	Femora upper spongiosa	186.87	3.3589×10^{-17}	2.12	–	3.3933×10^{-17}	3.1360×10^{-17}	–	0.9899	1.0711
3000	Femora upper medullary cavity	34.93	2.8231×10^{-17}	8.46	–	3.2738×10^{-17}	3.0557×10^{-17}	–	0.8623	0.9239
3100	Femora lower cortical	255.04	2.7417×10^{-17}	3.77	–	2.6887×10^{-17}	2.8321×10^{-17}	–	1.0197	0.9681
3200	Femora lower spongiosa	271.34	2.8367×10^{-17}	3.06	–	2.6880×10^{-17}	2.7477×10^{-17}	–	1.0553	1.0324
3300	Femora lower medullary cavity	38.34	2.9461×10^{-17}	7.92	–	3.3572×10^{-17}	3.1431×10^{-17}	–	0.8775	0.9373
3400	Tibiae cortical	325.55	2.6866×10^{-17}	2.25	–	2.5349×10^{-17}	2.6341×10^{-17}	–	1.0598	1.0199
3410	Fibulae cortical	54.33	1.9836×10^{-17}	11.07	–	2.4564×10^{-17}	2.3664×10^{-17}	–	0.8075	0.8382
3420	Patellae cortical	17.54	3.8258×10^{-17}	5.54	–	2.9372×10^{-17}	3.0240×10^{-17}	–	1.3025	1.2651
3500	Tibiae spongiosa	227.06	2.6431×10^{-17}	4.41	–	2.6755×10^{-17}	2.6888×10^{-17}	–	0.9879	0.9830
3510	Fibulae spongiosa	24.54	2.4231×10^{-17}	9.92	–	2.2193×10^{-17}	1.9485×10^{-17}	–	1.0918	1.2436
3520	Patellae spongiosa	17.83	3.8559×10^{-17}	14.49	–	3.5825×10^{-17}	3.8791×10^{-17}	–	1.0763	0.9940
3600	Tibiae medullary cavity	48.77	2.7523×10^{-17}	8.58	–	2.8606×10^{-17}	2.8255×10^{-17}	–	0.9621	0.9741
3610	Fibulae medullary cavity	5.81	3.0589×10^{-17}	19.30	–	2.3484×10^{-17}	2.3155×10^{-17}	–	1.3026	1.3211
3700	Ankles and foot cortical	176.95	2.2209×10^{-17}	3.35	–	2.2000×10^{-17}	2.0368×10^{-17}	–	1.0095	1.0904
3800	Ankles and foot spongiosa	306.52	2.4987×10^{-17}	3.23	–	2.1219×10^{-17}	2.0750×10^{-17}	–	1.1776	1.2042
3900	Mandible cortical	47.27	2.6697×10^{-17}	5.91	–	2.4418×10^{-17}	2.6225×10^{-17}	–	1.0933	1.0180
4000	Mandible spongiosa	28.85	3.2639×10^{-17}	12.05	–	2.6839×10^{-17}	3.0637×10^{-17}	–	1.2161	1.0653
4100	Pelvis cortical	215.14	3.0028×10^{-17}	4.20	–	2.8370×10^{-17}	3.0655×10^{-17}	–	1.0584	0.9796
4200	Pelvis spongiosa	553.25	3.1864×10^{-17}	2.25	–	3.0625×10^{-17}	3.1676×10^{-17}	–	1.0405	1.0059
4300	Ribs cortical	186.41	2.3403×10^{-17}	3.42	–	2.3653×10^{-17}	2.3719×10^{-17}	–	0.9895	0.9867
4400	Ribs spongiosa	208.05	2.4512×10^{-17}	2.70	–	2.3815×10^{-17}	2.4729×10^{-17}	–	1.0293	0.9912

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
4500	Scapulae cortical	172.60	1.5690×10^{-17}	3.70	–	1.4451×10^{-17}	1.5696×10^{-17}	–	1.0858	0.9996
4600	Scapulae spongiosa	147.47	1.6430×10^{-17}	8.30	–	1.4966×10^{-17}	1.5305×10^{-17}	–	1.0978	1.0735
4700	Cervical spine cortical	42.94	1.7423×10^{-17}	9.34	–	1.9810×10^{-17}	2.1216×10^{-17}	–	0.8795	0.8212
4800	Cervical spine spongiosa	68.33	2.4044×10^{-17}	4.33	–	2.3757×10^{-17}	2.2703×10^{-17}	–	1.0121	1.0591
4900	Thoracic spine cortical	84.06	1.4322×10^{-17}	6.10	–	1.3737×10^{-17}	1.8529×10^{-17}	–	1.0425	0.7729
5000	Thoracic spine spongiosa	230.52	1.4751×10^{-17}	4.38	–	1.5577×10^{-17}	1.7644×10^{-17}	–	0.9470	0.8360
5100	Lumbar spine cortical	41.25	2.2254×10^{-17}	8.30	–	2.3639×10^{-17}	2.2208×10^{-17}	–	0.9414	1.0021
5200	Lumbar spine spongiosa	270.61	2.2426×10^{-17}	3.41	–	2.2159×10^{-17}	2.2001×10^{-17}	–	1.0121	1.0193
5300	Sacrum cortical	86.54	2.1747×10^{-17}	5.61	–	2.3930×10^{-17}	2.3230×10^{-17}	–	0.9088	0.9362
5400	Sacrum spongiosa	140.21	2.2033×10^{-17}	4.95	–	2.3430×10^{-17}	2.4672×10^{-17}	–	0.9404	0.8931
5500	Sternum cortical	15.67	3.5799×10^{-17}	7.72	–	4.2185×10^{-17}	2.9205×10^{-17}	–	0.8486	1.2258
5600	Sternum spongiosa	28.43	3.6758×10^{-17}	5.26	–	4.2025×10^{-17}	3.2964×10^{-17}	–	0.8747	1.1151
5800	Cartilage trunk	161.94	3.1150×10^{-17}	2.51	–	3.1270×10^{-17}	3.0901×10^{-17}	–	0.9962	1.0080
6100	Brain	1348.99	1.6032×10^{-17}	1.57	–	1.7321×10^{-17}	1.6458×10^{-17}	–	0.9256	0.9741
6200	Breast left adipose tissue	77.62	4.5260×10^{-17}	2.05	–	4.1593×10^{-17}	3.8668×10^{-17}	–	1.0882	1.1705
6300	Breast left glandular tissue	51.75	3.8004×10^{-17}	7.43	–	3.5835×10^{-17}	4.0758×10^{-17}	–	1.0605	0.9324
6400	Breast right adipose tissue	77.62	3.9630×10^{-17}	5.97	–	3.8370×10^{-17}	3.7946×10^{-17}	–	1.0328	1.0444
6500	Breast right glandular tissue	51.75	3.8241×10^{-17}	5.37	–	4.1158×10^{-17}	4.1166×10^{-17}	–	0.9291	0.9290
6600	Eye lens sensitive left	0.0203	1.6073×10^{-17}	100.00	–	–	–	–	–	–
6601	Eye lens insensitive left	0.117	8.2113×10^{-17}	100.00	–	2.0583×10^{-17}	6.7533×10^{-17}	–	3.9894	1.2159
6700	Cornea left	0.996	2.2580×10^{-17}	37.75	–	2.9122×10^{-17}	2.1786×10^{-17}	–	0.7754	1.0364
6701	Aqueous left	0.344	9.2749×10^{-18}	90.20	–	1.4968×10^{-17}	–	–	0.6197	–
6702	Vitreous left	5.32	2.7980×10^{-17}	18.70	–	3.3015×10^{-17}	2.6892×10^{-17}	–	0.8475	1.0405
6800	Eye lens sensitive right	0.0203	2.8495×10^{-17}	100.00	–	–	–	–	–	–
6801	Eye lens insensitive right	0.117	1.1584×10^{-16}	81.52	–	–	1.2521×10^{-16}	–	–	0.9252
6900	Cornea right	0.996	1.5422×10^{-17}	49.13	–	3.9407×10^{-17}	3.0562×10^{-17}	–	0.3914	0.5046
6901	Aqueous right	0.344	2.9995×10^{-17}	93.78	–	4.4992×10^{-17}	1.6380×10^{-17}	–	0.6667	1.8312
6902	Vitreous right	5.32	1.7491×10^{-17}	21.15	–	3.2450×10^{-17}	2.8506×10^{-17}	–	0.5390	0.6136
7000	Gall bladder wall	7.56	3.6891×10^{-17}	17.12	–	2.8443×10^{-17}	3.1361×10^{-17}	–	1.2970	1.1763
7100	Gall bladder contents	42.00	2.9880×10^{-17}	6.35	–	2.5620×10^{-17}	2.8282×10^{-17}	–	1.1663	1.0565
7200	Stomach wall(0-60)	1.60	3.9117×10^{-17}	20.79	–	3.1998×10^{-17}	3.1111×10^{-17}	–	1.2225	1.2573
7201	Stomach wall(60-100)	1.07	3.5845×10^{-17}	19.17	–	2.8821×10^{-17}	3.2504×10^{-17}	–	1.2437	1.1028
7202	Stomach wall(100-300)	5.39	2.9223×10^{-17}	11.76	–	3.3759×10^{-17}	3.2989×10^{-17}	–	0.8656	0.8858
7203	Stomach wall(300-surface)	136.31	3.1512×10^{-17}	5.62	–	3.1853×10^{-17}	3.1733×10^{-17}	–	0.9893	0.9930
7300	Stomach contents	200.00	3.5532×10^{-17}	4.31	–	3.1053×10^{-17}	3.3694×10^{-17}	–	1.1442	1.0546
7400	Small intestine wall(0-130)	12.46	3.2206×10^{-17}	6.32	–	3.3901×10^{-17}	4.0522×10^{-17}	–	0.9500	0.7948
7401	Small intestine wall(130-150)	1.94	4.1614×10^{-17}	11.67	–	3.5281×10^{-17}	3.5234×10^{-17}	–	1.1795	1.1811
7402	Small intestine wall(150-200)	4.88	3.3719×10^{-17}	7.25	–	3.3833×10^{-17}	3.6060×10^{-17}	–	0.9966	0.9351

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
7403	Small intestine wall(200-surface)	616.55	3.5255×10^{-17}	2.38	–	3.5164×10^{-17}	3.5064×10^{-17}	–	1.0026	1.0054
7500	Small intestine contents(-500-0)	45.19	3.4318×10^{-17}	3.98	–	3.1671×10^{-17}	3.7140×10^{-17}	–	1.0836	0.9240
7501	Small intestine contents(centre–500)	234.81	3.3047×10^{-17}	3.21	–	3.4476×10^{-17}	3.3790×10^{-17}	–	0.9585	0.9780
7600	Ascending colon wall(0-280)	3.66	3.2172×10^{-17}	18.92	–	3.0064×10^{-17}	3.2106×10^{-17}	–	1.0701	1.0020
7601	Ascending colon wall(280-300)	0.265	3.7358×10^{-17}	32.38	–	4.0550×10^{-17}	2.2427×10^{-17}	–	0.9213	1.6658
7602	Ascending colon wall(300-surface)	68.67	3.7028×10^{-17}	3.67	–	3.8769×10^{-17}	3.8437×10^{-17}	–	0.9551	0.9633
7700	Ascending colon content	77.81	3.5141×10^{-17}	5.21	–	3.6183×10^{-17}	3.7215×10^{-17}	–	0.9712	0.9443
7800	Transverse colon wall right(0-280)	2.67	3.3495×10^{-17}	15.81	–	3.2989×10^{-17}	3.4657×10^{-17}	–	1.0154	0.9665
7801	Transverse colon wall right(280-300)	0.194	3.2351×10^{-17}	37.78	–	5.0063×10^{-17}	5.3745×10^{-17}	–	0.6462	0.6019
7802	Transverse colon wall right(300-surface)	72.88	3.7982×10^{-17}	5.43	–	4.1608×10^{-17}	3.9504×10^{-17}	–	0.9128	0.9615
7900	Transverse colon contents right	42.19	4.5842×10^{-17}	8.00	–	4.1095×10^{-17}	4.0515×10^{-17}	–	1.1155	1.1315
8000	Transverse colon wall left(0-280)	2.15	2.4295×10^{-17}	15.08	–	4.2830×10^{-17}	3.3784×10^{-17}	–	0.5672	0.7191
8001	Transverse colon wall left(280-300)	0.157	4.1339×10^{-17}	52.11	–	4.4484×10^{-17}	4.1208×10^{-17}	–	0.9293	1.0032
8002	Transverse colon wall left(300-surface)	67.63	3.9034×10^{-17}	7.45	–	4.1654×10^{-17}	4.1322×10^{-17}	–	0.9371	0.9446
8100	Transverse colon content left	27.01	4.6770×10^{-17}	5.66	–	3.6638×10^{-17}	3.6350×10^{-17}	–	1.2765	1.2867
8200	Descending colon wall(0-280)	2.69	3.8168×10^{-17}	8.86	–	2.7088×10^{-17}	3.0902×10^{-17}	–	1.4090	1.2351
8201	Descending colon wall(280-300)	0.196	4.6006×10^{-17}	27.48	–	3.6751×10^{-17}	3.2340×10^{-17}	–	1.2519	1.4226
8202	Descending colon wall(300-surface)	75.44	3.4106×10^{-17}	2.01	–	3.7538×10^{-17}	3.4241×10^{-17}	–	0.9086	0.9961
8300	Descending colon content	32.99	3.5414×10^{-17}	5.70	–	3.1578×10^{-17}	3.0575×10^{-17}	–	1.1215	1.1583
8400	Sigmoid colon wall(0-280)	3.22	3.2786×10^{-17}	21.98	–	2.9480×10^{-17}	2.2089×10^{-17}	–	1.1122	1.4843
8401	Sigmoid colon wall(280-300)	0.235	1.4263×10^{-17}	32.65	–	2.6717×10^{-17}	2.4613×10^{-17}	–	0.5339	0.5795
8402	Sigmoid colon wall(300-surface)	55.58	2.5952×10^{-17}	8.86	–	2.4883×10^{-17}	2.6215×10^{-17}	–	1.0430	0.9900
8500	Sigmoid colon contents	41.77	2.6402×10^{-17}	4.71	–	2.7757×10^{-17}	2.4260×10^{-17}	–	0.9512	1.0883
8600	Rectum wall(0-280)	1.44	3.8229×10^{-17}	20.55	–	3.4121×10^{-17}	1.6798×10^{-17}	–	1.1204	2.2758
8601	Rectum wall(280-300)	0.105	4.4897×10^{-17}	40.95	–	1.0561×10^{-17}	3.3833×10^{-17}	–	4.2514	1.3270
8602	Rectum wall(300-surface)	7.53	3.6362×10^{-17}	11.70	–	1.6832×10^{-17}	2.2772×10^{-17}	–	2.1602	1.5968
8603	Rectum contents	18.23	3.7083×10^{-17}	6.94	–	3.5354×10^{-17}	2.7837×10^{-17}	–	1.0489	1.3321
8700	Heart wall	251.49	2.6697×10^{-17}	3.20	–	2.9106×10^{-17}	2.9075×10^{-17}	–	0.9172	0.9182
8800	Blood in heart chamber	320.00	2.8471×10^{-17}	2.90	–	2.7853×10^{-17}	2.7427×10^{-17}	–	1.0222	1.0381
8900	Kidney left cortex	105.69	1.9420×10^{-17}	5.03	–	2.0715×10^{-17}	2.0407×10^{-17}	–	0.9375	0.9516
9000	Kidney left medulla	37.75	2.2475×10^{-17}	8.82	–	1.9560×10^{-17}	2.1820×10^{-17}	–	1.1490	1.0300
9100	Kidney left pelvis	7.55	1.6938×10^{-17}	17.61	–	1.8438×10^{-17}	2.0283×10^{-17}	–	0.9186	0.8351
9200	Kidney right cortex	105.69	2.0606×10^{-17}	8.14	–	1.8290×10^{-17}	1.9674×10^{-17}	–	1.1266	1.0474
9300	Kidney right medulla	37.75	1.8441×10^{-17}	10.26	–	1.9469×10^{-17}	1.7544×10^{-17}	–	0.9472	1.0511
9400	Kidney right pelvis	7.55	2.4741×10^{-17}	14.98	–	2.1656×10^{-17}	2.3939×10^{-17}	–	1.1425	1.0335
9500	Liver	1628.97	3.1501×10^{-17}	1.88	–	3.1560×10^{-17}	3.1355×10^{-17}	–	0.9981	1.0047

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
9700	Lung(AI) left	330.58	2.3350×10^{-17}	3.04	–	2.2336×10^{-17}	2.3029×10^{-17}	–	1.0454	1.0139
9900	Lung(AI) right	419.42	2.6322×10^{-17}	2.59	–	2.4518×10^{-17}	2.6070×10^{-17}	–	1.0736	1.0097
10000	Lymphatic nodes ET	11.52	2.8586×10^{-17}	15.61	–	2.6053×10^{-17}	2.0215×10^{-17}	–	1.0972	1.4141
10100	Lymphatic nodes thoracic	11.52	2.4479×10^{-17}	15.00	–	2.3455×10^{-17}	2.2387×10^{-17}	–	1.0436	1.0934
10200	Lymphatic nodes head	4.00	2.6921×10^{-17}	18.89	–	1.6956×10^{-17}	3.1669×10^{-17}	–	1.5878*	0.8501*
10300	Lymphatic nodes trunk	94.06	3.2573×10^{-17}	3.50	–	3.2950×10^{-17}	3.3160×10^{-17}	–	0.9886	0.9823
10400	Lymphatic nodes arms	7.96	2.1907×10^{-17}	21.61	–	2.6829×10^{-17}	3.3776×10^{-17}	–	0.8166	0.6486
10500	Lymphatic nodes legs	7.96	2.5022×10^{-17}	17.52	–	3.6123×10^{-17}	2.7835×10^{-17}	–	0.6927	0.8989
10600	Muscle head	416.65	2.5150×10^{-17}	3.49	–	2.5090×10^{-17}	2.5344×10^{-17}	–	1.0024*	0.9923*
10700	Muscle trunk	7978.95	2.7201×10^{-17}	0.58	–	2.7220×10^{-17}	2.6954×10^{-17}	–	0.9993*	1.0091*
10800	Muscle arms	1519.22	3.2481×10^{-17}	1.01	–	3.1750×10^{-17}	3.1699×10^{-17}	–	1.0230*	1.0247*
10900	Muscle legs	7446.05	2.7523×10^{-17}	0.51	–	2.6763×10^{-17}	2.7133×10^{-17}	–	1.0284	1.0144
11000	Oesophagus wall(0-190)	1.78	1.8781×10^{-17}	29.45	–	2.5926×10^{-17}	2.3319×10^{-17}	–	0.7244	0.8054
11001	Oesophagus wall(190-200)	0.0957	9.3898×10^{-18}	37.72	–	3.1408×10^{-17}	1.2104×10^{-17}	–	0.2990	0.7758
11002	Oesophagus wall(200-surface)	34.20	2.1238×10^{-17}	5.07	–	2.1336×10^{-17}	1.8502×10^{-17}	–	0.9954	1.1479
11003	Oesophagus contents	21.03	1.6297×10^{-17}	16.44	–	2.2103×10^{-17}	1.7944×10^{-17}	–	0.7373	0.9082
11100	Ovary left	3.19	2.5224×10^{-17}	25.46	–	4.0411×10^{-17}	3.0960×10^{-17}	–	0.6242	0.8147
11200	Ovary right	3.19	3.5395×10^{-17}	29.65	–	4.2508×10^{-17}	2.6601×10^{-17}	–	0.8327	1.3306
11300	Pancreas	117.63	2.4974×10^{-17}	6.18	–	2.5658×10^{-17}	2.6113×10^{-17}	–	0.9734	0.9564
11400	Pituitary gland	0.518	1.1517×10^{-17}	100.00	–	–	9.2820×10^{-18}	–	–	1.2408
11600	RST head	746.19	2.3832×10^{-17}	1.41	–	2.3634×10^{-17}	2.3449×10^{-17}	–	1.0084	1.0163
11700	RST trunk	10421.35	2.8558×10^{-17}	0.34	–	2.8590×10^{-17}	2.9555×10^{-17}	–	0.9989*	0.9662*
11800	RST arms	1774.29	3.3912×10^{-17}	1.43	–	3.2661×10^{-17}	2.9641×10^{-17}	–	1.0383*	1.1441*
11900	RST legs	5479.97	2.9877×10^{-17}	0.62	–	2.9785×10^{-17}	2.9568×10^{-17}	–	1.0031	1.0104
12000	Salivary glands left	33.65	2.1289×10^{-17}	9.82	–	2.5310×10^{-17}	2.1103×10^{-17}	–	0.8411	1.0088
12100	Salivary glandss right	33.65	2.0831×10^{-17}	8.34	–	2.9020×10^{-17}	2.6022×10^{-17}	–	0.7178	0.8005
12200	Skin head insensitive	130.08	1.9045×10^{-17}	4.58	–	1.7510×10^{-17}	1.7366×10^{-17}	–	1.0877	1.0967
12201	Skin head sensitive(50-100)	6.25	8.4924×10^{-18}	12.06	–	1.6869×10^{-17}	1.1584×10^{-17}	–	0.5034	0.7331
12300	Skin trunk insensitive	559.14	2.1682×10^{-17}	1.35	–	2.1475×10^{-17}	2.3209×10^{-17}	–	1.0096*	0.9342*
12301	Skin trunk sensitive(50-100)	26.45	1.3144×10^{-17}	6.35	–	1.3819×10^{-17}	1.4871×10^{-17}	–	0.9511*	0.8838*
12400	Skin arms insensitive	314.14	2.9105×10^{-17}	2.47	–	2.8827×10^{-17}	2.6523×10^{-17}	–	1.0097*	1.0974*
12401	Skin arms sensitive(50-100)	15.16	2.2128×10^{-17}	6.13	–	2.2109×10^{-17}	1.7735×10^{-17}	–	1.0009*	1.2477*
12500	Skin legs insensitive	691.14	2.2364×10^{-17}	1.37	–	2.2613×10^{-17}	2.2494×10^{-17}	–	0.9890*	0.9942*
12501	Skin legs sensitive(50-100)	33.06	1.4006×10^{-17}	5.28	–	1.6224×10^{-17}	1.5424×10^{-17}	–	0.8633	0.9080
12600	Spinal cord	54.22	1.9586×10^{-17}	7.53	–	1.4511×10^{-17}	1.9933×10^{-17}	–	1.3497	0.9826
12700	Spleen	179.32	2.5711×10^{-17}	4.87	–	2.5420×10^{-17}	2.4117×10^{-17}	–	1.0114	1.0661
12800	Erupted upper front buccal permanent ena	0.564	3.4615×10^{-17}	60.84	–	1.9609×10^{-17}	7.4722×10^{-18}	–	1.7653	4.6325

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
12801	Erupted upper front lingual permanent en	0.58	4.7781×10^{-18}	98.76	–	3.1966×10^{-17}	2.1824×10^{-17}	–	0.1495	0.2189
12804	Erupted upper front-left buccal permanen	0.219	3.9869×10^{-17}	58.13	–	–	2.5759×10^{-17}	–	–	1.5478
12805	Erupted upper front-left lingual permane	0.201	2.9465×10^{-17}	100.00	–	8.3771×10^{-18}	6.6622×10^{-19}	–	3.5173	44.2269
12806	Erupted upper front-right buccal permane	0.201	5.2593×10^{-17}	73.11	–	7.2795×10^{-17}	1.4570×10^{-19}	–	0.7225	360.9696
12807	Erupted upper front-right lingual perman	0.219	7.9011×10^{-18}	100.00	–	–	4.0180×10^{-17}	–	–	0.1966
12808	Erupted upper left buccal permanent enam	0.426	4.4960×10^{-17}	63.68	–	–	2.9399×10^{-17}	–	–	1.5293
12809	Erupted upper left lingual permanent ena	0.356	1.0015×10^{-17}	100.00	–	9.5953×10^{-18}	5.6878×10^{-18}	–	1.0437	1.7608
12812	Erupted upper right buccal permanent ena	0.381	2.5086×10^{-17}	91.32	–	9.8230×10^{-18}	7.7196×10^{-17}	–	2.5539	0.3250
12813	Erupted upper right lingual permanent en	0.4	2.2270×10^{-17}	68.27	–	5.7065×10^{-17}	1.7826×10^{-17}	–	0.3903	1.2493
12816	Erupted lower front buccal permanent ena	0.444	2.9918×10^{-17}	42.38	–	4.6314×10^{-17}	7.1735×10^{-17}	–	0.6460	0.4171
12817	Erupted lower front lingual permanent en	0.445	2.8393×10^{-17}	73.91	–	3.7608×10^{-17}	3.0071×10^{-17}	–	0.7550	0.9442
12820	Erupted lower front-left buccal permanen	0.181	6.8745×10^{-17}	94.52	–	8.2931×10^{-18}	3.2769×10^{-17}	–	8.2894	2.0979
12821	Erupted lower front-left lingual permane	0.182	0.0000×10^{0}	0.00	–	1.4977×10^{-17}	2.9392×10^{-18}	–	–	–
12822	Erupted lower front-right buccal permane	0.2	0.0000×10^{0}	0.00	–	8.4482×10^{-18}	1.0428×10^{-16}	–	–	–
12823	Erupted lower front-right lingual perman	0.163	9.5684×10^{-19}	100.00	–	2.1283×10^{-17}	4.7579×10^{-17}	–	0.0450	0.0201
12824	Erupted lower left buccal permanent enam	0.334	4.0368×10^{-17}	74.20	–	3.2370×10^{-17}	3.9270×10^{-17}	–	1.2471	1.0280
12825	Erupted lower left lingual permanent ena	0.466	1.7595×10^{-17}	57.44	–	3.2178×10^{-17}	2.6230×10^{-17}	–	0.5468	0.6708
12828	Erupted lower right buccal permanent ena	0.396	3.9833×10^{-18}	100.00	–	1.1348×10^{-17}	2.2945×10^{-17}	–	0.3510	0.1736
12829	Erupted lower right lingual permanent en	0.404	1.4203×10^{-18}	79.49	–	7.7621×10^{-18}	1.5164×10^{-17}	–	0.1830	0.0937
12832	Erupted upper front permanent dentin	3.83	2.3308×10^{-17}	31.26	–	1.7159×10^{-17}	4.2149×10^{-17}	–	1.3584	0.5530

continued

ID	Organ	Mass (g)	FLUKA	σ (%)	Geant4	MCNP6	PHITS	/G4	/MC	/PH
12834	Erupted upper front-left permanent denti	1.41	3.4452×10^{-17}	33.67	–	3.9978×10^{-18}	3.7105×10^{-17}	–	8.6176	0.9285
12835	Erupted upper front-right permanent dent	1.41	3.1363×10^{-17}	34.04	–	4.0337×10^{-17}	1.7369×10^{-17}	–	0.7775	1.8057
12836	Erupted upper left permanent dentin	2.62	2.1924×10^{-17}	23.18	–	2.9640×10^{-17}	1.9004×10^{-17}	–	0.7397	1.1537
12838	Erupted upper right permanent dentin	2.63	3.7248×10^{-17}	30.04	–	1.5433×10^{-17}	4.5886×10^{-17}	–	2.4136	0.8118
12840	Erupted lower front permanent dentin	2.93	2.8278×10^{-17}	25.98	–	2.9150×10^{-17}	2.4598×10^{-17}	–	0.9701	1.1496
12842	Erupted lower front-left permanent denti	1.21	2.6311×10^{-17}	50.81	–	1.3224×10^{-17}	4.2085×10^{-17}	–	1.9897	0.6252
12843	Erupted lower front-right permanent dent	1.21	3.5700×10^{-17}	44.92	–	4.2015×10^{-17}	2.8411×10^{-17}	–	0.8497	1.2566
12844	Erupted lower left permanent dentin	2.69	3.3829×10^{-17}	24.73	–	1.1168×10^{-17}	3.3052×10^{-17}	–	3.0290	1.0235
12846	Erupted lower right permanent dentin	2.68	1.2722×10^{-17}	38.75	–	2.8206×10^{-17}	2.6023×10^{-17}	–	0.4510	0.4889
12848	Unerupted permanent enamel	0.945	1.6017×10^{-17}	63.67	–	1.1946×10^{-17}	1.9960×10^{-17}	–	1.3408	0.8025
12850	Unerupted permanent dentin	3.18	2.9288×10^{-17}	25.31	–	3.0934×10^{-17}	2.2049×10^{-17}	–	0.9468	1.3283
12852	Permanent cementum	1.14	2.7225×10^{-17}	16.45	–	1.9996×10^{-17}	1.5394×10^{-17}	–	1.3615	1.7686
12854	Permanent pulp	0.771	2.1822×10^{-17}	41.78	–	1.7770×10^{-17}	2.2539×10^{-17}	–	1.2280	0.9682
12856	Teeth retention region	0.0412	1.9968×10^{-17}	29.16	–	2.3246×10^{-17}	2.3549×10^{-17}	–	0.8590	0.8479
13100	Thymus	31.05	2.4515×10^{-17}	10.82	–	2.5007×10^{-17}	2.5611×10^{-17}	–	0.9803	0.9572
13200	Thyroid	13.40	2.6479×10^{-17}	17.19	–	3.5023×10^{-17}	3.3883×10^{-17}	–	0.7560*	0.7815*
13300	Tongue upper(food)	18.08	2.4444×10^{-17}	11.85	–	2.1452×10^{-17}	2.8423×10^{-17}	–	1.1394	0.8600
13301	Tongue lower	36.65	2.7615×10^{-17}	10.34	–	2.6974×10^{-17}	2.7862×10^{-17}	–	1.0238	0.9911
13400	Tonsils	3.11	3.0097×10^{-17}	33.34	–	2.4258×10^{-17}	1.5023×10^{-17}	–	1.2407	2.0034
13500	Ureter left	6.21	3.4269×10^{-17}	20.54	–	3.2538×10^{-17}	2.4999×10^{-17}	–	1.0532	1.3708
13600	Ureter right	6.21	2.7273×10^{-17}	11.56	–	2.5674×10^{-17}	2.4203×10^{-17}	–	1.0623	1.1268
13700	Urinary bladder wall insensitive	33.93	3.4729×10^{-17}	9.74	–	3.7676×10^{-17}	3.1795×10^{-17}	–	0.9218	1.0923
13701	Urinary bladder wall sensitive(111-227)	1.66	3.6131×10^{-17}	16.70	–	3.0633×10^{-17}	2.4410×10^{-17}	–	1.1795	1.4802
13800	Urinary bladder content	140.00	3.8292×10^{-17}	4.10	–	3.5647×10^{-17}	3.5264×10^{-17}	–	1.0742	1.0859
13900	Uterus	31.06	3.3104×10^{-17}	8.11	–	3.2317×10^{-17}	3.0608×10^{-17}	–	1.0244	1.0815
14000	Air inside body	0.127	2.4684×10^{-17}	52.93	–	2.1538×10^{-17}	4.8718×10^{-17}	–	1.1461	0.5067

* For tissues bookkept as head, trunk, arm and leg pieces (muscle, residual soft tissue, skin, large vessels), the volumes tabulated in the distributed cell tables partition the tissue at different boundaries from the transported mesh, and the reference output is normalised by the tabulated volume. The marked ratio therefore compares two different partitions of the same tissue, and is left out of the summary statistics of Table S2.